Dell EMC PowerMax eNAS CLI Reference Guide

For Dell EMC PowerMax and VMAX All Flash

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Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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PREFACE

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Purpose

This reference guide provides man pages for all the eNAS CLI commands.

Audience

This manual provides reference information for command-line users and script programmers that focus on configuring and managing eNAS on VMAX or PowerMax arrays.

Related documentation

The following documents provide additional eNAS information:

- *PowerMax eNAS Release Notes*

  Describes new features and identifies any known functionality restrictions and performance issues that may exist with the current version and your specific storage environment.

- *PowerMax eNAS File Auto Recovery with SRDF/S*

  Describes how to install and use File Auto Recovery to failover/move eNAS Virtual Data Movers from source eNAS systems to destination eNAS systems using SRDF/S.

- *Using SRDF/S with VNX for Disaster Recovery*

  Explains how to configure and manage SRDF/S.

- *Dell EMC VNX Command Line Interface Reference for File*

  Explains the command used to configure and manage an Dell EMC file storage system.

- *Managing Volumes and File Systems on VNX Manually*

  Explains how to create and aggregate different volume types into usable file system storage.

- *Using VNX SnapSure*

  Explains how to use Dell EMC SnapSure to create and manage checkpoints.

- *Configuring Virtual Data Movers on VNX*

  Explains how to configure and manage VDMs on a file storage system.

- *Configuring CIFS on VNX*

  Explains how to configure and manage NFS.

- *Parameters Guide for VNX for File*

  Explains how to view and modify parameters and system settings.

Where to get help

Dell EMC support, product, and licensing information can be obtained as follows:

NOTE:

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  - EMEA: +353 (0) 21 4879862 and follow the voice prompts.

Your comments

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eNAS components

The following terminology is used throughout this document:

- Management Module Control Station (MMCS): Used by Dell EMC Customer Support to configure eNAS, if necessary.
- Network Address Translation (NAT) Gateway: Used to configure the external IP address of the Control Station.
- Control Station (CS): Provides management functions to the file-side components referred to as Data Movers.
- Data Mover (DM): Clients communicate with a Data Mover using either/both NFS and CIFS/SMB protocols. Clients are physically connected to the Data Mover through I/O modules on the storage array that are assigned to the Data Mover. The Data Mover accesses the client data by way of an internal interface to the storage array on which the Data Mover resides

Control station

The Control Station provides utilities for managing, configuring, and monitoring of the Data Movers in the eNAS system.

As the system administrator, you may type commands through the Control Station to perform tasks that include the following:

- Managing and Configuring the database and Data Movers
- Monitoring statistics of the eNAS components

Accessing the Control Station

You may use either local or remote access to the Control Station.

NOTE: To access locally a connection to serial port have to be established.

- Local access to the command line interface is available directly at the Control Station console.
- Remote access to the command line interface by using a secure, encrypted login application allows the use of the eNAS command set.

Accessing the command line interface

A description of how to gain local or remote access to the command line interface for the eNAS follows.

NOTE: For a local connection, connect a client to the Control Station serial port.

- For local access to the command line interface, at the prompt, log in with your administrative username and password.

  Establish the connection to the Control Station with the following settings:

  Table 1. Control Station serial port connection settings

Setting	Value
Bits per second	19200
Data bits	8
Parity	None
Stop bits	1
Flow control	None
Emulation	Auto Detect
Telnet terminal ID	ANSI

- For remote access to the command line interface:
  1. Use a secure, encrypted, remote login application capable of SSH. Type the IP address of the Control Station.
  2. Log in with your administrative username and password.

Role-Based access

The administrative user account you use to access the command line interface is associated with specific privileges, also referred to as roles. A role defines the privileges (operations) a user can perform on a particular eNAS object. The ability to select a predefined role or define a custom role that gives a user certain privileges is supported for users who access eNAS through the CLI, Dell EMC Unisphere™, and the XML API.

The *Security Configuration Guide for VNX* provides detailed information about how role-based access is used to determine the commands a particular user can execute. You create and manage user accounts and roles in Unisphere by using Settings > User Management.

Command set conventions

This manual uses commonly known command set conventions for the eNAS for file man pages. Each man page presents the command name at the top of the man page followed by a brief overview of what the command does. The synopsis contains the actual command usage. The description contains a more detailed breakdown of the features of the command, and the options describe what each switch or option does specifically.

The 'See Also' section refers to the technical modules that support the feature, in addition to any other commands that interact with the command.

The examples are at the end of the command.

The naming convention for the Data Mover variable in the command line interface is <movername> (default = server_2 to server_9).

The commands are prefixed, then appear in alphabetical order.

Synopsis

The synopsis is usage of each command. The synopsis appears in courier typeface, with variables such as movername, filename, and device name enclosed by angle brackets, with the command name appearing in bold. The switches and other options also appear in bold and, in most cases, are prefixed by a minus sign:

```
server_umount {<movername>|ALL}[-perm|-temp]{-all|<fs_name>|<pathname>}
```

Required entries

A switch or variable enclosed with curly brackets, or not enclosed at all, indicates a required entry:

```
{<movername>|ALL}
```

Optional entries

A switch or variable enclosed with square brackets indicates an optional entry:

```
[-perm|-temp]
```

Formatting

The variable name enclosed by angle brackets indicates the name of a specified object:

```
{<movername>|ALL}
```

Options

An option is prefixed with a minus (-) sign: -perm

If the option is spelled out, for example, -perm, in the command syntax, you may use just the first letter: -p

Options and names are case-sensitive. If an uppercase letter is specified in the syntax, a lowercase letter is not accepted.

The vertical bar symbol (|) represents or, meaning an alternate selection:

```
{-all|<fs_name>|<pathname>}
```

Command prefixes

Commands are prefixed depending on what they are administering. For example, commands prefixed with:

- `cel_ execute` to the remotely linked eNAS system.
- `cs_ execute` to the Control Station.
- `fs_ execute` to the specified file system.
- `nas_ execute` directly to the Control Station database.
- `server_ require` a movername entry and execute directly to a Data Mover.

General notes

Note the following:

- If a command is interrupted by using Ctrl-C, then the following messages or traces on the console are expected:
  - nas_cmd: system execution failed.
  - nas_cmd: PANIC: caught signal #11 (Segmentation fault) -- Giving up
- Use eNAS CLI for file to add IPv6 addresses to the NFS export host list. Enclose the IPv6 address in { } or square brackets in the CLI. The IPv6 addresses added to the NFS export list by using the CLI are displayed as read-only fields in the Unisphere software.

NASCLI Commands

This chapter lists the eNAS Command Set provided for managing, configuring, and monitoring of NAS database. The commands are prefixed with nas and appear alphabetically. The command line syntax (Synopsis), a description of the options, and an example of usage are provided for each command.

nas_acl	nas_autodiskmark	nas_automountmap
nas_ca_certificate	nas_cel	nas_checkup
nas_ckpt_schedule	nas_config	nas_connecthome
nas_copy	nas_cs	nas_dbtable
nas_devicegroup	nas_disk	nas_diskmark
nas_emailuser	nas_environment	nas_event
nas_fs	nas_fsck	nas_halt
nas_inventory	nas_license	nas_logviewer
nas_message	nas_migrate	nas_mview
nas_pool	nas_quotas	nas_rdf
nas_replicate	nas_server	nas_stats
nas_storage	nas_syncrep	nas_task
nas_tls	nas_version	nas_volume

nas_acl

Manages the access control level table.

SYNOPSIS

```
nas_acl
  -list
| -info {-user|-group|-owner} <numerical_id>
| -delete {-user|-group} <numerical_id>
|[-name <name>] -create {-user|-group} <numerical_id> level=<acl_level>
| -modify {-user|-group} <numerical_id>
     {[num_id=<numerical_id>][,level=<acl_level>]}
```

DESCRIPTION
----

nas_acl creates, lists, and displays information for access control level entries within the table, and deletes the specified group or entries.

The access control level table is created and recognized in the NAS database and contains assigned levels for users and groups. A user must be defined in the /etc/passwd file prior to being assigned an entry in the table. Creating an access control level entry defines the access level allowed for the user or group once a value has been established for an object.

Note: root privileges are required to create, modify, or delete the access control level table. The root user is permitted access to all objects.

OPTIONS

-list
Lists the access control level table.

-info {-user|-group|-owner} <numerical_id>
Displays information for the user, group, or index entry of the owner as specified by the <numerical_id>.

-delete {-user|-group} <numerical_id>
Deletes the entry for the specified user or group from the access control level table.

-create {-user|-group} <numerical_id> level= <acl_level>
Creates an access control level entry for the specified user or group. The <numerical_id> can be a user ID (UID) or group ID (GID).

Note: Before executing this command, the user or group must exist in the Control Station in the /etc/passwd file or the /etc/group file.

The <acl_level> is a single-digit (between numbers 2 and 9) input representing available access control levels. Levels 2, 3, and 4 which are established by default are:

Level 2. admin: Is the most privileged level and includes privileges allowed from the operator and observer levels.
Level 3. operator: Includes privileges from the observer level.
Level 4. observer: The least privileged.

Levels 5-9 are available for configuration.

[-name <name>]
The name is case-sensitive and indicates a name by which the entry is referred.

Once a value has been set, the level assigned the user or group is checked in the ACL table and the level of access to the object is determined.

-modify {-user|-group} <numerical_id> {[num_id=<numerical_id>] [,level=<acl_level>] }

Modifies the <numerical_id> and level for an access control level entry.

SEE ALSO

Controlling Access to System Objects on VNX, nas_fs, nas_volume, nas_rp, and nas_storage.

EXAMPLE #1

Before creating access control level entries, su to root. To create entries in the access control level table, type:

```
# nas_acl -name user1 -create -user 211 level=3
done

# nas_acl -name user2 -create -user 212 level=2
done

# nas_acl -name user3 -create -user 213 level=4
done

# nas_acl -name user4 -create -user 214 level=2
done

# nas_acl -name user5 -create -user 215 level=3
done

# nas_acl -name user6 -create -user 216 level=4
done
```

EXAMPLE #2

To display the access control level table, type:

```
$ nas_acl -list

index                   type            level        num_id       name
1                       user            admin        201       nasadmin
2                       user            operator     211          user1
3                       user            admin        212          user2
4                       user            observer     213          user3
5                       user            admin        214          user4
6                       user            operator     215          user5
7                       user            observer     216          user6
```

Where:

Value	Definition
index	Access control level table index entry number.
type	User or group for the entry.
level	Level of access permitted.
num_id	Numerical ID for identifying the entry.
name	Name given to the entry.

EXAMPLE #3

To display information for an access control level entry, type:

```
$ nas_acl -info -user 211
id        = 2
name      = user1
level     = operator
user_id   = 211
```

Where:

Value	Definition
id	Index entry.

name   Name given for the entry.
level   Level of access permitted.
user_id Also known as the num_id.

EXAMPLE #4

To modify an access control level entry, type:

nas_acl -modify -user 211 level=7
done

EXAMPLE #5

To delete an access control level entry, type:

nas_acl -delete -user 211
done

Last Modified: March 3, 2011 12:05 pm

nas_autodiskmark

enable/disable autodiskamrk feature.

SYNOPSIS
----
nas_autodiskmark
    -info
  | -modify -enabled { yes | no }

DESCRIPTION

This command is used to enable/disable autodiskamrk feature.

OPTIONS

-info
Displays whether autodiskmark feature is enabled or not.

-modify
    -enabled { yes | no }
    Enables or disables autodiskmark feature.

EXAMPLE #1

To check autodiskmark feature is enabled or not, type:

$ nas_autodiskmark -info
Feature Enabled = No

$ nas_autodiskmark -info
Feature Enabled = Yes

EXAMPLE #2

To enable/disable autodiskmark feature, type:

$ nas_autodiskmark -modify -enabled yes
OK

nas_automountmap

Manages the automount map file.

SYNOPSIS

```
nas_automountmap
  -list_conflict <infile> [-out <outfile>]
| -create [-in <infile>] [-out <outfile>]
```

DESCRIPTION
----

nas_automountmap creates and displays an automount map containing all permanently exported file systems used by the automount daemon.

OPTIONS

-list_conflict <infile>
Prints a list of the mount points that are used more than once.

[-out <outfile>]
Prints a conflicting list and saves it to an <outfile>.

-create
Creates an automount map and prints it to the screen only.

[-in <infile>] [-out <outfile>]
Merges an automount map with an existing map <infile> and outputs it to an <outfile>.

[-out <outfile>]
Creates an automount map and outputs it to an <outfile>.

EXAMPLE #1

To create an automount map, type:

```
$ nas_automountmap -create

ufs1 -rw,intr,nosuid 127.0.0.1,10.172.128.47,128.221.253.2,128.221.252.2:/ufs1
ufs2 -rw,intr,nosuid 127.0.0.1,10.172.128.47,128.221.253.2,128.221.252.2:/ufs2
```

EXAMPLE #2

To create an automount map and save it to a file, type:

```
$ nas_automountmap -create -out automountmap
$ more automountmap

ufs1 -rw,intr,nosuid 127.0.0.1,10.172.128.47,128.221.253.2,128.221.252.2:/ufs1
ufs2 -rw,intr,nosuid 127.0.0.1,10.172.128.47,128.221.253.2,128.221.252.2:/ufs2
```

EXAMPLE #3

To print a conflicting list, type:

```
$ nas_automountmap -list_conflict automountmap

Conflicting lists:
ufs1 -rw,intr,suid 172.16.21.202:/ufs1
ufs1_172.16.21.203 -rw,intr,suid 172.16.21.203:/ufs1
```

EXAMPLE #4

To merge an automount map file with an existing map file, type:

```
$ nas_automountmap -create -in automountmap -out automountmap1
```

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nas_ca_certificate

Manages the Control Station as a Certificate Authority (CA) for VNX's Public Key Infrastructure (PKI).

SYNOPSIS

```
nas_ca_certificate
  -display
| -generate
```

DESCRIPTION
----

nas_ca_certificate generates a public/private key set and a CA certificate for the Control Station. When the Control Station is serving as a CA, it must have a private key with which to sign the certificates it generates for the Data Mover. The Control Station CA certificate contains the corresponding public key, which is used by clients to verify the signature on a certificate received from the Data Mover.

nas_ca_certificate also displays the text of the CA certificate so you can copy it and distribute it to network clients. In order for a network client to validate a certificate sent by a Data Mover that has been signed by the Control Station, the client needs the Control Station CA certificate (specifically the public key from the CA certificate) to verify the signature of the Data Mover.s certificate.

The initial Control Station public/private key set and CA certificate are generated automatically during a VNX software 5.6 install or upgrade. A new Control Station public/private key set and CA certificate is not required unless the CA key set is compromised or the CA certificate expires. The Control Station CA certificate is valid for 5 years.

You must be root to execute the -generate option from the /nas/sbin directory.

Once a Control Station CA certificate is generated, you must perform several additional tasks to ensure that the new certificate is integrated into VNX.s PKI framework. The Security Configuration Guide for File and the Unisphere online help for the PKI interface explain these tasks.

OPTIONS

-display
Displays the Control Station CA certificate. The certificate text is displayed on the terminal screen. Alternatively, you can redirect it to a file.

-generate
Generates a new CA public/private key set and certificate for the Control Station. This certificate is valid for 5 years from the date it is generated.

SEE ALSO
----

server_certificate.

EXAMPLE #1

To generate a new Control Station CA certificate, type:

```
# /nas/sbin/nas_ca_certificate -generate
```

New keys and certificate were successfully generated.

EXAMPLE #2

To display the Control Station.s CA certificate, type:

```
# /nas/sbin/nas_ca_certificate -display

Clients need only the certificate text enclosed by BEGIN CERTIFICATE and END
CERTIFICATE although most clients can handle the entire output.
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number: 3 (0x3)
        Signature Algorithm: sha1WithRSAEncryption
        Issuer: O=Celerra Certificate Authority, CN=eng173100
        Validity
            Not Before: Mar 23 21:07:40 2007 GMT
            Not After : Mar 21 21:07:40 2012 GMT
        Subject: O=Celerra Certificate Authority, CN=eng173100
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
            RSA Public Key: (2048 bit)
                Modulus (2048 bit):
                    00:da:b2:37:86:05:a3:73:d5:9a:04:ba:db:05:97:
                    d2:12:fe:1a:79:06:19:eb:c7:2c:c2:51:93:7f:7a:
                    93:59:37:63:1e:53:b3:8d:d2:7f:f0:e3:49:42:22:
                    f4:26:9b:b4:e4:a6:40:6d:8d:e7:ea:07:8e:ca:b7:
                    7e:88:71:9d:11:27:5a:e3:57:16:03:a7:ee:19:25:
                    07:d9:42:17:b4:eb:e6:97:61:13:54:62:03:ec:93:
                    b7:e6:f1:7f:21:f0:71:2d:c4:8a:8f:20:d1:ab:5a:
                    6a:6c:f1:f6:2f:26:8c:39:32:93:93:67:bb:03:a7:
                    22:29:00:11:e0:a1:12:4b:02:79:fb:0f:fc:54:90:
                    30:65:cd:ea:e6:84:cc:91:fe:21:9c:c1:91:f3:17:
                    1e:44:7b:6f:23:e9:17:63:88:92:ea:80:a5:ca:38:
                    9a:b3:f8:08:cb:32:16:56:8b:c4:f7:54:ef:75:db:
                    36:7e:cf:ef:75:44:11:69:bf:7c:06:97:d1:87:ff:
                    5f:22:b5:ad:c3:94:a5:f8:a7:69:21:60:5a:04:5e:
                    00:15:04:77:47:03:ec:c5:7a:a2:bf:32:0e:4d:d8:
                    dc:44:fa:26:39:16:84:a7:1f:11:ef:a3:37:39:a6:
                    35:b1:e9:a8:aa:a8:4a:72:8a:b8:c4:bf:04:70:12:
                    b3:31
                Exponent: 65537 (0x10001)
X509v3 extensions:
            X509v3 Subject Key Identifier:
            35:06:F2:FE:CC:21:4B:92:DA:74:C9:47:CE:BB:37:21:5E:04:E2:E6
            X509v3 Authority Key Identifier:
keyid:35:06:F2:FE:CC:21:4B:92:DA:74:C9:47:CE:BB:37:21:5E:04:E2:E6
            DirName:/O=Celerra Certificate Authority/CN=eng173100
            serial:00
X509v3 Basic Constraints:
            CA:TRUE
            X509v3 Subject Alternative Name:
            DNS:eng173100
Signature Algorithm: sha1WithRSAEncryption
        09:c3:13:26:16:be:44:56:82:5d:0e:63:07:19:28:f3:6a:c4:
        f3:bf:93:25:85:c3:55:48:4e:07:84:1d:ea:18:cf:8b:b8:2d:
        54:13:25:2f:c9:75:c1:28:39:88:91:04:df:47:2c:c0:8f:a4:
        ba:a6:cd:aa:59:8a:33:7d:55:29:aa:23:59:ab:be:1d:57:f6:
        20:e7:2b:68:98:f2:5d:ed:58:31:d5:62:85:5d:6a:3f:6d:2b:
        2d:f3:41:be:97:3f:cf:05:8b:7e:f5:d7:e8:7c:66:b2:ea:ed:
        58:d4:f0:1c:91:d8:80:af:3c:ff:14:b6:e7:51:73:bb:64:84:
        26:95:67:c6:60:32:67:c1:f7:66:f4:79:b5:5d:32:33:3c:00:
        8c:75:7d:02:06:d3:1a:4e:18:0b:86:78:24:37:18:20:31:61:
        59:dd:78:1f:88:f8:38:a0:f4:25:2e:c8:85:4f:ce:8a:88:f4:
        4f:12:7e:ee:84:52:b4:91:fe:ff:07:6c:32:ca:41:d0:a6:c0:
        9d:8f:cc:e8:74:ee:ab:f3:a5:b9:ad:bb:d7:79:67:89:34:52:
        b4:6b:39:db:83:27:43:84:c3:c3:ca:cd:b2:0c:1d:f5:20:de:
        7a:dc:f0:1f:fc:70:5b:71:bf:e3:14:31:4c:7e:eb:b5:11:9c:
        96:bf:fe:6f
-----BEGIN CERTIFICATE-----
MIIDoDCCAoigAwIBAgIBAzANBgkqhkiG9w0BAQUFADA8MSYwJAYDVQQKEx1DZWxl
cnJhIENlcnRpZmljYXRlIEF1dGhvcml0eTESMBAGA1UEAxMJZW5nMTczMTAwMB4X
DTA3MDMyMzIxMDc0MFoXDTEyMDMyMTIxMDc0MFowPDEmMCQGA1UEChMdQ2VsZXJy
YSBDZXJ0aWZpY2F0ZSBBdXRob3JpdHkxEjAQBgNVBAMTCWVuZzE3MzEwMDCCASIw
DQYJKoZIhvcNAQEBBQADggEPADCCAQoCggEBANqyN4YFo3PVmgS62wWX0hL+GnkG
```

GevHLMJRk396k1k3Yx5Ts43Sf/DjSUIi9CabtOSmQG2N5+oHjsq3fohxnREnWuNX
FgOn7hklB9lCF7Tr5pdhE1RiA+yTt+bxfyHwcS3Eio8g0ataamzx9i8mjDkyk5Nn
uwOnIikAEeChEksCefsP/FSQMGXN6uaEzJH+IZzBkfMXHkR7byPpF2OIkuqApco4
mrP4CMsyFlaLxPdU73XbNn7P73VEEWm/fAaX0Yf/XyK1rcOUpfinaSFgWgReABUE
d0cD7MV6or8yDk3Y3ET6JjkWhKcfEe+jNzmmNbHpqKqoSnKKuMS/BHASszECAwEA
AaOBrDCBqTAdBgNVHQ4EFgQUNQby/swhS5LadMlHzrs3IV4E4uYwZAYDVR0jBF0w
W4AUNQby/swhS5LadMlHzrs3IV4E4uahQKQ+MDwxJjAkBgNVBAoTHUNlbGVycmEg
Q2VydGlmaWNhdGUgQXV0aG9yaXR5MRIwEAYDVQQDEwllbmcxNzMxMDCCAQAwDAYD
VR0TBAUwAwEB/zAUBgNVHREEDTALggllbmcxNzMxMDAwDQYJKoZIhvcNAQEFBQAD
ggEBAAnDEyYWvkRWgl0OYwcZKPNqxPO/kyWFw1VITgeEHeoYz4u4LVQTJS/JdcEo
OYiRBN9HLMCPpLqmzapZijN9VSmqI1mrvh1X9iDnK2iY8l3tWDHVYoVdaj9tKy3z
Qb6XP88Fi3711+h8ZrLq7VjU8ByR2ICvPP8UtudRc7tkhCaVZ8ZgMmfB92b0ebVd
MjM8AIx1fQIG0xpOGAuGeCQ3GCAxYVndeB+I+Dig9CUuyIVPzoqI9E8Sfu6EUrSR
/v8HbDLKQdCmwJ2PzOh07qvzpbmtu9d5Z4k0UrRrOduDJ0OEw8PKzbIMHfUg3nrc
8B/8cFtxv+MUMUx+67URnJa//m8=
-----END CERTIFICATE-----

Last Modified: March 3, 2011 12:37 pm

nas_cel

Performs management of remotely linked VNX or a linked pair of Data Movers.

SYNOPSIS

```
nas_cel
  -list
| -delete {<cel_name>|id=<cel_id>} [-Force]
| -info {<cel_name>|id=<cel_id>}
| -update {<cel_name>|id=<cel_id>}
| -modify {<cel_name>|id=<cel_id>}
        {[-passphrase <passphrase>][-name <new_name>][-ip <ipaddr>][,<ipaddr>,...
]}
| -create <cel_name> -ip <ipaddr>[,<ipaddr>,...] -passphrase <passphrase>
| -interconnect <interconnect_options>
| -syncrep <syncrep_options>
```

DESCRIPTION
----

nas_cel manages the linking of the remote VNX to the local VNX. nas_cel also creates the trusted relationship between source and destination eNAS Control Stations in configurations such as EMC eNAS Replicator.

For eNAS Replicator only, nas_cel -interconnect also builds the connection (interconnect) between a pair of Data Movers.

For VDM Sync Replication only, nas_cel -syncrep also sets up the RP/SRDF connection between a pair of EmbeddedNAS systems.

Linked VNX systems are acknowledged:
1. Automatically during the installation
2. When executing the nas_cel -create

When performing a nas_rdf -init to set up the SRDF relationship between two eNAS system.

OPTIONS

-list
Lists all VNX linked to the current VNX. The hostname of the Control Station active during installation appears as the <cel_name>.

The ID of the object is an integer and is assigned automatically. The name of the VNX might be truncated if it is too long for the display. To view the full name, use the -info option with the VNX ID.

-delete <cel_name>|id=<cel_id>} [-Force]
Deletes the relationship of the remote VNX, and removes its entry from the NAS database on the local VNX.

The -Force option applies to SRDF and EMC MirrorView./S configurations only. If the VNX to be deleted is part of an SRDF or MirrorView/S configuration, -delete must be specified with the -Force option; otherwise, an error is generated. You cannot use -Force if the specified VNX is also being used by VNX Replicator, file system copy (for example, with nas_copy), or TimeFinder/FS NearCopy or FarCopy. If the deletion is necessary, clean up these configurations before performing the forced deletion.

-info {<cel_name>|id=<cel_id>}
Displays information for the remote VNX. To view the <cel_id> of configured VNX, use -list.

-update {<cel_name>|id=<cel_id>}
Updates the local VNX entry with the local Control Station’s hostname and IP address configuration. It also updates the local Data Mover-to-Data Mover authentication setup.

For the remote VNX, updates all Data Movers that were down or experiencing errors during the -create or -modify and restores them to service by using the configuration required for Data Mover authentication.

Data Mover authentication is used in iSCSI replication as the mechanism enabling two Data Movers (local or remote) to authenticate themselves and perform the requested operations. The -update option communicates with each Data Mover and either updates the configuration, or creates the configuration if it is being done for the first time.

-modify {<cel_name>|id=<cel_id>}
{[-passphrase <passphrase>][-name <new_name>][-ip <ipaddr>]}
Changes the current passphrase, name, or IP address of the remote VNX to the new passphrase, name, or IP address in the local VNX database and modifies the remote Data Mover authentication setup by communicating with each Data Mover in the cabinet. The passphrase must have 6 to 15 characters.

-create <cel_name> -ip <ipaddr>[,<ipaddr,...] -passphrase <passphrase>
Builds the trusted relationship between one VNX and another VNX in a configuration such as VNX Replicator, SRDF, and MirrorView/S.

The -create must be executed twice to ensure communication from both sides, first on the source VNX (to identify the destination VNX) and then on the destination VNX (to identify the source VNX). You must assign a name when you create the relationship (for example, a name that identifies the remote VNX in a local entry). The IP address specified represents the appropriate remote VNX.s primary Control Station (in slot 0); the passphrase specified is used to manage the remote VNX. The passphrase must have 6 to 15 characters and be the same between the source and destination VNXs to enable communication.

INTERCONNECT OPTIONS

Type nas_cel -interconnect to display interconnect options:

```
-interconnect
  { -create <name>
    -source_server <movername>
    -destination_system {<cel_name>|id=<cel_id>}
    -destination_server <movername>
    -source_interfaces {<name_service_interface_name>|ip=<ipaddr>}
        [,{<name_service_interface_name>|ip=<ipaddr>},...]
    -destination_interfaces {<name_service_interface_name>|
        ip=<ipaddr>}[,{<name_service_interface_name>|ip=<ipaddr>},...]
        [-bandwidth <bandwidthSched>]
  | -modify {<name>|id=<interConnectId>}
        {[-source_interfaces {<name_service_interface_name>|ip=<ipaddr>},...]
        [-destination_interfaces {<name_service_interface_name>|ip=<ipaddr>},...]

        [-bandwidth <bandwidthSched>]
        [-name <newName>]}
  | -pause {<name>|id=<interConnectId>}
  | -resume {<name>|id=<interConnectId>}
  | -delete {<name>|id=<interConnectId>}
  | -info {<name>|id=<interConnectId>|-all}
  | -list [-destination_system {<cel_name>|id=<cel_id>}]
  | -validate {<name>|id=<interconnectId>}}
```

An interconnect supports VNX Replicator sessions by defining the communications path between a given Data Mover pair located on the same cabinet or different cabinets. The interconnect configures a list of local (source) and peer (destination) interfaces (using IP addresses and interface names), and a bandwidth schedule for all replication sessions using the interconnect. Only one interconnect can be established for a given Data Mover pair in any direction.

Note: You must delete all user-defined interconnects configured for a Data Mover before you can rename it. After you rename the Data Mover, you must re-create the source and peer interconnects with the new Data Mover name and then restart any associated replication sessions.

To fully establish an interconnect, nas_cel -interconnect must be issued

twice, once from each side (the local side and its peer side). Both sides of the interconnect must exist before VNX Replicator sessions for local or remote replication can use the interconnect. Only the local side of an interconnect on which the source replication object resides is specified when creating the replication session. Loopback interconnects are created and named automatically and can be viewed using nas_cel -interconnect -list. You cannot create, modify, or delete loopback interconnects.

-create <name>
Assigns a name, up to 255 characters, to the appropriate side of the interconnect. The name must be unique for each Data Mover. Make the name meaningful, identifying servers and, for remote replication, VNX names or sites.

Remote replication naming example:
s2CelA_s3CelB or NYs3_LAs4 (local side)
s3CelB_s2CelA or LAs4_NYs3 (peer side)

Local replication naming example:
s2_s3 (source side on local system)
s3_s2 (peer side on the same system)

-source_server <moverName>
Specifies the name of an available local Data Mover to use for the local side of the interconnect.

-destination_system {<cel_name>|id=<cel_id>}
Specifies the name or ID of the VNX where the peer Data Mover resides.

-destination_server <movername>
Specifies the name of an available Data Mover, on the same or different system, to use for the peer side of the interconnect.

-source_interfaces {<name_service_interface_name>|ip=<ipaddr>}
[,{<name_service_interface_name>|ip=<ipaddr>},.]
Configures a list of interfaces available for the local side of the interconnect. You can define the list by using IP addresses (IPv4 or IPv6
)
or name service interface names or a combination of both, but how you specify an interface determines how it must be specified by the replication session later (by name service interface name or IP address).

If you define an interface by using an IP address, make sure that the source interface list uses the same IPv4/IPv6 protocol. An IPv4 interface

cannot connect to an IPv6 interface and vice versa. Both sides of the connection must use the same protocol.

For each network protocol type (IPv4/IPv6) specified in the source interf
ace
list, at least one interface from the same type must be specified in the destination interfaces list and vice versa. For example, if the source interface list includes one or more IPv6 addresses, the destination inter
face
list must also include at least one IPv6 address.

The name service interface name is a fully qualified name given to a netw
ork
interface that must resolve to a single IP address (for example, using a
DNS
server).

Note: To prevent potential errors during interface selection (especially
after
a failover/switchover), it is highly recommended that you specify the sam
e
local and peer interface lists when configuring each side of the intercon
nect.

-destination_interfaces {<name_service_interface_name>| ip=<ipaddr>}

[,{<name_service_interface_name>|ip= <ipaddr>},.]
Configures a list of interfaces available on the peer side of the interconnect. You can define the list by using IP addresses (IPv4 or IPv6) or name service interface names or a combination of both, but how you specify each interface determines how it is specified by the replication session.

If you define an interface using an IP address, make sure that the source interface list uses the same IPv4/IPv6 protocol. An IPv4 interface cannot connect to an IPv6 interface and vice versa. Both sides of the connection must use the same protocol.

For each network protocol type (IPv4/IPv6) specified in the destination interface list, at least one interface from the same type must be specified in the source interfaces list and vice versa. For example, if the source interface list includes one or more IPv6 addresses, the destination interface list must also include at least one IPv6 address. The name service interface name is a fully qualified name given to a network interface that must resolve to a single IP address (for example, using a DNS server).

[-bandwidth <bandwidthSched>]
Specifies a schedule to control the interconnect bandwidth used on specific days, or times instead of using all available bandwidth at all times for the interconnect (the default).

Note: The bandwidth schedule executes based on Data Mover time, not Control Station time.

The schedule applies to all VNX Replicator sessions using the interconnect.

Specify a schedule with one or more comma-separated entries, most specific to least specific, as follows:

[{Su|Mo|Tu|We|Th|Fr|Sa}][HH:00-HH:00][/Kbps]
,[<next_entry>],[...]

Example:
MoTuWeThFr07:00-18:00/2000,/8000 means use a limit of 2000 Kb/s from 7 A.M. to 6 P.M. Monday through Friday; otherwise, use a bandwidth limit of 8000 Kb/s.

-interconnect -modify{<name>|id=<interConnectId>}
Modifies one or more of the following characteristics of an existing interconnect, as specified by the name or ID for the appropriate side of the interconnect.

Note: You cannot modify the peer side of an interconnect configured on a remote system; you must modify it from that system. Also, you cannot modify an interface in use by a replication session.

[-source_interfaces{<name_service_interface_name>| ip= <ipAddr>},.]
Modifies the list of interfaces (name service interface name or IP addresses or both) available for use on the local side of an interconnect. The new list of interfaces completely replaces the previous list.

Note: To avoid problems with interface selection, any changes made to the interface lists should be reflected on both sides of an interconnect.

[-destination_interfaces{<name_service_interface_name>|ip= <ipAddr>},.]
Modifies the list of interfaces (name service interface name or IP addresses or both) available for use on the peer side of an interconnect. The new list of interfaces completely replaces the previous list.

[-bandwidth <bandwidth>]
Modifies the existing bandwidth schedule for the specified interconnect, or creates a schedule if none existed previously. The schedule allocates the interconnect bandwidth for specific days or times or both instead of using all available bandwidth at all times for the interconnect (the default). The schedule applies to all replication sessions using the interconnect. Specify a schedule with one or more comma-separated entries, most specific to least specific, as follows:
[{Su|Mo|Tu|We|Th|Fr|Sa}][HH:00-HH:00][/Kbps],[<next_entry>],[...]

Example:
MoTuWeThFr07:00-18:00/2000,/8000 means use a limit of 2000 Kb/s from 7 A.M. to 6 P.M. Monday through Friday; otherwise, use a bandwidth limit of 8000 Kb/s.

[-name <newName>]
Changes the name of the specified interconnect to a new name.

-interconnect -pause {<name>|id=<interConnectId>}
Halts data transmission over the existing Data Mover interconnect until you resume transmission over the interconnect or delete the interconnect. This affects all replication sessions using the specified interconnect.

-interconnect -resume {<name>|id= <interConnectId>}
Resumes data transmission over the Data Mover interconnect, making the interconnect available for use by replication sessions.

-interconnect -delete {<name>|id= <interConnectId>}
Deletes the Data Mover interconnect, thereby making the interconnect unavailable for use by any replication sessions. You cannot delete an interconnect if it is in use by a replication session. You can delete a paused interconnect.

-interconnect -info {<name>|id=<interConnectId>| -all}
Displays information about the specified interconnect or about all interconnects known to the local system.

-interconnect -list [-destination_system <cel_name> |id=<cel_id>]
By default, lists the interconnects available on the local VNX. Specifying the name or ID of a remote VNX also lists the interconnects available on that VNX.

-interconnect -validate {<name>|id= <interconnectId>}
Verifies the interconnect, verifying that authentication is configured properly by opening the connection between the Data Mover pair. Validation is done for loopback, local, and remote configuration.

SYNCREP OPTIONS

Type nas_cel -syncrep to display syncrep options:
-syncrep
{ -enable { <cel_name> | id=<cel_id> }

```
            -local_fsidrange <from>,<to>
            -remote_fsidrange <from>,<to>
            -local_storage <symm_id> sym_dir=<director:port[,director:port,..]>
             rdf_group=<group_num>
            -remote_storage <symm_id> sym_dir=<director:port[,director:port,..]>
             rdf_group=<group_num>
          | -start { <cel_name> | id=<cel_id> }
          | -disable { <cel_name> | id=<cel_id> }
          | -info { <cel_name> | id=<cel_id> | -all } [-verbose]
          | -list
          }
```

The syncrep option is used for creating & deleting VDM Sync Replication Service RDF sessions between the source and remote Control Stations for Embedded NAS.

-enable { <cel_name> | id=<cel_id> }
Enables the VDM Sync Replication Service and creates the RDF session between the local and remote systems.

-disable { <cel_name> | id=<cel_id> }
Disables the VDM Sync Replication Service and deletes the RDF session(s) between the local and remote systems.

-start { <cel_name> | id=<cel_id> }
Starts the SRDF VDM Sync Replication Service.

-info { <cel_name> | id=<cel_id> | -all } [-verbose]
Displays information of local or remote or all the VDM Sync Replication Service(s).

-list
Lists the local and remote VDM Sync Replication Service

SEE ALSO

Using VNX Replicator, nas_copy, nas_replicate, and nas_task.

EXAMPLE #1

To create an entry for the remote VNX, type:

```
$ nas_cel -create cs110 -ip 172.24.102.240 -passphrase nasdocs
operation in progress (not interruptible)...

id         = 3
name       = cs110
owner      = 0
device     =
channel    =
net_path   = 172.24.102.240
VNX_id = APM000438070430000
passphrase = nasdocs
```

Where:

Value	Definition
id	ID of the remote VNX on the local VNX.
name	Name assigned in the local view to the remote VNX.
owner	ACL ID assigned automatically.
device	R2 device mounted by the local Control Station to read the database of the remote Control Station in the SRDF environment. This value is unique to the Symmetrix storage system.
channel	Pair of devices used in the rdf channel. One is used for writing

```
                messages to the remote (wdev), the other to read messages from them. Thi
s
                value is unique to the Symmetrix storage system.
        net_path IP address of the remote VNX.
        VNX_id   Unique VNX ID number.
        passphrase Used for authentication with a remote VNX.

        EXAMPLE #2
        ----------

        For the VNX for block, to list all remote VNXs, type:

        $ nas_cel -list

        id    name          owner mount_dev  channel    net_path        CMU
        0     cs100         0                           172.24.102.236  APM00042000818000
0
        3     cs110         0                           172.24.102.240  APM00043807043000
0

        For the VNX with a Symmetrix storage system, to list all remote VNXs, type:

        $ nas_cel -list

        id    name          owner mount_dev  channel    net_path        CMU
        0     cs30          0                           172.24.172.152  0028040001900006
        1     cs40          500   /dev/sdj1  /dev/sdg   172.24.172.151  0028040002180000

        Where:

        Value           Definition
        -----           ----------

        id              ID of the remote VNX on the local VNX.
        name            Name assigned in the local view to the remote VNX.
        owner           ACL ID assigned automatically.
        mount_dev       Mounted database from the remote VNX in the SRDF environment.
                        This value is unique to the Symmetrix storage system.
        channel         RDF channel from where information is read and written. This valu
e is
                        unique to the Symmetrix storage system.
        net_path        IP address of the remote VNX.
        CMU             VNX Management Unit (unique VNX ID number).


        EXAMPLE #3
        ----------

        To display information for the remote VNX, cs110, type:

        $ nas_cel -info cs110

        id         = 3
        name       = cs110
        owner      = 0
        device     =
        channel    =
        net_path   = 172.24.102.240
        VNX_id = APM000438070430000
        passphrase = nasdocs

        EXAMPLE #1 provides information for a description of command outputs.

        EXAMPLE #4
        ----------

        To update the Control Station entry for cs110, type:

        $ nas_cel -update cs110

        operation in progress (not interruptible)...
```

```
id         = 3
name       = cs110
owner      = 0
device     =
channel    =
net_path   = 172.24.102.240
VNX_id = APM000438070430000
passphrase = nasdocs
```

EXAMPLE #1 provides information for a description of command outputs.

EXAMPLE #5

To modify the passphrase and name for the remote Control Station cs110, type:

```
$ nas_cel -modify cs110 -passphrase nasdocs_replication -name cs110_target

operation in progress (not interruptible)...

id         = 3
name       = cs110_target
owner      = 0
device     =
channel    =
net_path   = 172.24.102.240
VNX_id = APM000438070430000
passphrase = nasdocs_replication
```

EXAMPLE #1 provides information for a description of command outputs.

EXAMPLE #6

To delete the Control Station entry of the remote VNX, cs110_target, type:

```
$ nas_cel -delete cs110_target

operation in progress (not interruptible)...

id         = 3
name       = cs110_target
owner      = 0
device     =
channel    =
net_path   = 172.24.102.240
VNX_id = APM000438070430000
passphrase = nasdocs_replication
```

EXAMPLE #1 provides information for a description of command outputs.

EXAMPLE #7

To create an interconnect NYs3_LAs2 between Data Mover server_3 and remote Data Mover server_2, and use a bandwidth limit of 2000 Kb/s from 7 A.M. to 6 P.M. Monday through Friday; otherwise, use a bandwidth limit of 8000 Kb/s, type:

```
$ nas_cel -interconnect -create NYs3_LAs2 -source_server server_3
-destination_system cs110 -destination_server server_2 -source_interfaces
ip=10.6.3.190 -destination_interfaces ip=10.6.3.173 -bandwidth
MoTuWeThFr07:00-18:00/2000,/8000

operation in progress (not interruptible)...

id                                  = 30003
name                                = NYs3_LAs2
source_server                       = server_3
source_interfaces                   = 10.6.3.190
```

```
destination_system                = cs110
destination_server                = server_2
destination_interfaces            = 10.6.3.173
bandwidth schedule                = MoTuWeThFr07:00-18:00/2000,/8000
crc enabled                       = yes
number of configured replications  = 0
number of replications in transfer = 0
status                            = The interconnect is OK.
```

Where:

Value	Definition
id	ID of the interconnect.
name	Name of the interconnect.
source_server	Name of an available local Data Mover to use for the local side of the interconnect.
source_interfaces	IP addresses available for the local side of the interconnect (at least one, or a name service interface name).
destination_system	Control Station names of the VNX systems available for use in a remote replication session. Local System is the default.
destination_server	Name of an available peer Data Mover to use for the peer side of the interconnect.
destination_interfaces	IP addresses available for the peer side of the interconnect (at least one, or a name service interface name). For loopback interconnects, the interface is fixed at 127.0.0.1.
bandwidth schedule	Bandwidth schedule with one or more comma-separated entries, most specific to least specific.
crc enabled	Indicates that the Cyclic Redundancy Check (CRC) method is in use for verifying the integrity of data sent over the interconnect. CRC is automatically enabled and cannot be disabled.
number of configured replications	Number of replication sessions currently configured.
number of replications in transfer	Number of replications are currently in transfer.
status	Status of the interconnect.

EXAMPLE #8

To modify the bandwidth schedule of the interconnect NYs3_LAs2, type:

```
$ nas_cel -interconnect -modify NYs3_LAs2 -bandwidth

MoWeFr07:00-18:00/2000,TuTh07:00-18:00/4000,/8000
operation in progress (not interruptible)...

id                                = 30003
name                              = NYs3_LAs2
source_server                     = server_3
source_interfaces                 = 10.6.3.190
destination_system                = cs110
destination_server                = server_2
destination_interfaces            = 10.6.3.173
bandwidth schedule                =
MoWeFr07:00-18:00/2000,TuTh07:00-18:00/4000,/8000
crc enabled                       = yes
number of configured replications  = 0
number of replications in transfer = 0
status                            = The interconnect is OK.
```

EXAMPLE #7 provides a description of the command outputs.

EXAMPLE #9

To list available interconnects, type:

```
$ nas_cel -interconnect -list

id     name               source_server   destination_system destination_server
20001  loopback           server_2        cs100              server_2
30001  loopback           server_3        cs100              server_3
30003  NYs3_LAs2          server_3        cs110              server_2
```

EXAMPLE #10

To pause the interconnect with id=30003, type:

```
$ nas_cel -interconnect -pause id=30003

done
```

EXAMPLE #11

To resume the interconnect NYs3_LAs2, type:

```
$ nas_cel -interconnect -resume NYs3_LAs2

done
```

EXAMPLE #12

To validate the interconnect NYs3_LAs2, type:

```
$ nas_cel -interconnect -validate NYs3_LAs2

NYs3_LAs2: validating 9 interface pairs: please wait...ok
```

EXAMPLE #13

To display the detailed information about the interconnect NYs3_LAs2, type:

```
$ nas_cel -interconnect -info NYs3_LAs2

id                                = 30003
name                              = NYs3_LAs2
source_server                     = server_3
source_interfaces                 = 10.6.3.190
destination_system                = cs110
destination_server                = server_2
destination_interfaces            = 10.6.3.173
bandwidth schedule                =
MoWeFr07:00-18:00/2000,TuTh07:00-18:00/4000,/8000
crc enabled                       = yes
number of configured replications  = 0
number of replications in transfer = 0
status                            = The interconnect is OK.
```

EXAMPLE #7 provides a description of the command outputs.

EXAMPLE #14

To delete interconnect NYs3_LAs2, type:

```
$ nas_cel -interconnect -delete NYs3_LAs2

operation in progress (not interruptible)...

id                                = 30003
name                              = NYs3_LAs2
```

```
source_server                       = server_3
source_interfaces                   = 10.6.3.190
destination_system                  = cs110
destination_server                  = server_2
destination_interfaces              = 10.6.3.173
bandwidth schedule                  =
MoWeFr07:00-18:00/2000,TuTh07:00-18:00/4000,/8000
crc enabled                         = no
number of configured replications  = 0
number of replications in transfer = 0
status                              = The interconnect is OK.
```

EXAMPLE #7 provides a description of the command outputs.

Last Modified Date: December 3, 2014 1:15 pm

EXAMPLE #15

To enable VDM syncrep service on local and remote Embedded NAS systems.

```
$ nas_cel -syncrep -enable  L9C26_CS0
-local_fsidrange 4096,12287 -remote_fsidrange 12288,24575 -local_storage
000196700261 sym_dir=1E:27 rdf_group=99 -remote_storage  000197100007
sym_dir=1E:27 rdf_group=99
Now saving FSID range [12288,24575] on remote system...     done
Now saving FSID range [4096,12287] on local system...       done
Now creating LUN mappings (may take several minutes)...     done
Now adding CTD access to local server server_2...           done
Now adding CTD access to local server server_3...           done
Now creating mountpoint for sync replica of NAS database... done
Now mounting sync replica of NAS database...                done
Now enabling sync replication service on remote system...   done
done
```

EXAMPLE #16

To disable a VDM syncrep service on local and remote Embedded NAS systems.

```
$ nas_cel -syncrep -disable L9C26_CS0
Now unmounting sync replica of NAS database...                                  done
Now deleting mountpoint for sync replica of NAS database...                     done
Now removing CTD access to local server server_2...                             done
Now removing CTD access to local server server_3...                             done
Now deleting local LUN mapping...                                               done
Now disabling service (including deleting LUN mapping) on remote system... done
Now removing FSID range [12288,24575] on remote system...                       done
Now removing FSID range [4096,12287] on local system...                         done
Now removing other sync replication service settings on local system...    done
done
```

nas_checkup

Provides a system health checkup for the VNX.

SYNOPSIS
----

nas_checkup
        [-version|-help|-rerun]

DESCRIPTION
----

nas_checkup runs scheduled and unscheduled health checks on the VNX and reports problems that are found, and the actions needed to fix the problem and acts as a system health monitor.

The scheduled run time for the nas_checkup command is every 2 weeks by default. If a warning or error is discovered during this time, an alert is posted on the Unisphere.

Set up email notification for warnings or errors in the Unisphere Notifications page, or modify and load the sample nas_checkup event configuration file.

If a problem is discovered that requires EMC Service Personnel assistance, nas_checkup will notify EMC.

OPTIONS

No arguments
Runs a series of system health checks on the VNX and reports the problems that are found and the actions needed to fix the problem.

No email, callhome, or Unisphere alert is posted when the health check is run unscheduled.

-version
Displays the version of health check that is run on the VNX.

-help
Provides help.

-rerun
Reruns the checks that produce error messages in the previous health checkup. It does not rerun the checks that produce warning or information messages. If there are no checks that produce error messages, then the -rerun switch generates a message that there is nothing to rerun.

CHECKS

Nas_checkup runs a subset of the available checks based on the configuration of your system. The complete list of available checks are:
Control Station Checks:
Check if minimum free space exists
Check if minimum free space exists ns
Check if enough free space exists
Check if enough free space exists ns
Check if NAS Storage API is installed correctly
Check if NAS Storage APIs match
Check if NBS clients are started
Check if NBS configuration exists
Check if NBS devices are accessible
Check if NBS service is started
Check if standby is up
Check if Symapi data is present
Check if Symapi is synced with Storage System
Check integrity of NASDB
Check if primary is active

```
        Check all callhome files delivered
        Check if NAS partitions are mounted

        Data Mover Checks:
        Check boot files
        Check if hardware is supported
        Check if primary is active
        Check if root filesystem has enough free space
        Check if using standard DART image
        Check MAC address
        Check network connectivity
        Check status

        Storage System Checks:
        Check disk emulation type
        Check disk high availability access
        Check disks read cache enabled
        Check disks and storage processors write cache enabled
        Check if access logix is enabled
        Check if FLARE is committed
        Check if FLARE is supported
        Check if microcode is supported
        Check no disks or storage processors are failed over
        Check that no disks or storage processors are faulted
        Check that no hot spares are in use
        Check that no hot spares are rebuilding
        Check control lun size
        Check if storage processors are read cache enabled

        FILES
        -----

        The files associated with system health checkups are:
        /nas/log/nas_ checkup-run.<timestamp>.log    Contains information about the check
s that
                                                     were run, problems found, and action
s needed
                                                     to fix the problem.
        /nas/log/nas_checkup.<timestamp>.log         Produced when a scheduled nas_checku
p
                                                     is run and contains the same informa
tion as
                                                     the nas_checkup-run.<timestamp>.log.
        /nas/log/syslog                              Contains the overall results of nas_
checkup.
        /nas/site/checkup_eventlog.cfg               Provides a sample nas_checkup event
                                                     configuration file. This is the file
 to be
                                                     modified to add email addresses and
load the                                               file.

        SEE ALSO
        --------

        Configuring Events and Notifications on VNX for File.

        EXAMPLE #1
        ----------

        To run a health check on the VNX, type:
        $ nas_checkup

        Check Version:  5.6.23.1
        Check Command:  /nas/bin/nas_checkup
        Check Log    :  /nas/log/checkup-run.070611-064115.log

        ------------------------------------Checks-------------------------------------
        Control Station: Checking if file system usage is under limit..............Pass
        Control Station: Checking if file systems have enough space to upgrade.....Pass
        Control Station: Checking if NAS Storage API is installed correctly........Pass
        Control Station: Checking if NBS clients are started.......................Pass
```

```
Control Station: Checking if NBS configuration exists.....................Pass
Control Station: Checking if NBS devices are accessible...................Pass
Control Station: Checking if NBS service is started.......................Pass
Control Station: Checking if standby is up................................N/A
Control Station: Checking if Symapi data is present.......................Pass
Control Station: Checking if Symapi is synced with Storage System.........Pass
Control Station: Checking integrity of NASDB..............................Pass
Control Station: Checking all callhome files delivered....................Pass
Control Station: Checking resolv conf.....................................Pass
Control Station: Checking if NAS partitions are mounted...................Pass
Control Station: Checking ipmi connection.................................Pass
Control Station: Checking nas site eventlog configuration.................Pass
Control Station: Checking nas sys mcd configuration.......................Pass
Control Station: Checking nas sys eventlog configuration..................Pass
Control Station: Checking logical volume status...........................Pass
Control Station: Checking ups is available................................Fail
Data Movers    : Checking boot files.....................................Pass
Data Movers    : Checking if primary is active...........................Pass
Data Movers    : Checking if root filesystem has enough free space.........Pass
Data Movers    : Checking if using standard DART image...................Pass
Data Movers    : Checking network connectivity...........................Pass
Data Movers    : Checking status.........................................Pass
Data Movers    : Checking dart release compatibility.....................Pass
Data Movers    : Checking dart version compatibility.....................Pass
Data Movers    : Checking server name....................................Pass
Data Movers    : Checking unique id......................................Pass
Data Movers    : Checking CIFS file server configuration..................N/A
Data Movers    : Checking domain controller connectivity and configuration.N/A
Data Movers    : Checking DNS connectivity and configuration..............N/A
Data Movers    : Checking connectivity to WINS servers...................N/A
Data Movers    : Checking connectivity to NTP servers....................N/A
Data Movers    : Checking connectivity to NIS servers....................Pass
Data Movers    : Checking virus checker server configuration..............N/A
Data Movers    : Checking if workpart is OK..............................Pass
Data Movers    : Checking if free full dump is available..................?
Data Movers    : Checking if each primary data mover has standby..........Fail
Storage System : Checking disk emulation type............................Pass
Storage System : Checking disk high availability access..................Pass
Storage System : Checking disks read cache enabled.......................Pass
Storage System : Checking disks and storage processors write cache enabled.Pass
Storage System : Checking if access logix is enabled.....................Pass
Storage System : Checking if FLARE is committed..........................Pass
Storage System : Checking if FLARE is supported..........................Pass
Storage System : Checking if microcode is supported......................Pass
Storage System : Checking no disks or storage processors are failed over..Pass
Storage System : Checking that no disks or storage processors are faulted..Pass
Storage System : Checking that no hot spares are in use..................Pass
Storage System : Checking that no hot spares are rebuilding..............Pass
Storage System : Checking minimum control lun size.......................Pass
Storage System : Checking maximum control lun size.......................Fail
Storage System : Checking system lun configuration.......................Pass
Storage System : Checking if storage processors are read cache enabled.....Pass
Storage System : Checking if auto assign are disabled for all luns.........Pass
Storage System : Checking if auto trespass are disabled for all luns.......Pass
Storage System : Checking backend connectivity...........................Pass
--------------------------------------------------------------------------------

One or more warnings are shown below. It is recommended that you follow the
instructions below to correct the problem then try again.



-----------------------------------Information----------------------------------
Control Station: Check ups is available
Symptom: The following UPS emcnasUPS_i0 emcnasUPS_i1 is(are)
         not available

Data Movers: Check if each primary data mover has standby
Symptom: The following primary Data Movers server_2, server_3 does
         not have a standby Data Mover configured. It is recommended that each
         primary Data Mover have a standby configured for it with automatic
```

```
         failover policy for high availability.

Storage System: Check maximum control lun size
Symptom:
         * The size of control LUN 5 is 32 GB. It is larger than the
           recommended size of 14 GB. The additional space will be reserved by
           the system.

--------------------------------------------------------------------------------

-----------------------------------Warnings-------------------------------------
Data Movers: Check if free full dump is available
Symptom: Cannot get workpart structure. Command failed.
         * Command: /nas/sbin/workpart -r
         * Command output: open: Permission denied

         * Command exit code: 2
Action : Contact EMC Customer Service and refer to EMC Knowledgebase
         emc146016. Include this log with your support request.
--------------------------------------------------------------------------------

EXAMPLE #2
----------

To display help for nas_checkup, type:

$ nas_checkup -help

Check Version:  5.6.23.1
Check Command:  /nas/bin/nas_checkup

usage: nas_checkup
          [ -help | -version ]

EXAMPLE #3
----------

To display the version of nas_checkup utility, type:

$ nas_checkup -version

Check Version:  5.6.23.1
Check Command:  /nas/bin/nas_checkup

DIAGNOSTICS
-----------

nas_checkup returns one of the following exit statuses:
0 . No problems found
1 . nas_checkup posted information
2 . nas_checkup discovered a warning
3 . nas_checkup discovered an error
255 . Any other error

Examples of errors that could cause a 255 exit status include, but are not
limited to:

-If nas_checkup is run when another instance of nas_checkup is running
-If nas_checkup is run by someone other than root or the administrator group
 (generally nasadmin)
-If nas_checkup is run on the standby Control Station

---------------------------------------
Last Modified: March 3, 2011 1:30 pm
```

nas_ckpt_schedule

Manages SnapSure checkpoint scheduling for the VNX.

SYNOPSIS

```
nas_ckpt_schedule
  -list
| -info {-all|<name>|id=<id>}
| -create <name>
    -filesystem {<name>|id=<id>} [-description <description>]
    -recurrence {
    once [-start_on <YYYY-MM-DD>] -runtimes <HH:MM>
    [-ckpt_name <ckpt_name>]
    | daily [-every <number_of_days>]
            [-start_on <YYYY-MM-DD>][-end_on <YYYY-MM-DD>]
             -runtimes <HH:MM>[,...]
            {-keep <number_of_ckpts>|-ckpt_names <ckpt_name>[,...]}
    | weekly [-every <number_of_weeks>]
             -days_of_week {Mon|Tue|Wed|Thu|Fri|Sat|Sun}[,...]
            [-start_on <YYYY-MM-DD>][-end_on <YYYY-MM-DD>]
             -runtimes <HH:MM>[,...]
            {-keep <number_of_ckpts>|-ckpt_names <ckpt_name>[,...]}
    | monthly [-every <number_of_months>] -days_of_month <1-31>[,...]
            [-start_on <YYYY-MM-DD>][-end_on <YYYY-MM-DD>]
             -runtimes <HH:MM>[,...]
            {-keep <number_of_ckpts>|-ckpt_names <ckpt_name>[,...]}}
    [{-cvfsname_prefix <prefix>|-time_based_cvfsname }]
| -modify {<name>|id=<id>}
    [-name <new_name>]
    [{-cvfsname_prefix <prefix>| -time_based_cvfsname}]
    [-description <description>]
    [-recurrence {daily|weekly|monthly}]
    [-every {number_of_days|number_of_weeks|<number_of_months}]
    [-days_of_week {Mon|Tue|Wed|Thu|Fri|Sat|Sun}[,...]]
    [-days_of_month <1-31>[,...]][ -start_on <YYYY-MM-DD>]
    [-end_on <YYYY-MM-DD>][ -runtimes <HH:MM>[,...]]
| -delete {<name>|id=<id>}
| -pause {<name>|id=<id>}
| -resume {<name>|id=<id>}
```

DESCRIPTION
----

nas_ckpt_schedule creates and lists the schedules for the SnapSure checkpoints. Schedules can be run once, daily, weekly, or monthly and can be modified, paused, resumed, and deleted.

OPTIONS

-list
Lists all checkpoint schedules on the system, the name of the schedule, the next run date, the state, and the description.

-info {-all|<name>|id=<id>}
Lists detailed information for all schedules or for the specified schedule.

-create <name> -filesystem {<name>|id=<id>}
[-description <description>] -recurrence {
Creates a checkpoint schedule for the file system that is specified by <name> or <id>. The schedule name in -create <name> must be unique. The -description option provides a label for the schedule. The -recurrence option specifies if the checkpoint operation occurs once, daily, weekly, or monthly.

Note: It is recommended that a time interval of at least 15 minutes in between the creation of two checkpoints on the same production file system. Using VNX SnapSure provides information on checkpoint scheduling.

once [-start_on <YYYY-MM-DD>] -runtime <HH:MM> [-ckpt_name <ckpt_name>]

If once is specified, the hours and minutes for the snapshot to be run must be specified. A start date and name may be optionally assigned to the checkpoint.

For a one-time checkpoint schedule, only one runtime can be provided. For one-time schedules, the option -ckpt_name can specify a name for the single checkpoint; if omitted, the default naming is used (<schedule_name>_<fs_name>_<num>) where <num> is a four digit integer beginning with 0001.

|daily [-every <number_of_days>] [-start_on <YYYY-MM-DD>][-end_on <YYYY-MM-DD>] -runtimes <HH:MM>[,...]
{-keep <number_of_ckpts>|-ckpt_names <ckpt_name>[,...]}
If daily is specified, the checkpoint is taken every day unless -every is specified indicating the number of days between runs. The -start_on option indicates the day when the checkpoints will start and -end_on indicates the day when they end.

The -runtimes option specifies one or more times to take a checkpoint on each scheduled day. The -keep option specifies the maximum number of checkpoints to be kept at any one time (using default checkpoint naming). <number_of_ckpts> should be equal to the number of checkpoint names specified for a schedule. Th
e
-ckpt_name option assigns one or more specific names to each checkpoint as it is taken.

|weekly [-every <number_of_weeks>] -days_of_week {Mon|Tue|Wed|Thu|Fri|Sat|Sun} [,...][-start_on <YYYY-MM-DD>][-end_on <YYYY-MM-DD>]-runtimes <HH:MM>[,...]
{-keep <number_of_ckpts>|-ckpt_names <ckpt_name>[,...]}
If weekly is specified, the checkpoint is taken every week unless the -every option is specified indicating the number of weeks between runs. The -days_of_week option specifies one or more days during the week on which to run the schedule. The -start_on option indicates the day when the checkpoints will start and -end_on indicates the day when they end.

The -runtimes option specifies one or more times to take a checkpoint on each scheduled day. The -keep option specifies the maximum number of checkpoints to be kept at any one time (using default checkpoint naming). The -ckpt_name option assigns one or more specific names to each checkpoint as it is taken.

|monthly [-every <number_of_months>] -days_of_month <1-31>[,...][-start_on <YYYY-MM-DD>][-end_on <YYYY-MM-DD>]-runtimes <HH:MM>[,...]
{-keep <number_of_ckpts> |-ckpt_names <ckpt_name>[,...]}}
If monthly is specified, the checkpoint is taken every month unless the -every is specified indicating the number of months between runs. The -days_of_month option specifies one or more days during the month on which to run the schedule. <days> is specified as an integer 1 through 31. The -start_on option indicates the day when the checkpoints will start and -end_on indicates the day when they end.

The -runtimes option specifies one or more times to take a checkpoint on each scheduled day. The -keep option specifies either the maximum number of checkpoints to be kept at any one time (using default checkpoint naming) or using the -ckpt_name option, one or more specific names to assign each checkpoint as it is taken.

The schedule that is set takes effect immediately unless -start_on is specified. Daily, weekly, and monthly schedules run indefinitely unless -end_on is included.

The -cvfsname_prefix option specifies the customized prefix of a CVFS name. This prefix along with the cvfsname_delimiter and the cvfs_starting_index make up the CVFS name. The -time_based_cvfsname option specifies the CVFS name based on the creation time of the CVFS. It is the default method for generating CVFS names and will be used if the prefix is not specified.

Note: The prefix must be a PFS-wide unique string and can contain up to 20 ASC
II
characters. The prefix must not include intervening spaces, colons (:), or slashes (/).

-modify {<name>|id=<id>} [-name <new_name>] [{-cvfsname_prefix <prefix>|
-time_based_cvfsname}] [-description <description>] [-recurrence
{daily|weekly|monthly}] [-every <number_of_days>| <number_of_weeks>
|<number_of_months>] [-days_of_week {Mon|Tue|Wed|Thu|Fri|Sat|Sun}[,...]]
[-days_of_month <1-31>[,...]][-start_on <YYYY-MM-DD>]
[-end_on <YYYY-MM-DD>][-runtimes <HH:MM>[,...]]
Modifies the scheduled checkpoint entry as specified.

-delete {<name>|id=<id>}
Deletes the specified checkpoint schedule. This operation does not delete any
checkpoints.

-pause {<name>|id=<id>}
Pauses the specified checkpoint schedule, including checkpoint creations.

-resume {<name>|id=<id>}
Resumes a paused checkpoint schedule.

SEE ALSO

Using VNX SnapSure.

EXAMPLE #1

To create a checkpoint schedule that creates a checkpoint of the file system
ufs1 daily at 8 A.M. and 8 P.M. starting on 11/13/06 with the last run on
12/13/07, and keep 7 checkpoints, type:

$ nas_ckpt_schedule -create ufs1_ckpt_sched1 -filesystem ufs1 -description
"Daily
Checkpoint schedule for ufs1" -recurrence daily -every 1 -start_on 2006-11-13
-end_on 2007-12-13 -runtimes 8:00,20:00 -keep 7 -cvfsname_prefix daily

This command returns no output.

EXAMPLE #2

To create a checkpoint schedule that creates a checkpoint of the file system
ufs1 weekly on Mondays at 6 P.M., starting on 11/13/06 with the last run on
12/13/07, and name new checkpoints ufs1_ckpt_mon1, ufs1_ckpt_mon2,
ufs1_ckpt_mon3, ufs1_ckpt_mon4, type:

$ nas_ckpt_schedule -create ufs1_ckpt_sched2 -filesystem ufs1 -description
"Weekly Checkpoint schedule for ufs1" -recurrence weekly -every 1
-days_of_week Mon -start_on 2006-11-13 -end_on 2007-12-13 -runtimes 18:00
-ckpt_names ufs1_ckpt_mon1,ufs1_ckpt_mon2,ufs1_ckpt_mon3,ufs1_ckpt_mon4
-cvfsname_prefix weekly

This command returns no output.

EXAMPLE #3

To create a checkpoint schedule that creates a checkpoint of the file system
ufs1 every other 15th of the month at 7 P.M., and keep 12 checkpoints, type:

$ nas_ckpt_schedule -create ufs1_ckpt_sched3 -filesystem ufs1 -description
"Monthly Checkpoint schedule for ufs1" -recurrence monthly -every 2
-days_of_month
15 -runtimes 19:00 -keep 12 -cvfsname_prefix monthly

This command returns no output.

EXAMPLE #4

To create a checkpoint schedule that creates a checkpoint of the file system

ufs1 once at 3:09 P.M., type:

```
$ nas_ckpt_schedule -create ufs1_ckpt_sched4 -filesystem ufs1 -description
"One-time Checkpoint Schedule for ufs1" -recurrence once -runtimes 15:09
```

This command returns no output.

EXAMPLE #5

To list all checkpoint schedules, type:

```
$ nas_ckpt_schedule -list

id          = 6
name        = ufs1_ckpt_sched2
description = Weekly Checkpoint schedule for ufs1
state       = Pending
next run    = Mon Nov 13 18:00:00 EST 2006

id          = 80
name        = ufs1_ckpt_sched4
description = One-time Checkpoint Schedule for ufs1
state       = Pending
next run    = Tue Nov 14 15:09:00 EST 2006

id          = 5
name        = ufs1_ckpt_sched1
description = Daily Checkpoint schedule for ufs1
state       = Pending
next run    = Mon Nov 13 20:00:00 EST 2006

id          = 7
name        = ufs1_ckpt_sched3
description = Monthly Checkpoint schedule for ufs1
state       = Pending
next run    = Wed Nov 15 19:00:00 EST 2006
```

EXAMPLE #6

To modify the recurrence of the checkpoint schedule ufs1_ckpt_sched3 to run every 10th of the month, type:

```
$ nas_ckpt_schedule -modify ufs1_ckpt_sched3 -recurrence monthly -every 1
-days_of_month 10
```

This command returns no output.

EXAMPLE #7

To get detailed information about checkpoint schedule, type:

```
$ nas_ckpt_schedule -info ufs1_ckpt_sched3

id = 7
name = ufs1_ckpt_sched3
description = Monthly Checkpoint schedule for ufs1
CVFS name prefix = monthly
tasks = Checkpoint ckpt_ufs1_ckpt_sched3_001 on filesystem id=25, Checkpoint
ckpt_ufs1_ckpt_sched3_002 on filesystem id=25, Checkpoint
ckpt_ufs1_ckpt_sched3_003 on filesystem id=25, Checkpoint
ckpt_ufs1_ckpt_sched3_004 on filesystem id=25, Checkpoint
ckpt_ufs1_ckpt_sched3_005 on filesystem id=25, Checkpoint
ckpt_ufs1_ckpt_sched3_006 on filesystem id=25, Checkpoint
ckpt_ufs1_ckpt_sched3_007 on filesystem id=25, Checkpoint
ckpt_ufs1_ckpt_sched3_008 on filesystem id=25, Checkpoint
ckpt_ufs1_ckpt_sched3_009 on filesystem id=25, Checkpoint
ckpt_ufs1_ckpt_sched3_010 on filesystem id=25, Checkpoint
ckpt_ufs1_ckpt_sched3_011 on filesystem id=25, Checkpoint
```

```
ckpt_ufs1_ckpt_sched3_012 on filesystem id=25
next run = Sun Dec 10 19:00:00 EST 2006
state = Pending
recurrence = every 1 months
start on = Mon Nov 13 16:47:51 EST 2006
end on =
at which times = 19:00
on which days of week  =
on which days of month = 10
```

EXAMPLE #8

To pause a checkpoint schedule, type:

```
$ nas_ckpt_schedule -pause ufs1_ckpt_sched1
```

This command returns no output.

EXAMPLE #9

To resume a checkpoint schedule, type:

```
$ nas_ckpt_schedule -resume ufs1_ckpt_sched1
```

This command returns no output.

EXAMPLE #10

To delete a checkpoint schedule, type:

```
$ nas_ckpt_schedule -delete ufs1_ckpt_sched2
```

This command returns no output.

Last Modified: March 4 2011, 11:20 am

nas_config

Manages a variety of configuration settings on the Control Station, some of which are security based.

SYNOPSIS
----

nas_config

```
  -IPalias {-list
        | -create [-name <device_name>] <numeric_id>
        | -delete [-name <device_name>] <numeric_id>}
| -ssl
| -sessiontimeout [<number_in_minutes>|off]
| -password [-min <6..15>] [-retries <max_allowed>] [-newchars <min_num>]
[-digits <min_num>] [-spechars <min_num>] [-lcase <min_num>] [-ucase <min_num>]
| -password -default
```

DESCRIPTION
----

nas_config -IPalias configures different IP addresses to point to the same network device allowing use of IP aliasing to manage the Control Station. This enables communication with the primary Control Station using a single IP address regardless of whether the primary Control Station is running in slot 0 or slot 1.

NOTE: Creation and Deletion of aliases for control station IPs are not supported in Embedded NAS. These operations are only used for engineering purpose and end users are not allowed to do such operations. User will see the below message when create or delete operation is performed.

 "Creation or Deletion of IP alias is not supported for Embedded NAS"

nas_config -ssl generates an X.509 digital certificate on the Control Station. Unisphere uses the Secure Socket Layer (SSL) protocol to create a secure connection between a user.s Web browser and the Control Station.s Apache Web server. When a VNX system is initially installed, a generic certificate is generated. After configuring the Control Station’s network configuration (hostname, DNS domain name, or IP address) and before using the Unisphere, a new certificate should be generated.

nas_config -sessiontimeout sets a system-wide value that automatically times out a Control Station shell session after a specified period of inactivity. nas_config -password supports a password quality policy by requiring that passwords chosen by VNX users adhere to certain rules.

You must be root to execute this command from the /nas/sbin directory.

OPTIONS

-IPalias -list
Lists IP aliases configured on the VNX.

-IPalias -create [-name <device_name>] <numeric_id>
Creates an IP alias for the Control Station.

<device_name> is the name for a specified device:
1. If you specify a device name that device must have an IP address.
2. If you do not specify a device name, the system uses the external network interface.

<numeric_id> is a user-defined number, and can be an integer between 0 and 255. The system allows up to 256 aliases for any device.

-delete [-name <device_name>] <numeric_id>
Deletes an IP alias for the Control Station.

-ssl

Installs a SSL certificate on the Control Station and restarts the HTTP server.

-sessiontimeout [<number_in_minutes>|off]
Displays the current session timeout value in minutes. <number_in_minutes> sets the number of minutes a Control Station shell sesssion can be inactive before it is timed out. Possible values are 5 to 240 minutes. The default value is 60 minutes. Session timeout is enabled by default. To disable session timeout, type off or 0 to indicate zero minutes.

The -sessiontimeout option enables the native timeout properties of the underlying shells on the Control Station. The relevant shell man page provides a description of how the mechanism works.

-password
Prompts for specific password policy definitions. The current value for each policy definition is shown in brackets.

[-min <6..15>] defines the minimum length of the new password. The default length is eight characters. The length has to be a value between 6 and 15 characters.

[-retries <max_allowed>] defines the number of attempts a user can make to define an acceptable new password before the command fails. The default value is three attempts.

[-newchars <min_num>] defines the minimum number of characters that must be in the new password that were not included in the old password. The default value is three characters.

[-digits <min_num>] defines the minimum number of digits that must be included in the new password. The default value is one digit.

[-spechars <min_num>] defines the minimum number of special characters (such as ! @ # $ % & ^ and *) that must be included in the new password. The default value is 0.

[-lcase <min_num>] defines the minimum number of lowercase characters that must be included in the new password. The default value is 0.

[-ucase <min_num>] defines the minimum number of uppercase characters that must be included in the new password. The default value is 0.

-password -default
Resets the password policy definitions to their default values.

SEE ALSO

Security Configuration Guide for File.

EXAMPLE #1

To create an IP alias for the Control Station, type:

```
# /nas/sbin/nas_config -IPalias -create 0

Do you want slot_0 IP address <1.2.3.4> as your alias [yes or no]: no
Please enter an IP address to use as an alias: 1.2.3.6
Do you want slot_0 IP address <1.2.3.4> as your alias [yes or no]: yes
Please enter a new IP address for slot_0: 1.2.3.6
```

EXAMPLE #2

To view the IP alias that you created, type:

```
# /nas/sbin/nas_config -IPalias -list

alias IPaddress state
```

```
eth2:0 1.2.3.6 UP

EXAMPLE #3
----------

To delete an IP alias, type:

# /nas/sbin/nas_config -IPalias -delete 0

All current sessions using alias eth2:0 will terminate
Do you want to continue [yes or no]: yes
done

EXAMPLE #4
----------

To generate and install a certificate for the Apache Web server on the Control
Station, type:

# /nas/sbin/nas_config -ssl

Installing a new SSL certificate requires restarting the Apache web server.
Do you want to proceed? [y/n]: y
New SSL certificate has been generated and installed successfully.

EXAMPLE #5
----------

To change the session timeout value from the default value of 60 minutes to
100 minutes, type:

# /nas/sbin/nas_config -sessiontimeout 100

done

EXAMPLE #6
----------

To disable session timeout, type:

# /nas/sbin/nas_config -sessiontimeout 0

done

or

# /nas/sbin/nas_config -sessiontimeout off

done

EXAMPLE #7
----------

To set specific password policy definitions, type:

# /nas/sbin/nas_config -password

Minimum length for a new password (Between 6 and 15): [8]
Number of attempts to allow before failing: [3]
Number of new characters (not in the the old password): [3]
Number of digits that must be in the new password: [1]
Number of special characters that must be in a new password: [0]
Number of lower case characters that must be in password: [0]
Number of upper case characters that must be in password: [0]

EXAMPLE #8
----------

To set the minimum length of a new password to 10 characters, type:

# /nas/sbin/nas_config -password -min 10
```

EXAMPLE #9

To reset the current password policy definitions to their default values, type:

```
# /nas/sbin/nas_config -password -default
```

Last Modified: March 4, 2011 12:45 pm

nas_connecthome

Configures email, FTP, modem, HTTPS and ESRS transport mechanisms for transportin
g
Callhome event files to user-configured destinations.

SYNOPSIS

```
nas_connecthome
  -info
| -test {-email_1|-email_2|-ftp_1|-ftp_2|-modem_1|-modem_2|-https|-esrs}
| -modify [-modem_priority {Disabled|1|2|3}]
        [-modem_number <phone_number>]
        [-modem_number_2 <phone_number>]
        [-ftp_priority {Disabled|1|2|3}]
        [-ftp_server {<hostname>|<ip_addr>}]
        [-ftp_port <port>]
        [-ftp_user <username>]
        [-ftp_passwd [<passwd>]]
        [-ftp_folder <path>]
        [-ftp_ipprotocol {IPV4|IPV6}]
        [-ftp_mode {active|passive}]
        [-ftp_server_2 {<hostname>|<ip_addr>}]
        [-ftp_port_2 <port>]
        [-ftp_user_2 <username>]
        [-ftp_passwd_2 [<passwd>]]
        [-ftp_folder_2 <path>]
        [-ftp_ipprotocol_2 {IPV4|IPV6}]
        [-ftp_mode_2 {active|passive}]
        [-email_priority {Disabled|1|2|3}]
        [-email_from <email_addr>]
        [-email_to {<email_addr>[,<email_addr>]}
        [-email_subject <email_subject>]
        [-email_server {<hostname>|<ip_addr>}]
        [-email_ipprotocol {IPV4|IPV6}]
        [-email_server_2 {<hostname>|<ip_addr>}]
        [-email_ipprotocol_2 {IPV4|IPV6}]
        [-esrs_priority {Disabled|1|2|3}]
        [ -https_priority {Disabled|1|2|3}]
        [ -https_url {url}
        [ -https_ipprotocol {IPv4|IPv6}
        [-dial_in_number <phone_number>]
        [-serial_number <serial_number>]
        [-site_id <site_id>]
        [-encryption_enabled {yes|no}]
        [-dial_in_enabled {yes|no}]
| -help
```

DESCRIPTION
----

nas_connecthome pauses and resumes the ConnectHome service,
displays and configures parameters for email, FTP, modem, HTTPS,and ESRS,
which are mechanisms used for transmitting event files.

nas_connecthome enables a user to configure primary and optional
secondary destinations for each transport mechanism.

nas_connecthome also tests connectivity to the destination
configured for a transport mechanism.

This command must be executed from /nas/sbin/.

OPTIONS

-info

Displays the enabled and disabled configuration parameters for all transport mechanisms.

-test {-email_1|-email_2|-ftp_1|-ftp_2|-modem_1|-modem_2|-https|-esrs}
Tests connectivity to the destination configured and enabled for the specified transport mechanism.

-modify
Modifies the following configuration parameters for any or all transport mechanisms:

[-modem_priority {Disabled|1|2|3}]
Enables modem as a Primary, Secondary, or Tertiary transport mechanism. Specifying Disabled removes modem as a transport mechanism.

[-modem_number <phone_number>]
Sets or modifies the primary phone number of the modem.

Note: Specifying "" (empty double quotes) disables the use of the existing phone number.

[-modem_number_2 <phone_number>]
Sets or modifies the secondary phone number of the modem.

Note: Specifying "" (empty double quotes) disables the use of the existing phone number for this transport mechanism.

[-ftp_priority {Disabled|1|2|3}]
Enables FTP as a Primary, Secondary, or Tertiary transport mechanism. Specifying Disabled removes FTP as a transport mechanism.

[-ftp_server {<hostname>|<ip_addr>}]
Sets or modifies the hostname or IP address of the primary FTP server and corresponding port. The allowable input is IPv4 address, IPv6 address, or domain name.

[-ftp_port <port>]
Sets or modifies the port of the primary FTP server and corresponding port. The valid input is an integer between 1 and 65535. If an empty string " " is provided for this option, the port number is reset to the default value 21.

[-ftp_user <username>]
Sets or modifies the username of the login account on the primary FTP server.

Note: Specifying "" (empty double quotes) reverts to the default value of onalert.

[-ftp_passwd [<passwd>]]
Sets or modifies the password of the login account on the primary FTP server.

Note: Specifying "" (empty double quotes) reverts to the default value of EMCCONNECT.

[-ftp_folder <path>]
Sets or modifies the path to the folder on the primary FTP server where the event files have to be deposited.

Note: Specifying "" (empty double quotes) reverts to the default value of incoming.

[-ftp_ipprotocol {IPV4|IPV6}]
Sets or modifies the transfer mode of the primary FTP transport mechanism. If an IPv4 address is provided to FTP server, the corresponding IP protocol is changed to IPv4 automatically. If an IPv6 address is used, the IP protocol is changed to IPv6. When hostname is specified, no IP protocol change is made.

[-ftp_mode {active|passive}]
Sets or modifies the transfer mode of the primary FTP transport mechanism.

Note: Specifying "" (empty double quotes) reverts to the default value of active.

[-ftp_server_2 <hostname>[<ip_addr>]]
Sets or modifies the hostname or IP address of the secondary FTP server and corresponding port. The allowable input is IPv4 address, IPv6 address, or domain name.

[-ftp_port_2 <port>]
Sets or modifies the port of the secondary FTP server and corresponding port. The valid input is an integer between 1 and 65535. If an empty string "" is provided for this option, the port number is reset to the default value of 21.

[-ftp_user_2 <username>]
Sets or modifies the username of the login account on the secondary FTP server.

Note: Specifying "" (empty double quotes) reverts to the default value of onalert.

[-ftp_passwd_2 [<passwd>]]
Sets or modifies the password of the login account on the secondary FTP server.

Note: Specifying "" (empty double quotes) reverts to the default value of EMCCONNECT.

[-ftp_folder_2 <path>]
Sets or modifies the path of the folder on the secondary FTP server where the event files have to be deposited.

Note: Specifying "" (empty double quotes) reverts to the default value of incoming.

[-ftp_ipprotocol_2 {IPv4|IPv6}]
Sets or modifies the transfer mode of the secondary FTP transport mechanism.

[-ftp_mode_2 { active|passive}]
Sets or modifies the transfer mode of the secondary FTP transport mechanism.

Note: Specifying "" (empty double quotes) reverts to the default value of active.

[-email_priority {Disabled|1|2|3 }]
Enables email as a Primary, Secondary, or Tertiary transport mechanism. Specifying Disabled removes email as a transport mechanism.

[-email_from <email_addr>]
Sets or modifies the sender’s email address. The maximum number of characters that can be specified is 63.

Note: Specifying "" (empty double qoutes) reverts to the default value of connectemc@emc.com.

[-email_to <email_addr>[,<email_addr>]]
Sets or modifies the destination email addresses that receive the event files. Multiple email addresses can be specified with a comma separating each address. The maximum number of characters that can be specified is 255.

Note: Specifying "" (empty double quotes) reverts to the default value of emailalert@emc.com.

[-email_subject <email_subject>]
Sets or modifies the subject of the email message.

Note: Specifying "" (empty double quotes) reverts to the default value of CallHome Alert.

[-email_server {<hostname>|<ip_addr>}]
Sets or modifies the primary email server that accepts and routes email messages.

Note: Specifying "" (empty double quotes) disables the use of the existing email server for this transport mechanism.

[-email_ipprotocol {IPv4|IPv6}]
Sets or modifies the secondary email server that accepts and routes email messages.

[-email_server_2 {<hostname>|<ip_addr>}]
Sets or modifies the secondary email server that accepts and routes email messages.

Note: Specifying "" (empty double quotes) disables the use of the existing email server for this transport mechanism.

[-email_ipprotocol_2 {IPv4|IPv6}]
Sets or modifies the secondary email server that accepts and routes email messages.

[-esrs_priority {Disabled|1|2|3}]
Enables ESRS as a Primary, Secondary, or Tertiary transport mechanism. Specifying Disabled removes ESRS as a transport mechanism.

[-https_priority {Disabled|1|2|3}]
Enables HTTPS as a Primary, Secondary, or Tertiary transport mechanism. Specifying Disabled removes HTTPS as a transport mechanism.

[-https_url]
The url of the monitoring station.

[-https_ipprotocol {IPv4|IPv6}]
Sets or modifies the transfer mode of the secondary HTTPS transport mechanism.

[-dial_in_number <phone_number>]
Sets or modifies the dial-in phone number of the modem.

Note: Specifying "" (empty double quotes) does not disable the number or restore its default value. The empty string is stored as is.

[-serial_number <serial_number>]
Sets or modifies the VNX serial number, if it was not automatically detected.

Note: Specifying "" (empty double quotes) does not disable the number or restore its default value. The empty string is stored as is.

[-site_id <site_id>]
Sets or modifies the site ID.

Note: Specifying "" (empty double quotes) does not disable the number or restore its default value. The empty string is stored as is.

[-encryption_enabled {yes|no}]
Enables or disables the encryption of the CallHome payload during transmission.

Note: Specifying "" (empty double quotes) reverts to the default value

of yes.

[-dial_in_enabled {yes|no}]
Enables or disables dial-in login sessions.

Note: Specifying "" (empty double quotes) reverts to the default value of yes.

SEE ALSO

Configuring Events and Notifications on VNX for File.

EXAMPLE #1

To display configuration information, type:

```
# /nas/sbin/nas_connecthome -info

ConnectHome Configuration:
Encryption Enabled     = yes
Dial In :
    Enabled            = yes
    Modem phone number = 9123123123
    Site ID            = MY SITE
    Serial number      = APM00054703223
ESRS :
    Priority           = 1
Email :
    Priority           = 1
    Sender Address     = admin@yourcompany.com
    Recipient Address(es) = emailalert@emc.com
    Subject            = CallHome Alert
    Primary :
    Email Server       = backup.mailhub.company.com
    Secondary :
     Email Server =
FTP :
        Priority       = 2
        Primary :
          FTP Server   = 1.2.3.4
          FTP Port     = 22
            FTP User Name = onalert
            FTP Password = **********
            FTP Remote Folder = incoming
            FTP Transfer Mode = active
        Secondary :
            FTP Server = 1.2.4.4
            FTP Port = 22
            FTP User Name = onalert
            FTP Password = **********
            FTP Remote Folder = incoming
            FTP Transfer Mode = active
Modem :
        Priority        = Disabled
        Primary :
           Phone Number =
           BT Tymnet    = no
        Secondary :
           Phone Number =
           BT Tymnet    = no
```

EXAMPLE #2

To test the primary email server, type:

```
# /nas/sbin/nas_connecthome -test -email_1
```

```
        ------------------------------------------------------
        ConnectEMC 2.0.27-bl18 Wed Aug 22 10:24:32 EDT 2007
        RSC API Version: 2.0.27-bl18
        Copyright (C) EMC Corporation 2003-2007, all rights reserved.
        ------------------------------------------------------
Reading configuration file: ConnectEMC.ini.
Run Service begin...
Test succeeded for Primary Email.

EXAMPLE #3
----------

To modify the configuration information, type:

# /nas/sbin/nas_connecthome -modify -esrs_priority 1

       ------------------------------------------------------
       ConnectEMC 2.0.27-bl18 Wed Aug 22 10:24:32 EDT 2007
       RSC API Version: 2.0.27-bl18
       Copyright (C) EMC Corporation 2003-2007, all rights reserved.
       ------------------------------------------------------
Reading configuration file: ConnectEMC.ini.
Run Service begin...
Modify succeeded for Primary ESRS.

--------------------------------------
Last Modified: September 26, 2012 11:15a.m
```

nas_copy

Creates a replication session for a one-time copy of a file system. This command is available with VNX Replicator.

SYNOPSIS

```
nas_copy
        -name <sessionName>
-source
        {-fs {<name>|id=<fsId>}|-ckpt {<ckptName>|id=<ckptId>}
-destination
        {-fs {id=<dstFsId>|<existing_dstFsName>}
        |-pool {id=<dstStoragePoolId>}|<dstStoragePool>}}
        [-storageSystem <dstStorageSerialNumber>]}
[-from_base {<ckpt_name>|id=<ckptId>}]
-interconnect {<name>|id=<interConnectId>}
[-source_interface {<nameServiceInterfaceName>|ip=<ipaddr>}]
[-destination_interface {<nameServiceInterfaceName>|ip=<ipaddr>}]
[-overwrite_destination]
[-refresh]
[-full_copy]
[-background]
```

DESCRIPTION
----

nas_copy from the Control Station on the source side, performs a one-time copy of a source read-only file system or a checkpoint file system.

Note: Depending on the size of the data in the source, this command may take some time to complete. Once a copy session begins, you can monitor it or interrupt it if necessary using the nas_task command. You can list all replication sessions, including copy sessions, using the nas_replicate -list command.

OPTIONS

-name <sessionName> -source -fs {<name>|id=<fsId>|-ckpt { <ckptName>|id=<ckptId>} -destination {-fs {<existing_dstFsName>|id=<dstFsId>[-pool <dstStoragePool>| id=<dstStoragePoolId>}] [-from_base {<ckpt_Name>|id=<ckptId>}] -interconnect {<name>|id=<interConnectId>}
Creates a VNX Replicator session that performs a one-time copy of a source read-only file system or a checkpoint file system.

The session name assigned must be unique for the Data Mover pair as defined by the interconnect. The naming convention <source_fs_or_ckpt_name>_replica<#> is used if a read-only file system or checkpoint at the destination already has the same name as the source. An integer between 1 and 4 is assigned according to how many replicas of that file system or checkpoint already exist.

The -source specifies the name or ID of an existing read-only file system or checkpoint file system as the source for this copy session. This is to be used as a common base for the initial transfer. The checkpoint is identified by checkpoint name or checkpoint file system ID. This option is intended to accommodate upgrade situations to VNX Replicator.

The -destination specifies either an existing destination file system or the storage needed to create the destination file system automatically, as part of the copy operation. An existing destination file system must be read-only and the same size as the source. Specifying a storage pool or ID creates the destination file system automatically, as read-only, using the same name and size as the source file system.

[-storageSystem <dstStorageSerialNumber>]
When the destination file system is to be created from a pool, this

specifies the storage system for the destination file system to reside. Use the nas_storage -list command to obtain the serial number of the storage system.

[-from_base {ckpt_name>|id=<ckptId>}]
Specifies an existing source file system checkpoint to be used as a common base for the initial data transfer. The checkpoint is identified by the checkpoint name or ID.

The -interconnect specifies the local (source) side of an established Data Mover interconnect to use for this copy session. Use the nas_cel -interconnect -list command on the source VNX to list the interconnects available to VNX Replicator sessions.

[-source_interface {<nameServiceInterfaceName>|ip= <ipAddr>}]
Instructs the copy session to use a specific local interface defined for the interconnect on the source VNX instead of selecting the local interface supporting the lowest number of sessions (the default). If this local interface was defined for the interconnect using a name service interface name, specify the name service interface name; if it was defined using an IP address, specify the IP address. The source_interfaces field of the output from the nas_cel -interconnect -info command shows how the source interface was defined. This option does not apply to a loopback interconnect, which always uses 127.0.0.1.

[-destination_interface {<nameServiceInterfaceName>|ip=<ipaddr>}]
Instructs the copy session to use a specific interface defined for the interconnect on the destination VNX instead of selecting the peer interface supporting the lowest number of sessions (the default). If this peer interface was defined for the interconnect using a name service interface name, specify the name service interface name; if it was defined using an IP address, specify the IP address. The destination_interfaces field of the output from the nas_cel -interconnect -info command shows how the peer interface was defined. This option does not apply to a loopback interconnect, which always uses 127.0.0.1.

[-overwrite_destination]
For an existing destination, discards any changes made to the destination object and restores it from the established common base (differential copy). If this option is not specified and an existing destination object contains different content than the established common base, an error is returned.

[-refresh {<name>|id=<session_id>}
Updates a destination checkpoint that has the same name as the copied checkpoint. This option does not refresh the source object; it refreshes only the destination for a existing checkpoint. If you specify this option and no checkpoint exists with the same name, the command returns an error.

[-full_copy]
For an existing destination object, if a common base checkpoint exists, this performs a full copy of the source checkpoint to the destination, instead of a differential copy. If this option is not specified and an existing destination object has different content than the established common base, an error is returned.

[-background]
Executes the command in asynchronous mode. Use the nas_task command to check the status of the command.

SEE ALSO
----

nas_cel, nas_replicate, nas_task.

EXAMPLE #1

To create a one-time copy of a checkpoint file system with session name ufs1_replica1 with the source checkpoint ufs_ckpt1 and destination pool clar_r5_performance on the interconnect NYs3_LAs2, source interface 10.6.3.190, and destination interface 10.6.3.173, type:

```
$ nas_copy -name ufs1_replica1 -source -ckpt ufs1_ckpt1 -destination
-pool clar_r5_performance -interconnect NYs3_LAs2 -source_interface
10.6.3.190 -destination_interface 10.6.3.173

OK
```

EXAMPLE #2

To create a one-time copy of a read-only file system for the session ufs1_replica1 with source file system ufs1 and overwrite an existing destination file system ufs1 on the interconnect NYs3_LAs2, source interface 10.6.3.190, and destination interface 10.6.3.173, type:

```
$ nas_copy -name ufs1_replica1 -source -fs ufs1 -destination -fs ufs1
-interconnect NYs3_LAs2 -source_interface 10.6.3.190 -destination_interface
10.6.3.173 -overwrite_destination

OK
```

EXAMPLE #3

To initiate a differential copy of ufs_ckpt2 to the ufs1_destination file system using ufs1_ckpt1 as the common base, using the -from_base option, type:

```
$ nas_copy -name ufs1_replica1 -source -ckpt -ufs1_ckpt2
-destination -fs ufs1_destination -from_base ufs1_ckpt1
-interconnect NYs3_LAs2

OK
```

Caution: Using the -from_base option overrides any common base that may exist.Ensure that the specified checkpoint represents the correct state of the destination file system.

EXAMPLE #4

To refresh the destination of the replication session ufs1_replica1 for the source checkpoint ufs1_ckpt1 and destination file system ufs1 on the interconnect NYs3_LAs2, type:

```
$ nas_copy -name ufs1_replica1 -source -ckpt ufs1_ckpt1 -destination -fs
ufs1 -interconnect NYs3_LAs2 -refresh

OK
```

EXAMPLE #5

To perform a full copy of the source checkpoint to the destination for the replication session ufs1_replica1 with the source file system ufs1 and destination file system ufs1 on the interconnect NYs3_LAs2, type:

```
$ nas_copy -name ufs1_replica1 -source -fs ufs1 -destination -fs ufs1
-interconnect NYs3_LAs2 -overwrite_destination -full_copy -background

Info 26843676673: In Progress: Operation is still running. Check task id 4177
on the Task Status screen for results.
```

Last Modified: July 13, 2011 11:00 am

nas_cs

Manages the configuration properties of the Control Station.

SYNOPSIS

```
nas_cs
  -info [-timezones]
| -set [-hostname <hostname>]
       [ -nat1_ip4address <ipv4_address> ]
       [ -nat1_ip4netmask <ipv4_netmask> ]
       [ -nat1_ip6address <ipv6_address[/prefix_length]> ]
       [ -nat2_ip4address <ipv4_address> ]
       [ -nat2_ip4netmask <ipv4_netmask> ]
       [ -nat2_ip6address <ipv6_address[/prefix_length]> ]
       [-dns_domain <dns_domain_name>]
       [-search_domain <domain_name>[,...]]
       [-dns_servers <dns_server>[,...]]
       [-session_monitor_timeout <days>]
       [-session_idle_timeout <minutes>]
       [-time <yyyymmddhhmm [ss]>]
       [-timezone <time_zone_str>]
       [-ntp_servers <ntp_server>[,...]]
| -natsync [ -dnssync ]
| -reboot
```

DESCRIPTION
----

nas_cs sets, clears, and lists the Control Station configuration. nas_cs can be used to reboot the Control Station.

OPTIONS

-info [-timezones]
Displays the Control Station configuration. The -timezones option displays all supported time zones on the Control Station.

-set [-hostname <hostname>]
Sets the user configurable parameters of the Control Station configuration. Sets the hostname of the primary Control Station. To specify a hostname, the maximum number of characters is 64, excluding white spaces and dot characters.

[-ip4address {<ipv4_address>]
Sets the IPv4 network address of the primary Control Station. The IPv4 address must be a valid address.

[-ip4netmask <ipv4_netmask>]
Sets the subnet mask for a valid IPv4 network address on the primary Control Station.

[-ip4gateway <ipv4_gateway>]
Sets the IPv4 network address of the gateway machine for the primary Control Station on the network. The IPv4 address must be a valid address.

[-ip6address <ipv6_addr[/prefix_length]>]
Sets the IPv6 network address of the primary Control Station. The IPv6 address must be a valid address. The /prefix_length option sets the integer value, between 8 and 128, for the prefix length of the IPv6 address of the primary Control Station.

[-ip6gateway <ipv6_gateway>]
Sets the IPv6 network address of the gateway machine for the primary Control Station on the network. The IPv6 address must be a valid address.

[-dns_domain <dns_domain_name>]

Sets the Domain Name System of which the primary Control Station is a member. It can accept valid domain names.

[-search_domain <domain_name>[,...]]
Sets the order in which DNS domains are searched to resolve a comma-separated list of valid domain names.

[-dns_servers <ip_addr>[,.]]
Sets the IP addresses of the DNS servers of the domain. It is a comma-separated list of valid IPv4 or IPv6 addresses and can have a maximum of three DNS addresses.

[-session_idle_timeout <minutes>]
Sets the timeout period in minutes for an inactive administrator session to become invalid.

[-session_monitor_timeout <days>]
Sets the limit for the number of days until when a valid login is allowed to run queries on the primary Control Station. Any active management session requires a login on the primary Control Station.

[-time <yyyymmddhhmm [ss]>]
Sets the current system date and time in the format <yyyymmddhhmm [ss]>].

[-timezone <time_zone_str>]
Sets a valid time zone value on the primary Control Station.

[-ntp_servers <ip_addr>[,.]]
Sets the IP addresses of the NTP server used by the primary Control Station. It is a comma-separated list of valid IPv4 or IPv6 addresses and can have a maximum of four NTP addresses.

-natsync
[-dnssync]
This operation is added to sync-up NAT1A/2A and DNS information after OCC.

-reboot
Reboots the primary Control Station.

EXAMPLE #1

To display the configuration properties of the primary Control Station, type:

```
$ nas_cs -info

Host name = eng24416
Version = 6.0
Location = system:NS40G:HK1908075100410000|controlStation::0
Status = Ok
Standby location = system:NS40G:HK1908075100410000|controlStation::1
Stand by status = Ok
IPv4 address = 172.24.250.26
IPv4 gateway = 172.24.250.10
IPv4 netmask = 255.255.255.0
IPv6 address = 2002:ac18:af02:f4:20e:cff:fe6e:d524/64
IPv6 gateway = 2002:ac18:af02:f4:20e:cff:fe6e:d527
DNS Domain = eng.lss.emc.com
DNS Domain search order = eng.lss.emc.com,rtp.lab.emc.com
DNS servers = 2002:ac18:af02:f4:20e:cff:fe6e:d526
Session idle timeout = 10 Minutes
Session moniotor timeout = 10 Days
Current Time = Thu Nov 6 07:54:52 EST 2008
NTP Servers = 2002:ac18:af02:f4:20e:cff:fe6e:d529
```

EXAMPLE #2

To set the hostname,nat1_ip4address for the primary Control Station, type:

```
$ nas_cs -set -hostname Ml9q26-cs0 -nat1_ip4address
10.246.124.63
OK
```

EXAMPLE #3

To set the nat1_ip6address for the primary Control Station, type:

```
$ nas_cs -set -nat1_ip6address
2620:0:170:260:16ff:fe5d:535c:2467/64
OK
```

EXAMPLE #4

To set the DNS domain, search domains, and DNS servers for the primary Control Station, type:

```
$ nas_cs -set -dns_domain eng.lss.emc.com -search_domain
lss.emc.com,rtp.lab.emc.com -dns_servers
172.24.175.172,172.24.175.173

OK
```

EXAMPLE #5

To set the session monitor timeout and session idle timeout for the primary Control Station, type:

```
$ nas_cs -set -session_monitor_timeout 2 -session_idle_timeout 30

OK
```

EXAMPLE #6

To set the date, time, timezone, and NTP servers for the primary Control Station, type:

```
$ nas_cs -set -time 200811070205 -timezone
America/New_York -ntp_server 128.221.252.0

OK
```

EXAMPLE #7

To reboot the primary Control Station, type:

```
$ nas_cs -reboot

OK
```

Last modifed: May 14, 2012 11:45 am

nas_dbtable

Displays the table records of the Control Station.

SYNOPSIS
----
nas_dbtable

To execute the command against a database that is on the Data Mover area:

```
        -info -mover <movername> -db <dbname>
        -query <tablename> -mover <movername> -db <dbname>
        -filter {(<fieldname> <operator> <value> [{-and|-or}
         <fieldname>{<|<=|>|>=|=|.CONTAIN.};<value>]...] ]
        -list -mover <movername>
```

DESCRIPTION

Displays the table records of the specified Data Mover. It also filters the records of a particular field, and lists those records by using primary or secondary key values.

To execute the command against a database that is on the Control Station area:

```
        -info -cs_path <cs_pathname> -db <dbname>
        -query <tablename> -cs_path <cs_pathname> -db <dbname>
        -filter {(<fieldname> <operator> <value> [{-and|-or}
                  <fieldname>{<|<=|>|>=|=|.CONTAIN.};<value>]...] ]
        -list -cs_path <cs_pathname>
```

DESCRIPTION

Displays the table records of the Control Station. It also filters the records of a particular field, and lists those records by using primary or secondary key values.

The database located in the Data Mover can be read directly. The backup of the database is read on the Control Station. If the database is inconsistent, the nas_dbtable command allows you to manually verify the backup of the database before restoring it.

The Data Mover table uses the standard XML interface of the administration commands. The application can structure each table data and keys as a set of fields. Each field has a unique name, type, and size.

The table structure is stored in the db.<base name> file. It is backed up and restored with the database. The DBMS reader uses this description of the table structure to read and display the records from the backup database.

DATA MOVER OPTIONS

-info -mover <movername> -db <dbname>
Displays the schema of a table or the list of fields and keys. It also displays the number of records of the table so that the user can know if it is reasonable to dump the entire table.

-query <tablename> -mover <movername> -db <dbname>
Displays the records of a table. Selects the records to display on the value of some fields or secondary keys.

-filter { (<fieldname><operator><value> [{-and|-or}
<fieldname>{<|<=|>|>=|=|.CONTAIN.};<value>]...]]
Filters the records of a particular field, and lists the records using primary or secondary key values. The default with multiple filters is the -and option. Only the = operator is supported in the first implementation.

NOTE: The keys are used when the -and option is used. Multiple fields with the -or option parses the table, and applies a filter on each record.

The <fieldname> argument is the name of a secondary key or field. If the secondary key is declared as a sequence of fields, it is used by specifying either the value of its fields or value. If the secondary key is not declared in the schema, then rename the key and its value as filter.

The <value> argument is the value of the field encoded in character.

CONTROL STATION OPTIONS

-info -cs_path <cs_pathname> -db <dbname>
Displays the schema of a table or the list of fields and keys. It also displays the number of records of the table so that the user can know if it is reasonable to dump the entire table.

-query <tablename> -cs_path <cs_pathname> -db <dbname>
Displays the records of the table. Selects the records to display on the value of some fields or secondary keys.

-filter { (<fieldname><operator><value> [{-and|-or}
<fieldname>{<|<=|>|>=|=|.CONTAIN.};<value>]...]]
Filters the records of a particular field, and lists the records using primary or secondary key values. The default with multiple filters is the -and option. Only the = operator is supported in the first implementation.

NOTE: Keys are used when the -filter option contains all components of the key, and the -and option is used. With the -or option, it is necessary to parse all the records.

The <fieldname> argument is the name of a secondary key or field. If the secondary key is declared as a sequence of fields, it is used by specifying either the value of its fields or the secondary key value. If the secondary key is not declared in the schema, rename the key and its value as filter.

The <value> argument is the value of the field encoded in character.

-list -cs_path <cs_pathname>
Displays the list of databases and tables within a particular directory of the Control Station area.

SEE ALSO
----

server_dbms

EXAMPLE #1

To display the Secmap schema of the Data Mover, type:

$ nas_dbtable -info -mover <movername> -db Secmap

Database identification
=======================

Base Name   = Secmap
Table Name   = Mapping

Primary Key Schema
==================

sid   = SID

Secondary Key Components
========================

xid   = xidType, fxid

Data Schema

```
==========

origin               = Enumeration
                       Unknown    : 0
                       Secmap     : 16
                       Localgroup : 32
                       Etc        : 48
                       Nis        : 64
                       AD         : 80
                       Usrmap     : 96
                       Ldap       : 112
                       Ntx        : 128
xidType              = Enumeration
                       unknown_name : -2
                       unknown_sid  : -1
                       unknown_type : 0
                       user         : 1
                       group        : 2
fxid                 = Unsigned Integer size : 4
cdate                = Date
gid                  = Unsigned Integer size : 4
name                 = String, length container size : 2

EXAMPLE #2
----------

To filter the records of the Secmap schema, type:

$ nas_dbtable -query Mapping -mover <movername> -db Secmap -filter fxid=10011

sid                  = S-1-5-15-2b3be507-6bc5c62-3f32a78a-8cc

origin               = Nis
xidType              = user
fxid                 = 10011
cdate                = Fri Sep 11 17:39:09 2009
gid                  = 107
name                 = DVT2KA\MaxUsers00000011

Record count         = 1
Last key             = 10500000000000051500000007e53b2b625cbc068aa7323fcc080000

------------------------------------------------------------------
Last modified: July 13, 2011 12:55pm
```

nas_devicegroup

Manages an established MirrorView/Synchronous (MirrorView/S) consistency group, also known as a device group.

SYNOPSIS

```
nas_devicegroup
  -list
| -info {<name>|id=<id>|-all} [-sync [yes|no]]
| -acl <acl_value> {<name>|id=<id>}
| -suspend {<name>|id=<id>}
| -resume {<name>|id=<id>}
```

DESCRIPTION
----

nas_devicegroup lists the device group information for a MirrorView/S configuration, gets detailed information about a consistency group, specifies an access control level value for the group, suspends MirrorView/S operations, or resumes operations of the device group.

A MirrorView/S with a VNX configuration involves source and destination VNXs attached to old versions of storage systems. MirrorView/S performs synchronous mirroring of source storage logical units (LUNs) representing production images, where the mirrored LUNs are part of a MirrorView/S consistency group.

On the source VNX, you must be root to issue the -acl, -suspend, and -resume options.

nas_devicegroup must be run from a Control Station in slot 0; it will report an error if run from a Control Station in slot 1.

OPTIONS

-list
Displays a list of available configured MirrorView/S device groups.

-info {<name>|id=<id>|-all} [-sync [yes|no]]
Displays detailed information about the MirrorView/S configuration for a specific device group or for all groups.

[-sync [yes|no]]
The -sync option first synchronizes the Control Station's view with the VNX for block before displaying configuration information. The default is yes.

-acl <acl_value> {<name>|id=<id>}
Sets an access control level value that defines the owner of the storage system, and the level of access allowed for users and groups defined in the access control level table. The nas_acl command provides more information).

CAUTION
The access control level value for the group should not be changed from the default setting. A change in access control level value can prevent MirrorView/S from functioning properly.

-suspend {<name>|id=<id>}
Temporarily halts mirroring from the source to the destination, thereby suspending the link. Changes can still be made to the source LUNs, but are not applied to the destination LUNs until operations are resumed.

-resume {<name>|id=<id>}
Resumes device group operations and restarts mirroring, synchronizing the destination LUNs with the source LUNs.

SEE ALSO

Using MirrorView/Synchronous with VNX for Disaster Recovery, nas_acl, and nas_logviewer.

STORAGE SYSTEM OUTPUT

The number associated with the storage device is dependent on the attached storage system of the system; for MirrorView/S, some VNX for block display a prefix of APM before a set of integers, for example, APM00033900124-0019. The VNX for block supports the following system-defined AVM storage pools for MirrorView/S only: cm_r1, cm_r5_performance, cm_r5_economy, cmata_archive, and cmata_r3.

EXAMPLE #1

To list the configured MirrorView/S device groups that are available, type:

```
$ nas_devicegroup -list

ID name owner storage ID acl type
2 mviewgroup 500 APM00053001549 0 MVIEW
```

EXAMPLE #2

To display detailed information for a MirrorView/S device group, type:

```
$ nas_devicegroup -info mviewgroup

Sync with CLARiiON backend ..... done
name = mviewgroup
description =
uid = 50:6:1:60:B0:60:27:20:0:0:0:0:0:0:0:0
state = Synchronized
role = Primary
condition = Active
recovery policy = Automatic
number of mirrors = 16
mode = SYNC
owner = 500
mirrored disks =
local clarid = APM00053001549
remote clarid = APM00053001552
mirror direction = local -> remote
```

Where:

Value	Definition
Sync with CLARiiON storage system	Indicates that a sync with the VNX for block was performed to retrieve the most recent information. This does not appear if you specify -info -sync no.
name	Name of the device group.
description	Brief description of device group.
uid	UID assigned, based on the system.
state	State of the device group (for example, Consistent, Synchronized, Out-of-Sync, Synchronizing, Scrambled, Empty, Incomplete, or Local Only).
role	Whether the current system is the Primary (source) or Secondary (destination).
condition	Whether the group is functioning (Active), Inactive, Admin Fractured (suspended), Waiting on Sync, System Fractured (which indicates link down), or Unknown.

recovery policy	Type of recovery policy (Automatic is the default and recommended value for group during storage system configuration; if Manual is set, use -resume after a link down failure).
number of mirrors	Number of mirrors in group.
mode	MirrorView mode (always SYNC in this release).
owner	User whom the object is assigned to, and is indicated by the index number in the access control level table. nas_acl provides information.
mirrored disks	Comma-separated list of disks that are mirrored.
local clarid	APM number of local VNX for block storage array.
remote clarid	APM number of remote VNX for block storage array.
mirror direction	On primary system, local to remote (on primary system); on destination system, local from remote.

EXAMPLE #3

To display detailed information about a MirrorView/S device group without synchronizing the Control Station.s view with the VNX for block, type:

```
$ nas_devicegroup -info id=2 -sync no

name = mviewgroup
description =
uid = 50:6:1:60:B0:60:27:20:0:0:0:0:0:0:0:0
state = Consistent
role = Primary
condition = Active
recovery policy = Automatic
number of mirrors = 16
mode = SYNC
owner = 500
mirrored disks =
local clarid = APM00053001549
remote clarid = APM00053001552
mirror direction = local -> remote
```

EXAMPLE #4

To halt operation of the specified device group, as root user, type:

```
# nas_devicegroup -suspend mviewgroup

Sync with CLARiiON backend ..... done
STARTING an MV 'SUSPEND' operation.
Device group: mviewgroup ............ done
The MV 'SUSPEND' operation SUCCEEDED.
done
```

EXAMPLE #5

To resume operations of the specified device group, as root user, type:

```
# nas_devicegroup -resume mviewgroup

Sync with CLARiiON backend ..... done
STARTING an MV 'RESUME' operation.
Device group: mviewgroup ............ done
The MV 'RESUME' operation SUCCEEDED.
done
```

Last modified: May 11, 2011 10:00 am.

nas_disk

Manages the disk table.

SYNOPSIS

```
nas_disk
          -list
        | -delete <disk_name> [[-perm]|[-unbind]]
        | -info {<disk_name>|id=<disk_id>}
        | -rename <old_name> <new_name>
```

DESCRIPTION
----

nas_disk displays a list of known disks and renames, deletes, or
displays information for the specified disk.

OPTIONS

-list
Lists the disk table.
It also displays the new type of device. This device is having the default storag
e
group without any SLO or SRP set.
Example - DSL,R1DSL,R2DSL,R1BDSL,R2BDSL

Note: The ID of the object is an integer and is assigned automatically.
The name of the disk might be truncated if it is too long for the display.
To display the full name, use the -info option with the disk ID.

-delete <disk_name> [[-perm]|[-unbind]
Deletes an entry from the disk table. In a VNX, restores the VNX for
block LUN name to its default value.

Unless -perm is specified, the disk is still identified as a VNX disk
and can be discovered and marked again using server_devconfig.
The -perm option removes the entry from the disk table and deletes
the diskmark. The disk is then available to be deployed for use by
another platform. The -unbind option removes the LUN from the
VNX Storage group (if EMC Access Logix. is enabled). The -unbind
option permanently destroys the LUN and its contents. If this is the
last LUN using a RAID group, then the RAID group will be deleted.

-info {<disk_name>|id=<disk_id>}
Displays information for a specific <disk_name> or <disk_id> such
as size, type, and Access control level (ACL).

-rename <old_name> <new_name>
Renames a disk to <new_name>.

Note: If a VNX for block LUN uses the default name, renames it in the format
VNX_<VNX-hostname>_<lun-id>_<VNX-dvol-name>.

SEE ALSO

VNX System Operations and server_devconfig.

SYSTEM OUTPUT

The number associated with the storage device is dependent on the attached
system. VNX for block display a prefix of alphabetic characters before a
set of integers, for example, FCNTR074200038-0019. Symmetrix systems
display as a set of integers, for example, 002804000190-003C.

EXAMPLE #1

EXAMPLE #1 To list the disk table, type:

```
$ nas_disk -list
id inuse sizeMB storageID-devID type name servers
1 y 22874 000197100127-00001 STD root_disk 1,2
2 y 11619 000197100127-00002 STD root_ldisk 1,2
3 y 2077 000197100127-00008 STD d3 1,2
4 y 2077 000197100127-00009 STD d4 1,2
5 y 4154 000197100127-00006 STD d5 1,2
6 y 65542 000197100127-00007 STD d6 1,2
7 y 17261 000197100127-00021 DSL d7 1,2
8 n 17261 000197100127-00022 DSL d8 1,2
9 n 17261 000197100127-00023 DSL d9 1,2
10 n 17261 000197100127-00024 DSL d10 1,2
11 n 17261 000197100127-00025 DSL d11 1,2
12 n 17261 000197100127-00026 DSL d12 1,2
13 n 17261 000197100127-00027 DSL d13 1,2
14 n 17261 000197100127-00028 DSL d14 1,2
15 y 17261 000197100127-00029 DSL d15 1,2
17 y 17261 000197100127-0002A DSL d17 1,2
```

Where:

Value	Definition
id	ID of the disk (assigned automatically).
inuse	Used by any type of volume or file system.
sizeMB	Total size of disk.
storageID-devID	ID of the system and device associated with the disk.
type	Type of disk contingent on the system attached; CLSTD, CLATA, CMSTD, CLEFD, CMEFD, CMATA, MIXED (indicates tiers used in the pool contain multiple disk types), Performance, Capacity, Extreme_performance, Mirrored_mixed, Mirrored_performance, Mirrored_capacity, and Mirrored_extreme_performance are VNX disk types and STD, BCV, R1BCV, R2BCV, R1STD, R2STD, ATA, R1ATA, R2ATA,BCVA, R1BCA, R2BCA, EFD, FTS, R1FTS, R2FTS, BCVF, R1BCF, R2BCF, BCVMIXED, R1MIXED, R2MIXED, R1BCVMIXED, and R2BCVMIXED are Symmetrix disk types.
name	Name of the disk; 'dd' in a disk name indicates a remote disk.
servers	Servers that have access to this disk.

EXAMPLE #2

To list the disk table for the system with a Symmetrix system, type:

```
$ nas_disk -list

id inuse sizeMB storageID-devID type name servers
1 y 11507 000190100530-00FB STD root_disk 1,2,3,4,5,6,7,8
2 y 11507 000190100530-00FC STD root_ldisk 1,2,3,4,5,6,7,8
3 y 2076 000190100530-00FD STD d3 1,2,3,4,5,6,7,8
4 y 2076 000190100530-00FE STD d4 1,2,3,4,5,6,7,8
5 y 2076 000190100530-00FF STD d5 1,2,3,4,5,6,7,8
6 y 65536 000190100530-04D4 STD d6 1,2,3,4,5,6,7,8
7 n 28560 000190100530-0102 STD d7 1,2,3,4,5,6,7,8
8 n 28560 000190100530-0103 STD d8 1,2,3,4,5,6,7,8
9 n 28560 000190100530-0104 STD d9 1,2,3,4,5,6,7,8
10 n 28560 000190100530-0105 STD d10 1,2,3,4,5,6,7,8
11 n 28560 000190100530-0106 STD d11 1,2,3,4,5,6,7,8
12 n 28560 000190100530-0107 STD d12 1,2,3,4,5,6,7,8
13 n 28560 000190100530-0108 STD d13 1,2,3,4,5,6,7,8
14 n 28560 000190100530-0109 STD d14 1,2,3,4,5,6,7,8
15 n 28560 000190100530-010A STD d15 1,2,3,4,5,6,7,8
16 n 28560 000190100530-010B STD d16 1,2,3,4,5,6,7,8
```

```
17 n 28560 000190100530-010C STD d17 1,2,3,4,5,6,7,8
18 n 28560 000190100530-010D STD d18 1,2,3,4,5,6,7,8
19 n 28560 000190100530-010E STD d19 1,2,3,4,5,6,7,8
20 n 28560 000190100530-010F STD d20 1,2,3,4,5,6,7,8
21 n 28560 000190100530-0110 STD d21 1,2,3,4,5,6,7,8
22 n 28560 000190100530-0111 STD d22 1,2,3,4,5,6,7,8
23 n 28560 000190100530-0112 STD d23 1,2,3,4,5,6,7,8
24 n 28560 000190100530-0113 STD d24 1,2,3,4,5,6,7,8
[....]
155 n 28560 000190100530-0196 STD d155 1,2,3,4,5,6,7,8
156 n 28560 000190100530-0197 STD d156 1,2,3,4,5,6,7,8
157 n 28560 000190100530-0198 BCV rootd157 1,2,3,4,5,6,7,8
158 n 28560 000190100530-0199 BCV rootd158 1,2,3,4,5,6,7,8
159 n 28560 000190100530-019A BCV rootd159 1,2,3,4,5,6,7,8
160 n 28560 000190100530-019B BCV rootd160 1,2,3,4,5,6,7,8
161 n 28560 000190100530-019C BCV rootd161 1,2,3,4,5,6,7,8
162 n 28560 000190100530-019D BCV rootd162 1,2,3,4,5,6,7,8
163 n 28560 000190100530-019E BCV rootd163 1,2,3,4,5,6,7,8
164 n 28560 000190100530-019F BCV rootd164 1,2,3,4,5,6,7,8
165 n 28560 000190100530-01A0 BCV rootd165 1,2,3,4,5,6,7,8
166 n 28560 000190100530-01A1 BCV rootd166 1,2,3,4,5,6,7,8
167 n 28560 000190100530-01A2 BCV rootd167 1,2,3,4,5,6,7,8
168 n 28560 000190100530-01A3 BCV rootd168 1,2,3,4,5,6,7,8
169 n 28560 000190100530-01A4 BCV rootd169 1,2,3,4,5,6,7,8
170 n 28560 000190100530-01A5 BCV rootd170 1,2,3,4,5,6,7,8
171 n 28560 000190100530-01A6 BCV rootd171 1,2,3,4,5,6,7,8
172 n 28560 000190100530-01A7 BCV rootd172 1,2,3,4,5,6,7,8
173 n 28560 000190100530-01A8 BCV rootd173 1,2,3,4,5,6,7,8
174 n 28560 000190100530-01A9 BCV rootd174 1,2,3,4,5,6,7,8
175 n 28560 000190100530-01AA BCV rootd175 1,2,3,4,5,6,7,8
176 n 28560 000190100530-01AB BCV rootd176 1,2,3,4,5,6,7,8
177 n 28560 000190100530-01AC BCV rootd177 1,2,3,4,5,6,7,8
178 n 28560 000190100530-01AD BCV rootd178 1,2,3,4,5,6,7,8
179 n 28560 000190100530-01AE BCV rootd179 1,2,3,4,5,6,7,8
180 n 28560 000190100530-01AF BCV rootd180 1,2,3,4,5,6,7,8
181 n 28560 000190100530-01B0 BCV rootd181 1,2,3,4,5,6,7,8
182 n 28560 000190100530-01B1 BCV rootd182 1,2,3,4,5,6,7,8
183 n 28560 000190100530-01B2 BCV rootd183 1,2,3,4,5,6,7,8
184 n 28560 000190100530-01B3 BCV rootd184 1,2,3,4,5,6,7,8
185 n 28560 000190100530-01B4 BCV rootd185 1,2,3,4,5,6,7,8
186 n 28560 000190100530-01B5 BCV rootd186 1,2,3,4,5,6,7,8
187 n 11507 000190100530-051D EFD d187 1,2,3,4,5,6,7,8
188 n 11507 000190100530-051E EFD d188 1,2,3,4,5,6,7,8
189 n 11507 000190100530-051F EFD d189 1,2,3,4,5,6,7,8
190 n 11507 000190100530-0520 EFD d190 1,2,3,4,5,6,7,8
191 n 11507 000190100530-0521 EFD d191 1,2,3,4,5,6,7,8
192 n 11507 000190100530-0522 EFD d192 1,2,3,4,5,6,7,8
193 n 11507 000190100530-0523 EFD d193 1,2,3,4,5,6,7,8
194 n 11507 000190100530-0524 EFD d194 1,2,3,4,5,6,7,8
195 n 11507 000190100530-0525 EFD d195 1,2,3,4,5,6,7,8
196 n 11507 000190100530-0526 EFD d196 1,2,3,4,5,6,7,8
197 n 11507 000190100530-0527 EFD d197 1,2,3,4,5,6,7,8
198 n 11507 000190100530-0528 EFD d198 1,2,3,4,5,6,7,8
199 n 11507 000190100530-0529 EFD d199 1,2,3,4,5,6,7,8
200 n 11507 000190100530-052A EFD d200 1,2,3,4,5,6,7,8
201 n 11507 000190100530-052B EFD d201 1,2,3,4,5,6,7,8
202 n 11507 000190100530-052C EFD d202 1,2,3,4,5,6,7,8
203 n 11507 000190100530-052D EFD d203 1,2,3,4,5,6,7,8
204 y 11507 000190100530-052E EFD d204 1,2,3,4,5,6,7,8
```

Note: This is a partial listing due to the length of the output.

EXAMPLE #1 provides a description of command outputs.

EXAMPLE #3

To view information for disk d7 for a system with a VNX for block, type:

```
$ nas_disk -info d7
```

```
id = 7
name = d7
acl = 0
in_use = True
pool = TP1
size (MB) = 273709
type = Mixed
protection= RAID5(4+1)
stor_id = FCNTR074200038
stor_dev = 0012
volume_name = d7
storage_profiles = TP1
thin = True
tiering_policy = Auto-tier
compressed= False
mirrored = False
servers = server_2,server_3,server_4,server_5
server = server_2 addr=c0t1l2
server = server_2 addr=c32t1l2
server = server_2 addr=c16t1l2
server = server_2 addr=c48t1l2
server = server_3 addr=c0t1l2
server = server_3 addr=c32t1l2
server = server_3 addr=c16t1l2
server = server_3 addr=c48t1l2
server = server_4 addr=c0t1l2
server = server_4 addr=c32t1l2
server = server_4 addr=c16t1l2
server = server_4 addr=c48t1l2
server = server_5 addr=c0t1l2
server = server_5 addr=c32t1l2
server = server_5 addr=c16t1l2
server = server_5 addr=c48t1l2
```

Where:

Value	Definition
id	ID of the disk (assigned automatically).
name	Name of the disk.
acl	Access control level value of the disk.
in_use	Used by any type of volume or file system.
pool	Name of the storage pool in use.
size (MB)	Total size of the disk.
type	Type of disk contingent on the system attached; VNX for block disk types are CLSTD, CLATA, CMSTD, CLEFD, CLSAS, CMEFD, CMATA, MIXED (indicates tiers used in the pool contain multiple disk types), Performance, Capacity, Extreme_performance, Mirrored_mixed, Mirrored_performance, Mirrored_capacity, and Mirrored_extreme_performance.
protection	The type of disk protection that has been assigned.
stor_id	ID of the system associated with the disk.
stor_dev	ID of the device associated with the disk.
volume_name	Name of the volume residing on the disk.
storage_profiles	The storage profiles to which the disk belongs.
thin	Indicates whether the block system uses thin provisioning. Values are: True, False.
tiering_policy	Indicates the tiering policy in effect. If the initial tier and the tiering policy are the same, the values are: Auto-Tier, Highest Available Tier, Lowest Available Tier. If the initial tier and the tiering policy are not the same, the values are: Auto-Tier/No Data Movement, Highest Available Tier/No Data Movement, Lowest Available Tier/No Data Movement.

```
        compressed              For VNX for block, indicates whether data is compressed.
                                Values are: True, False, Mixed (indicates some of the LU
Ns,
                                but not all, are compressed).
        mirrored                Indicates whether the disk is mirrored.
        servers                 Lists the servers that have access to this disk.
        addr                    Path to system (SCSI address).

        EXAMPLE #4
        ----------

        To view information for disk d205 for the system with a Symmetrix system, type:

        $ nas_disk -info d205

        id = 205
        name = d205
        acl = 0
        in_use = True
        pool = SG0
        size (MB) = 28560
        type = Mixed
        protection= RAID1
        symm_id = 000190100530
        symm_dev = 0539
        volume_name = d205
        storage_profiles = SG0_000192601245
        thin = True
        tiering_enabled = True
        compression = True
        mirrored = False
        servers =
        server_2,server_3,server_4,server_5,server_6,server_7,server_8,server_9
        server = server_2 addr=c0t14l0 FA=03A FAport=0
        server = server_2 addr=c16t14l0 FA=04A FAport=0
        server = server_3 addr=c0t14l0 FA=03A FAport=0
        server = server_3 addr=c16t14l0 FA=04A FAport=0
        server = server_4 addr=c0t14l0 FA=03A FAport=0
        server = server_4 addr=c16t14l0 FA=04A FAport=0
        server = server_5 addr=c0t14l0 FA=03A FAport=0
        server = server_5 addr=c16t14l0 FA=04A FAport=0
        server = server_6 addr=c0t14l0 FA=03A FAport=0
        server = server_6 addr=c16t14l0 FA=04A FAport=0
        server = server_7 addr=c0t14l0 FA=03A FAport=0
        server = server_7 addr=c16t14l0 FA=04A FAport=0
        server = server_8 addr=c0t14l0 FA=03A FAport=0
        server = server_8 addr=c16t14l0 FA=04A FAport=0
        server = server_9 addr=c0t14l0 FA=03A FAport=0
        server = server_9 addr=c16t14l0 FA=04A FAport=0

        Where:

        Value                   Definition
        -----                   ----------

        id                      ID of the disk (assigned automatically).
        name                    Name of the disk.
        acl                     Access control level value of the disk.
        in_use                  Used by any type of volume or file system.
        pool                    Name of the storage pool in use.
        size (MB)               Total size of disk.
        type                    Type of disk contingent on the system attached;
                                Symmetrix disk types are STD, BCV, R1BCV, R2BCV,
                                R1STD, R2STD, ATA, R1ATA, R2ATA, BCVA, R1BCA, R2BCA,
                                EFD, FTS, R1FTS, R2FTS, BCVF, R1BCF, R2BCF, BCVMIXED,
                                R1MIXED, R2MIXED, R1BCVMIXED, and R2BCVMIXED. If
                                multiple disk volumes are used, the type is Mixed.
        protection              The type of disk protection that has been assigned.
        symm_id                 ID of the Symmetrix system associated with the disk.
        symm_dev                ID of the Symmetrix device associated with the disk.
        volume_name             Name of the volume residing on the disk.
```

storage_profiles	The storage profiles to which the disk belongs.
thin	Indicates whether the system uses thin provisioning. Values are: True, False, Mixed.
tiering_enabled	Indicates whether a tiering policy is being used.
compressed	For VNX with Symmetrix backend, indicates whether data is compressed. Values are:True, False, Mixed (indicates some of the LUNs, but not all, are compressed).
mirrored	Indicates whether the disk is mirrored.
servers	Lists the servers that have access to this disk.
addr	Path to system (SCSI address).

EXAMPLE #5

To view information for disk d17 (FTS device created using eDisk configured in external provisioning mode) for the system with a Symmetrix system, type:

```
$ nas_disk -info id=17
id = 17
name = d17
acl = 0
in_use = True
pool = user_pool
size (MB) = 17261
type = DSL
protection= TDEV
symm_id = 000197100127
symm_dev = 0002A
volume_name = d17
storage_profiles = symm_dsl
thin = True
compressed= False
mirrored = False
servers = server_2,server_3
server = server_2 addr=c0t1l9
server = server_2 addr=c16t1l9
server = server_2 addr=c32t1l9
server = server_2 addr=c48t1l9
server = server_3 addr=c0t1l9
server = server_3 addr=c16t1l9
server = server_3 addr=c32t1l9
server = server_3 addr=c48t1l9
```

EXAMPLE #4 provides a description of command outputs.

EXAMPLE #6

To rename a disk in the system with a VNX for block, type:

```
$ nas_disk -rename d7 disk7

id = 7
name = disk7
acl = 0
in_use = True
size (MB) = 273709
type = CLSTD
protection= RAID5(4+1)
stor_id = FCNTR074200038
stor_dev = 0012
volume_name = disk7
storage_profiles = clar_r5_performance
virtually_provisioned = False
mirrored = False
servers = server_2,server_3,server_4,server_5
server = server_2 addr=c0t1l2
server = server_2 addr=c32t1l2
server = server_2 addr=c16t1l2
server = server_2 addr=c48t1l2
server = server_3 addr=c0t1l2
```

```
server = server_3 addr=c32t1l2
server = server_3 addr=c16t1l2
server = server_3 addr=c48t1l2
server = server_4 addr=c0t1l2
server = server_4 addr=c32t1l2
server = server_4 addr=c16t1l2
server = server_4 addr=c48t1l2
server = server_5 addr=c0t1l2
server = server_5 addr=c32t1l2
server = server_5 addr=c16t1l2
server = server_5 addr=c48t1l2
```

EXAMPLE #4 provides a description of command outputs.

EXAMPLE #7

To delete a disk entry from the disk table for the system with a VNX for block, type:

```
$ nas_disk -delete d24

id = 24
name = d24
acl = 0
in_use = False
size (MB) = 456202
type = CLATA
protection= RAID5(6+1)
stor_id = FCNTR074200038
stor_dev = 0023
storage_profiles = clarata_archive
virtually_provisioned = False
mirrored = False
servers = server_2,server_3,server_4,server_5
```

EXAMPLE #4 provides a description of command outputs.

Last Modified: Jan 11, 2013 3:17 pm

nas_diskmark

Queries the system, manages and lists the SCSI devices configuration.

SYNOPSIS

```
nas_diskmark
          -mark {-all|<movername>} [-discovery {y|n}] [-monitor {y|n}]
              [-Force {y|n}]
        | -list {-all|<movername>}
```

DESCRIPTION
----

nas_diskmark queries the available system device and tape device configuration; saves the device configuration into the Data Movers database, and lists SCSI devices. This command also manages NAS database configuration related to advanced data services from back-end storage system.

CAUTION
It is recommended that all Data Movers have the same device configuration. When adding devices to the device table for a single Data Mover only, certain actions such as standby failover are not successful unless the standby Data Mover has the same disk device configuration as the primary Data Mover.

The -all option executes the command for all Data Movers.

LUN migration for VNX Symmetrix systems

When a newly created LUN having an ID same as that of some removed device is detected, the command may report a conflict error. After a LUN is removed at the backend and then a new LUN is created with the same ID, Control Station cannot be aware of its deletion at first. The error occurs because the new LUN has duplicate storage ID and device ID with stale disk volume. This case only applies in Symmetrix backend.
For example, During LUN migration, where a Symmetrix device is moved from source storage group (SG) to destination SG, the LUN ID of this device in the source SG should be maintained even in the destination SG. Else, this will reflect in a conflict error on the Control Station during running nas_diskmark.

OPTIONS

-mark {-all|<movername>}
Queries SCSI devices and saves them into the device table database on the Data Mover.

Modifies VNX for block LUN names to the VNX_<VNX-hostname>_<lun-id>_<VNX-dvol-name> format, if the LUNs use the default Unisphere name.

CAUTION
The time taken to complete this command may be lengthy, depending on the number and type of attached devices.

[-discovery {y|n}]
Enables or disables the storage discovery operation.

CAUTION
Disabling the -discovery option should be done only under the direction of an EMC Customer Service Engineer.

[-monitor {y|n}]
Displays the progress of the query and discovery operations.

[-Force {y|n}]
Overrides the health check failures and changes the storage configuration.

CAUTION
Use the -Force option only under the direction of an EMC Customer Service Engineer, as high availability can be lost when changing storage configuration.

-list {-all|<movername>}
Lists the SCSI devices for the specified Data Mover or all Data Movers.

EXAMPLE #1

To query SCSI devices on server_2 and display the progress of the query operation, type:

```
$ nas_diskmark -mark server_2 -monitor y

Discovering storage (may take several minutes)
server_2:
chain 0 ..........
chain 16 ........
chain 32 ........
chain 48 ..........
chain 96 ..........
chain 112 ..........
Verifying disk reachability
Verifying file system reachability
Verifying local domain
Verifying disk health
Verifying gate keepers
Verifying device group
done
```

EXAMPLE #2

To list the SCSI devices for server_2, type:

```
$ nas_diskmark -list server_2

server_2 : chain 0 :
chain= 0, scsi-0
stor_id= HK190807090011 VNX_id= HK1908070900110032
tid/lun= 0/0 type= disk sz= 11263 val= 1 info= DGC RAID 5 03243200000032NI
tid/lun= 0/1 type= disk sz= 11263 val= 2 info= DGC RAID 5 03243300010033NI
tid/lun= 0/2 type= disk sz= 2047 val= 3 info= DGC RAID 5 03243400020034NI
tid/lun= 0/3 type= disk sz= 2047 val= 4 info= DGC RAID 5 03243500030035NI
tid/lun= 0/4 type= disk sz= 2047 val= 5 info= DGC RAID 5 03243600040036NI
tid/lun= 0/5 type= disk sz= 32767 val= 6 info= DGC RAID 5 03243700050037NI
tid/lun= 1/0 type= disk sz= 274811 val= 7 info= DGC RAID 5 03244400100044NI
tid/lun= 1/1 type= disk sz= 274811 val= -5 info= DGC RAID 5 03244500110045NI
tid/lun= 1/2 type= disk sz= 274811 val= 8 info= DGC RAID 5 03244600120046NI
tid/lun= 1/3 type= disk sz= 274811 val= -5 info= DGC RAID 5 03244700130047NI
tid/lun= 1/4 type= disk sz= 274811 val= 9 info= DGC RAID 5 03245600140056NI
tid/lun= 1/5 type= disk sz= 274811 val= -5 info= DGC RAID 5 03245700150057NI
tid/lun= 1/6 type= disk sz= 274811 val= 10 info= DGC RAID 5 03245800160058NI
tid/lun= 1/7 type= disk sz= 274811 val= -5 info= DGC RAID 5 03245900170059NI
tid/lun= 1/8 type= disk sz= 274811 val= 99 info= DGC RAID 5 03245A0018005ANI
tid/lun= 1/9 type= disk sz= 274811 val= -5 info= DGC RAID 5 03245B0019005BNI
tid/lun= 1/10 type= disk sz= 274811 val= 97 info= DGC RAID 5 03245C001A005CNI
tid/lun= 1/11 type= disk sz= 274811 val= -5 info= DGC RAID 5 03245D001B005DNI
tid/lun= 1/12 type= disk sz= 274811 val= 13 info= DGC RAID 5 03245E001C005ENI
```

```
tid/lun= 1/13 type= disk sz= 274811 val= -5 info= DGC RAID 5 03245F001D005FNI
tid/lun= 1/14 type= disk sz= 274811 val= 14 info= DGC RAID 5 032460001E0060NI
tid/lun= 1/15 type= disk sz= 274811 val= -5 info= DGC RAID 5 032461001F0061NI
server_2 : chain 1 :
no drives on chain
server_2 : chain 2 :
no drives on chain
server_2 : chain 3 :
no drives on chain
server_2 : chain 4 :
no drives on chain
server_2 : chain 5 :
no drives on chain
server_2 : chain 6 :
no drives on chain
server_2 : chain 7 :
no drives on chain
server_2 : chain 8 :
no drives on chain
server_2 : chain 9 :
no drives on chain
server_2 : chain 10 :
no drives on chain
server_2 : chain 11 :
no drives on chain
server_2 : chain 12 :
no drives on chain
server_2 : chain 13 :
no drives on chain
server_2 : chain 14 :
no drives on chain
server_2 : chain 15 :
no drives on chain
Note: This is a partial listing due to the length of the output.
```

Last Modified: Feb 21, 2013 11:00 am

nas_emailuser

Manages email notifications for serious system events.

SYNOPSIS

```
nas_emailuser
-info
| -test
| -modify
   [-enabled {yes|no}]
   [-to <email_addr> [,...]]
   [-cc <email_addr> [,...]]
   [-email_server <email_server>]
   [-subject_prefix <email_subject>]
   [-from <email_addr>]
| -init
```

DESCRIPTION

nas_emailuser enables, configures, and tests email notifications for serious system events.

OPTIONS

-info
Displays the configuration for email notifications.

-test
Generates a test event that sends a test email notification to the email addresses configured in -to and -cc. The recipient email address must be configured prior to testing email notification.

Note: After the -test option is run, all the configured recipients must be asked to confirm whether they received the test email with the correct system identification information.

-modify
Modifies one or more of the following configuration parameters:

[-enabled {yes|no}]
Enables email notification if yes is specified. The recipient email address must be configured prior to enabling email notification. Disables email notification if no is specified.

[-to <email_addr> [,...]]
Configures one or more recipient email addresses. The email addresses are comma-separated, enclosed in single-quotes, and follow the mailbox@fully_qualified_domain_name format. For example, storage_admin@yourcompany.com, backup_admin@yourcompany.com.

Refer the following email address format guidelines for configuring email addresses. An email address can contain:

* A maximum of 63 characters; the field can contain a maximum of 255 characters:

* ASCII characters: a through z, A through Z, 0 through 9, ! # $ % & * + - / = ? ^ _ ` {|,} ~ are allowed; a period, if it is not the first or last character in the mailbox

* Alphanumeric strings

* Single quotes, if they are escaped in the format:
  - your\'email@yourcompany.com

-'first'\''email@yourcompany.com,second'\''email@yourcompany.com'

[-cc <email_addr> [,...]]
Configures a list of carbon-copy recipients. The email addresses are comma-separated, enclosed in single-quotes, and follow the mailbox@fully_qualified_domain_name format. For example, 'storage_admin@yourcompany.com'. For the email address character set and format guidelines, refer the -to option.

[-email_server <email_server>]
Configures the email server that accepts and routes the email notifications. <email_server> specifies an IP address or the fully qualified domain name, which can have 1 to 63 characters. The IP addresses 0.0.0.0 and 255.255.255.255 are not allowed.

[-subject_prefix <email_subject>]
Specifies the email subject prefix. The subject prefix for the email notification can be from 1 to 63 characters long, is enclosed in quotes, and should contain printable ASCII characters. You can customize the subject prefix for specific needs like email filtering. The default subject is "System Notification."

[-from <email_addr>]
Configures the sender's email address. If the sender's email address is not specfied, a default email address of the format root@<hostname> is configured. The email address follows the mailbox@fully_qualified_domain_name format. For example, 'storage_admin@yourcompany.com'. For the email address character set and format guidelines, refer the -to option.

-init
Initializes the default state; displays a status message if the feature has already been initialized. The -init option must be used only when directed.

SEE ALSO

Configuring Events and Notifications on VNX for File.

EXAMPLE #1

To configure email notifications using email server 10.6.50.122 from administrator to support, while copying engineering and documentation, type:

```
$ nas_emailuser -modify -to
szg30@fire2.hosts.pvt.dns,support1@nasdocs.emc.com,documentation@nasdocs.emc.com
OK
```

EXAMPLE #2

To display information on email notifications, type:

```
$ nas_emailuser -info
Service Enabled         = Yes
Recipient Address(es)   =
szg30@fire2.hosts.pvt.dns,support1@nasdocs.emc.com,documentation@nasdocs.emc.com
Carbon copy Address(es) =
Email Server            = 10.241.168.23
Subject Prefix          = System Notification
Sender Address          =
```

EXAMPLE #3

To test email notifications, type:

```
$ nas_emailuser -test
```

```
OK

EXAMPLE #4
----------

To disable email notification, type:

$ nas_emailuser -modify -enabled no
OK

----------------------------------------
Last Modified: May 14, 2012 1:00 pm
```

nas_environment

[Nas_environment command is not supported by Embedded NAS]
Reports the inlet air temperatures and input power to the user.

SYNOPSIS

```
nas_environment -info
{
| -system [-present|-average]
| -dme [enclosure_id] [-intemp [f|c]|-power] [-present]|[-average]
| -array [-present|-average]
| -shelf {<shelf_id>|<-all>}[-intemp [f|c]|-power][-present|-average]
| -battery [a|b] [-present|-average]
| -spe [-intemp [f|c]|-power] [-present|-average]
| -all
}
```

DESCRIPTION
----

nas_environment -info displays the inlet air temperatures of the data mover enclosures, disk array enclosures, the input power of the data mover enclosures, disk array enclosures, and standby power supply through the CLI and Unisphere GUI.

OPTIONS

-system
Displays the present or average input power information of the system, which includes file and block on VNX systems, and file only on gateway systems.

-present
Displays the current value, which is a sum of the present input power for all supported systems. The current value is computed as the 30 second average of the power consumption sampled every three seconds.

-average
Displays the average value. It requires an hour to calculate the correct value. N/A is displayed if there is less than one hour worth of data. The average value is computed as the 60 minute rolling average of the present power consumption values.

-dme
Displays the present or average inlet air temperature and/or input power information on a specified data mover enclosure. If a specific enclosure_id is not specified, all data mover enclosure information is displayed.

enclosure_id
Specifies a data mover enclosure id on which to display information.

-intemp [f|c]
Displays the inlet air temperature information. The f flag indicates Fahrenheit. The default value or c flag indicates Celsius.

-power
Displays the input power information.

-present
Displays the current value.

-average
Displays the average value. It requires an hour to calculate the correct value. N/A is displayed if

there is less than one hour worth of data.

-array
Displays the present or average input power information on the array.

-present
Displays the current value.

-average
Displays the average value. It requires an hour to calculate the correct value. N/A is displayed if there is less than one hour worth of data.

-shelf
Allows to input a value for a selected enclosure. It displays the present and average inlet air temperature and input power information on a specified disk array enclosure. If a specific enclosure_id is not specified, all disk array enclosure information is displayed.

<shelf_id>
Specifies a disk array enclosure id on which to display information.

-intemp f|c
Displays the inlet air temperature information. The f flag indicates Fahrenheit. The default value or c flag indicates Celsius.

-power
Displays the input power information.

-present
Displays the current value.

-average
Displays the average value. It requires an hour to calculate the correct value. N/A is displayed if there is less than one hour worth of data.

-battery
Displays the present and average input power information on a specified standby power supply. If no -a or -b is specified, then the information is displayed on both standby power supplies.

-a
Specifies a standby power supply A on which to display information.

-b
Specifies a standby power supply B on which to display information.

-present
Displays the current value.

-average
Displays the average value. It requires an hour to calculate the correct value. N/A is displayed if there is not one hour worth of data.

-spe
Displays the present and average input power information on a specified standby power supply.

-intemp [f|c]
Displays the inlet air temperature information. The f flag indicates Fahrenheit. The default value or c flag indicates Celsius.

-power
Displays the input power information.

-present
Displays the current value.

-average

Displays the average value. It requires an hour to calculate the correct value. N/A is displayed if there is less than one hour worth of data.

-all
Displays the following:
* System input power
* Data mover enclosure inlet air temperatures and input power
* Array input power
* Disk array enclosure inlet air temperatures and input power
* Storage processor enclosure inlet air temperatures and input power
* Standby power supply input power

Expected Output For Embedded Nas

```
[nasadmin@CS-0 ~]$ nas_environment -info -system
Component Name            = VMAX IN-EE-NAS-SN 00019710012200013
Power Status              = Error 13690667103: Unsupported
Present (watts)           = N/A
Rolling Average (watts)   = N/A

[nasadmin@CS-0 ~]$ nas_environment -info -all
Component Name            = VMAX IN-EE-NAS-SN 00019710012200013
Power Status              = Error 13690667103: Unsupported
Present (watts)           = N/A
Rolling Average (watts)   = N/A

Component Name            = Symmetrix VMAX200K 000197100122
Power Status              = Error 13690667103: Unsupported
Present (watts)           = N/A
Rolling Average (watts)   = N/A

[nasadmin@CS-0 ~]$ nas_environment -info -array -present
Component Name            = Symmetrix VMAX200K 000197100122
Power Status              = Error 13690667103: Unsupported
Present (watts)           = N/A

[nasadmin@CS-0 ~]$ nas_environment -info -spe -present
Error 14764736517: No SPE found.
```

nas_event

Provides a user interface to system-wide events.

SYNOPSIS

```
nas_event
  -Load {-info|<file_name>}
| -Unload <file_name>
| -list
         -action {-info|{trap|logfile|mail|callhome|exec|udprpc|tcprpc|terminate}
           [-component {<component>|<id>}][-facility {<facility>
           | <id>}][-severity {<severity>|<id>}]]|[-id]}
        |-component {-info|{<component>|<id>} [-facility {<facility>|<id>}]|[-id]
}
        |-severity {-info|<severity>|<id>} [-component {<component>|<id>}
           [-facility {<facility>|<id>}]]|[-id]
        |-keyword <keyword> [-component {<component>|<id>}
           [-facility{<facility>|<id>}][-severity {<severity>|<id>}]]|[-id]
```

DESCRIPTION
----

nas_event provides a user interface to system-wide events within the VNX. The VNX includes a default event configuration file that contains a mapping of facilities that generate events, and the associated actions triggered by those events.

To list the default configuration files, type:

```
$ nas_event -Load -info
```

Using a text editor, a new event configuration file can be created and loaded into the database to add an event, or change an action.

VNX facilities generate events that trigger specific actions. An event consists of:

. An assigned ID for the event and the facility issuing the event
. The high water mark for the severity of the event
. A description of the event
. The system-defined action to take when the event occurs

CAUTION
The callhome events in the system are carefully reviewed and configured to provide the right level of support. Do not add, delete, or modify lines that specify the callhome action in the event configuration files in the /nas/sys directory. User-defined event configuration files should not use the callhome action.

OPTIONS

-Load {-info|<file_name>}
Loads the event configuration file <file_name> into the system. The -info option displays the currently loaded configuration files.

-Unload <file_name>
Unloads the configuration file <file_name> from the system.

CAUTION
The /nas/sys/nas_eventlog.cfg configuration file must not be deleted, as this can lead to data loss. Unloading or modifying configuration files that are loaded by default can affect logging, alerts, notifications, and system operations.

-list
The -list option displays components and facilities that generate events, and the actions that are taken as a result. You can search for an event, facility, or action by using a keyword. Component, facility, and

severity can be specified by either the text name or ID. The output is displayed with parameter names in the form $(paraname, typeIndicator, fmtStr).

-action
{-info|{trap|logfile|mail|callhome|exec|udprpc|tcprpc|terminate}
With the -info option, lists all the possible actions associated with events. If one of the actions trap, logfile, mail, callhome, exec, udprpc, tcprpc, or terminate is specified, lists the possible events that trigger the specified action. These events are categorized by component and facility:

[-component {<component>|<id>}][-facility {<facility>|<id>}]
Lists the possible events in the specified component that trigger the given action. If facility is specified, lists the events in the specified component and facility that trigger the given action.

[-severity {<severity>|<id>}]
Lists the possible events with the specified severity that trigger the given action.

[-id]
Lists the output with the MessageID number in addition to BaseID, Severity, and Brief_Description.

-component {-info|{<component>|<id>}[-facility{<facility>
|<id>}]
With the -info option, lists the ids and names of all the components. If the component is specified, lists the ids and names of all the facilities under that component. Specifying facility lists the events that can be generated by the specified facility and component.

[-id]
Lists the output with the message ID number in addition to BaseID and Brief_Description.

-severity {-info|<severity>|<id>}]
With the -info option, lists the severity levels. If severity is specified, lists the events with the specified severity level.

[-component {<component>|<id>} [-facility <facility>|<id>]
Lists the events filtered by the given severity and component.

If facility is specified lists the events further filtered by the given facility.

[-id]
Lists the output with the MessageID number in addition to BaseID, Severity, and Brief_Description.

Note: To receive email notifications sent to multiple recipients, specify the email addresses within the quotes and separate them w
ith
a comma.

-keyword <keyword>
Lists all events that match the specified keyword.

[-component {<component>|<id>}][-facility{<facility> |<id>}]
Lists events filtered by the specified keyword and component.

If facility is specified, lists the events further filtered by the given facility.

[-severity {<severity>|<id>}]
Lists events filtered by the specified severity.

[-id]
Lists the output with the MessageID number in addition to

BaseID, Severity, and Brief_Description.

SEE ALSO

Configuring Events and Notifications on VNX for File.

EXAMPLE #1

After using a text editor to create an event configuration file, to load the new configuration file into the NAS database, type:

```
$ nas_event -Load /nas/site/new_eventlog.cfg

EventLog : will load /nas/site/new_eventlog.cfg...done
```

EXAMPLE #2

To verify that the configuration file was loaded, type:

```
$ nas_event -Load -info

Loaded config. files:
1: /nas/sys/nas_eventlog.cfg
2: /nas/http/webui/etc/web_client_eventlog.cfg
3: /nas/site/new_eventlog.cfg
```

EXAMPLE #3

To list actions, type:

```
$ nas_event -list -action -info
action
terminate
trap
exec
mail
callhome
logfile
```

EXAMPLE #4

To list the events that trigger the mail action, type:

```
$ nas_event -list -action mail
CS_PLATFORM(6)
|--> EventLog(130)
BaseID Severity Brief_Description
50 EMERGENCY(0) ${text,8,%s}
51 ALERT(1) ${text,8,%s}
52 CRITICAL(2) ${text,8,%s}
```

EXAMPLE #5

To list the components, type:

```
$ nas_event -list -component -info
Id       Component
1        DART
2        CS_CORE
5        XML_API
6        CS_PLATFORM
```

EXAMPLE #6

To list the facilities under the component DART, type:

```
$ nas_event -list -component DART -info

DART(1)
|->Id Facility
24 ADMIN
26 CAM
27 CFS
36 DRIVERS
40 FSTOOLS
43 IP
45 KERNEL
46 LIP
51 NDMP
52 NFS
54 SECURITY
56 SMB
58 STORAGE
64 UFS
68 LOCK
70 SVFS
72 XLT
73 NETLIB
75 MGFS
77 VRPL
78 LDAP
81 VC
83 RCPD
84 VMCAST
86 CHAMII
93 USRMAP
101 ACLUPD
102 FCP
108 REP
111 DPSVC
115 SECMAP
117 WINS
118 DNS
122 DBMS
144 PERFSTATS
146 CEPP
148 DEDUPE
```

EXAMPLE #7

To list the events generated by DART in the facility with the ID 146, type:

```
$ nas_event -list -component DART -facility 146

DART(1)
|--> CEPP(146)
BaseID Severity Brief_Description
1 NOTICE(5) CEPP server ${ipaddr,8,%s} of pool ${pool,8,%s} is
${status,8,%s}. Vendor ${vendor,8,%s}, ntStatus
0x${ntstatus,2,%x}.
2 ERROR(3) Error on CEPP server ${ipaddr,8,%s} of pool
${pool,8,%s}: ${status,8,%s}. Vendor ${vendor,8,%s},
ntStatus 0x${ntstatus,2,%x}.
3 NOTICE(5) The CEPP facility is started.
4 NOTICE(5) The CEPP facility is stopped.
```

EXAMPLE #8

To list events with severity 4 generated by component CS_CORE and facility DBMS, and to display the MessageID in the output, type:

```
$ nas_event -list -severity 4 -component CS_CORE -facility DBMS -id
```

```
CS_CORE(2)
|--> DBMS(122)
MessageID BaseID Brief_Description
86444212226 2 Db: Compact${compact_option,8,%s}: ${db_name,8,%s}:
Failed: ${db_status,8,%s}.
86444212227 3 Db Env: ${db_env,8,%s}: Log Remove: Failed:
${db_status,8,%s}.
```

EXAMPLE #9

To list events filtered by the keyword freeblocks, type:

```
$ nas_event -list -keyword freeblocks

DART(1)
|--> DBMS(122)
BaseID Severity Brief_Description
2 CRITICAL(2) Only ${freeblocks,3,%llu} free blocks in the root
file system (fsid ${fsid,2,%u}) of the VDM
${vdm,8,%s}.
3 ALERT(1) The root file system (fsid ${fsid,2,%u}) of the
VDM ${vdm,8,%s} is full. There are only
${freeblocks,3,%llu} free blocks.
```

EXAMPLE #10

To list events with the keyword data generated in DART with the severity level 6, type:

```
$ nas_event -list -keyword data -component DART -severity 6

DART(1)
|--> USRMAP(93)
BaseID Severity Brief_Description
1 INFO(6) The Usermapper database has been created.
4 INFO(6) The Usermapper database has been destroyed.
8 INFO(6) The migration of the Usermapper database to the
VNX version 5.6 format has
started.
9 INFO(6) The Usermapper database has been successfully
migrated.
DART(1)
|--> SECMAP(115)
BaseID Severity Brief_Description
1 INFO(6) The migration of the secmap database to the VNX version
5.6 format has started.
2 INFO(6) The secmap database has been successfully migrated.
```

EXAMPLE #11

To unload the event configuration file, type:

```
$ nas_event -Unload /nas/site/new_eventlog.cfg

EventLog : will unload /nas/site/new_eventlog.cfg... done
```

EXAMPLE #12

To receive email notifications that are sent to multiple recipients, add the following line to your /nas/sys/eventlog.cfg file:

```
disposition severity=0-3, mail "nasadmin@nasdocs.emc.com,
helpdesk@nasdocs.emc.com"
```

EXAMPLE #13

To list the events that trigger a particular trap action, type:

```
$ nas_event -l -a trap | more
CS_PLATFORM(6)
|--> BoxMonitor(131)
BaseID Severity Brief_Description
1 CRITICAL(2) EPP failed to initialize.
3 CRITICAL(2) Failed to create ${threadname,8,%s} thread.
4 CRITICAL(2) SIB Read failure: ${string,8,%s}
..
CS_PLATFORM(6)
|--> SYR(143)
BaseID Severity Brief_Description
5 INFO(6) The SYR file ${src_file_path,8,%s} with
${dest_extension,8,%s} extension is attached.
```

Last modified: May 14, 2012 1:35 pm

nas_fs

Manages local file systems for the VNX.

SYNOPSIS

```
nas_fs
  -list [-all]
| -delete <fs_name> [-option <options>][-Force]
| -info [-size] {-all|<fs_name>|id=<fs_id>} [-Ads] [-option <options>]
| -rename <old_name> <new_name> [-Force]
| -size <fs_name>
| -acl <acl_value> <fs_name>
| -translate <fs_name> -access_policy start
  -to {MIXED} -from {NT|NATIVE|UNIX|SECURE}
| -translate <fs_name> -access_policy status
| -xtend <fs_name> {<volume_name>|size=<integer>[T|G|M|%][pool=<pool>]
   [storage=<system_name>]} [-option <options>]
| -modify <fs_name> -auto_extend {no|yes [-thin {no|yes}]}
   [-hwm <50-99>%][-max_size <integer>[T|G|M]]
| -modify <fs_name> -worm [-default_retention {<integer>{Y|M|D}|infinite}]
   [-min_retention {<integer>{Y|M|D}|infinite}]
   [-max_retention {<integer>{Y|M|D}|infinite}]
| -modify <fs_name> -worm [-auto_lock {enable[-policy_interval
  <integer>{M|D|H}]|disable}]
   [-auto_delete {enable|disable}]
   [-policy_interval <integer>{M|H|D}]
| -modify <fs_name> -worm -reset_epoch <year>
| -Type <type> <fs_name> -Force
| [-name <name>][-type <type>] -create <volume_name>
   [samesize=<fs_name>[:cel=<cel_name>]]
   [worm={enterprise|compliance|off}]
   [-default_retention {<integer>{Y|M|D} |infinite}] [-min_retention
   {<integer>{Y|M|D}|infinite}]
   [-max_retention {<integer>{Y|M|D}|infinite}]]
  [log_type={common|split}][fast_clone_level={1|2}] [-option <options>]
| [-name <name>][-type <type>] -create {size=<integer>[T|G|M]
   | samesize=<fs_name>[:cel=<cel_name>]}
   pool=<pool> [storage=<system_name>][worm={enterprise|compliance|off}]
   [-default_retention {<integer>{Y|M|D}|infinite}]
   [-min_retention {<integer>{Y|M|D}|infinite}]
   [-max_retention {<integer>{Y|M|D}|infinite}]]
  [log_type={common|split}][fast_clone_level={1|2}]
   [-auto_extend {no|yes} [-thin {no|yes}]
   [-hwm <50-99>%][-max_size <integer>[T|G|M]]}]
   [-option <options>]
| [-name <name>] -type nmfs -create
```

DESCRIPTION
----

nas_fs creates, deletes, extends, and lists file systems. nas_fs displays the attributes of a file system, translates the access policy, enables, automatic file system extension and thin provisioning capabilities, manages retention periods, enables automatic file locking and automatic file deletion, and manages access control level values.

OPTIONS

-list [-all]
Displays a list of file systems and their attributes such as the name, ID, usage, type, access control level setting, the residing volume, and the server. The -all option displays all file systems including system-generated internal file systems. For example, Replicator internal checkpoints.

Note: The ID is an integer and is assigned automatically, but not always sequentially, depending on ID availability. The name of a file system might be truncated if it is more than 19 characters. To display the full file system name, use the -info option with a file system ID.

The file system types are:

1=uxfs (default)
5=rawfs (unformatted file system)
6=mirrorfs (mirrored file system)
7=ckpt (checkpoint)
8=mgfs (migration file system)
100=group file system
102=nmfs (nested mount file system)

Note: The file system types uxfs, mgfs, nmfs, and rawfs are created by using nas_fs. Other file system types are created either automatically or with their specific commands.

-delete <fs_name>
Deletes the file system specified by file system name or ID. A file system cannot be deleted when it is mounted or part of a group.

[-option <options>]
Specifies the following comma-separated options:

volume
Deletes the file system's underlying volume structure.

Note: If a checkpoint is created with a volume that has been specified by size, the underlying volume is deleted when the checkpoint is deleted. If a file system, using a storage pool is deleted, the underlying volume structure is also deleted.

[-Force]
Forces the deletion of a file system with SnapSure checkpoints known as the PFS, when a task scheduler such as an automated scheduler for SnapSure is running or is enabled.

-info [-size] [-Ads] {-all|<fs_name>|id=<fs_id>}
Displays the attributes of a single file system, or all file systems, including the configuration of associated disks and replication sessions that are stopped or configured on the file system. If a file system is mounted, data is reported from the NAS database and the Data Mover. If a file system is unmounted, data is reported from the NAS database only.

The -size option also displays the total size of the file system and the block count in megabytes.

The -Ads option displays the advanced data service properties of the file system.

[-option <options>]
Specifies the following comma-separated options:

mpd
Displays the current directory type and translation status for the specified Multi-Protocol Directory (MPD) file system.

-rename <old_name> <new_name>
Changes the file system name from <old_name> to <new_name>.

[-Force]
Forces the rename of the file system with SnapSure checkpoints known as the PFS.

-size <fs_name>
Displays the total size of the file system and the block count in megabytes. The total size of a file system relates to the mounted or unmounted status of a file system.

-acl <acl_value> <fs_name>
Sets an access control level value that defines the owner of a file system, and the level of access allowed for users and groups defined in the access control level table. The nas_acl command provides more information.

-translate <fs_name> -access_policy start -to{MIXED}
-from {NT |NATIVE|UNIX|SECURE}
Synchronizes the UNIX and Windows permissions on the specified file system. Prior to executing the -translate option by using server_mount, mount the specified file system with the MIXED access-checking policy. The <fs_name> must be a uxfs file system type mounted as read/write.

The policy specified in the -from option instructs the VNX about which operating system (UNIX or Windows) to derive permissions from, when migrating to the MIXED or MIXED_COMPAT access-checking policy (set with server_mount). For example, if you type UNIX in the -from option, all ACLs are regenerated from the UNIX mode bits. The policy typed in the -from option does not relate to the policy previously used by the file system object.

-translate <fs_name> -access_policy status
Prints the status of the access policy translation for the specified file system.

-xtend <fs_name> <volume_name>
Adds the specified volume to the mounted file system.

-xtend <fs_name> size=<integer>[T|G|M|%]
Adds the volume as specified by its desired size to the file system or checkpoint. Type an integer within the range of 1 to 1024, then specify T for terabytes, G for gigabytes (default), M for megabytes, or type an integer representing the percentage of a file system's size followed by the percent sign. The extended volume added to the file system by the system will have a size equal to or greater than the total size specified.

Caution: When executing this command, extends should be performed incrementally by using like volumes to reduce time consumption.

[pool=<pool>]
Applies the specified storage pool rule set to the volume that has been added to the mounted file system.

Note: The storage pool is a rule set that contains automatically created volumes and defines the type of disk volumes used and how they are aggregated.

[storage=<system_name>]
Specifies the storage system on which the checkpoint resides. If a storage system is not specified, the default storage system is the one on which the file system resides. If the file system spans multiple storage systems, the default is to use all the storage systems on which the file system resides. Use nas_storage -list to obtain attached storage system names.

[-option <options>]
Specifies the following comma-separated options:

slice={y|n}
Specifies whether the disk volumes used by the file system may be shared with other file systems that use a slice. The slice=y option allows the file system to share disk volumes with other file systems. The slice=n option gives the new filesystem exclusive access to the disk volumes it uses, and is relevant when using TimeFinder/FS.

When symm_std, symm_std_rdf_src, symm_ata, symm_ata_rdf_src, symm_ata_rdf_tgt, and symm_std_rdf_tgt, symm_fts, symm_fts_rdf_tgt, symm_dsl and symm_fts_rdf_src pools are specified,

the default is not to slice the volumes, which is overridden with slice=y. For symm_efd, the default is slice=y,because TimeFinder/FS is not supported with Flash(EFD) disk types.

When clar_r1, clar_r5_performance, clar_r5_economy, clar_r6, clarata_r3, clarata_r6, clarata_r10, clarata_archive, cm_r1, cm_r5_performance, cm_r5_economy, cm_r6, cmata_r3, cmata_archive, cmata_r6, cmata_r10, clarsas_archive, clarsas_r6, clarsas_r10, clarefd_r5, clarefd_r10, cmsas_archive, cmsas_r6, cmsas_r10, and cmefd_r5 pools are specified, the default for standard AVM pools is to slice the volumes (slice=y), which is overridden by using slice=n. The default for mapped pools is not to slice the volumes (slice=n). Use nas_pool to change the default slice option.

-modify <fs_name> -auto_extend {no|yes [-thin{yes|no}]}
[-hwm <50-99>%][-max_size <integer>[T|G|M]]]
For an AVM file system, turns automatic file system extension and thin provisioning on or off, and sets a high water mark and maximum size for the file system. When file system extension is turned on, the file system is automatically extended up to the maximum size specified when the high water mark is reached. The default for -auto_extend is no.

Thin provisioning reports the maximum file system size to the CIFS, NFS, and FTP users, even if the actual size of the file system is smaller. If thin provisioning is disabled, the true file system size and maximum file system sizes are reported to the system administrator. Thin provisioning requires that a maximum file system size also be set. If a file system is created by using a virtual storage pool that contains:

* Only thick LUNs, use the nas_fs command's -thin option to enable thin provisioning on the file system.
* Only thin LUNs, using the nas_fs command's -thin option is not recommended. It is redundant, but allowed, for a thin file system to be built on thin LUNs.
* Both thick and thin LUNs, the file system may be built on either thick LUNs, thin LUNs, or both thick and thin LUNs. Using the nas_fs command's -thin option may be redundant if the file system uses thin LUNs.

Automatic file system extension cannot be used for any file system that is part of an RDF configuration (for example, file systems on Data Movers configured with an RDF standby). Do not use the nas_fs command with the -auto_extend option for file systems associated with RDF configurations.

[-hwm <50-99>%]
Specifies the size threshold that must be reached before the file system is automatically extended. Type an integer between the range of 50 and 99 to represent the percentage of file system usage. The default is 90.

[-max_size <integer> [T|G|M]]
Sets the maximum file system size to which a file system can be extended. Type an integer and specify T for terabytes, G for gigabytes (default), or M for megabytes. If the -max_size option is not specified, then it defaults to the maximum limit of the file system size which is 16 terabytes.

-modify <fs_name> -worm [-default_retention {<integer>{Y|M|D}|infinite}]
[-min_retention {<integer>{Y|M|D}|infinite}]
[-max_retention {<integer>{Y|M|D}|infinite}]
For an FLR-enabled file system, manages retention periods.

[-default_retention {<integer>{Y|M|D}|infinite}]
Sets a default retention period that is used in an FLR-enabled file system when a file is locked and a retention period is not specified. This value must be greater than or equal to the -min_retention option, and less than or equal to the -max_retention option. Type an integer and specify Y for years, M for months, or D for days. The default value is infinite. Setting infinite means that the files can never be deleted.

[-min_retention {<integer>{Y|M|D}|infinite}]
Sets the minimum retention period that files on an FLR-enabled filesystem can be locked and protected from deletion. This value must be less than or equal to the -max_retention option. Type an integer and specify Y for years, M for months, or D for days. The default value is one day. Setting infinite means that the files can never be deleted.

[-max_retention {<integer>{Y|M|D}|infinite}]
Sets the maximum retention period that files on an FLR-enabled filesystem can be locked and protected from deletion. Type an integer and specify Y for years, M for months, or D for days. The default value is infinite. Setting infinite means that the files can never be deleted.

-modify <fs_name> -worm [-auto_lock {enable[-policy_interval <integer>{M|D|H}]|disable}]
[-auto_delete {enable|disable}][-policy_interval <integer>{M|D|H}]
For an FLR-enabled filesystem, manages automatic file locking and automatic file deletion.

[-auto_lock {enable|disable}]
Specifies whether automatic file locking for all files in an FLR-enabled file system is on or off. When enabled, auto-locked files are set with the default retention period value.

[-policy_interval <integer>{M|D|H}]
Specifies an interval for how long to wait after the files are modified before

the files are automatically locked in an FLR-enabled file system. Type an integer and

specify M for minutes, D for days, or H for hours. The policy interval has a minimum value of one minute and a maximum value of 366 days. The default value is one hour.

[-auto_delete {enable|disable}]
Specifies whether automatically deleting locked files from an FLR-enabled file system once the retention period has expired is on or off.

-modify <fs_name> -worm -reset_epoch <year>
For an FLR-enabled file system, specifies the base year used for calculating the retention date of a file beyond 2038. Type an integer between the range of 2000 and 2037. The default value is 2003. The maximum value for the retention period is December 31, 2104 11:59:59 p.m. Trying to set a date beyond this value generates an error. Refer to Using VNX File-Level Retention for additional information.

-Type <type> <fs_name> -Force
Changes the file system type from the one of <fs_name> to the new specified <type>.

Caution: Converting uxfs to rawfs is prevented.

Caution: The conversion from rawfs to uxfs will fail with "Error 3105: invalid filesystem specified" because a uxfs is not available on the rawfs. However, if the user initially creates a rawfs, and restores an NDMP volume backup on the rawfs, then the conversion from the rawfs to a uxfs will be successful.

CREATING A FILE SYSTEM

File systems can be created by using:

* A volume specified by name
* A volume specified by its size and desired storage pool
* An existing local or remote filesystem with the samesize option
* An existing local or remote filesystem with the samesize option and by using space from the available storage pool

[-name <name>][-type <type>] -create <volume_name>

Creates a file system on the specified volume and assigns an optional name to the file system. If a name is not specified, one is assigned automatically.

A file system name cannot:

* Begin with a dash (-)
* Be comprised entirely of integers
* Be a single integer
* Contain the word root or contain a colon (:)

The -type option assigns the file system type to be uxfs (default), mgfs, or rawfs.

[samesize=<fs_name>[:cel=<cel_name>]]
Specifies that the new file system must be created with the same size as the specified local or remote file system. When using the samesize option by using the options described below, the slice= must be set to y.

Note: The specified file system must be mounted.

[worm={enterprise|compliance|off}]
Enables storage capability on a new file system. The option can only be specified when creating a new file system; existing file systems cannot be converted. After a file system is enabled, it is persistently marked as such until the time when it is deleted.

Note: The compliance file system cannot be deleted if it has protected files.

Caution: The Enterprise version of this feature is intended for self-regulated archiving. The administrator is considered a trusted user and the capability does not protect the archived data from the administrator's actions. If the administrator attempts to delete the file system, the file system issues a warning message and prompts the administrator to confirm the operation. This version is not intended for high-end compliance applications such as pharmaceuticals, aerospace, or finance.

As part of enabling file-level retention (worm) on a new file system, you can also set these retention period options:

[-default_retention {<integer>{Y|M|D}|infinite}]
Sets a default retention period that is used in an FLR-enabled filesystem when a file is locked and a retention period is not specified. This value must be greater than or equal to the -min_retention option, and less than or equal to the -max_retention option. Type an integer and specify Y for years, M for months, D for days, or infinite. The default value is infinite which means that the files can never be deleted.

[-min_retention {<integer>{Y|M|D}|infinite}]
Sets the minimum retention period that files on an FLR-enabled file system can be locked and protected from deletion. This value must be less than or equal to the -max_retention option. Type an integer and specify Y for years, M for months, D for days, or infinite. The default value is one day. Setting infinite means that the files can never be deleted.

[-max_retention {<integer>{Y|M|D}|infinite}]
Sets the maximum retention period that files on an FLR-enabled file system can

be locked and protected from deletion. Type an integer and specify Y for years, M for months, D for days, or infinite. The default value is infinite which means that the files can never be deleted.

log_type={common|split}
Specifies the type of log file associated with the file system. Log files can be either shared (common) or uniquely assigned to individual file systems(split). For SRDF Async or STAR feature, split option is strongly recommended to avoid fsck before mounting a BCV file system on SiteB or SiteC.

[fast_clone_level={1|2}]
fast_clone_level=2 enables ability to create fast clone of a fast clone (also called as the second level fast clone) on the file system. fast_clone_level=1 enables ability to create a fast clone. File level retention and fast clone creation cannot be enabled together on a file system. Enabling split log implies fast_clone_level=2, if file level retention is not enabled on the filesystem. Replication sessions cannot be created between two different fast_clone_level capable filesystems.

Note: fast_clone_level=1 indicates that a fast clone can be created on the filesystem and it is the default option if nothing is specified.

[-option <options>]
Specifies the following comma-separated options:

nbpi=<number>
The number of bytes per inode block. The default is 8192 bytes.

mover=<movername>
Assigns an optional Data Mover to build a file system. If no Data Mover is assigned, the system will automatically pick the first available Data Mover to build the file system.

slice={y|n}
Specifies whether the disk volumes used by the new file system may be shared with other file systems by using a slice. The slice=y option allows the file system to share disk volumes with other file systems. The slice=n option ensures that the new file system has exclusive access to the disk volumes it uses, and is relevant when using TimeFinder/FS.

When symm_std, symm_std_rdf_src, symm_ata, symm_ata_rdf_src, symm_ata_rdf_tgt, symm_std_rdf_tgt, symm_fts, symm_fts_rdf_tgt, and symm_fts_rdf_src pools are specified, the default is not to slice the volumes. When slice=y is specified, it overrides the default. For symm_efd, the default is slice=y, because TimeFinder/FS is not supported with Flash disk types.

When clar_r1, clar_r5_performance, clar_r5_economy, clar_r6, clarata_r3, clarata_r6, clarata_r10, clarata_archive, cm_r1, cm_r5_performance, cm_r5_economy, cm_r6, cmata_r3, cmata_archive, cmata_r6, cmata_r10, clarsas_archive, clarsas_r6, clarsas_r10, clarefd_r5, clarefd_r10, cmsas_archive, cmsas_r6, cmsas_r10, and cmefd_r5 pools are specified, the default for standard AVM pools is to slice the volumes (slice=y), which is overridden by using slice=n. The default for mapped pools is not to slice the volumes (slice=n). Use nas_pool to change the default slice option.

id=<desired_id>
Specifies the ID to be assigned to the new file system. If a file system already exists with the specified ID, a warning is displayed indicating that the ID is not available, and the new file system is assigned the next available ID.

[-name <name>][-type <type>] -create {size=
<integer>[T|G|M]|samesize=<fs_name>[:cel=<cel_name>]} pool=<pool>
Creates a file system on the volume specified by its desired size and storage pool or by using the same size as a specified local or remote file system. Also assigns an optional name and file system type to a file system. If a name is not specified, one is assigned automatically. A file system name can be up to 240 characters, but cannot begin with a dash (-), be comprised entirely of integers or be a single integer, contain the word root or contain a colon (:). Available file system types are uxfs (default), mgfs, or rawfs.

When using the samesize option by using the options described below, the slice= should be set to y. The new file system is created with the same size as the specified file system.

The pool option specifies a rule set for the new file system that contains automatically created volumes and defines the type of disk volumes used and how they are aggregated. Storage pools are system defined (storage pool description provides more information) or user defined.

[worm={enterprise|compliance|off}]
Enables the storage capability on the new file system. The capability can only be specified when creating a new file system; existing file systems cannot be converted. After a file system is enabled, it is persistently marked as such until the time when it is deleted.

Caution: The Enterprise version of this feature is intended for self-regulated archiving. The administrator is considered a trusted user and feature does not protect the archived data from the administrator's actions. If the administrator attempts to delete a file system, the file system issues a warning message and prompts the administrator to confirm the operation. This version of this feature is not intended for high-end compliance applications such as pharmaceuticals, aerospace, or finance.

As part of enabling file-level retention (worm) on a new file system, you can also set these retention period options:

[-default_retention {<integer>{Y|M|D}|infinite}]
Sets a default retention period that is used in an FLR-enabled file system when a file is locked and a retention period is not specified. This value must be greater than or equal to the -min_retention option, and less than or equal to the -max_retention option. Type an integer and specify Y for years, M for months, D for days, or infinite. The default value is infinite which means that the files can never be deleted.

[-min_retention {<integer>{Y|M|D}|infinite}]
Setsthe minimum retention period that files on an FLR-enabled file system can be locked and protected from deletion. This value must be less than or equal to the -max_retention option. Type an integer and specify Y for years, M for months, D for days, or infinite. The default value is 1 day. Setting infinite means that the files can never be deleted.

[-max_retention {<integer>{Y|M|D}|infinite}]
Sets the maximum retention period that files on an FLR-enabled file system can be locked and protected from deletion. Type an integer and specify Y for years, M for months, D for days, or infinite. The default value is infinite

which means that the files can never be deleted.

[storage=<system_name>]
Specifies the system on which the file system resides. Use
nas_storage -list to obtain a list of the available system names.

[-auto_extend {no|yes} [-thin {no|yes}]
For an AVM file system, turns automatic file system extension
and thin provisioning on or off, and sets a high water mark and
maximum size for the file system. When automatic file system
extension is turned on, the file system is automatically extended
up to the maximum size specified when the high water mark is
reached. The default for -auto_extend is no.

Thin provisioning reports the maximum file system size to the
CIFS, NFS, and FTP users, even if the actual size of the file system
is smaller. If disabled, the true file system size and maximum file syste
m
sizes are reported to the system administrator. Thin provisioning require
s
that a maximum file system size also be set.

If a file system is created in a storage pool that contains:

* Only thick LUNs, use the nas_fs command's -thin option to enable thin
provisioning on the file system.
* Only thin LUNs, using the nas_fs command's -thin option is not recomme
nded. It
is redundant, but allowed, for a thin file system to be built on thin
LUNs.
* Both thick and thin LUNs, the file system may be built on either thick
LUNs,
thin LUNs, or both thick and thin LUNs. Using the nas_fs command's -th
in
option may be redundant if the file system uses thin LUNs.

Note: SRDF pools are not supported.

[-hwm <50-99>%]
Specifies the size threshold that must be reached before the file system
is automatically extended. Type an integer between the range of 50 and 99

to represent the percentage of file system usage. The default is 90.

[-max_size <integer> [T|G|M]]
Sets the maximum file system size to which a file system can be
extended. Type an integer and specify T for terabytes, G for
gigabytes (default), or M for megabytes. If the -max_size option is
not specified, then it defaults to the maximum limit of the file system
size which is 16 terabytes. Maximum size must be set to
enable thin provisioning. The maximum size is what is presented
to users as the file system size through thin provisioning.

[-option <options>]
Specifies the following comma-separated options:

nbpi=<number>
The number of bytes per inode block. The default is 8192 bytes.

mover=<movername>
Assigns an optional Data Mover on which to build a file system.
If no Data Mover is assigned, the system will automatically pick
the first available Data Mover to build the file system.

slice={y|n}
Specifies whether the disk volumes used by the new file system
may be shared with other file systems by using a slice. The
slice=y option allows the file system to share disk volumes with
other file systems. The slice=n option ensures that the new file system

has exclusive access to the disk volumes it uses, and is

relevant when using TimeFinder/FS.

When symm_std, symm_std_rdf_src, symm_ata, symm_ata_rdf_src, symm_ata_rdf_tgt, and symm_std_rdf_tgt, symm_fts, symm_fts_rdf_tgt, and symm_fts_rdf_src pools are specified, the default

is not to slice the volumes, which is overridden with slice=y. For symm_efd, the default is slice=y, because TimeFinder/FS is not supported with Flash disk types.

When clar_r1, clar_r5_performance, clar_r5_economy, clar_r6, clarata_r3, clarata_r6, clarata_r10, clarata_archive, cm_r1, cm_r5_performance, cm_r5_economy, cm_r6, cmata_r3, cmata_archive, cmata_r6, cmata_r10, clarsas_archive, clarsas_r6, clarsas_r10, clarefd_r5, clarefd_r10, cmsas_archive, cmsas_r6, cmsas_r10, and cmefd_r5 pools are specified, the default for standard AVM pools is to slice the volumes (slice=y), which is overridden by using slice=n. The default for mapped pools is not to slice the volumes (slice=n). Use nas_pool to change the default slice option.

[-name <name>] -type nmfs -create
Creates a nested mount file system (NMFS) that can be used to combine multiple uxfs file systems into a single virtual file system. The NMFS can then be mounted and exported as a single share or mount point.

SEE ALSO

Managing Volumes and File Systems with VNX Automatic Volume Management, Managing Volumes and FileSystems for VNX Manually, Using VNX File-Level Retention, Controlling Access to System Objects on VNX, Using VNX Replicator, fs_ckpt, fs_timefinder, nas_acl, nas_rdf, nas_volume, server_export, server_mount, fs_dedupe, and server_mountpoint.

STORAGE SYSTEM OUTPUT

The number associated with the storage device is dependent on the attached storage system. VNX for Block displays a prefix of APM before a set of integers, for example, APM00033900124-0019. Symmetrix storage systems appear as 002804000190-003C. The outputs displayed in the examples use a VNX for Block.

VNX for Block supports the following system-defined storage pools: clar_r1, clar_r5_performance, clar_r5_economy, clar_r6, clarata_r3, clarata_r6, clarata_r10, clarata_archive, cm_r1, cm_r5_performance, cm_r5_economy, cm_r6, cmata_r3, cmata_archive, cmata_r6, cmata_r10, clarsas_archive, clarsas_r6, clarsas_r10, clarefd_r5, clarefd_r10, cmsas_archive, cmsas_r6, cmsas_r10, and cmefd_r5.

VNXs with a Symmetrix storage system support the following system-defined storage pools: symm_std_rdf_src, symm_std, symm_ata, symm_ata_rdf_src, symm_ata_rdf_tgt, symm_std_rdf_tgt, symm_efd, symm_fts, symm_fts_rdf_tgt, and symm_fts_rdf_src.

For user-defined storage pools, the difference in output is in the disk type. Disk types when using a Symmetrix are: STD, R1STD, R2STD, BCV, R1BCV, R2BCV, ATA, R1ATA, R2ATA, BCVA, R1BCA, R2BCA, EFD, FTS, R1FTS, R2FTS, R1BCF, R2BCF, and BCVF.

Disk types when using VNX for Block are: CLSTD, CLEFD, CLATA, MIXED (indicates that tiers used in the pool contain multiple disk types), Performance, Capacity, and Extreme_performance and for VNX for block involving mirrored disks are: CMEFD, CMSTD, CMATA, Mirrored_mixed, Mirrored_performance,

Mirrored_capacity, and Mirrored_extreme_performance.

EXAMPLE #1

To create a file system named ufs1 on metavolume mtv1, type:

```
$ nas_fs -name ufs1 -create mtv1
id                        = 37
name                      = ufs1
acl                       = 0
in_use                    = False
type                      = uxfs
worm                      = enterprise with no protected files
worm_clock                = Clock not initialized
worm Max Retention Date = NA
worm Default Retention Period = infinite
worm Minimum Retention Period = 1 Day
worm Maximum Retention Period = infinite
FLR Auto_lock             = off
FLR Policy Interval       = 3600 seconds
FLR Auto_delete           = off
FLR Epoch Year            = 2003
volume                    = mtv1
pool                      =
rw_servers                =
ro_servers                =
rw_vdms                   =
ro_vdms                   =
auto_ext                  = no,thin=no
deduplication             = off
stor_devs                 = APM00042000818-0012,APM00042000818-0014
disks                     = d7,d9
```

Where:

Value	Definition
id	Automatically assigned ID of a file system.
name	Name assigned to a file system.
acl	Access control value assigned to the file system.
in_use	If a file system is registered into the mount table of a Data Mover.
type	Type of file system. See -list for a description of the types.
volume	Volume on which a file system resides.
worm	Write Once Read Many (WORM) state of the file system.It states whether the file-level retention is disabled or set to either compliance or enterprise.
pool	Storage pool for the file system.
rw_servers	Servers with read/write access to a file system.
ro_servers	Servers with read-only access to a file system.
rw_vdms	VDM servers with read/write access to a file system.
ro_vdms	VDM servers with read-only access to a file system.
worm_clock	Software clock maintained by the file system. The clock functions only when the file system is mounted read/write.
worm Max Retention Date	Time when the protected files expire. The file system can be deleted only after this date. The special values returned are: * 3 - The file system is is set to File-Level retention enterprise with protected files. * 2 - The file system is scanning for max_retention period. * 1 - The default value (No protected files created). * 0 - Infinite retention period (if the server is up and running.)
worm Default Retention Period	Specifies a default retention period that files on an FLR-enabled filesystem will be locked and protected from deletion . If you do not set either a minimum retention period or a maximum retention period, this default value is used when the file-level retention is enabled.
worm Minimum	Specifies the minimum retention period that files on an

Retention Period — FLR-enabled file system will be locked and protected from deletion.

worm Maximum Retention Period — Specifies the maximum retention period that files on an FLR-enabled file system will be locked and protected from deletion.

FLR Auto_Lock — Specifies whether automatic file locking for all files in an FLR-enabled file system is on or off.

FLR Policy Interval — Specifies an interval for how long to wait after files are modified before the files are automatically locked and protected from deletion.

FLR Auto_delete — Specifies whether locked files are automatically deleted once the retention period has expired.

FLR Epoch Year — Specifies the base year used for calculating the retention date of a file beyond 2038. When a file is locked with its atime set to a value greater than the FLR Epoch Year value, the file's retention date is set to the file's atime value. When a file is locked with its atime set to a value less than the FLR Epoch Year value, the file's retention date is set to 2038 + (YEAR(atime) - 1970).

volume — Volume on which a file system resides.

pool — Storage pool for the file system.

rw_servers — Servers with read/write access to a file system.

ro_servers — Servers with read-only access to a file system.

rw_vdms — VDM servers with read/write access to a file system.

ro_vdms — VDM servers with read-only access to a file system.

auto_ext — Indicates whether auto-extension and thin provisioning are enabled.

deduplication — Deduplication state of the file system. The file data is transferred to the storage which performs the deduplication and compression on the data. The states are:

* On - deduplication on the file system is enabled.
* Suspended - Deduplication on the file system is suspended. Deduplication does not perform any new space reduction but the existing files that were reduced in space remain the same.
* Off - Deduplication on the file system is disabled. Deduplication does not perform any new space reduction and the data is now reduplicated.

stor_devs — Storage system devices associated with a file system.

disks — Disks on which the metavolume resides.

Note: The Deduplication state is unavailable when the file system is unmounted.

EXAMPLE #2

To display information about a file system using the file system ID 14, using the clar_mapped_pool VNX mapped pool, type:

```
$ nas_fs -info id=14
id              = 14
name            = ufs2_flre
acl             = 0
in_use          = True
type            = uxfs
worm            = enterprise with no protected files
worm_clock      = Fri Jul 29 07:56:42 EDT 2011
worm Max Retention Date= No protected files created
worm Default Retention Period= 10 Years
worm Minimum Retention Period= 30 Days
worm Maximum Retention Period= 10 Years
FLR Auto_lock   = off
FLR Policy Interval= 3600 seconds
FLR Auto_delete = off
FLR Epoch Year  = 2003
volume          = v117
pool            = clar_mapped_pool
member_of       = root_avm_fs_group_50
rw_servers      = server_2
ro_servers      =
```

```
        rw_vdms         =
        ro_vdms         =
        auto_ext        = no,thin=no
        deduplication   = Off
        thin_storage    = True
        tiering_policy  = Auto-tier
        compressed      = False
        mirrored        = False
        stor_devs       =
        stor_devs =
        BB005056830430-0019,BB005056830430-0016,BB005056830430-0015,BB005056830430-0010
        disks     = d16,d13,d12,d7
         disk=d16   stor_dev=BB005056830430-0019 addr=c0t1l9         server=server_2
         disk=d16   stor_dev=BB005056830430-0019 addr=c16t1l9        server=server_2
         disk=d13   stor_dev=BB005056830430-0016 addr=c0t1l6         server=server_2
         disk=d13   stor_dev=BB005056830430-0016 addr=c16t1l6        server=server_2
         disk=d12   stor_dev=BB005056830430-0015 addr=c0t1l5         server=server_2
         disk=d12   stor_dev=BB005056830430-0015 addr=c16t1l5        server=server_2
         disk=d7    stor_dev=BB005056830430-0010 addr=c0t1l0         server=server_2
         disk=d7    stor_dev=BB005056830430-0010 addr=c16t1l0        server=server_2

        Where:

        Value           Definition

        thin_storage    Indicates whether the VNX for Block storage system uses thin
                        provisioning. Values are: True, False, Mixed.
        tiering_policy  Indicates the tiering policy in effect. If the initial tier
                        and the tiering policy are the same, the values are: Auto-Tier, H
ighest
                        Available Tier, Lowest Available Tier. If the initial tier and th
e tiering
                        policy are not the same, the values are: Auto-Tier/No Data Moveme
nt, Highest
                        Available Tier/No Data Movement, Lowest Available Tier/No Data Mo
vement.
        compressed      Indicates whether data is compressed. Values are True, False, Mix
ed                                          (indicates some of the LUNs, but not all,
 are compressed).
        mirrored        Indicates whether the disk is mirrored.


        EXAMPLE #3
        ----------

        To display a list of file systems, type:

        $ nas_fs -list
        id      inuse type acl   volume    name            server
        1         n    1   0     20        root_fs_1
        2         y    1   0     50        root_fs_common     1
        3         n    5   0     83        root_fs_ufslog
        5         n    5   0     103       root_fs_d3
        6         n    5   0     104       root_fs_d4
        7         n    5   0     105       root_fs_d5
        8         n    5   0     106       root_fs_d6
        9         y    1   0     22        root_fs_2          1
        10        n    5   0     108       root_panic_reserve
        11        y    1   0     112       ufs1               1
        13        y    1   0     115       ufs1_flr           1
        14        y    1   0     117       ufs2_flre          1

        EXAMPLE #4
        ----------

        To list all the file systems including internal checkpoints, type:

        $ nas_fs -list -all
        id      inuse type acl   volume    name           server
        1         n    1   0     24        root_fs_1
        2         y    1   0     26        root_fs_2          1
```

```
3         y    1   0     28        root_fs_3           2
4         n    1   0     30        root_fs_4
5         n    1   0     32        root_fs_5
6         n    1   0     34        root_fs_6
7         n    1   0     36        root_fs_7
8         n    1   0     38        root_fs_8
9         n    1   0     40        root_fs_9
10        n    1   0     42        root_fs_10
11        n    1   0     44        root_fs_11
12        n    1   0     46        root_fs_12
13        n    1   0     48        root_fs_13
14        n    1   0     50        root_fs_14
15        n    1   0     52        root_fs_15
16        y    1   0     54        root_fs_common      2,1
17        n    5   0     87        root_fs_ufslog
18        n    5   0     90        root_panic_reserve
212       y    1   0     315       v2src1              1
213       y    101 0     0         root_avm_fs_group_3
214       n    1   0     318       v2dst1
230       y    1   0     346       v2srclun1           1
231       y    1   0     349       v2dstlun1           2
342       y    1   0     560       root_fs_vdm_srcvdm1 1
343       y    1   0     563       root_fs_vdm_srcvdm2 1
986       n    11  0     0         vpfs986
987       y    7   0     1722      gstest              1
988       y    1   0     1725      src1                1
989       y    5   0     1728      dst1                1
1343      n    11  0     0         vpfs1343
1344      y    7   0     2351      root_rep_ckpt_342_2 1
1345      y    7   0     2351      root_rep_ckpt_342_2 1
1346      y    1   0     2354      root_fs_vdm_srcvdm1 1
1347      n    11  0     0         vpfs1347
1348      y    7   0     2358      root_rep_ckpt_1346_ 1
1349      y    7   0     2358      root_rep_ckpt_1346_ 1
1350      y    1   0     2367      fs1                 v9
1354      n    1   0     2374      fs1_replica1
1358      n    11  0     0         vpfs1358
1359      y    7   0     2383      root_rep_ckpt_1350_ v9
1360      y    7   0     2383      root_rep_ckpt_1350_ v9
1361      n    1   0     2385      fs1_replica2
1362      n    11  0     0         vpfs1362
1363      n    7   0     2388      root_rep_ckpt_1361_
1364      n    7   0     2388      root_rep_ckpt_1361_
1365      y    1   0     2392      fs1365              1
1366      y    7   0     2383      root_rep_ckpt_1350_ v9
1367      y    7   0     2383      root_rep_ckpt_1350_ v9
1368      n    11  0     0         vpfs1368
1369      n    7   0     2395      root_rep_ckpt_1354_
1370      n    7   0     2395      root_rep_ckpt_1354_
1371      y    1   0     2399      root_fs_vdm_v1      1
1372      y    1   0     2401      f1                  v40
1376      y    1   0     2406      root_fs_vdm_v1_repl 2
1380      n    11  0     0         vpfs1380
1381      y    7   0     2414      root_rep_ckpt_1372_ v40
1382      y    7   0     2414      root_rep_ckpt_1372_ v40
1383      y    1   0     2416      f1_replica1         v41
1384      n    11  0     0         vpfs1384
1385      y    7   0     2419      root_rep_ckpt_1383_ v41
1386      y    7   0     2419      root_rep_ckpt_1383_ v41
1387      y    1   0     2423      cworm               1
1388      n    1   0     2425      cworm1
1389      y    1   0     2427      fs2                 2
1390      y    1   0     2429      fs3                 2
1391      n    11  0     0         vpfs1391
1392      y    7   0     2432      root_rep_ckpt_1389_ 2
1393      y    7   0     2432      root_rep_ckpt_1389_ 2
1394      n    11  0     0         vpfs1394
1395      y    7   0     2435      root_rep_ckpt_1390_ 2
1396      y    7   0     2435      root_rep_ckpt_1390_ 2
1397      y    7   0     2432      fs2_ckpt1           2
1398      y    1   0     2439      fs4                 2
```

```
1399      y    1   0     2441      fs5                2
1400      n    11  0     0         vpfs1400
1401      y    7   0     2444      root_rep_ckpt_1398_ 2
1402      y    7   0     2444      root_rep_ckpt_1398_ 2
1403      n    11  0     0         vpfs1403
1404      y    7   0     2447      root_rep_ckpt_1399_ 2
1405      y    7   0     2447      root_rep_ckpt_1399_ 2
1406      y    7   0     2444      fs4_ckpt1          2
```

Note: NDMP and Replicator internal checkpoints can be identified by specific prefixes in the filename. Using VNX SnapSure provides more information for internal checkpoints naming formats.

EXAMPLE #5

To create a uxfs file system named ufs20 on storage system BB005056830430, with a size of 1 GB, using the clar_r5_performance pool and allowing the file system to share disk volumes with other file systems, type:

```
$ nas_fs -name ufs20 -type uxfs -create size=1G pool=clar_r5_performance
storage=BB005056830430 -option slice=y
id        = 15
name      = ufs20
acl       = 0
in_use    = False
type      = uxfs
worm      = off
volume    = v119
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers=
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication = unavailable
stor_devs =
BB005056830430-0018,BB005056830430-0017,BB005056830430-0014,BB005056830430-0011
disks     = d15,d14,d11,d8
```

Where:

Value	Definition
member_of	Filesystem group to which the filesystem belongs.

EXAMPLE #1 provides a description of command output.

EXAMPLE #6

To create a rawfs file system named ufs3 with the same size as the file system ufs1 using the clar_r5_performance pool and allowing the file system to share disk volumes with other filesystems, type:

```
$ nas_fs -name ufs3 -type rawfs -create samesize=ufs1 pool=clar_r5_performance
storage=APM00042000818 -option slice=y
id        = 39
name      = ufs3
acl       = 0
in_use    = False
type      = rawfs
worm      = off
volume    = v173
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers=
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication = unavailable
```

```
stor_devs =
APM00042000818-001F,APM00042000818-001D,APM00042000818-0019,APM00042
000818-0016
disks     = d20,d18,d14,d11

EXAMPLE #1 and EXAMPLE #3 provide for a description of
command outputs.

EXAMPLE #7
----------

To create a uxfs file system named ufs4, with a size of 100 GB, using the
clar_r5_performance pool, with file-level retention set to enterprise, 4096 bytes
per
inode, and server_3 for file system building, type:

$ nas_fs -name ufs4 -create size=100G pool=clar_r5_performance worm=enterprise
-option nbpi=4096,mover=server_3
id        = 16
name      = ufs4
acl       = 0
in_use    = False
type      = uxfs
worm      = enterprise with no protected files
worm_clock= Clock not initialized
worm Max Retention Date= NA
worm Default Retention Period= infinite
worm Minimum Retention Period= 1 Day
worm Maximum Retention Period= infinite
FLR Auto_lock= off
FLR Policy Interval= 3600 seconds
FLR Auto_delete= off
FLR Epoch Year= 2003
volume    = v121
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers=
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication   = unavailable
stor_devs =
BB005056830430-0019,BB005056830430-0016,BB005056830430-0015,BB005056830430-0010
disks     = d16,d13,d12,d7

To ensure retention of protected files, it can also be set to compliance by
typing:

$ nas_fs -name ufs4 -create size=100G pool=clar_r5_performance worm=compliance
-option nbpi=4096,mover=server_3
id        = 17
name      = ufs4
acl       = 0
in_use    = False
type      = uxfs
worm      = compliance with no protected files
worm_clock= Clock not initialized
worm Max Retention Date= NA
worm Default Retention Period= infinite
worm Minimum Retention Period= 1 Day
worm Maximum Retention Period= infinite
FLR Auto_lock= off
FLR Policy Interval= 3600 seconds
FLR Auto_delete= off
FLR Epoch Year= 2003
volume    = v123
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers=
ro_servers=
```

```
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication   = unavailable
stor_devs =
BB005056830430-0018,BB005056830430-0017,BB005056830430-0014,BB005056830430-0011
disks     = d15,d14,d11,d8
```

EXAMPLE #1 provides a description of command outputs.

EXAMPLE #8

To create a file system named ufs30, with a size of 1 GB, by using the clar_r5_performance pool, with file-level retention set to enterprise, a minimum retention period of 30 days, and a maximum retention period of 10 years, type:

```
$ nas_fs -name ufs30 -create size=1G pool=clar_r5_performance worm=enterprise
-min_retention 30D -max_retention 10Y

id        = 18
name      = ufs30
acl       = 0
in_use    = False
type      = uxfs
worm      = enterprise with no protected files
worm_clock= Clock not initialized
worm Max Retention Date= NA
worm Default Retention Period= 10 Years
worm Minimum Retention Period= 30 Days
worm Maximum Retention Period= 10 Years
FLR Auto_lock= off
FLR Policy Interval= 3600 seconds
FLR Auto_delete= off
FLR Epoch Year= 2003
volume    = v125
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers=
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication   = unavailable
stor_devs =
BB005056830430-0019,BB005056830430-0016,BB005056830430-0015,BB005056830430-0010
disks     = d16,d13,d12,d7
```

EXAMPLE #1 provides a description of command outputs.

EXAMPLE #9

To display information about file system ufs4, type:

```
$ nas_fs -info ufs4

id        = 16
name      = ufs4
acl       = 0
in_use    = False
type      = uxfs
worm      = enterprise with no protected files
worm_clock= Clock not initialized
worm Max Retention Date= NA
worm Default Retention Period= infinite
worm Minimum Retention Period= 1 Day
worm Maximum Retention Period= infinite
FLR Auto_lock= off
FLR Policy Interval= 3600 seconds
FLR Auto_delete= off
FLR Epoch Year= 2003
```

```
volume    = v121
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers=
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication   = unavailable
stor_devs =
BB005056830430-0019,BB005056830430-0016,BB005056830430-0015,BB005056830430-0010
disks     = d16,d13,d12,d7
```

EXAMPLE #1 provides a description of command outputs.

EXAMPLE #10

To create a uxfs file system named ufs40, with a size of 10 GB, by using the clar_r5_performance pool, and an ID of 8000 assigned to ufs1, type:

```
$ nas_fs -name ufs40 -type uxfs -create size=10G pool=clar_r5_performance
-option slice=y,id=8000
id = 8000
name = ufs40
acl = 0
in_use = False
type = uxfs
worm = off
volume = v127
pool = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers=
ro_servers=
rw_vdms =
ro_vdms =
auto_ext = no,thin=no
deduplication = unavailable
stor_devs =
BB005056830430-0018,BB005056830430-0017,BB005056830430-0014,BB005056830430-0011
disks     = d15,d14,d11,d8
```

EXAMPLE #11

To create a uxfs file system named ufs41, with a size of 10 GB, by using the clar_r5_performance pool, and an ID of 8000 assigned to ufs1, type:

```
$ nas_fs -name ufs41 -type uxfs -create size=10G pool=clar_r5_performance
-option slice=y,id=8000

id        = 8001
name      = ufs41
acl       = 0
in_use    = False
type      = uxfs
worm      = off
volume    = v129
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers=
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication   = unavailable
stor_devs =
BB005056830430-0019,BB005056830430-0016,BB005056830430-0015,BB005056830430-0010
disks     = d16,d13,d12,d7

Warning 17716815881: unavailable id : 8000.
```

Note: The warning output is displayed if the desired ID is not available. Because id=8000 was used in Example 10, the system set the id to 8001 instead.

EXAMPLE #12

To view the size of ufs1, type:

```
$ nas_fs -size ufs1
total = 945 avail = 945 used = 1 ( 0% ) (sizes in MB) ( blockcount = 2097152 )
volume: total = 1024 (sizes in MB) ( blockcount = 2097152 ) avail = 944 used =
80 ( 8% )
```

When a file system is mounted, the size info for the volume and a file system, as well as the number of blocks that are used are displayed.

Where:

Value	Definition
total	Total size of the file system.
blockcount	Total number of blocks used.

EXAMPLE #13

To rename a file system from ufs1 to ufs5, type:

```
$ nas_fs -rename ufs1 ufs5
id        = 11
name      = ufs5
acl       = 0
in_use    = True
type      = uxfs
worm      = off
volume    = v112
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers= server_2
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication = Off
stor_devs =
BB005056830430-0019,BB005056830430-0016,BB005056830430-0015,BB005056830430-0010
disks     = d16,d13,d12,d7
 disk=d16   stor_dev=BB005056830430-0019 addr=c0t1l9         server=server_2
 disk=d16   stor_dev=BB005056830430-0019 addr=c16t1l9        server=server_2
 disk=d13   stor_dev=BB005056830430-0016 addr=c0t1l6         server=server_2
 disk=d13   stor_dev=BB005056830430-0016 addr=c16t1l6        server=server_2
 disk=d12   stor_dev=BB005056830430-0015 addr=c0t1l5         server=server_2
 disk=d12   stor_dev=BB005056830430-0015 addr=c16t1l5        server=server_2
 disk=d7    stor_dev=BB005056830430-0010 addr=c0t1l0         server=server_2
 disk=d7    stor_dev=BB005056830430-0010 addr=c16t1l0        server=server_2
```

EXAMPLE #1 and EXAMPLE #3 provide a description of command outputs.

EXAMPLE #14

To extend the file system, ufs1, with the volume, emtv2b, type:

```
$ nas_fs -xtend ufs1 emtv2b

id        = 38
name      = ufs1
acl       = 0
in_use    = True
type      = uxfs
worm      = off
volume    = v171
pool      = clar_r5_performance
```

```
member_of = root_avm_fs_group_3
rw_servers= server_2
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication = off
stor_devs =
APM00042000818-001F,APM00042000818-001D,APM00042000818-0019,APM00042
000818-0016,APM00042000818-001C
disks     = d20,d18,d14,d11,d17
 disk=d20   stor_dev=APM00042000818-001F addr=c0t1l15        server=server_2
 disk=d20   stor_dev=APM00042000818-001F addr=c32t1l15       server=server_2
 disk=d18   stor_dev=APM00042000818-001D addr=c0t1l13        server=server_2
 disk=d18   stor_dev=APM00042000818-001D addr=c32t1l13       server=server_2
 disk=d14   stor_dev=APM00042000818-0019 addr=c0t1l9         server=server_2
 disk=d14   stor_dev=APM00042000818-0019 addr=c32t1l9        server=server_2
 disk=d11   stor_dev=APM00042000818-0016 addr=c0t1l6         server=server_2
 disk=d11   stor_dev=APM00042000818-0016 addr=c32t1l6        server=server_2
 disk=d17   stor_dev=APM00042000818-001C addr=c0t1l12        server=server_2
 disk=d17   stor_dev=APM00042000818-001C addr=c32t1l12       server=server_2
```

EXAMPLE #1 provides a description of command outputs.

EXAMPLE # 15

To extend the file system named ufs5, with the specified size of 1 GB, by using clar_r5_performance pool, type:

```
$ nas_fs -xtend ufs5 size=1G pool=clar_r5_performance

id        = 11
name      = ufs5
acl       = 0
in_use    = True
type      = uxfs
worm      = off
volume    = v112
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers= server_2
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication   = Off
stor_devs =
BB005056830430-0019,BB005056830430-0016,BB005056830430-0015,BB005056830430-0010
disks     = d16,d13,d12,d7
 disk=d16   stor_dev=BB005056830430-0019 addr=c0t1l9         server=server_2
 disk=d16   stor_dev=BB005056830430-0019 addr=c16t1l9        server=server_2
 disk=d13   stor_dev=BB005056830430-0016 addr=c0t1l6         server=server_2
 disk=d13   stor_dev=BB005056830430-0016 addr=c16t1l6        server=server_2
 disk=d12   stor_dev=BB005056830430-0015 addr=c0t1l5         server=server_2
 disk=d12   stor_dev=BB005056830430-0015 addr=c16t1l5        server=server_2
 disk=d7    stor_dev=BB005056830430-0010 addr=c0t1l0         server=server_2
 disk=d7    stor_dev=BB005056830430-0010 addr=c16t1l0        server=server_2
```

EXAMPLE #1 provides a description of command outputs.

EXAMPLE #16

To set the access control level to 1432 for the file system ufs5, type:

```
$ nas_fs -acl 1432 ufs5

id        = 11
name      = ufs5
acl       = 1432, owner=nasadmin, ID=201
in_use    = True
type      = uxfs
```

```
worm      = off
volume    = v112
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers= server_2
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication   = Off
stor_devs =
BB005056830430-0019,BB005056830430-0016,BB005056830430-0015,BB005056830430-0010
disks     = d16,d13,d12,d7
 disk=d16   stor_dev=BB005056830430-0019 addr=c0t1l9         server=server_2
 disk=d16   stor_dev=BB005056830430-0019 addr=c16t1l9        server=server_2
 disk=d13   stor_dev=BB005056830430-0016 addr=c0t1l6         server=server_2
 disk=d13   stor_dev=BB005056830430-0016 addr=c16t1l6        server=server_2
 disk=d12   stor_dev=BB005056830430-0015 addr=c0t1l5         server=server_2
 disk=d12   stor_dev=BB005056830430-0015 addr=c16t1l5        server=server_2
 disk=d7    stor_dev=BB005056830430-0010 addr=c0t1l0         server=server_2
 disk=d7    stor_dev=BB005056830430-0010 addr=c16t1l0        server=server_2

Note: The value 1432 specifies nasadmin as the owner and gives users with an acce
ss
level of at least observer read access only, users with an access level of at
least operator read/write access, and users with an access level of at least
admin read/write/delete access.

EXAMPLE #1 provides a description of command outputs.

EXAMPLE #17
-----------

To set the maximum retention period for file system ufs2_flre to 11 years,
type:
$ nas_fs -modify ufs2_flre -worm -max_retention 11Y

id        = 14
name      = ufs2_flre
acl       = 0
in_use    = True
type      = uxfs
worm      = enterprise with no protected files
worm_clock= Fri Jul 29 11:14:27 EDT 2011
worm Max Retention Date= No protected files created
worm Default Retention Period= 10 Years
worm Minimum Retention Period= 30 Days
worm Maximum Retention Period= 11 Years
FLR Auto_lock= off
FLR Policy Interval= 3600 seconds
FLR Auto_delete= off
FLR Epoch Year= 2003
volume    = v117
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers= server_2
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication   = Off
stor_devs =
BB005056830430-0019,BB005056830430-0016,BB005056830430-0015,BB005056830430-0010
disks     = d16,d13,d12,d7
 disk=d16   stor_dev=BB005056830430-0019 addr=c0t1l9         server=server_2
 disk=d16   stor_dev=BB005056830430-0019 addr=c16t1l9        server=server_2
 disk=d13   stor_dev=BB005056830430-0016 addr=c0t1l6         server=server_2
 disk=d13   stor_dev=BB005056830430-0016 addr=c16t1l6        server=server_2
 disk=d12   stor_dev=BB005056830430-0015 addr=c0t1l5         server=server_2
 disk=d12   stor_dev=BB005056830430-0015 addr=c16t1l5        server=server_2
 disk=d7    stor_dev=BB005056830430-0010 addr=c0t1l0         server=server_2
```

```
 disk=d7    stor_dev=BB005056830430-0010 addr=c16t1l0        server=server_2
```

EXAMPLE #1 provides a description of command outputs.

EXAMPLE #17

To set the maximum retention period for file system ufs2_flre to 11 years, type:

```
$ nas_fs -modify ufs2_flre -worm -max_retention 11Y

id        = 14
name      = ufs2_flre
acl       = 0
in_use    = True
type      = uxfs
worm      = enterprise with no protected files
worm_clock= Fri Jul 29 11:14:27 EDT 2011
worm Max Retention Date= No protected files created
worm Default Retention Period= 10 Years
worm Minimum Retention Period= 30 Days
worm Maximum Retention Period= 11 Years
FLR Auto_lock= off
FLR Policy Interval= 3600 seconds
FLR Auto_delete= off
FLR Epoch Year= 2003
volume    = v117
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers= server_2
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication   = Off
stor_devs =
BB005056830430-0019,BB005056830430-0016,BB005056830430-0015,BB005056830430-0010
disks     = d16,d13,d12,d7
 disk=d16   stor_dev=BB005056830430-0019 addr=c0t1l9         server=server_2
 disk=d16   stor_dev=BB005056830430-0019 addr=c16t1l9        server=server_2
 disk=d13   stor_dev=BB005056830430-0016 addr=c0t1l6         server=server_2
 disk=d13   stor_dev=BB005056830430-0016 addr=c16t1l6        server=server_2
 disk=d12   stor_dev=BB005056830430-0015 addr=c0t1l5         server=server_2
 disk=d12   stor_dev=BB005056830430-0015 addr=c16t1l5        server=server_2
 disk=d7    stor_dev=BB005056830430-0010 addr=c0t1l0         server=server_2
 disk=d7    stor_dev=BB005056830430-0010 addr=c16t1l0        server=server_2
```

EXAMPLE #1 provides a description of command outputs.

EXAMPLE #18

To reset the FLR epoch year for file system ufs2_flre to 2000, type:

```
$ nas_fs -modify ufs2_flre -worm -reset_epoch 2000
id        = 14
name      = ufs2_flre
acl       = 0
in_use    = True
type      = uxfs
worm      = enterprise with no protected files
worm_clock= Fri Jul 29 11:18:36 EDT 2011
worm Max Retention Date= No protected files created
worm Default Retention Period= 10 Years
worm Minimum Retention Period= 30 Days
worm Maximum Retention Period= 11 Years
FLR Auto_lock= off
FLR Policy Interval= 3600 seconds
FLR Auto_delete= off
FLR Epoch Year= 2000
volume    = v117
pool      = clar_r5_performance
```

```
member_of = root_avm_fs_group_3
rw_servers= server_2
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication   = Off
stor_devs =
BB005056830430-0019,BB005056830430-0016,BB005056830430-0015,BB005056830430-0010
disks     = d16,d13,d12,d7
disk=d16   stor_dev=BB005056830430-0019 addr=c0t1l9         server=server_2
 disk=d16   stor_dev=BB005056830430-0019 addr=c16t1l9        server=server_2
 disk=d13   stor_dev=BB005056830430-0016 addr=c0t1l6         server=server_2
 disk=d13   stor_dev=BB005056830430-0016 addr=c16t1l6        server=server_2
 disk=d12   stor_dev=BB005056830430-0015 addr=c0t1l5         server=server_2
 disk=d12   stor_dev=BB005056830430-0015 addr=c16t1l5        server=server_2
 disk=d7    stor_dev=BB005056830430-0010 addr=c0t1l0         server=server_2
 disk=d7    stor_dev=BB005056830430-0010 addr=c16t1l0        server=server_2

EXAMPLE #19
-----------

To enable FLR automatic file locking with a policy interval of 30 minutes for
file system ufs2_flre, type:

$ nas_fs -modify ufs2_flre -worm -auto_lock enable -policy_interval 30M
id        = 14
name      = ufs2_flre
acl       = 0
in_use    = True
type      = uxfs
worm      = enterprise with no protected files
worm_clock= Fri Jul 29 12:14:44 EDT 2011
worm Max Retention Date= No protected files created
worm Default Retention Period= 10 Years
worm Minimum Retention Period= 30 Days
worm Maximum Retention Period= 11 Years
FLR Auto_lock= on
FLR Policy Interval= 1800 seconds
FLR Auto_delete= off
FLR Epoch Year= 2000
volume    = v117
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers= server_2
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication   = Off
stor_devs =
BB005056830430-0019,BB005056830430-0016,BB005056830430-0015,BB005056830430-0010
disks     = d16,d13,d12,d7
 disk=d16   stor_dev=BB005056830430-0019 addr=c0t1l9         server=server_2
 disk=d16   stor_dev=BB005056830430-0019 addr=c16t1l9        server=server_2
 disk=d13   stor_dev=BB005056830430-0016 addr=c0t1l6         server=server_2
 disk=d13   stor_dev=BB005056830430-0016 addr=c16t1l6        server=server_2
 disk=d12   stor_dev=BB005056830430-0015 addr=c0t1l5         server=server_2
 disk=d12   stor_dev=BB005056830430-0015 addr=c16t1l5        server=server_2
 disk=d7    stor_dev=BB005056830430-0010 addr=c0t1l0         server=server_2
 disk=d7    stor_dev=BB005056830430-0010 addr=c16t1l0        server=server_2

EXAMPLE #20
-----------

To enable FLR automatic file deletion for file system ufs2_flre, type:
$ nas_fs -modify ufs2_flre -worm -auto_delete enable

id        = 40
name      = ufs4
acl       = 0
```

```
in_use    = True
type      = uxfs
worm      = enterprise with no protected files
worm_clock= Wed Jul  6 11:11:13 UTC 2011
worm Max Retention Date= No protected files created
worm Default Retention Period= 1 Year
worm Minimum Retention Period= 1 Day
worm Maximum Retention Period= 1 Year
FLR Auto_lock= on
FLR Policy Interval= 1800 seconds
FLR Auto_delete= on
FLR Epoch Year= 2000
volume    = v175
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers=
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication = Off
stor_devs =
APM00042000818-001F,APM00042000818-001D,APM00042000818-0019,APM00042
000818-0016
disks     = d20,d18,d14,d11
```

EXAMPLE #21

To start the conversion of the file system, ufs2, and to conform to the MIXED access policy mode, type:

```
$ nas_fs -translate ufs2 -access_policy start -to MIXED -from NT
id        = 38
name      = ufs2
acl       = 1432, owner=nasadmin, ID=201
in_use    = True
type      = uxfs
worm      = off
volume    = v171
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers= server_2
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication = off
stor_devs =
APM00042000818-001F,APM00042000818-001D,APM00042000818-0019,APM00042
000818-0016,APM00042000818-001C
disks     = d20,d18,d14,d11,d17
disk=d20   stor_dev=APM00042000818-001F addr=c0t1l15        server=server_2
disk=d20   stor_dev=APM00042000818-001F addr=c32t1l15       server=server_2
disk=d18   stor_dev=APM00042000818-001D addr=c0t1l13        server=server_2
disk=d18   stor_dev=APM00042000818-001D addr=c32t1l13       server=server_2
disk=d14   stor_dev=APM00042000818-0019 addr=c0t1l9         server=server_2
disk=d14   stor_dev=APM00042000818-0019 addr=c32t1l9        server=server_2
disk=d11   stor_dev=APM00042000818-0016 addr=c0t1l6         server=server_2
disk=d11   stor_dev=APM00042000818-0016 addr=c32t1l6        server=server_2
disk=d17   stor_dev=APM00042000818-001C addr=c0t1l12        server=server_2
disk=d17   stor_dev=APM00042000818-001C addr=c32t1l12       server=server_2
```

EXAMPLE #1 provides a description of command outputs.

EXAMPLE #22

To display the status of access policy conversion for ufs2, type:

```
$ nas_fs -translate ufs2 -access_policy status
status=In progress
```

```
percent_inode_scanned=90

EXAMPLE #23
-----------

To create a nested mount file system, nmfs1, type:

$ nas_fs -name nmfs1 -type nmfs -create
id        = 8002
name      = nmfs1
acl       = 0
in_use    = False
type      = nmfs
worm      = off
volume    = 0
pool      =
rw_servers=
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication   = unavailable
stor_devs =
disks     =

EXAMPLE #1 provides a description of command outputs.

EXAMPLE #24
-----------

To delete ufs1, type:

$ nas_fs -delete ufs41
name      = ufs41
acl       = 0
in_use    = False
type      = uxfs
worm      = off
volume    = v129
rw_servers=
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication   = unavailable
stor_devs =
BB005056830430-0019,BB005056830430-0016,BB005056830430-0015,BB005056830430-0010
disks     = d16,d13,d12,d7

EXAMPLE #1 provides a description of command outputs.

EXAMPLE #25
-----------

To create a file system named ufs3, with a size of 1 GB, by using the
clar_r5_performance pool, a maximum size of 10 GB and with auto-extend and
thin provisioning enabled, type:

$ nas_fs -name ufs3 -create size=1G pool=clar_r5_performance -auto_extend yes
-max_size 10G -thin yes
id        = 8003
name      = ufs3
acl       = 0
in_use    = False
type      = uxfs
worm      = off
volume    = v133
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers=
ro_servers=
```

```
rw_vdms   =
ro_vdms   =
auto_ext  = hwm=90%,max_size=10240M,thin=yes
deduplication   = unavailable
stor_devs =
BB005056830430-0018,BB005056830430-0017,BB005056830430-0014,BB005056830430-0011
disks     = d15,d14,d11,d8
```

EXAMPLE #1 provides a description of command outputs.

EXAMPLE # 26

To disable thin provisioning on ufs3, type:

```
$ nas_fs -modify ufs3 -thin no
id        = 8003
name      = ufs3
acl       = 0
in_use    = False
type      = uxfs
worm      = off
volume    = v133
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers=
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = hwm=90%,max_size=10240M,thin=no
deduplication   = unavailable
stor_devs =
BB005056830430-0018,BB005056830430-0017,BB005056830430-0014,BB005056830430-0011
disks     = d15,d14,d11,d8
```

EXAMPLE #1 provides a description of command outputs.

EXAMPLE # 27

To query the current directory type and translation status for MPD, type:

```
$ nas_fs -info ufs5 -option mpd
id        = 11
name      = ufs5
acl       = 1432, owner=nasadmin, ID=201
in_use    = True
type      = uxfs
worm      = off
volume    = v112
pool      = clar_r5_performance
member_of = root_avm_fs_group_3
rw_servers= server_2
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
deduplication   = Off
stor_devs =
BB005056830430-0019,BB005056830430-0016,BB005056830430-0015,BB005056830430-0010
disks     = d16,d13,d12,d7
 disk=d16   stor_dev=BB005056830430-0019 addr=c0t1l9         server=server_2
 disk=d16   stor_dev=BB005056830430-0019 addr=c16t1l9        server=server_2
 disk=d13   stor_dev=BB005056830430-0016 addr=c0t1l6         server=server_2
 disk=d13   stor_dev=BB005056830430-0016 addr=c16t1l6        server=server_2
 disk=d12   stor_dev=BB005056830430-0015 addr=c0t1l5         server=server_2
 disk=d12   stor_dev=BB005056830430-0015 addr=c16t1l5        server=server_2
 disk=d7    stor_dev=BB005056830430-0010 addr=c0t1l0         server=server_2
 disk=d7    stor_dev=BB005056830430-0010 addr=c16t1l0        server=server_2
```

Multi-Protocol Directory Information

```
        Default_directory_type    = DIR3
        Needs_translation         = False
        Translation_state         = Never
        Has_translation_error     = False
```

where:

Value	Definition
Default_directory_type	The default directory type for the file system. Available types are: DIR3 and COMPAT.
Needs_translation	If true, then the file system may contain more than one directory type. If false, then all directories are of the file system default directory type.
Translation_state	The current state of the translation thread. Available states are: never, not requested, pending, queued, running, paused, completed, and failed.
Has_translation_error	Indicated if the most recent translation encountered any errors.

Default_directory_type	Needs_translation state	File system
DIR3	False	Is MPD. No action required.
DIR3	True	Requires translation or file system maintenance. Contact EMC Customer Service.
COMPAT	False	Is COMPAT and requires translation. Contact EMC Customer Service.
COMPAT	True	Requires translation. Contact EMC Customer Service.

The state where both Default_directory_type=DIR3 and Needs_transalation=False assure that this filesystem's directories are all in MPD format, and there are no directories of the obsolete single-protocol format.

Any other combination of states, for example, Needs_transalation=True, indicates that there could be non-MPD directories in the filesystem which may not be compatible with a future release.

EXAMPLE #28

To display the information about the file system ufs3 and a valid fast_clone_level of 1 or 2, type:

```
$ nas_fs -info ufs3
id        = 478
name      = ufs2_flre
acl       = 0
in_use    = False
type      = uxfs
worm      = off
volume    = v1168
pool      = clarsas_archive
member_of = root_avm_fs_group_32
rw_servers=
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
fast_clone_level= unavailable
deduplication   = unavailable
stor_devs =
APM00112101832-0019,APM00112101832-0028,APM00112101832-0027,APM00112101832-0022
disks     = d25,d19,d32,d16
```

EXAMPLE #29

To display the information about a file system using the file system ufs4 using Symmetrix backend mapped pool, type:

```
$ nas_fs -info ufs4

id = 32
name = ufs4
acl = 0
in_use = True
type = uxfs
worm = off
volume = v644
pool = symm_mapped_pool
member_of = root_avm_fs_group_21
rw_servers= server_2
ro_servers=
rw_vdms =
ro_vdms =
auto_ext = hwm=50%,max_size=1024M,thin=yes
fast_clone_level = 1
deduplication = Off
compressed= Mixed
frontend_io_quota = maxiopersec 500,maxmbpersec 500
stor_devs = 000196900016-0553
disks = d524
disk=d524 stor_dev=000196900016-0553 addr=c4t3l5-0-0 server=server_2
disk=d524 stor_dev=000196900016-0553 addr=c20t3l5-0-0 server=server_2
disk=d524 stor_dev=000196900016-0553 addr=c36t3l5-0-0 server=server_2
disk=d524 stor_dev=000196900016-0553 addr=c52t3l5-0-0 server=server_2
disk=d524 stor_dev=000196900016-0553 addr=c68t3l5-0-0 server=server_2
disk=d524 stor_dev=000196900016-0553 addr=c84t3l5-0-0 server=server_2
disk=d524 stor_dev=000196900016-0553 addr=c100t3l5-0-0 server=server_2
disk=d524 stor_dev=000196900016-0553 addr=c116t3l5-0-0 server=server_2
```

where:

Value	Definition
compressed	For VNX with Symmetrix backend, indicates whether data is compressed. Values are: True, False, Mixed (indicates some of the LUNs, but not all, are compressed).
frontend_io_quota	For VNX with Symmetrix backend, indicates if Frot end IO Quota is configured on this mapped pool, could also have v alue as False (indicates Frontend IO Quota is not configured on mapped SG in Symmetrix backend).

Last Modified: Jan 11, 2013 4:12 pm

nas_fsck

Manages fsck and aclchk utilities on specified file systems.

SYNOPSIS

```
nas_fsck
  -list
| -info {-all|<fs_name>|id=<fs_id>}
| -start {<fs_name>|id=<fs_id>} [-aclchkonly][-monitor][-mover <mover_name>]
    [-Force]
```

DESCRIPTION
----

nas_fsck uses the fsck and aclchk utilities to perform a check for consistency and errors on the specified file system. nas_fsck also lists and displays the status of the fsck and aclchk utilities. File systems must be mounted read-write to use these utilities.

Depending on the size of the file system, the FSCK utility may use a significant portion of the system's memory and may affect overall system performance. Hence, it should not be run on a server under heavy load as it is possible that the server may run out of resources. In most cases, the user will be notified if sufficient memory is not available to run a file system check. In these cases, one of the following can be done:

. Start the file system during off-peak hours.
. Reboot the server and let nas_fsck run on reboot.
. Run nas_fsck on a different server if the file system is unmounted.

OPTIONS

-list
Displays a list of all the file systems undergoing fsck or aclchk.

-info {-all|<fs_name>|id=<fs_id>}
Queries the Data Mover and displays information about the status of the fsck or aclchk utilities for the specified file system.

-start {<fs_name>|id=<fs_id>}
Starts the fsck and the aclchk utilities on the specified file system.

CAUTION
If file system check is started on a mounted file system, the file system will be unavailable for the duration of the check. NFS clients will display the message NFS server not responding and CIFS clients will lose connectivity with the server and will have to remap shares.

[-aclchkonly]
Initiates the aclchk utility only, which checks and corrects any errors in the ACL database and removes duplicate ACL information stored on the specified file system. The aclchkonly option can only be used on a file system that is not exported. The default is for both fsck and aclchk.

Note: The NDMP backup process must be stopped on the Data Mover before using the nas_fsck -aclchkonly command.

[-monitor]
Displays the status of fsck and aclchk until the command completes.

Note: For a mounted file system, a <movername> is not required since the fsck and aclchk utilities are run on the Data Mover where the file system is mounted.

[-Force]
Forces a fsck or aclchk to be run on a enabled file system.

SEE ALSO

Managing Volumes and File Systems for VNX Manually and nas_fs.

EXAMPLE #1

To start file system check on ufs1 and monitor the progress, type:

```
$ nas_fsck -start ufs1 -monitor

id = 27
name = ufs1
volume = mtv1
fsck_server = server_2
inode_check_percent = 10..20..30..40..60..70..80..100
directory_check_percent = 0..0..100
used_ACL_check_percent = 100
free_ACL_check_status = Done
cylinder_group_check_status = In Progress..Done
```

Where:

Value	Definition
id	Automatically assigned ID of a file system.
name	Name assigned to the file system.
volume	Volume on which the file system resided.
fsck_server	Name of the Data Mover where the utility is being run.
inode_check_percent	Percentage of inodes in the file system checked and fixed.
directory_check_percent	Percentage of directories in the file system checked and fixed.
used_ACL_check_percent	Percentage of used ACLs that have been checked and fixed.
free_ACL_check_status	Status of the ACL check.
cylinder_group_check_status	Status of the cylinder group check.

EXAMPLE #2

To start ACL check on ufs1, type:

```
$ nas_fsck -start ufs1 -aclchkonly

ACLCHK: in progress for file system ufs1
```

EXAMPLE #3

To start a file system check on ufs2 using Data Mover server_5,
type:

```
$ nas_fsck -start ufs2 -mover server_5

name = ufs2
id = 23
volume = v134
fsck_server = server_5
inode_check_percent = 40
directory_check_percent = 0
```

```
used_ACL_check_percent = 0
free_ACL_check_status = Not Started
cylinder_group_check_status = Not Started
```

EXAMPLE #1 provides a description of command outputs.

EXAMPLE #4

To list all current file system checks, type:

```
$ nas_fsck -list

id type state volume name server
23 1 FSCK 134 ufs2 4
27 1 ACLCHK 144 ufs1 1
```

Where:

Value	Definition
id	Automatically assigned ID of a file system.
type	Type of file system.
state	Utility being run.
volume	Volume on which the file system resided.
name	Name assigned to the file system.
server	Server on which fsck is being run.

EXAMPLE #5

To display information about file system check for ufs2 that is currently running, type:

```
$ nas_fsck -info ufs2

name = ufs2
id = 23
volume = v134
fsck_server = server_5
inode_check_percent = 100
directory_check_percent = 100
used_ACL_check_percent = 100
free_ACL_check_status = Done
cylinder_group_check_status = In Progress
```

EXAMPLE #1 provides a description of command outputs.

EXAMPLE #6

To display information about all file system checks that are currently running, type:

```
$ nas_fsck -info -all

name = ufs2
id = 23
volume = v134
fsck_server = server_5
inode_check_percent = 30
directory_check_percent = 0
used_ACL_check_percent = 0
free_ACL_check_status = Not Started
cylinder_group_check_status = Not Started
name = ufs1
id = 27
volume = mtv1
fsck_server = server_2
inode_check_percent = 100
```

```
directory_check_percent = 0
used_ACL_check_percent = 0
free_ACL_check_status = Not Started
cylinder_group_check_status = Not Started

EXAMPLE #1 provides a description of command outputs.

-----------------------------------------------------------------------
Last modified: May 11, 2011 9:30 am.
```

nas_halt

Performs a controlled halt of all Control Stations and Data Movers in the VNX.

SYNOPSIS
----

nas_halt now

DESCRIPTION
----

nas_halt performs an orderly shutdown of the Control Stations and Data Movers in the VNX. nas_halt must be executed from the /nas/sbin directory.

OPTIONS
----
now
Performs an immediate halt for the VNX.

SEE ALSO

VNX System Operations and server_cpu.

EXAMPLE #1

To perform an immediate halt of the VNX, type:

/nas/sbin/nas_halt now

usage: nas_halt now
Perform a controlled halt of the Control Stations and Data Movers
/nas/sbin/nas_halt now
****************************** WARNING! ******************************
You are about to HALT this system including all of its Control Stations
and Data Movers. DATA will be UNAVAILABLE when the system is halted.
Note that this command does *not* halt the storage array.

ARE YOU SURE YOU WANT TO CONTINUE? [yes or no] : yes
Sending the halt signal to the Master Control Daemon...: Done
May 3 11:12:54 cs100 EMCServer: nas_mcd: Check and halt other CS...: Done
May 3 11:13:26 cs100 JSERVER: *** Java Server is exiting ***
May 3 11:13:31 cs100 ucd-snmp[11218]: Received TERM or STOP signal... shutting down...
May 3 11:13:31 cs100 snmpd: snmpd shutdown succeeded
May 3 11:13:32 cs100 setup_enclosure: Executing -dhcpd stop option
May 3 11:13:32 cs100 snmptrapd[11179]: Stopping snmptrapd
May 3 11:13:32 cs100 EV_AGENT[13721]: Signal TERM received
May 3 11:13:32 cs100 EV_AGENT[13721]: Agent is going down
May 3 11:13:40 cs100 DHCPDMON: Starting DHCPD on CS 0
May 3 11:13:41 cs100 setup_enclosure: Executing -dhcpd start option
May 3 11:13:41 cs100 dhcpd: Internet Software Consortium DHCP Server V3.0pl1
May 3 11:13:41 cs100 dhcpd: Copyright 1995-2001 Internet Software Consortium.
May 3 11:13:41 cs100 dhcpd: All rights reserved.
May 3 11:13:41 cs100 dhcpd: For info, please visit http://www.isc.org/products/DHCP
May 3 11:13:41 cs100 dhcpd: Wrote 0 deleted host decls to leases file.
May 3 11:13:41 cs100 dhcpd: Wrote 0 new dynamic host decls to leases file.
May 3 11:13:41 cs100 dhcpd: Wrote 0 leases to leases file.
May 3 11:13:41 cs100 dhcpd: Listening on LPF/eth2/00:00:f0:9d:04:13/128.221.253.0/24
May 3 11:13:41 cs100 dhcpd: Sending on LPF/eth2/00:00:f0:9d:04:13/128.221.253.0/24
May 3 11:13:41 cs100 dhcpd: Listening on LPF/eth0/00:00:f0:9d:01:e5/128.221.252.0/24
May 3 11:13:41 cs100 dhcpd: Sending on LPF/eth0/00:00:f0:9d:01:e5/128.221.252.0/24

```
May 3 11:13:41 cs100 dhcpd: Sending on Socket/fallback/fallback-net
May 3 11:13:59 cs100 mcd_helper: : Failed to umount /nas (0)
May 3 11:13:59 cs100 EMCServer: nas_mcd: Failed to gracefully shutdown MCD and
halt servers. Forcing halt and reboot...
May 3 11:13:59 cs100 EMCServer: nas_mcd: Halting all servers...
May 3 11:15:00 cs100 get_datamover_status: Data Mover server_5: COMMAND doesnt
match.
```

Last modified: May 10, 2011 5:25 pm.

nas_inventory

Provides detailed information about hardware components in the system.

SYNOPSIS

```
nas_inventory
{
        -list [-location]
      | {-info <location>|-all}
      | -tree
}
```

DESCRIPTION
----

nas_inventory displays detailed information about the hardware components that are configured on a system.

OPTIONS

-list
Displays a list of all hardware components and their associated name, type, status, and system ID.

[-location]
Displays the location string for each component in the output. The location string is a unique identifier for the component.

Specifies the location string with enclosed double quotes (" ") and displays a list of detailed information for the specific component for which the string is the unique ID.

-info <location_string>|-all
Displays a list of all the properties for a component, including the component name, type, status, variant, associated system, serial number, part number, and history.

The -all option lists detailed information for all components in the system.

-tree
Displays a hierarchical tree of components, including the status of each component.

EXAMPLE #1

To display a list of components on the system, type:

```
$ nas_inventory -list

Component          Type          Status          System ID
Battery A          Battery       OK              CLARiiON CX4-240FCNTR083000055
VNX NS40G
FCNTR083000055001A VNX           Warning         VNX NS40GFCNTR083000055001A
CLARiiON CX4-240
FCNTR083000055     CLARiiON      OK              CLARiiON CX4-240 FCNTR083000055
DME 0 Data Mover 2 Data Mover    OK              VNX NS40G FCNTR083000055001A
DME 0 Data Mover 2
Ethernet Module    Module        OK              VNX NS40G FCNTR083000055001A
DME 0 Data Mover 2
SFP BE0            SFP           OK              VNX NS40G FCNTR083000055001A
DME 0 Data Mover 2
SFP BE1            SFP           OK              VNX NS40G FCNTR083000055001A
DME 0 Data Mover 2
SFP FE0            SFP           OK              VNX NS40G FCNTR083000055001A
```

Where:

Value	Definition
Component	Description of the component.
Type	The type of component. Possible types are: battery, blower, VNX, Control Station, Data Mover, and disk.
Status	The current status of the component. Status is component type specific. There are several possible status values, each of which is associated with a particular component type.
System ID	The identifier for the VNX or the storage ID of the system containing the component.

EXAMPLE #2

To display a list of components and component locations, type:

```
$ nas_inventory -list -location

Component Type Status System ID
Location
Battery A Battery OK CLARiiON CX4-240 FCNTR083000055
system:NS40G:FCNTR083000055001A|clariionSystem:CX4-240:FCNTR083000055|sps::A
Celerra NS40G FCNTR083000055001A Celerra Warning Celerra NS40G
FCNTR083000055001A system:NS40G:FCNTR083000055001A
CLARiiON CX4-240 FCNTR083000055 CLARiiON OK CLARiiON CX4-240 FCNTR083000055
system:NS40G:FCNTR083000055001A|clariionSystem:CX4-240:FCNTR083000055
DME 0 Data Mover 2 Data Mover OK Celerra NS40G FCNTR083000055001A
system:NS40G:FCNTR083000055001A|enclosure:xpe:0|mover:NS40:2
DME 0 Data Mover 2 Ethernet Module Module OK Celerra NS40G FCNTR083000055001A
system:NS40G:FCNTR083000055001A|enclosure:xpe:0|mover:NS40:2|module:ethernet:
DME 0 Data Mover 2 SFP BE0 SFP OK Celerra NS40G FCNTR083000055001A
system:NS40G:FCNTR083000055001A|enclosure:xpe:0|mover:NS40:2|sfp::BE0
DME 0 Data Mover 2 SFP BE1 SFP OK Celerra NS40G FCNTR083000055001A
system:NS40G:FCNTR083000055001A|enclosure:xpe:0|mover:NS40:2|sfp::BE1
DME 0 Data Mover 2 SFP FE0 SFP OK Celerra NS40G FCNTR083000055001A
system:NS40G:FCNTR083000055001A|enclosure:xpe:0|mover:NS40:2|sfp::FE0
```

EXAMPLE #3

To list information for a specific component, type:

```
$ nas_inventory -info "system:NS40G:FCNTR083000055001A|
clariionSystem:CX4-240:FCNTR083000055|iomodule::B0"

Location = system:NS40G:FCNTR083000055001A|clariionSystem:CX4-240:
FCNTR083000055|iomodule::B0
Component Name = IO Module B0
Type = IO Module
Status = OK
Variant = 4 PORT FIBRE IO MODULE
Storage System = CLARiiON CX4-240 FCNTR083000055
Serial Number = CF2YW082800426
Part Number = 103-054-100C
History = EMC_PART_NUMBER:103-054-100C
EMC_ARTWORK_REVISION:C01
EMC_ASSEMBLY_REVISION:C03
EMC_SERIAL_NUMBER:CF2YW082800426
VENDER_PART_NUMBER:N/A
VENDER_ARTWORK_NUMBER:N/A
VENDER_ASSEMBLY_NUMBER:N/A
VENDER_SERIAL_NUMBER:N/A
VENDOR_NAME:N/A
LOCATION_OF_MANUFACTURE:N/A
YEAR_OF_MANUFACTURE:N/A
MONTH_OF_MANUFACTURE:N/A
```

DAY_OF_MONTH_OF_MANUFACTURE:N/A
ASSEMBLY_NAME:4 PORT FIBRE IO MODULE

Note: The location string must be enclosed in double quotes.

Where:

Value	Definition
Location	The unique identifier of the component and where the component is located in the component hierarchy.
Component	The description of the component.
Type	The type of component. Possible types are: battery, blower, VNX for file, VNX for block, Control Station, Data Mover, and disk.
Status	The current condition of the component. Status is component type specific. There are several possible status values, each of which is associated with a particular component type.
Variant	The specific type of hardware.
Storage System	The model and serial number of the system.
Serial Number	The serial number of the hardware component.
Part Number	The part number of the hardware component.
History	If available, the history information of the component. Possible values are: part number, serial number, vendor, date of manufacture, and CPU information.

EXAMPLE #4

To display components in a tree structure, type:

$ nas_inventory -tree

Component	Type	Status
Celerra NS40G FCNTR083000055001A	Celerra	Warning
CLARiiON CX4-240 FCNTR083000055	CLARiiON	OK
Battery A	Battery	OK
IO Module A0	IO Module	OK
IO Module A1	IO Module	OK
IO Module A2	IO Module	Empty
IO Module A3	IO Module	Empty
IO Module A4	IO Module	Empty
IO Module B0	IO Module	OK
IO Module B1	IO Module	OK
IO Module B2	IO Module	Empty
IO Module B3	IO Module	Empty
IO Module B4	IO Module	Empty
Power Supply A0	Power Supply	OK
Power Supply A1	Power Supply	OK
Power Supply B0	Power Supply	OK
Power Supply B1	Power Supply	OK

EXAMPLE #5

To list information for a specific component, type:

$ nas_inventory -info "system:EA-NAS-SN:00019670026100013|enclosure:SYMM:Eng 3 Dir A|mover:EA-NAS-SN:3|iomodule::3"
Location = system:EA-NAS-SN:00019670026100013|enclosure:SYMM:Eng 3 Dir A|mover:EA-NAS-SN:3|iomodule::3
Component Name = SYMM Eng 3 Dir A Data Mover 3 IO Module 3
Type = IO Module
Status = OK
Variant = 4 PORT CU GIGE
History = FIRMWARE_VERSION:3.28
ASSEMBLY_NAME:4 PORT CU GIGE

Note: The location string must be enclosed in double quotes.

Where:

Value	Definition
Location	The unique identifier of the component and where the component is located in the component hierarchy.
Component	The description of the component.
Type	The type of component. Possible types are: battery, blower, VNX for file, VNX for block, Control Station, Data Mover, and disk.
Status	The current condition of the component. Status is component type specific. There are several possible status values, each of which is associated with a particular component type.
Variant	The specific type of hardware.
Storage System	The model and serial number of the system.
Serial Number	The serial number of the hardware component.
Part Number	The part number of the hardware component.
History	If available, the history information of the component. Possible values are: part number, serial number, vendor, date of manufacture, firmware version and CPU information.

FIRMWARE_VERSION: Displays firmware version of iomodule component

---

Last modified: May 11, 2011 10:00 am.

nas_license

Enables software packages.

SYNOPSIS

```
nas_license
  -list
| -create <package_name>[|<key_code>]
| -delete <package_name>
| -init
```

DESCRIPTION
----

nas_license enables software packages that are available for use with the system. The <key_code> is supplied by EMC.

All entries are case-sensitive.

OPTIONS

No arguments
Displays a usage message containing all available and valid software packages that can be installed.

-list
Displays the site_key as a string and any software packages for which a license has been installed. The site_key is a permanent license and cannot be deleted.

Note: Licenses installed on the Control Station are read by the system. The site_key is a unique identifier which gets generated the first time nas_license is run. The site_key is also used to decode the key_code supplied by EMC personnel for special packages.

-create <package_name>[=<key_code>]
Installs the license for the indicated <package_names>. Valid <package_names> are:

site key
nfs
cifs
snapsure
advancedmanager
replicator
filelevelretention

Note: These packages do not require key_code as they can be enabled from the GUI. Special packages are supplied along with the required Key_code by the EMC Customer Service Representative.

-delete <package_name>
Deletes the license for the specified <package_name>.

-init
Initializes the database and re-creates the license file by using the site_key that is already installed. The license file is located at /nas/site as nas_license. It contains license keys in an encrypted format. The -init option should be run only if the license file containing all the license information has been lost and the following error message is received:

license table is not initialized

Once the license file has been re-created, the rest of the entries, if present, should be re-added by using the -create option.

EXAMPLE #1

To install a license for the snapsure software package, type:

```
$ nas_license -create snapsure

done
```

EXAMPLE #2

To display all software packages with currently installed licenses, type:

```
$ nas_license -list

key                 status                value
site_key            online          42 de 6f d1
advancedmanager     online
nfs                 online
cifs                online
snapsure            online
replicator          online
filelevelretention online
```

EXAMPLE #3

To delete a license for specified software package, type:

```
$ nas_license -delete snapsure

done
```

EXAMPLE #4

To initialize the database and re-create the license file, type:

```
$ nas_license -init

done
```

Last modified: Jan 15, 2013 4:25 pm

nas_logviewer

Displays the content of nas_eventlog generated log files.

SYNOPSIS
----
nas_logviewer <file_name>
        [-f][-v|-t]

DESCRIPTION
----

nas_logviewer displays the event log and other logs created by
nas_eventlog. The log files may be system generated, or created by
the user. Information in the log file is read from oldest to newest.

OPTIONS

No arguments
Displays the contents of the specified logfile.

-f
Monitors the growth of the log by entering into an endless loop,
pausing and reading the log as it is being generated. To exit, press
Ctrl-C together.

[-v|-t]
Displays the log files in verbose or terse format.

SEE ALSO

Configuring Events and Notifications on VNX for File and server_log.

EXAMPLE #1

To view the contents of the sys_log file, type:

$ nas_logviewer /nas/log/sys_log|more

May 12 18:01:57 2007:CS_PLATFORM:NASDB:INFO:300:::::nasdb_backup: NAS_DB
checkpoint in progress
May 12 18:02:59 2007:CS_PLATFORM:NASDB:INFO:305:::::nasdb_backup: NAS_DB
Checkpoint done
May 12 18:03:00 2007:CS_PLATFORM:NASDB:ERROR:202:::::NAS database error
detectedMay 12 18:03:12 2007
:CS_PLATFORM:NASDB:INFO:306:::::nasdb_backup: NAS DB Backup done
May 12 19:01:52 2007:CS_PLATFORM:NASDB:INFO:300:::::nasdb_backup: NAS_DB
checkpoint in progress
May 12 19:02:50 2007:CS_PLATFORM:NASDB:INFO:305:::::nasdb_backup: NAS_DB
Checkpoint done
May 12 19:02:51 2007:CS_PLATFORM:NASDB:ERROR:202:::::NAS database error
detectedMay 12 19:03:02 2007
:CS_PLATFORM:NASDB:INFO:306:::::nasdb_backup: NAS DB Backup done
May 12 20:01:57 2007:CS_PLATFORM:NASDB:INFO:300:::::nasdb_backup: NAS_DB
checkpoint in progress
May 12 20:02:58 2007:CS_PLATFORM:NASDB:INFO:305:::::nasdb_backup: NAS_DB
Checkpoint done
May 12 20:02:59 2007:CS_PLATFORM:NASDB:ERROR:202:::::NAS database error
detectedMay 12 20:03:10 2007
:CS_PLATFORM:NASDB:INFO:306:::::nasdb_backup: NAS DB Backup done
May 12 21:01:52 2007:CS_PLATFORM:NASDB:INFO:300:::::nasdb_backup: NAS_DB
checkpoint in progress
May 12 21:02:51 2007:CS_PLATFORM:NASDB:INFO:305:::::nasdb_backup: NAS_DB
Checkpoint done
May 12 21:02:52 2007:CS_PLATFORM:NASDB:ERROR:202:::::NAS database error
detectedMay 12 21:03:03 2007
:CS_PLATFORM:NASDB:INFO:306:::::nasdb_backup: NAS DB Backup done

Note: This is a partial listing due to the length of the outputs.

EXAMPLE #2

To display the contents of the log files in terse format, type:

```
$ nas_logviewer -t /nas/log/sys_log

May 12 18:01:57 2007:96108871980:nasdb_backup: NAS_DB checkpoint in progress
May 12 18:02:59 2007:96108871985:nasdb_backup: NAS_DB Checkpoint done
May 12 18:03:00 2007:83223969994:NAS database error detected
May 12 18:03:12 2007:96108871986:nasdb_backup: NAS DB Backup done
May 12 19:01:52 2007:96108871980:nasdb_backup: NAS_DB checkpoint in progress
May 12 19:02:50 2007:96108871985:nasdb_backup: NAS_DB Checkpoint done
May 12 19:02:51 2007:83223969994:NAS database error detected
May 12 19:03:02 2007:96108871986:nasdb_backup: NAS DB Backup done
May 12 20:01:57 2007:96108871980:nasdb_backup: NAS_DB checkpoint in progress
May 12 20:02:58 2007:96108871985:nasdb_backup: NAS_DB Checkpoint done
May 12 20:02:59 2007:83223969994:NAS database error detected
May 12 20:03:10 2007:96108871986:nasdb_backup: NAS DB Backup done
May 12 21:01:52 2007:96108871980:nasdb_backup: NAS_DB checkpoint in progress
May 12 21:02:51 2007:96108871985:nasdb_backup: NAS_DB Checkpoint done
```

EXAMPLE #3

To display the contents of the log files in verbose format, type:

```
$ nas_logviewer -v /nas/log/sys_log|more

logged time = May 12 18:01:57 2007
creation time = May 12 18:01:57 2007
slot id =
id = 96108871980
severity = INFO
component = CS_PLATFORM
facility = NASDB
baseid = 300
type = EVENT
brief discription = nasdb_backup: NAS_DB checkpoint in progress
full discription = The Celerra configuration database is being checkpointed.
recommended action = No action required.
logged time = May 12 18:02:59 2007
creation time = May 12 18:02:59 2007
slot id =
id = 96108871985
severity = INFO
component = CS_PLATFORM
facility = NASDB
baseid = 305
type = EVENT
brief description = nasdb_backup: NAS_DB Checkpoint done
full description = The NAS DB backup has completed a checkpoint of the current
reparation for performing a backup of NAS system data.
recommended action = No action required.
```

EXAMPLE #4

To monitor the growth of the current log, type:

```
$ nas_logviewer -f /nas/log/sys_log|more

May 12 18:01:57 2007:CS_PLATFORM:NASDB:INFO:300:::::nasdb_backup: NAS_DB
checkpoint in progress
May 12 18:02:59 2007:CS_PLATFORM:NASDB:INFO:305:::::nasdb_backup: NAS_DB
Checkpoint done
May 12 18:03:00 2007:CS_PLATFORM:NASDB:ERROR:202:::::NAS database error
detectedMay 12 18:03:12 2007
:CS_PLATFORM:NASDB:INFO:306:::::nasdb_backup: NAS DB Backup done
```

May 12 19:01:52 2007:CS_PLATFORM:NASDB:INFO:300:::::nasdb_backup: NAS_DB checkpoint in progress
May 12 19:02:50 2007:CS_PLATFORM:NASDB:INFO:305:::::nasdb_backup: NAS_DB Checkpoint done
May 12 19:02:51 2007:CS_PLATFORM:NASDB:ERROR:202:::::NAS database error detectedMay 12 19:03:02 2007
:CS_PLATFORM:NASDB:INFO:306:::::nasdb_backup: NAS DB Backup done
May 12 20:01:57 2007:CS_PLATFORM:NASDB:INFO:300:::::nasdb_backup: NAS_DB checkpoint in progress
May 12 20:02:58 2007:CS_PLATFORM:NASDB:INFO:305:::::nasdb_backup: NAS_DB Checkpoint done
May 12 20:02:59 2007:CS_PLATFORM:NASDB:ERROR:202:::::NAS database error detectedMay 12 20:03:10 2007
:CS_PLATFORM:NASDB:INFO:306:::::nasdb_backup: NAS DB Backup done
May 12 21:01:52 2007:CS_PLATFORM:NASDB:INFO:300:::::nasdb_backup: NAS_DB checkpoint in progress
May 12 21:02:51 2007:CS_PLATFORM:NASDB:INFO:305:::::nasdb_backup: NAS_DB Checkpoint done
May 12 21:02:52 2007:CS_PLATFORM:NASDB:ERROR:202:::::NAS database error detectedMay 12 21:03:03 2007
:CS_PLATFORM:NASDB:INFO:306:::::nasdb_backup: NAS DB Backup done

---

Last modified: May 10, 2011 1:00 pm.

nas_message

Displays message description.

SYNOPSIS
----

nas_message
  -info <MessageId>

DESCRIPTION
----

nas_message provides detailed descriptions to a dedicated message.
A brief description, full description, and recommended user action of
the message are displayed.

OPTIONS

-info <MessageId>
Displays detailed descriptions of the error message, including
severity, component, facility, BaseID, and recommended user action.
The message parameters are displayed in the form ${stateDesc,8,%s}
and not as parameter values. The <MessageId> must be a positive
integer.

SEE ALSO

Celerra Network Server Error Messages Guide.

EXAMPLE #1

To display detailed descriptions for error message 13421838337, type:

$ nas_message -info 13421838337

MessageID = 13421838337
BaseID = 1
Severity = ERROR
Component = CS_CORE
Facility = default
Type = STATUS
Brief_Description = Operation not permitted${arg0,8,%s}
Full_Description = The operation is not permitted due to an ACL or ownership
issue on the specified object.
Recommended_Action = Check ownership or ACL of the object in question.If
appropriate change the setting to resolve the conflict. Refer to the nas_acl
and chmod man page.

Last modified: May 10, 2011 5:25 pm.

nas_migrate

Plans migrations for Virtual Data Mover (VDM) level, and
manages migrations for both VDM and File system (FS) level.

SYNOPSIS

```
nas_migrate
          -list [{-all|-mover <movername>}][-id]
        | -info [{-all|-mover <movername>|id=<migId>|<migName>}]
        | -plan {
                -list [{-all|-mover <movername>}][-id]
              | -info [{-all|-mover <movername>|id=<planId>|<planName>}]
              | -create <planName>
                        -source <vdmName>
                        -destination
                        {<existing_dstVdmName>
                        |-pool{id=<dstStoragePoolId>|<dstStoragePoolName>}
                        }
                        -interconnect {<interConnectName>|id=<interConnectId>}
                        [-storage_pools
                        {<<srcStoragePoolName>:<dstStoragePoolName>[,...]>
                        |-id<<srcStoragePoolId>:<dstStoragePoolId>[,...]>
                        }]
                        [-take_over_ips
                          [-network_devices <<srcDeviceName>:<dstDeviceName>[,...
]>]]
                        [-checkpoint_excluded]
                        [-background]
              | -modify {<planName>|id=<planId>}
                        [-name <newPlanName>]
                        [-filesystems
                        {<srcFs=<name>[:dstFs=<id>][:dstPool=<name>]
                         [:srcSavPool=<name>][:dstSavPool=<name>][,...]>
                        |-id<srcFs=<id>[:dstFs=<id>][:dstPool=<id>]
                         [:srcSavPool=<id>][:dstSavPool=<id>][,...]>
                        }]
                        [-interfaces <name=<infName>:dstDevice=<devName>[,...] >]
                        [-background]
              | -delete {<planName>|id=<planId>} [-background]
              }
        | -create <migName> -vdm
                -plan {<planName>|id=<planId>}[-background]
        | -create <migName> -fs
                -source {id=<fsId>|<fsName>}
                        [-sav {id=<srcSavVolStoragePoolId>
                        |<srcSavVolStoragePoolName>}]
                -destination
                        {{<existing_dstFsName>|id=<existing_dstFsId>}
                        |-pool{id=<dstStoragePoolId>|<dstStoragePoolName>}
                        }
                        [-sav{id=<dstSavVolStoragePoolId>
                         |<dstSavVolStoragePoolName>}]
                        [-vdm <dstMountVdmName>]
                -interconnect {<interConnectName>|id=<interConnectId>}
                [-checkpoint_excluded]
                [-background]
        | -complete {<migName>|id=<migId>}
                [-checkpoint_mismatch_ignored] [-background]
        | -delete {<migName>|id=<migId>} [-background]
        | -stop {<migName>|id=<migId>} [-background]
        | -start {<migName>|id=<migId>} [-background]
```

DESCRIPTION

The nas_migrate command manages the migration of VDMs, and FS or
checkpoints mounted to VDMs. It is also used for migration of FS and
its mounted checkpoints.

OPTIONS

-list [{-all|-mover <movername>}][-id]
Lists summary of all the migrations for the cabinet regarded as destination cabinet, or for the specified mover regarded as destination mover. The -id option shows migration ID in the summary; by default, the system-generated migration ID isn't shown.

-info [{-all|-mover <movername>|id=<migId>|<migName>}]
Displays detailed information of all the migrations for the cabinet regarded as destination cabinet, for the specified mover regarded as destination mover, or for one migration with a specified ID or name.

-plan
Generates a migration plan for VDM migration.

-list [{-all|-mover <movername>}][-id]
Lists summary information for all VDM migration plans with cabinet as destination cabinet if -all option is specified or no option is specified by default, or a specific data mover as destination mover. The -id option shows migration plan ID in the summary; by default, the system-generated migration plan ID isn't shown.

-info [{-all|-mover <movername>|id=<planId>|<planName>}]
Displays detailed information for all VDM migration plans with cabinet as destination cabinet if -all option is specified or no option is specified by default. Also, displays detailed information for a specific data mover as destination mover, or for one migration plan with specific ID or name.

-create <planName>
Creates a VDM migration plan.

-source <vdmName>
Specifies the name of the source VDM to migrate. The source system information is implied in the interconnect option of VNX for File Operating Environment. See the usage of -interconnect option.

-destination {<existing_dstVdmName>
|-pool {id=<dstStoragePoolId>|<dstStoragePoolName>}}
Specifies the name of an existing destination VDM, or the pool name or ID to create a new destination VDM.

-interconnect {<interConnectName>|id=<interConnectId>}
Specifies the name or ID of the local VNX for File Operating Environment interconnect configured on the destination. The mutual VNX for File Operating Environment interconnects are supposed to be configured between source and destination. The source system information is implied in the interconnect option of VNX for File Operating Environment.

[-storage_pools {<<srcStoragePoolName>:<dstStoragePoolName>[,...]>
|-id <<srcStoragePoolId>:<dstStoragePoolId[,...]>}]
Indicates mapping of source and destination storage pools, and SavVol pools. These pools must exist on the source or the destination
.
This will guide the migration to create file systems in the storage pools on the destination, based on the mapping relationship of the source storage pools where their source file system lie. Either pool name or pool ID can be specified, with commas separating
each
mapping.If no mappings are specified, destination storage pools will
be
automatically selected to create each file systems.

[-take_over_ips][-network_devices <<srcDeviceName>:<dstDeviceName [,.
..]>]
Takes over source network interfaces if -take_over_ips option is

specified. Indicates mapping of source and destination network device
s.
This will guide the migration to choose the network devices, which
will be used when creating destination network interfaces when
option -take_over_ips is specified. Without -take_over_ips, the
destination interfaces must be manually created on the destination mo
ver
by the user, with names identical to the source interfaces.

No matter whether the source network interfaces are to be
taken-over or not, the interfaces attached to the source VDM
will be turned down after migration is completed.

Note: To take over IPs, the interfaces must be in the same subnet and
have the
same VLAN settings at the source and the destination. Also, the inter
faces must
be IPv4. IPv6 interfaces cannot be taken over.

Note: To exclude a File System, it must be unexported or unshared and
unmounted
before creating a VDM level migration plan.

[-checkpoint_excluded]
Excludes all the existing read-only user checkpoints from the migrati
on.

[-background]
Runs the task in background.

Note: When -background is specified, a task ID will be returned, and
the user can check nas_task -i <taskId> to see the result of the task
:
succeeded, failed or running. Otherwise, if -background is not specif
ied,
the command will return "OK" till the task is finished or succeeded,
or an error message if failed.

-modify {<planName>| id=<planId>}
Modifies a VDM migration plan of specific name or ID.

[-name <newPlanName>]
Renames the migration plan.

[-filesystems {<srcFs=<name>[:dstFs=<id>][:dstPool=<name>]
[:srcSavPool=<name>][:dstSavPool=<name>][,...]>
|-id <srcFs=<id>[:dstFs=<id>][:dstPool=<id>][:srcSavPool=<id>]
[:dstSavPool=<id>][,...]>]}]
Updates source SavVol, recommended destination FSID, destination pool
,
or destination SavVol for a specified source file system. Only needs
to specify the file system(s) to be reconfigured, especially
srcFs=<name> or srcFs=<id>. Source file system that is not in
the current migration plan cannot be specified.

[-interfaces <name=<infName>:dstDevice=<devName> [,...]>]
Updates network devices on which to create network interfaces.
Both the device name <devName> and interface name(s) <infName>
should be specifed, especially the interface name(s). The interface
name(s) to be reconfigured are key. Interface name(s) that are not
in the current migration plan cannot be specified.

[-background]
Runs the task in background.

Note: When -background is specified, a task ID will be returned, and
the user can check nas_task -i <taskId> to see the result of the task
:
succeeded, failed or running. Otherwise, if -background is not specif
ied,
the command will return "OK" till the task is finished or succeeded,

or an error message if failed.

-delete {<planName>| id=<planId>}
Deletes a VDM migration plan of specific name or ID.The migration plan cannot
be deleted when a migration exists that references the migration plan.

[-background]
Runs the task in background.

Note: When -background is specified, a task ID will be returned, and
the user can check nas_task -i <taskId> to see the result of the task:
succeeded, failed or running. Otherwise, if -background is not specified,
the command will return "OK" till the task is finished or succeeded,
or an error message if failed.

-create <migName> -vdm
Creates a VDM level migration after creating a migration plan. The <migName> specifies
the name of migration session, which is unique per destination cabinet.

-plan {<planName>|id=<planId>}
Specifies the name or ID of the VDM migration plan. The <planName> is
created by the user and the <planId> is generated by the system, both of them
are unique per destination cabinet.

[-background]
Runs the task in background.

Note: When -background is specified, a task ID will be returned, and the user
can check nas_task -i <taskId> to see the result of the task: succeeded, failed
or running. Otherwise, if -background is not specified, the command will return
"OK" till the task is finished or succeeded, or an error message if failed.

-create <migName> -fs
Creates a File System level migration after creating a migration plan. The <migName>
specifies the name of migration session, which is unique per destination
cabinet and unchangeable.

-source {id=<fsId>|<fsName>}
Specifies the source File System name or ID.

[-sav{id=<srcSavVolStoragePoolId>|<srcSavVolStoragePoolName>}]
Specifies the SavVol pool used by all subsequent checkpoints of the source
file system. If it is not specified, VNX File Migration applies the same
storage pool of the source file system as the SavVol pool.

Note: This option is only valid when the source file system has
no checkpoints before the migration.

-destination {<existing_dstFsName>|id=<existing_dstFsId>}
Specifies the destination file system, either from a name or ID of an existing
file system name as the destination. An existing destination file system
must be mounted as read-only and have the same size and configuration as the
source.

-pool {id=<dstStoragePoolId>|<dstStoragePoolName>}}
Specifies a storage pool to create the destination file system automatically,
using the same size as the source file system.

[-sav{id=<dstSavVolStoragePoolId>|<dstSavVolStoragePoolName>}]
Specifies the SavVol pool used by all subsequent checkpoints of the destination
file system. If it is not specified, VNX File Migration applies the same
storage pool of the destination file system as the SavVol pool.

Note: All the checkpoints for a file system share the same SavVol.

[-vdm <dstMountVdmName>]}
Specifies a VDM to mount the newly created destination file system. If the
destination file system is mounted to a VDM, this option is mandatory.
By default, without this option, the newly-created destination file system
will be mounted to the destination Data Mover specified in the VNX for File
Operating Environment interconnect.

-interconnect {<interConnectName>|id=<interConnectId>}
Specifies the name or ID of the local VNX for File Operating Environment
interconnect configured on the destination. The mutual VNX for File Operating
Environment interconnects are supposed to be configured
between source and destination data movers.

[-checkpoint_excluded]
Excludes all the existing read-only user checkpoints from the migration.

[-background]
Runs the task in background.

Note: When -background is specified, a task ID will be returned, and the user
can check nas_task -i <taskId> to see the result of the task: succeeded, failed
or running. Otherwise, if -background is not specified, the command will return
"OK" till the task is finished or succeeded, or an error message if failed.

-complete {<migName>|id=<migId>} [-checkpoint_mismatch_ignored] [-background]
Completes a migration when (1)the migration state is READY_TO_COMPLETE or COMPLETE_FAILED,
(2)names of the destination network interfaces must be configured the same as the source
for VDM level migration, and (3)the migration has globalsystem/mover configuration migrated
with the system configuration migration script, or manually by an administrator.

Completes a migration with a specified name or ID. The -checkpoint_mismatch_ignored
option forcefully completes the migration, ignoring any mismatching checkpoints.The
-background option means the task can be run in the background.

-delete {<migName>|id=<migId>} [-background]
Deletes an existing migration with a specified name or ID, when no migration commands
are running. Executes the commands when the migration state is not INITIAL_COPYING,
STARTING, STOPPING, or DELETING. The source and destination VDM, file systems, checkpoints and
interfaces will not be deleted. The -background option means the task can be run in
the background.

-stop {<migName>|id=<migId>} [-background]
Stops a migration with a specified name or ID, when the migration state is
READY_TO_COMPLETE, STOP_FAILED or START_FAILED. The -background option means
the task can be run in the background.

```
        -start {<migName>|id=<migId>} [-background]
        Starts a migration with a specified name or ID, when the migration state is STOPP
ED,
        STOP_FAILED, START_FAILED or INITIAL_COPY_FAILED. The -background option means th
e task can be run
        in the background.

        SEE ALSO
        --------

        migrate_system_conf, nas_replicate, nas_fs, and fs_ckpt.

        SYSTEM OUTPUT
        -------------
        The migration states that can appear in the output include CREATING, INITIAL_COPY
ING,
        INITIAL_COPY_FAILED, READY_TO_COMPLETE, COMPLETING, COMPLETE_FAILED, COMPLETED, S
TOPPING,
        STOP_FAILED, STARTING, STOPPED, START_FAILED, DELETING and DELETE_FAILED.

        EXAMPLE #1
        ----------

        To list summary information of all the migrations, type:

        $ nas_migrate -list -all -id
        ID          Name      Type       State             Source Celerra/VNX Source VDM/
FS DestVDM/FS   Network Status
        20000010804 vdmMigEx1 VDM        READY_TO_COMPLETE spring vdmEx1      vdmEx1
   vdmEx1      OK
        20000000877 fsMigEx1  FILESYSTEM READY_TO_COMPLETE spring fs3         fs3
   fs3         OK

        EXAMPLE #2
        ----------
        To view detailed information of all the migrations whose destination are current
cabinet,
        type:

        $ nas_migrate -info -all

        ID                                      =       20000010804
        Name                                    =       vdmMigEx1
        Type                                    =       VDM
        State                                   =       READY_TO_COMPLETE
        Network Status                          =       OK
        Source Celerra/VNX Network Server       =       spring
        Peer Dart Interconnect                  =       spring_winter
        Dart Interconnect                       =       winter_spring
        Source VDM                              =       vdmEx1
        Destination VDM                         =       vdmEx1
        Vdm Migration Plan                      =       planEx1
        File Systems                            =       fs1->fs1;Checkpoints:fs1_ckpt1->f
s1_ckp
                                                t1(*Mismatched),fs1_ckpt2->fs1_ckpt2
                                                       fs2->fs2;Checkpoints:fs2_ckpt1->fs
2_ckpt1
        Source Mover                            =       server_2
        Destination Mover                       =       server_3
        Read-Only User Checkpoints Excluded     =       No
        Takeover IP Addresses                   =       Yes
        Interfaces to Takeover                  =       eth1, eth2
        Replications                            =       3093_BB005056903C71_0000_2926_BB0
050569059F6_0000:VDM
                                                        3095_BB005056903C71_0000_2931_BB0
050569059F6_0000:Filesystem
                                                        3097_BB005056903C71_0000_2933_BB0
050569059F6_0000:Filesystem

        ID                                      =       20000000877
```

```
        Name                                    =       fsMigEx1
        Type                                    =       FILESYSTEM
        State                                   =       READY_TO_COMPLETE
        Network Status                          =       OK
        Source Celerra/VNX Network Server       =       spring
        Peer Dart Interconnect                  =       spring_summer
        Dart Interconnect                       =       summer_spring
        File Systems                            =       fs3->fs3
        Source Mover                            =       server_2
        Destination Mover                       =       server_4
        Read-Only User Checkpoints Excluded     =       Yes
        Replications                            =       337_BB005056903C71_0000_2951_BB00
50569059F6_0000 : Filesystem

        EXAMPLE #3
        ----------

        To list summary information of all VDM migration plans, type:

        $ nas_migrate -plan -list -id

        ID          Name     Source Celerra/VNX  Source VDM Destination VDM  Destination
Pool
        20000034500 PlanEx1  spring              vdmEx1     N/A              dstpool3
        20000035780 PlanEx2  spring              vdmEx2     vdmEx2           N/A

        Note: Either Destination VDM or Pool is N/A because the user can specify a pool t
o create
        destination VDM root File system, or the existing destination VDM.

        EXAMPLE #4
        ----------

        To display detailed information about migration plan PlanEx1, type:

        $ nas_migrate -plan -info PlanEx1

        ID                                  = 20000034500
        Name                                = planEx1
        Source Celerra/VNX Network Server   = spring
        Peer Dart Interconnect              = spring_winter
        Dart Interconnect                   = winter_spring
        Source VDM                          = vdmEx1
        Destination VDM                     = N/A
        Destination Pool (for VDM)          = dstpool3
        File Systems                        = srcFs                     =       fs1
                                            |-- dstFs(Recommended ID)   =       1001,NOT
PRESERVED
                                            |-- dstPool                 =       dstpool1
                                            |-- srcSavPool              =       srcpool1
                                            ï¿½-- dstSavPool            =       dstpool1
                                            = srcFs                     =       fs2
                                            |-- dstFs(Recommended ID )  =       1002, PRE
SERVED
                                            |-- dstPool                 =       dstpool2
                                            |-- srcSavPool              =       srcpool2
                                            ï¿½-- dstSavPool              =     Dstpool2

        Read-Only User Checkpoints Excluded = No
        Takeover IP Addresses               = Yes
        Interfaces - Devices                = name=eth1:dstDevice=cge1
                                              name=eth2:dstDevice=cge20
        Where:

        Value           Definition
        -----           ----------

        NOT PRESERVED   The source file system ID cannot be preserved, then the NFS clien
ts
                        have to remount this file system after VDM migration completes.
```

EXAMPLE #5

To create a VDM migration plan with the default setting when IP-takeover applies, type:

```
$ nas_migrate -plan -create planEx1 -source vdmEx1 -destination -pool dstpool3
-interconnect winter_spring -take_over_ips

Info 26843676673: In Progress: Operation is still running. Check task id 24416 on

    the Background Tasks screen for results.
Validate plan name ... succeeded
Create plan ...
Validate destination system licenses ... succeeded
Validate interconenct ... succeeded
Validate source system licenses ... succeeded
Validate system versions ... succeeded
Validate I18N and CIFS service ... succeeded
Validate source VDM ... succeeded
Make migration plan for VDM ... succeeded
Validate source file system(s) ... succeeded
Make migration plan for file system(s) ... succeeded
Make migration plan for interface(s) ... succeeded
Create plan ... succeeded
Save plan ... succeeded

OK
```

EXAMPLE #6

To create a VDM migration plan with storage pool mapping and IP-takeover, type:

```
$ nas_migrate -plan -create planEx1 -source vdmEx1 -destination -pool dstpool3
-interconnect winter_spring -storage_pools srcpool1:dstpool1,srcpool2:dstpool2
-take_over_ips
```

Output omitted for brevity.

Where:

Value	Definition
storage_pools	Specifies the storage pool mapping. When not specified, the default matching rules are: auto-select a storage pool on the destination for each file system by (in the priority order) storage pool profile, disk type, then size.

EXAMPLE #7

To create a VDM migration plan with network device mapping, type:

```
$ nas_migrate -plan -create planEx1 -source vdmEx1 -destination -pool dstpool3
-interconnect winter_spring -take_over_ips -network_devices cge_src1:cge_dst1,
cge_src2:cge_dst2
```

Output omitted for brevity.

Where:

Value	Definition
network_devices	Specifies the network device mapping to create destination interf

aces
with the exact same IP addresses as source interfaces. When not specified, the default matching rule is to use the network devices with identical names as those of the source network devices. This option is a sub-option for "-take_over_ips."

EXAMPLE #8

To create a VDM migration plan with storage pool mapping, IP-takeover, network device mapping, and file systems excluded, type:

```
$ nas_migrate -plan -create planEx1 -source vdmEx1 -destination -pool dstpool3
-interconnect winter_spring -storage_pools srcpool1:dstpool1,srcpool2:dstpool2
-take_over_ips -network_devices cge1:cge1,cge2:cge20
```

Output omitted for brevity.

EXAMPLE #6 and EXAMPLE #7 provide descriptions of storage pool and netork device mapping.

EXAMPLE #9

To modify a VDM migration plan, type:

```
$ nas_migrate -plan -modify plan001 -name plan001_New -filesystems -id
srcFs=100:dstFs=100,srcFs=300:dstPool=3 -interfaces name=eth10:dstDevice=cge10
```

Output omitted for brevity.

EXAMPLE #10

To delete a VDM migration plan, type:

```
$ nas_migrate -plan -delete plan001
```

Output omitted for brevity.

EXAMPLE #11

To create a VDM level migration, type:

```
$ nas_migrate -create vdmMigEx1 -vdm -plan planEx1
```

Output omitted for brevity.

EXAMPLE #12

To create a file system level migration, type:

```
$ nas_migrate -create fsMigEx1 -fs -source fs3 -destination -pool dstpool3
-interconnect summer_spring

Info 26843676673: In Progress: Operation is still running. Check task id 63654 on
   the Background Tasks screen for results.
Validate migration name <fsmigEx1> ... succeeded
Query migration plan ... succeeded
Validate migration ... succeeded
Create migration session ... succeeded
Create FS [<fs name>] ... succeeded
Create interfaces ... succeeded
Create FS replication [<fs name>] ... succeeded
Initial Copy FS [<fs name>] ... succeeded.
```

```
Create destination file systems ...
Create destination file systems: <#created>/<#total>(updated per 2 minutes)
Create destination file systems... succeeded
Create checkpoints ...
Create checkpoints: <#created>/<#total>(updated per 2 minutes)
Create checkpoints ... succeeded
Create replications ...
Create replications: <#created>/<#total>(updated per 2 minutes)
Create replications ... succeeded
Update Migration State [INITIAL_COPYING] ... succeeded
Initial Copy ...
Initial Copy: Total=50000(M): Copied=10000(M): Transfer Rate=2000(KB/s)(updated
  per 10 minutes)
Initial Copy: Total=50000(M): Copied=20000(M): Transfer Rate=3000(KB/s)(updated
  per 10 minutes)
Initial Copy ... succeeded
Modify RPO of replications ... succeeded
Update migration state to [READY_TO_COMPLETE] ... succeeded

OK
```

EXAMPLE #13

To complete a migration with the background flag, type:

```
$ nas_migrate -complete fsMigEx1 -checkpoint_mismatch_ignored -background

Info 26843676432: In Progress: Operation is still running. Check task id 134227
on the Background Tasks screen for results.
```

EXAMPLE #14

To delete a migration with the background flag, type:

```
$ nas_migrate -delete fsMigEx1 -background

Info 26843676556: In Progress: Operation is still running. Check task id 142811
on the Background Tasks screen for results.
```

EXAMPLE #15

To stop a migration with the background flag, type:

```
$ nas_migrate -stop fsMigEx1 -background

Info 26843676556: In Progress: Operation is still running. Check task id 144511
on the Background Tasks screen for results.
```

EXAMPLE #16

To stop a migration, type:

```
$ nas_migrate -stop id=20002224601

Info 26843676673: In Progress: Operation is still running. Check task id 17919 on

   the Background Tasks screen for results.
Check migration state  ... succeeded
Change migration state to STOPPING  ... succeeded
Check local replication state  ... succeeded
Check remote replication state  ... succeeded
Stop replication in parallel  ...
Stop replication task state:  Total=10 Succeeded=0 Failed=0
Stop replication task state:  Total=10 Succeeded=5 Failed=0
Stop replication task state:  Total=10 Succeeded=6 Failed=0
Stop replication task state:  Total=10 Succeeded=10 Failed=0
Stop replication in parallel succeeded
```

```
Change migration state to STOPPED  ... succeeded

EXAMPLE #17
-----------

To start a migration, type:

$ nas_migrate -start id=20002224601 -background

Info 26843676673: In Progress: Operation is still running. Check task id 144527
on the Background Tasks screen for results.

------------------------------------------------------------------------

Last modified: Feb 22 2013, 4:34 pm
```

nas_mview

Performs MirrorView/Synchronous (MirrorView/S) operations on a system attached to an older version of VNX for block.

SYNOPSIS

```
nas_mview
  -info
| -init <cel_name>
| -activate
| -restore
```

DESCRIPTION
----

nas_mview retrieves MirrorView/S cabinet-level information, initializes the source and destination systems for MirrorView/S, activates a failover to a destination VNX for file, or restores the source site after a failover.

MirrorView/S is supported on a system attached to an older version of VNX for block array serving as the boot storage, not the secondary storage. nas_mview must be run from a Control Station in slot 0; it will report an error if run from a Control Station in slot 1.

nas_mview must be issued as root from the /nas/sbin directory. For the -init and -info options, log in with your administrative username and use the su root command to log in as root. For the -activate and -restore options, you must log in to the destination system using the remote administration account (for example, dradmin) and log in as root.

OPTIONS

-info
Displays disaster recovery information such as the MirrorView/S device group eligible, displays the MirrorView/S Data Mover configuration for the current system.

-init <cel_name>
Initializes the MirrorView/S relationship between the source and destination systems based on if the configuration is active/passive (unidirectional) or active/active' (bidirectional).

Note: The apostrophe in active/active' indicates that both sites have source LUNs mirrored at the other site.

The passphrase-protected relationship between the source and destination systems in the MirrorView/S configuration must be built prior to initialization using the nas_cel -create command:

. On the destination Control Station in a MirrorView/S active/passive configuration, use the -init option to specify the name of the source system.

. On the Control Station of each system in a MirrorView/S active/active. configuration, use the -init option to specify the name of the remote system. The active/active configuration is a bidirectional configuration in which a VNX for file can serve both as source and destination for another system.

-activate
Executed from the destination system using the remote administration account, initiates a failover from the source to the destination system. The activation works as follows:

. If the source is available, the -activate option swaps the primary-secondary role for all mirrors in the MirrorView/S device group and makes the destination LUNs read/write. The

standby Data Movers acquire the IP and MAC addresses, file systems, and export tables of their source counterparts.

. If the original source site is unavailable, the destination LUNs are promoted to the primary role, making them visible to the destination VNX for file. The original source LUNs cannot be converted to backup images; they stay visible to the source VNX for file, and the original destination site is activated with new source (primary) LUNs only. If the source cannot be shut down in a disaster scenario, any writes occurring after the forced activation will be lost during a restore.

-restore
Issued from the destination system using the remote administration account, restores a source system after a MirrorView/S failover, and fails back the device group to the source system.

The restore process begins by checking the state of the device group. If the device group state is Local Only (where each mirror has only the source LUN), the device group will be fully synchronized and rebuilt before the failback can occur. If the device group condition is fractured, an incremental synchronization is performed before the failback occurs. Source devices are then synchronized with the data on the original destination devices, I/O access is shut down, the original destination Data Movers are rebooted as remote standbys, and the mirrored devices are failed back. When the source side is restored, the source Data Movers and their services are restarted.

If the restore fails, the source Control Station is not reachable on the data network. To complete the restore, access the source, log in as root, and type /nasmcd/sbin/nas_mview -restore.

SEE ALSO

Using MirrorView/Synchronous with VNX for Disaster Recovery, nas_cel, and nas_checkup.

STORAGE SYSTEM OUTPUT

The number associated with the storage device reflects the attached storage system; for MirrorView/S, VNX for block displays a prefix of APM before a set of integers, for example, APM00033900124-0019.

The VNX for block supports the following system-defined AVM storage pools for MirrorView/S only: cm_r1, cm_r5_performance, cm_r5_economy, cmata_archive, cmata_r3, cm_r6, and cmata_r6.

EXAMPLE #1

To initialize a destination VNX for file in an active/passive configuration to communicate with source site source_cs, from the destination Control Station, type:

```
# /nas/sbin/nas_mview -init source_cs

Celerra with MirrorView/Synchronous Disaster Recovery

Initializing source_cs --> target_cs

Contacting source_cs for remote storage info

Local storage system: APM00053001549
Remote storage system: APM00053001552

Enter the Global CLARiiON account information
Username: emc
Password: *** Retype your response to validate
Password: ***
```

```
Discovering storage on source_cs (may take several minutes)
Setting security information for APM00053001549
Discovering storage APM00053001552 (may take several minutes)

Discovering storage (may take several minutes)

Contacting source_cs for remote storage info
Gathering server information...
Contacting source_cs for server capabilities...
Analyzing server information...

Source servers available to be configured for remote DR
------------------------------------------------------
1. server_2:source_cs
2. server_3:source_cs [ local standby ]
v. Verify standby server configuration
q. Quit initialization process
c. Continue initialization
Select a source_cs server: 1

Destination servers available to act as remote standby
------------------------------------------------------
1. server_2:target_cs [ unconfigured standby ]
2. server_3:target_cs [ unconfigured standby ]
b. Back
Select a target_cs server: 1
Source servers available to be configured for remote DR
------------------------------------------------------
1. server_2:source_cs [ remote standby is server_2:target_cs ]
2. server_3:source_cs [ local standby ]
v. Verify standby server configuration
q. Quit initialization process
c. Continue initialization
Select a source_cs server: 2
Destination servers available to act as remote standby
------------------------------------------------------
server_2:target_cs [ is remote standby for server_2:source_cs ]
2. server_3:target_cs [ unconfigured standby ]
b. Back
Select a target_cs server: 2
Source servers available to be configured for remote DR
------------------------------------------------------
1. server_2:source_cs [ remote standby is server_2:target_cs ]
2. server_3:source_cs [ remote standby is server_3:target_cs ]
v. Verify standby server configuration
q. Quit initialization process
c. Continue initialization
Select a source_cs server: c

Standby configuration validated OK

Enter user information for managing remote site source_cs
Username: dradmin
Password: ******* Retype your response to validate
Password: *******

Active/Active configuration
Initializing (source_cs-->target_cs)
Do you wish to continue? [yes or no] yes
Updating MirrorView configuration cache
Setting up server_3 on source_cs
Setting up server_2 on source_cs
Creating user account dradmin
Setting acl for server_3 on target_cs
Setting acl for server_2 on target_cs
Updating the Celerra domain information
Creating device group mviewgroup on source_cs
done

EXAMPLE #2
```

```
----------

To get information about a source MirrorView configuration (for
example, on new_york configured as active/passive), type:

# /nas/sbin/nas_mview -info

***** Device Group Configuration *****

name = mviewgroup
description =
uid = 50:6:1:60:B0:60:26:BC:0:0:0:0:0:0:0:0
state = Consistent
role = Primary
condition = Active
recovery policy = Automatic
number of mirrors = 16
mode = SYNC
owner = 0
mirrored disks =
root_disk,root_ldisk,d5,d8,d10,d11,d24,d25,d26,d27,d29,d30,d31,d32,d33,d39,
local clarid = APM00053001552
remote clarid = APM00053001549
mirror direction = local -> remote

***** Servers configured with RDFstandby *****

id = 1
name = server_2
acl = 1000, owner=nasadmin, ID=201
type = nas
slot = 2
member_of =
standby = server_3, policy=auto
RDFstandby= slot=2
status :
defined = enabled
actual = online, active

id = 2
name = server_3
acl = 1000, owner=nasadmin, ID=201
type = standby
slot = 3
member_of =
standbyfor= server_2
RDFstandby= slot=3
status :
defined = enabled
actual = online, ready

***** Servers configured as standby *****

No servers configured as standby

Where:

Value                           Definition
-----                           ----------

Device group configuration:

name                            Name of the consistency (device) group.
description                     Brief description of device group.
uid                             UID assigned, based on the system.
state                           State of the device group (for example,
                                Consistent, Synchronized, Out-of-Sync,
                                Synchronizing, Scrambled, Empty, Incomplete,
                                or Local Only).
role                            Whether the current system is the Primary
                                (source) or Secondary (destination) for this
```

	group.
condition	Whether the group is functioning (Active), Inactive, Admin Fractured (suspended), Waiting on Sync, System Fractured (which indicates link down), or Unknown.
recovery policy	Type of recovery policy (Automatic is the default and recommended value for group during storage system configuration; if Manual is set, you must use -resume after a link down failure).
number of mirrors	Number of mirrors in group.
mode	MirrorView mode (always SYNC in this release).
owner	ACL ID assigned (0 indicates no control). nas_acl provides information.
mirrored disks	Comma-separated list of disks that are mirrored.
local clarid	APM number of local VNX for block storage array.
remote clarid	APM number of remote VNX for block storage array.
mirror direction	On primary system, local to remote (on primary system); on destination system, local from remote.

Servers configured with RDFstandby/ Servers configured as standby:

id	Server ID
name	Server name
acl	ACL value and owner
type	Server type (for example, nas or standby)
slot	Slot number for this Data Mover
member_of	If applicable, shows membership information.
standby	If this Data Mover is configured with local standbys, the server that is the local standby and any policy information.
RDFstandby	If this Data Mover is configured with a remote RDF standby, the slot number of the destination Data Mover that serves as the RDF standby.
standbyfor	If this Data Mover is also configured as a local standby, the server numbers for which it is a local standby.
status	Indicates whether the Data Mover is defined and online/ready.

EXAMPLE #3

To activate a failover, log in to destination Control Station using dradmin account, su to root, and type:

```
# /nas/sbin/nas_mview -activate

Sync with CLARiiON backend ...... done
Validating mirror group configuration ...... done
Is source site source_cs ready for complete shut down (power OFF)? [yes or no]
yes
Contacting source site source_cs, please wait... done
Shutting down remote site source_cs .....................................
done
Sync with CLARiiON backend ...... done
STARTING an MV 'FAILOVER' operation.
Device group: mviewgroup ............ done
The MV 'FAILOVER' operation SUCCEEDED.
Failing over Devices ... done
Adding NBS access for server_2 ........ done
Adding NBS access for server_3 ........ done
Activating the target environment ... done
```

```
server_2 : going offline
rdf : going active
replace in progress ...done
failover activity complete

server_3 : going offline
rdf : going active
replace in progress ...done
failover activity complete
commit in progress (not interruptible)...done
commit in progress (not interruptible)...done
commit in progress (not interruptible)...done
commit in progress (not interruptible)...done

done
```

EXAMPLE #4

To restore, log in to the destination Control Station using dradmin account, as root user, and type:

```
# /nas/sbin/nas_mview -restore

Sync with CLARiiON backend ..... done
Validating mirror group configuration ..... done
Contacting source site source_cs, please wait... done
Running restore requires shutting down source site source_cs.
Do you wish to continue? [yes or no] yes
Shutting down remote site source_cs ..... done
Is source site source_cs ready for storage restoration ? [yes or no] yes
Sync with CLARiiON backend ..... done
STARTING an MV 'RESUME' operation.
Device group: mviewgroup ........... done
The MV 'RESUME' operation SUCCEEDED.
Percent synchronized: 100
Updating device group ... done
Is source site ready for network restoration ? [yes or no] yes
Restoring servers ..... done
Waiting for servers to reboot ..... done
Removing NBS access for server_2 .. done
Removing NBS access for server_3 .. done
Waiting for device group ready to failback .... done
Sync with CLARiiON backend ..... done
STARTING an MV 'FAILBACK' operation.
Device group: mviewgroup ........... done
The MV 'FAILBACK' operation SUCCEEDED.
Restoring remote site source_cs ..... failed
Error 5008: -1:Cannot restore source_cs. Please run restore on site source_cs.
```

Then on the Source Control Station, as the root user, type:

```
# /nasmcd/sbin/nas_mview -restore

Stopping NAS services. Please wait...

Powering on servers ( please wait ) ..... done
Sync with CLARiiON backend ..... done
STARTING an MV 'SUSPEND' operation.
Device group: mviewgroup ........... done
The MV 'SUSPEND' operation SUCCEEDED.
server_2 : going standby
rdf : going active
replace in progress ...done
failover activity complete
server_3 : going standby
rdf : going active
replace in progress ...done
failover activity complete
commit in progress (not interruptible)...done
```

```
commit in progress (not interruptible)...done
Sync with CLARiiON backend ..... done
STARTING an MV 'RESUME' operation.
Device group: mviewgroup ............ done
The MV 'RESUME' operation SUCCEEDED.
Restarting NAS services ..... done
commit in progress (not interruptible)...done
commit in progress (not interruptible)...done
done
```

Last modified: May 11, 2011 11:25 am.

nas_pool

Manages the user-defined and system-defined storage pools for the system.

SYNOPSIS

```
nas_pool
          -list
        | -info {<name>|id=<id>|-all} [-Ads] [-storage <system_name>]
        | -size {<name>|id=<id>|-all} [-mover <mover>][-slice {y|n}]
               [-storage <system_name>]
        | -create [-name <name>][-acl <acl>][-description <desc>]
               [-volumes <volume_name>[,<volume_name>,...]]
               [-default_slice_flag {y|n}] [-is_greedy {y|n}]
        | -create [-name <name>][-acl <acl>][-description <desc>]
               [-default_slice_flag {y|n}] [-is_greedy {y|n}]
                -size <integer>[M|G|T][-storage <system_name>]
                -template <system_pool_name> [-num_stripe_members <num>]
                       [-stripe_size <num>]
        | -modify {<name>|id=<id>} [-storage <system_name>] [-name <name>]
               [-acl <acl>] [-description <desc>][-default_slice_flag {y|n}]
               [-is_dynamic {y|n}][-is_greedy {y|n}]
        | -delete {<name>|id=<id>} [-deep] [-storage <system_name>]
        | -xtend {<name>|id=<id>} [-storage <system_name>]
              -volumes <volume_name>[,<volume_name>,...]
        | -xtend {<name>|id=<id>} -size <integer> [M|G|T][-storage <system_name>]
        | -shrink {<name>|id=<id>} [-storage <system_name>] -volumes
               <volume_name>[,<volume_name>,...][-deep]
```

DESCRIPTION
----

nas_pool creates, deletes, extends, shrinks, lists, displays, manages the access control level, and modifies a user-defined storage pool.

nas_pool extends, shrinks, lists, displays, and modifies system-defined storage pools.

OPTIONS

-list
Lists all storage pools on the system.

-info {<name>|id=<id>|-all} [-Ads] [-storage <system_name>]
Displays detailed information for the specified storage pool, or all storage pools. The -storage option can be used to differentiate pools when the same pool name is used in multiple storage systems.

The -Ads option displays the advanced data service properties of the file system.

-size {<name>|id=<id>|-all}
Displays the size for the specified storage pool, or all storage pools.

[-mover <mover>]
Displays size information that is visible to the physical Data Mover or the virtual Data Mover (VDM).

[-slice {y|n}]
If y is typed, displays size information when volumes in the storage pool are sliced. If n is typed, displays size information when volumes in the storage pool are not sliced. The -slice option defaults to the value of default_slice_flag for the storage pool.

[-storage <system_name>]
Displays size information for members that reside on a specified

storage system.

-create
Creates a user-defined storage pool.

[-name <name>]
Assigns a name to the new storage pool. If no name is specified, assigns one by default.

[-acl <acl>]
Sets an access control level value that defines the owner of the storage pool, and the level of access allowed for users and groups defined in the access control level table. The nas_acl command provides more information.

[-description <desc>]
Assigns a comment to the storage pool.

[-volumes <volume_name>[,<volume_name>,...]
Designates the members to be added to the storage pool. The members can be any meta, slice, stripe, or disk volumes.

[-default_slice_flag {y|n}]
If set to y (default), then members of the storage pool might be sliced when space is allocated from the storage pool. If set to n, members of the storage pool will not be sliced when space is dispensed from the storage pool and the volumes specified cannot be built on a slice.

[-is_greedy {y|n}]
If set to n (default), the system uses space from the user-defined storag
e
pool's existing member volumes in the order that the volumes were added t
o the
pool to create a new file system or extend an existing file system.

If set to y, the user-defined storage pool uses space from the least-used member volume to create a new file system. When there is more than one least-used member volume available, AVM selects the member volume that contains the most disk volumes. For example, if one member volume contain
s
four disk volumes and another member volume contains eight disk volumes,
AVM
selects the one with eight disk volumes. If there are two or more member volumes that have the same number of disk volumes, AVM selects the one wi
th
the lowest ID.

[-size <integer> {M|G|T}]
Creates a storage pool with the size specified. M specifies megabytes, G specifies gigabytes (default), and T specifies terabytes. The maximum siz
e
that you can specify for a storage pool is the maximum supported storage capacity for the system.

[-storage <system_name>]
Specifies the storage system on which one or more volumes will be created, to be added to the storage pool.

[-template <system_pool_name>]
Specifies a system pool name, required when the -size option is specified. The user pool will be created using the profile attributes of the specified system pool template.

[-num_stripe_members <num>]
Specifies the number of stripe members for user pool creation by size. The -num_stripe_members option works only when both -size and -template options are specified. It overrides the number of stripe members attribute of the specified system pool template.

[-stripe_size <num>]

Specifies the stripe size for user pool creation by size. The -stripe_size option works only when both -size and -template options are specified. It overrides the stripe size attribute of the specified system pool template.

-modify {<name>|id=<id>} [-storage <system_name>]
Modifies the attributes of the specified user-defined or system-defined storage pool. The -storage option can be used to differentiate pools when the same pool name is used in multiple storage systems.

Managing Volumes and File Systems with VNX Automatic Volume Management lists the available system-defined storage pools.

[-name <name>]
Changes the name of the storage pool to the new name.

[-acl <acl>]
Sets an access control level value that defines the owner of the storage pool, and the level of access allowed for users and groups defined in the access control level table. The nas_acl command provides more information.

[-description <desc>]
Changes the comment for the storage pool.

[-default_slice_flag {y|n}]
If set to y (default), then members of the storage pool might be sliced when space is dispensed from the storage pool. If set to n, members of the storage pool will not be sliced when space is dispensed from the storage pool and the volumes specified cannot be built on a slice.

[-is_dynamic {y|n}]
Allows a system-defined storage pool to automatically extend or shrink member volumes.

Note: The -is_dynamic option is for system-defined storage pools only.

[-is_greedy {y|n}]
For system-defined storage pools, if set to y, then the storage pool attempts to create new member volumes before using space from existing member volumes.

A system-defined storage pool that is not greedy (set to n), consumes all the space existing in the storage pool before trying to add additional member volumes. A y or n value must be specified when modifying a system-defined storage pool.

For user-defined storage pools, if set to n (default), the system uses space from the user-defined storage pool's existing member volumes in the order that the volumes were added to the pool to create a new file system.

For user-defined storage pools, if set to y, the system uses space from the least-used member volume in the user-defined storage pool to create a new file system. When there is more than one least-used member volume available, AVM selects the member volume that contains the most disk volumes. For example, if one member volume contains four disk volumes and another member volume contains eight disk volumes, AVM selects the one with eight disk volumes. If there are two or more member volumes that have the same number of disk volumes, AVM selects the one with the lowest ID.

For both system-defined and user-defined pools when extending a file syst

em,
the is_greedy attribute is ignored unless there is not enough free space
on
the existing volumes that the file system is using to meet the requested extension size.

-delete {<name>|id=<id>} [-storage <system_name>]
Deletes a storage pool. Storage pools cannot be deleted if any members are in use. After deletion, the storage pool no longer exists on the system, however, members of the storage pool are not deleted. The -storage option can be used to differentiate pools when the same pool name is used in multiple storage systems.

[-deep]
Deletes the storage pool and also recursively deletes each member of the storage pool. Each storage pool member is deleted unless it is in use or is a disk volume.

-xtend {<name>|id=<id>} [-storage <system_name>]
-volumes <volume_name>[, <volume_name>,...]
Adds one or more unused volumes to a storage pool. The -storage option can be used to differentiate pools when the same pool name is used in multiple storage systems. If the default_slice_value is set to n, member volumes cannot contain slice volumes (for compatibility with TimeFinder/FS).

Note: Extending a storage pool by volume is for user-defined storage pools only.

-xtend {<name>|id=<id>} -size <integer> [M|G|T]
Extends the specified storage pool with one or more volumes of the size equal to or greater than the size specified. When specifying the volume by size, type an integer between 1 and 1024, then specify T for terabytes, G for gigabytes (default), or M for megabytes.

[-storage <system_name>]
Specifies the storage system on which one or more volumes will be created, to be added to the storage pool.

Note: To successfully extend a system-defined storage pool by size, the is_dynamic attribute must be set to n, and there must be enough available disk volumes to satisfy the request.

-shrink {<name>|id=<id>} [-storage <system_name>]
-volumes <volume_name>[,<volume_name>,...][-deep]
Shrinks the storage pool by the specified unused volumes. The -storage option can be used to differentiate pools when the same pool name is used in multiple storage systems. When the -deep option is used to shrink a user-defined storage pool, it removes the specified member volumes from the pool, and recursively deletes any unused volumes unless it is a disk volume. If the -deep option is not used to shrink a user-defined storage pool, the member volumes are left intact so that they can be reused. The is_dynamic option must be set to n before shrinking system-defined storage pools.

Note: Shrinking of a system-defined storage pool by default deletes member volumes automatically. Specifying the -deep option on the system-defined storage pool shrink does not make any difference.

SEE ALSO

Managing Volumes and File Systems with VNX Automatic Volume Management, Managing Volumes and File Systems for VNX Manually, Controlling Access to System Objects on VNX, Using TimeFinder/FS, NearCopy, and FarCopy on VNX for File, fs_timefinder, nas_fs, nas_volume, and nas_slice.

STORAGE SYSTEM OUTPUT

VNX for block supports the following traditional system-defined storage pools: clar_r1, clar_r5_performance, clar_r5_economy, clar_r6, clarata_r3, clarata_r6, clarata_r10, clarata_archive, cm_r1, cm_r5_performance, cm_r5_economy, cm_r6, cmata_r3, cmata_archive, cmata_r6, cmata_r10, clarsas_archive, clarsas_r6, clarsas_r10, clarefd_r5, clarefd_r10, cmsas_archive, cmsas_r6, cmsas_r10, and cmefd_r5.

A mapped pool was formerly called a thin or virtual pool. Disk types when using VNX for block are CLSTD, CLEFD, CLATA, MIXED (indicates that tiers used in the pool contain multiple disk types), Performance, Capacity, and Extreme_performance and for VNX for block involving mirrored disks are: CMEFD, CMSTD, CMATA, Mirrored_mixed, Mirrored_performance, Mirrored_capacity, and Mirrored_extreme_performance. Disk types when using VNX for block are CLSTD, CLEFD, and CLATA, and for VNX for block involving mirrored disks are: CMEFD, CMSTD, and CMATA.

VNX with a Symmetrix storage system support the following system-defined storage pools: symm_std, symm_std_rdf_src, symm_ata, symm_ata_rdf_src, symm_ata_rdf_tgt, symm_std_rdf_tgt, symm_efd, symm_fts, symm_fts_rdf_tgt, and symm_fts_rdf_src.

For user-defined storage pools, the difference in output is in the disk type. Disk types when using a Symmetrix are STD, R1STD, R2STD, BCV, R1BCV, R2BCV, ATA, R1ATA, R2ATA, BCVA, R1BCA, R2BCA, EFD, FTS, R1FTS, R2FTS, R1BCF, R2BCF, BCVF, BCVMIXED, R1MIXED, R2MIXED, R1BCVMIXED, and R2BCVMIXED.

EXAMPLE #1

To create a storage pool with the name, marketing, with a description, with the following disk members, d12, d13, and with the default slice flag set to y, type:

```
$ nas_pool -create -name marketing -description 'Storage
Pool' -volumes d12,d13 -default_slice_flag y

id                  = 20
name                = marketing
description         = Storage Pool
acl                 = 0
in_use              = False
clients             =
members             = d12,d13
storage_system(s)   = FNM00105000212
default_slice_flag  = True
is_user_defined     = True
thin                = False
disk_type           = CLSTD
server_visibility   = server_2,server_3,server_4,server_5
is_greedy           = False
template_pool       = N/A
num_stripe_members  = N/A
stripe_size         = N/A
```

Where:

Value	Definition
id	ID of the storage pool.
name	Name of the storage pool.
description	Comment assigned to the storage pool.
acl	Access control level value assigned to the storage pool.
in_use	Whether the storage pool is being used by a file system.
clients	File systems using the storage pool.
members	Volumes used by the storage pool.

```
        storage_systems(s) Storage systems used by the storage pool.
        default_slice_flag Allows slices from the storage pool.
        is_user_defined    User-defined as opposed to system-defined.
        thin               Indicates whether thin provisioning is enabled or disabled.
        disk_type          Type of disk contingent on the storage system attached. CLSTD,
                           CLATA, CMSTD, CLEFD, CMEFD, CMATA, MIXED (indicates tiers used

                           in the pool contain multiple disk
                           types), Performance, Capacity, Extreme_performance, Mirrored_m
ixed,
                           Mirrored_performance, Mirrored_capacity, and Mirrored_extreme_
performance
                           are for VNX for block, and STD, BCV, R1BCV, R2BCV, R1STD, R2ST
D,
                           ATA, R1ATA, R2ATA, BCVA, R1BCA, R2BCA, EFD, BCVMIXED, R1MIXED,
 R2MIXED,
                           R1BCVMIXED, and R2BCVMIXED are for Symmetrix.
        server_visibility  Storage pool is visible to the physical Data Movers
                           specified.
        is_greedy          Indicates whether the system-defined storage pool will use new
                           member volumes as needed.
        template_pool      System pool template used to create the user pool. Only
                           applicable to user pools created by size or if the last member
 volume
                           is a stripe or both.
        num_stripe_members Number of stripe members used to create the user pool. Applica
ble to
                           system pools and user pools created by size or if the last mem
ber volume
                           is a stripe or both.
        stripe_size        Stripe size used to create the user pool. Applicable to system
                           pools and user pools created by size or if the last member vol
ume is a
                           stripe or both.

        EXAMPLE #2
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        To change the description for the marketing storage pool to include a descriptive
 comment,
        type:

        $ nas_pool -modify marketing -description 'Marketing Storage Pool'

        id                  = 20
        name                = marketing
        description         = Marketing Storage Pool
        acl                 = 0
        in_use              = False
        clients             =
        members             = d12,d13
        storage_system(s)   = FNM00105000212
        default_slice_flag  = True
        is_user_defined     = True
        thin                = False
        disk_type           = CLSTD
        server_visibility   = server_2,server_3,server_4,server_5
        is_greedy           = False
        template_pool       = N/A
        num_stripe_members  = N/A
        stripe_size         = N/A

                        EXAMPLE #1 provides a description of command output.

        EXAMPLE #3
        ----------

        To view the size information for the FP1 mapped pool, type:

        $ nas_pool -size FP1
```

```
id                  = 40
name                = FP1
used_mb             = 0
avail_mb            = 0
total_mb            = 0
potential_mb        = 2047
```

Where:

Value	Definition
used_mb	Space in use by the storage pool specified.
avail_mb	Unused space still available in the storage pool.
total_mb	Total space in the storage pool (total of used and unused).
potential_mb	Available space that can be added to the storage pool.

Note: Each of the options used with the command nas_pool - size is filters for the output of the command. For example, if you specify a Data Mover, the output will reflect only the space to which the specified Data Mover has visibility. Physical used_mb, Physical avail_mb, and Physical total_mb are applicable for system-defined virtual AVM pools only.

EXAMPLE #4

To view the size information for the TP1 mapped pool which contains only virtual LUNs, type:

```
$ nas_pool -size TP1

id                  = 40
name                = TP1
used_mb             = 0
avail_mb            = 0
total_mb            = 0
potential_mb        = 2047
Physical storage usage in tp1 on FCNTR074200038:
used_mb             = 0
avail_mb            = 20470
```

Where:

Value	Definition
Physical used_mb	Used physical size of a storage system mapped pool in MB (some may be used by non-VNX hosts).
Physical avail_mb	Available physical size of a storage system mapped pool in MB.

Note: Physical used_mb and Physical avail_mb are applicable for system-defined AVM pools that contain virtual LUNs only.

EXAMPLE #5

For VNX system, to change the -is_greedy and -is_dynamic options for the system defined, clar_r5_performance storage pool, type:

```
$ nas_pool -modify clar_r5_performance -is_dynamic n -is_greedy y

id                  = 3
name                = clar_r5_performance
description         = CLARiiON RAID5 4plus1
acl                 = 421
in_use              = False
clients             =
members             = v120
storage_system(s)   =
```

```
        default_slice_flag   = True
        is_user_defined      = False
        thin                 = False
        disk_type            = CLSTD
        server_visibility    = server_2,server_3,server_4,server_5
        volume_profile       = clar_r5_performance_vp
        is_dynamic           = False
        is_greedy            = True
        num_stripe_members   = 4
        stripe_size          = 32768
```

EXAMPLE #1 provides a description of command output.

EXAMPLE #6

For VNX for file with a Symmetrix system, to change the -is_greedy and -is_dynamic options for the system-defined, symm_std storage pool, type:

```
$ nas_pool -modify symm_std -is_dynamic y -is_greedy y

id = 1
name = symm_std
description = Symmetrix STD
acl = 1421, owner=nasadmin, ID=201
in_use = True
clients = ufs3
members = v169,v171
default_slice_flag = False
is_user_defined = False
thin = False
disk_type = STD
compressed = True
server_visibility = server_2,server_3,server_4,server_5
volume_profile = symm_std_vp
is_dynamic = True
is_greedy = True
num_stripe_members = 8
stripe_size = 32768
```

Where:

Value	Definition
id	ID of the storage pool.
name	Name of the storage pool.
description	Comment assigned to the storage pool.
acl	Access control level value assigned to the storage pool.
in_use	Whether the storage pool is being used by a filesystem.
clients	File systems using the storage pool.
members	Disks used by the storage pool.
default_slice_flag	Allows slices from the storage pool.
is_user_defined	User-defined as opposed to system-defined.
thin	Indicates whether thin provisioning is enabled or disabled.
disk_type	Contingent on the storage system attached.
compressed	For VNX with Symmetrix backend, indicates whether data is compressed. Values are: True, False, Mixed (indicates some of the LUNs, but not all, are compressed).
server_visibility	Storage pool is visible to the physical Data Movers specified.
volume_profile	Volume profile used.
is_dynamic	Whether the system-defined storage pool can add or remove volumes.
is_greedy	Indicates whether the system-defined storage pool will use new member volumes as needed.

```
        template_pool           System pool template used to create the user pool. Only
                                applicable to user pools created
                                by size or if the last member volume is a stripe or both
.
        num_stripe_members      Number of stripe members used to create the user pool.
                                Applicable to system pools and user pools created by siz
e
                                or if the last member volume is a stripe or both.
        stripe_size             Stripe size used to create the user pool. Applicable to
system
                                pools and user pools created by size or if the last memb
er
                                volume is a stripe or both.
```

EXAMPLE #7

To change the -is_greedy option for the user-defined, user_pool storage pool, type:

```
$ nas_pool -modify user_pool -is_greedy y

id                  = 58
name                = user_pool
description         =
acl                 = 0
in_use              = False
clients             =
members             = d21,d22,d23,d24
storage_system(s)   = FNM00105000212
default_slice_flag  = True
is_user_defined     = True
thin                = False
disk_type           = CLSTD
server_visibility   = server_2
is_greedy           = True
template_pool       = N/A
num_stripe_members  = N/A
stripe_size         = N/A
```

EXAMPLE #1 provides a description of command output.

EXAMPLE #8

To add the volumes, d7 and d8, to the marketing storage pool, type:

```
$ nas_pool -xtend marketing -volumes d7,d8

id                = 20
name              = marketing
description       = Marketing Storage Pool
acl               = 0
in_use            = False
clients           =
members           = d12,d13,d7,d8
default_slice_flag = True
is_user_defined   = True
thin              = True
disk_type         = CLSTD
server_visibility = server_2,server_3,server_4,server_5
template_pool     = N/A
num_stripe_members = N/A
stripe_size       = N/A
```

EXAMPLE #1 provides a description of command output.

EXAMPLE #9

For a VNX system, to extend the system-defined storage pool

by a specified size with a specified system, type:

```
$ nas_pool -xtend clar_r5_performance -size 128M -storage APM00042000818

id                = 3
name              = clar_r5_performance
description       = CLARiiON RAID5 4plus1
acl               = 1421, owner=nasadmin, ID=201
in_use            = False
clients           =
members           = v120
default_slice_flag = True
is_user_defined   = False
thin              = False
disk_type         = CLSTD
server_visibility  = server_2,server_3,server_4,server_5
volume_profile    = clar_r5_performance_vp
is_dynamic        = False
is_greedy         = True
num_stripe_members = 4
stripe_size       = 32768
```

EXAMPLE #1 provides a description of command output.

EXAMPLE #10

For a VNX system, to remove d7 and d8 from the marketing storage pool, type:

```
$ nas_pool -shrink marketing -volumes d7,d8

id                = 20
name              = marketing
description       = Marketing Storage Pool
acl               = 0
in_use            = False
clients           =
members           = d12,d13
default_slice_flag = True
is_user_defined   = True
thin              = True
disk_type         = CLSTD
server_visibility  = server_2,server_3,server_4,server_5
template_pool     = N/A
num_stripe_members = N/A
stripe_size       = N/A
```

EXAMPLE #1 provides a description of command output.

EXAMPLE #11

To list the storage pools, type:

```
$ nas_pool -list

id   inuse   acl    name               storage_system
2      n     421   clar_r1               N/A
3      n     421   clar_r5_performance  FCNTR074200038
4      n     421   clar_r5_economy         N/A
10     n     421   clarata_archive      FCNTR074200038
11     n     421   clarata_r3              N/A
20     n      0    marketing            FCNTR074200038
40     y      0    TP1                  FCNTR074200038
41     y      0    FP1                  FCNTR074200038
```

Where:

Value	Definition

```
id                      ID of the storage pool.
inuse                   Whether the storage pool is being used by a filesystem.
acl                     Access control level value assigned to the storage pool.
name                    Name of the storage pool.
storage_system          Name of the storage system where the storage pool resides
.

EXAMPLE #12
-----------

To display information about the user-defined storage pool called
marketing, type:

$ nas_pool -info marketing

id                  = 20
name                = marketing
description         = Marketing Storage Pool
acl                 = 0
in_use              = False
clients             =
members             = d12,d13
storage_system(s)   =
default_slice_flag  = True
is_user_defined     = True
thin                = True
disk_type           = CLSTD
server_visibility   = server_2,server_3,server_4,server_5
is_greedy           = False
template_pool       = N/A
num_stripe_members  = N/A
stripe_size         = N/A

                EXAMPLE #1 provides a description of command output.

EXAMPLE #13
-----------

To display information about the system-defined clar_r5_performance storage pool,
type:

$ nas_pool -info clar_r5_performance

id                  = 3
name                = clar_r5_performance
description         = CLARiiON RAID5 4plus1
acl                 = 1421, owner=nasadmin, ID=201
in_use              = False
clients             =
members             = v120
default_slice_flag  = True
is_user_defined     = False
thin                = False
disk_type          = CLSTD
server_visibility  = server_2,server_3,server_4,server_5
volume_profile     = clar_r5_performance_vp
is_dynamic         = False
is_greedy          = True
num_stripe_members = 4
stripe_size        = 32768

                EXAMPLE #1 provides a description of command output.

EXAMPLE #14
-----------

To display information about the system-defined engineer virtual
pool, type:

$ nas_pool -info engineer
```

```
id                  = 40
name                = engineer
description         = Mapped Pool engineer on APM00084401666
acl                 = 0
in_use              = True
clients             = DA_BE_VIRT_FS,vp_test,vp_test1,vp_test12,cvpfs1,cvpfs3
members             = v363
default_slice_flag = True
is_user_defined     = False
thin                = True
disk_type           = CLSTD
server_visibility   = server_2,server_3
volume_profile      = engineer_APM00084401666_vp
is_dynamic          = True
is_greedy           = True
num_stripe_members = N/A
stripe_size         = N/A
```

EXAMPLE #1 provides a description of command output.

EXAMPLE #15

To display information about the mapped storage pool called FP1 from a VNX for block, type:

```
$ nas_pool -info FP1

id                    = 40
name                  = FP1
description           = Mapped Pool on FCNTR074200038
acl                   = 0
in_use                = False
clients               =
members               =
default_slice_flag    = True
is_user_defined       = False
thin                  = True
tiering_policy        = Auto-tier
compressed            = False
mirrored              = False
disk_type             = Mixed
volume_profile        = FP1
is_dynamic            = True
is_greedy             = True
```

Where:

Value	Definition
tiering_policy	Indicates the tiering policy in effect. If the initial tier and the tiering policy are the same, the values are: Auto-Tier, Highest Available Tier, Lowest Available Tier. If the initial tier and the tiering policy are not the same, the values are: Auto-Tier/No Data Movement, Highest Available Tier/No Data Movement, Lowest Available Tier/No Data Movement.
compressed	For VNX for block, indicates whether data is compressed. Values are: True, False, Mixed (indicates some of the LUNs, but not all, are compressed).
mirrored	Indicates whether the disk is mirrored.

EXAMPLE #16

To display information about the mapped storage pool called SG0 from a Symmetrix storage system, type:

```
$ nas_pool -info SG0

id                   = 40
name                 = SG0
description          = Symmetrix Mapped Pool on 000192601245
acl                  = 0
in_use               = False
clients              =
members              =
default_slice_flag   = True
is_user_defined      = False
thin                 = True
tiering_policy       = symm_policy_1
compressed           = True
frontend_io_quota    = maxiopersec 500,maxmbpersec 500
disk_type            = Mixed
volume_profile       = True
is_dynamic           = True
is_greedy            = N/A
```

Where:

Value	Definition
id	ID of the storage pool.
name	Name of the storage pool.
description	Comment assigned to the storage pool.
acl	Access control level value assigned to the storage pool.
in_use	Whether the storage pool is being used by a file system.
clients	File systems using the storage pool.
members	Volumes used by the storage pool.
default_slice_flag	Allows slices from the storage pool.
is_user_defined	User-defined as opposed to system-defined.
thin	Indicates whether thin provisioning is enabled or disabled.
tiering_policy	Indicates the tiering policy in effect. If the initial tier and the tiering policy are the same, the values are: Auto-Tier, Highest Available Tier, Lowest Available Tier. If the initial tier and the tiering policy are not the same, the values are: Auto-Tier/No Data Movement,Highest Available Tier/No Data Movement, Lowest Available Tier/No Data Movement.
compressed	For VNX with Symmetrix backend, indicates whether data is compressed. Values are: True, False, Mixed (indicates some of the LUNs, but not all, are compressed).
frontend_io_quota	For VNX with Symmetrix backend, indicates if Frotend IO Quota is configured on this mapped pool, could also have value as False (indicates Frontend IO Quota is not configured on mapped SG in Symmetrix backend).
disk_type	Type of disk contingent on the system attached. CLSTD, CLATA, CMSTD, CLEFD, CMEFD, CMATA, MIXED (indicates tiers used in the pool contain multiple disk types), Performance, Capacity, Extreme_performance, Mirrored_mixed, Mirrored_performance, Mirrored_capacity, and Mirrored_extreme_performance are for VNX for block, and STD, BCV, R1BCV, R2BCV, R1STD, R2STD, ATA, R1ATA, R2ATA, BCVA, R1BCA, R2BCA, EFD, BCVMIXED, R1MIXED, R2MIXED, R1BCVMIXED, and R2BCVMIXED are for Symmetrix.
volume_profile	Volume profile used.

is_dynamic Whether the system-defined storage pool can add or remove volumes.

is_greedy Indicates whether the system-defined storage pool will use new member volumes as needed.

EXAMPLE #17

To delete the storage pool, marketing, and each of the storage pool member volumes recursively, type:

```
$ nas_pool -delete marketing -deep

id                  = 20
name                = marketing
description         = Marketing Storage Pool
acl                 = 0
in_use              = False
clients             =
members             =
storage_system(s)   =
default_slice_flag  = True
is_user_defined     = True
is_greedy           = True
thin                = True
template_pool       = N/A
num_stripe_members  = N/A
stripe_size         = N/A
```

EXAMPLE #1 provides a description of command output.

Last modified: January 11 2013, 4:31 pm.

nas_quotas

Manages quotas for mounted file systems.

SYNOPSIS

```
nas_quotas
  -edit [-user|-group] {-mover <movername>|-fs <fs_name>} [-path
    <pathname>]}[[-proto <proto_id>]|[-block <hard_limit>[:<soft_limit>]]
    [-inode <hard_limit>[:<soft_limit>]]] <id> [<id>...]
| -edit -config {-mover <movername>|-fs <fs_name>}[-path <pathname>]}
    [-option <options>]
| -edit -tree -fs <fs_name>
    [[-proto <proto_id>]|[-block <hard_limit>[:<soft_limit>]]
    [-inode <hard_limit>[:<soft_limit>]]]
    [-comment <comment>] <id> [<id>...]

| -report [-user|-group]{-mover <movername>|-fs <fs_name>}
    [-path <pathname>]}[<id> <id>...]
| -report -config {-mover <movername>|-fs <fs_name>} [-path <pathname>]}
| -report -tree -fs <fs_name> [<id> <id>...]

| {-on|-off|-clear} [-user|-group|-both]
   {-mover <movername>|-fs <fs_name>|[-path <pathname>] -all}
| -on -tree -fs <fs_name> -path <pathname> [-comment <comment>]
| -off -tree -fs <fs_name> -path <pathname>
| {-list|-clear} -tree -fs <fs_name>
| -check -start [-mode online|offline] [-tree] -fs <fs_name> [-path
<pathname>]
| -check {-stop|-status} -fs <fs_name> [-path <pathname>]
| -quotadb {-info|-upgrade [-Force]} {-mover <movername>|-fs <fs_name>}
```

DESCRIPTION
----

nas_quotas edits quotas for mounted file systems, and displays a listing of quotas and disk usage at the file system level (by the user, group, or tree), or at the quota-tree level (by the user or group).

nas_quotas also turns quotas on and off, and clears quotas records for a file system, quota tree, or a Data Mover. When a Data Mover is specified, the action applies to all mounted file systems on the Data Mover.

nas_quotas also starts and stops quota database checks either online or offline for quota trees and file systems, and allows you to upgrade the quota database limits to the maximum limit value for a file system. When a Data Mover is specified, the action applies to all mounted file systems on the Data Mover.

Caution: Quotas should be turned on (enabled) before file systems go into a production environment. Enabling (or disabling, or clearing) quotas in a production environment is time consuming and the process may disrupt file system operation. CIFS clients are disconnected during these events and NFS clients receive a message that the server is not responding. However, once enabled, quotas can be changed at any time without impact.

OPTIONS

-edit [-user|-group] {-mover <movername>|-fs <fs_name>
[-path <pathname>]} [<id> [<id> ...]
Sets the quota limits for users or groups on a specified Data Mover, mounted file system, or directory tree.

For a user, the ID can be a user ID or UID, however, if NIS or the local password file on the Data Mover is available, a username can also be used.

For a group, the ID can be a group ID or GID, however, if NIS or the local password file is available, a group name can also be used.

Upon execution, a vi session (unless the EDITOR environment variable specifies otherwise) is opened to edit the quota configuration file. Changes to the file are applied when the vi session is saved and exited.

[-proto <proto_id>]|[-block <hard_limit>[:<soft_limit>]]
Applies the quota configuration defined for the prototype user for each specified ID, and sets a hard and soft limit for storage (block) usage in kilobytes.

[-inode <hard_limit>[:<soft_limit>]][<id> [<id>...]
[-block <hard_limit>[:<soft_limit>]]
Edits the inode (file count) limits and the block (storage in KBs) limits directly into the quota configuration file without opening an editing session.

-edit -config {-mover <movername>|-fs <fs_name>}
[-path <pathname>]}
Edits the default quota configuration for all users/groups currently without quotas or subsequently added to the specified Data Mover or file system or quota tree. Also edits the grace periods for soft quotas, and the conditions upon which to generate a quotas-event message to the system log.

[-option <options>]
Specifies the following comma-separated options:

BGP=<integer>
Sets the block grace period in seconds.

IGP=<integer>
Sets the inode grace period in seconds.

DUBSL=<integer>
Sets the default user block soft limit in KB.

DUBHL=<integer>
Sets the default user block hard limit in KB.

DUISL=<integer>
Sets the default user inode soft limit.

DUIHL=<integer>
Sets the default user inode hard limit.

DGBSL=<integer>
Sets the default group block soft limit in KB.

DGBHL=<integer>
Sets the default group block hard limit in KB.

DGISL=<integer>
Sets the default group inode soft limit.

DGIHL=<integer>
Sets the default group inode hard limit.

HLE={True|False}
Specifies whether the hard limit is enforced.

ESFCS={True|False}
Specifies the event for check start has been sent.

ESFCE={True|False}
Specifies the event for check end has been sent.

ESFBSL={True|False}
Specifies that the event for block soft limits has been sent.

ESFBHL={True|False}

Specifies that the event for block hard limits has been sent.

-edit -tree -fs <fs_name> [[-proto <proto_id>]|
[-block <hard_limit> [:<soft_limit>]][-inode
<hard_limit>[:<soft_limit>]]][-comment <comment>]
<id> [<id>...]
Edits the quota limits for trees (inodes or blocks used by a tree directory) where the <id> is the tree ID. This option can only be applied on each file system basis. The -list option to display the tree IDs.

The -proto option applies the quota configuration of the prototype tree for each specified tree ID, or sets a hard and soft limit for blocks. The <proto_id> must be a tree ID.

The -inode and -block options edit the inode/block limits for the tree directly in the quota configuration file without opening an editing session.

The -comment option associates a comment with the quota tree. The comment is delimited by single quotes. Comment length is limited to 256 bytes (represented as 256 ASCII characters or a variable number of Unicode multibyte characters) and cannot include single quotes (. .), double quotes (" "), semicolons (;), NL (New Line), or FF (Form Feed).

-report [-user|-group] {-mover <movername>|-fs
<fs_name>} [-path <pathname>]} [<id> <id> ...]
Displays a summary of disk usage and quotas for the user or group, including the number of files and space in kilobytes for the specified <fs_name>, or all file systems mounted on the specified <movername>, or for the specified quota tree. The -edit option provides more information for the usage of UIDs and GIDs.

Note: The nas_quotas can show report for maximum 1024 IDs at a time.

-report -config {-mover <movername>|-fs <fs_name>}
[-path <pathname>]}
Displays quota configuration information as viewed from the specified Data Mover, file system, or quota-tree level, including:

* Active quota policy
* Quota status (user/group quotas enabled or disabled)
* Grace period
* Default limits currently set for users/groups
* Hard-quota enforcement option setting (deny disk space enabled or disabled)
* Quota conditions that trigger event-logging

-report -tree -fs <fs_name>[<id> <id>...]
Displays the quota limits for a specified quota tree in a file system. The <id> is a tree ID.

Note: The <id> is either a user ID, a group ID, or a tree ID. If the quota type is not specified, the default is set to the '-user' ID.

{-on|-off|-clear} [-user|-group|-both] {-mover
<movername>|-fs <fs_name>|[-path <pathname>]|-all}
Turns quotas on, off, and clears quotas for the user, group, or both (users and groups at once) on the <movername>, <fs_name>, <pathname>, for all users, or groups on all file systems on all Data Movers in the cabinet.

The -clear option permanently removes all quota records, deletes the quota configuration file, and turns quotas off.

Caution: While quotas are being turned on, off, or cleared, other operations to a file system may be disrupted. CIFS clients are disconnected during this execution.

-on -tree -fs <fs_name> -path <pathname>

Turns on (enables) tree quotas so that quota tracking and hard-limit enforcement (if enabled) can occur. When enabling tree quotas, the directory must not exist; it is created in this tree-quota-enabling process.

Note: The quota path length (which VNX for file calculates as including the file system mountpoint) must be less than 1024 bytes. If Unicode is enabled on the selected Data Mover, -path accepts any characters defined by the Unicode 3.0 standard. Otherwise, it accepts only ASCII characters.

[-comment <comment>]
The -comment option associates a comment with the quota tree. The comment is delimited by single quotes. Comment length is limited to 256 bytes (represented as 256 ASCII characters or a variable number of Unicode multibyte characters) and cannot include single quotes (. .), double quotes (" "), semicolons (;), NL (New Line), or FF (Form Feed).

-off -tree -fs <fs_name> -path <pathname>
Turns tree quotas off. When turning tree quotas off, the tree directory must be empty.

{-list|-clear} -tree -fs <fs_name>
The -list option displays all active quota trees and their respective tree IDs used by -edit and -report with the specified file system. Use the -tree -clear option to clear all the information from the database after you disable (turn off) quotas for all trees within a file system. Once cleared, the database information is not recoverable.

Caution: The -clear option deletes the usage and the limit information for tree quotas. The limits cannot be recovered.

-check -start [-mode online|offline] [-tree] -fs <fs_name> [-path <pathname>]
Starts a check of a quota database in online or offline mode for a tree quota or a file system quota. The default mode is online if the -mode option is not specified, and a quota check is run while the file system remains online.

-check {-stop|-status} -fs <fs_name> [-path <pathname>]
Stops or provides status of a file system quota database check that is in progress.

-quotadb {-info|-upgrade [-Force]} {-mover <movername>|-fs <fs_name>}
Either displays status related to the quota database upgrade or starts an upgrade of the quota database for a specific file system or all file systems on a Data Mover.

The -info option displays the status related to the quota database limits upgrade.

Use the -upgrade option to perform an upgrade of the quota database. If the -Force option is not specified, you are in interactive mode while upgrading the quota database. If the -Force option is specified, you are in non-interactive mode while upgrading the quota database.

Use -mover <movername> to upgrade all mounted file systems.
quota databases on a Data Mover.

Use -fs <fs_name> to upgrade a specific file system.s quota database.

Note: Before the upgrade process runs, the Control Station displays the estimated upgrade time on the file system whose quota database will be upgraded, and also displays a warning message to notify users that the file system will be unavailable during the upgrade process. If users are in interactive mode, a dialog displays letting users choose whether they want to continue. If users are in non-interactive mode, after displaying the estimated upgrade time message and warning message, the upgrade process starts immediately.

SEE ALSO

Using Quotas on VNX.

EXAMPLE# 1

To enable quotas for users and groups of a file system, type:

```
$ nas_quotas -on -both -fs ufs1
done
```

EXAMPLE #2

To open a vi session to edit file system quotas on ufs1 for the specified user, 1000, type:

```
$ nas_quotas -edit -user -fs ufs1 1000

Userid : 1000
fs ufs1 blocks (soft = 2000, hard = 3000) inodes (soft = 0, hard = 0)
~
~
~
~
"/tmp/EdP.agGQuIz" 2L, 84C written
done
```

EXAMPLE #3

To change the block limit and inode limit for a file without opening up a vi session, type:

```
$ nas_quotas -edit -user -fs ufs1 -block 7000:6000 -inode 700:600 2000
done
```

EXAMPLE #4

To view a report of user quotas for ufs1, type:

```
$ nas_quotas -report -user -fs ufs1

        Report for user quotas on filesystem ufs1 mounted on /ufs1
        +-----------+--------------------------------+--------------------------
---+
        |User       |        Bytes Used  (1K)        |           Files
   |
        +-----------+-------+-------+-------+--------+-------+------+------+-----
---+
        |           |  Used |  Soft |  Hard |Timeleft|  Used | Soft | Hard |Timel
eft|
        +-----------+-------+-------+-------+--------+-------+------+------+-----
---+
        |#1000      |   1328|   2000|   3000|        |     54|     0|     0|
   |
        |#2000      |   6992|   6000|   7000| 7.0days|     66|   600|   700|
   |
        |#5000      | 141592|      0|      0|        |    516|     0|     0|
   |
        +-----------+-------+-------+-------+--------+-------+------+------+-----
---+

        done
```

EXAMPLE #5

```
----------

To select user 300 as prototype user for ufs1, and assign other users the
same limits, type:

$ nas_quotas -group -edit -fs ufs1 -proto 300 301 302 303
done

EXAMPLE #6
----------

To display the group quotas information for ufs1, type:

$ nas_quotas -report -group -fs ufs1

        Report for group quotas on filesystem ufs1 mounted on /ufs1
        +-----------+--------------------------------+--------------------------
---+
        | Group     |       Bytes Used  (1K)         |           Files
   |
        +-----------+-------+-------+-------+--------+-------+------+------+-----
---+
        |           |  Used |  Soft |  Hard |Timeleft|  Used | Soft | Hard |Timel
eft|
        +-----------+-------+-------+-------+--------+-------+------+------+-----
---+
        |#1         |    296|      0|      0|        |     12|     0|     0|
   |
        |#300       |   6992|   6000|   7000| 7.0days|     67|   600|   700|
   |
        |#301       |      0|   6000|   7000|        |      0|   600|   700|
   |
        |#302       |      0|   6000|   7000|        |      0|   600|   700|
   |
        |#303       |      0|   6000|   7000|        |      0|   600|   700|
   |
        |#32772     |  22296|      0|      0|        |    228|     0|     0|
   |
        +-----------+-------+-------+-------+--------+-------+------+------+-----
---+

        done

EXAMPLE #7
----------

To edit the default quota configuration for server_2, type:

$ nas_quotas -edit -config -mover server_2

File System Quota Parameters:
fs "ufs1"
Block Grace: (1.0 weeks)
Inode Grace: (1.0 weeks)
* Default Quota Limits:
User: block (soft = 5000, hard = 8000) inodes (soft = 100, hard= 200)
Group: block (soft = 6000, hard = 9000) inodes (soft = 200, hard= 400)
Deny disk space to users exceeding quotas: (yes)
* Generate Events when:
Quota check starts: (no)
Quota check ends: (no)
soft quota crossed: (no)
hard quota crossed: (no)
fs "ufs2"
Block Grace: (1.0 weeks)
Inode Grace: (1.0 weeks)
* Default Quota Limits:
User: block (soft = 0, hard = 0) inodes (soft = 0, hard= 0)
Group: block (soft = 0, hard = 0) inodes (soft = 0, hard= 0)
Deny disk space to users exceeding quotas: (yes)
* Generate Events when:
```

```
Quota check starts: (no)
Quota check ends: (no)
soft quota crossed: (no)
hard quota crossed: (no)
~
~
~
~
"/tmp/EdP.ahCPdAB" 25L, 948C written
done
```

EXAMPLE #8

To open a vi session and edit the quota configuration for a file system, type:

```
$ nas_quotas -edit -config -fs ufs1

File System Quota Parameters:
fs "ufs1"
Block Grace: (1.0 weeks)
Inode Grace: (1.0 weeks)
* Default Quota Limits:
User: block (soft = 5000, hard = 8000) inodes (soft = 100, hard= 200)
Group: block (soft = 6000, hard = 9000) inodes (soft = 200, hard= 400)
Deny disk space to users exceeding quotas: (yes)
* Generate Events when:
Quota check starts: (no)
Quota check ends: (no)
soft quota crossed: (yes)
hard quota crossed: (yes)
~
~
~
~
"/tmp/EdP.a4slhyg" 13L, 499C written
done
```

EXAMPLE #9

To view the quota configuration for the file system, ufs1, type:

```
$ nas_quotas -report -config -fs ufs1

        +-----------------------------------------------------------+
        | Quota parameters for filesystem ufs1 mounted on /ufs1:
        +-----------------------------------------------------------+
        | Quota Policy: blocks
        | User Quota: ON
        | Group Quota: ON
        | Block grace period: (1.0 weeks)
        | Inode grace period: (1.0 weeks)
        | Default USER quota limits:
        | Block Soft: ( 5000), Block Hard: ( 8000)
        | Inode Soft: ( 100), Inode Hard: ( 200)
        | Default GROUP quota limits:
        | Block Soft: ( 6000), Block Hard: ( 9000)
        | Inode Soft: ( 200), Inode Hard: ( 400)
        | Deny Disk Space to users exceeding quotas: YES
        | Log an event when ...
        | Block hard limit reached/exceeded: YES
        | Block soft limit (warning level) crossed: YES
        | Quota check starts: NO
        | Quota Check ends: NO
        +-----------------------------------------------------------+

        done
```

EXAMPLE #10

```
----------

To enable tree quotas for ufs1, type:

$ nas_quotas -on -tree -fs ufs1 -path /tree1 -comment 'Tree #1'
done

EXAMPLE #11
-----------

To create a tree quota with multibyte character support, type:

$ nas_quotas -on -tree -fs fs_22 -path /<path_in_local_language_text>
-comment <comment_in_local_language_text>
done

EXAMPLE #12
----------

To list the tree quotas for ufs1, type:

$ nas_quotas -list -tree -fs ufs1

        +-------------------------------------------------------------------------
----+
        | Quota trees for filesystem ufs1 mounted  on /ufs1:
        +------+------------------------------------------------------------------
----+
        |TreeId| Quota tree path (Comment)
    |
        +------+------------------------------------------------------------------
----+
        |    1 | /tree1 (Tree #1)
    |
        |    2 | /tree2 (Tree #2)
    |
        |    3 | /<tree_path_in_local_language_text> (Tree #3)
    |
        +------+------------------------------------------------------------------
----+

        done

EXAMPLE #13
-----------

To edit tree quotas for ufs1 and add a comment, type:

$ nas_quotas -edit -tree -fs ufs1 -comment 'Quota for Tree1' 1
done

EXAMPLE #14
-----------

To edit tree quotas for ufs1, type:

$ nas_quotas -edit -tree -fs ufs1 1

treeid : 1
fs ufs1 blocks (soft = 6000, hard = 8000) inodes (soft = 200, hard = 300)
~
~
~
~
"/tmp/EdP.aiHKgh5" 2L, 85C written
done

EXAMPLE #15
-----------

To edit tree quotas for ufs1 and change the block and inodes, type:
```

```
$ nas_quotas -edit -tree -fs ufs1 -block 8000:6000 -inode 900:800 1
done

EXAMPLE #16
-----------

To edit tree quotas for ufs1 and apply the quota configuration of the
prototype tree, type:

$ nas_quotas -edit -tree -fs ufs1 -proto 1 2
done

EXAMPLE #17
-----------

To display any currently active trees on a file system, type:

$ nas_quotas -report -tree -fs ufs1

        Report for tree quotas on filesystem ufs1 mounted on /ufs1
        +-----------+---------------------------------+---------------------------
---+
        | Tree      |       Bytes Used  (1K)          |           Files
   |
        +-----------+-------+-------+-------+--------+-------+------+------+-----
---+
        |           |  Used |  Soft |  Hard |Timeleft|  Used | Soft | Hard |Timel
eft|
        +-----------+-------+-------+-------+--------+-------+------+------+-----
---+
        |#1         |    384|   6000|   8000|        |      3|   800|   900|
   |
        |#2         |   7856|   6000|   8000| 7.0days|     60|   800|   900|
   |
        +-----------+-------+-------+-------+--------+-------+------+------+-----
---+

        done

EXAMPLE #18
-----------

To disable tree quotas, type:

$ nas_quotas -tree -off -fs ufs1 -path /tree1
done

EXAMPLE #19
-----------

To enable quotas for users and groups on tree quota, /tree3, of a file
system, ufs1, type:

$ nas_quotas -on -both -fs ufs1 -path /tree3
done

EXAMPLE #20
----------

To open a vi session to edit file system quotas on quota tree, /tree3, on
ufs1 for the specified user, 1000, type:

$ nas_quotas -edit -user -fs ufs1 -path /tree3 1000

Userid : 1000
fs ufs1 tree "/tree3" blocks (soft = 4000, hard = 6000) inodes (soft = 30,
hard = 50)
~
~
~
```

```
~
"/tmp/EdP.aMdtIQR" 2L, 100C written
done

EXAMPLE #21
-----------

To change the block limit and inode limit on quota tree, /tree3, on ufs1 for
the specified user, 1000, without opening up a vi session, type:

$ nas_quotas -edit -user -fs ufs1 -path /tree3 -block 6000:4000 -inode

300:200 1000
done

EXAMPLE #22
-----------

To view a report of user quotas on tree quota, /tree3, for ufs1, type:

$ nas_quotas -report -user -fs ufs1 -path /tree3

        Report for user quotas on quota tree /tree3 on filesystem ufs1 mounted on
        /ufs1

        +-----------+--------------------------------+--------------------------
---+
        |User       |        Bytes Used  (1K)        |           Files
   |
        +-----------+-------+-------+-------+--------+-------+------+------+-----
---+
        |           |  Used |  Soft |  Hard |Timeleft|  Used | Soft | Hard |Timel
eft|
        +-----------+-------+-------+-------+--------+-------+------+------+-----
---+
        |#1000      |   2992|   4000|   6000|        |     34|   200|   300|
   |
        |#32768     |   9824|      0|      0|        |     28|     0|     0|
   |
        +-----------+-------+-------+-------+--------+-------+------+------+-----
---+

        done

EXAMPLE #23
-----------

To open a vi session and edit the quota configuration for tree quota,
/tree3, on a file system, ufs1, type:

$ nas_quotas -edit -config -fs ufs1 -path /tree3

Tree Quota Parameters:
fs "ufs1"
tree "/tree3"
Block Grace: (1.0 weeks)
Inode Grace: (1.0 weeks)
* Default Quota Limits:
User: block (soft = 8000, hard = 9000) inodes (soft = 200, hard= 300)
Group: block (soft = 8000, hard = 9000) inodes (soft = 300, hard= 400)
Deny disk space to users exceeding quotas: (yes)
* Generate Events when:
Quota check starts: (no)
Quota check ends: (no)
soft quota crossed: (yes)
hard quota crossed: (yes)
~
~
~
~
"/tmp/EdP.aDTOKeU" 14L, 508C written
```

```
done

EXAMPLE #24
-----------

To view the quota configuration for tree quota, /tree3, on file system,
ufs1, type:

$ nas_quotas -report -config -fs ufs1 -path /tree3

        +-----------------------------------------------------------------+
        | Quota parameters for tree quota /tree3 on filesystem ufs1 mounted
        | on /ufs1:
        +-----------------------------------------------------------------+
        | Quota Policy: blocks
        | User Quota: ON
        | Group Quota: ON
        | Block grace period: (1.0 weeks)
        | Inode grace period: (1.0 weeks)
        | Default USER quota limits:
        | Block Soft: ( 8000), Block Hard: ( 9000)
        | Inode Soft: ( 200), Inode Hard: ( 300)
        | Default GROUP quota limits:
        | Block Soft: ( 8000), Block Hard: ( 9000)
        | Inode Soft: ( 300), Inode Hard: ( 400)
        | Deny Disk Space to users exceeding quotas: YES
        | Log an event when ...
        | Block hard limit reached/exceeded: YES
        | Block soft limit (warning level) crossed: YES
        | Quota check starts: NO
        | Quota Check ends: NO
        +-----------------------------------------------------------------+

        done

EXAMPLE #25
-----------

To disable user quota and group quota on tree quota, /tree3, type:

$ nas_quotas -off -both -fs ufs1 -path /tree3
done

EXAMPLE #26
-----------

To disable group quotas for ufs1, type:

$ nas_quotas -off -group -fs ufs1
done

EXAMPLE #27
-----------

To clear all tree quotas for ufs1, type:

$ nas_quotas -clear -tree -fs ufs1
done

EXAMPLE #28
-----------

To clear quotas for users and groups of a Data Mover, type:

$ nas_quotas -clear -both -mover server_2
done

EXAMPLE #29
-----------

To start a tree quota check in quota tree /mktg-a/dir1 in file system ufs1
```

with the file system online, type:

```
$ nas_quotas -check -start -mode online -tree -fs ufs1 /mktg-a/dir1
done
```

EXAMPLE #30

To stop a tree quota check in file system ufs1, type:

```
$ nas_quotas -check -stop -fs ufs1
done
```

EXAMPLE #31

To view the status of a tree quota check in quota tree /mktg-a/dir1 in file system ufs1, type:

```
$ nas_quotas -check -status -tree -fs ufs1 -path /mktg-a/dir1

Tree quota check on filesystem ufs1 and path /mktg-a/dir is running and is 60% complete.
Done
```

EXAMPLE #32

To list quota database limits for all file systems on a Data Mover, type:

```
$ nas_quotas -quotadb -info -mover server_2

Info 13421850365 : The quota limit on ufs0 is at 4TB. The upgrade to 256 TB is estimated to take 5 seconds.

A total number of 1500 data blocks in the quota database will be converted at a speed of 300 blocks per second.

Info 13421850365 : The quota limit on ufs1 is at 4TB. The upgrade to 256 TB is estimated to take 5 seconds.

A total number of 1500 data blocks in the quota database will be converted at a speed of 300 blocks per second.

Info 13421850365 : The quota limit on ufs2 is at 4TB. The upgrade to 256 TB is estimated to take 5 seconds.

A total number of 1500 data blocks in the quota database will be converted at a speed of 300 blocks per second.

Info 13421850365 : The quota limit on ufs3 is at 4TB. The upgrade to 256 TB is estimated to take 5 seconds.

A total number of 1500 data blocks in the quota database will be converted at a speed of 300 blocks per second.

Info 13421850366 : The quota limit on ufs4 is at 256 TB
```

EXAMPLE #33

To list quota database limits for file system ufs4, type:

```
$ nas_quotas -quotadb -info -fs ufs4

Info 13421850366 : The quota limit on ufs4 is at 256 TB
```

EXAMPLE #34

To upgrade all file systems on a Data Mover, in interactive mode, type:

```
$ nas_quotas -quotadb -upgrade -mover server_2

Info 13421850365 : The quota limit on ufs0 is at 4TB. The upgrade to 256 TB is
estimated to take 5 seconds.

A total number of 1500 data blocks in the quota database will be converted at
a speed of 300 blocks per second.

Info 13421850365 : The quota limit on ufs1 is at 4TB. The upgrade to 256 TB is
estimated to take 5 seconds.

A total number of 1500 data blocks in the quota database will be converted at
a speed of 300 blocks per second.

Info 13421850365 : The quota limit on ufs2 is at 4TB. The upgrade to 256 TB is
estimated to take 5 seconds.

A total number of 1500 data blocks in the quota database will be converted at
a speed of 300 blocks per second.

Info 13421850365 : The quota limit on ufs3 is at 4TB. The upgrade to 256 TB is
estimated to take 5 seconds.

A total number of 1500 data blocks in the quota database will be converted at
a speed of 300 blocks per second.

Info 13421850366 : The quota limit on ufs4 is at 256 TB

Warning 17716861297: The file systems specified in the list above will not be
accessible during the quota database upgrade, and a file system's CIFS share
and NFS export also will not be accessible during the upgrade. The file
systems shown above are listed in the order that the quota database conversion
is performed, one by one sequentially. The estimated time ( shown above )
needed to upgrade the quota database may change based on the file system's
quota configuration and I/O performance when the conversion is running.
Do you really want to upgrade the file system quota database now[Y/N]: Y
Info 13421850367 : quota db upgraded on ufs0
Info 13421850367 : quota db upgraded on ufs1
Info 13421850367 : quota db upgraded on ufs2
Error 13421850368 :  Timeout occurred when upgrading quota db on ufs3. The
Quota db upgrade may still be in progress. Use the "-info" option to check
status.
Info 13421850369 :  quota db already upgraded on ufs4
```

EXAMPLE #35

To list quota database limits for file system ufs3 after an upgrade has timed out, type:

```
$ nas_quotas -quotadb -info -fs ufs3

Info 13421850370 : The quota limit on ufs3 is at 4TB. Upgrade is 48% complete.
```

EXAMPLE #36

To upgrade all file systems on a Data Mover, in non-interactive mode, type:

```
$ nas_quotas -quotadb -upgrade -Force -mover server_2

Info 13421850365 : The quota limit on ufs0 is at 4TB. The upgrade to 256 TB is
estimated to take 5 seconds.

A total number of 1500 data blocks in the quota database will be converted at
a speed of 300 blocks per second.

Info 13421850365 : The quota limit on ufs1 is at 4TB. The upgrade to 256 TB is
estimated to take 5 seconds.
```

A total number of 1500 data blocks in the quota database will be converted at a speed of 300 blocks per second.

Info 13421850365 : The quota limit on ufs2 is at 4TB. The upgrade to 256 TB is estimated to take 5 seconds.

A total number of 1500 data blocks in the quota database will be converted at a speed of 300 blocks per second.

Info 13421850365 : The quota limit on ufs3 is at 4TB. The upgrade to 256 TB is estimated to take 5 seconds.

A total number of 1500 data blocks in the quota database will be converted at a speed of 300 blocks per second.

Info 13421850366 : The quota limit on ufs4 is at 256 TB

Warning 17716861297: The file systems specified in the list above will not be accessible during the quota database upgrade, and a file system's CIFS share and NFS export also will not be accessible during the upgrade. The file systems shown above are listed in the order that the quota database conversion is performed, one by one sequentially. The estimated time (shown above) needed to upgrade the quota database may change based on the file system's quota configuration and I/O performance when the conversion is running.
Info 13421850367 : quota db upgraded on ufs0
Info 13421850367 : quota db upgraded on ufs1
Info 13421850367 : quota db upgraded on ufs2
Error 13421850368 : Timeout occurred when upgrading quota db on ufs3. The Quota db upgrade may still be in progress. Use the "-info" option to check status.
Info 13421850369 : quota db already upgraded on ufs4

EXAMPLE #37

To upgrade file system ufs3, in interactive mode, type:

$ nas_quotas -quotadb -upgrade -fs ufs3

Info 13421850365 : The quota limit on ufs3 is at 4TB. The upgrade to 256 TB is estimated to take 5 seconds.
A total number of 1500 data blocks in the quota database will be converted at a speed of 300 blocks per second.

Warning 17716861297: The file systems specified in the list above will not be accessible during the quota database upgrade, and a file system's CIFS share and NFS export also will not be accessible during the upgrade. The file systems shown above are listed in the order that the quota database conversion is performed, one by one sequentially. The estimated time (shown above) needed to upgrade the quota database may change based on the file system's quota configuration and I/O performance when the conversion is running.
Do you really want to upgrade the file system quota database now[Y/N]: Y
Info 13421850367 : quota db upgraded on ufs3
done

EXAMPLE #38

To upgrade file system ufs3, in non-interactive mode, type:

$ nas_quotas -quotadb -upgrade -Force -fs ufs3

Info 13421850365 : The quota limit on ufs3 is at 4TB. The upgrade to 256 TB is estimated to take 5 seconds.
A total number of 1500 data blocks in the quota database will be converted at a speed of 300 blocks per second.

Warning 17716861297: The file systems specified in the list above will not be accessible during the quota database upgrade, and a file system's CIFS share and NFS export also will not be accessible during the upgrade. The file systems shown above are listed in the order that the quota database conversion

is performed, one by one sequentially. The estimated time (shown above) needed to upgrade the quota database may change based on the file system's quota configuration and I/O performance when the conversion is running.
Info 13421850367 : quota db upgraded on ufs3

done

Last Modified: May 12, 2011 3:15 pm

nas_rdf

Facilitates communication between two VNX systems. Its primary use is to manage VNX for file systems and define the relationships needed for disaster recovery in a SRDF environment.

SYNPOSIS

```
nas_rdf
  -init
| -activate [-reverse]|-skip_rdf_operations][-skip_SiteA_shutdown][-nocheck]
| -restore [-skip_rdf_operations [-skip_SiteA_shutdown]][-nocheck]
  -check {-all|<test>,...}
```

DESCRIPTION

nas_rdf establishes and manages relationships for Control Stations and Data Movers that physically reside in separate VNX for file cabinets.

For SRDF, nas_rdf initializes the VNX, activates a failover to a destination VNX for file, or restores a source VNX. For Dynamic SRDF, nas_rdf activates a failover and reverses the system from a destination volume (R2) to a source volume (R1). Configuration details depend on the type of SRDF: active/passive or active/active'. SRDF/S for synchronous replication with disaster recovery, or active/passive SRDF/A for extended-distance, or asynchronous replication with a point-in-time replica.

Note: The apostrophe in active/active' indicates that both sites have a source volume mirrored at the other site.

SRDF is supported only on a VNX attached to a Symmetrix system. Also, this command must be run from a primary Control Station in slot 0; it will report an error if run from a Control Station in slot 1.

Note: This command must be executed from the /nas/sbin directory, unless otherwise directed. Log in with your administrative username and password, and execute this functionality from root.

OPTIONS

-init
Initializes a source or destination (target) VNX for SRDF/S or SRDF/A.

-activate [-reverse]
Initiates an SRDF failover from the source VNX for file to the destination. The -activate option is executed on the destination VNX at the discretion of the user. The -activate option sets each SRDF-protected volume on the source VNX as read-only, and each mirrored volume on the destination VNX is set as read-write. The SRDF standby Data Movers acquire the IP and MAC addresses, file systems, and export tables of their source counterparts. The -reverse option reverses SRDF direction by converting R2 volumes at destination site to R1 and synchronizing the destination and source sites. The -reverse option adds SYMCLI swap and establishes operations on the system after the normal activate operation is performed. When the -activate option is executed, an automatic, internal SRDF health check is performed before activating a failover. The -nocheck option allows you to skip this health check.

-activate -skip_rdf_operations
Skips RDF backend operations like symrdf failover. The backend operations must be done using Solution Enablers or Mainframe host component prior to this

command. SiteA shutdown (Data Mover shutdown and reboot Control Station) will be skipped all the time when this option is specified. However Control Station reboot is sent to SiteA at the end of the activate operation when the backend RDF status is not "Split" to clean up old processes. (The "Split" status means SiteA is read write, and the production site is up and running). For failover from SiteB to SiteC or SiteC to SiteB, the Control Station reboot is sent to SiteB or SiteC. SiteB/SiteC must be read write before starting this operation. The -activate -skip_rdf_operations -skip_SiteA_shutdown will do the same operation.

-activate -skip_SiteA_shutdown
Skips SiteA shutdown (Data Mover shutdown and reboot Control Station) operation. However the SiteA shutdown is sent to SiteA at the end of the activate operation. This option is mainly used to minimize the failover time.

-restore -skip_rdf_operations
Skips RDF backend operations like symrdf failback. This option also completes only SiteB/SiteC restore operations and skip SiteA restore operation. The SiteA restore operation must be done separately at SiteA after the SiteB/SiteC restore operation completes. SiteB/SiteC must be read write before starting this operation.

-restore -skip_rdf_operations -skip_SiteA_shutdown
Skips RDF backend operations like symrdf failback and also skip SiteA shutdown operation. This is mainly used to failover from SiteB to SiteC or from SiteC to SiteB.

-restore
Restores a source VNX after a failover. The -restore option is initially executed on the destination VNX. The data on each destination volume is copied to the corresponding volume on the source VNX. On the destination VNX, services on each SRDF standby Data Mover are stopped. (NFS clients connected to these Data Movers see a "server unavailable" message; CIFS client connections time out.) Each volume on the source VNX is set as read-write, and each mirrored volume on the destination VNX is set as read-only.

Finally, nas_rdf -restore can be remotely executed on the source VNX to restore the original configuration. Each primary Data Mover reacquires its IP and MAC addresses, file systems, and export tables. When the -restore option is executed, an automatic, internal SRDF health check is performed before restoring source and destination VNX systems. The -nocheck option allows you to skip this health check.

-check { -all|<test>,...}
Runs SRDF health checks on the VNX. The -check option can be executed either by using the -all option or by specifying one or more of the following individual checks: SRDF standby Data Mover configuration check (r1_dm_config, r2_dm_config), SRDF session state check (r1_session, r2_session), Device group configuration check (r1_dev_group, r2_dev_group), Data Mover mirrored device state check (dev_not_normal), and SRDF restored state check (restored). In these checks, r1 represents the source side and r2 represents the destination side.

When the -all option is used, all the checks are performed automatically. If the -check option detects invalid configurations or state issues, it prints relevant warning messages with recommended actions so that the issues can be resolved before running the activate or restore options. You can use the -check option to perform health checks at any time.

Note: To run the -check option, you must log in to the VNX either as nasadmin and then switch (su) to root, or as rdfadmin and then switch (su) to root.

SEE ALSO

Using SRDF/S with VNX for Disaster Recovery, Using SRDF/S with
VNX, and nas_cel.

EXAMPLE #1

To start the initialization process on a destination VNX in an active/passive
SRDF/S configuration, as a nasadmin su to root user, type:

```
# /nas/sbin/nas_rdf -init
Discover local storage devices ...

Discovering storage on eng564168 (may take several minutes)
done

Start R2 dos client ...
done
Start R2 nas client ...
done

Contact CS_A ... is alive
Create a new login account to manage the RDF site CELERRA
Caution: For an active-active configuration, avoid using the same UID
that was used for the rdfadmin account on the other side.
New login username and UID (example: rdfadmin:500): rdfadmin:600
done
New UNIX password:
BAD PASSWORD: it is based on a dictionary word
Retype new UNIX password:
Changing password for user rdfadmin.
passwd: all authentication tokens updated successfully.
done
operation in progress (not interruptible)...
id = 1
name = CS_A
owner = 600
device = /dev/ndj1
channel = rdev=/dev/ndg, off_MB=391; wdev=/dev/nda, off_MB=391
net_path = 10.245.64.169
celerra_id = 0001949004310028
passphrase = nasadmin
Discover remote storage devices ...done
The following servers have been detected on the system (CS_B):
id type acl slot groupID state name
1 4 2000 2 0 server_2
2 1 0 3 0 server_3
Please enter the id(s) of the server(s) you wish to reserve
(separated by spaces) or "none" for no servers.
Select server(s) to use as standby: 1
operation in progress (not interruptible)...
id = 1
name = CS_A
owner = 600
device = /dev/ndj1
channel = rdev=/dev/ndg, off_MB=391; wdev=/dev/nda, off_MB=391
net_path = 10.245.64.169
celerra_id = 0001949004310028
passphrase = nasadmin
```

EXAMPLE #2

To initiate an SRDF failover from the source VNX to the destination, as a
rdfadmin su to root, type:

```
# /nas/sbin/nas_rdf -activate
Is remote site CELERRA completely shut down (power OFF)?
Do you wish to continue? [yes or no]: yes
Successfully pinged (Remotely) Symmetrix ID: 000187430809
Successfully pinged (Remotely) Symmetrix ID: 000190100559
Successfully pinged (Remotely) Symmetrix ID: 000190100582
    Write Disable device(s) on SA at source (R1).............Done.
```

```
    Suspend RDF link(s).....................................Done.
    Read/Write Enable device(s) on RA at target (R2).......Done.
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done
fsck 1.35 (28-Feb-2004)
/dev/ndj1: recovering journal
/dev/ndj1: clean, 13780/231360 files, 233674/461860 blocks
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done

id      type  acl  slot groupID  state  name
1        1    1000  2              0    server_2
2        4    1000  3              0    server_3
3        1    1000  4              0    server_4
4        4    1000  5              0    server_5

server_2 :
server_2 : going offline
rdf : going active
replace in progress ...done
failover activity complete
commit  in progress (not interruptible)...done
done

server_3 :
server_3 : going offline
 rdf : going active
 replace in progress ...done
 failover activity complete
 commit  in progress (not interruptible)...done
done

server_4 :
Error 4003: server_4 : standby is not configured

server_5 :
Error 4003: server_5 : standby is not configured
    Suspend RDF link(s).....................................Done.
    Merge device track tables between source and target.....Started.
    Device: 045A in (0557,005)............................. Merged.
    Merge device track tables between source and target.....Done.
    Resume RDF link(s)......................................Started.
    Resume RDF link(s)......................................Done.

EXAMPLE #3
----------

To initiate an SRDF failover from the source VNX to the destination,
without the SRDF health check, as rdfadmin su to root user, type:

# /nas/sbin/nas_rdf -activate -nocheck
Skipping SRDF health check ....

Is remote site CELERRA completely shut down (power OFF)?The nas Commands
Do you wish to continue? [yes or no]: yes
Successfully pinged (Remotely) Symmetrix ID: 000187430809
Successfully pinged (Remotely) Symmetrix ID: 000190100559
Successfully pinged (Remotely) Symmetrix ID: 000190100582
Write Disable device(s) on SA at source (R1).............Done.
Suspend RDF link(s).....................................Done.
Read/Write Enable device(s) on RA at target (R2).......Done.
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done
fsck 1.35 (28-Feb-2004)
/dev/ndj1: recovering journal
/dev/ndj1: clean, 13780/231360 files, 233674/461860 blocks
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done
id type acl slot groupID state name
1 1 1000 2 0 server_2
2 4 1000 3 0 server_3
```

```
3 1 1000 4 0 server_4
4 4 1000 5 0 server_5
server_2 :
server_2 : going offline
rdf : going active
replace in progress ...done
failover activity complete
commit in progress (not interruptible)...done
done
server_3 :
server_3 : going offline
rdf : going active
replace in progress ...done
failover activity complete
commit in progress (not interruptible)...done
done
server_4 :
Error 4003: server_4 : standby is not configured
server_5 :
Error 4003: server_5 : standby is not configured
Suspend RDF link(s).....................................Done.
Merge device track tables between source and target.......Started.
Device: 045A in (0557,005)............................... Merged.
Merge device track tables between source and target.......Done.
Resume RDF link(s)......................................Started.
Resume RDF link(s)......................................Done.

EXAMPLE #4
----------

To initiate a Dynamic SRDF failover from the source VNX to the
destination, as rdfadmin su to root user, type:

#/nas/sbin/nas_rdf -activate -reverse
Is remote site CELERRA completely shut down (power OFF)?
Do you wish to continue? [yes or no]: yes
Successfully pinged (Remotely) Symmetrix ID: 000280600118
    Write Disable device(s) on SA at source (R1)..............Done.
    Suspend RDF link (s)....................................Done.
    Read/Write Enable device(s) on RA at target (r2)..........Done.
fsck 1.35 (28-Feb-2004)
/dev/sdj1: recovering journal
Clearing ophaned inode 37188 (uid0, gid=0, mode=0100644, size=0)
/dev/sdj1: clean, 12860/219968 files, 194793/439797 blocks
id      type  acl  slot groupID  state  name
1        1    1000  2             0    server_2
2        4    1000  3             0    server_3
3        4    2000  4             0    server_4
4        4    2000  5             0    server_5

server_2 :
 server_2 : going offline
 rdf : going active
 replace in progress ...done
 failover activity complete
 commit         in progress (not interruptible)...done
done
server_3 :
 server_3 : going offline
 rdf : going active
 replace in progress ...done
 failover activity complete
 commit         in progress (not interruptible)...done
done
An RDF 'Swap Personality' operation execution is
in progress for device group '1R2_500_1'. Please wait...

   Swap RDF Personality....................................Started.
   Swap RDF Personality....................................Done.

The RDF 'Swap Personality' operation successfully executed for
```

```
device group '1R2_500_1'.

An RDF 'Incremental Establish' operation execution is
in progress for device group '1R2_500_1'. Please wait...

   Suspend RDF link(s).....................................Done.
   Resume RDF link(s)......................................Started.
   Merge device track tables between source and target.....Started.
   Devices: 0009-000B ..................................... Merged.
   Devices: 0032-0034 ..................................... Merged.
   Devices: 0035-0037 ..................................... Merged.
   Devices: 0038-003A ..................................... Merged.
   Devices: 003B-003D ..................................... Merged.
   Devices: 003E-0040 ..................................... Merged.
   Devices: 0041-0043 ..................................... Merged.
   Devices: 0044-0046 ..................................... Merged.
   Devices: 0047-0049 ..................................... Merged.

   Merge device track tables between source and target.....Done.
   Resume RDF link(s)......................................Done.

The RDF 'Incremental Establish' operation successfully initiated for
device group '1R2_500_1'.

EXAMPLE #5
----------

To restore a source VNX after failover, as rdfadmin su to root user, type:

# /nas/sbin/nas_rdf -restore
Is remote site CELERRA ready for Storage restoration?
Do you wish to continue? [yes or no]: yes
Contact Joker_R1_CS0 ... is alive

Restore will now reboot the source site control station.
Do you wish to continue? [yes or no]: yes

Device Group (DG) Name             : 1R2_500_5
DG's Type                          : RDF2
DG's Symmetrix ID                  : 000190100557

       Target (R2) View                 Source (R1) View     MODES
--------------------------------    ------------------------ ----- ------------
             ST                  LI      ST
Standard      A                   N       A
Logical       T  R1 Inv   R2 Inv  K       T  R1 Inv   R2 Inv       RDF Pair
Device  Dev   E  Tracks   Tracks  S Dev   E  Tracks   Tracks MDA   STATE
--------------------------------- -- ------------------------ ----- ------------

DEV001  045A RW      10        0 RW 045A WD       0        0 S..   R1 Updated
DEV002  045B RW    2054        0 NR 045B WD       0        0 S..   Failed Over
DEV003  045C RW       0        0 NR 045C WD       0        0 S..   Failed Over
DEV004  045D RW       0        0 NR 045D WD       0        0 S..   Failed Over
DEV005  045E RW    1284        0 NR 045E WD       0        0 S..   Failed Over
DEV006  045F RW       0        0 NR 045F WD       0        0 S..   Failed Over
DEV007  0467 RW       0        0 NR 0467 WD       0        0 S..   Failed Over
DEV008  0468 RW       2        0 NR 0468 WD       0        0 S..   Failed Over
DEV009  0469 RW       0        0 NR 0469 WD       0        0 S..   Failed Over
DEV010  046A RW       0        0 NR 046A WD       0        0 S..   Failed Over
DEV011  046B RW       2        0 NR 046B WD       0        0 S..   Failed Over
DEV012  046C RW       0        0 NR 046C WD       0        0 S..   Failed Over
DEV013  046D RW       0        0 NR 046D WD       0        0 S..   Failed Over
DEV014  046E RW       0        0 NR 046E WD       0        0 S..   Failed Over
DEV015  046F RW       2        0 NR 046F WD       0        0 S..   Failed Over
DEV016  0470 RW       0        0 NR 0470 WD       0        0 S..   Failed Over
DEV017  0471 RW       2        0 NR 0471 WD       0        0 S..   Failed Over
DEV018  0472 RW       0        0 NR 0472 WD       0        0 S..   Failed Over
DEV019  0473 RW       0        0 NR 0473 WD       0        0 S..   Failed Over
DEV020  0474 RW       0        0 NR 0474 WD       0        0 S..   Failed Over
DEV021  0475 RW       0        0 NR 0475 WD       0        0 S..   Failed Over
DEV022  0476 RW       0        0 NR 0476 WD       0        0 S..   Failed Over
```

```
DEV023  0477 RW       2        0 NR 0477 WD       0        0 S..   Failed Over
DEV024  0478 RW       2        0 NR 0478 WD       0        0 S..   Failed Over
DEV025  0479 RW       0        0 NR 0479 WD       0        0 S..   Failed Over
DEV026  047A RW       0        0 NR 047A WD       0        0 S..   Failed Over
DEV027  047B RW       0        0 NR 047B WD       0        0 S..   Failed Over
DEV028  047C RW       0        0 NR 047C WD       0        0 S..   Failed Over
DEV029  047D RW       0        0 NR 047D WD       0        0 S..   Failed Over
DEV030  047E RW       0        0 NR 047E WD       0        0 S..   Failed Over
DEV031  047F RW       0        0 NR 047F WD       0        0 S..   Failed Over
DEV032  0480 RW       0        0 NR 0480 WD       0        0 S..   Failed Over
DEV033  0481 RW       0        0 NR 0481 WD       0        0 S..   Failed Over
DEV034  0482 RW       0        0 NR 0482 WD       0        0 S..   Failed Over
DEV035  0483 RW       0        0 NR 0483 WD       0        0 S..   Failed Over
DEV036  0484 RW       0        0 NR 0484 WD       0        0 S..   Failed Over
DEV037  0485 RW       0        0 NR 0485 WD       0        0 S..   Failed Over
DEV038  0486 RW       0        0 NR 0486 WD       0        0 S..   Failed Over
DEV039  0487 RW       0        0 NR 0487 WD       0        0 S..   Failed Over
DEV040  0488 RW       0        0 NR 0488 WD       0        0 S..   Failed Over
DEV041  0489 RW       0        0 NR 0489 WD       0        0 S..   Failed Over
DEV042  048A RW       0        0 NR 048A WD       0        0 S..   Failed Over
DEV043  048B RW       0        0 NR 048B WD       0        0 S..   Failed Over
DEV044  048C RW       0        0 NR 048C WD       0        0 S..   Failed Over
DEV045  048D RW       0        0 NR 048D WD       0        0 S..   Failed Over
DEV046  048E RW       0        0 NR 048E WD       0        0 S..   Failed Over
DEV047  048F RW       2        0 NR 048F WD       0        0 S..   Failed Over
DEV048  0490 RW       0        0 NR 0490 WD       0        0 S..   Failed Over
DEV049  0491 RW       0        0 NR 0491 WD       0        0 S..   Failed Over
DEV050  0492 RW       0        0 NR 0492 WD       0        0 S..   Failed Over
DEV051  0493 RW       0        0 NR 0493 WD       0        0 S..   Failed Over
DEV052  0494 RW       0        0 NR 0494 WD       0        0 S..   Failed Over
DEV053  0495 RW       0        0 NR 0495 WD       0        0 S..   Failed Over
DEV054  0496 RW       0        0 NR 0496 WD       0        0 S..   Failed Over
DEV055  0497 RW       2        0 NR 0497 WD       0        0 S..   Failed Over
DEV056  0498 RW       2        0 NR 0498 WD       0        0 S..   Failed Over
DEV057  0499 RW       0        0 NR 0499 WD       0        0 S..   Failed Over
DEV058  049A RW       0        0 NR 049A WD       0        0 S..   Failed Over
DEV059  049B RW       0        0 NR 049B WD       0        0 S..   Failed Over
DEV060  049C RW       0        0 NR 049C WD       0        0 S..   Failed Over
DEV061  049D RW       0        0 NR 049D WD       0        0 S..   Failed Over
DEV062  049E RW       0        0 NR 049E WD       0        0 S..   Failed Over
DEV063  049F RW       0        0 NR 049F WD       0        0 S..   Failed Over
DEV064  04A0 RW       0        0 NR 04A0 WD       0        0 S..   Failed Over
DEV065  04A1 RW       0        0 NR 04A1 WD       0        0 S..   Failed Over
DEV066  04A2 RW       0        0 NR 04A2 WD       0        0 S..   Failed Over
DEV067  04A3 RW       0        0 NR 04A3 WD       0        0 S..   Failed Over
DEV068  04A4 RW       0        0 NR 04A4 WD       0        0 S..   Failed Over
DEV069  04A5 RW       0        0 NR 04A5 WD       0        0 S..   Failed Over
DEV070  04A6 RW       0        0 NR 04A6 WD       0        0 S..   Failed Over
Total          -------- --------           -------- --------
  Track(s)         3366        0                  0        0
  MB(s)           105.2      0.0                0.0      0.0
Legend for MODES:
 M(ode of Operation): A = Async, S = Sync, E = Semi-sync, C = Adaptive Copy
 D(omino)           : X = Enabled, . = Disabled
 A(daptive Copy)    : D = Disk Mode, W = WP Mode, . = ACp off
    Suspend RDF link(s).....................................Done.
    Merge device track tables between source and target.....Started.
    Devices: 045A-045F, 0467-0477 in (0557,005)............. Merged.
    Devices: 0478-0489 in (0557,005)........................ Merged.
    Devices: 048A-049B in (0557,005)........................ Merged.
    Devices: 049C-04A6 in (0557,005)........................ Merged.
    Merge device track tables between source and target.....Done.
    Resume RDF link(s)......................................Started.
    Resume RDF link(s)......................................Done.
Is remote site CELERRA ready for Network restoration?
Do you wish to continue? [yes or no]: yes
server_2 : done
server_3 : done
server_4 :
Error 4003: server_4 : standby is not configured
server_5 :
```

```
Error 4003: server_5 : standby is not configured
fsck 1.35 (28-Feb-2004)
/dev/ndj1: clean, 13836/231360 files, 233729/461860 blocks
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done
Waiting for 1R2_500_5 access ...done
    Write Disable device(s) on RA at target (R2).............Done.
    Suspend RDF link(s).....................................Done.
    Merge device track tables between source and target.....Started.
    Devices: 045A-045F, 0467-0477 in (0557,005)..............  Merged.
    Devices: 0478-0489 in (0557,005)......................... Merged.
    Devices: 048A-049B in (0557,005)......................... Merged.
    Devices: 049C-04A6 in (0557,005)......................... Merged.
    Merge device track tables between source and target.....Done.
    Resume RDF link(s)......................................Started.
    Resume RDF link(s)......................................Done.
    Read/Write Enable device(s) on SA at source (R1).........Done.
Waiting for 1R2_500_5 sync ...done
Starting restore on remote site CELERRA ...
Waiting for nbs clients to start ... done
Waiting for nbs clients to start ... done
    Suspend RDF link(s).....................................Done.
server_2 :
 server_2 : going standby
 rdf : going active
 replace in progress ...done
 failover activity complete
 commit  in progress (not interruptible)...done
done
server_3 :
 server_3 : going standby
 rdf : going active
 replace in progress ...done
 failover activity complete
 commit  in progress (not interruptible)...done
done
server_4 :
Error 4003: server_4 : standby is not configured
server_5 :
Error 4003: server_5 : standby is not configured
    Resume RDF link(s)......................................Started.
    Resume RDF link(s)......................................Done.
If the RDF device groups were setup to operate in ASYNCHRONOUS ( SRDF/A ) mode,
now would be a good time to set it back to that mode.
Would you like to set device group 1R2_500_5 to ASYNC Mode ? [yes or no]: no
done

EXAMPLE #6
-----------

To restore a source VNX after failover, without the SRDF health
check, as rdfadmin su to root user, type:

# /nas/sbin/nas_rdf -restore -nocheck
Skipping SRDF health check ....

Is remote site CELERRA ready for Storage restoration?
Do you wish to continue? [yes or no]: yes
Contact Joker_R1_CS0 ... is alive
Restore will now reboot the source site control station.
Do you wish to continue? [yes or no]: yes
Device Group (DG) Name : 1R2_500_5
DG's Type : RDF2
DG's Symmetrix ID : 000190100557
Target (R2) View Source (R1) View MODES
-------------------------------- ------------------------ ----- ------------
ST              LI      ST
Standard    A                N      A
Logical     T R1 Inv  R2 Inv  K      T R1 Inv R2 Inv     RDF Pair
```

Device	Dev	E	Tracks	Tracks	S	Dev	E	Tracks	Tracks	MDA	STATE
DEV001	045A	RW	10	0	RW	045A	WD	0	0	S..	R1 Updated
DEV002	045B	RW	2054	0	NR	045B	WD	0	0	S..	Failed Over
DEV003	045C	RW	0	0	NR	045C	WD	0	0	S..	Failed Over
DEV004	045D	RW	0	0	NR	045D	WD	0	0	S..	Failed Over
DEV005	045E	RW	1284	0	NR	045E	WD	0	0	S..	Failed Over
DEV006	045F	RW	0	0	NR	045F	WD	0	0	S..	Failed Over
DEV007	0467	RW	0	0	NR	0467	WD	0	0	S..	Failed Over
DEV008	0468	RW	2	0	NR	0468	WD	0	0	S..	Failed Over
DEV009	0469	RW	0	0	NR	0469	WD	0	0	S..	Failed Over
DEV010	046A	RW	0	0	NR	046A	WD	0	0	S..	Failed Over
DEV011	046B	RW	2	0	NR	046B	WD	0	0	S..	Failed Over
DEV012	046C	RW	0	0	NR	046C	WD	0	0	S..	Failed Over
DEV013	046D	RW	0	0	NR	046D	WD	0	0	S..	Failed Over
DEV014	046E	RW	0	0	NR	046E	WD	0	0	S..	Failed Over
DEV015	046F	RW	2	0	NR	046F	WD	0	0	S..	Failed Over
DEV016	0470	RW	0	0	NR	0470	WD	0	0	S..	Failed Over
DEV017	0471	RW	2	0	NR	0471	WD	0	0	S..	Failed Over
DEV018	0472	RW	0	0	NR	0472	WD	0	0	S..	Failed Over
DEV019	0473	RW	0	0	NR	0473	WD	0	0	S..	Failed Over
DEV020	0474	RW	0	0	NR	0474	WD	0	0	S..	Failed Over
DEV021	0475	RW	0	0	NR	0475	WD	0	0	S..	Failed Over
DEV022	0476	RW	0	0	NR	0476	WD	0	0	S..	Failed Over
DEV023	0477	RW	2	0	NR	0477	WD	0	0	S..	Failed Over
DEV024	0478	RW	2	0	NR	0478	WD	0	0	S..	Failed Over
DEV025	0479	RW	0	0	NR	0479	WD	0	0	S..	Failed Over
DEV026	047A	RW	0	0	NR	047A	WD	0	0	S..	Failed Over
DEV027	047B	RW	0	0	NR	047B	WD	0	0	S..	Failed Over
DEV028	047C	RW	0	0	NR	047C	WD	0	0	S..	Failed Over
DEV029	047D	RW	0	0	NR	047D	WD	0	0	S..	Failed Over
DEV030	047E	RW	0	0	NR	047E	WD	0	0	S..	Failed Over
DEV031	047F	RW	0	0	NR	047F	WD	0	0	S..	Failed Over
DEV032	0480	RW	0	0	NR	0480	WD	0	0	S..	Failed Over
DEV033	0481	RW	0	0	NR	0481	WD	0	0	S..	Failed Over
DEV034	0482	RW	0	0	NR	0482	WD	0	0	S..	Failed Over
DEV035	0483	RW	0	0	NR	0483	WD	0	0	S..	Failed Over
DEV036	0484	RW	0	0	NR	0484	WD	0	0	S..	Failed Over
DEV037	0485	RW	0	0	NR	0485	WD	0	0	S..	Failed Over
DEV038	0486	RW	0	0	NR	0486	WD	0	0	S..	Failed Over
DEV039	0487	RW	0	0	NR	0487	WD	0	0	S..	Failed Over
DEV040	0488	RW	0	0	NR	0488	WD	0	0	S..	Failed Over
DEV041	0489	RW	0	0	NR	0489	WD	0	0	S..	Failed Over
DEV042	048A	RW	0	0	NR	048A	WD	0	0	S..	Failed Over
DEV043	048B	RW	0	0	NR	048B	WD	0	0	S..	Failed Over
DEV044	048C	RW	0	0	NR	048C	WD	0	0	S..	Failed Over
DEV045	048D	RW	0	0	NR	048D	WD	0	0	S..	Failed Over
DEV046	048E	RW	0	0	NR	048E	WD	0	0	S..	Failed Over
DEV047	048F	RW	2	0	NR	048F	WD	0	0	S..	Failed Over
DEV048	0490	RW	0	0	NR	0490	WD	0	0	S..	Failed Over
DEV049	0491	RW	0	0	NR	0491	WD	0	0	S..	Failed Over
DEV050	0492	RW	0	0	NR	0492	WD	0	0	S..	Failed Over
DEV051	0493	RW	0	0	NR	0493	WD	0	0	S..	Failed Over
DEV052	0494	RW	0	0	NR	0494	WD	0	0	S..	Failed Over
DEV053	0495	RW	0	0	NR	0495	WD	0	0	S..	Failed Over
DEV054	0496	RW	0	0	NR	0496	WD	0	0	S..	Failed Over
DEV055	0497	RW	2	0	NR	0497	WD	0	0	S..	Failed Over
DEV056	0498	RW	2	0	NR	0498	WD	0	0	S..	Failed Over
DEV057	0499	RW	0	0	NR	0499	WD	0	0	S..	Failed Over
DEV058	049A	RW	0	0	NR	049A	WD	0	0	S..	Failed Over
DEV059	049B	RW	0	0	NR	049B	WD	0	0	S..	Failed Over
DEV060	049C	RW	0	0	NR	049C	WD	0	0	S..	Failed Over
DEV061	049D	RW	0	0	NR	049D	WD	0	0	S..	Failed Over
DEV062	049E	RW	0	0	NR	049E	WD	0	0	S..	Failed Over
DEV063	049F	RW	0	0	NR	049F	WD	0	0	S..	Failed Over
DEV064	04A0	RW	0	0	NR	04A0	WD	0	0	S..	Failed Over
DEV065	04A1	RW	0	0	NR	04A1	WD	0	0	S..	Failed Over
DEV066	04A2	RW	0	0	NR	04A2	WD	0	0	S..	Failed Over
DEV067	04A3	RW	0	0	NR	04A3	WD	0	0	S..	Failed Over
DEV068	04A4	RW	0	0	NR	04A4	WD	0	0	S..	Failed Over
DEV069	04A5	RW	0	0	NR	04A5	WD	0	0	S..	Failed Over

```
DEV070 04A6 RW 0      0      NR 04A6 WD 0      0     S.. Failed Over
Total -------- -------- -------- --------
Track(s) 3366 0 0 0he nas Commands
MB(s) 105.2 0.0 0.0 0.0
Legend for MODES:
M(ode of Operation): A = Async, S = Sync, E = Semi-sync, C = Adaptive Copy
D(omino) : X = Enabled, . = Disabled
A(daptive Copy) : D = Disk Mode, W = WP Mode, . = ACp off
Suspend RDF link(s).....................................Done.
Merge device track tables between source and target.....Started.
Devices: 045A-045F, 0467-0477 in (0557,005)..... Merged.
Devices: 0478-0489 in (0557,005)..... Merged.
Devices: 048A-049B in (0557,005)..... Merged.
Devices: 049C-04A6 in (0557,005)..... Merged.
Merge device track tables between source and target.....Done.
Resume RDF link(s)......................................Started.
Resume RDF link(s)......................................Done.
Is remote site CELERRA ready for Network restoration?
Do you wish to continue? [yes or no]: yes
server_2 : done
server_3 : done
server_4 :
Error 4003: server_4 : standby is not configured
server_5 :
Error 4003: server_5 : standby is not configured
fsck 1.35 (28-Feb-2004)
/dev/ndj1: clean, 13836/231360 files, 233729/461860 blocks
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done
Waiting for 1R2_500_5 access ...done
Write Disable device(s) on RA at target (R2).............Done.
Suspend RDF link(s).....................................Done.
Merge device track tables between source and target.....Started.
Devices: 045A-045F, 0467-0477 in (0557,005)..... Merged.
Devices: 0478-0489 in (0557,005)..... Merged.
Devices: 048A-049B in (0557,005)..... Merged.
Devices: 049C-04A6 in (0557,005)..... Merged.
Merge device track tables between source and target.....Done.
Resume RDF link(s)......................................Started.
Resume RDF link(s)......................................Done.
Read/Write Enable device(s) on SA at source (R1).........Done.
Waiting for 1R2_500_5 sync ...done
Starting restore on remote site CELERRA ...
Waiting for nbs clients to start ... done
Waiting for nbs clients to start ... done
Suspend RDF link(s).....................................Done.
server_2 :
server_2 : going standby
rdf : going active
replace in progress ...done
failover activity complete
commit in progress (not interruptible)...done
doneThe nas Commands
server_3 :
server_3 : going standby
rdf : going active
replace in progress ...done
failover activity complete
commit in progress (not interruptible)...done
done
server_4 :
Error 4003: server_4 : standby is not configured
server_5 :
Error 4003: server_5 : standby is not configured
Resume RDF link(s)......................................Started.
Resume RDF link(s)......................................Done.
If the RDF device groups were setup to operate in ASYNCHRONOUS ( SRDF/A )
mode,
now would be a good time to set it back to that mode.
```

```
Would you like to set device group 1R2_500_5 to ASYNC Mode ? [yes or no]: no
done

EXAMPLE #7
----------

To restore a source VNX after failover, when using Dynamic SRDF, rdfadmin su
to root user, type:

# /nas/sbin/nas_rdf -restore
Is remote site CELERRA ready for Storage restoration?
Do you wish to continue? [yes or no]: yes
Contact eng17335 ... is alive

Restore will now reboot the source site control station.
Do you wish to continue? [yes or no]: yes

Device Group (DG) Name             : 1R2_500_1
DG's Type                          : RDF1
DG's Symmetrix ID                  : 000280600187    (Microcode Version: 5568)
Remote Symmetrix ID                : 000280600118    (Microcode Version: 5568)
RDF (RA) Group Number              : 1 (00)

     Source (R1) View               Target (R2) View  MODES
------------------------------    -------------------- ---- ------------
             ST                LI     ST
Standard      A                 N      A
Logical       T R1 Inv R2 Inv   K      T R1 Inv R2 Inv     RDF Pair
Device  Dev   E Tracks Tracks   S Dev  E Tracks Tracks MDA STATE
------------------------------ -- -------------------- --- ------------

DEV001  0056 RW      0      0 RW 0030 WD      0      0 S.. Synchronized
DEV002  0057 RW      0      0 RW 0031 WD      0      0 S.. Synchronized
DEV003  0032 RW      0      0 RW 000C WD      0      0 S.. Synchronized
..............
BCV008  0069 RW      0      0 RW 005F WD      0      0 S.. Synchronized
BCV009  006A RW      0      0 RW 0060 WD      0      0 S.. Synchronized
BCV010  006B RW      0      0 RW 0061 WD      0      0 S.. Synchronized

Total            ------ ------           ------ ------
  Track(s)           0      0                0      0
  MB(s)            0.0    0.0              0.0    0.0

Legend for MODES:
 M(ode of Operation): A = Async, S = Sync, E = Semi-sync, C = Adaptive Copy
 D(omino)           : X = Enabled, . = Disabled
 A(daptive Copy)    : D = Disk Mode, W = WP Mode, . = ACp off

Is remote site CELERRA ready for Network restoration?
Do you wish to continue? [yes or no]: yes
server_2 : done
server_3 : done
server_4 :
Error 4003: server_4 : standby is not configured
server_5 :
Error 4003: server_5 : standby is not configured

fsck 1.35 (28-Feb-2004)
/dev/sdj1: clean, 12956/219968 files, 188765/439797 blocks

An RDF 'Failover' operation execution is
in progress for device group '1R2_500_1'. Please wait...

    Write Disable device(s) on SA at source (R1).............Done.
    Suspend RDF link(s).....................................Done.
    Swap RDF Personality....................................Started.
    Swap RDF Personality....................................Done.
    Suspend RDF link(s).....................................Done.
    Read/Write Enable device(s) on SA at source (R1).........Done.
    Resume RDF link(s)......................................Started.
    Resume RDF link(s)......................................Done.
```

```
    Read/Write Enable device(s) on SA at target (R2).........Done.

The RDF 'Failover' operation successfully executed for
device group '1R2_500_1'.

Waiting for 1R2_500_1 sync ...done

Starting restore on remote site CELERRA ...
Suspend RDF link(s).....................................Done.
server_2 :
server_2 : going standby
 rdf : going active
 replace in progress ...done
 failover activity complete
 commit in progress (not interruptible)...done
done
server_3 :
server_3 : going standby
 rdf : going active
 replace in progress ...done
 failover activity complete
 commit in progress (not interruptible)...done
done
server_4 :
Error 4003: server_4: standby is not configured
server_5 :
Error 4003: server_5: standby is not configured

done
```

EXAMPLE #8

To run all available checks on a source VNX, as a nasadmin su to root user, type:

```
# /nas/sbin/nas_rdf -check -all
--------------------- SRDF Health Checks ---------------------
SRDF: Checking system is restored..................... Pass
SRDF: Checking device is normal....................... Pass
SRDF: Checking R1 SRDF session is Synch or Consistent.... Pass
SRDF: Checking R1 Data Mover configuration is valid...... Pass
SRDF: Checking R1 devices are available.................. Pass
SRDF: Checking R1 device group has all devices........... Pass
SRDF: Checking R2 SRDF session is Synch or Consistent.... Pass
SRDF: Checking R2 Data Mover configuration is valid...... Pass
SRDF: Checking R2 devices are available.................. Pass
SRDF: Checking R2 device group has all devices........... Pass
```

EXAMPLE #9

To run one or more specific available checks on a source VNX, as a nasadmin su to root user, type:

```
# /nas/sbin/nas_rdf -check r1_dev_group,r2_dev_group

--------------------- SRDF Health Checks ---------------------
SRDF: Checking R1 device group has all devices........... Pass
SRDF: Checking R2 device group has all devices........... Pass
```

EXAMPLE #10

To initiate an SRDF failover from the source VNX to the destination, without the SRDF health check for the following use cases, a rdfadmin su to root user, type:

```
# /nas/sbin/nas_rdf -activate -skip_rdf_operations -nocheck
```

* SRDF STAR concurrent or cascaded

```
* SRDF concurrent or cascaded
* SRDF R2 enable (Split)

SiteA to SiteB/SiteC failover case
Skipping SRDF health check ....
Skipping Site A shutdown process for the skip_rdf_opertaions option ....

Successfully pinged (Remotely) Symmetrix ID: 000194900462

Successfully pinged (Remotely) Symmetrix ID: 000194900546
Skipping symrdf failover process ....
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done
fsck 1.39 (29-May-2006)
/dev/ndj1: recovering journal
/dev/ndj1: clean, 15012/252928 files, 271838/516080 blocks
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done
id      type  acl  slot groupID  state  name
1        1    0     2             0    server_2
2        1    0     3             0    server_3

server_2 :
 server_2 : going offline
rdf : going active
replace in progress ...done
failover activity complete
commit  in progress (not interruptible)...done
done

Skipping symrdf update process ....
A reboot Control Station request was sent to Site A to clean up old processes
....


SiteB to SiteC failover case
[root@CS_C rdfadmin]# /nas/sbin/nas_rdf -activate
-skip_rdf_operations -nocheck
Skipping Site A shutdown process ....

For Site B to Site C failover or Site C to Site B failover, nas_rdf -restore
-skip_rdf_operations -skip_SiteA_shutdown and reboot -f -n operations must be
done on the source side Control Station (with read write backend) to clean up
old processes before continue this activate operation unless the source side
is not reachable or destroyed.
Do you wish to continue? [yes or no]: yes

Successfully pinged (Remotely) Symmetrix ID: 000194900431

Successfully pinged (Remotely) Symmetrix ID: 000194900546

Successfully pinged (Remotely) Symmetrix ID: 000194900673
Skipping symrdf failover process ....
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done
fsck 1.39 (29-May-2006)
/dev/ndj1: clean, 14717/252928 files, 279439/516080 blocks
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done
server_2 :
server_2 : going standby
rdf : going active
replace in progress ...done
failover activity complete
commit  in progress (not interruptible)...done
done

Skipping symrdf update process ....
A reboot Control Station request was sent to 10.245.64.168 to clean up old
processes ....
```

EXAMPLE # 11

To initiate an SRDF failover from the source VNX to the destination, without the SRDF health check for the case SiteA Data Movers are already shutdown and the Control Station is already rebooted, type:

```
# /nas/sbin/nas_rdf -activate -skip_SiteA_shutdown -nocheck

Skipping SRDF health check ....
Skipping Site A shutdown process ....

This skip_SiteA_shutdown option is only for the case the Site A Data Movers
have been already shutdown and the Site A Control Station has been already
rebooted to clean up old processes.
Do you wish to continue? [yes or no]: yes

Successfully pinged (Remotely) Symmetrix ID: 000194900431

Successfully pinged (Remotely) Symmetrix ID: 000194900462

Successfully pinged (Remotely) Symmetrix ID: 000194900673
    Write Disable device(s) on SA at source (R1).............Done.
    Suspend RDF link(s).....................................Done.
    Read/Write Enable device(s) on RA at target (R2).........Done.
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done
fsck 1.39 (29-May-2006)
/dev/ndj1: recovering journal
/dev/ndj1: clean, 14237/252928 files, 297432/516080 blocks
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done
id      type  acl  slot groupID  state  name
1        4    2000  2             0    server_2
2        1    1000  3             0    server_3

server_3 :
server_3 : going offline
rdf : going active
replace in progress ...done
failover activity complete
commit  in progress (not interruptible)...done
done

    Suspend RDF link(s).....................................Done.
    Merge device track tables between source and target.....Started.
    Devices: 0078-0078 in (0546,011)........................Merged.
    Merge device track tables between source and target.....Done.
    Resume RDF link(s)......................................Started.
    Resume RDF link(s)......................................Done.
A shutdown request was sent to Site A to clean up old processes ....
```

EXAMPLE #12

To restore a source VNX after failover for the following use cases, as a nasadmin su to root user, type:

```
# /nas/sbin/nas_rdf -restore -skip_rdf_operations

SRDF STAR concurrent or cascaded
SRDF concurrent or cascaded
SRDF R2 enable (Split)
Restore on SiteB/SiteC
Skipping session check  ....

Is remote site CELERRA ready for Storage restoration?
Do you wish to continue? [yes or no]: yes
Contact eng564169 ... is alive

Restore will now reboot the source site control station. This process may take
```

```
several minutes.
Do you wish to continue? [yes or no]: yes
Halting SiteA Data Movers and rebooting SiteA Control Station ....
Checking SiteA Data Mover halt status ....
Skipping symrdf update operation ....

Is remote site CELERRA ready for Network restoration?
Do you wish to continue? [yes or no]: yes
server_2 : done
server_3 :
Error 4003: server_3 : standby is not configured
fsck 1.39 (29-May-2006)
/dev/ndj1: clean, 14716/252928 files, 279441/516080 blocks
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done
Skipping symrdf failback operation & Site A restore ....

Restore on SiteA
To restore on siteA as a nasadmin su to root user, type:
[root@CS_A nasadmin]# /nasmcd/sbin/nas_rdf -restore
-skip_rdf_operations
Waiting for NAS services to finish starting..... Done

Ensure that SiteA is currently write-enabled to continue this restore
operation.

Do you wish to continue? [yes or no]: yes
Waiting for nbs clients to start ... done
Waiting for nbs clients to start ... done
server_2 :
server_2 : going standby
rdf : going active
replace in progress ...done
failover activity complete
commit  in progress (not interruptible)...done
done

server_3 :
Error 4003: server_3 : standby is not configured
Skipping symrdf set async operation ....
Run 'nas_diskmark -mark -all' on all Control Stations in the SRDF
configuration to make sure the SRDF configuration and nasdb are restored
completely.
Starting Services ...done

EXAMPLE #13
------------

To disable SiteB for failover from SiteB to SiteC, as a rdfadmin su to root
user, type:

# /nas/sbin/nas_rdf -restore -skip_rdf_operations -skip_SiteA_shutdown

Skipping session check  ....
Skipping Site A shutdown process ....
Skipping symrdf update operation ....

Is remote site CELERRA ready for Network restoration?
Do you wish to continue? [yes or no]: yes
server_2 : done

server_3 :
Error 4003: server_3 : standby is not configured
fsck 1.39 (29-May-2006)
/dev/ndj1: clean, 14717/252928 files, 279439/516080 blocks
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done
Waiting for nbs clients to die ... done
Waiting for nbs clients to start ... done
```

Skipping symrdf failback operation & Site A restore ....

Last Modified: May 28, 2012 3:45 p.m.

nas_replicate

Manages loopback, local, and remote VNX Replicator sessions.

SYNOPSIS

```
nas_replicate
          -list [-id]
        | -info {-all|id=<sessionId>|<name>}
        | -create <name>
              -source -fs {<fsName>|id=<fsId>}
                [-sav {<srcSavVolStoragePool>|id=<srcSavVolStoragePoolId>}
                [-storageSystem <srcSavStorageSerialNumber>]]
              -destination {-fs {id=<dstFsId>|<existing_dstFsName>}
               | -pool {id=<dstStoragePoolId>|<dstStoragePool>}
                [-storageSystem <dstStorageSerialNumber> ] }
                [-vdm <dstVdmName>]}
                [-sav {id=<dstSavVolStoragePoolId>|<dstSavVolStoragePool>}
                [-storageSystem <dstSavStorageSerialNumber> ] ]
              -interconnect {<name>|id=<interConnectId>}
                [-source_interface {ip=<ipAddr>|<nameServiceInterfaceName>}]
                [-destination_interface {ip=<ipAddr>|<nameServiceInterfaceName>
}]
                [{-max_time_out_of_sync <maxTimeOutOfSync>|-manual_refresh}]
                [-overwrite_destination][-tape_copy][-background]
        | -create <name>
              -source -vdm <vdmName>
              -destination {-vdm <existing_dstVdmName>|-pool
                {id=<dstStoragePoolId>|<dstStoragePool>}[-storageSystem
                <dstStorageSerialNumber> ]}
              -interconnect {<name>|id=<interConnectId>}
                [-source_interface {ip=<ipAddr>|<nameServiceInterfaceName>}]
                [-destination_interface {ip=<ipAddr>|<nameServiceInterfaceName>
}]
                [{-max_time_out_of_sync <maxTimeOutOfSync>|-manual_refresh}]
                [-overwrite_destination][-background]
        | -start {<name>|id=<sessionId>}
                [-interconnect {<name>|id=<interConnectId>}]
                [-source_interface {ip=<ipAddr>|<nameServiceInterfaceName>}]
                [-destination_interface {ip=<ipAddr>|<nameServiceInterfaceName>
}]
                [{-max_time_out_of_sync <maxTimeOutOfSync>|-manual_refresh}]
                [-overwrite_destination][-reverse][-full_copy][-background]
        | -modify {<name>|id=<sessionId>} [-name <new name>]
                [-source_interface {ip=<ipAddr>|<nameServiceInterfaceName>}]
                [-destination_interface {ip=<ipAddr>|<nameServiceInterfaceName>
}]
                [{-max_time_out_of_sync <maxTimeOutOfSync>|-manual_refresh}]
        | -stop {<name>|id=<sessionId>} [-mode {source|destination|both}]
                [-background]
        | -delete {<name>|id=<sessionId>} [-mode {source|destination|both}]
                [-background]
        | -failover {<name>|id=<sessionId>} [-background]
        | -switchover {<name>|id=<sessionId>}
        | -reverse {<name>|id=<sessionId>} [-background]
        | -refresh {<name>|id=<sessionId>} [-source {<ckptName>|id=<ckptId>}
                 -destination {<ckptName>|id=<ckptId>}] [-background]
```

Note: This command manages replication sessions using VNX Replicator. For a one-time file system copy using VNX Replicator, use the nas_copy command. For ongoing file system copy, use the nas_replicate command.

DESCRIPTION
----

nas_replicate creates, manages, or displays session information for ongoing VNX Replicator replication of a file system, Virtual Data Mover (VDM) at a destination using an existing Data Mover interconnect. Each session handles a single source object and destination, and is assigned a globally unique ID, fixed for the life of the session.

In response to a potential disaster scenario, use nas_replicate to perform a failover of a specified replication session with possible data loss. The -switchover option switches over a replication relationship and performs synchronization of the source and destination without data loss. Use nas_replicate to also reverse the direction of a replication session or refresh the destination side with updates to the source based on a time-out of synchronization value or on demand.

OPTIONS

-list [-id]
Displays all configured (or stopped) replication sessions on each Data Mover in the VNX for file cabinet. Each session is represented by either a name or a session ID that is generated automatically whenever a session is configured and is globally unique.

Use this option to obtain the session ID needed for another command. Since session IDs are lengthy, the session ID obtained from this command can be copied and pasted into the command.

-info {-all|id=<sessionId>|<name>}
Displays the status of a specific configured (or stopped) replication session or copy session, or the status of all replication sessions.

CREATING FILE SYSTEM REPLICATION

-create <name>
Assigns a name to the file system replication session. The name must be unique for each Data Mover pair, which is defined by the interconnect.

-source -fs {<fsName>|id=<fsId>} [-sav<srcSavVolStoragePool>| id=<srcSavVolStoragePoolId>}[-storageSystem<srcSavStorageSerialNumber>]]

Specifies the name or ID of the existing source file system to replicate.

The source file system must be mounted as read-only or read and write.

Note: If the source file system is mounted to a VDM and the goal is to replicate a CIFS environment for disaster recovery (that is, replicate a VDM and the file systems mounted to the VDM), create a session to replicate the VDM first, before replicating a file system mounted to the VDM.

The -sav option allocates a storage pool for all subsequent checkpoints for the file system. By default, if checkpoint storage (the checkpoint SavVol) needs to be allocated for checkpoints of the file system, the command uses the same storage pool used to create the source file system.

The -storageSystem option identifies the storage system on which all subsequent checkpoionts for the source file system resides. For RAID group-based pools, specifies the backend storage system when there are multiple systems attached. For mapped pools, specify the pool ID or the pool ID and storage system serial number to uniquely identify a pool.

-destination {-fs {<existing_dstFsName>| id=<dstFsId>}| -pool {<dstStoragePool>| id=<dstStoragePoolId>}[-storageSystem <dstStorageSerialNumber>]}
Specifies an existing destination file system or the storage needed to create the destination file system. An existing destination file system must be mounted as read-only and the same size as the source. Specifying a storage pool or ID creates the read-only, destination file system automatically, using the same name and size as the source file system.

The -storageSystem option identifies the storage system on which the destination file system will reside. This is necessary when there are

multiple back-end systems attached. Use nas_storage -list to obtain attached storage system serial numbers.

-vdm <dstVdmName>]} [-sav {id=<dstSavVolStoragePoolId>|<dstSavVolStoragePool>}[-storageSystem <dstStorageSerialNumber>]
Specifying a pool with the -vdm option mounts the destination file system to an existing VDM as part of replication in a CIFS environment.

The -sav option allocates a storage pool for all subsequent checkpoints of the destination file system. By default, if destination checkpoint storage needs to be allocated for checkpoints, the command uses the same storage pool used to create the destination file system. The -storageSyst
em
option identifies the storage system on which the destination checkpoint will reside. This is necessary when there are multiple back-end systems attached. Use nas_storage -list to obtain attached storage system serial numbers.

By default, the destination file system name will be the same as the sour
ce
file system name. If a file system with the same name as the source file system already exists on the destination, the naming convention <source_fs_name>_replica<#> will be used. A number 1-4 is assigned accord
ing
to how many replicas of that file system already exist.

-interconnect {<name>|id=<interConnectId>}
Specifies the local (source) side of an established Data Mover interconnect to use for this replication session.

Use the nas_cel -interconnect -list command on the source VNX for file to list the interconnects available to the replication sessions.

[-source_interface{<nameServiceInterfaceName>|ip=<ipAddr>}]
Instructs the replication session to use a specific local interface defined for the interconnect on the source VNX instead of selecting the local interface supporting the lowest number of sessions (the default). If this local interface was defined for the interconnect using a name service interface name, specify the name service interface name; if it was defined using an IP address, specify the IP address. If you define an interface using an IP address, make sure that the destination interface uses the same IPv4/IPv6 protocol. An IPv4 interface cannot connect to an IPv6 interface and vice versa. Both sides of the connection must use the same protocol.

The source_interfaces field of the output from the nas_cel -interconnect -info command shows how the source interface was defined. This option does not apply to a loopback interconnect, which always uses 127.0.0.1.

If no source interface is specified, the system will select an interface. This ensures that the interface selected can communicate with the destination interface.

[-destination_interface{<nameServiceInterfaceName>|ip=<ipaddr>}]
Instructs the replication session to use a specific peer interface defined for the interconnect on the destination VNX instead of selecting the peer interface supporting the lowest number of sessions (the default). If this peer interface was defined for the interconnect using a name service interface name, specify the name service interface name; if it was defined using an IP address, specify the IP address. If you define an interface using an IP address, make sure that the source interface uses the same IPv4/IPv6 protocol. An IPv4 interface cannot connect to an IPv6 interface and vice versa. Both sides of the connection must use the same protocol.

The destination_interfaces field of the output from the nas_cel -interconnect -info command shows how the peer

interface was defined. This option does not apply to a loopback interconnect, which always uses 127.0.0.1.

If no destination interface is specified, the system will select an interface. This ensures that the interface selected can communicate with the source interface.

[{-max_time_out_of_sync <maxTimeOutOfSync>|
-manual_refresh}]
Specifies the time, in 1.1440 minutes (up to 24 hours), that the source and destination can be out of synchronization before an update occurs. If you do not specify a max_time_out_of_sync value, use the -manual_refresh option to indicate that the destination will be updated on demand using the nas_replicate -refresh command. If no option is selected, the refresh default time for a file system replication is 10 minutes.

[-overwrite_destination]
For an existing destination object, discards any changes made to the destination object and restores it from the established common base, thereby starting the replication session from a differential copy. If this option is not specified and the destination object contains different content than the established common base, an error is returned.

[-tape_copy]
For file system replication only, creates and stops the replication session to enable an initial copy using the physical tape backup and process instead of an initial copy over the network. Using VNX Replicator describes the procedures for performing a tape copy, which involves a manually issued backup to tape from the source file system, a restore from tape to the destination file system, and a start of the replication session.

[-background]
Executes the command in asynchronous mode. Use the nas_task command to check the status of the command.

CREATING VDM REPLICATION

-create <name>
Assigns a name to the VDM replication session. The name must be unique for each Data Mover pair, which is defined by the interconnect.

-source -vdm <vdmName>|[id=<VdmId>]
Specifies the name or ID of an existing VDM to replicate. This replicates the CIFS working environment information contained in the root file system of the VDM. The source VDM must be in a loaded read/write or mounted read-only state. The source VDM can be the source or destination VDM of another replication session.

Note: Any file system mounted to a VDM must be replicated using file system replication. VDM replication affects the VDM only.

-destination {-vdm {<existing_dstVdmName>| id=<dstVdmId>|-pool {id=<dstStoragePoolId>| <dstStoragePool>}[-storageSystem <dstStorageSerialNumber>]}
Specifies either an existing destination VDM or the storage needed to create the destination VDM. An existing destination VDM must be mounted as read-only, the same size as the source, and not loaded. The destinatio
n
VDM can be the source of another replication but cannot be the destinatio
n
of another replication. Specifying a storage pool creates the destination
VDM
automatically, as read-only, using the same name and size as the source V
DM.

The -storageSystem option identifies the storage system on which the destination VDM will reside. This is necessary when there are multiple back-end systems attached. Use nas_storage -list to obtain attached stora
ge
system serial numbers.

-interconnect {<name>|id=<interConnectId>}
Specifies the local (source) side of an established Data Mover interconne
ct
to use for this replication session.

Use the nas_cel -interconnect -list command on the source VNX to list the interconnects available to replication sessions. The nas_cel -interconnect -create command is executed twice, one from each side, to create an interconnect between a pair of Data Movers (two local Data Movers for local replication, or one local and one remote, for remote replication). Loopback interconnects are created for each Data Mover and are named automatically.

[-source_interface {<nameServiceInterfaceName>|ip=<ipAddr>}]
Instructs the replication session to use a specific local interface defined for the interconnect on the source VNX instead of selecting the local interface supporting the lowest number of sessions (the default). If this local interface was defined for the interconnect using a name service interface name, specify the name service interface name; if it was defined using an IP address, specify the IP address. If you define an interface using an IP address, make sure that the destination interface uses the same IPv4/IPv6 protocol. An IPv4 interface cannot connect to an IPv6 interface and vice versa. Both sides of the connection must use the same protocol.

The source_interfaces field of the output from the nas_cel -interconnect -info command shows how the source interface was defined. This option does not apply to a loopback interconnect, which always uses 127.0.0.1.

If no source interface is specified, the system will select an interface. This ensures that the interface selected can communicate with the destination interface.

[-destination_interface{<nameServiceInterfaceName>|ip=<ipaddr>}]
Instructs the replication session to use a specific peer interface defined for the interconnect on the destination VNX instead of selecting the peer interface supporting the lowest number of sessions (the default). If this peer interface was defined for the interconnect using a name service interface name, specify the name service interface name; if it was defined using an IP address, specify the IP address. If you define an interface using an IP address, make sure that the source interface uses the same IPv4/IPv6 protocol. An IPv4 interface cannot connect to an IPv6 interface and vice versa. Both sides of the connection must use the same protocol.

The destination_interfaces field of the output from the nas_cel -interconnect -info command shows how the peer interface was defined. This option does not apply to a loopback interconnect, which always uses 127.0.0.1.

If no destination interface is specified, the system will select an interface. This ensures that the interface selected can communicate with the source interface.

[{-max_time_out_of_sync <maxTimeOutOfSync>|-manual_refresh}]
Specifies the time, in 1.1440 minutes (up to 24 hours), that the source and destination can be out of synchronization before an update occurs. If you do not specify a max_time_out_of_sync value, use the -manual_refresh option to indicate that the destination will be updated on demand using the nas_replicate -refresh command. If no option is selected, the refresh default

time for a VDM replication is 5 minutes.

[-overwrite_destination]
For an existing destination object, discards any changes made to the destination object and restores it from the established common base, thereby starting the replication session from a differential copy. If this option is not specified, and the destination object contains different content than the established common base, an error is returned.

[-background]
Executes the command in asynchronous mode. Use the nas_task command to check the status of the command.

START OPTIONS

-start {<name>|id=<sessionId>}
From the source side only, specifies the name or session ID needed to start the replication session. A replication name is unique for each Data Mover pair; if a duplicate name is detected on the system, the session ID is required. To get the session ID, use nas_replicate -list.

[-interconnect {<name>|id=<interConnectId>}]
Specifies an established source-side (local) Data Mover interconnect to use for the replication session. Use the nas_cel -interconnect -list command to list the interconnects available to replication sessions. The nas_cel -interconnect -create command creates an interconnect between a pair of Data Movers (two local Data Movers for local replication, or one local and one remote, for remote replication). Loopback interconnects are created and named automatically, and always use IP address 127.0.0.1.

[-source_interface {<nameServiceInterfaceName>|ip=<ipaddr>}]
As the source interface for the replication session, uses a specific local interface defined for the interconnect instead of any local interface defined for the interconnect (the default, which enables the software to select the interface supporting the lowest number of sessions). If this interface was defined for the interconnect using a name service interface name, specify the name service interface name; if it was defined using an IP address, specify the IP address (IPv4 or IPv6). If you define an interface using an IP address, make sure that the destination interface uses the same IPv4/IPv6 protocol. An IPv4 interface cannot connect to an IPv6 interface and vice versa. Both sides of the connection must use the same protocol.

[-destination_interface{<nameServiceInterfaceName>|ip=<ipaddr>}]
As the destination interface for the replication session, uses a specific peer interface defined for the interconnect instead of any peer interface defined for the interconnect (the default, which enables the software to select the interface supporting the lowest number of sessions). If this interface was defined for the interconnect using a name service interface name, specify the name service interface name; if it was defined using an IP address, specify the IP address (IPv4 or IPv6). If you define an interface using an IP address, make sure that the source interface uses the same IPv4/IPv6 protocol. An IPv4 interface cannot connect to an IPv6 interface and vice versa. Both sides of the connection must use the same protocol.

[{-max_time_out_of_sync <maxtimeOutOfSync>|-manual_refresh}]
Specifies the time, in 1.1440 minutes (up to 24 hours), that the source and destination can be out of synchronization before an update occurs. If you do not specify a max_time_out_of_sync value, use the -manual_refresh option to indicate that the destination will be updated on demand using the nas_replicate -refresh command. If no option is selected, the refresh default time for file system replication is 10 minutes, and 5 minutes for VDM replication sessions.

[-overwrite_destination]
For an existing destination object, discards any changes made to the destination object and restores the destination object from the established, internal common base checkpoint, thereby starting the replication session from a differential copy. If this option is not specified and the destination object has different content than the established common base, an error is returned.

[-reverse]
Reverses the direction of the replication session when invoked from the new source side (the original destination). A reverse operation continues to use the established replication name or replication session ID. Use this option to restart replication after a failover or switchover.

[-full_copy]
For an existing destination object that contains content changes, performs a full copy of the source object to the destination object. If replication cannot be started from a differential copy using the -overwrite_destination option, omitting this option causes the command to return an error.

[-background]
Executes the command in asynchronous mode. Use the nas_task command to check the status of the command.

MODIFY OPTIONS

-modify {<name>|id=<sessionId>}
From the source side only, specifies the name or session ID of the replication session to be modified. If a duplicate name is detected on the system, the session ID (fixed for the life of the session) is required. To get the session ID, use nas_replicate -list.

Note: A session cannot be modified if a -stop, -delete, -reverse, -failover, -switchover, -create, or -start operation is running. However, once a -modify operation is running, any other operation is permitted.

[-name <newName>]
Renames the replication session to the new name specified. When renaming a session, note that the name must be unique for each Data Mover pair.

[-source_interface {<nameServiceInterfaceName>|ip=<ipaddr>}]
Changes the source interface used for the session to another local interface from the list defined for the interconnect. If this interface was defined for the interconnect using a name service interface name, specify the name service interface name; if it was defined using an IP address, specify the IP address (IPv4 or IPv6). If you change an IP address, make sure that the destination interface uses the same IPv4/IPv6 protocol. An IPv4 interface cannot connect to an IPv6 interface and vice versa. Both sides of the connection must use the same protocol.

[-destination_interface{<nameServiceInterfaceName>|ip=<ipaddr>}]
Changes the destination interface used for the session to another peer interface from the list defined for the interconnect. If this interface was defined for the interconnect using a name service interface name, specify the name service interface name; if it was defined using an IP address, specify the IP addres (IPv4 or IPv6).

If you change an IP address, make sure that the source interface uses the same IPv4/IPv6 protocol. An IPv4 interface cannot connect to an IPv6 interface and vice versa. Both sides of the connection must use the same protocol.

[-max_time_out_of_sync <maxTimeOutOfSync>|-manual_refresh]
Specifies the time, from 1.1440 minutes (up to 24 hours), that the source and destination can be out of synchronization before an

update occurs. If you do not specify a max_time_out_of_sync value, use the -manual_refresh option to indicate that the destination will be updated on demand using the nas_replicate -refresh command. If no option is selected, the refresh default time for file system replication is 10 minutes,and 5 minutes for VDM replication sessions.

STOP OPTIONS

-stop {<name>|id=<session_id>}
Executed from the Control Station on the source VNX, stops the specified replication session but retains the session.s configuration information. Any data transfer in progress is terminated immediately and the destination object is restored to a consistent state.

Note: A session cannot be stopped if the -delete option is already running for the session. Once a stop operation is in progress, only the options -list, -info, and the nas_task command are permitted.

[-mode {source|destination|both}]
When stopping a session handling a local or remote replication from the source side, the -mode both option immediately stops both sides of the replication session. The -mode source option stops only the replication session on the source and ignores the other side of the replication relationship. If the destination side is not operational, the -mode source option is required to stop the session. From the destination side, only the -mode destination option can be issued. When stopping a session handling a loopback replication, you can specify any -mode option to stop the session.

[-background]
Executes the command in asynchronous mode. Use the nas_task command to check progress.

DELETE OPTIONS

-delete {<name>|id=<session_id>}
Executed from the Control Station on the source VNX, cancels replication data transfer if it is in progress, performs an internal checkpoint restore of the latest destination checkpoint to bring the file system back to a consistent state and then deletes the replication session specified by the -mode options.

[-mode {source|destination|both}]
When deleting a local or remote replication session from the source side, the -mode both option deletes both sides of the replication session. The -mode source option immediately aborts only the replication session on the source and ignores the other side of the replication relationship. If the destination side is not operational, the -mode source option is required to delete the session. From the destination side, only the -mode destination option can be issued. When deleting a loopback replication, you can specify any -mode option to stop the session.

[-background]
Executes the command in asynchronous mode. Use the nas_task command to check progress.

The execution of the -delete option is asynchronous and can be delayed if there is a network problem. During the delete process, other operations on the replication session are not allowed.

FAILOVER OPTIONS

-failover {<name>|id=<session_id>}
In response to a potential disaster scenario, performs a failover of the specified replication session with possible data loss. Execute this

command from the Control Station on the destination VNX only. This command cancels any data transfer that is in process and marks the destination object as read-write so that it can serve as the new source object. When the original source Data Mover becomes reachable, the source object is changed to read-only.

CAUTION: The execution of the failover operation is asynchronous and results in data loss if all the data was not transferred to the destination site prior to issuing the failover.

If there are multiple sessions using the same source object, only one replication session can be failed over. After the selected session is failed over, the other sessions become inactive until the session is restarted or failed back.

[-background]
Executes the command in asynchronous mode. Use the nas_task command to check progress.

SWITCHOVER OPTIONS

-switchover {<name>|id=<session_id>}
For test or migration purposes, switches over the specified replication relationship and performs synchronization of the source and destination without data loss. Execute this command from the Control Station on the source VNX only. This command stops replication, mounts the source object as read-only, and marks the destination object as read-write so that it can act as the new source object.

Unlike a reverse operation, a switchover operation does not restart replication.

[-background]
Executes the command in asynchronous mode. Use the nas_task command to check progress.

REVERSE OPTIONS

-reverse {<name>|id=<session_id>}
If executed from the source side of a replication session, reverses the direction of the specified replication session without data loss. A reverse synchronizes the destination with the source, mounts the source object as read-only, stops replication, marks the destination object as read-write so that it can act as the new source object, then restarts replication in the reverse direction from a differential copy (using the same configuration parameters established originally for the session).

[-background]
Executes the command in asynchronous mode. Use the nas_task command to check progress.

REFRESH OPTIONS

REFRESH OPTIONS
-refresh{<name>|id=<session_id>}
Updates the destination side of the specified replication session based on changes to the source side. Execute this command from the Control Station on the source side only. A refresh operation handles updates on demand; as an alternative, the -max_time_out_of_sync option performs an update automatically after a specified number of minutes.

If the data changes on the source are large, this command can take a long time to complete. Consider running this command in background mode.

[-source{<ckptName>|id=<ckptId>} -destination{<ckptName>|id=<ckptId>}]

Instructs the replication -refresh option to use a specific checkpoint on the source side and a specific checkpoint on the destination side. Specifying source and destination checkpoints for the -refresh option is optional. However, if you specify a source checkpoint, you must also specify a destination checkpoint. Replication transfers the contents of the user-specified source checkpoint to the destination file system. This transfer can be either a full copy or a differential copy depending on the existing replication semantics. After the transfer, the replication internally refreshes the user specified destination checkpoint and marks the two checkpoints as common bases.

After the replication refresh operation completes successfully, both the source and destination checkpoints have the same view of their file systems. The replication continues to use these checkpoints as common bases until the next transfer is completed. After a user checkpoint is marked with a common base property, the property is retained until the checkpoint is refreshed or deleted. A checkpoint that is already paired as a common base with another checkpoint propagates its common base property when it is specified as the source in a replication refresh operation. This propagation makes it possible for file systems without a direct replication relationship to have common base checkpoints.

[-background]
Executes the command in asynchronous mode. Use the nas_task command to check progress.

STORAGE SYSTEM OUTPUT

The number associated with the storage device is dependent on the attached storage system. VNX for block displays a prefix of APM before a set of integers, for example, APM00033900124-0019. For example, Symmetrix storage systems appear as 002804000190-003C. The outputs displayed in the examples use a VNX for block.

EXAMPLE #1

To list all the VNX Replicator sessions, type:

$ nas_replicate -list

Name Type Local Mover Interconnect Celerra Status

ufs1_rep1 filesystem server_3 -->NYs3_LAs2 cs110 OK

vdm1_rep1 vdm server_3 -->NYs3_LAs2 cs110 OK

Where:

Value	Definition
Name	Either the name of the session or the globally unique session ID for the session, if there are duplicate names on the system.
Type	The type of replication session (ongoing file system (fs), copy, or VDM).
Source Mover	The source Data Mover for the session.
Interconnect	The name of the source-side interconnect used for the session.
Celerra	The name of the VNX system.
Status	The status of the session (OK, Active, Idle, Stopped, Error, Waiting) Info, Critical.

EXAMPLE #2

To create a file system replication session ufs1_rep1 on the source file

system ufs1 and destination pool clar_r5_performance on the
interconnect NYs3_LAs2 using the specified source and destination
IP addresses to be updated automatically every 5 minutes, type:

```
$ nas_replicate -create ufs1_rep1 -source -fs ufs1 -destination -pool clar_r5_per
formance
-interconnect NYs3_LAs2 -source_interface ip=10.6.3.190 -destination_interface ip
=10.6.3.173
-max_time_out_of_sync 5

OK
```

EXAMPLE #3

To display information for a replication session ufs1_rep1, type:

```
$ nas_replicate -info ufs1_rep1

ID = 184_APM00064600086_0000_173_APM00072901601_0000
Name = ufs1_rep1
Source Status = OK
Network Status = OK
Destination Status = OK
Last Sync Time = Thu Dec 13 14:47:16 EST 2007
Type = filesystem
Celerra Network Server = cs110
Dart Interconnect = NYs3_LAs2
Peer Dart Interconnect = 20004
Replication Role = source
Source Filesystem = ufs1
Source Data Mover = server_3
Source Interface = 10.6.3.190
Source Control Port = 0
Source Current Data Port = 0
Destination Filesystem = ufs1_replica3
Destination Data Mover = server_2
Destination Interface = 10.6.3.173
Destination Control Port = 5081
Destination Data Port = 8888
Max Out of Sync Time (minutes) = 5
Next Transfer Size (Kb) = 0
Latest Snap on Source =
Latest Snap on Destination =
Current Transfer Size (KB) = 0
Current Transfer Remain (KB) = 0
Estimated Completion Time =
Current Transfer is Full Copy = No
Current Transfer Rate (KB/s) = 76
Current Read Rate (KB/s) = 11538
Current Write Rate (KB/s) = 580
Previous Transfer Rate (KB/s) = 0
Previous Read Rate (KB/s) = 0
Previous Write Rate (KB/s) = 0
Average Transfer Rate (KB/s) = 6277
Average Read Rate (KB/s) = 0
Average Write Rate (KB/s) = 0
```

EXAMPLE #4

To create a VDM replication session vdm_rep1 on source VDM vdm1 and destination p
ool
clar_r5_performance on the interconnect NYs3_LAs2 with the given source and desti
nation
IP addresses to be updated automatically every 5 minutes, type:

```
$ nas_replicate -create vdm1_rep1 -source -vdm vdm1 -destination -pool clar_r5_pe
rformance
-interconnect NYs3_LAs2 -source_interface ip=10.6.3.190 -destination_interface ip
=10.6.3.173
```

```
-max_time_out_of_sync 5

OK

EXAMPLE #5
----------

To list existing replication sessions, type:

$ nas_replicate -list

Name              Type   Local Mover   Interconnect    Celerra    Status
ufs1_rep1   filesystem      server_3   -->NYs3_LAs2    cs110          OK
vdm1_rep1          vdm      server_3   -->NYs3_LAs2    cs110          OK

EXAMPLE #6
----------

To manually synchronize source and destination for the replication session
ufs1_rep1, type:

$ nas_replicate -refresh ufs1_rep1

OK

EXAMPLE #7
----------

To manually synchronize source and destination for the replication session
ufs1_rep1 by using user checkpoints on the source and the destination, type:

$ nas_replicate -refresh ufs1_rep1 -source id=101 -destination id=102

OK

EXAMPLE #8
----------

To stop replication on both source and destination for the replication session
ufs1_rep1, type:

$ nas_replicate -stop ufs1_rep1 -mode both

OK

EXAMPLE #9
-----------

To start stopped replication session ufs1_rep1 on interconnect NYs3_LAs2,
specify manual refresh and to overwrite the destination file system performing
a full copy, type:

$ nas_replicate -start ufs1_rep1 -interconnect NYs3_LAs2 -manual_refresh
-overwrite_destination -full_copy

OK

EXAMPLE #10
-----------

To display information for the VDM replication session vdm_rep1, type:

$ nas_replicate -info vdm1_rep1

ID = 278_APM00064600086_0000_180_APM00072901601_0000
Name = vdm1_rep1
Source Status = OK
Network Status = OK
Destination Status = OK
Last Sync Time = Fri Dec 14 16:49:54 EST 2007
Type = vdm
```

```
Celerra Network Server = cs110
Dart Interconnect = NYs3_LAs2
Peer Dart Interconnect = 20004
Replication Role = source
Source VDM = vdm1
Source Data Mover = server_3
Source Interface = 10.6.3.190
Source Control Port = 0
Source Current Data Port = 0
Destination VDM = vdm1
Destination Data Mover = server_2
Destination Interface = 10.6.3.173
Destination Control Port = 5081
Destination Data Port = 8888
Max Out of Sync Time (minutes) = 5
Next Transfer Size (Kb) = 0
Latest Snap on Source =
Latest Snap on Destination =
Current Transfer Size (KB) = 0
Current Transfer Remain (KB) = 0
Estimated Completion Time =
Current Transfer is Full Copy = No
Current Transfer Rate (KB/s) = 313
Current Read Rate (KB/s) = 19297
Current Write Rate (KB/s) = 469
Previous Transfer Rate (KB/s) = 0
Previous Read Rate (KB/s) = 0
Previous Write Rate (KB/s) = 0
Average Transfer Rate (KB/s) = 155
Average Read Rate (KB/s) = 0
Average Write Rate (KB/s) = 0
```

EXAMPLE #11

To change the session name vdm1_rep1 to vdm1_rep2, and to change max time out of sync value to 90, type:

```
$ nas_replicate -modify vdm1_rep1 -name vdm1_rep2 -max_time_out_of_sync 90

OK
```

EXAMPLE #12

To failover the replication session ufs1_rep1, type on destination:

```
$ nas_replicate -failover ufs1_rep1

OK
```

EXAMPLE #13

To start failed over replication in the reverse direction, type:

```
$ nas_replicate -start ufs1_rep1 -interconnect LAs2_NYs3 -reverse -overwrite_dest
ination

OK
```

EXAMPLE #14

To reverse direction of the replication session ufs1_rep1, type:

```
$ nas_replicate -reverse ufs1_rep1

OK
```

EXAMPLE #15

To switch over the replication session ufs1_rep1 using the background option, type:

$ nas_replicate -switchover ufs1_rep1 -background

Info 26843676673: In Progress: Operation is still running. Check task id 4058 on the
Task Status screen for results.

*** Comment: Use nas_task -info command to find out the status of background task.

EXAMPLE #16

To delete the replication session fs1_rep1 on both source and destination, type:

$ nas_replicate -delete fs1_rep1 -mode both

OK

Last modified: Feb 21 2013, 2:34 pm

nas_server

Manages the Data Mover (server) table.

SYNOPSIS

```
nas_server
          -list [-all|-vdm]
        | -delete <movername>
        | -info {-all|<movername>|<slot_number>|id=<mover_id>
                | -vdm {-all|<vdm_name>|id=<vdm_id>}}
        | -rename <old_movername> <new_movername>
        | -acl <acl_value> <movername>
        | [-name <name>][-type <type>] -create <movername> [-setstate <state>]
          [-fs <fs_name>|pool=<pool> [storage=<system_name>]][-option <options>]
        | -vdm <vdm_name> -attach <interface> [,<interface2>...]
        | -vdm <vdm_name> -detach <interface> [,<interface2>...]
        | -vdm <vdm_name> -setstate <state> [<movername>][-ConvertI18N]
        | -vdm <vdm_name> -move <movername> [-ConvertI18N]
```

DESCRIPTION
----

nas_server manages the server tables for both physical and virtual Data Movers (VDMs), creates a VDM, sets an access control value for a physical Data Mover or VDM, renames a Data Mover and displays attributes for a specified Data Mover or all Data Movers, deletes a physical Data Mover entry from the server table, deletes the VDM configuration for a Data Mover, and attaches or detaches the network interface to a VDM.

OPTIONS

-list
Lists the Data Mover server table. The server table displays the ID, type, access control level value, slot number, group ID, state, and name of a Data Mover. VDMs have their own server table and do not have a numeric reference in the general server table.

Note: The ID of the object is an integer and is assigned automatically. The name of the Data Mover might be truncated if it is too long for the display. To display the full name, use the -info option with the Data Mover ID.

Valid Data Mover types are:

1=nas
2=not used
3=not used
4=standby
5=not used
6=rdf

Note: The nas type is set automatically, vdm is set using nas_rp, rdf and standby are set up using server_ssh.

[-all|-vdm]
The -all option displays the physical Data Mover and VDMs. The -vdm option lists the VDMs only.

-delete <movername>
Deletes the specified physical Data Mover entry from the server table or deletes the VDM configuration. A Data Mover that is being deleted cannot contain mounted filesystems.

Deleting a physical Data Mover requires the root command. Use /nas/sbin/rootnas_server to execute a delete.

-info {-all|<movername>|<slot_number>|id=<mover_id>}
Displays attributes for all physical Data Movers, or a Data Mover specified by its <movername>, <slot_number>, or <mover_id>.

-info -vdm {-all|<vdm_name>|id=<vdm_id>}
Displays attributes for all VDMs, or a specified VDM, including the network interfaces that are being used by the CIFS servers.

-rename <old_movername> <new_movername>
Changes the name of the physical Data Mover or the VDM to the specified name. The -create option provides more information for rules applicable to naming a Data Mover.

-acl <acl_value> <movername>
Sets an access control level value that defines the owner of the physical Data Mover or the Virtual Data Mover, and the level of access allowed for users and groups defined in the access control level table. The nas_acl command provides more information.

[-name <name>][-type vdm] -create <movername>
Creates a VDM with an optional name for the specified physical or VDM. The movername is case-sensitive and supports the following characters: a through z, A through Z, 0 through 9, _(underscore), -(hyphen) though names may not start with a hyphen. The default type is nas.

[-setstate <state>]
Sets the Data Mover to loaded or mounted.

The loaded option installs the image of the VDM onto the physical Data Mover, but does not mount the non-root filesystems. The mounted option mounts the root_fs as read-only, but the VDM image is not installed. The -setstate option is for use with replication.

Note: Before a VDM image is loaded, the image must be unloaded from the previous Data Mover, and the CIFS server must be joined using server_cifs. The network interfaces used by the CIFS servers on the VDM must be available on the destination Data Mover.

[-fs <fs_name>|pool=<pool>]
Specifies a filesystem or assigns a rule set known as a storage pool for the VDM root filesystem.

[storage=<system_name>]
For the -fs option, the filesystem must be unmounted, clean (nas_fsck provides more information), and be either of type uxfs or rawfs. For a loaded state VDM, only an uxfs filesystem type can be used, but for mounted state VDM, both uxfs and rawfs can be used. The filesystem to be provided as the VDM root file system is renamed to root_fs_vdm_<vdm_name>. This is deleted when the VDM is deleted.

The storage pool option assigns a rule set for the root filesystem of the VDM that contains automatically created volumes and defines the type of disk volumes used and how they are aggregated. Storage pools are system defined (storage pool description provides more information) or user defined. nas_pool provides a description of pool types.

[-option <options>]
Specifies the following comma-separated list of options:

fstype={rawfs|uxfs}
Specifies the filesystem type of the root file system for the server. It can be either rawfs or uxfs type.

log_type={common|split}
Specifies the type of log file associated with the file system. Log files can
be either shared (common) or uniquely assigned to individual file systems(split). For SRDF Async or STAR feature, split option is strongly

recommended to avoid fsck before mounting a BCV file system on SiteB or SiteC.

-vdm <vdm_name> -attach <interface> [,<interface2>...]
Allows the user to manage the network interface(s) for a VDM. The interfaces are attached to a VDM when the VDM state is loaded. When an interface is attached to a VDM, the NFS clients connecting the Data Mover through this interface have access to the filesystem exported by the VDM configuration.

-vdm <vdm_name> -detach <interface> [,<interface2>...]
An attempt to delete an interface attached to the VDM with the server_ifconfig command fails with an error message. It indicates that the interface is currently used by the VDM <vdm_name>. The user must detach the interface from the VDM before deleting it.

Note: If the user wants to share a VDM interface for both CIFS and NFSv3 or NFSv4 clients, the administrator must create a CIFS server and assign it to the interface by using the server_cifs command.

-vdm <vdm_name> -setstate <state>
Sets the state of the VDM to loaded, mounted, tempunloaded, or permunloaded.

The loaded option installs the image of the VDM onto the physical Data Mover, but does not mount the non-root filesystems. The mounted option mounts the root_fs read-only, but the VDM image is not installed.

The tempunloaded option, temporarily unloads the VDM image, while the permunloaded option permanently unloads the image.

[<movername>]
Specifies a physical Data Mover for the VDM.

[-ConvertI18N]
When loading the VDM image, forces the conversion of the I18N mode of the VDM's root filesystem from ASCII to UNICODE. The I18N mode of the Data Mover can be either ASCII or UNICODE. The mode of the VDM must be the same as the physical Data Mover, for example, when performing the -move option, or when replicating.

This mode is used when the mode of the VDM root filesystem is different from that of the physical Data Mover.

Default states are:

-vdm <vdm_name> -move <movername>
Moves the image of the VDM onto the physical Data Mover, and mounts the non-root filesystems.

Note: Before a VDM image is loaded, the image must be unloaded from the previous Data Mover, and the CIFS server must be joined using server_cifs. The network interfaces used by the CIFS servers on the VDM must be available on the destination Data Mover.

[-ConvertI18N]
When loading the VDM image, forces the conversion of the I18N mode of the VDM's root filesystem from ASCII to UNICODE. The I18N mode of the Data Mover can be either ASCII or UNICODE. The mode of the VDM must be the same as the physical Data Mover, for example, when performing the -move option, or when replicating.

This mode is used when the mode of the VDM root filesystem is different from that of the physical Data Mover.

SEE ALSO

Configuring Virtual Data Mover on VNX, Using International Character Sets for File, nas_fs, nas_volume, and server_cifs.

SYSTEM OUTPUT

VNX systems support the following system-defined storage pools: clar_r1, clar_r5_performance, clar_r5_economy, clar_r6, clarata_r3, clarata_r6, clarata_r10, clarata_archive, cm_r1, cm_r5_performance, cm_r5_economy, cm_r6, cmata_r3, cmata_archive, cmata_r6, cmata_r10, clarsas_archive, clarsas_r6, clarsas_r10, clarefd_r5, clarefd_r10, cmsas_archive, cmsas_r6, cmsas_r10, and cmefd_r5.

Disk types when using VNX for block are CLSTD, CLEFD, and CLATA, and for VNX for block involving mirrored disks are CMEFD, CMSTD, and CMATA.

VNX with a Symmetrix system supports the following system-defined storage pools: symm_std, symm_std_rdf_src, symm_ata, symm_ata_rdf_src, symm_ata_rdf_tgt, symm_std_rdf_tgt, symm_ata_rdf_tgt, symm_std_rdf_tgt, and symm_efd.

For user-defined storage pools, the difference in output is in the disk type. Disk types when using a Symmetrix are STD, R1STD, R2STD, BCV, R1BCV, R2BCV, ATA, R1ATA, R2ATA, BCVA, R1BCA, R2BCA, and EFD.

EXAMPLE #1

To list the physical Data Mover table, type:

```
$ nas_server -list

id      type    acl   slot groupID state name
1       1       1000   2      0      server_2
2       1       1000   3      0      server_3
3       1       1000   4      0      server_4
4       4       1000   5      0      server_5
```

Where:

Value	Definition
id	ID of the Data Mover.
type	Type assigned to Data Mover.
acl	Access control level value assigned to the Data Mover or VDM.
slot	Physical slot in the cabinet where the Data Mover resides.
groupID	ID of the Data Mover group.
state	Whether the Data Mover is enabled=0, disabled=1, failed over=2.
name	Name given to the Data Mover.

EXAMPLE #2

To list the physical Data Mover and VDM table, type:

```
$ nas_server -list -all

id      type    acl     slot  groupID state name
1       1       1000      2     0       server_2
2       1       1000      3     0       server_3
3       1       1000      4     0       server_4
4       4       1000      5     0       server_5

id      acl     server  mountedfs  rootfs name
```

```
3       0       1          31        vdm_1
```

EXAMPLE #1 provides a description of outputs for the physical Data Movers. The following table provides a description of the command output for the VDM table.

Where:

Value	Definition
id	ID of the Data Mover.
acl	Access control level value assigned to the Data Mover or VDM.
server	Server on which the VDM is loaded on.
mountedfs	Filesystems that are mounted on this VDM.
rootfs	ID number of the root file system.
name	Name given to the Data Mover or VDM.

EXAMPLE #3

To list the VDM server table, type:

```
$ nas_server -list -vdm

id      acl     server   mountedfs   rootfs   name
3       0         1                    31     vdm_1
```

EXAMPLE #4

To list information for a Data Mover, type:

```
$ nas_server -info server_2

id = 1
name = server_2
acl = 1000, owner=nasadmin, ID=201
type = nas
slot = 2
member_of =
standby = server_5, policy=auto
status :
defined = enabled
actual = online, ready
```

Where:

Value	Definition
id	ID of the Data Mover
name	Name given to the Data Mover
acl	Access control level value assigned to the Data Mover or VDM.
type	Type assigned to Data Mover
slot	Physical slot in the cabinet where the Data Mover resides.
member_of	Group to which the Data Mover is a member.
standby	If the Data Mover has a local standby associated with it.
status	Whether the Data Mover is enabled or disabled, and whether it is active.

EXAMPLE #5

To display detailed information for all servers, type:

```
$ nas_server -info -all

id = 1
```

```
name = server_2
acl = 1000, owner=nasadmin, ID=201
type = nas
slot = 2
member_of =
standby = server_5, policy=auto
status :
defined = enabled
actual = online, active
id = 2
name = server_3
acl = 1000, owner=nasadmin, ID=201
type = nas
slot = 3
member_of =
standby = server_5, policy=auto
status :
defined = enabled
actual = online, ready
id = 3
name = server_4
acl = 1000, owner=nasadmin, ID=201
type = nas
slot = 4
member_of =
standby = server_5, policy=auto
status :
defined = enabled
actual = online, ready
id = 4
name = server_5
acl = 1000, owner=nasadmin, ID=201
type = standby
slot = 5
member_of =
standbyfor= server_4,server_2,server_3
status :
defined = enabled
actual = online, ready
```

EXAMPLE #4 provides a description of command outputs.

EXAMPLE #6

To display information for all VDMs, type:

```
$ nas_server -info -vdm -all

id = 3
name = vdm_1
acl = 0
type = vdm
server = server_2
rootfs = root_fs_vdm_1
I18N mode = UNICODE
mountedfs =
member_of =
status :
defined = enabled
actual = mounted
Interfaces to services mapping:
```

Where:

Value	Definition
id	ID of the Data Mover.
name	Name of the Data Mover.
acl	Access control level value assigned to the VDM.

```
type                    For VDM server, the type is always VDM.
server                  Server on which the VDM loaded.
rootfs                  Root filesystem of the VDM.
I18N mode               L18N mode of the VDM. I18N mode is either ASCII
                        or UNICODE.
mountedfs               Filesystems that are mounted on this VDM.
member_of               If it is a member of a cluster, then this field will
show
                        the cluster name.
status                  Whether the VDM is enabled or disabled, and whether
it
                        can be loaded ready, loaded active, mounted,
temporarily
                        unloaded, and permanently unloaded.
Interfaces to services
mapping                 List of interfaces that are used for the services
                        configured on this VDM. Currently, only CIFS service
is
                        provided, so this field lists all the interfaces used
in
                        the CIFS servers configured on this VDM.
EXAMPLE #7
----------

To create a mounted VDM named vdm_1 on server_2 using the storage pool,
clar_r5_performance with a rawfs, type:

$ nas_server -name vdm_1 -type vdm -create server_2 -setstate mounted
pool=clar_r5_performance -option fstype=uxfs

id = 3
name = vdm_1
acl = 0
type = vdm
server = server_2
rootfs = root_fs_vdm_1
I18N mode = UNICODE
mountedfs =
member_of =
status :
defined = enabled
actual = mounted
Interfaces to services mapping:

EXAMPLE #6 provides a description of command outputs.

EXAMPLE #8
----------

To set the state of a vdm_1 to mounted, type:

$ nas_server -vdm vdm_1 -setstate mounted

id = 3
name = vdm_1
acl = 0
type = vdm
server = server_2
rootfs = root_fs_vdm_1
I18N mode = UNICODE
mountedfs =
member_of =
status :
   defined = enabled
   actual = mounted
Interfaces to services mapping:

EXAMPLE #6 provides a description of command outputs.

EXAMPLE #9
----------
```

To move the image of vdm_1 onto server_4, type:

```
$ nas_server -vdm vdm_1 -move server_4

id = 3
name = vdm_1
acl = 0
type = vdm
server = server_4
rootfs = root_fs_vdm_1
I18N mode = UNICODE
mountedfs =
member_of =
status :
defined = enabled
actual = loaded, ready
Interfaces to services mapping:
```

EXAMPLE #6 provides a description of command outputs.

EXAMPLE #10

To rename a Data Mover entry from server_2 to dm2, type:

```
$ nas_server -rename server_2 dm2

id = 1
name = dm2
acl = 1000, owner=nasadmin, ID=201
type = nas
slot = 2
member_of =
standby = server_5, policy=auto
status :
defined = enabled
actual = online, active
```

EXAMPLE #4 provides a description of command outputs.

EXAMPLE #11

To set the access control level for server_2, type:

```
$ nas_server -acl 1432 server_2

id = 1
name = server_2
acl = 1432, owner=nasadmin, ID=201
type = nas
slot = 2
member_of =
standby = server_5, policy=auto
status :
defined = enabled
actual = online, ready
```

Note: The value 1432 specifies nasadmin as the owner, gives users with an access level of at least observer read-only access, users with an access level of at least operator read/write access, and users with an access level of at least admin read/write/delete access.

EXAMPLE #4 provides a description of command outputs.

EXAMPLE #12

To delete vdm_1, type:

```
$ nas_server -delete vdm_1

id = 3
name = vdm_1
acl = 0
type = vdm
server =
rootfs = root_fs_vdm_1
I18N mode = UNICODE
mountedfs =
member_of =
status :
defined = enabled
actual = permanently unloaded
Interfaces to services mapping:

EXAMPLE #6 provides a description of command outputs.

EXAMPLE #13
-----------
To delete a physical Data Mover using root command, type:

$ /nas/sbin/rootnas_server -delete server_3

id = 2
name = server_3
acl = 0
type = nas
slot = 3
member_of =
standby = server_5, policy=auto
status :
defined = disabled
actual = boot_level=0

EXAMPLE #6 provides a description of command outputs.

EXAMPLE #14
-----------

To create a VDM named vdm1 on the server 3, type:

$ nas_server -name vdm1 -type vdm -create server_3
id = 43
name = vdm1
acl = 0
type = vdm
server = server_3
rootfs = root_fs_vdm_vdm1
I18N mode = UNICODE
mountedfs =
member_of =
status :
    defined = enabled
    actual = loaded, ready
Interfaces to services mapping:

EXAMPLE #15
-----------

To assign the network interface to vdm1, assuming vdm1if1 and
vdm1if2 exist and are not attached to another vdm, type:

$ nas_server -vdm vdm1 -attach vdm1if1, vdm1if2
id = 43
name = vdm1
acl = 0
type = vdm
server = server_2
rootfs = root_fs_vdm_vdm1
I18N mode = UNICODE
```

```
mountedfs =
member_of =
status :
    defined = enabled
    actual = loaded, ready
    Interfaces to services mapping:
    interface=vdm1if1 :vdm
    interface=vdm1if2 :vdm


EXAMPLE #16
-----------

To query the vdm1 state, type:

$ nas_server -info -vdm vdm1
id = 43
name = vdm1
acl = 0
type = vdm
server = server_2
rootfs = root_fs_vdm_vdm1
I18N mode = UNICODE
mountedfs =
member_of =
status :
    defined = enabled
           actual = loaded, ready
Interfaces to services mapping:
  interface=vdm1if2 :cifs vdm
  interface=vdm1if1 :vdm

EXAMPLE #17
-----------
To create a VDM named vdm2 on the server_3 using split ufs log type, type:
$ nas_server -name vdm2 -type vdm -create server_3 -setstate loaded
pool=symm_std_rdf_src -o log_type=split

id        = 2
name      = vdm2
acl       = 0
type      = vdm
server    = server_3
rootfs    = root_fs_vdm_vdm2
I18N mode = ASCII
mountedfs =
member_of =
status    :
  defined = enabled
   actual = loaded, ready
Interfaces to services mapping:

To confirm a VDM ufs log type, type:
/nas/sbin/rootnas_fs -i root_fs_vdm_vdm2
id        = 49
name      = root_fs_vdm_vdm2
acl       = 0
in_use    = True
type      = uxfs
worm      = off
volume    = v1260
pool      = symm_std_rdf_src
member_of = root_avm_fs_group_8
rw_servers= server_3
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,thin=no
log_type  = split
fast_clone_level = 2
deduplication   = Off
```

```
stor_devs =
000194900462-10C6,000194900462-10CE,000194900462-10D6,000194900462-10DE,
000194900462-10E6,000194900462-10EE,000194900462-10F6,000194900462-10FE
disks     = d1102,d1103,d1104,d1105,d1106,d1107,d1108,d1109
disk=d1102 stor_dev=000194900462-10C6   addr=c4t3l4-72-0    server=server_3
disk=d1102 stor_dev=000194900462-10C6   addr=c20t3l4-71-0   server=server_3
disk=d1102 stor_dev=000194900462-10C6   addr=c36t3l4-71-0   server=server_3
disk=d1102 stor_dev=000194900462-10C6   addr=c52t3l4-72-0   server=server_3
disk=d1103 stor_dev=000194900462-10CE   addr=c4t3l5-72-0    server=server_3
disk=d1103 stor_dev=000194900462-10CE   addr=c20t3l5-71-0   server=server_3
disk=d1103 stor_dev=000194900462-10CE   addr=c36t3l5-71-0   server=server_3
disk=d1103 stor_dev=000194900462-10CE   addr=c52t3l5-72-0   server=server_3
disk=d1104 stor_dev=000194900462-10D6   addr=c4t3l6-72-0    server=server_3
disk=d1104 stor_dev=000194900462-10D6   addr=c20t3l6-71-0   server=server_3
disk=d1104 stor_dev=000194900462-10D6   addr=c36t3l6-71-0   server=server_3
disk=d1104 stor_dev=000194900462-10D6   addr=c52t3l6-72-0   server=server_3
disk=d1105 stor_dev=000194900462-10DE   addr=c4t3l7-72-0    server=server_3
disk=d1105 stor_dev=000194900462-10DE   addr=c20t3l7-71-0   server=server_3
disk=d1105 stor_dev=000194900462-10DE   addr=c36t3l7-71-0   server=server_3
disk=d1105 stor_dev=000194900462-10DE   addr=c52t3l7-72-0   server=server_3
disk=d1106 stor_dev=000194900462-10E6   addr=c4t3l8-72-0    server=server_3
disk=d1106 stor_dev=000194900462-10E6   addr=c20t3l8-71-0   server=server_3
disk=d1106 stor_dev=000194900462-10E6   addr=c36t3l8-71-0   server=server_3
disk=d1106 stor_dev=000194900462-10E6   addr=c52t3l8-72-0   server=server_3
disk=d1107 stor_dev=000194900462-10EE   addr=c4t3l9-72-0    server=server_3
disk=d1107 stor_dev=000194900462-10EE   addr=c20t3l9-71-0   server=server_3
disk=d1107 stor_dev=000194900462-10EE   addr=c36t3l9-71-0   server=server_3
disk=d1107 stor_dev=000194900462-10EE   addr=c52t3l9-72-0   server=server_3
disk=d1108 stor_dev=000194900462-10F6   addr=c4t3l10-72-0   server=server_3
disk=d1108 stor_dev=000194900462-10F6   addr=c20t3l10-71-0  server=server_3
disk=d1108 stor_dev=000194900462-10F6   addr=c36t3l10-71-0  server=server_3
disk=d1108 stor_dev=000194900462-10F6   addr=c52t3l10-72-0  server=server_3
disk=d1109 stor_dev=000194900462-10FE   addr=c4t3l11-72-0   server=server_3
disk=d1109 stor_dev=000194900462-10FE   addr=c20t3l11-71-0  server=server_3
disk=d1109 stor_dev=000194900462-10FE   addr=c36t3l11-71-0  server=server_3
disk=d1109 stor_dev=000194900462-10FE   addr=c52t3l11-72-0  server=server_3
```

Last modified: December 3, 2014 12:20 p.m.

nas_stats

Manages Statistics Groups.

SYNOPSIS

```
nas_stats
-groups
        { -list
        | -info [-all|<statgroup_name>[,...]]
        | -create <statgroup_name>
                [-description "<description_line>"]
                  {<statpath_name>|<statgroup_name>}[,...]
        | -modify <statgroup_name>
                {[-rename <new_statgroup_name>]
                 [-description "<description_line>"]
                 [{<statpath_name>|<statgroup_name>}[,...]]}
        | -add <statgroup_name>
                {<statpath_name>|<statgroup_name>}[,...]
        | -remove <statgroup_name>
                {<statpath_name>|<statgroup_name>}[,...]
        | -delete <statgroup_name> [-Force]
        | -database
                { -recover [-Force]
                 |-verify }
```

DESCRIPTION
----

nas_stats allows the user to manage Statistics Groups. There are two types of Statistics Groups: System-defined and User-defined groups.

System-defined statistics groups are created as part of the installation (or upgrade) process and include the following statistics, which correspond to the summary and table collections used by server_stats:

System-defined statistics group name	server_stats collection name
basic-std	-summary basic
caches-std	-summary caches
cifs-std	-summary cifs
nfs-std	-summary nfs
cifsOps-std	-table cifs
diskVolumes-std	-table dvol
metaVolumes-std	-table fsvol
netDevices-std	-table net
nfsOps-std	-table nfs

Note: server_stats collection names are deprecated and will not be supported in future releases.

Statistics Groups can be created to include any combination of statpath names, displayed through the server_stats command, as well as other existing statgroup names.

Any Statistics Group name can be used with server_stats -monitor to collect statistics as defined in its member_stats list.

OPTIONS

-list
Lists system and user-defined Statistics Groups.

-info
Provides detailed information on all (or specified) Statistics Groups.

-create
Creates a statistics group and specifies what statpath names it includes. It also allows the nesting of statgroups by adding existing statgroups to new statgroups.

Statgroup names can be used with the -info request. A statgroup name is limited to 255 characters. Space, slash, back slash, quote, double quote, and comma are the illegal characters in it.

[-description]
The -description option is optional and defaults to the statgroup name. If the -description option is used, its argument must be enclosed in quotation marks.

-modify
Allows you to modify a statgroup.s member_stats list by specifying the new member statistics of the group, overriding the previous contents.

-add
Allows you to add statpath and existing statgroup names to a statgroup by specifying additional items to be appended to the statgroup.s list member_stats list.

-remove
Allows you to remove member statpath and statgroup names from a statgroup by specifying the items to remove from the statgroup.s member_stats list.

-delete
Allows you to delete a statgroup. However, this option does not delete any statsgroups that are members of the statgroup.

-recover
Attempts to recover the latest uncorrupted copy of the Statistics Groups database from the NAS database backups. nas_stats searches through the available backups and restores the latest copy. In this event, NAS database backups do not contain a healthy version of the Statistics Groups; a new Statistics Groups database is installed. In the case of a new Statistics Groups database, all user-defined information is lost. NAS database backups run hourly and VNX maintains the last 12 backups.

[-Force]
Use the -Force option with the -recover option to skip the warning prompt.

-verify
Checks the health status of the Statistics Groups database.

SEE ALSO
----

server_stats

EXAMPLE #1

To list the system-defined and user-defined Statistics Groups, type:

```
$ nas_stats -groups -list

Type             Name
System           basic-std
System           basicCifs-std
...              ...
User             basic
User             nfsNet
...              ...
```

EXAMPLE #2

To provide detailed information on all (or specified) Statistics Groups, type:

```
$ nas_stats -groups -info
```

```
name = basic-std
description = The basic system-defined group.
type = System-defined
member_stats =
kernel.cpu.utilization.cpuUtil,net.basic.inBytes,net.basic.outBytes,store.readBy
tes,store.writeBytes
member_elements =
member_of =

name = basic3
description = CPU and Memory
type = User-defined
member_stats = kernel.cpu.utilization.cpuUtil,kernel.memory.freeBytes
member_elements =
member_of =

name = caches-std
description = The caches system-defined group.
type = System-defined
member_stats =
fs.dnlc.hitRatio,fs.ofCache.hitRatio,kernel.memory.bufferCache.hitRatio
member_elements =
member_of =

name = cifs-std
description = The cifs system-defined group.
type = System-defined
member_stats =
cifs.global.basic.totalCalls,cifs.global.basic.reads,cifs.global.basic.readBytes
,cifs.global.basic.readAvgSize,cifs.global.basic.writes,cifs.global.basic.writeB
ytes,cifs.global.basic.writeAvgSize,cifs.global.usage.currentConnections,cifs.gl
obal.usage.currentOpenFiles
member_elements =
member_of = newSG

name = cifsOps-std
description = The cifs table system-defined group.
type = System-defined
member_stats = cifs.smb1.op,cifs.smb2.op
member_elements =
member_of =

name = diskVolumes-std
description = The disk volume table system-defined group.
type = System-defined
member_stats = store.diskVolume
member_elements =

name = metaVolumes-std
description = The meta volume table system-defined group.
type = System-defined
member_stats = store.logicalVolume.metaVolume
member_elements =
member_of =

name = netDevices-std
description = The net table system-defined group.
type = System-defined
member_stats = net.device
member_elements =
member_of =

name = newSG
description = newSG
type = User-defined
member_stats = cifs-std,nfs.v3.op,nfs.v4.op
member_elements =
member_of =

name = nfs-std
```

```
Description = The nfs system-defined group.
type = System-defined
member_stats =
nfs.totalCalls,nfs.basic.reads,nfs.basic.readBytes,nfs.basic.readAvgSize,nfs.bas
ic.writes,nfs.basic.writeBytes,nfs.basic.writeAvgSize,nfs.currentThreads
member_elements =
member_of =

name = nfsOps-std
description = The nfs table system-defined group.
type = System-defined
member_stats = nfs.v2.op,nfs.v3.op,nfs.v4.op
member_elements =
member_of =

name = statgroup1
description = My first group
type = User-defined
member_stats =
net.basic.inBytes,net.basic.outBytes,store.readBytes,store.writeBytes
member_elements =
member_of = statgroup2

name = statgroup2
description = My first group
type = User-defined
member_stats =
net.basic.inBytes,net.basic.outBytes,store.readBytes,store.writeBytes,kernel.cpu
.utilization.cpuUtil,statgroup1
member_elements =
member_of =

EXAMPLE #3
----------

To provide detailed information on all (or specified) Statistics Groups, type:

$ nas_stats -groups -info statsA

name = statsA
description = My group # 2
type = user-defined
member_stats = statpath1, statpath2, statpath3, statsC
member_elements =
member_of = statsB

EXAMPLE #4
----------

To create a statistics group called basic3, type:

$ nas_stats -groups -create basic3 -description "CPU and Memory"
kernel.cpu.utilization.cpuUtil,kernel.memory.freeBytes

'basic3' created successfully.

EXAMPLE #5
----------

To create a statistics group called statgroup2, type:

$ nas_stats -groups -create statgroup2 statgroup1,nfs,net

'statgroup2' created successfully.

EXAMPLE #6
----------

To use an existing statgroup, type:

$ nas_stats -groups -create statgroup1 -description "My
```

```
first group" kernel.cpu.utilization.cpuUtil,
net.basic.inBytes,net.basic.outBytes,store.readBytes,
store.writeBytes

ERROR (13421969439): 'statgroup1' already exists.

EXAMPLE #7
----------

To modify a statgroup by specifying the new contents of the group,
overriding the previous contents, type:

$ nas_stats -groups -modify statgroup2 cifs,nfs-std

'statgroup2' modified successfully.

EXAMPLE #8
----------

To modify the description of a statgroup, type:

$ nas_stats -groups -modify basic1 -description "My basic
group"

'basic1' modified successfully.

EXAMPLE #9
----------

To rename a user-defined statgroup, type:

$ nas_stats -groups -modify statgroup2 -rename basic2

'statgroup2' modified successfully.

EXAMPLE #10
-----------

To add to the member_stats list of a statgroup, type:

$ nas_stats -groups -add statgroup2
kernel.cpu.utilization.cpuUtil,statgroup1

Adding the following statistics:
... kernel.cpu.utilization.cpuUtil
... statgroup1

Statistics added to 'statgroup2' successfully.

EXAMPLE #11
-----------

To remove from the member_stats list of a statgroup, type:

$ nas_stats -groups -remove statgroup1 kernel.cpu.utilization.cpuUtil

Removing the following statistics:
... kernel.cpu.utilization.cpuUtil
Statistics removed from 'statgroup1' successfully.

EXAMPLE #12
-----------

To delete a statgroup, type:

$ nas_stats -groups -delete statgroup1

'statgroup1' deleted successfully.

EXAMPLE #13
-----------
```

To delete reference from other groups using statgroupA, type:

```
$ nas_stats -groups -delete statgroupA

'statgroupA' is used in group (s): mystats1, mystats2.
Clear 'statgroupA' from other groups? [Y/N] Y
'statgroupA' deleted successfully.
```

EXAMPLE #14

To delete reference from other groups using statgroupA and the -Force option to skip the warning prompt, type:

```
$ nas_stats -groups -delete statgroupA -F

'statgroupA' is used in group (s): mystats1, mystats2.
'statgroupA' deleted successfully.
```

EXAMPLE #15

To recover the latest healthy (uncorrupted) copy of a statgroup database from the NAS database backups, type:

```
$ nas_stats -groups -database -recover

Latest healthy database modified last on Tue Apr 7 17:29:06 EDT 2009.
Any updates performed after the latest backup will be lost. Continue? [Y/N] Y
The nas_stats command recover operation is completed successfully.
```

EXAMPLE #16

To recover the latest healthy (uncorrupted) copy of the statgroup database from the NAS database backups using the -Force option to skip the warning prompt, type:

```
$ nas_stats -groups -database -recover -Force

Latest healthy database modified last on Tue Apr 7 17:29:06 EDT 2009.

The nas_stats command recover operation is completed successfully.
```

EXAMPLE #17

To check the health status of the Statistics Groups database, type:

```
$ nas_stats -groups -database -verify

Database is healthy.
```

Last modified: May 10, 2011 4:30 pm.

nas_storage

Controls storage system access and performs some management tasks.

SYNOPSIS

```
nas_storage
  -list
| -info {-all|<name>|id=<storage_id>} [-option <options>]
| -rename <old_name> <new_name>
| -acl <acl_value> <name>
| -delete {<name>|id=<storage_id>} [-spare <spindle-id>|-group
        <diskgroup-id>]
| -failback {<name>|id=<storage_id>}
| -sync {-all|<name>|id=<storage_id>}
| -check {-all|<name>|id=<storage_id>}
| -modify {<name>|id=<storage_id>} -network
        {-spa|-spb} <IP>
| -modify {<name>|id=<storage_id>}
        -security [-username <username>][-password <password>]
        [-newpassword <new_password>]]
```

Note: Output from this command is determined by the type of storage system attached to the VNX.

DESCRIPTION
----

nas_storage sets the name for a storage system, assigns an access control value, displays attributes, performs a health check, synchronizes the storage system with the Control Station, and performs a failback for VNX for block systems.

OPTIONS

-list
Displays a list of all attached storage systems available for the VNX.

Note: The ID of the object is an integer and is assigned automatically. The name of the storage system may be truncated if it is too long for the display. To display the full name, use the -info option with the storage system ID.

-info {-all|<name>|id=<storage_id>}
Displays the configuration of the attached storage system.

[-option <options>]
Specifies a comma-separated list of options.

sync={yes|no}
Synchronizes the Control Station.s view with that of the storage system before displaying configuration information. Default=yes.

-rename <old_name> <new_name>
Renames the current storage system name to a new name. By default, the storage system name is its serial number.

-acl <acl_value> <name>
Sets an access control level value that defines the owner of the storage system, and the level of access allowed for users and groups defined in the access control level table (nas_acl provides information).

-delete {<name>|id=<storage_id> [-spare<spindle-id>|-group <diskgroup-id>
]
Deletes an entry from the storage system table. The storage system can only be deleted after all disks on the storage system have been deleted using nas_disk. The storage system and disks can be rediscovered using the server_devconfig command. The -spare option deletes the hot spare disk from the hot spare pool on the VNX

for block storage used by NAS. The -group option deletes the disk group specified. This deletes and unbinds the LUNs in the RAID groups used by VNX for file. If there are other LUNs in the RAID group not allocated to the VNX, the RAID group is not unbound. After removing the VNX LUNs, the RAID group is empty and it will be destroyed.

-sync {-all|<name>|id=<storage_id>}
Synchronizes the Control Station's view with that of the storage system.

-check {-all|<name>|id=<storage_id>}
Performs a health check on the storage system to verify if it is configured for, and in a state to provide the level of high availability that is required.

Use this option after making any management changes to your storage system (for example, changes to VNX for block array properties, such as enabling/disabling statistics polling).

Note: This option does not support remote storage. For example, for recoverpoint configurations where remote storage is listed, the check wil
l

only run on first listed storage system.

For VNX for Block only

-failback {<name>|id=<storage_id>}
Returns the storage system.s normal operating state by returning ownership of all disk volumes to their default storage processor.

To verify that the storage system failed over, type the -info option. If the value appears as failed_over=True, then the system has failed over.

-modify {<name>|id=<storage_id>} -network {-spa|-spb} <IP>
Modifies the IP address of the VNX for block in the VNX database.

-modify {<name>|id=<storage_id>} -security [-username <username>] [-password <password>]
Updates the login information the VNX for file uses to authenticate with the VNX and changes the VNX username, or password if the VNX account is changed or the following error is reported:

Error 5010: APM00055105668: Storage API code=4651: SYMAPI_C_CLAR_NOT_PRIVILEGED

Operation denied by Clariion array - you are not privileged to perform the requested operation

[-newpassword <new_password]
Assigns a new password to the username on the VNX for block.

Note: This operation is not supported for Symmetrix storage systems.

-resetssv
Resets hostname. For this SE lock box also need to be updated to take the modified host name.

SEE ALSO

VNX System Operations, nas_rdf, nas_disk, and server_devconfig.

STORAGE SYSTEM OUTPUT

The number associated with the storage device is dependent on the attached storage system. VNX for block displays a prefix of APM before a set of integers, for example, APM00033900124-0019. For

example, Symmetrix storage systems appear as 002804000190-003C.

EXAMPLE #1

For the VNX storage system, to list all attached storage systems, type:

```
$ nas_storage -list

id       acl               name          serial_number
1          0     APM00042000818         APM00042000818
```

For the VNX with a Symmetrix storage system, to list all attached storage systems, type:

```
$ nas_storage -list

id       acl             name            serial_number
1          0     000187940260             000187940260
```

Where:

Value	Definition
id	ID number of the attached storage system.
acl	Access control level value assigned to the attached storage system.
name	Name assigned to the attached storage system.
serial_number	Serial number of the attached storage system.

EXAMPLE #2

For the VNX storage system, to display information for the attached storage system, type:

```
$ nas_storage -info APM00042000818

id = 1
arrayname = APM00042000818
name = APM00042000818
type = Clariion
model_type = RACKMOUNT
model_num = 700
db_sync_time = 1131986667 == Mon Nov 14 11:44:27 EST 2005
API_version = V6.0-629
num_disks = 60
num_devs = 34
num_pdevs = 8
num_storage_grps = 1
num_raid_grps = 16
cache_page_size = 8
wr_cache_mirror = True
low_watermark = 60
high_watermark = 80
unassigned_cache = 0
is_local = True
failed_over = False
captive_storage = False

Active Software

-AccessLogix = -
FLARE-Operating-Environment= 02.16.700.5.004
-NavisphereManager = -

Storage Processors

SP Identifier = A
```

```
signature = 1057303
microcode_version = 2.16.700.5.004
serial_num = LKE00040201171
prom_rev = 3.30.00
agent_rev = 6.16.0 (4.80)
phys_memory = 3967
sys_buffer = 773
read_cache = 122
write_cache = 3072
free_memory = 0
raid3_mem_size = 0
failed_over = False
hidden = False
network_name = spa
ip_address = 172.24.102.5
subnet_mask = 255.255.255.0
gateway_address = 172.24.102.254
num_disk_volumes = 20 - root_disk root_ldisk d3 d4 d5 d6 d7 d8 d9 d10 d11 d12
d13 d14 d15 d16 d17 d18 d19 d20

Port Information

Port 1

uid = 50:6:1:60:B0:60:1:CC:50:6:1:61:30:60:1:CC
link_status = UP
port_status = ONLINE
switch_present = True
switch_uid = 10:0:8:0:88:A0:36:F3:20:42:8:0:88:A0:36:F3
sp_source_id = 6373907

<...removed...>

Port 2

uid = 50:6:1:60:B0:60:1:CC:50:6:1:62:30:60:1:CC
link_status = UP
port_status = ONLINE
switch_present = True
switch_uid = 10:0:8:0:88:A0:36:F3:20:41:8:0:88:A0:36:F3
sp_source_id = 6373651

SP Identifier = B
signature = 1118484
microcode_version = 2.16.700.5.004
serial_num = LKE00041700812
prom_rev = 3.30.00
agent_rev = 6.16.0 (4.80)
phys_memory = 3967
sys_buffer = 773
read_cache = 122
write_cache = 3072
free_memory = 0
raid3_mem_size = 0
failed_over = False
hidden = False
network_name = spb
ip_address = 172.24.102.6
subnet_mask = 255.255.255.0
gateway_address = 172.24.102.254
num_disk_volumes = 0

Port Information

Port 1

uid = 50:6:1:60:B0:60:1:CC:50:6:1:69:30:60:1:CC
link_status = UP
port_status = ONLINE
switch_present = True
switch_uid = 10:0:8:0:88:A0:36:F3:20:3E:8:0:88:A0:36:F3
```

```
sp_source_id = 6372883

<...removed...>

Port 2

uid = 50:6:1:60:B0:60:1:CC:50:6:1:6A:30:60:1:CC
link_status = UP
port_status = ONLINE
switch_present = True
switch_uid = 10:0:8:0:88:A0:36:F3:20:3D:8:0:88:A0:36:F3
sp_source_id = 6372627

Storage Groups

id = A4:74:8D:50:6E:A1:D9:11:96:E1:8:0:1B:43:5E:4F
name = ns704g-cs100
num_hbas = 18
num_devices = 24
shareable = True
hidden = False

Hosts

uid = 50:6:1:60:90:60:3:49:50:6:1:60:10:60:3:49
storage_processor = B
port = 1
server = server_4
uid = 50:6:1:60:90:60:3:49:50:6:1:60:10:60:3:49
storage_processor = A

port = 0
server = server_4
uid = 50:6:1:60:80:60:4:F0:50:6:1:61:0:60:4:F0
storage_processor = B
port = 0
server = server_2

<...removed...>

uid = 50:6:1:60:80:60:4:F0:50:6:1:68:0:60:4:F0
storage_processor = B
port = 1
server = server_3

uid = 20:0:0:0:C9:2B:98:77:10:0:0:0:C9:2B:98:77
storage_processor = B
port = 0

uid = 20:0:0:0:C9:2B:98:77:10:0:0:0:C9:2B:98:77
storage_processor = A
port = 0

ALU HLU
------------
0000 -> 0000
0001 -> 0001
0002 -> 0002
0003 -> 0003
0004 -> 0004
0005 -> 0005
0018 -> 0018
0019 -> 0019
0020 -> 0020
0021 -> 0021
0022 -> 0022
0023 -> 0023
0024 -> 0024
0025 -> 0025
0026 -> 0026
0027 -> 0027
```

```
0028 -> 0028
0029 -> 0029
0030 -> 0030
0031 -> 0031
0032 -> 0032
0033 -> 0033
0034 -> 0034
0035 -> 0035

Disk Groups

id = 0000
storage profiles = 2 - clar_r5_performance,cm_r5_performance
raid_type = RAID5
logical_capacity = 1068997528
num_spindles = 5 - 0_0_0 0_0_1 0_0_2 0_0_3 0_0_4
num_luns = 6 - 0000 0001 0002 0003 0004 0005
num_disk_volumes = 6 - root_disk root_ldisk d3 d4 d5 d6
spindle_type = FC
bus = 0
raw_capacity = 1336246910
used_capacity = 62914560
free_capacity = 1006082968
hidden = False

<...removed...>

id = 2_0_14
product = ST314670 CLAR146
revision = 6A06
serial = 3KS02RHM
capacity = 280346624
used_capacity = 224222822
disk_group = 0014
hidden = False
type = FC
bus = 2
enclosure = 0
slot = 14
vendor = SEAGATE
remapped_blocks = -1
state = ENABLED

For the VNX with a Symmetrix storage system, to display
information for the attached storage system, type:

$ nas_storage -info 000187940260

id = 1
serial_number = 000187940260
name = 000187940260
type = Symmetrix
ident = Symm6
model = 800-M2
microcode_version = 5670
microcode_version_num = 16260000
microcode_date = 03012004
microcode_patch_level = 69
microcode_patch_date = 03012004
symmetrix_pwron_time = 1130260200 == Tue Oct 25 13:10:00 EDT 2005
db_sync_time = 1133215405 == Mon Nov 28 17:03:25 EST 2005
db_sync_bcv_time = 1133215405 == Mon Nov 28 17:03:25 EST 2005
db_sync_rdf_time = 1133215405 == Mon Nov 28 17:03:25 EST 2005
last_ipl_time = 1128707062 == Fri Oct 7 13:44:22 EDT 2005
last_fast_ipl_time = 1130260200 == Tue Oct 25 13:10:00 EDT 2005
API_version = V6.0-629
cache_size = 32768
cache_slot_count = 860268
max_wr_pend_slots = 180000
max_da_wr_pend_slots = 90000
max_dev_wr_pend_slots = 6513
```

```
permacache_slot_count = 0
num_disks = 60
num_symdevs = 378
num_pdevs = 10
sddf_configuration = ENABLED
config_checksum = 0x01ca544
num_powerpath_devs = 0
config_crc = 0x07e0ba1e6
is_local = True

Physical Devices
/nas/dev/c0t0l15s2
/nas/dev/c0t0l15s3
/nas/dev/c0t0l15s4
/nas/dev/c0t0l15s6
/nas/dev/c0t0l15s7
/nas/dev/c0t0l15s8
/nas/dev/c16t0l15s2
/nas/dev/c16t0l15s3
/nas/dev/c16t0l15s4
/nas/dev/c16t0l15s8
```

Director Table

type	num	slot	ident	stat	scsi	vols	ports	p0_stat	p1_stat	p2_stat	p3_stat
DA	1	1	DF-1A	On	NA	21	2	On	On	NA	NA
DA	2	2	DF-2A	On	NA	8	2	On	On	NA	NA
DA	15	15	DF-15A	On	NA	21	2	On	On	NA	NA
DA	16	16	DF-16A	On	NA	8	2	On	On	NA	NA
DA	17	1	DF-1B	On	NA	8	2	On	On	NA	NA
DA	18	2	DF-2B	On	NA	21	2	On	On	NA	NA
DA	31	15	DF-15B	On	NA	152	2	On	On	NA	NA
DA	32	16	DF-16B	On	NA	165	2	On	On	NA	NA
FA	33	1	FA-1C	On	NA	0	2	On	On	NA	NA
FA	34	2	FA-2C	On	NA	0	2	On	On	NA	NA
FA	47	15	FA-15C	On	NA	0	2	On	On	NA	NA
FA	48	16	FA-16C	On	NA	0	2	On	On	NA	NA
FA	49	1	FA-1D	On	NA	0	2	On	On	NA	NA

Note: This is a partial listing due to the length of the outputs.

EXAMPLE #3

To rename a storage system, type:

```
$ nas_storage -rename APM00042000818 cx700_1

id = 1
serial_number = APM00042000818
name = cx700_1
acl = 0
```

EXAMPLE #4

To set the access control level for the storage system cx700_1, type:

```
$ nas_storage -acl 1000 cx700_1

id = 1
serial_number = APM00042000818
name = cx700_1
acl = 1000, owner=nasadmin, ID=201
```

Note: The value 1000 specifies nasadmin as the owner and gives read, write, and delete access only to nasadmin.

EXAMPLE #5

```
----------

To change the existing password on the VNX for block, type:

$ nas_storage -modify APM00070204288 -security -username

nasadmin -password nasadmin -newpassword abc
Changing password on APM00070204288

EXAMPLE #6
----------

To avoid specifiying passwords in clear text on the command line,
type:

$ nas_storage -modify APM00070204288 -security
-newpassword

Enter the Global CLARiiON account information
Username: nasadmin
Password: *** Retype your response to validate
Password: ***
New Password
Password: ******** Retype your response to validate
Password: ********
Changing password on APM00070204288
Done

EXAMPLE #7
----------

To failback a VNX for block, type:

$ nas_storage -failback cx700_1

id = 1
serial_number = APM00042000818
name = cx700_1
acl = 1000, owner=nasadmin, ID=201
EXAMPLE #8 To display information for a VNX for block and turn synchronization
off, type:
$ nas_storage -info cx700_1 -option sync=no
id = 1
arrayname = APM00042000818
name = cx700_1
type = Clariion
model_type = RACKMOUNT
model_num = 700
db_sync_time = 1131986667 == Mon Nov 14 11:44:27 EST 2005
API_version = V6.0-629
num_disks = 60
num_devs = 34
num_pdevs = 8
num_storage_grps = 1
num_raid_grps = 16
cache_page_size = 8
wr_cache_mirror = True
low_watermark = 60
high_watermark = 80
unassigned_cache = 0
is_local = True
failed_over = False
captive_storage = False

Active Software

-AccessLogix = -
FLARE-Operating-Environment= 02.16.700.5.004
-NavisphereManager = -

Storage Processors
```

```
SP Identifier = A
signature = 1057303
microcode_version = 2.16.700.5.004
serial_num = LKE00040201171
prom_rev = 3.30.00
agent_rev = 6.16.0 (4.80)
phys_memory = 3967
sys_buffer = 773
read_cache = 122
write_cache = 3072
free_memory = 0
raid3_mem_size = 0
failed_over = False
hidden = False
network_name = spa
ip_address = 172.24.102.5
subnet_mask = 255.255.255.0
gateway_address = 172.24.102.254
num_disk_volumes = 20 - root_disk root_ldisk d3 d4 d5 d6 d7 d8 d9 d10
d11 d12 d13 d14 d15 d16 d17 d18 d19 d20

Port Information

Port 1

uid = 50:6:1:60:B0:60:1:CC:50:6:1:61:30:60:1:CC
link_status = UP
port_status = ONLINE
switch_present = True
switch_uid = 10:0:8:0:88:A0:36:F3:20:42:8:0:88:A0:36:F3
sp_source_id = 6373907

<...removed...>

Port 2

uid = 50:6:1:60:B0:60:1:CC:50:6:1:62:30:60:1:CC
link_status = UP
port_status = ONLINE
switch_present = True
switch_uid = 10:0:8:0:88:A0:36:F3:20:41:8:0:88:A0:36:F3
sp_source_id = 6373651
SP Identifier = B
signature = 1118484
microcode_version = 2.16.700.5.004
serial_num = LKE00041700812
prom_rev = 3.30.00
agent_rev = 6.16.0 (4.80)
phys_memory = 3967
sys_buffer = 773
read_cache = 122
write_cache = 3072
free_memory = 0
raid3_mem_size = 0
failed_over = False
hidden = False
network_name = spb
ip_address = 172.24.102.6
subnet_mask = 255.255.255.0
gateway_address = 172.24.102.254
num_disk_volumes = 0

Port Information

Port 1

uid = 50:6:1:60:B0:60:1:CC:50:6:1:69:30:60:1:CC
link_status = UP
port_status = ONLINE
switch_present = True
```

```
switch_uid = 10:0:8:0:88:A0:36:F3:20:3E:8:0:88:A0:36:F3
sp_source_id = 6372883

<...removed...>

Port 2

uid = 50:6:1:60:B0:60:1:CC:50:6:1:6A:30:60:1:CC
link_status = UP
port_status = ONLINE
switch_present = True
switch_uid = 10:0:8:0:88:A0:36:F3:20:3D:8:0:88:A0:36:F3
sp_source_id = 6372627

Storage Groups

id = A4:74:8D:50:6E:A1:D9:11:96:E1:8:0:1B:43:5E:4F
name = ns704g-cs100
num_hbas = 18
num_devices = 24
shareable = True
hidden = False

Hosts

uid = 50:6:1:60:90:60:3:49:50:6:1:60:10:60:3:49
storage_processor = B
port = 1
server = server_4

uid = 50:6:1:60:90:60:3:49:50:6:1:60:10:60:3:49
storage_processor = A
port = 0
server = server_4

uid = 50:6:1:60:80:60:4:F0:50:6:1:61:0:60:4:F0
storage_processor = B
port = 0
server = server_2

<...removed...>

uid = 50:6:1:60:80:60:4:F0:50:6:1:68:0:60:4:F0
storage_processor = B
port = 1
server = server_3

uid = 20:0:0:0:C9:2B:98:77:10:0:0:0:C9:2B:98:77
storage_processor = B
port = 0

uid = 20:0:0:0:C9:2B:98:77:10:0:0:0:C9:2B:98:77
storage_processor = A
port = 0

ALU HLU
------------
0000 -> 0000
0001 -> 0001
0002 -> 0002
0003 -> 0003
0004 -> 0004
0005 -> 0005
0018 -> 0018
0019 -> 0019
0020 -> 0020
0021 -> 0021
0022 -> 0022
0023 -> 0023
0024 -> 0024
0025 -> 0025
```

```
0026 -> 0026
0027 -> 0027
0028 -> 0028
0029 -> 0029
0030 -> 0030
0031 -> 0031
0032 -> 0032
0033 -> 0033
0034 -> 0034
0035 -> 0035

Disk Groups

id = 0000
storage profiles = 2 - clar_r5_performance,cm_r5_performance
raid_type = RAID5
logical_capacity = 1068997528
num_spindles = 5 - 0_0_0 0_0_1 0_0_2 0_0_3 0_0_4
num_luns = 6 - 0000 0001 0002 0003 0004 0005
num_disk_volumes = 6 - root_disk root_ldisk d3 d4 d5 d6
spindle_type = FC
bus = 0
raw_capacity = 1336246910
used_capacity = 62914560
free_capacity = 1006082968
hidden = False

<...removed...>

id = 0205
storage profiles = 0
raid_type = SPARE
logical_capacity = 622868992
num_spindles = 1 - 0_1_0
num_luns = 1 - 0205
num_disk_volumes = 0
spindle_type = ATA
bus = 0
raw_capacity = 622868992
used_capacity = 622868992
free_capacity = 0
hidden = False

Spindles

id = 0_0_0
product = ST314670 CLAR146
revision = 6A06
serial = 3KS088SQ
capacity = 280346624
used_capacity = 12582912
disk_group = 0000
hidden = False
type = FC
bus = 0
enclosure = 0
slot = 0
vendor = SEAGATE
remapped_blocks = -1
state = ENABLED

<...removed...>

id = 2_0_14
product = ST314670 CLAR146
revision = 6A06
serial = 3KS02RHM
capacity = 280346624
used_capacity = 224222822
disk_group = 0014
hidden = False
```

```
type = FC
bus = 2
enclosure = 0
slot = 14
vendor = SEAGATE
remapped_blocks = -1
state = ENABLED

Note: This is a partial display due to the length of the outputs.

EXAMPLE #9
----------

To delete a storage system with no attached disks, type:

$ nas_storage -delete APM00035101740

id = 0
serial_number = APM00035101740
name = APM00035101740
acl = 0

EXAMPLE #10
-----------

To turn synchronization on for all systems, type:

$ nas_storage -sync -all

done

EXAMPLE #11
-----------

To perform a health check on the storage system, type:

$ nas_storage -check -all

Discovering storage (may take several minutes)

done

EXAMPLE #12
-----------

To set the access control level for the storage system
APM00042000818, type:

$ nas_storage -acl 1432 APM00042000818

id = 1
serial_number = APM00042000818
name = APM00042000818
acl = 1432, owner=nasadmin, ID=201

Note: The value 1432 specifies nasadmin as the owner and gives users with
an access level of at least observer read access only, users with an access
level of at least operator read/write access, and users with an access level
of at least admin read/write/delete access.

EXAMPLE #13
-----------

To modify the IP address of the VNX for block, type:

$ nas_storage -modify APM00072303347 -network -spa
10.6.4.225

Changing IP address for APM00072303347
Discovering storage (may take several minutes)
done
```

EXAMPLE #14

To reset hostname.

```
$ nas_storage -resetssv
done
```

Last modified: July 26, 2011 12:35 pm.

nas_syncrep

Manages Virtual Data Mover (VDM) synchronous replication sessions. The list, info, and create switches of this command can be executed on both the active and standby systems. Execute the delete switch of this command on the active system. Execute the reverse, failover, and Clean switches of this command on the standby system.

SYNOPSIS

```
nas_syncrep
          -list
        | -info { -all | <name> | id=<id> } [-verbose]
        | -create <name>
              -vdm <vdm_name>
              -remote_system <cel_name>
              -remote_pool <pool_name>
              -remote_mover <mover_name>
              -network_devices <local_device_name>:<remote_device_name>[,...]
        | -start { -all | <name> | id=<id> }
        | -delete { <name> | id=<id> }
        | -reverse { <name> | id=<id> }
        | -failover { <name> | id=<id> }
        | -Clean { -all | <name> | id=<id> } [-Force]
        | -Refresh_pairs { -all | <name> | id=<id> }
        | -health_check
        | -modify { <name> | id=<id> }
              -network_devices <local_device_name>:<remote_device_name>[,...]
```

DESCRIPTION
----

nas_syncrep creates, manages, or displays session information for ongoing VDM synchronous replication sessions. Each session handles a single object between the active and standby systems.

OPTIONS

-list
Displays all configured synchronous replication sessions on the local system's NAS database and those having the local system as the standby system in the remote system's replicated NAS database.

-info { -all | <name> | id=<id> } [-verbose]
Displays the status of a specific configured synchronous replication session, or the status of all synchronous replication sessions.

-create <name>
Assigns a name to the synchronous replication session. The session name is case-sensitive and supports the following characters: a through z, A through Z, 0 through 9, _(underscore), -(hyphen) though names may not start with a hyphen. The maximum length of the name is 128 characters.

The following items will need to be manually migrated using the migrate_system_conf command after the creation of a synchronous replication session and any time this data changes:

 DNS
 NIS
 NTP
 Local passwd and group
 Usermapper client
 FTP/SFTP, LDAP, HTTP, CEPP, CAVA, Server Parameters
 Netgroup
 Nsswitch
 Hosts

-vdm <vdm_name>

Specifies the name of an existing source sync-replicable VDM to replicate.

-remote_system <cel_name>
Specifies the name of an existing remote VNX system.

-remote_pool <pool_name>
Specifies the name of an existing remote user-defined pool.

-remote_mover <mover_name>
Specifies the name of the existing remote Data Mover.

-network_devices <local_device_name>:<remote_device_name>[,...]
Specifies the mappings of the local and remote network devices.

-local_storage journal=<alu>
Specifies the assigned system LUN for the local journal volume.

-remote_storage journal=<alu>
Specifies the assigned system LUN for the remote journal volume.

-start { -all | <name> | id=<id> }
Starts all SRDF synchronous replication sessions or a specified
synchronous replication session. Execute this switch on the standby system.

-delete { <name> | id=<id> }
Deletes a synchronous replication session of specific name or ID
with local system as active. Execute this switch on the active system.

-reverse { <name> | id=<id> }
Switches the active/standby role of the two VNX systems in a
synchronous replication session when both are up.
Execute this switch on the standby system.

-failover { <name> | id=<id> }
Fails over the specified VDM to the standby system to make it active.
Execute this switch on the standby system.

-Clean { -all | <name> | id=<id> } [-Force]
Cleans all synchronous replication sessions or a specified
synchronous replication session. Execute this switch on the standby system.

-Refresh_pairs { -all | <name> | id=<id> }
Refreshs all synchronous replication session or a specified
synchronous replication session to establish RDF pairing for any new devices
added. Execute this switch on the active system.

-health_check
Performs a health check of the VDM sync sessions, checking for conditions that
could possibly cause a failover or a reverse operation to fail.

-modify { <name> | id=<id> }
    -network_devices <local_device_name>:<remote_device_name>[,...]
Reflects the change of existing network devices with new network devices for a
specific session.

Note: After failover, the LUNs on the standby system under synchronous
replication are Read Only and the original VDM/File Systems/checkpoints remain
on them. If any write operation occurs on those objects, such as mount a
File System or write I/O to a File System, the Data Mover will run into panic.
The Clean operation removes those obsoleted objects from the failed system
for the specified synchronous session or all synchronous replication sessions
on the standby system so that the Data Mover can be returned to use.

EXAMPLE #1

To list synchronous replication sessions, type:

```
$ nas_syncrep -list
id      name                    vdm_name      remote_system    session_status
5020    my_syncrep1             my_vdm1       -->my_system1    sync_in_progress
```

```
10030   my_syncrep2             my_vdm2       <--my_system1    in_sync
```

EXAMPLE #2

To display information of a synchronous replication session by name, type:

```
$ nas_syncrep -i id=4096
id                  = 4096
name                = LY2E6_session1
vdm_name            = LY2E6_vdm1
syncrep_role        = active
local_system        = LY2E6_CS0
local_pool          = src_sg_1
local_mover         = server_2
remote_system       = L9P36_CS0
remote_pool         = dst_sg_1
remote_mover        = server_2
device_group        = 61_260_60_125
session_status      = in_sync
```

EXAMPLE #3

To create a synchronous replication session, type:

```
$ nas_syncrep -create LY2E6_session1 -vdm LY2E6_vdm1 -remote_system L9P36_CS0
-remote_pool l9p36_marketing_sg -remote_mover server_2 -network_devices cge0:cge0
Now validating params...                                          done
Now creating LUN mapping...                                       done
Now creating remote network interface(s)...                       done
Now marking remote pool as standby pool...                        done
Now updating local disk type...                                   done
Now updating remote disk type...                                  done
Now generating session entry...                                   done
done
```

EXAMPLE #4

To delete a synchronous replication session, type:

```
$ nas_syncrep -delete my_syncrep1
WARNING: Please do not perform any operation on my_syncrep1 on standby
system until delete is done.
Deleting...                                                       done
done
```

EXAMPLE #5

To reverse a synchronous replication session, type:

```
$ nas_syncrep -reverse id=4315
WARNING: There will be a period of Data Unavailabilty during the reverse
operation, and, after the reverse operation, the VDM/FS(s)/checkpoint(s)
protected by the sync replication session will be reversed to the local site.
Are you sure you want to proceed?  [yes or no] yes
Now doing precondition check...                        done: 19 s
Now doing health check...                              done: 11 s
Now cleaning local...                                  done: 1  s

                                  Service outage start......
Now turning down remote network interface(s)...        done: 8  s
Now switching the session (may take several minutes)...  done: 7  s
Now importing sync replica of NAS database...          done: 16 s
Now creating VDM...                                    done: 5  s
Now importing VDM settings...                          done: 0  s
Now mounting exported FS(s)/checkpoint(s)...           done: 13 s
Now loading VDM...                                     done: 3  s
Now turning up local network interface(s)...           done: 0  s
```

```
                                    Service outage end: 52 s

Now mounting unexported FS(s)/checkpoint(s)...          done: 0  s
Now importing schedule(s)...                            done: 0  s
Now unloading remote VDM/FS(s)/checkpoint(s)...         done: 16 s
Now cleaning remote...                                  done: 17 s
                                          Elapsed time: 116s
done

EXAMPLE #6
----------

To failover a synchronous replication session, type:

$ nas_syncrep -failover id=4560
WARNING: You have just issued the nas_syncrep -failover command.
Verify whether the peer system or any of its file storage resources
are accessible. If they are, then you should issue the nas_syncrep
-reverse command instead. Running the nas_syncrep -failover command
while the peer system is still accessible could result in Data
Unavailability or Data Loss. Are you sure you want to proceed?
[yes or no] yes
Now doing precondition check...                         done: 30 s
Now doing health check...                               done: 7  s
Now cleaning local...                                   done: 1  s
Now switching the session (may take several minutes)...  done: 4  s
Now importing sync replica of NAS database...           done: 15 s
Now creating VDM...                                     done: 5  s
Now importing VDM settings...                           done: 0  s
Now mounting exported FS(s)/checkpoint(s)...            done: 3  s
Now loading VDM...                                      done: 4  s
Now turning up local network interface(s)...            done: 0  s
                                    Service outage end: 69 s

Now mounting unexported FS(s)/checkpoint(s)...          done: 0  s
Now importing schedule(s)...                            done: 0  s
                                          Elapsed time: 69 s
done

EXAMPLE #7
----------

To clean a synchronous replication session, type:

[nasadmin@L9P36_CS0 ~]$ nas_syncrep -Clean LY2E6_session1
WARNING: You have just issued the nas_syncrep -Clean command. This may result
in a reboot of the original source Data Mover that the VDM was failed over
from. Verify whether or not you have working VDM(s)/FS(s)/checkpoint(s) on
this Data Mover and plan for this reboot accordingly. Running the nas_syncrep
-Clean command while you have working VDM(s)/FS(s)/checkpoint(s) on this Data
Mover will result in Data Unavailability during the reboot. Are you sure you
want to proceed?  [yes or no] yes
Now cleaning session LY2E6_session1 (may take several minutes)...  done
Now starting session LY2E6_session1...                             done
done

EXAMPLE #8
----------

To refresh a synchronous replication session, type:

$ nas_syncrep -Refresh_pairs LY2E6_session1
WARNING: You have just issued the nas_syncrep -Refresh_pairs command.
Please do not perform any operation(s) on the remote (R2) side during
the same. Also note that the operation cannot be reverted. Are you
sure you want to proceed?  [yes or no] yes

Now refreshing session LY2E6_session1...                done

EXAMPLE #9
----------
```

To perform a health check of the VDM synchronous replication session, type:

```
$ nas_syncrep -health_check
Health check starting ...
Initializing ...

Check No.       | Check Name                                | Message ID  |
Status |  Brief Description
Check (  1/12 ) | Check SRDF Group State                    | 34906964006 |
PASS   |  SRDF Group online.
Check (  2/12 ) | Check SRDF session(s) status              | 34906964010 |
PASS   |  SRDF session in sync.
Check (  3/12 ) | Check VDM Sync session(s) disktype(s)     | 34906964014 |
PASS   |  VDM Sync session disktype ok.
Check (  4/12 ) | File system has 128K free space           | 34906964018 |
PASS   |  File system has at least 128K free space.
Check (  5/12 ) | Network Configuration Check               | 34906963984 |
PASS   |  Interfaces correct.
Check (  6/12 ) | Check SRDF link status                    | 34906963989 |
PASS   |  Remote SRDF is pingable.
Check (  7/12 ) | Data Mover status                         | 34906963992 |
PASS   |  Data movers are in the correct state.
Check (  8/12 ) | eNAS, SE, Enginuity version check         | 34906963993 |
PASS   |  eNAS, SE and microcode version check passed.
Check (  9/12 ) | Check for filesystem ID consistency       | 34906963997 |
PASS   |  No conflict in file system IDs.
Check ( 10/12 ) | Pool to SRDF session mapping              | 34906964014 |
PASS   |  VDM Sync session disktypes in correct state.
Check ( 11/12 ) | Check for director ports online/offline   | 34906963999 |
PASS   |  Directors and ports online.
Check ( 12/12 ) | Check for Equivalent Data Services        | 34906964026 |
PASS   |  Data services match.

Health check complete.
Check /nas/log/nas_syncrep.log for more details.
Use 'nas_message -i <messageID>' to view detailed information about the
message IDs.
```

EXAMPLE #10

Reflects the change of existing network devices with new network devices for a specific session.

```
$ nas_syncrep -modify cyp_ssn1 -network_devices cge-8-0:cge-3-0
WARNING: You have just issued the nas_syncrep -Sync_interfaces command. Please
do not perform any operation(s) on the remote (R2) side during the same, In
case of the command failure please make sure to create the interfaces manually
on the remote (R2) side. Are you sure you want to proceed? [yes or no] yes
done
```

Last modified: December 5, 2014 11:20 a.m.

nas_task

Manages in-progress or completed tasks.

SYNOPSIS

```
nas_task
  -list [-remote_system {<remoteSystemName>|id=<id>}]
| -info {-all|<taskId>}
    [-remote_system {<remoteSystemName>|id=<id>}]
| -abort <taskId>
    [-mover <moverName>][-remote_system {<remoteSystemName>|id=<id>}]
| -delete <taskId>
     -remote_system {<remoteSystemName>|id=<id>}]
```

DESCRIPTION
----

nas_task lists the tasks associated with commands currently in progress or completed, reports information about a particular task, aborts a task, or deletes a task. Each task can be uniquely identified by its task ID and the remote VNX system name or ID.

Use the nas_task command to monitor, abort, and delete long running tasks and tasks started in asynchronous mode.

OPTIONS

-list
Lists all local tasks that are in progress, or completed tasks that have not been deleted. For each task, lists the task ID, remote system name, a description of the task, and the task state (running, recovering, succeeded, or failed).

        -remote_system {<remoteSystemName>|id=<id>}
        Lists local tasks initiated by the specified remote VNX system. Specify the remote system name or ID.

-info {-all|<taskId>}
Provides more detailed status information for all tasks or for a particular task. Displays the run time status, estimated completion time, and percent complete for running tasks. Displays the completion status and actual end time for completed tasks. The taskID is the ID returned from a command run in the background mode or from the nas_task -list command.

Note: The ID of the task is an integer and is assigned automatically. The task ID is unique to the VNX.

        [-remote_system {<remoteSystemName>|id=<id>}]
        Provides more detailed status information of local tasks initiated by the specified remote VNX system. Specify the remote system name or remote system ID. The remote system name is returned from the nas_task -list command.

-abort <taskId>
Aborts the specified task leaving the system in a consistent state. For example, it aborts a one-time copy in progress. This might take a long time to complete because a remote system may be unavailable or the network may be down. You should check the status of the task to verify that the task was aborted. This command can be executed from the source only.

CAUTION
This option might leave the system in an inconsistent state. Use caution when using this option.

        [-mover <moverName>]
        Aborts a task running locally on the specified Data Mover.

This command can be executed from the source or destination side. Use this command when the source and destination VNX systems cannot communicate. You should run this command on both sides.

[-remote_system {<remoteSystemName>|id=<id>]
Aborts a task that was initiated on a remote VNX leaving the source side intact. Specify the Data Mover to abort a task from the destination side. Specify the Data Mover and remote system name or remote system id along with the task id.

-delete id <taskId>
Based on the task ID, deletes a completed task from the database on the Control Station.

[-remote_system {<remoteSystemName>|id=<id>]
Deletes a task that was initiated on a remote VNX. Specify the remote system name or remote system id along with the task id.

SEE ALSO

Using VNX Replicator, nas_copy, nas_replicate, and nas_cel.

EXAMPLE #1

To display detailed information about the task with taskID 4241, type:

```
$ nas_task -info 4241

Task Id = 4241
Celerra Network Server = cs100
Task State = Running
Percent Complete = 95
Description = Create Replication ufs1_replica1.
Originator = nasadmin@cli.localhost
Start Time = Mon Dec 17 14:21:35 EST 2007
Estimated End Time = Mon Dec 17 19:24:21 EST 2007
Schedule = n/a
```

Where:

Value	Definition
Task Id	Globally unique character string used as the identifier of the task.
VNX	When set, local.
Remote Task Id	When set, identifies a remote task.
State	Running, Recovering, Completed, or Failed. Running could be a combination of completed and failed.
Current Activity	Displays state property when available.
Percent Completed	Appears only when set and not complete.
Description	Appears if details are set.
Originator	User or host that initiated the task.
Start Time/End Time	The starting time and ending time (or status) for the task.
Estimated End Time	Appears instead of previous line when available and task is incomplete.
Schedule	The schedule in effect, or n/a for a task that is not a scheduled checkpoint refresh.
Response Statuses	Displayed list of messages, if any. A completed task should always have one.

EXAMPLE #2

To display the list of all tasks, type:

```
$ nas_task -list
```

```
ID       Task    State   Originator      Start   Time   Description      Schedule
Remote System
4241 Running nasadmin@cli+ Mon Dec 17 14:21:35 EST 2007 Create Replication
ufs1_r+ cs100
4228 Succeeded nasadmin@cli+ Mon Dec 17 14:04:02 EST 2007 Delete task NONE:
4214.
cs100
4177 Failed nasadmin@cli+ Mon Dec 17 13:59:26 EST 2007 Create Replication
ufs1_r+ cs100
4150 Succeeded nasadmin@cli+ Mon Dec 17 13:55:39 EST 2007 Delete task NONE:
4136.
cs100
4127 Succeeded nasadmin@cli+ Mon Dec 17 11:38:32 EST 2007 Delete task NONE:
4113.
cs100
4103 Succeeded nasadmin@cli+ Mon Dec 17 11:21:00 EST 2007 Delete task NONE:
4098.
cs100
4058 Succeeded nasadmin@cli+ Fri Dec 14 16:43:23 EST 2007 Switchover
Replication
NONE. cs100
2277 Succeeded nasadmin@cli+ Fri Dec 14 16:42:08 EST 2007 Reverse Replication
NONE. cs110
2270 Succeeded nasadmin@cli+ Fri Dec 14 16:40:29 EST 2007 Start Replication
NONE.
cs110
2265 Failed nasadmin@cli+ Fri Dec 14 16:40:11 EST 2007 Start Replication NONE.
cs110
```

EXAMPLE #1 provides a description of the outputs.

EXAMPLE #3

To abort task 4267 running locally on server_3, type:

```
$ nas_task -abort 4267 -mover server_3

OK
```

EXAMPLE #4

To delete the existing task 4267, type:

```
$ nas_task -delete 4267

OK
```

Last Modified: May 10, 2011 5:00 pm

nas_tls

Manages the TLS settings.

SYNOPSIS

```
nas_tls
    -info
  | -set { TLSv1.0 | TLSv1.1 | TLSv1.2 } [-Force]
  | -help
```

DESCRIPTION
----

nas_tls configures and displays TLS settings on eNAS. It can configure
 and display TLS versions supported by Apache, ECOM, Data Movers on the File side
.
Note: root privileges are required to set or get the TLS protocols supported
by different components.

OPTIONS

-info
List TLS versions supported by Apache, ECOM, Data Movers, CS LDAP client on the
File side.

-set { TLSv1.0 | TLSv1.1 | TLSv1.2 } [-Force]
Set the lowest version of the TLS protocol the system supports for SSL communicat
ion.

[-Force]
 No warning message promote to user when setting TLS protocol.

-help
Help message of nas_tls command.

EXAMPLE #1

To check TLS settings, su to root and type:

```
# nas_tls -info
TLS versions supported on File side
Apache                    : TLSv1.0    TLSv1.1    TLSv1.2
ECOM                      : TLSv1.0    TLSv1.1    TLSv1.2
server_2                  : TLSv1.0    TLSv1.1    TLSv1.2
server_3                  : TLSv1.0    TLSv1.1    TLSv1.2
```

EXAMPLE #2

To set lowest version of TLS protocal, su to root and type:

```
# nas_tls -set TLSv1.1
WARNING: This operation will restart the following services and temporarily termi
nate its interactive sessions!
Apache service
CS ECOM service
Data Mover service (IO will be affected and may result in Data Unavailability dur
ing the reboot!)
Do you wish to continue? [y/n]:Y
Reboot server_2
```

EXAMPLE #3

To show more TLS setting options, su to root and type:

```
# nas_tls -help
usage: nas_tls
      -info
    | -set { TLSv1.0 | TLSv1.1 | TLSv1.2 } [-Force]
```

-info: List the versions of the TLS protocol the system supports for SSL communication.

-set: Set the lowest version of the TLS protocol the system supports for SSL communication.

nas_version

Displays the software version running on the Control Station.

SYNOPSIS
----

nas_version
        [-h|-l]

DESCRIPTION
----

nas_version displays the Control Station version in long form or
short form. When used during a software upgrade, informs the user
about the upgrade in progress.

OPTIONS

No arguments
Displays the software version running on the Control Station.

-h
Displays command usage.

-l
Displays detailed software version information for the Control
Station.

EXAMPLE #1

To display the software version running on the Control Station
during a software upgrade, type:

$ nas_version

5.6.25-0

EXAMPLE #2

To display the system output during a software upgrade, type:

$ nas_version

5.6.19-0
Warning!!Upgrade is in progress from 5.6.19-0 to 5.6.20-0
Warning!!Please log off IMMEDIATELY if you are not upgrading the system

EXAMPLE #3

To display the usage for nas_version, type:

$ nas_version -h

usage: /nas/bin/nas_version [-h|-l]
-h help
-l long_format

EXAMPLE #4

To display detailed software version information for the Control
Station, type:

$ nas_version -l

Name : emcnas Relocations: /nas

```
Version : 5.6.19 Vendor: EMC
Release : 0 Build Date: Tue 19 Dec 2006 08:53:31 PM EST
Size : 454239545 License: EMC Copyright
Signature : (none)
Packager : EMC Corporation
URL : http://www.emc.com
Summary : EMC nfs base install
Description : EMC nfs base install
```

EXAMPLE #5

To display detailed software version information for the Control Station during a software upgrade, type:

```
$ nas_version -l

Name : emcnas Relocations: /nas
Version : 5.6.19 Vendor: EMC
Release : 0 Build Date: Wed 14 Mar 2007 12:36:55 PM EDT
Size : 500815102 License: EMC Copyright
Signature : (none)
Packager : EMC Corporation
URL : http://www.emc.com
Summary : EMC nfs base install
Description : EMC nfs base install
Warning!!Upgrade is in progress from 5.6.19-0 to 5.6.20-0
Warning!!Please log off IMMEDIATELY if you are not upgrading the system
```

Last modified: May 10, 2011 5:15 pm.

nas_volume

Manages the volume table.

SYNOPSIS

```
nas_volume
          -list
         | -delete <volume_name>
         | -info [-size] {-all|<volume_name>} [-tree]
         | -rename <old_name> <new_name>
         | -size <volume_name>
         | -acl <acl_value> <volume_name>
         | -xtend <volume_name> {<volume_name>,...}
         | [-name <name>] -create [-Stripe [<stripe_size>]|-Meta]
                [-Force] {<volume_name>,...}
         | -Clone <volume_name> [{<svol>:<dvol>,...}][-option <options>]
```

DESCRIPTION
----

nas_volume creates metavolumes and stripe volumes and lists, renames, extends, clones, and deletes metavolumes, stripe, and slice volumes. nas_volume sets an access control value for a volume, and displays detailed volume attributes, including the total size of the volume configuration.

OPTIONS

-list
Displays the volume table.
Note: The ID of the object is an integer and is assigned automatically. The name of the volume may be truncated if it is more than 17 characters. To display the full name, use the -info option with the volume ID.

-delete <volume_name>
Deletes the specified volume.

-info [-size] {-all|<volume_name>} [-tree]
Displays attributes and the size for all volumes, or the specified <volume_name>. The -tree option recursively displays the volume set, that is, the list of component volumes for the specified volume or all volumes.

-rename <old_name> <new_name>
Changes the current name of a volume to a new name.

-size <volume_name>
Displays the total size in MB of the <volume_name>, including used and available space.

-acl <acl_value> <volume_name>
Sets an access control level value that defines the owner of the volume, and the level of access allowed for users and groups defined in the access control level table. The nas_acl command provides information.

-xtend <volume_name> {<volume_name>,...}
Extends the specified metavolume by adding volumes to the configuration. The total size of the metavolume increases by the sum of all the volumes added.

Note: Only metavolumes can be extended. The volume that was added remains in use until the original metavolume is deleted. Volumes containing mounted file systems cannot be extended using this option. The nas_fs command provides information to extend a volume that is hosting a mounted file system.

-create {<volume_name>,...}

Creates a volume configuration from the specified volumes. Unless otherwise specified, volumes are automatically created as metavolumes.

[-name <name>]
Assigns a <name> to volume. If a name is not specified, one is assigned automatically. The name of a volume is case-sensitive.

[-Stripe <stripe_size>|-Meta]
Sets the type for the volume to be either a stripe volume or metavolume (default). If -Stripe is specified, type a stripe size in multiples of 8192 bytes with a recommended size of 262,144 bytes (256 KB) for all environments and drive types. If a stripe size is not specified, the
system creates a 256 KB stripe by default.

nas_slice provides information to create a slice volume.

[-Force] {<volume_name>,...}
Forces the creation of a volume on a mixed storage system.

-Clone <volume_name>
Creates an exact clone of the specified <volume_name>. Volumes can be cloned from slice, stripe, or metavolumes. The name automatically assigned to the clone is derived from the ID of the volume.

[{<svol>:<dvol>,...}]
Sets a specific disk volume set for the source volume and the destination volume. The size of the destination volume must be the same as the source volume.

-option disktype=<type>
Specifies the type of disk to be created.

Disk types when using VNX for block are CLSTD, CLEFD, and CLATA, and for VNX for block involving mirrored disks are: CMEFD, CMSTD, and CMATA.

Disk types when using a Symmetrix are STD, R1STD, R2STD, BCV, R1BCV, R2BCV, ATA, R1ATA, R2ATA, BCVA, R1BCA, R2BCA, and EFD.

SEE ALSO

Managing Volumes and File Systems with VNX Automatic Volume Management, Managing Volumes and File Systems for VNX Manually, Using TimeFinder/FS, NearCopy, and FarCopy on VNX for File, Controlling Access to System Objects on VNX, nas_slice, nas_disk, nas_acl, and nas_fs.

EXAMPLE #1

To list all volumes, type:

```
$ nas_volume -list

id inuse type acl name cltype clid
1 y 4 0 root_disk 0 1,2,3,4,5,6,7,8,9,10,11,
12,13,14,15,16,17,18,19,20,
21,22,23,24,25,26,27,28,29,
30,31,32,33,34,51
2 y 4 0 root_ldisk 0 35,36,37,38,39,40,41,42,
43,44,45,46,47,48,49,50,52
3 y 4 0 d3 1 76
4 y 4 0 d4 1 77
5 y 4 0 d5 1 78
6 y 4 0 d6 1 79
7 n 1 0 root_dos 0
8 n 1 0 root_layout 0
```

```
9 y 1 0 root_slice_1 1 10
10 y 3 0 root_volume_1 2 1
11 y 1 0 root_slice_2 1 12
12 y 3 0 root_volume_2 2 2
13 y 1 0 root_slice_3 1 14
...
```

Note: This is a partial listing due to the length of the outputs.

Where:

Value	Definition
id	ID of the volume.
inuse	Whether the volume is used.
type	Type assigned to the volume. Available types are: 1=slice, 2=stripe, 3=meta, 4=disk, and 100=pool.
acl	Access control level assigned the volume.
name	Name assigned to the volume.
cltype	The client type of the volume. Available values are: 0 - If the clid field is not empty then the client is a slice. 1 - The client is another volume (meta, stripe, volume_pool). 2 - The client is a file system.
clid	ID of the client.

EXAMPLE #2

To create a metavolume named, mtv1, on disk volume, d7, type:

```
$ nas_volume -name mtv1 -create d7

id = 146
name = mtv1
acl = 0
in_use = False
type = meta
volume_set = d7
disks = d7
```

Where:

Value	Definition
id	ID of the volume.
name	Name assigned to the volume.
acl	Access control level value assigned to the volume.
in_use	Whether the volume is used.
type	Type assigned to the volume. Types are meta, stripe, slice, disk, and pool.
volume_set	Name assigned to the volume.
disks	Disks used to build a file system.

EXAMPLE #3

To display configuration information for mtv1, type:

```
$ nas_volume -info mtv1

id = 146
name = mtv1
acl = 0
in_use = False
type = meta
volume_set = d7
```

```
disks = d7

EXAMPLE #4
----------

To rename a mtv1 to mtv2, type:

$ nas_volume -rename mtv1 mtv2

id = 146
name = mtv2
acl = 0
in_use = False
type = meta
volume_set = d7
disks = d7

EXAMPLE #5
-----------

To create a stripe volume named, stv1, with a size of 32768 bytes
on disk volumes d10, d12, d13, and d15, type:

$ nas_volume -name stv1 -create -Stripe 32768 d10,d12,d13,d15

id = 147
name = stv1
acl = 0
in_use = False
type = stripe
stripe_size = 32768
volume_set = d10,d12,d13,d15
disks = d10,d12,d13,d15

Where:

Value           Definition
-----           ----------

stripe_size     Specified size of the stripe volume.

EXAMPLE #6
----------

To clone mtv1, type:

$ nas_volume -Clone mtv1

id = 146
name = mtv1
acl = 0
in_use = False
type = meta
volume_set = d7
disks = d7
id = 148
name = v148
acl = 0
in_use = False
type = meta
volume_set = d8
disks = d8

EXAMPLE #7
----------

To clone the volume mtv1 and set the disk type to BCV, type:

$ /nas/sbin/rootnas_volume -Clone mtv1 -option disktype=BCV
id = 322
name = mtv1
```

```
acl = 0
in_use = False
type = meta
volume_set = d87
disks = d87
id = 323
name = v323
acl = 0
in_use = False
type = meta
volume_set = rootd99
disks = rootd99
```

EXAMPLE #8

To extend mtv1 with mtv2, type:

```
$ nas_volume -xtend mtv1 mtv2

id = 146
name = mtv1
acl = 0
in_use = False
type = meta
volume_set = d7,mtv2
disks = d7,d8
```

EXAMPLE #9

To display the size of mtv1, type:

```
$ nas_volume -size mtv1

total = 547418 avail = 547418 used = 0 ( 0% ) (sizes in MB)
```

Where:

Value	Definition
total	Total size of the volume.
avail	Amount of unused space on the volume.
used	Amount of space used on the volume.

EXAMPLE #10

To set the access control level for the metavolume mtv1, type:

```
$ nas_volume -acl 1432 mtv1

id = 125
name = mtv1
acl = 1432, owner=nasadmin, ID=201
in_use = False
type = meta
volume_set = d7,mtv2
disks = d7,d8
```

Note: The value 1432 specifies nasadmin as the owner and gives users with an access level of at least observer read access only, users with an access level of at least operator read/write access, and users with an access level of at least admin read/write/delete access.

EXAMPLE #11

To delete mtv2, type:

```
$ nas_volume -delete mtv1

id = 146
name = mtv1
acl = 1432, owner=nasadmin, ID=201
in_use = False
type = meta
volume_set = d7,mtv2
disks = d7,d8

--------------------------------------------------------------------------
Last modified: April 29 2011, 3:15 pm.
```

FS CLI Commands

This chapter lists the eNAS Command Set provided for managing, configuring, and monitoring the specified file system. The commands are prefixed with fs and appear alphabetically. The command line syntax (Synopsis), a description of the options, and examples of usage are provided for each command.

fs_ckpt	fs_dedupe	fs_dhsm
fs_group	fs_rdf	fs_timefinder

fs_ckpt

Manages checkpoints using the EMCSnapSure functionality.

SYNOPSIS

```
fs_ckpt {<fs_name>|id=<fs_id>}
  -list [-all]
| [-name <name>] -Create [-readonly {y|n}][<volume_name>]
   [-option <options>]
| [-name <name>] -Create [-readonly {y|n}][size=<integer>[T|G|M|%]]
   [pool=<pool>][storage=<system_name>][-option <options>]
| -refresh [-option <options>]
| [-name <name>] -Restore [-Force][-option <options>]
| -modify [%full=<value>][maxsavsize=<integer>[T|G|M]]
```

DESCRIPTION

The fs_ckpt command creates a checkpoint of a Production File System (PFS), lists associated checkpoints, refreshes a checkpoint to the current time, and restores a PFS back to a specific point in time using a checkpoint. Checkpoints are deleted using nas_fs

What is a checkpoint file system?

A PFS is made up of blocks. When a block within a PFS is modified, a copy containing the original contents of that block is saved to a metavolume called the SavVol. Subsequent changes made to the same block in the PFS are not copied into the SavVol. The original blocks from the PFS (in the SavVol) and the unchanged PFS blocks (that remain in the PFS) are read according to a bitmap and blockmap data tracking structure. These blocks combine to provide a complete point-in-time file system image which is called a checkpoint.

OPTIONS

-list [all]

Displays all of the associated checkpoints for the specified file system. The -all option displays system-generated Replication checkpoints in addition to checkpoints created by the user.

[-name <name>] -Create
Creates, mounts, and optionally assigns a name to the checkpoint of the PFS. The checkpoint must be unmounted prior to unmounting the PFS. Names assigned to a checkpoint cannot be all numeric. If a name is not chosen, one is assigned by default.

[-readonly {y|n}]
Specifies whether a checkpoint is read only or not. y (default) sets the checkpoint as read only; n sets the checkpoint as writeable.

[<volume_name>]
Specifies an unused metavolume for the checkpoint.

Note: A volume can be specified for only the first checkpoint of a PFS as all of the subsequent checkpoints share the same SavVol. The minimum size required for a SavVol is 64 MB. The volume size is 10 GB. However, if the PFS is less than 10 GB, the volume is the same size as the file system.

[-option <options>]
Specifies the following comma-separated options:

%full=<value>
Specifies a value as the percentage threshold permitted for the SavVol. When that value is reached, a warning is sent to the server_log and the syslog files. The Control Station acknowledges the event and automatically extends the checkpoint. The SavVol is automatically extended by 10 GB if its default %full value is reached. If the %full value is set to zero, the option is

disabled.

maxsavsize=<integer>[T|G|M]
Limits the final size to which the SavVol can be automatically extended when the high watermark value specified in %full has been reached. Automatic extension of the SavVol stops when the size of the SavVol reaches the value specified in maxsavsize. The range for maxsavsize is 64 MB to 16 TB.

automount=no
Stops the checkpoint from being automatically mounted.

[-name <name>] -Create

Creates, mounts, and optionally assigns a name to the checkpoint of the PFS. The checkpoint must be unmounted prior to unmounting the PFS. Names assigned to a checkpoint cannot be all numeric. If a name is not chosen, one is assigned by default.

[-readonly {y|n}]
Specifies whether a checkpoint is read-only or not. The default option is y. It sets the checkpoint as read-only; n sets the checkpoint as writeable.

[size=<integer>[T|G|M|%]]
Specifies a size for the checkpoint file system. Type an integer between 1 and 1024, specify T for terabytes, G for gigabytes (default) or M for megabytes. An integer representing the percentage of a file systems size can also be typed, followed by the percent sign.

[pool=<pool>]
Specifies the storage pool to be used for the checkpoint. Storage pools can either be user-defined or system-defined. The nas_pool -list command displays a listing of available pool types.

[storage=<system_name>]
Specifies the attached storage system where the checkpoint SavVol will reside.

[-option <options>]
Specifies the following comma-separated options:

%full=<value>

Specifies a value as the percentage threshold permitted for the SavVol. When that value is reached, a warning is sent to the server_log and the syslog files. The Control Station acknowledges the event and automatically extends the checkpoint. The SavVol is automatically extended by 10 GB if its default %full value is reached. If the %full value is set to zero, the option is %disabled. The default for <value> is 90 and it can be within the range of 10 %to 99.

automount=no
Stops the checkpoint from being automatically mounted.

-refresh
Initiates an immediate update of a checkpoint, thereby allowing the SavVol space to be reused. Refreshing a checkpoint does not add to the number of checkpoints of the PFS.

[-option <options>] %full=<value>
Specifies a value as the percentage threshold permitted for the metavolume. When that value is reached, a warning is sent to the server_log and the syslog files. The Control Station acknowledges the event and automatically extends the checkpoint. The SavVol is automatically extended by 10 GB if its default %full value is reached. If the %full value is set to zero, the option is

%disabled. The default for <value> is 90.

-modify
Modifies one or all of the following options:

Note: The -modify action works only on the PFS and not on the checkpoint.

[%full=<value>]
Modifies the value of the percentage threshold permitted for the metavolume.

[maxsavsize=<integer>[T|G|M]]
Modifies the final size to which the SavVol can be automatically extended, when the size specified in %full is reached.

[-name <name>] -Restore
Restores the PFS from the specified checkpoint and optionally assigns a name to the automatically created checkpoint. If a name is not chosen, one is assigned by default.

Note: As part of the restore, a new checkpoint is automatically created to capture the latest point-in-time image of the PFS. This is for protection in the event that the restored image is discarded.

[-Force]
The -Force option must be used when restoring a production file system with File-Level Retention enabled.

Caution: Forcing a restore of a production file system with File-Level Retention enabled from a checkpoint will delete or overwrite files that were written after this checkpoint was created or refreshed.

[-option <options>]
Specifies the following comma-separated option(s):

%full=<value>
Specifies a value as the percentage threshold permitted for the SavVol. When that value is reached, a warning is sent to the server_log and the syslog file. The Control Station acknowledges the event and automatically extends the checkpoint. The SavVol is automatically extended by 10 GB if its default %full value is reached. If the %full value is set to zero, the option is disabled. The <value> can be an integer between 10 and 75 (default).

automount=no
Stops the checkpoint from being automatically mounted.

SEE ALSO

Using VNX Snapsure, nas_fs, and nas_pool

STORAGE SYSTEM OUTPUT

The number associated with the storage device is dependent on the attached storage system. VNX for block displays a prefix of APM before a set of integers, for example, APM00033900124-0019. For example, EMC Symmetrix storage sy
stems display as 002804000190-003C. The outputs displayed in the examples use a VNX for block.

EXAMPLE #1

To display the checkpoint for the file system fs4, type:
$ fs_ckpt fs4 -list

id ckpt_name creation_time inuse fullmark total_savvol_used
ckpt_usage_on_savvol
1406 fs4_ckpt1 05/26/2008-16:22:19-EDT y 0% 51%
0%

```
        id    ckpt_name                          inuse  fullmark  total_savvol_used  base
ckpt_usage_on_savvol

        EXAMPLE #2
        ----------

        To display all the checkpoints including internal checkpoints
        for the file system fs4, type:

        $ fs_ckpt fs4 -list -all
        id                    ckpt_name            creation_time  inuse  fullmark  total
_savvol_used        ckpt_usage_on_savvol
        1401  root_rep_ckpt_1398_21625_1  05/26/2008-16:11:10-EDT      y       90%
         51%                          0%
        1402  root_rep_ckpt_1398_21625_2  05/26/2008-16:11:22-EDT      y       90%
         51%                          0%
        1406                   fs4_ckpt1  05/26/2008-16:22:19-EDT      y       90%
         51%                          0%
        id                   wckpt_name                            inuse  fullmark  total
_savvol_used  base  ckpt_usage_on_savvol

        EXAMPLE #3
        ----------

        To create a checkpoint of ufs1, on the volume, ssmtv1, type:

        $ fs_ckpt ufs1 -Create ssmtv1
        operation in progress (not interruptible)...id = 22
        name      = ufs1
        acl       = 0
        in_use    = True
        type      = uxfs
        worm      = off
        volume    = mtv1
        pool      =
        rw_servers= server_2
        ro_servers=
        rw_vdms   =
        ro_vdms   =
        ckpts     = ufs1_ckpt1
        stor_devs = APM00043807043-0010,APM00043807043-0014
        disks     = d7,d9
         disk=d7   stor_dev=APM00043807043-0010 addr=c0t1l0        server=server_2
         disk=d7   stor_dev=APM00043807043-0010 addr=c16t1l0       server=server_2
         disk=d9   stor_dev=APM00043807043-0014 addr=c0t1l4        server=server_2
         disk=d9   stor_dev=APM00043807043-0014 addr=c16t1l4       server=server_2

        id        = 24
        name      = ufs1_ckpt1
        acl       = 0
        in_use    = True
        type      = ckpt
        worm      = off
        volume    = vp132
        pool      =
        member_of =
        rw_servers=
        ro_servers= server_2
        rw_vdms   =
        ro_vdms   =
        checkpt_of= ufs1 Wed Oct 13 18:01:04 EDT 2004
        used      = 0%
        full(mark)= 90%
        stor_devs = APM00043807043-0011,APM00043807043-0017
        disks     = d12,d15
         disk=d12   stor_dev=APM00043807043-0011 addr=c16t1l1      server=server_2
         disk=d12   stor_dev=APM00043807043-0011 addr=c0t1l1       server=server_2
         disk=d15   stor_dev=APM00043807043-0017 addr=c16t1l7      server=server_2
         disk=d15   stor_dev=APM00043807043-0017 addr=c0t1l7       server=server_2

        Where:
```

Value	Definition
id	Automatically assigned ID of a file system or the checkpoint.
name	Name assigned to the file system or the checkpoint
acl	Access control value for a file system. See nas_acl.
in_use	If a file system is registered into the mount table of a Data Mover.
type	Type of file system. See -list for a description of the types.
worm	Whether the File-Level Retention feature is enabled.
volume	Volume on which a file system resides.
pool	Storage pool for the file system.
member_of	Group to which the file system belongs.
rw_servers	Servers with read-write access to a file system.
ro_servers	Servers with read-only access to a file system.
rw_vdms	VDM servers with read-write access to a file system.
ro_vdms	VDM servers with read-only access to a file system.
ckpts	Associated checkpoints for the file system.
checkpt_of	Name of the PFS related to the existing checkpoints.
used	Percentage of SavVol space used by the checkpoints of the PFS.
full(mark)	SavVol usage point which, when reached, sends a warning message to the system log, and auto-extends the SavVol as system space permits.
stor_devs	Storage system devices associated with a file system.
disks	Disks on which the metavolume resides.

EXAMPLE #4

To create a checkpoint of ufs1 named ufs1_ckpt2 with a size of 2 GB using the clar_r5_performance pool, with the specified storage system, with the %full set to 95, type:

```
$ fs_ckpt ufs1 -name ufs1_ckpt2 -Create size=2G pool=clar_r5_performance
storage=APM00043807043 -option %full=95
operation in progress (not interruptible)...id = 27
name      = ufs1
acl       = 0
in_use    = True
type      = uxfs
worm      = off
volume    = mtv1
pool      =
rw_servers= server_2
ro_servers=
rw_vdms   =
ro_vdms   =
ckpts     = ufs1_ckpt1,ufs1_ckpt2
stor_devs = APM00043807043-0010,APM00043807043-0014
disks     = d7,d9
disk=d7     stor_dev=APM00043807043-0010 addr=c0t1l0         server=server_2
disk=d7     stor_dev=APM00043807043-0010 addr=c16t1l0        server=server_2
disk=d9     stor_dev=APM00043807043-0014 addr=c0t1l4         server=server_2
disk=d9     stor_dev=APM00043807043-0014 addr=c16t1l4        server=server_2

id        = 30
name      = ufs1_ckpt2
acl       = 0
in_use    = True
type      = ckpt
worm      = off
volume    = vp145
pool      =
member_of =
rw_servers=
ro_servers= server_2
rw_vdms   =
ro_vdms   =
checkpt_of= ufs1 Wed Nov 10 14:00:20 EST 2004
used      = 0%
full(mark)= 95%
```

```
stor_devs = APM00043807043-0011,APM00043807043-0017
disks     = d12,d15
disk=d12    stor_dev=APM00043807043-0011 addr=c16t1l1       server=server_2
disk=d12    stor_dev=APM00043807043-0011 addr=c0t1l1        server=server_2
disk=d15    stor_dev=APM00043807043-0017 addr=c16t1l7       server=server_2
disk=d15    stor_dev=APM00043807043-0017 addr=c0t1l7        server=server_2
```

EXAMPLE #3 provides a description of command output.

EXAMPLE #5

To create a checkpoint of ufs2 named ufs2_ckpt1 with a size of 2 GB by using the clar_mapped_pool VNX mapped pool, with the specified system, with the %full set to 95, type:

```
$ fs_ckpt ufs2 -name ufs2_ckpt1 -Create size=2G pool=clar_mapped_pool
storage=APM00043807043 -option %full=95

operation in progress (not interruptible)...id = 435
name = ufs2
acl = 0
in_use = True
type = uxfs
worm = off
volume = v731
pool = clar_mapped_pool
member_of = root_avm_fs_group_50
rw_servers= server_2
ro_servers=
rw_vdms =
ro_vdms =
auto_ext = no,thin=no
fast_clone_level = 1
deduplication = Off
thin_storage = False
tiering_policy = N/A/Optimize Pool
compressed= False
mirrored = False
ckpts = ufs2_ckpt1
stor_devs =
FNM00103400314-0036,FNM00103400314-0037,FNM00103400314-0038,FNM00103400314-0039
disks = d60,d61,d62,d63
disk=d60 stor_dev=FNM00103400314-0036 addr=c0t1l0 server=server_2
disk=d60 stor_dev=FNM00103400314-0036 addr=c16t1l0 server=server_2
disk=d61 stor_dev=FNM00103400314-0037 addr=c0t1l1 server=server_2
disk=d61 stor_dev=FNM00103400314-0037 addr=c16t1l1 server=server_2
disk=d62 stor_dev=FNM00103400314-0038 addr=c0t1l2 server=server_2
disk=d62 stor_dev=FNM00103400314-0038 addr=c16t1l2 server=server_2
disk=d63 stor_dev=FNM00103400314-0039 addr=c0t1l3 server=server_2
disk=d63 stor_dev=FNM00103400314-0039 addr=c16t1l3 server=server_2

id = 438
name = ufs2_ckpt1
acl = 0
in_use = True
type = ckpt
worm = off
volume = vp735
pool = clar_mapped_pool
member_of =
rw_servers=
ro_servers= server_2
rw_vdms =
ro_vdms =
checkpt_of= ufs2 Fri Jan 4 01:43:20 EST 2013
deduplication = Off
thin_storage = False
tiering_policy = N/A/Optimize Pool
compressed= False
mirrored = False
```

```
used = 13%
full(mark)= 95%
stor_devs =
FNM00103400314-0036,FNM00103400314-0037,FNM00103400314-0038,FNM00103400314-0039
disks = d60,d61,d62,d63
disk=d60 stor_dev=FNM00103400314-0036 addr=c0t1l0 server=server_2
disk=d60 stor_dev=FNM00103400314-0036 addr=c16t1l0 server=server_2
disk=d61 stor_dev=FNM00103400314-0037 addr=c0t1l1 server=server_2
disk=d61 stor_dev=FNM00103400314-0037 addr=c16t1l1 server=server_2
disk=d62 stor_dev=FNM00103400314-0038 addr=c0t1l2 server=server_2
disk=d62 stor_dev=FNM00103400314-0038 addr=c16t1l2 server=server_2
disk=d63 stor_dev=FNM00103400314-0039 addr=c0t1l3 server=server_2
disk=d63 stor_dev=FNM00103400314-0039 addr=c16t1l3 server=server_2
```

Where:

Value	Definition
thin_storage	Indicates whether the VNX for block system uses thin provisioning. Values are: True, False, Mixed.
tiering_policy	Indicates the tiering policy is in effect. If the initial tier and the tiering policy are the same, the values are: Auto-Tier, Highest Available Tier, Lowest Available Tier. If the initial tier and the tiering policy are not the same, the values are: Auto-Tier/No Data Movement, Highest Available Tier/No Data Movement, Lowest Available Tier/No Data Movement.
compressed	Indicates whether data is compressed. Values are True, False, Mixed. (indicates some of the LUNs, but not all, are compressed)
mirrored	Indicates whether the disk is mirrored.

EXAMPLE #6

To create a writeable checkpoint of baseline checkpoint ufs1_ckpt1, type:

```
$ fs_ckpt ufs1_ckpt1 -Create -readonly n
operation in progress (not interruptible)...id = 45
name      = ufs1_ckpt1
acl       = 0
in_use    = False
type      = ckpt
worm      = off
volume    = vp145
pool      = clar_r5_performance
member_of =
rw_servers=
ro_servers=
rw_vdms   =
ro_vdms   =
checkpt_of= ufs1 Tue Nov  6 14:56:43 EST 2007
ckpts     = ufs1_ckpt1_writeable1
used      = 38%
full(mark)= 90%
stor_devs =
APM00042000814-0029,APM00042000814-0024,APM00042000814-0021,APM000420
00814-001C
disks     = d34,d17,d30,d13
id        = 46
name      = ufs1_ckpt1_writeable1
acl       = 0
in_use    = True
type      = wckpt
worm      = off
volume    = vp145
```

```
pool      = clar_r5_performance
member_of =
rw_servers= server_2
ro_servers=
rw_vdms   =
ro_vdms   =
checkpt_of= ufs1
baseline_ckpt = ufs1_ckpt1 Tue Nov  6 14:56:43 EST 2007
used      = 38%
full(mark)= 90%
stor_devs =
APM00042000814-0029,APM00042000814-0024,APM00042000814-0021,APM000420
00814-001C
disks     = d34,d17,d30,d13
 disk=d34   stor_dev=APM00042000814-0029 addr=c16t219        server=server_2
 disk=d34   stor_dev=APM00042000814-0029 addr=c32t219        server=server_2
 disk=d34   stor_dev=APM00042000814-0029 addr=c0t219         server=server_2
 disk=d34   stor_dev=APM00042000814-0029 addr=c48t219        server=server_2
 disk=d17   stor_dev=APM00042000814-0024 addr=c0t214         server=server_2
 disk=d17   stor_dev=APM00042000814-0024 addr=c48t214        server=server_2
 disk=d17   stor_dev=APM00042000814-0024 addr=c16t214        server=server_2
 disk=d17   stor_dev=APM00042000814-0024 addr=c32t214        server=server_2
 disk=d30   stor_dev=APM00042000814-0021 addr=c16t211        server=server_2
 disk=d30   stor_dev=APM00042000814-0021 addr=c32t211        server=server_2
 disk=d30   stor_dev=APM00042000814-0021 addr=c0t211         server=server_2
 disk=d30   stor_dev=APM00042000814-0021 addr=c48t211        server=server_2
 disk=d13   stor_dev=APM00042000814-001C addr=c0t1l12        server=server_2
 disk=d13   stor_dev=APM00042000814-001C addr=c48t1l12       server=server_2
 disk=d13   stor_dev=APM00042000814-001C addr=c16t1l12       server=server_2
 disk=d13   stor_dev=APM00042000814-001C addr=c32t1l12       server=server_2
```

Where:

Value	Definition
baseline_ckpt	Name of the read-only checkpoint from which the writeable checkpoint is created.

EXAMPLE #3 provides a description of command output.

EXAMPLE #7

To list checkpoints for ufs1, type:

```
$ fs_ckpt ufs1 -list
id  ckpt_name  creation_time           inuse full(mark) used
29  ufs1_ckpt1 11/04/2004-14:54:06-EST  n     95%        0%
30  ufs1_ckpt2 11/10/2004-14:00:20-EST  y     95%        0%
```

Where:

Value	Definition
id	Automatically assigned ID of a file system or checkpoint.
ckpt_name	Name assigned to the checkpoint.
creation_time	Date and time the checkpoint was created.
inuse	If a checkpoint is registered into the mount table of a Data Mover.
full(mark)	SavVol-usage point which, when reached, sends a warning message to the system log, and auto-extends the SavVol as system space permits.
used	Percentage of SavVol space used by checkpoints of the PFS.

EXAMPLE #8

To refresh ufs1_ckpt2 using the %full at 85, type:

```
$ fs_ckpt ufs1_ckpt2 -refresh -option %full=85
operation in progress (not interruptible)...id = 30
name      = ufs1_ckpt2
acl       = 0
```

```
in_use    = True
type      = ckpt
worm      = off
volume    = vp145
pool      =
member_of =
rw_servers=
ro_servers= server_2
rw_vdms   =
ro_vdms   =
checkpt_of= ufs1 Wed Nov 10 14:02:59 EST 2004
used      = 0%
full(mark)= 85%
stor_devs = APM00043807043-0011,APM00043807043-0017
disks     = d12,d15
disk=d12    stor_dev=APM00043807043-0011 addr=c16t1l1       server=server_2
disk=d12    stor_dev=APM00043807043-0011 addr=c0t1l1        server=server_2
disk=d15    stor_dev=APM00043807043-0017 addr=c16t1l7       server=server_2
disk=d15    stor_dev=APM00043807043-0017 addr=c0t1l7        server=server_2
```

EXAMPLE #3 provides a description of command output.

EXAMPLE #9

Using root command, to restore ufs1_ckpt2 and capture the latest point-in-time image of the PFS on ufs1_ckpt3, type:

```
$ /nas/sbin/rootfs_ckpt ufs1_ckpt2 -name ufs1_ckpt3 -Restore
operation in progress (not interruptible)...id = 30
name      = ufs1_ckpt2
acl       = 0
in_use    = True
type      = ckpt
worm      = off
volume    = vp145
pool      =
member_of =
rw_servers=
ro_servers= server_2
rw_vdms   =
ro_vdms   =
checkpt_of= ufs1 Wed Nov 10 14:02:59 EST 2004
used      = 0%
full(mark)= 90%
stor_devs = APM00043807043-0011,APM00043807043-0017
disks     = d12,d15
disk=d12    stor_dev=APM00043807043-0011 addr=c16t1l1       server=server_2
disk=d12    stor_dev=APM00043807043-0011 addr=c0t1l1        server=server_2
disk=d15    stor_dev=APM00043807043-0017 addr=c16t1l7       server=server_2
disk=d15    stor_dev=APM00043807043-0017 addr=c0t1l7        server=server_2
```

EXAMPLE #3 provides a description of command output.

EXAMPLE #10

To modify the %full value of the SavVol associated with the file system ufs1 and set it to 95, type:

```
$ fs_ckpt ufs1 -modify %full=95
operation in progress (not interruptible)...id        = 33
name      = ufs1
acl       = 0
in_use    = True
type      = uxfs
worm      = off
volume    = vp145
pool      =
rw_servers= server_2
ro_servers=
```

```
rw_vdms   =
ro_vdms   =
auto_ext  = no,virtual_provision=no
ckpts     = wipckpt
stor_devs = APM00062400708-0014,APM00062400708-0016
disks     = d26,d27
 disk=d26   stor_dev=APM00062400708-0014 addr=c0t1l4         server=server_2
 disk=d26   stor_dev=APM00062400708-0014 addr=c16t1l4        server=server_2
 disk=d27   stor_dev=APM00062400708-0016 addr=c0t1l6         server=server_2
 disk=d27   stor_dev=APM00062400708-0016 addr=c16t1l6        server=server_2
```

EXAMPLE #11

To modify the maxsavsize value of the SavVol associated with the file system ufs1 and set it to 65 GB, type:

```
$ fs_ckpt ufs1 -modify maxsavsize=65G
operation in progress (not interruptible)...id        = 33
name      = ufs1
acl       = 0
in_use    = True
type      = uxfs
worm      = off
volume    = vp145
pool      =
rw_servers= server_2
ro_servers=
rw_vdms   =
ro_vdms   =
auto_ext  = no,virtual_provision=no
ckpts     = wipckpt
stor_devs = APM00062400708-0014,APM00062400708-0016
disks     = d26,d27
 disk=d26   stor_dev=APM00062400708-0014 addr=c0t1l4         server=server_2
 disk=d26   stor_dev=APM00062400708-0014 addr=c16t1l4        server=server_2
 disk=d27   stor_dev=APM00062400708-0016 addr=c0t1l6         server=server_2
 disk=d27   stor_dev=APM00062400708-0016 addr=c16t1l6        server=server_2
```

DIAGNOSTICS

fs_ckpt returns one of the following return codes:

```
        0 - Command completed successfully
        1 - Usage error
        2 - Invalid object error
        3 - Unable to acquire lock
        4 - Permission error
        5 - Communication error
        6 - Transaction error
        7 - Dart error
        8 - Backend error
```

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fs_dedupe
Manages filesystem deduplication state.

SYNOPSIS
----

fs_dedupe {

-list
 | -info {-all|<fs_name>|id=<fs_id>}
 | -modify {<fs_name>|id=<fs_id>} [-state
   {off|suspended|on}][-minimum_scan_interval <days>][-minimum_size <KB>]
  [-maximum_size <MB>][-access_time <days>][-modification_time <days>]
  [-case_sensitive {yes|no}][-pathname_exclude_list <path_list>]
  [-file_ext_exclude_list <ext_list>][-duplicate_detection_method
  {sha1|byte|off}][-savvol_threshold <percent>][-backup_data_threshold
  <percent>][-cifs_compression_enabled {yes|no}] [-compression_method{fast|deep}]
 | -clear {<fs_name>|id=<fs_id>}[-minimum_scan_interval][-minimum_size]
  [-maximum_size][-access_time][-modification_time][-case_sensitive]
  [-pathname_exclude_list][-file_ext_exclude_list]
  [-duplicate_detection_method][-savvol_threshold]
  [-backup_data_threshold][-cifs_compression_enabled][-compression_method]
 | -default {
   -info {<mover_name>|-all}
 | -set {<mover_name>|-all}[-minimum_scan_interval <days>]
      [-minimum_size<KB>][-maximum_size <MB>][-access_time
      <days>][-modification_time <days>][-case_sensitive
      {yes|no}][-file_ext_exclude_list <ext_list>] [-duplicate_detection_method
      {sha1|byte|off}][-savvol_threshold <percent>][-cpu_usage_low_watermark
      <percent>] [-cpu_usage_high_watermark <percent>][-backup_data_threshold
      <percent>] [-cifs_compression_enabled {yes|no}]
 | -clear {<mover_name>|-all}
   [-minimum_scan_interval][-minimum_size][-maximum_size][-access_time]
   [-modification_time][-case_sensitive][-file_ext_exclude_list]
   [-duplicate_detection_method][-savvol_threshold]
   [-cpu_usage_low_watermark][-cpu_usage_high_watermark] [-backup_data_threshold
   <percent>][-cifs_compression_enabled]
   }
 }

DESCRIPTION
----

fs_dedupe allows the VNX administrator to enable, suspend, and undo all deduplication processing on a filesystem or a Data Mover. The Data Mover settings are the global settings that can be used for both the Data Mover and the filesystem. If a user sets a value for a specific filesystem, then that value overrides the Data Mover global value. If a user clears a value set for a specific filesystem, then that value is reset to the Data Mover global value.

OPTIONS

-list
Lists all deduplication-enabled filesystems on the VNX.

-info {-all|<fs_name>|id=<fs_id>}
Lists the existing filesystems and provides information on the state of deduplication processing.

-all
Lists all filesystems and provides detailed information on the state of deduplication processing.

<fs_name>
Lists the filesystem information for the specified filesystem name.

id=<fs_id>
Lists the filesystem information for the specified identifier.

The filesystem state and status information displayed includes:

If the state is off and the status is not reduplicating:
- ID
- Name
- Deduplication state

If the state is off and the status is reduplicating:
- ID
- Name
- Deduplication state
- Progress information (the percentage of files scanned)

If the state of the filesystem is on or suspended, and the status is Idle or Scanning:
- ID
- Name
- Reduplication state
- Status
- The percentage of files scanned
- Last system scan time
- Number of files scanned
- Number of files deduplicated
- The percentage of files deduplicated
- File system capacity
- Logical data size
- Percentage of filesystem usage
- Space saved (in MB and percent)

-modify {<fs_name>|id=<fs_id>} [-state {off|suspended|on}]
Modifies the deduplication state of the filesystem for each specified filesystem identifier or filesystem name. The state can be set to off, on, or suspended.

[-minimum_scan_interval <days>]
Defines the minimum number of days between completing one scan of a filesystem and before scanning the same filesystem again. The values range from 1 to 365 and the default value is 7 days.

[-minimum_size <KB>]
Defines the file size in KB that limits deduplication. File sizes equal to this value or smaller will not be deduplicated. Setting this value to zero disables it. This value should not be set lower than 24 KB. The values range from 0 to 1000 and the default value is 24 KB.

[-maximum_size <MB>]
Defines the file size in MB of the largest file to be processed for deduplication. Files larger than this size in MB will not be deduplicated. Setting this value to zero disables it. The values range from 0 to 8388608 and the default value is 8388608 MB.

[-access_time <days>]
Defines the minimum required file age in days based on read access time. Files that have been read within the specified number of days will not be deduplicated. This setting does not apply to files with an FLR locked state. Setting this value to zero disables it. The values range from 0 to 365 and the default value is 15 days.

[-modification_time <days>]
Defines the minimum required file age in days based on modification time. Files updated within the specified number of days will not be deduplicated. Setting this value to zero disables it. The values range from 0 to 365 and the default value is 15 days.

[-case_sensitive {yes|no}]
Defines whether case-sensitive (for NPS environments) or case-insensitive (for CIFS environments) string comparisons will be used during scans. By default, case-insensitive comparisons will be done to be consistent for CIFS

environments. The default value is zero (false).

[-pathname_exclude_list <path_list>]
This is a filesystem setting only (no global setting). It is empty by default.

Defines a semicolon-delimited list of relative pathnames, in UTF-8 format, to be excluded from deduplication. Any directory below a specified pathname will be excluded from deduplication. You can specify a maximum of 10 pathnames and each one can be up to 1024 bytes. The default value is ' ' (empty).

[-file_ext_exclude_list <ext_list>]
Specifies a colon-delimited list of filename extensions to be excluded from deduplication. Each extension must include the leading dot. The default value is ' ' (empty).

[-duplicate_detection_method {sha1|byte|off}]
- 0 (off) - This means that duplicate data detection is disabled. With this setting, every deduplicated file is considered unique and the only space savings made are accomplished with compression.
- 1 (sha1) - The SHA-1 hash is used to detect duplicate data. It is faster than a byte comparison. This is the default method.
- 2 (byte) - This will use a byte-by-byte comparison to detect duplicate data. This adds considerable overhead especially for large files.

[-savvol_threshold <percent>]
Represents the percentage of the configured save volume (SavVol) auto extension threshold that can be used during deduplication. When the specified amount of SavVol is used, deduplication stops on this filesystem. By default, this value is 90 percent and the SavVol auto extension is also 90 percent; this option will apply when the SavVol is 81 percent full (90 * 90). Setting this value to zero disables it. The values range from 0 to 100.

Warning: If you set the SavVol threshold option to 0 to disable it, be aware that the SavVol may grow up to the size of the compressed version of the data, consuming disk space that cannot be reclaimed unless you delete all checkpoints.

[-backup_data_threshold <percent>]
Indicates the full percentage that a deduplicated file has to be below in order to trigger space-reduced backups for NDMP. For example, when set to 90, any deduplicated file whose physical size (compressed file plus changed blocks) is greater than 90 percent of the logical size of the file will have the entire file data backed up without attempting to back it up in a space-reduced format. Setting this value to zero disables it. The values range from 0 to 200 and the default value is 90 percent.

[-cifs_compression_enabled {yes|no}]
This option controls whether CIFS compression is allowed. When the default is yes, enable CIFS compression is allowed. When set to yes and the deduplication state of the filesystem is either on or suspended, then CIFS compression is enabled. If the deduplication state is either off or in the process of being turned off, then CIFS compression is not allowed, regardless of whether this option is set to yes.

[-compression_method {fast|deep}]
Indicates whether the compression algorithm is set to fast (default setting) or deep. This option is valid for VNX systems that use version 7.1 and later. You can set this value for filesystems only. You cannot set it as a Data Mover global value.
The fast option is the default compression algorithm that achieves the original compression ratios and performance.
The deep option is the compression algorithm that achieves space savings up to 30% greater than the fast method. For example, if a file is 50% compressible, then the deep algorithm can compress the same file up to 65%. However, the compression and decompression time when using this deep option is longer than when using the fast option. You obtain more storage space at the cost of slower access. Selecting this deep compression method applies only to new files that are subsequently compressed, and not to existing compressed files.

When using VNX Replicator, VNX systems that use version 7.0 and earlier cannot read the deep compression format and will return an I/O error if a read operation is attempted. Select the deep compression format only if downstream replication sessions are using compatible software or are scheduled to be upgraded soon.

-clear {<fs_name>|id=<fs_id>}
Sets the filesystem setting back to the Data Mover setting, which is the default setting.

[-minimum_scan_interval]
Defines the minimum number of days between completing one scan of a filesystem and before scanning the same filesystem again. The values range from 1 to 365 and the default value is 7 days.

[-minimum_size]
Defines the file size in KB that limits deduplication. File sizes equal to this value or smaller will not be deduplicated. File sizes greater than this value will be candidates for deduplication. Setting this value to zero disables it. This value should not be set lower than 24 KB. The values range from 0 to 1000 and the default value is 24 KB.

[-maximum_size]
Defines the file size in MB of the largest file to be processed for deduplication. Files larger than this size in MB will not be deduplicated. Setting this value to zero disables it. The values range from 0 to 8388608 and the default value is 8388608 MB.

[-access_time]
Defines the minimum required file age in days based on read access time. Files that have been read within the specified number of days will not be deduplicated. This setting does not apply to files with an FLR locked state. Setting this value to zero disables it. The values range from 0 to 365 and the default value is 15 days.

[-modification_time]
Defines the minimum required file age in days based on modification time. Files updated within the specified number of days will not be deduplicated. Setting this value to zero disables it. The values range from 0 to 365 and the default value is 15 days.

[-case_sensitive]
Defines whether case-sensitive (for NPS environments) or case-insensitive (for CIFS environments) string comparisons will be used during scans. By default, case insensitive comparisons will be done to be consistent for CIFS environments. The default value is zero (false).

[-pathname_exclude_list]
This is a filesystem setting only (no global setting).
Specifies a semicolon-delimited list of relative path names, in UTF-8 format, to be excluded from deduplication. Any directory below a specified path name will be excluded from deduplication. You can specify a maximum of 10 path names and each one can be up to 1024 bytes. The default value is ' ' (empty).

[-file_ext_exclude_list]
Specifies a colon-delimited list of filename extensions to be excluded from deduplication. Each extension must include the leading dot. The default value is ' ' (empty).

[-duplicate_detection_method {sha1|byte|off}]
- 0 (off) - This means that duplicate data detection is disabled. With this setting, every deduplicated file is considered unique and the only space savings made are accomplished with compression.
- 1 (sha1) - The SHA-1 hash is used to detect duplicate data. It is faster than a byte comparison. This is the default method.
- 2 (byte) - This will use a byte-by-byte comparison to detect duplicate data. This adds considerable overhead especially for large files.

[-savvol_threshold]

Represents the percentage of the configured save volume (SavVol) auto extension threshold that can be used during deduplication. After the specified amount of SavVol is used, deduplication stops on this filesystem. By default, this value is 90 percent and the SavVol auto extension is also 90 percent; this option will apply when the SavVol is 81 percent full (90 * 90). Setting this value to zero disables it. The values range from 0 to 100.

[-backup_data_threshold]
Indicates the full percentage that a deduplicated file has to be below in order to trigger space-reduced backups for NDMP. For example, when set to 90, any deduplicated file whose physical size (compressed file plus changed blocks)

is greater than 90 percent of the logical size of the file will have the entire

file data backed up without attempting to back it up in a space-reduced format.

Setting this value to zero disables it. The values range from 0 to 200 and the default value is 90 percent.

[-cifs_compression_enabled]
This option controls whether CIFS compression is allowed. The default is yes, enable CIFS compression. When set to yes and the deduplication state of the filesystem is either on or suspended, then CIFS compression is allowed. If the deduplication state is either off or in the process of being turned off, then CIFS compression is not allowed, regardless of whether this option is set to yes.

[-compression_method]
This is a filesystem setting only (no global setting). Identifies the compression algorithm: fast (default) or deep.

| -default {-info {<mover_name>|-all}|-set {<mover_name>|-all}
Manages the Data Mover settings. The -set option determines the Data Mover settin
gs.

[-minimum_scan_interval <days>]
Defines the minimum number of days between completing one scan of a file system and before scanning the same filesystem again. The values range from 1 to 365 and the default value is 7 days.

[-minimum_size <KB>]
Defines the file size in KB that limits deduplication. File sizes equal to this value or smaller will not be deduplicated. File sizes greater than this value will be candidates for deduplication. Setting this value to zero disables it. This value should not be set lower than 24 KB. The values range from 0 to 1000 and the default value is 24 KB.

[-maximum_size <MB>]
Defines the file size in MB of the largest file to be processed for deduplication. Files larger than this size in MB will not be deduplicated. Setting this value to zero disables it. The values range from 0 to 8388608 and the default value is 8388608 MB.

[-access_time <days>]
Defines the minimum required file age in days based on read access time. Files that have been read within the specified number of days will not be deduplicated. This setting does not apply to files with an FLR locked state. Setting this value to zero disables it. The values range from 0 to 365 and the default value is 15 days.

[-modification_time <days>]
The minimum required file age in days based on modification time. Files updated within the specified number of days will not be deduplicated. Setting this value to zero disables it. The values range from 0 to 365 and the default value is 15 days.

[-case_sensitive {yes|no}]
Defines whether case-sensitive (for NPS environments) or case-insensitive (for CIFS environments) string comparisons will be used during scans. By default, case insensitive comparisons will be done to be consistent for CIFS environments. The default value is zero (false).

[-file_ext_exclude_list <ext_list>]
Specifies a colon-delimited list of filename extensions to be excluded from deduplication. Each extension must include the leading dot.
The default value is ' ' (empty).

[-duplicate_detection_method {sha1|byte|off}]
0 (off) - This means that duplicate data detection is disabled. With this setting, every deduplicated file is considered unique and the only space savings made are accomplished with compression.
1 (sha1) - The SHA-1 hash is used to detect duplicate data. It is faster than a byte comparison. This is the default method.
2 (byte) - This will use a byte-by-byte comparison to detect duplicate data. This adds considerable overhead especially for large files.

[-savvol_threshold <percent>]
Represents the percentage of the configured save volume (SavVol) auto extension threshold that can be used during deduplication. Once the specified amount of SavVol is used, deduplication stops on this filesystem. By default, this value is 90 percent and the SavVol auto-extension is also 90 percent; this option will apply when the SavVol is 81 percent full (90 * 90). Setting this value to zero disables it. The values range from 0 to 100.

Warning: If you set the SavVol threshold option to 0 to disable it, be aware that the SavVol may grow up to the size of the compressed version of the data, consuming disk space that cannot be reclaimed unless you delete all checkpoints.

[-cpu_usage_low_watermark <percent>]
Defines the average percent of CPU usage that can be used during the deduplication process at which full throttle mode is re-entered. The values range from 0 to 100 and the default value is 40 percent. This is a global setting only.

[-cpu_usage_high_watermark <percent>]
Defines the average percent of CPU usage that can be used during the deduplication process which should trigger a slow throttle mode. The system starts in full throttle mode. The values range from 0 to 100 and the default value is 75 percent. This is a global setting only.

[-backup_data_threshold <percent>]
Defines the full percentage that a deduplicated file has to be below in order to trigger space-reduced backups for NDMP. For example, when set to 90, any deduplicated file whose physical size (compressed file plus changed blocks) is greater than 90 percent of the logical size of the file will have the entire file data backed up without attempting to back it up in a space-reduced format. Setting this value to zero disables it. The values range from 0 to 200 and the default value is 90 percent.

[-cifs_compression_enabled {yes|no}]
This option controls whether CIFS compression is allowed. The default is yes, enable CIFS compression. When set to yes and the deduplication state of the filesystem is either on or suspended, then CIFS compression is allowed. If the deduplication state is either off or in the process of being turned off, then CIFS compression is not allowed, regardless of whether this option is set to yes.

| -clear {<mover_name>|-all}
The -clear option sets the global setting back to the default value.

[-minimum_scan_interval]
Defines the minimum number of days between completing one scan of a file system and before scanning the same file system again. The values range from

to 365 and the default value is 7 days.

[-minimum_size]
Defines the file size in KB that limits deduplication. File sizes equal to this value or smaller will not be deduplicated. File sizes greater than this

value will be candidates for deduplication. Setting this value to zero disables it. This value should not be set lower than 24 KB. The values range from 0 to 1000 and the default value is 24 KB.

[-maximum_size]
Defines the file size in MB of the largest file to be processed for deduplication. Files larger than this size in MB will not be deduplicated. Setting this value to zero disables it. The values range from 0 to 8388608 and the default value is 8388608 MB.

[-access_time]
Defines the minimum required file age in days based on read access time. Files that have been read within the specified number of days will not be deduplicated. This setting does not apply to files with an FLR locked state. Setting this value to zero disables it. The values range from 0 to 365 and the default value is 15 days.

[-modification_time]
Defines the minimum required file age in days based on modification time. Files updated within the specified number of days will not be deduplicated. Setting this value to zero disables it. The values range from 0 to 365 and the default value is 15 days.

[-case_sensitive]
Defines whether case-sensitive (for NPS environments) or case-insensitive (for CIFS environments) string comparisons will be used during scans. By default, case insensitive comparisons will be done to be consistent for CIFS environments. The default value is zero (false).

[-file_ext_exclude_list]
Specifies a colon-delimited list of filename extensions to be excluded from deduplication. Each extension must include the leading dot.
The default value is ' ' (empty).

[-duplicate_detection_method]
- 0 (off) - This means that duplicate data detection is disabled. With this setting, every deduplicated file is considered unique and the only space savings made are accomplished with compression.
- 1 (sha1) - The SHA-1 hash is used to detect duplicate data. It is faster than a byte comparison. This is the default method.
- 2 (byte) - This will use a byte-by-byte comparison to detect duplicate data. This adds considerable overhead especially for large files.

[-savvol_threshold]
Represents the percentage of the configured save volume (SavVol) auto extension threshold that can be used during deduplication. After the specified amount of SavVol is used, deduplication stops on this filesystem. By default, this value is 90 percent and the SavVol auto extension is also 90 percent; this option will apply when the SavVol is 81 percent full (90 * 90). Setting this value to zero disables it. The values range from 0 to 100.

[-cpu_usage_low_watermark]
Specifies the average percent of CPU usage that can be used during the deduplication process at which full throttle mode is re-entered. The values range from 0 to 100 and the default value is 25 percent. This is a global setting only.

[-cpu_usage_high_watermark]
Specifies the average percent of CPU usage that can be used during the deduplication process which should trigger a slow throttle mode. The system starts in full throttle mode. The values range from 0 to 100 and the default value is 75 percent. This is a global setting only.

[-backup_data_threshold <percent>]

Specifies the full percentage that a deduplicated file has to be below in order to trigger space-reduced backups for NDMP. For example, when set to 90, any deduplicated file whose physical size (compressed file plus changed block
s)
is greater than 90 percent of the logical size of the file will have the enti
re
file data backed up without attempting to back it up in a space-reduced forma
t.
Setting this value to zero disables it. The values range from 0 to 200 and th
e
default value is 90 percent.

[-cifs_compression_enabled]
This option controls whether CIFS compression is allowed. When the default is yes, enable CIFS compression. When set to yes and the deduplication state of the filesystem is either on or suspended, then CIFS compression is allowed. If the deduplication state is either off or in the process of being turned off, then CIFS compression is not allowed, regardless of whether this option is set to yes.

SEE ALSO: nas_fs

EXAMPLE #1

To list the filesystems and their deduplication states, type:

```
$ fs_dedupe -list
id   name    state    status time_of_last      original_data  usage space_saved
                             _scan             _size

141 ranap1-  Suspended       Wed Nov 12        5 MB           0%     0 MB(0%)
    replica                  09:04:45 EST 2008

104 ds850gb  On       Idle   Fri Nov 21        875459MB       84%    341590 MB
   _replica1                 10:31:15 EST 2008                       (39%)

495 cworm    On       Idle   Thu Nov 20        3 MB           0%     0 MB(0%)
                             09:14:09 EST 2008

33 chrisfs1  On       Idle   Sat Nov 22        1100 MB        18%    424 MB
                             10:04:33 EST 2008                        (38%)
```

Where:

Value	Definition
id	Filesystem identifier
name	Name of the filesystem
state	Deduplication state of the filesystem. The file data is transferred to the storage which performs the deduplication and compression on the data The states are: On-- Deduplication on the filesystem is enabled. Suspended-- Deduplication on the filesystem is suspended. Deduplication does not perform any new space reduction but the existing files that were reduced in space remain the same. Off-- Deduplication on the filesystem is disabled. Deduplication does not perform any new space reduction and the data is now reduplicated.
status	Current state of the deduplication enabled file system. The progress statuses are: Idle-- Deduplication process is currently idle. Scanning-- Filesystem is being scanned for deduplication. It displays the percentage of scanned files in the filesystem. Reduplicating-- Filesystem files are being reduplicated from the deduplicated files. It displays the percentage of reduplicated files.
time_of_last_scan	Time when the filesystem was last scanned

```
original_data_size      Original size of the filesystem before deduplication
usage                   Current space usage of the filesystem
space_saved             Filesystem space saved after deduplication

EXAMPLE #2
----------

To list the filesystems and provide detailed reports on the state of the
deduplication processing, type:

$ fs_dedupe -info -all
Id                            = 53
Name                          = svr2fs1
Deduplication                 = Off
File system parameters:
   Case Sensitive             = no
   Duplicate Detection Method  = sha1
   Access Time                = 15
   Modification Time          = 15
   Minimum Size               = 24 KB
   Maximum Size               = 8388608 MB
   File Extension Exclude List =
   Minimum Scan Interval      = 7
   Savevol Threshold          = 90
   Backup Data Threshold      = 90
   Cifs Compression Enabled   = yes
   Pathname Exclude List      =
   Compression Method         = fast

Id                            = 2040
Name                          = server_2_fsltest2
Deduplication                 = Suspended
As of the last file system scan (Mon Aug 17 11:33:38 EDT 2009):
   Files scanned              = 4
   Files deduped              = 3 (75% of total files)
   File system capacity       = 2016 MB
   Original data size         = 6 MB (0% of current file system capacity)
   Space saved                = 0 MB (0% of original data size)
File system parameters:
   Case Sensitive             = no
   Duplicate Detection Method  = sha1
   Access Time                = 15
   Modification Time          = 15
   Minimum Size               = 24 KB
   Maximum Size               = 8388608 MB
   File Extension Exclude List =
   Minimum Scan Interval      = 7
   Savevol Threshold          = 90
   Backup Data Threshold      = 90
   Cifs Compression Enabled   = yes
   Pathname Exclude List      =
   Compression Method         = fast

Id                            = 506
Name                          = demofs
Deduplication                 = Off
File system parameters:
   Case Sensitive             = no
   Duplicate Detection Method  = sha1
   Access Time                = 15
   Modification Time          = 15
   Minimum Size               = 24 KB
   Maximum Size               = 8388608 MB
   File Extension Exclude List =
   Minimum Scan Interval      = 7
   Savevol Threshold          = 90
   Backup Data Threshold      = 90
   Cifs Compression Enabled   = yes
   Pathname Exclude List      =

Id                            = 2113
```

```
Name                            = testrdefs
Deduplication                   = Suspended
As of the last file system scan (Thu Aug 13 14:22:31 EDT 2009):
   Files scanned                = 1
   Files deduped                = 0 (0% of total files)
   File system capacity         = 1008 MB
   Original data size           = 0 MB (0% of current file system capacity)
   Space saved                  = 0 MB (0% of original data size)
File system parameters:
   Case Sensitive               = no
   Duplicate Detection Method   = sha1
   Access Time                  = 15
   Modification Time            = 15
   Minimum Size                 = 24 KB
   Maximum Size                 = 8388608 MB
   File Extension Exclude List  =
   Minimum Scan Interval        = 7
   Savevol Threshold            = 90
   Backup Data Threshold        = 90
   Cifs Compression Enabled     = yes
   Pathname Exclude List        =
   Compression Method           = fast

Id                              = 2093
Name                            = kfs_ckpt1
Deduplication                   = Off
File system parameters:
   Case Sensitive               = no
   Duplicate Detection Method   = sha1
   Access Time                  = 15
   Modification Time            = 15
   Minimum Size                 = 24 KB
   Maximum Size                 = 8388608 MB
   File Extension Exclude List  =
   Minimum Scan Interval        = 7
   Savevol Threshold            = 90
   Backup Data Threshold        = 90
   Cifs Compression Enabled     = yes
   Pathname Exclude List        =
   Compression Method           = fast

Id                              = 2095
Name                            = ranap-test3
Deduplication                   = On
Status                          = Idle
As of the last file system scan (Tue Aug 11 17:37:58 EDT 2009):
   Files scanned                = 30
   Files deduped                = 2 (7% of total files)
   File system capacity         = 5041 MB
   Original data size           = 1109 MB (22% of current file system capacity)
   Space saved                  = 0 MB (0% of original data size)
File system parameters:
   Case Sensitive               = no
   Duplicate Detection Method   = sha1
   Access Time                  = 15
   Modification Time            = 15
   Minimum Size                 = 24 KB
   Maximum Size                 = 8388608 MB
   File Extension Exclude List  =
   Minimum Scan Interval        = 7
   Savevol Threshold            = 90
   Backup Data Threshold        = 90
   Cifs Compression Enabled     = yes
   Pathname Exclude List        =
   Compression Method           = deep

Where:
Value                       Definition
Deduplication               Current deduplication state of the filesystem.
Status                      Progress status of the files being scanned.
Name                        Name of the filesystem.
```

```
        Id                          Filesystem identifier.
        Files scanned               Number of files scanned.
        Files deduped               Number of files in the filesystem that has been dedu
plicated.
        Original data size          Proportion of space in use with respect to the file
                                    system capacity.
        File system capacity        Current space usage of the filesystem.
        Space saved                 Proportion of space saved with respect to the origin
al data
                                    size.
        Case Sensitive              Method of string comparison: case sensitive or case
                                    insensitive.
        Duplicate Detection Method  Method of duplication detection: 0, sha-1, or
                                    byte-by-byte.
        Access Time                 Minimum required file age in days based on read acce
ss time.
        Modification Time           Minimum required file age in days based on
                                    modification time.
        Minimum Size                Minimum file size to be processed for deduplication.
        Maximum Size                Maximum file size to be processed for deduplication.
        File Extension Exclude List Lists filename extensions to be excluded from
                                    the deduplication.
        Minimum Scan Interval       Minimum number of days between completing one scan o
f
                                    a filesystem and before scanning the same file syste
m again.
        SavVol Threshold            Percentage of SavVol space that can be used during
                                    deduplication.
        Backup Data Threshold       Percentage below which a deduplicated file has to be
                                    in order to trigger space-reduced NDMP backups.
        Cifs Compression Enabled    Controls whether CIFS permission is enabled.
        Pathname Exclude List       Lists relative path names to be excluded from the
                                    deduplication.
        Compression Method          Compression algorithm used: fast or deep.

        Note: If reduplication fails, then the state transitions to the suspended state a
nd
        a CCMD message will be sent to the server's event log. If reduplication
        succeeds, then it remains in the off state.

        EXAMPLE #3
        ----------

        To list the filesystems for a given filesystem name, type:

        $ fs_dedupe -info server3_fs3
        Id                                = 98
        Name                              = server3_fs3
        Deduplication                     = On
        Status                            = Idle
        As of the last filesystem scan on Tue Sep 23 13:28:01 EDT 2008:
           Files deduped                  = 30 (100%)
           Filesystem capacity            = 413590 MB
           Original data size             = 117 MB (0% of current filesystem
                                            capacity)
           Space saved                    = 106 MB (90% of original data size)
        Filesystem parameters:
           Case Sensitive                 = yes
           Duplicate Detection Method     = sha1
           Access Time                    = 30
           Modification Time              = 30
           Minimum Size                   = 20
           Maximum Size                   = 200
           File Extension Exclude List    = .jpg:.db:.pst
           Minimum Scan Interval          = 1
           SavVol Threshold               = 90
           Backup Data Threshold          = 90
           Pathname Exclude List          = root;etc
           Compression Method             = fast

        EXAMPLE #2 provides a description of command output.
```

EXAMPLE #6

To list the duplication properties of a given Data Mover, type:

```
$ fs_dedupe -default -info server_2
Server parameters:
   Case Sensitive                  = yes
   Duplicate Detection Method      = sha1
   Access Time                     = 30
   Modification Time               = 30
   Minimum Size                    = 20
   Maximum Size                    = 200
   File Extension Exclude List     = .jpg:.db:.pst
   Minimum Scan Interval           = 1
   SavVol Threshold                = 90
   Backup Data Threshold           = 90
   CPU % Usage Low Water Mark      = 25
   CPU % Usage High Water Mark     = 90
   Cifs Compression Enabled        = yes
```

Where:

Value	Definition
Deduplication	Current deduplication state of the filesystem.
Status	Progress status of the files being scanned.
Name	Name of the filesystem.
Id	Filesystem identifier.
Files scanned	Number of files scanned.
Files deduped	Number of files in the filesystem that has been deduplicated.
Original data size	Proportion of space in use with respect to the file system capacity.
File system capacity	Current space usage of the filesystem.
Space saved	Proportion of space saved with respect to the original data size.
Case Sensitive	Method of string comparison - case sensitive or case insensitive.
Duplicate Detection Method	Method of duplication detection : 0, sha-1, or byte-by-byte.
Access Time	Minimum required file age in days based on read access time.
Modification Time	Minimum required file age in days based on modification time.
Mininum Size	Minimum file size to be processed for deduplication.
Maximum Size	Maximum file size to be processed for deduplication.
File Extension Exclude List	Lists filename extensions to be excluded from the deduplication.
Mininum Scan Interval	Minimum number of days between completing one scan of a filesystem and before scanning the same file system again.
SavVol Threshold	Percentage of SavVol space that can be used during deduplication.
Backup Data Threshold	Percentage below which a deduplicated file has to be in order to trigger space-reduced NDMP backup.
CPU % Usage Low Water Mark	Average percentage of CPU usage which should trigger full throttle mode.
CPU % Usage High Water Mark	Average percentage of CPU usage which should trigger slow throttle mode.

EXAMPLE #5

To modify the filesystem, type:

```
$ fs_dedupe -modify testrdefs -state on
Done
```

EXAMPLE #6

```
----------

To modify the filesystem settings to the user specified values, type:

$ fs_dedupe -modify testrdefs -maximum_size 100 -file_extension_exclude_list
.jpg:.db:.pst
Done

EXAMPLE #7
----------

To modify specific Data Mover settings, type:

$ fs_dedupe -default -set server_2 -maximum_size 100 -minimum_size 20
-duplicate_detection_method sha1
Done

EXAMPLE #8
-----------

To reset the filesystem settings to the default settings (which are the Data
Mover settings) type:

$ fs_dedupe -clear testrdefs -maximum_size -minimum_size -duplicate_detection_met
hod
Done

EXAMPLE #9
-----------

To reset specific Data Mover settings to the default settings, type:

$ fs_dedupe -default -clear server_2 -maximum_size -minimum_size
-duplicate_detection_method
Done

EXAMPLE #10
-----------

To reset all options for a specific Data Mover to the default settings, type:

$ fs_dedupe -default -clear server_2
Done

EXAMPLE #11
-----------

To reset all options on all Data Movers to the default settings, type:

$ fs_dedupe -default -clear -all
Done

--------------------------------------------------------------------------------

Last modified: April 13, 2012 1:00 p.m.
```

fs_dhsm

Manages the VNX FileMover file system connections.

SYNOPSIS
----

fs_dhsm

```
  -list
| -info [<fs_name>|id=<fs_id>]
| -modify {<fs_name>|id=<fs_id>}[-state enabled]
   [-popup_timeout <sec>][-backup {offline|passthrough}]
   [-log {on|off}][-max_log_size <mb>][-offline_attr {on|off}]
   [-read_policy_override {none|full|passthrough|partial}]}
| -modify {<fs_name>|id=<fs_id>}[-state disabled]
| -connection {<fs_name>|id=<fs_id>}
    -list
| -info [<cid>]
| -create -type {nfsv3|nfsv2} -secondary <nfs_server>:/<path>
   [-read_policy_override {full|passthrough|partial|none}]
   [-useRootCred {true|false}][-proto {UDP|TCP}][-nfsPort <port>]
   [-mntPort <port>][-mntVer {3|2|1}][-localPort <port>]
| -create -type cifs -admin [<fqdn>\]<admin_name>
   -secondary \\<fqdn>\<share>[\<path>]
   -local_server <host_name> [-wins <address>][-password <password>]
   [-read_policy_override {full|passthrough|partial|none}]
| -create -type http -secondary http://<host><url_path>
   [-read_policy_override {full|passthrough|partial|none}]
   [-httpPort <port>][-localPort <port>]
   [-user <username> [-password <password>]]
   [-timeout <seconds>][-cgi {y|n}]
| -create -type https -secondary https://<host><url_path>
   [-read_policy_override {full|passthrough|partial|none}]
   [-httpsPort <port>][-localPort <port>]
   [-user <username> [-password <password>]]
   [-timeout <seconds>][-cgi {y|n}]
| -delete {-all|<cid>[,<cid>...]} [-recall_policy {check|no|yes}]
| -modify {-all|<cid>[,<cid>...]} [-state {enabled|disabled|recallonly}]
   [-read_policy_override {full|passthrough|partial|none}]
   [{[-nfs_server <address>] [-localPort <port>]
   [-proto {TCP|UDP}] [-useRootCred {true|false}]}
| {[-cifs_server <fqdn>][-local_server <host_name>]
   [-password <password>][-admin [<fqdn>\]<admin_name>]
   [-wins <address>]}
| {[-http_server <host>][-httpPort <port>][-httpsPort <port>]
   [-localPort <port>][-user <username>]
   [-password <password>][-timeout <seconds>]}
```

DESCRIPTION

The fs_dhsm command modifies the properties on file systems enabled for VNX FileMover. The fs_dhsm command creates, deletes, and modifies NFS, CIFS, and HTTP connections to remote hosts, lists VNX FileMover file systems, and provides information on the connections.

OPTIONS

-list
Lists all file systems enabled with the VNX FileMover.

-info [<fs_name>|id=<fs_id>]
Displays information for the specified VNX FileMover file systems.

-modify {<fs_name>|id=<fs_id>}
Sets VNX FileMover parameters for the specified file system.

Note: When specifying the -modify option on a disabled file system, the state

is automatically changed to enabled. When specifying the -state disabled option, it is not possible to specify any other parameter to modify.

[-state enabled]
Enables VNX FileMover operations on the specified file system. The file system must be enabled to accept other options.

[-state disabled]
Disables VNX FileMover operations on the specified file system. New FileMover attributes cannot be specified as part of a disable command, nor can be specified for a file system that is in the disabled state. The attributes persist. If the file system is enabled after a disable command, then the attributes prior to the disable command take effect.

[-popup_timeout <sec>]
Specifies the Windows popup timeout value in seconds. If a CIFS I/O request cannot be processed within the specified time, then a popup notification of the delay is sent to the CIFS client. The default for <sec> is 0 (zero) which disables Windows popups.

Note: It may take up to 10 seconds before the popup is displayed.

[-backup {offline|passthrough}]
Specifies the nature of CIFS network backups. The offline option backs up the stub file only. The passthrough (default) option backs up all of the file data by using passthrough read.

[-log {on|off}]
Enables or disables VNX FileMover logging. The default log filename is dhsm.log; it resides in the /.etc directory on the FileMover-enabled file system.

[-max_log_size <mb>]
Specifies the maximum size of the log file. The current log file, in addition to four old log files, is saved. The minimum log file size is 10 MB.

[-offline_attr {on|off}]
Specifies whether the Data Mover should set the CIFS offline file attributes on the stub files. The default is on.

Caution: It is recommended that you do not disable the CIFS offline attributes.

[-read_policy_override {none|full|passthrough|partial}]
Specifies the migration method option used by the VNX, in the connection level or file system level, to override the migration method specified in the stub file. none (default) specifies no override, full recalls the whole file to the VNX on a read request before the data is returned, passthrough retrieves data without recalling the data to the VNX, and partial recalls only the blocks required to satisfy the client read request.

Note: The full migration may take several minutes or hours if the file is very large.

-connection {<fs_name>|id=<fs_id>} -list
Lists all connections for the specified file system.

-connection {<fs_name>|id=<fs_id>} -info [<cid>]
Displays details on all connections for the specified file system. If the <cid> is specified, only information for that connection is displayed.

Note: A connection ID is automatically created when a connection is established. The connection ID is displayed using the -list and is referred to as the <cid> in other commands.

NFS CONNECTIONS

-connection {<fs_name>|id=<fs_id>} -create -type

{nfsv3|nfsv2} -secondary <nfs_server>:/<path>
Creates a connection using the NFS protocol between the specified file system and the secondary file system. The secondary file system stores migrated data. The -type option specifies the NFS version that the Data Mover should use when connecting to the secondary server.

Note: VNX FileMover does not currently support NFSv4 protocol.

The -secondary option specifies the location of the remote file system.

Note: Although an IP address can be specified for an <nfs_server>, EMC strongly suggests using the hostname of the server, which allows you to take advantage of Domain Name System (DNS) failover capability.

[-read_policy_override {full|passthrough|partial|none}]
Specifies the migration method for data recall in response to client read requests. full migrates the whole file before it returns the requested blocks. passthrough leaves the stub file, but retrieves the requested data from the secondary file system. partial migrates only the blocks required to satisfy the client read request. none (default) defaults to the read method option specified in the stub file.

Note: The full migration may take minutes or hours if the file is very large.

[-useRootCred {true|false}]
Specifies the user credentials that the Data Mover uses when requesting data from the secondary VNX. When set to true, the Data Mover requests data as the root user (UID 0). When set to false (default), the Data Mover requests data as the owner of the file as specified in the stub file.

Note: If the -useRootCred option is set to true, the secondary storage NFS server must grant the Data Mover root privilege for NFS traffic.

[-proto {TCP|UDP}]
Specifies the protocol for the Data Movers to use for communication to the secondary <nfs_server>. TCP is the default.

[-nfsPort <port>]
Specifies an NFS port on the secondary <nfs_server>. A default port is discovered automatically.

[-mntPort <port>]
Specifies a mount port on the secondary <nfs_server>. A default mount port is discovered automatically.

Note: The -nfsPort and the -mntPort options are used for secondary servers which do not have the Portmapper running. The admin starts the nfsd and mountd daemons on specific ports to avoid hackers.

[-mntVer {1|2|3}]
Specifies the mount version for the NFS connection. If the -type is nfsv3, then the -mntVer must be 3. If the -type is nfsv2, then 1 or 2 can be specified. The default for nfsv2 is 2.

[-localPort <port>]
Overrides the default port that the Data Mover uses during connection to be compatible with firewalls. The default for UDP is 1020. By default, TCP uses a random port over 1024 to make the connection.

-connection {<fs_name>|id=<fs_id>} -modify {-all|<cid>[,<cid>...]}
Changes parameters on an existing NFS VNX FileMover connection. Either all connections can be removed or just the specified <cid> connection can be removed.

[-state {enabled|disabled|recallonly}]
Sets the state of VNX FileMover operations on the specified file

system. enabled (default) allows both the creation of stub files and data migration through reads and writes. If the state is disabled, neither stub files nor data migration is possible. Data currently on the VNX can be read and written to in the disabled state.

If the state is recallonly, the policy engine is not allowed to create stub files, but the user is still able to trigger data migration using a read or write request from the secondary file system to the VNX.

[-read_policy_override {full|passthrough|partial|none}]
Specifies the migration method option used by the VNX, in the connection level or file system level, to override the migration method specified in the stub file. none (default) specifies no override, full recalls the whole file to the VNX on read request before the data is returned, passthrough retrieves data without recalling the data to the VNX, and partial recalls only the blocks required to satisfy the client read request.

Note: The full migration may take minutes or hours if the file is very large.

[-nfs_server <address>]
Specifies the name or IP address of the secondary NFS server.

Note: Although an IP address can be specified for the <nfs_server>, EMC strongly suggests using the hostname of the server, which allows use of the DNS failover capability.

[-localPort <port>]
Specifies a port to override the default port used by the Data Mover during connection for compatibility with firewalls.

[-proto {TCP|UDP}]
Specifies the protocol for the Data Mover to use for NFS communications to the secondary <nfs_server>. TCP is the default.

[-useRootCred {true|false}]}
Specifies the user credentials that the Data Mover uses when requesting data from the secondary VNX. When set to true, the Data Mover requests data as the root user (UID 0). When set to false (default), the Data Mover requests data as the owner of the file as specified in the stub file.

Note: If the -useRootCred option is set to true, the secondary storage NFS server must grant the Data Mover root privilege for NFS traffic.

-connection <fs_name> -delete {-all|<cid>[,<cid>...]}
Removes an existing NFS connection between the file system and the secondary file system. Either all connections can be removed or just the specified <cid> connection can be removed.

[-recall_policy {check|no|yes}]
Specifies the recall policy for any migrated file during the -delete. check (default) scans the file system for stub files that depend on the connection and fails on the first one. no deletes the connection without checking for stub files that depend on the connection, and yes migrates the files back to the VNX before the connection is removed. If no is specified and stub files exist, an I/O error appears when the file is read because the connection no longer exists.

CIFS CONNECTIONS

-connection {<fs_name>|id=<fs_id>} -create -type cifs
Creates a connection using the CIFS protocol between the specified file system and a secondary file system. A connection ID is automatically created when a connection is established. The connection ID is seen using the -list and is referred to as the <cid> in other commands.

-admin [<fqdn>\]<admin_name>
Specifies the <admin_name> used to make the CIFS connection. If an optional <fqdn> is specified, it must be a fully qualified domain name. The [<fqdn>\]<admin_name> entry must be enclosed within quotes as shown in EXAMPLE #2. If the <fqdn> is not specified, the -local_server domain is used.

-secondary \\<fqdn>\<share>[\<path>]
Specifies the CIFS server, the share, and path for the secondary server for connection. The <fqdn>\<share>[\<path>] entry must be enclosed within quotes. The domain must be fully qualified; an IP address will not work.

-local_server <host_name>
Specifies the NetBIOS name or computer name of the local CIFS server on the Data Mover.

[-wins <address>]
Specifies a WINS server to resolve names in a Windows domain.

[-password <password>]
Allows the user to specify the admin password. The password is not recorded in the command log. If the -password option is given but no password is specified, the user is prompted interactively.

Caution: When specifying the password with this option, be aware it is unmasked, and visible to other users. The command may also be read from the log of the shell.

[-read_policy_override {full|passthrough|partial|none}]
Specifies the migration method for data recall in response to client read requests. full migrates the whole file before it returns the requested blocks. passthrough leaves the stub file, but retrieves the requested data from the secondary file system. partial migrates only the blocks required to satisfy the client read request. none (default) defaults to the read method option specified in the stub file.

Note: The full migration may take several minutes or hours if the file is very large.

-connection {<fs_name>|id=<fs_id>} -modify {-all|<cid>[,<cid>...]}
Changes parameters on an existing NFS VNX FileMover connection.

[-state {enabled|disabled|recallonly}]
Sets the state of VNX FileMover operations on the specified file system. enabled (default) allows both the creation of stub files and data migration through reads and writes. If the state is disabled, neither stub files nor data migration is possible. Data currently on the VNX can be read and written to in the disabled state.

If the state is recallonly, the policy engine is not allowed to create stub files, but the user is still able to trigger data migration using a read or write request from the secondary file system to the VNX.

[-read_policy_override {full|passthrough|partial|none}]
Specifies the migration method option used by the VNX, in the connection level or file system level, to override the migration method specified in the stub file. none (default) specifies no override, full recalls the whole file to the VNX on read request before the data is returned, passthrough retrieves data without recalling the data to the VNX, and partial recalls only the blocks required to satisfy the client read request.

Note: The full migration may take minutes or hours if the file is very large.

[-cifs_server <fqdn>]

Specifies the fully qualified domain name of the secondary CIFS server.

[-local_server <host_name>]
Specifies the NetBIOS name or computer name of the local CIFS server on the Data Mover.

[-password <password>]
Allows the user to specify the admin password. The password is not recorded in the command log. If the -password option is given but no password is specified, the user is prompted interactively.

When specifying the password with this option, be aware it is unmasked, and visible to other users. The command may also be read from the log of the shell.

[-admin [<fqdn>\]<admin_name>]
Specifies the <admin_name> used to make the CIFS connection. If an optional <fqdn> is specified, it must be a fully qualified domain name. If the <fqdn> is not specified, the -local_server domain is used.

[-wins <address>]}
Specifies a WINS server to resolve names in a Windows domain.

-connection <fs_name> -delete {-all|<cid> [,<cid>...]}
Removes an existing CIFS connection between the file system and the secondary file system.

[-recall_policy {check|no|yes}]
Specifies the recall policy for any migrated file during the -delete option. check (default) scans the file system for stub files that depend on the connection and fails on the first one. no deletes the connection without checking for stub files that depend on the connection, and yes migrates the files back to the VNX before the connection is removed. If no is specified and stub files exist, an I/O error appears when the file is read because the connection no longer exists.

HTTP CONNECTIONS

-connection {<fs_name>|id=<fs_id>} -create -type http -secondary http://<host><url_path>
Creates a connection using the HTTP protocol between the specified primary file system and a secondary file system. There are two types of HTTP connections: CGI and non-CGI. For CGI connections, the value of the -secondary option specifies the hostname of the server running the secondary storage HTTP server and the location of the CGI application that provides access to a storage system. For non-CGI connections, the value for the -secondary option specifies the hostname and, optionally, a portion of the hierarchical namespace published by the web server.

Note: Although an IP address can be specified for a <host>, EMC strongly suggests using the hostname of the server, which allows the DNS failover capability.

[-read_policy_override {full|passthrough|partial|none}]
Specifies the migration method option used by the VNX, in the connection level or file system level, to override the migration method specified in the stub file. none (default) specifies no override, full recalls the whole file to the VNX on read request before the data is returned, passthrough retrieves data without recalling the data to the VNX, and partial recalls only the blocks required to satisfy the client read request.

Note: The full migration may take several minutes or hours if the file is very large.

-httpPort <port>
Specifies the remote port number that the Data Mover delivers the HTTP request to. If not specified, the Data Mover issues HTTP requests to port 80 on the secondary storage HTTP server.

-localPort <port>
Specifies the local port number the Data Mover uses to issue HTTP requests to the web server running on the secondary storage. The <port> specified should be an integer no less than 1024. If not specified, the Data Mover selects a port to issue the HTTP requests.

Note: The two end points of an HTTP connection are specified by the file system name and the value specified for the -secondary option. If multiple connections are created using identical end points with different attributes such as, -cgi, -user, -password, -localPort, -httpPort, the connection will fail.

[-user <username>]
Defines the username the HTTP client uses if digest authentication is required by the secondary storage HTTP server.

[-password <password>]
Allows the user to specify the admin password. The password is not recorded in the command log. If the -password option is given but no password is specified, the user is prompted interactively.

Use the -password option when digest authentication is required by the secondary storage HTTP server.

[-timeout <seconds>
Specifies the timeout value in seconds. By default, the Celerra HTTP client waits 30 seconds for a reply from the HTTP server and then retries the operation once.

[-cgi {y|n}
Specifies the HTTP connection type: CGI or non-CGI. By default, FileMover assumes that the web server is using CGI connections to access migrated file data using a CGI application. For non-CGI connections, set the -cgi option to n; FileMover then assumes the web server has direct access to migrated file content on secondary storage.

-connection {<fs_name>|id=<fs_id>} -modify {-all| <cid>[,<cid>...]}
Changes parameters on an existing NFS VNX FileMover connection.

[-state {enabled|disabled|recallonly}]
Sets the state of VNX FileMover operations on the specified file system. enabled (default) allows both the creation of stub files and data migration through reads and writes. If the state is disabled, neither stub files nor data migration is possible. Data currently on the VNX can be read and written to in the disabled state.

If the state is recallonly, the policy engine is not allowed to create stub files, but the user is still able to trigger data migration by using a read or write request from the secondary file system to the VNX.

[-read_policy_override {full|passthrough|partial|none}]
Specifies the migration method option used by the VNX, in the connection level or file system level, to override the migration method specified in the stub file. none (default) specifies no override, full recalls the whole file to the VNX on read request before the data is returned, passthrough retrieves data without recalling the data to the VNX, and partial recalls only the blocks required to satisfy the client read request.

Note: The full migration may take minutes or hours if the file is very large.

[-http_server <host>]
Specifies the hostname of the secondary storage HTTP server.

-httpPort <port>
Specifies the remote port number that the Data Mover delivers the HTTP request to. If not specified, the Data Mover issues HTTP requests to port 80 on the secondary storage HTTP server.

-localPort <port>
Specifies the local port number the Data Mover uses to issue HTTP requests to the web server active on the secondary storage. The <port> specified should be an integer no less than 1024. If not specified, the Data Mover selects a port to issue the HTTP requests.

Note: If you attempt to create multiple HTTP connections by using identical end points with different attributes such as -cgi, -user, -password, -localPort, -httpPort, the connection will fail.

[-user <username>]
An optional attribute used to define the username the HTTP client uses if digest authentication is required by the secondary storage HTTP server.

[-password <password>]
Allows the user to specify the admin password. The password is not recorded in the command log. If the -password option is given but no password is specified, the user is prompted interactively.

[-timeout <sec>]
Specifies the timeout value in seconds. By default, VNX.s HTTP client waits 30 seconds for a reply from the HTTP server and then retries the operation once before commencing the failover operation.

-connection <fs_name> -delete {-all|<cid>[,<cid>...]}
Removes an existing HTTP connection between the file system and the secondary file system. Either all connections can be removed or just the specified <cid> connection can be removed.

[-recall_policy {check|no|yes}]
Specifies the recall policy for any migrated file during the -delete option. The check (default) argument scans the file system for stub files that depend on the connection and fails on the first one. no deletes the connection without checking for stub files that depend on the connection, and yes migrates the files back to the VNX before the connection is removed. If no is specified and stub files exist, an I/O error appears when the file is read because the connection no longer exists.

HTTPS CONNECTIONS

-connection {<fs_name>|id=<fs_id>} -create -type
https -secondary https://<host><url_path>
Creates a connection by using the HTTPS protocol between the specified primary file system and a secondary file system. There are two types of HTTPS connections: CGI and non-CGI. For CGI connections, the value of the -secondary option specifies the hostname of the server running the secondary storage HTTPS server and the location of the CGI application that provides access to a storage system. For non-CGI connections, the value for the -secondary option specifies the hostname and, optionally, a portion of the hierarchical namespace published by the web server.

Note: Although an IP address can be specified for a <host>, EMC strongly suggests using the hostname of the server, which allows the DNS failover capability.

[-read_policy_override {full|passthrough|partial|none}]
Specifies the migration method option used by the VNX, in the connection level or file system level, to override the migration

method specified in the stub file. none (default) specifies no override, full recalls the whole file to the VNX on read request before the data is returned, passthrough retrieves data without recalling the data to the VNX, and partial recalls only the blocks required to satisfy the client read request.

Note: The full migration may take several minutes or hours if the file is very large.

[-httpsPort <port>]
Specifies the remote port number that the Data Mover delivers the HTTPS request to. If not specified, the Data Mover issues HTTPS requests to port 443 on the secondary storage HTTPS server.

[-localPort <port>]
Specifies the local port number the Data Mover uses to issue HTTPS requests to the web server active on the secondary storage. The <port> specified should be an integer no less than 1024. If not specified, the Data Mover selects a port to issue the HTTPS requests.

Note: The two end points of an HTTPS connection are specified by the file system name and the value specified for the -secondary option. If multiple connections are created by using identical end points with different attributes such as -cgi, -user, -password, -localPort, -httpsPort, the connection will fail.

[-user <username>]
Defines the username the HTTPS client uses if digest authentication is required by the secondary storage HTTPS server.

[-password <password>]
Allows the user to specify the admin password. The password is not recorded in the command log. If the -password option is given but no password is specified, the user is prompted interactively.

Use the -password option when digest authentication is required by the secondary storage HTTPS server.

[-timeout <seconds>
Specifies the timeout value in seconds. By default, the VNX HTTPS client waits 30 seconds for a reply from the HTTPS server and then retries the operation once.

[-cgi {y|n}
Specifies the HTTPS connection type: CGI or non-CGI. By default, FileMover assumes that the web server is using CGI connections to access migrated file data by using a CGI application. For non-CGI connections, set the -cgi option to n; FileMover then assumes the web server has direct access to migrated file content on secondary storage.

-connection {<fs_name>|id=<fs_id>} -modify {-all|<cid>[,<cid>...]}
Changes parameters on an existing NFS VNX FileMover connection.

[-state {enabled|disabled|recallonly}]
Sets the state of VNX FileMover operations on the specified file system. enabled (default) allows both the creation of stub files and data migration through reads and writes. If the state is disabled, neither stub files nor data migration is possible. Data currently on the VNX can be read and written to in the disabled state.

If the state is recallonly, the policy engine is not allowed to create stub files, but the user is still able to trigger data migration by using a read or write request from the secondary file system to the VNX.

[-read_policy_override {full|passthrough|partial|none}]

Specifies the migration method option used by the VNX, in the connection level or file system level, to override the migration method specified in the stub file. none (default) specifies no override, full recalls the whole file to the VNX on read request before the data is returned, passthrough retrieves data without recalling the data to the VNX, and partial recalls only the blocks required to satisfy the client read request.

Note: The full migration may take minutes or hours if the file is very large.

[-http_server <host>]
Specifies the hostname of the secondary storage HTTPS server.

-httpsPort <port>
Specifies the remote port number that the Data Mover delivers the HTTPS request to. If not specified, the Data Mover issues HTTPS requests to port 443 on the secondary storage HTTPS server.

Note: Although the -http_server option is used to modify the name of the secondary storage HTTPS server, files that can be converted into a stub by using an HTTPS connection can be brought back online using only HTTPS and not using NFS, CIFS, or even HTTP.

-localPort <port>
Specifies the local port number the Data Mover uses to issue HTTPS requests to the web server active on the secondary storage. The <port> specified should be an integer no less than 1024. If not specified, the Data Mover selects a port to issue the HTTPS requests.

Note: If you attempt to create multiple HTTPS connections by using identical end points with different attributes such as -cgi, -user, -password, -localPort, -httpsPort, the connection will fail.

[-user <username>]
An optional attribute used to define the username the HTTPS client uses if digest authentication is required by the secondary storage HTTPS server.

[-password <password>]
Allows the user to specify the admin password. The password is not recorded in the command log. If the -password option is given but no password is specified, the user is prompted interactively.

[-timeout <sec>]
Specifies the timeout value in seconds. By default, VNX.s HTTPS client waits 30 seconds for a reply from the HTTPS server and then retries the operation once before commencing the failover operation.

-connection <fs_name> -delete {-all|<cid>[,<cid>...]}
Removes an existing HTTPS connection between the file system and the secondary file system. Either all connections can be removed or just the specified <cid> connection can be removed.

[-recall_policy {check|no|yes}]
Specifies the recall policy for any migrated file during the -delete. check (default) scans the file system for stub files that depend on the connection and fails on the first one. no deletes the connection without checking for stub files that depend on the connection,and yes migrates the files back to the VNX before the connection is removed. If no is specified and stub files exist, an I/O error appears when the file is read because the connection no longer exists.

SEE ALSO

Using VNX FileMover, server_cifs, server_http, and server_nfs.

EXAMPLE #1

To enable VNX FileMover on a file system, type:

```
$ fs_dhsm -modify ufs1 -state enabled
ufs1:
state = enabled
offline attr = on
popup timeout = 0
backup = passthrough
read policy override = none
log file = on
max log size = 10MB

Done
```

Where:

Value	Definition
State	Whether VNX FileMover is enabled or disabled on the file system
offline attr	Whether CIFS clients should be notified that a file is migrated
popup timeout	Timeout value in seconds, before Windows popup notification is sent to the CIFS client
backup	Nature of CIFS network backups
read policy override	Migration method option used to override the read method specified in the stub file
log file	Whether FileMover logging is enabled or disabled
max log size	Maximum size of the log file

EXAMPLE #2

To create a CIFS connection for ufs1 to the secondary file system \\winserver2.nasdocs.emc.com\dhsm1 with a specified administrative account nasdocs.emc.com\Administrator and local server dm102-cge0:

```
$ fs_dhsm -connection ufs1 -create -type cifs -admin
'nasdocs.emc.com\Administrator' -secondary
'\\winserver2.nasdocs.emc.com\dhsm1' -local_server dm102-cge0
Enter Password:********
ufs1:
state = enabled
offline attr = on
popup timeout = 0
backup = passthrough
read policy override = none
log file = on
max log size = 10MB
cid = 0
type = CIFS
secondary = \\winserver2.nasdocs.emc.com\dhsm1\
state = enabled
read policy override = none
write policy = full
local_server = DM102-CGE0.NASDOCS.EMC.COM
admin = nasdocs.emc.com\Administrator
wins =

Done
```

Where:

Value	Definition
state	Whether VNX FileMover is enabled or disabled on the file system
offline attr	Whether CIFS clients should be notified that a file is migrated
popup timeout	Timeout value, in seconds, before a popup notification is sent to CIFS client
backup	Nature of CIFS network backups
read policy override	Migration method option used to override the read method specified in the stub file
log file	Whether FileMover logging is enabled or disabled
max log size	Maximum size of the log file
cid	Connection ID
type	Type of file system See -list for a description of the types
secondary	Hostname or IP address of the remote file system
state	Specifies whether VNX FileMover is enabled or disabled on the file system
read policy override	Migration method option used to override the read method specified in the stub file
write policy	Write policy option used to recall data from secondary storage
local_server	Name of the local CIFS server used to authenticate the CIFS connection

EXAMPLE #3

To create a CIFS connection for ufs1 to the secondary file system \\winserver2.nasdocs.emc.com\dhsm2 with a specified administrative account nasdocs.emc.com\Administrator, local server dm102-cge0, WINS server, and with the migration method set to full, type:

```
$ fs_dhsm -connection ufs1 -create -type cifs -admin '
nasdocs.emc.com\Administrator' -secondary
'\\winserver2.nasdocs.emc.com\dhsm1'
-local_server dm102-cge0 -wins 172.24.102.25 -read_policy_override full
Enter Password:********
ufs1:
state = enabled
offline attr = on
popup timeout = 0
backup = passthrough
read policy override = none
log file = on
max log size = 10MB
cid = 0
type = CIFS
secondary = \\winserver2.nasdocs.emc.com\dhsm1\
state = enabled
read policy override = full
write policy = full
local_server = DM102-CGE0.NASDOCS.EMC.COM
admin = nasdocs.emc.com\Administrator
wins = 172.24.102.25
```

```
Done

EXAMPLE #2 provides a description of command output.

EXAMPLE #4
----------

To display connection information for ufs1, type:

$ fs_dhsm -connection ufs1 -info 1
ufs1:
state = enabled
offline attr = on
popup timeout = 0
backup = passthrough
read policy override = none
log file = on
max log size = 10MB

EXAMPLE #2 provides a description of command output.

EXAMPLE #5
----------

To modify the read_policy_override setting for connection 0 for ufs1, type:

$ fs_dhsm -connection ufs1 -modify 0 -read_policy_override passthrough
ufs1:
state = enabled
offline attr = on
popup timeout = 0
backup = passthrough
read policy override = none
log file = on
max log size = 10MB
cid = 0
type = CIFS
secondary = \\winserver2.nasdocs.emc.com\dhsm1\
state = enabled
read policy override = pass
write policy = full
local_server = DM102-CGE0.NASDOCS.EMC.COM
admin = nasdocs.emc.com\Administrator
wins = 172.24.102.25

Done

EXAMPLE #2 provides a description of command output.

EXAMPLE #6
----------

To modify the VNX FileMover connection for ufs1, type:

$ fs_dhsm -connection ufs1 -modify 0 -nfs_server 172.24.102.115 -proto TCP
ufs1:
state = enabled
offline attr = on
popup timeout = 10
backup = offline
read policy override = full
log file = on
max log size = 25MB
cid = 0
type = NFSV3
secondary = 172.24.102.115:/export/dhsm1
state = enabled
read policy override = full
write policy = full
options = useRootCred=true proto=TCP
cid = 1
```

```
type = CIFS
secondary = \\winserver2.nasdocs.emc.com\dhsm1\
state = enabled
read policy override = none
write policy = full
local_server = DM102-CGE0.NASDOCS.EMC.COM
admin = nasdocs.emc.com\Administrator
wins = 172.24.102.25
cid = 2
type = HTTP
secondary = http://172.24.102.115/export/dhsm1
state = enabled
read policy override = none
write policy = full
user =
options = cgi=n

Done
```

EXAMPLE #2 provides a description of command output.

EXAMPLE #7

To create the NFSv3 connection for ufs1 to the secondary file system 172.24.102.115:/export/dhsm1 with the migration method set to full, the -useRootCred set to true, and the protocol set to UDP, type:

```
$ fs_dhsm -connection ufs1 -create -type nfsv3 -secondary
172.24.102.115:/export/dhsm1 -read_policy_override full -useRootCred true
-proto UDP
ufs1:
state = enabled
offline attr = on
popup timeout = 0
backup = passthrough
read policy override = none
log file = on
max log size = 10MB
cid = 0
type = CIFS
secondary = \\winserver2.nasdocs.emc.com\dhsm1\
state = enabled
read policy override = pass
write policy = full
local_server = DM102-CGE0.NASDOCS.EMC.COM
admin = nasdocs.emc.com\Administrator
wins = 172.24.102.25
cid = 1
type = NFSV3
secondary = 172.24.102.115:/export/dhsm1
state = enabled
read policy override = full
write policy = full
options = useRootCred=true proto=UDP

Done
```

EXAMPLE #2 provides a description of command output.

EXAMPLE #8

To modify the VNX FileMover connection for ufs1, type:

```
$ fs_dhsm -connection ufs1 -modify 1 -proto TCP
ufs1:
state = enabled
offline attr = on
popup timeout = 0
backup = passthrough
```

```
read policy override = none
log file = on
max log size = 10MB
cid = 0
type = CIFS
secondary = \\winserver2.nasdocs.emc.com\dhsm1\
state = enabled
read policy override = pass
write policy = full
local_server = DM102-CGE0.NASDOCS.EMC.COM
admin = nasdocs.emc.com\Administrator
wins = 172.24.102.25
cid = 1
type = NFSV3
secondary = 172.24.102.115:/export/dhsm1
state = enabled
read policy override = full
write policy = full
options = useRootCred=true proto=TCP
Done
```

EXAMPLE #2 provides a description of command output.

EXAMPLE #9

To display VNX FileMover connection information for ufs1, type:

```
$ fs_dhsm -info ufs1
ufs1:
state = enabled
offline attr = on
popup timeout = 0
backup = passthrough
read policy override = none
log file = on
max log size = 10MB
```

EXAMPLE #1 provides a description of command output.

EXAMPLE #10

To list VNX FileMover connections, type:

```
$ fs_dhsm -connection ufs1 -list
id name cid
29 ufs1 0
29 ufs1 1
29 ufs1 2
```

EXAMPLE #11

To modify the VNX FileMover connection for ufs1, type:

```
$ fs_dhsm -modify ufs1 -popup_timeout 10 -backup offline -log on
-max_log_size 25 -offline_attr on -read_policy_override full
ufs1:
state = enabled
offline attr = on
popup timeout = 10
backup = offline
read policy override = full
log file = on
max log size = 25MB
cid = 0
type = CIFS
secondary = \\winserver2.nasdocs.emc.com\dhsm1\
state = enabled
read policy override = pass
```

```
write policy = full
local_server = DM102-CGE0.NASDOCS.EMC.COM
admin = nasdocs.emc.com\Administrator
wins = 172.24.102.25
cid = 1
type = NFSV3
secondary = 172.24.102.115:/export/dhsm1
state = enabled
read policy override = full
write policy = full
options = useRootCred=true proto=TCP

Done
```

EXAMPLE #2 provides a description of command output.

EXAMPLE #12

To modify the state of the VNX FileMover connection 0 for ufs1, type:

```
$ fs_dhsm -connection ufs1 -modify 0 -state disabled
ufs1:
state = enabled
offline attr = on
popup timeout = 10
backup = offline
read policy override = full
log file = on
max log size = 25MB
cid = 0
type = CIFS
secondary = \\winserver2.nasdocs.emc.com\dhsm1\
state = disabled
read policy override = pass
write policy = full
local_server = DM102-CGE0.NASDOCS.EMC.COM
admin = nasdocs.emc.com\Administrator
wins = 172.24.102.25
cid = 1
type = NFSV3
secondary = 172.24.102.115:/export/dhsm1
state = enabled
read policy override = full
write policy = full
options = useRootCred=true proto=TCP

Done
```

EXAMPLE #2 provides a description of command output.

EXAMPLE #13

To modify the state of the VNX FileMover connection 1 for ufs1, type:

```
$ fs_dhsm -connection ufs1 -modify 1 -state recallonly
ufs1:
state = enabled
offline attr = on
popup timeout = 10
backup = offline
read policy override = full
log file = on
max log size = 25MB
cid = 0
type = CIFS
secondary = \\winserver2.nasdocs.emc.com\dhsm1\
state = enabled
read policy override = pass
write policy = full
```

```
local_server = DM102-CGE0.NASDOCS.EMC.COM
admin = nasdocs.emc.com\Administrator
wins = 172.24.102.25
cid = 1
type = NFSV3
secondary = 172.24.102.115:/export/dhsm1
state = recallonly
read policy override = full
write policy = full
options = useRootCred=true proto=TCP

Done
```

EXAMPLE #2 provides a description of command output.

EXAMPLE #14

To delete the VNX FileMover connections 1 and 2 for ufs1, and specify the recall policy for any migrated files during the delete, type:

```
$ fs_dhsm -connection ufs1 -delete 0,1 -recall_policy
no
ufs1:
state = enabled
offline attr = on
popup timeout = 10
backup = offline
read policy override = full
log file = on
max log size = 25MB

Done
```

EXAMPLE #2 provides a description of command output.

EXAMPLE #15

To change the state of the VNX FileMover connection for ufs1 to disabled, type:

```
$ fs_dhsm -modify ufs1 -state disabled
ufs1:
state = disabled
offline attr = on
popup timeout = 10
backup = offline
read policy override = full
log file = on
max log size = 25MB

Done
```

EXAMPLE #1 provides a description of command output.

EXAMPLE #16

To create an HTTP connection for ufs1 to the secondary file system /export/dhsm1 on the web server http://172.24.102.115 which has direct access to the storage, type:

```
$ fs_dhsm -connection ufs1 -create -type http -secondary
http://172.24.102.115/export/dhsm1 -cgi n
ufs1:
state = enabled
offline attr = on
popup timeout = 10
backup = offline
read policy override = full
```

```
log file = on
max log size = 25MB
cid = 2
type = HTTP
secondary = http://172.24.102.115/export/dhsm1
state = enabled
read policy override = none
write policy = full
user =
options = cgi=n

Done
```

EXAMPLE #2 provides a description of command output.

EXAMPE #17

To create an HTTP connection for ufs1 to the secondary file system using CGI connections to access migrated file data using a CGI application, type:

```
$ fs_dhsm -connection ufs1 -create -type http -secondary
http://www.nasdocs.emc.com/cgi-bin/access.sh
ufs1:
state = enabled
offline attr = on
popup timeout = 0
backup = passthrough
read policy override = none
log file = on
max log size = 10MB
cid = 0
type = HTTP
secondary = http://www.nasdocs.emc.com/cgi-bin/access.sh
state = enabled
read policy override = none
write policy = full
user =
options =
Done
```

EXAMPLE #2 provides a description of command output.

EXAMPLE #18

To create an HTTPS connection for server2_fs1 on the web server https://int16543 with read_policy_override set to full, type:

```
$ fs_dhsm -connection server2_fs1 -create -type https -secondary
https://int16543 -read_policy_override full -cgi n
server2_fs1:
state                = enabled
offline attr         = on
popup timeout        = 0
backup               = passthrough
read policy override = passthrough
log file             = on
max log size         = 10MB
 cid                 = 0
   type                 = HTTPS
   secondary            = https://int16543
   state                = enabled
   read policy override = full
   write policy         =        full
   user                 =
   options              =

Done
```

EXAMPLE #2 provides a description of command output.

EXAMPLE #19

To create an HTTPS connection for ufs1 to the secondary file system using CGI connections to access migrated file data using a CGI application, type:

```
$ fs_dhsm -connection ufs1 -create -type https .secondary
https://www.nasdocs.emc.com/cgi-bin/access.sh
ufs1:
state = enabled
offline attr = on
popup timeout = 0
backup = passthrough
read policy override = none
log file = on
max log size = 10MB
cid = 0
type = HTTPS
secondary = https://www.nasdocs.emc.com/cgi-bin/access.sh
state = enabled
read policy override = none
write policy = full
user =
options =
Done
```

EXAMPLE #2 provides a description of command output.

EXAMPLE #20

To create an HTTPS connection on httpsPort 443 for server2_ufs1 on the web server https://int16543 with read_policy_override set to passthrough, type:

```
$ fs_dhsm -connection server2_fs1 -create -type https -secondary
https://int16543 -read_policy_override passthrough -httpsPort 443 -cgi n
server2_fs1:
state                = enabled
offline attr         = on
popup timeout        = 0
backup               = passthrough
read policy override = passthrough
log file             = on
max log size         = 10MB
 cid                 = 1
   type                 = HTTPS
   secondary            = https://int16543
   state                = enabled
   read policy override = pass
   write policy         =        full
   user                 =
   options              =

Done
```

EXAMPLE #2 provides a description of command output.

EXAMPLE #21

To create an HTTPS connection on localPort 80 for server2_ufs1 on the web server https://int16543 with read_policy_override set to passthrough, type:

```
$ fs_dhsm -connection server2_fs1 -create -type https -secondary
https://int16543 -read_policy_override passthrough -localPort 80 -cgi n
server2_fs1:
state                = enabled
offline attr         = on
popup timeout        = 0
backup               = passthrough
```

```
read policy override = passthrough
log file             = on
max log size         = 10MB
 cid                 = 0
   type                 = HTTPS
   secondary            = https://int16543
   state                = enabled
   read policy override = pass
   write policy         =        full
   user                 =
   options              =

Done
```

EXAMPLE #2 provides a description of command output.

EXAMPLE #22

To create an HTTPS connection on httpsPort 443 for server2_ufs1 on the web server https://int16543 with a specified user dhsm_user, type:

```
$ fs_dhsm -connection server2_fs1 -create -type https -secondary
https://int16543 -read_policy_override full -httpsPort 443 .user dhsm_user
-password dhsm_user -cgi n
server2_fs1:
state                = enabled
offline attr         = on
popup timeout        = 0
backup               = passthrough
read policy override = passthrough
log file             = on
max log size         = 10MB
 cid                 = 1
   type                 = HTTPS
   secondary            = https://int16543
   state                = enabled
   read policy override = full
   write policy         =        full
   user                 = dhsm_user
   options              =

Done
```

EXAMPLE #2 provides a description of command output.

EXAMPLE #23

To modify the read_policy_override setting for connection 1 from server2_fs1, type:

```
$ fs_dhsm -connection server2_fs1 -modify 1 -read_policy_override passthrough
server2_fs1:
state                = enabled
offline attr         = on
popup timeout        = 0
backup               = passthrough
read policy override = passthrough
log file             = on
max log size         = 10MB
 cid                 = 1
   type                 = HTTPS
   secondary            = https://int16543
   state                = enabled
   read policy override = pass
   write policy         =        full
   user                 = dhsm_user
   options              =

Done
```

EXAMPLE #2 provides a description of command output.

EXAMPLE #24

To delete the VNX FileMover connection 0 for ufs1, type:

```
$ fs_dhsm -connection ufs1 -delete 0
ufs1:
state                = enabled
offline attr         = on
popup timeout        = 0
backup               = passthrough
read policy override = none
log file             = on
max log size         = 10MB

Done
```

EXAMPLE #1 provides a description of command output.

Last Modified: March 29, 2011 05:00 Pm

fs_group

Creates a file system group from the specified file systems or a single file system.

SYNOPSIS

```
fs_group
  -list
| -delete <fs_group_name>
| -info {<fs_group_name>|id=<fs_group_id>}
| [-name <name>] -create {<fs_name>,...}
| -xtend <fs_group_name> {<fs_name>,...}
| -shrink <fs_group_name> {<fs_name>,...}
```

DESCRIPTION

The fs_group command combines file systems to be acted upon simultaneously as a single group for TimeFinder/FS.

OPTIONS

-list
Displays a listing of all file system groups.

Note: The ID of the object is an integer and is assigned automatically. The name of a file system may be truncated if it is too long for the display. To display the full name, use the -info option with a file system ID.

-delete <fs_group_name>
Deletes the file system group configuration. Individual file systems are not deleted.

-info {<fs_group_name>|id=<fs_group_id>}
Displays information about a file system group, either by name or group ID.

[-name <name>] -create {<fs_name>,...}
Creates a file system group from the specified file systems. If a name is not specified, one is assigned by default.

-xtend <fs_group_name> {<fs_name>,...}
Adds the specified file systems or group, to a file system group.

-shrink <fs_group_name> {<fs_name>,...}
Removes the specified file systems or group from a file system group. Individual file systems are not deleted.

SEE ALSO

Managing Volumes and File Systems for VNX Manually and Using TimeFinder/FS, NearCopy, and FarCopy on VNX for File, fs_timefinder, and nas_fs.

STORAGE SYSTEM OUTPUT

The number associated with the storage device is dependent on the attached storage system. VNX for block displays a prefix of APM before a set of integers, for example, APM00033900124-0019. Symmetrix storage storage systems appear as 002804000190-003C.

EXAMPLE #1

To create a file system group named, ufsg1, and add ufs1, type:

```
$ fs_group -name ufsg1 -create ufs1
```

```
id = 22
name = ufsg1
acl = 0
in_use = False
type = group
fs_set = ufs1
pool =
stor_devs =
000187940268-0006,000187940268-0007,000187940268-0008,000187940268-0009
disks = d3,d4,d5,d6

Where:
 Value      Indicates:

 id         ID of the group that is automatically assigned

 name       Name assigned to the group

 acl        Access control value for the group

 in_use     Whether a file system is used by a group

 type       Type of file system

 fs_set     File systems that are part of the group

 pool       Storage pool given to the file system group

 stor_devs  Storage system devices associated with the group

 disks      Disks on which the metavolume resides

EXAMPLE #2
----------

To list all file system groups, type:

$ fs_group -list
id name acl in_use type member_of fs_set
20 ufsg1 0   n     100              18

Where:
 Value      Indicates:

 member_of  Groups which the file system group belong to

EXAMPLE #3
----------

To display information for the file system group, ufsg1, type:

$ fs_group -info ufsg1
id = 22
name = ufsg1
acl = 0
in_use = False
type = group
fs_set = ufs1
pool =
stor_devs =
000187940268-0006,000187940268-0007,000187940268-0008,000187940268-0009
disks = d3,d4,d5,d6

EXAMPLE #1 provides a description of command output.

EXAMPLE #4
----------

To add file system, ufs2, to the file system group, ufsg1, type:

$ fs_group -xtend ufsg1 ufs2
```

```
        id = 22
        name = ufsg1
        acl = 0
        in_use = False
        type = group
        fs_set = ufs1,ufs2
        pool =
        stor_devs =
        000187940268-0006,000187940268-0007,000187940268-0008,000187940268-0009,000187940
268-000A,000187940268-000B,000187940268-000C,000187940268-000D

        disks = d3,d4,d5,d6,d7,d8,d9,d10

        EXAMPLE #1 provides a description of command output.

        EXAMPLE #5
        ----------

        To remove file system, ufs2, from the file system group, ufsg1, type:

        $ fs_group -shrink ufsg1 ufs2
        id = 22
        name = ufsg1
        acl = 0
        in_use = False
        type = group
        fs_set = ufs1
        pool =
        stor_devs =
        000187940268-0006,000187940268-0007,000187940268-0008,000187940268-0009
        disks = d3,d4,d5,d6

        EXAMPLE #1 provides a description of command output.

        EXAMPLE #6
        ----------

        To delete file system group, ufsg1, type:

        $ fs_group -delete ufsg1
        id = 22
        name = ufsg1
        acl = 0
        in_use = False
        type = group
        fs_set =
        stor_devs =
        disks =

        EXAMPLE #1 provides a description of command output.

        ---------------------------------------
        Last Modified: March 29, 2010 6:00 pm
```

fs_rdf

Manages the Remote Data Facility (RDF) functionality for a file system residing on RDF drives.

SYNOPSIS

```
fs_rdf {<fs_name>|id=<fs_id>}
  -Mirror {on|off|refresh}
| -Restore [-Force]
| -info
```

Note: RDF is supported only on a VNX attached to a Symmetrix.

DESCRIPTION

The fs_rdf command turns mirroring on and off for an RDF file system and displays information about RDF relationships.

OPTIONS

-Mirror {on|off|refresh}
The on option resumes the link between the RDF drives of a file system thereby enabling mirroring for the RDF file system. The off option halts mirroring between the file systems, and the refresh option does an immediate mirror on then off which refreshes the file system image.

-Restore [-Force]
Restores a file system from the R2 side (remote) when remote TimeFinder/FS FarCopy is used. The -Restore can only be executed on the R1 side. The -Force option must be used when restoring a file system with enabled.

-info
Displays information about RDF relationships.

SEE ALSO

Using SRDF/S with VNX for Disaster Recovery, Using TimeFinder/FS, NearCopy, and FarCopy on VNX for File, and Using VNX File-Level Retention.

EXAMPLE #1

To turn on mirroring for ufs1_snap1 from the R1 Control Station, type:

```
$ fs_rdf ufs1_snap1 -Mirror on
id = 20
name = ufs1_snap1
acl = 0
in_use = False
type = uxfs
volume = v168
pool =
rw_servers=
ro_servers=
rw_vdms =
ro_vdms =
backup_of = ufs1 Fri Apr 23 16:29:23 EDT 2004
stor_devs =
002804000190-0052,002804000190-0053,002804000190-0054,002804000190-0055
disks = rootd33,rootd34,rootd35,rootd36

RDF Information:
remote_symid = 002804000218
```

```
        remote_sym_devname =
        ra_group_number = 2
        dev_rdf_type = R1
        remote_symid = 002804000218
        remote_sym_devname =
        ra_group_number = 2
        dev_rdf_type = R1
        dev_ra_status = READY
        dev_link_status = READY
        rdf_mode = SYNCHRONOUS
        rdf_pair_state = SYNCINPROG
        rdf_domino = DISABLED
        adaptive_copy = DISABLED
        adaptive_copy_skew = 65535
        num_r1_invalid_tracks = 0
        num_r2_invalid_tracks = 736440
        dev_rdf_state = READY
        remote_dev_rdf_state = WRITE_DISABLED
        rdf_status = 0
        link_domino = DISABLED
        prevent_auto_link_recovery = DISABLED
        link_config =
        suspend_state = NA
        consistency_state = DISABLED
        adaptive_copy_wp_state = NA
        prevent_ra_online_upon_pwron = ENABLED
```

Where:

Value	Definition
id	ID of a file system that is assigned automatically
name	Name assigned to a file system
acl	Access control value for a file system
in_use	Whether a file system is registered into the mount table
type	Type of file system See nas_fs for a description of the types
volume	Volume on which a file system resides
pool	Storage pool for the file system
rw_servers	Servers with read-write access to a file system
ro_servers	Servers with read-only access to a file system
rw_vdms	VDM servers with read-write access to a file system
ro_vdms	VDM servers with read-only access to a file system
backup_of	The remote RDF file system
stor_devs	The storage system devices associated with a file system
disks	The disks on which the metavolume resides
remote_symid	The serial number of the storage system containing the target volume
remote_sym_devname	The storage system device name of the remote device in an RDF pair
ra_group_number	The RA group number (1-n)

dev_rdf_type	The type of RDF device Possible values are: R1 and R2
dev_ra_status	RA status. Possible values are: READY, NOT_READY, WRITE_DISABLED, STATUS_NA, STATUS_MIXED
dev_link_status	Link status Possible values are: READY, NOT_READY, WRITE_DISABLED, NA, MIXED
rdf_mode	The RDF mode.Possible values are: SYNCHRONOUS, SEMI_SYNCHRONOUS, ADAPTIVE_COPY, MIXED
rdf_pair_state	Composite state of the RDF pair Possible values are: INVALID, SYNCINPROG, SYNCHRONIZED, SPLIT,SUSPENDED, FAILED_OVER, PARTITIONED, R1_UPDATED, R1_UPDINPROG, MIXED
rdf_domino	The RDF device domino Possible values are: ENABLED, DISABLED, MIXED
adaptive_copy	Possible values are: DISABLED, WP_MODE, DISK_MODE, MIXED
adaptive_copy_skew	Number of invalid tracks when in Adaptive copy mode
num_r1_invalid_tracks	Number of invalid tracks on the source (R1) device
num_r2_invalid_tracks	Number of invalid tracks on the target (R2) device
dev_rdf_state	Specifies the composite RDF state of the RDF device Possible values are: READY, NOT_READY, WRITE_DISABLED, NA, MIXED
remote_dev_rdf_state	Specifies the composite RDF state of the remote RDF device Possible values are: READY, NOT_READY, WRITE_DISABLED, NA, MIXED
rdf_status	Specifies the RDF status of the device Possible values are: READY,NOT_READY,WRITE_DISABLED,NA,MIXED
link_domino	RDF link domino Possible values are: ENABLED, DISABLED
prevent_auto_link_recovery	When enabled, prevents the automatic resumption of data copy across the RDF links as soon as the links have recovered Possible values are: ENABLED, DISABLED
link_config	Possible values are: CONFIG_ESCON, CONFIG_T3
suspend_state	Specifies the status of R1 devices in a consistency group Possible states are: NA, OFFLINE, OFFLINE_PEND, ONLINE_MIXED
consistency_state	Specifies state of an R1 device related to consistency groups Possible states are: ENABLED, DISABLED
adaptive_copy_wp_state	Specifies state of the adaptive copy mode Possible states are: NA, OFFLINE, OFFLINE_PEND, ON

LINE_MIXED

prevent_ra_online_upon_pwron	Specifies the state of the RA director coming online after power on Possible states are: ENABLED, DISABLED

EXAMPLE #2

To display RDF-related information for ufs1_snap1 from the R2 Control Station, type:

```
$ fs_rdf ufs1_snap1 -info
id = 20
name = ufs1_snap1
acl = 0
in_use = False
type = uxfs
volume = v168
pool =
rw_servers=
ro_servers=
rw_vdms =
ro_vdms =
backup_of = ufs1 Fri Apr 23 16:29:23 EDT 2004
stor_devs =
002804000190-0052,002804000190-0053,002804000190-0054,002804000190-0055
disks = rootd33,rootd34,rootd35,rootd36

RDF Information:
remote_symid = 002804000218
remote_sym_devname =
ra_group_number = 2
dev_rdf_type = R1
dev_ra_status = READY
dev_link_status = READY
rdf_mode = SYNCHRONOUS
rdf_pair_state = SYNCINPROG
rdf_domino = DISABLED
adaptive_copy = DISABLED
adaptive_copy_skew = 65535
num_r1_invalid_tracks = 0
num_r2_invalid_tracks = 696030
dev_rdf_state = READY
remote_dev_rdf_state = WRITE_DISABLED
rdf_status = 0
link_domino = DISABLED
prevent_auto_link_recovery = DISABLED
link_config =
suspend_state = NA
consistency_state = DISABLED
adaptive_copy_wp_state = NA
prevent_ra_online_upon_pwron = ENABLED
```

EXAMPLE #1 provides a description of command output.

EXAMPLE #3

To turn the mirroring off for ufs1_snap1 on the R1 Control Station, type:

```
$ fs_rdf ufs1_snap1 -Mirror off
remainder(MB) = 20548..17200..13110..8992..4870..746 0
id = 20
name = ufs1_snap1
remainder(MB) = 20548..17200..13110..8992..4870..746 0
id = 20
name = ufs1_snap1
acl = 0
in_use = False
type = uxfs
```

```
volume = v168
pool =
rw_servers=
ro_servers=
rw_vdms =
ro_vdms =
backup_of = ufs1 Fri Apr 23 16:29:23 EDT 2004
stor_devs =
002804000190-0052,002804000190-0053,002804000190-0054,002804000190-0055
disks = rootd33,rootd34,rootd35,rootd36

RDF Information:
remote_symid = 002804000218
remote_sym_devname =
ra_group_number = 2
dev_rdf_type = R1
dev_ra_status = READY
dev_link_status = NOT_READY
rdf_mode = SYNCHRONOUS
rdf_pair_state = SUSPENDED
rdf_domino = DISABLED
adaptive_copy = DISABLED
adaptive_copy_skew = 65535
num_r1_invalid_tracks = 0
num_r2_invalid_tracks = 0
dev_rdf_state = READY
remote_dev_rdf_state = WRITE_DISABLED
rdf_status = 0
link_domino = DISABLED
prevent_auto_link_recovery = DISABLED
link_config =
suspend_state = OFFLINE
consistency_state = DISABLED
adaptive_copy_wp_state = NA
prevent_ra_online_upon_pwron = ENABLED
```

EXAMPLE #1 provides a description of command output.

EXAMPLE #4

To perform a mirror refresh for ufs1_snap1 on the R1 Control Station, type:

```
$ fs_rdf ufs1_snap1 -Mirror refresh
remainder(MB) = 1 0

id = 20
name = ufs1_snap1
acl = 0
in_use = False
type = uxfs
volume = v168
pool =
rw_servers=
ro_servers=
rw_vdms =
ro_vdms =
backup_of = ufs1 Fri Apr 23 16:29:23 EDT 2004
stor_devs =
002804000190-0052,002804000190-0053,002804000190-0054,002804000190-0055
disks = rootd33,rootd34,rootd35,rootd36

RDF Information:
remote_symid = 002804000218
remote_sym_devname =
ra_group_number = 2
dev_rdf_type = R1
dev_ra_status = READY
dev_link_status = NOT_READY
rdf_mode = SYNCHRONOUS
rdf_pair_state = SUSPENDED
```

```
rdf_domino = DISABLED
adaptive_copy = DISABLED
adaptive_copy_skew = 65535
num_r1_invalid_tracks = 0
num_r2_invalid_tracks = 0
dev_rdf_state = READY
remote_dev_rdf_state = WRITE_DISABLED
rdf_status = 0
link_domino = DISABLED
prevent_auto_link_recovery = DISABLED
link_config =
suspend_state = OFFLINE
consistency_state = DISABLED
adaptive_copy_wp_state = NA
prevent_ra_online_upon_pwron = ENABLED
```

EXAMPLE #1 provides a description of command output.

EXAMPLE #5

To restore the file system ufs1_snap1 from the R1 Control Station, type:

```
$ /nas/sbin/rootfs_rdf ufs1_snap1 -Restore
remainder(MB) = 1 0

id = 20
name = ufs1_snap1
acl = 0
in_use = False
type = uxfs
volume = v168
pool =
rw_servers=
ro_servers=
rw_vdms =
ro_vdms =
backup_of = ufs1 Fri Apr 23 16:29:23 EDT 2004
stor_devs =
002804000190-0052,002804000190-0053,002804000190-0054,002804000190-0055
disks = rootd33,rootd34,rootd35,rootd36

RDF Information:
remote_symid = 002804000218
remote_sym_devname =
ra_group_number = 2
dev_rdf_type = R1
dev_ra_status = READY
dev_link_status = READY
rdf_mode = SYNCHRONOUS
rdf_pair_state = SYNCHRONIZED
rdf_domino = DISABLED
adaptive_copy = DISABLED
adaptive_copy_skew = 65535
num_r1_invalid_tracks = 0
num_r2_invalid_tracks = 0
dev_rdf_state = READY
remote_dev_rdf_state = WRITE_DISABLED
rdf_status = 0
link_domino = DISABLED
prevent_auto_link_recovery = DISABLED
link_config =
suspend_state = NA
consistency_state = DISABLED
adaptive_copy_wp_state = NA
prevent_ra_online_upon_pwron = ENABLED
```

EXAMPLE #1 provides a description of command output.

Last Modified: March 29, 2010 06:15 pm

fs_timefinder

Manages the TimeFinderTM/FS functionality for the specified filesystem or filesystem group.

SYNOPSIS

```
fs_timefinder {<fs_name>|id=<fs_id>}
-Mirror {on|off|refresh [-Force]}[-star]
| [-name <name>] -Snapshot [-volume <volume_name>][-option <options>][-star]
| -Restore [-Force][-option <options>][-star]
```

Note: TimeFinder/FS is supported only on a VNX attached to a Symmetrix.

DESCRIPTION

The fs_timefinder command creates a copy of a file system or file system group that can be placed into a mirrored mode with its original file system. The Symmetrix must already have business continuance volumes (BCVs) configured to the same size as the volumes on the VNX. After the copy of the filesystem has been made, it can be mounted on any Data Mover.

OPTIONS

-Mirror {on|off|refresh}
on places the unmounted filesystem copy, created by using the -Snapshot option, into mirrored mode with its original filesystem. The file system copy is frozen and remains unavailable to users until mirrored mode is turned off.

The refresh option initiates an immediate -Mirror on then off for the unmounted file system copy, thereby refreshing the filesystem copy.

[-Force]
The file system copy should not be mounted read-write when placed into mirrored mode or when refreshed. If the filesystem copy is mounted read-write, the -Force option can be used to force a refresh if the metavolume is an STD type. The -Force option requires root command and must be executed by using /nas/sbin/rootfs_timefinder.

[-star]
The -star option allows the fs_timefinder command to run on STAR SRDF configuration.

Caution: Performing a mirror refresh may be time consuming, relative to the amount of data that has changed in the file system.

[-name <name>] -Snapshot
Creates a copy of a file system and assigns an optional name to the file system copy. If a name is not specified, one is assigned by default. If no options are provided, a name and metavolume are automatically assigned. Use nas_fs to delete the copy of the file system.

Caution: Creating a copy by using -Snapshot may be time consuming, relative to the size of a file system.

[-volume <volume_name>]
Assigns a metavolume to a file system copy. The metavolume must be created by using the nas_volume -Clone command prior to executing this option. The metavolume must be a BCV type and have the same characteristics as the metavolume of the original file system.

[-option <options>]
Specifies the following comma-separated options:

mirror=on
Leaves the file system copy in mirrored mode.

disktype=<disktype>
For systems with both local and R1BCVs, specifies the type of volume to use when creating a snapshot. In a TimeFinder/FS FarCopy configuration, use disktype=R1BCV for creating a snapshot of the PFS on the local VNX for file. For creating a snapshot of an imported FarCopy snapshot on the remote VNX for file, use disktype=STD. This option is supported for only RAID group based disk volumes and cannot be combined with the "mapped_pool=" option.

By default, the system uses the first available R1BCV or BCV, or R1STD or STD device.
Use the disktype= option to designate which to use if there are R1 devices in your configuration.

pool=<mapped_pool>
Specifies the mapped pool to use when creating a snapshot from that pool. This option is supported only for mapped pool disk volumes and cannot be combined with the disktype= option.

A mapped pool is a VNX for file storage pool that is dynamically generated when diskmark is run. It is a one-to-one mapping with either a VNX for block storage pool or a Symmetrix Storage Group.

Note: If the pool= option is used when creating a snapshot, the disk volume will be selected from only this pool. If the pool does not have enough disk volumes to create a snapshot for the source file system, the fs_timefinder command reports an error.

[-star]
The -star option allows the fs_timefinder command to run on STAR SRDF configuration.

-Restore
Restores a file system to its original location by using the unmounted file system copy created with the -Snapshot option. The original file system must not have any associated SnapSure checkpoints.

Caution: Restoring a file system may be time consuming, relative to the amount of data that has changed in the file system.

[-Force]
Forces a restore of a file system copy that is mounted on the metavolume as read-only, or if the volume is an STD type.

[-option <options>]
Specifies the following comma-separated options:

mirror=on
Places the file system copy in mirrored mode.

[-star]
The -star option allows the fs_timefinder command to run on STAR SRDF configuration.

SEE ALSO

Using TimeFinder/FS, NearCopy, and FarCopy on VNX for File, fs_ckpt, fs_group, and nas_fs.

EXAMPLE #1

To create a TimeFinder/FS copy of the PFS, type:

```
$ fs_timefinder ufs1 -Snapshot
operation in progress (not interruptible)...
remainder(MB) =
43688..37205..31142..24933..18649..12608..7115..4991..4129..3281..2457..1653..81
5..0
operation in progress (not interruptible)...id = 18
name = ufs1
acl = 0
in_use = True
type = uxfs
worm = off
volume = mtv1
pool =
rw_servers= server_2
ro_servers=
rw_vdms =
ro_vdms =
backups = ufs1_snap1
auto_ext  = no,thin=no
fast_clone_level = 1
deduplication   = Off
stor_devs =
000187940268-0006,000187940268-0007,000187940268-0008,000187940268-0009
disks = d3,d4,d5,d6
disk=d3 stor_dev=000187940268-0006 addr=c0t1l0-48-0 server=server_2
disk=d3 stor_dev=000187940268-0006 addr=c16t1l0-33-0 server=server_2
disk=d4 stor_dev=000187940268-0007 addr=c0t1l1-48-0 server=server_2
disk=d4 stor_dev=000187940268-0007 addr=c16t1l1-33-0 server=server_2
disk=d5 stor_dev=000187940268-0008 addr=c0t1l2-48-0 server=server_2
disk=d5 stor_dev=000187940268-0008 addr=c16t1l2-33-0 server=server_2
disk=d6 stor_dev=000187940268-0009 addr=c0t1l3-48-0 server=server_2
disk=d6 stor_dev=000187940268-0009 addr=c16t1l3-33-0 server=server_2

id = 19
name = ufs1_snap1
acl = 0
in_use = False
type = uxfs
worm = off
volume = v456
pool =
rw_servers=
ro_servers=
rw_vdms =
ro_vdms =
backup_of = ufs1 Thu Oct 28 14:13:30 EDT 2011
auto_ext  = no,thin=no
fast_clone_level = unavailable
deduplication   = unavailable
stor_devs =
000187940268-0180,000187940268-0181,000187940268-0182,000187940268-0183
disks = rootd378,rootd379,rootd380,rootd381
```

Where:

Value	Definition
name	Name assigned to the file system.
acl	Access control value for a file system. nas_ac provides informatio n.
in_use	If a file system is registered into the mount table of a Data Mover.
type	Type of file system.-list option provides a description of the types.
worm	Whether is enabled.
volume	Volume on which the file system resides.
pool	Storage pool for the file system.

```
        rw_servers      Servers with read-write access to a file system.
        ro_servers      Servers with read-only access to a file system.
        rw_vdms         VDM servers with read-write access to a file system.
        ro_vdms         VDM servers with read-only access to a file system.
        backups         Name of associated backups.
        backup_of       File system that the file system copy is made from.
        auto_ext        Indicates whether auto-extension and thin provisioning are
                        enabled.
        fast_clone_     fast_clone_level=1 enables ability to create a fast clone. File le
vel
        level           retention and fast clone creation cannot be enabled together on a
                        filesystem.
                        fast_clone_level=2 enables ability to create fast clone of a fast
clone (also
                        called as the second level fast clone) on the filesystem.
        deduplication   Deduplication state of the file system. The file data is
                        transferred to the storage which performs the deduplication
                        and compression on the data. The states are:
                        On - Deduplication on the file system is enabled.
                        Suspended - Deduplication on the file system is suspended.
                        Deduplication does not perform any new space reduction but the exi
sting files
                        that were reduced in space remain the same.
                        Off - Deduplication on the file system is disabled.
                        Deduplication does not perform any new space reduction and the dat
a is now
                        reduplicated.
        stor_devs       Storage system devices associated with a file system. The
                        storage device output is the result of the Symmetrix hardware
                        storage system.
        disks           Disks on which the metavolume resides.

        EXAMPLE #2
        ----------

        To create a TimeFinder/FS copy of the PFS, ufs1, and leave a file system
        copy in mirrored mode, type:

        $ fs_timefinder ufs1 -Snapshot -option mirror=on
        operation in progress (not interruptible)...id = 18
        name = ufs1
        acl = 0
        in_use = True
        type = uxfs
        worm = off
        volume = mtv1
        pool =
        rw_servers= server_2
        ro_servers=
        rw_vdms =
        ro_vdms =
        backups = ufs1_snap1
        auto_ext  = no,thin=no
        fast_clone_level = 1
        deduplication   = Off
        stor_devs =
        000187940268-0006,000187940268-0007,000187940268-0008,000187940268-0009
        disks = d3,d4,d5,d6
        disk=d3 stor_dev=000187940268-0006 addr=c0t1l0-48-0 server=server_2
        disk=d3 stor_dev=000187940268-0006 addr=c16t1l0-33-0 server=server_2
        disk=d4 stor_dev=000187940268-0007 addr=c0t1l1-48-0 server=server_2
        disk=d4 stor_dev=000187940268-0007 addr=c16t1l1-33-0 server=server_2
        disk=d5 stor_dev=000187940268-0008 addr=c0t1l2-48-0 server=server_2
        disk=d5 stor_dev=000187940268-0008 addr=c16t1l2-33-0 server=server_2
        disk=d6 stor_dev=000187940268-0009 addr=c0t1l3-48-0 server=server_2
        disk=d6 stor_dev=000187940268-0009 addr=c16t1l3-33-0 server=server_2

        id = 19
        name = ufs1_snap1
        acl = 0
        in_use = False
```

```
type = mirrorfs
worm = off
volume = v456
pool =
rw_servers=
ro_servers=
rw_vdms =
ro_vdms =
backup_of = ufs1 Thu Oct 28 14:19:03 EDT 2012
auto_ext  = no,thin=no
fast_clone_level = unavailable
deduplication   = unavailable
remainder = 0 MB (0%)
stor_devs =
000187940268-0180,000187940268-0181,000187940268-0182,000187940268-0183
disks = rootd378,rootd379,rootd380,rootd381

EXAMPLE #1 provides a description of command output.

EXAMPLE #3
----------

To turn mirroring off for a file system copy, ufs1_snap1, type:

$ fs_timefinder ufs1_snap1 -Mirror off

operation in progress (not interruptible)...
remainder(MB) = 0
operation in progress (not interruptible)...id = 18
name = ufs1
acl = 0
in_use = True
type = uxfs
worm = off
volume = mtv1
pool =
rw_servers= server_2
ro_servers=
rw_vdms =
ro_vdms =
backups = ufs1_snap1
auto_ext  = no,thin=no
fast_clone_level = 1
deduplication   = Off
stor_devs =
000187940268-0006,000187940268-0007,000187940268-0008,000187940268-0009
disks = d3,d4,d5,d6
disk=d3 stor_dev=000187940268-0006 addr=c0t1l0-48-0 server=server_2
disk=d3 stor_dev=000187940268-0006 addr=c16t1l0-33-0 server=server_2
disk=d4 stor_dev=000187940268-0007 addr=c0t1l1-48-0 server=server_2
disk=d4 stor_dev=000187940268-0007 addr=c16t1l1-33-0 server=server_2
disk=d5 stor_dev=000187940268-0008 addr=c0t1l2-48-0 server=server_2
disk=d5 stor_dev=000187940268-0008 addr=c16t1l2-33-0 server=server_2
disk=d6 stor_dev=000187940268-0009 addr=c0t1l3-48-0 server=server_2
disk=d6 stor_dev=000187940268-0009 addr=c16t1l3-33-0 server=server_2

id = 19
name = ufs1_snap1
acl = 0
in_use = False
type = uxfs
worm = off
volume = v456
pool =
rw_servers=
ro_servers=
rw_vdms =
ro_vdms =
backup_of = ufs1 Thu Oct 28 14:21:50 EDT 2011
auto_ext  = no,thin=no
fast_clone_level = unavailable
```

```
deduplication   = unavailable
stor_devs =
000187940268-0180,000187940268-0181,000187940268-0182,000187940268-0183
disks = rootd378,rootd379,rootd380,rootd381

EXAMPLE #1 provides a description of command output.

EXAMPLE #4
----------

To turn mirroring on for a file system copy, ufs1_snap1, type:

$ fs_timefinder ufs1_snap1 -Mirror on
operation in progress (not interruptible)...id = 18
name = ufs1
acl = 0
in_use = True
type = uxfs
worm = off
volume = mtv1
pool =
rw_servers= server_2
ro_servers=
rw_vdms =
ro_vdms =
backups = ufs1_snap1
auto_ext  = no,thin=no
fast_clone_level = 1
deduplication   = Off
stor_devs =
000187940268-0006,000187940268-0007,000187940268-0008,000187940268-0009
disks = d3,d4,d5,d6
disk=d3 stor_dev=000187940268-0006 addr=c0t1l0-48-0 server=server_2
disk=d3 stor_dev=000187940268-0006 addr=c16t1l0-33-0 server=server_2
disk=d4 stor_dev=000187940268-0007 addr=c0t1l1-48-0 server=server_2
disk=d4 stor_dev=000187940268-0007 addr=c16t1l1-33-0 server=server_2
disk=d5 stor_dev=000187940268-0008 addr=c0t1l2-48-0 server=server_2
disk=d5 stor_dev=000187940268-0008 addr=c16t1l2-33-0 server=server_2
disk=d6 stor_dev=000187940268-0009 addr=c0t1l3-48-0 server=server_2
disk=d6 stor_dev=000187940268-0009 addr=c16t1l3-33-0 server=server_2

id = 19
name = ufs1_snap1
acl = 0
in_use = False
type = mirrorfs
worm = off
volume = v456
pool =
rw_servers=
ro_servers=
rw_vdms =
ro_vdms =
backup_of = ufs1 Thu Oct 28 14:21:50 EDT 2011
auto_ext  = no,thin=no
fast_clone_level = unavailable
deduplication   = unavailable
remainder = 0 MB (0%)
stor_devs =
000187940268-0180,000187940268-0181,000187940268-0182,000187940268-0183
disks = rootd378,rootd379,rootd380,rootd381

EXAMPLE #1 provides a description of command output.

EXAMPLE #5
----------

To perform a mirror refresh on ufs1_snap1, type:

$ fs_timefinder ufs1_snap1 -Mirror refresh
operation in progress (not interruptible)...
```

```
remainder(MB) = 4991..4129..3281..2457..1653..815..0
operation in progress (not interruptible)...id = 18
name = ufs1
acl = 0
in_use = True
type = uxfs
worm = off
volume = mtv1
pool =
rw_servers= server_2
ro_servers=
rw_vdms =
ro_vdms =
backups = ufs1_snap1
auto_ext  = no,thin=no
fast_clone_level = 1
deduplication   = Off
stor_devs =
000187940268-0006,000187940268-0007,000187940268-0008,000187940268-0009
disks = d3,d4,d5,d6
disk=d3 stor_dev=000187940268-0006 addr=c0t1l0-48-0 server=server_2
disk=d3 stor_dev=000187940268-0006 addr=c16t1l0-33-0 server=server_2
disk=d4 stor_dev=000187940268-0007 addr=c0t1l1-48-0 server=server_2
disk=d4 stor_dev=000187940268-0007 addr=c16t1l1-33-0 server=server_2
disk=d5 stor_dev=000187940268-0008 addr=c0t1l2-48-0 server=server_2
disk=d5 stor_dev=000187940268-0008 addr=c16t1l2-33-0 server=server_2
disk=d6 stor_dev=000187940268-0009 addr=c0t1l3-48-0 server=server_2
disk=d6 stor_dev=000187940268-0009 addr=c16t1l3-33-0 server=server_2

id = 19
name = ufs1_snap1
acl = 0
in_use = False
type = uxfs
worm = off
volume = v456
pool =
rw_servers=
ro_servers=
rw_vdms =
ro_vdms =
backup_of = ufs1 Thu Oct 28 14:25:21 EDT 2011
auto_ext  = no,thin=no
fast_clone_level = unavailable
deduplication = unavailable
stor_devs =
000187940268-0180,000187940268-0181,000187940268-0182,000187940268-0183
disks = rootd378,rootd379,rootd380,rootd381
```

Example #1 provides a description of command output.

EXAMPLE #6

To restore the file system copy, ufs1_snap1, to its original location, type:

```
$ /nas/sbin/rootfs_timefinder ufs1_snap1 -Restore -Force
operation in progress (not interruptible)...
remainder(MB) = 0
operation in progress (not interruptible)...id = 19
name = ufs1_snap1
acl = 0
in_use = False
type = uxfs
worm = off
volume = v456
pool =
rw_servers=
ro_servers=
rw_vdms =
ro_vdms =
```

```
backup_of = ufs1 Thu Oct 28 14:25:21 EDT 2011
auto_ext  = no,thin=no
fast_clone_level = unavailable
deduplication   = unavailable
stor_devs =
000187940268-0180,000187940268-0181,000187940268-0182,000187940268-0183
disks = rootd378,rootd379,rootd380,rootd381

id = 18
name = ufs1
acl = 0
in_use = True
type = uxfs
worm = off
volume = mtv1
pool =
rw_servers= server_2
ro_servers=
rw_vdms =
ro_vdms =
backups = ufs1_snap1
auto_ext  = no,thin=no
fast_clone_level = 1
deduplication   = Off
stor_devs =
000187940268-0006,000187940268-0007,000187940268-0008,000187940268-0009
disks = d3,d4,d5,d6
disk=d3 stor_dev=000187940268-0006 addr=c0t1l0-48-0 server=server_2
disk=d3 stor_dev=000187940268-0006 addr=c16t1l0-33-0 server=server_2
disk=d4 stor_dev=000187940268-0007 addr=c0t1l1-48-0 server=server_2
disk=d4 stor_dev=000187940268-0007 addr=c16t1l1-33-0 server=server_2
disk=d5 stor_dev=000187940268-0008 addr=c0t1l2-48-0 server=server_2
disk=d5 stor_dev=000187940268-0008 addr=c16t1l2-33-0 server=server_2
disk=d6 stor_dev=000187940268-0009 addr=c0t1l3-48-0 server=server_2 disk=d6
stor_dev=000187940268-0009 addr=c16t1l3-33-0 server=server_2

EXAMPLE #7
----------

To create a snapshot for a mapped pool, type:
$ fs_timefinder ufs1 -name ufs1_snap1 -Snapshot -option pool=bcv_sg
operation in progress (not interruptible)...
remainder(MB) = ..14184..0
operation in progress (not interruptible)...id        = 87
name      = ufs1
acl       = 0
in_use    = False
type      = uxfs
worm      = off
volume    = mtv1
pool      =
rw_servers=
ro_servers=
rw_vdms   =
ro_vdms   =
backups   = ufs1_snap1
fast_clone_level = 1
deduplication   = Off
deduplication   = unavailable
auto_ext  = no,thin=no
deduplication   = unavailable
stor_devs = 000194900546-0037
disks     = d11

id        = 88
name      = ufs1_snap1
acl       = 0
in_use    = False
type      = uxfs
worm      = off
volume    = v456
```

```
        pool      = bcv_sg
        member_of = root_avm_fs_group_49
        rw_servers=
        ro_servers=
        rw_vdms   =
        ro_vdms   =
        backup_of = ufs1 Fri Oct  1 12:03:10 EDT 2011
        auto_ext  = no,thin=no
        fast_clone_level = unavailable
        deduplication   = unavailable
        thin_storage    = False
        tiering_policy  = thickfp2
        mirrored  = False
        stor_devs = 000194900546-003C
        disks     = rootd16
```

Where:

Value	Definition
auto_ext	Indicates whether auto-extension and thin provisioning are enabled.
deduplication	Deduplication state of the file system. The file data is transferred to the storage which performs the deduplication and compression on the data. The states are: On - Deduplication on the file system is enabled. Suspended - Deduplication on the file system is suspended. Deduplication does not perform any new space reduction but the existing files that were reduced in space remain the same. Off - Deduplication on the file system is disabled. Deduplication does not perform any new space reduction and the data is now reduplicated.
thin_storage	Indicates whether the block storage system uses thin provisioning. Values are: True, False, Mixed.
tiering_policy	Indicates the tiering policy in effect. If the initial tier and the tiering policy are the same, the values are: Auto-Tier, Highest Available Tier, Lowest Available Tier. If the initial tier and the tiering policy are not the same, the values are: Auto-Tier/No Data Movement, Highest Available Tier/No Data Movement, Lowest Available Tier/No Data Movement.
mirrored	Indicates whether the disk is mirrored.

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Server CLI Commands

This chapter lists the eNAS Command Set provided for managing, configuring, and monitoring Data Movers. The commands are prefixed with server and appear alphabetically. The command line syntax (Synopsis), a description of the options, and an example of usage are provided for each command.

server_archive	server_arp	server_cdms
server_cepp	server_certificate	server_checkup
server_cifs	server_cifssupport	server_cpu
server_date	server_dbms	server_devconfig
server_df	server_dns	server_export
server_file	server_fileresolve	server_ftp
server_http	server_ifconfig	server_ip
server_kerberos	server_ldap	server_log
server_mount	server_mountpoint	server_mpfs
server_mt	server_name	server_netstat
server_nfs	server_nis	server_nsdomains
server_param	server_pax	server_ping
server_ping6	server_rip	server_route
server_security	server_setup	server_snmpd
server_ssh	server_standby	server_stats
server_sysconfig	server_sysstat	server_tftp
server_umount	server_uptime	server_user
server_usermapper	server_version	server_viruschk

server_vtlu	

server_archive

Reads and writes file archives, and copies directory hierarchies.

SYNOPSIS

```
server_archive <movername> [-cdnvN] -f <archive_file> [-J [p][w|d|u]]
[-I <client_dialect>]
        [-e <archive_name>][-s <replstr>] ...
        [-T [<from_date>][,<to_date>]][<pattern>] ...
server_archive <movername> -r [-cdiknuvDNYZ][-E <limit>]
        [-J [w|d|u]][-C d|i|m][-I <client_dialect>]
        [-f <file_name>][-e <archive_name>]
        [-p <string>] ... [-s <replstr>] ...
        [-T [<from_date>][,<to_date>]] ... [<pattern> ...]
server_archive <movername> -w [-dituvLNPX]
        [-J [w|d|u]][-I <client_dialect>]
        [-b <block_size>][-f <file_name>][-e <archive_name>]
        [-x <format>][-B bytes][-s <replstr>] ...
        [-T [<from_date>][,<to_date>][/[c][m]]] ...
        [[-0]|[-1]][<file> ...]
server_archive -r -w [-diklntuvDLPXYZ]
        [-J [w|d|u]][-C d|i|m]
        [-p <string>] ... [-s <replstr>] ...
        [-T [<from_date>][,<to_date>][/[c][m]]] ...
        [<file> ...] <directory>
```

DESCRIPTION
----

server_archive reads, writes, and lists the members of an archive file, and copies directory hierarchies. The server_archive operation is independent of the specific archive format, and supports a variety of different archive formats.

Note: A list of supported archive formats can be found under the description of the -x option.

The presence of the -r and the -w options specifies the following functional modes: list, read, write, and copy.

-List (no arguments)
server_archive writes to standard output a table of contents of the members of the archive file read from archive, whose pathnames match the specified patterns.

Note: If no options are specified, server_archive lists the contents of the archive.

Read (-r)
server_archive extracts the members of the archive file read from the archive, with pathnames matching the specified patterns. The archive format and blocking are automatically determined on input. When an extracted file is a directory, the entire file hierarchy rooted at that directory is extracted.

Note: Ownership, access, and modification times, and file mode of the extracted files are discussed in more detail under the -p option.

Write (-w)
server_archive writes an archive containing the file operands to archive using the specified archive format. When a file operand is also a directory, the entire file hierarchy rooted at that directory is included.

Copy (-r -w)
server_archive copies the file operands to the destination directory. When a file operand is also a directory, the entire file hierarchy rooted at that directory is included. The effect of the copy is as if the copied files were written to an archive file and then subsequently extracted,

except that there may be hard links between the original and the copied files. The -l option provides more information.

CAUTION

The destination directory must exist and must not be one of the file operands or a member of a file hierarchy rooted at one of the file operands. The result of a copy under these conditions is unpredictable.

While processing a damaged archive during a read or list operation, server_archive attempts to recover from media defects and searches through the archive to locate and process the largest number of archive members possible (the -E option provides more details on error handling).

OPERANDS

The directory operand specifies a destination directory pathname. If the directory operand does not exist, or it is not writable by the user, or it is not a directory name, server_archive exits with a non-zero exit status.

The pattern operand is used to select one or more pathnames of archive members. Archive members are selected using the pattern matching notation described by fnmatch 3. When the pattern operand is not supplied, all members of the archive are selected.

When a pattern matches a directory, the entire file hierarchy rooted at that directory is selected. When a pattern operand does not select at least one archive member, server_archive writes these pattern operands in a diagnostic message to standard error and then exits with a non-zero exit status.

The file operand specifies the pathname of a file to be copied or archived. When a file operand does not select at least one archive member, server_archive writes these file operand pathnames in a diagnostic message to standard error and then exits with a non-zero exit status.

The archive_file operand is the name of a file where the data is stored (write) or read (read/list). The archive_name is the name of the streamer on which the data will be stored (write) or read (read/list).

Note: To obtain the device name, you can use server_devconfig -scsi.

OPTIONS

The following options are supported:

-r
Reads an archive file from archive and extracts the specified files. If any intermediate directories are needed to extract an archive member, these directories will be created as if mkdir 2 was called with the bit-wise inclusive OR of S_IRWXU, S_IRWXG, and S_IRWXO, as the mode argument. When the selected archive format supports the specification of linked files and these files cannot be linked while the archive is being extracted, server_archive writes a diagnostic message to standard error and exits with a non-zero exit status at the completion of operation.

-w
Writes files to the archive in the specified archive format.

-0 (zero)
With this option, a full referenced backup is performed with the time and date of launching put in a reference file. This reference file is an ASCII file and is located in /.etc/BackupDates. The backup is

referenced by the pathname of the files to back up and the time and date when the backup was created. This file is updated only if the backup is successful.

Backup files can be copied using the server_file command.

-<x>
Level x (x=1.9) indicates a backup of all files in a file system that have been modified since the last backup of a level smaller than the previous backup.

For example, a backup is performed for:

```
Monday: level 0 = full backup
Tuesday: level 3 = files modified since Monday
Friday: level 5 = files modified since Tuesday
Saturday: level 4 = files modified since Tuesday
Sunday: level 4 = files modified since Tuesday
```

Note: If the backup type is not indicated, a full backup is performed automatically.

-b <block_size>
When writing an archive, blocks the output at a positive decimal integer number of bytes per write to the archive file. The <block_size> must be a multiple of 512 bytes with a maximum size of 40 kilobytes.

Note: To remain POSIX-compatible, do not exceed 32256 Bytes.

A <block_size> can end with k or b to specify multiplication by 1024 (1K) or 512, respectively. A pair of <block_size> can be separated by x to indicate a product. A specific archive device may impose additional restrictions on the size of blocking it will support. When blocking is not specified, the default for <block_size> is dependent on the specific archive format being used. The -x option provides more information.

-c
Matches all file or archive members except those specified by the pattern and file operands.

-d
Causes files of type directory being copied or archived, or archive members of type directory being extracted, to match only the directory file or archive member, and not the file hierarchy rooted at the directory.

-e <archive_name>
Specifies the archive name when it is streamed.
Note: To prevent the tape from rewinding at the end of command execution, use the -N option with the -e <archive_name> option.

-f <archive_file>
Specifies the archive name when it is a file.
Note: A single archive may span multiple files and different archive devices. When required, server_archive prompts for the pathname of the file or device of the next volume in the archive.

-i
Interactively renames files or archive members. For each archive member matching a pattern operand, or each file matching a file operand, server_archive prompts to /dev/tty giving the name of the file, its file mode, and its modification time. Then server_archive reads a line from /dev/tty. If this line is blank, the file or archive member is skipped. If this line consists of a single period, the file or archive member is processed with no modification to its name.

Otherwise, its name is replaced with the contents of the line. Then server_archive immediately exits with a non-zero exit status if <EOF> is encountered when reading a response, or if /dev/tty cannot

be opened for reading and writing.

-k
Does not allow overwriting existing files.

-l
Links files. In the copy mode (-r, -w), hard links are made between the source and destination file hierarchies whenever possible.

-I <client_dialect>
Allows filename information recovered from an archive to be translated into UTF-8.

-n
Selects the first archive member that matches each pattern operand. No more than one archive member is matched for each pattern. When members of type directory are matched, the file hierarchy rooted at that directory is also matched (unless -d is also specified).

-p <string>
Specifies one or more file characteristic options (privileges). The <string> option-argument is a string specifying file characteristics to be retained or discarded on extraction. The string consists of the specification characters a, e, m, o, and p. Multiple characteristics can be concatenated within the same string and multiple -p options can be specified. The meaning of the specification characters is as follows:

a
Do not preserve file access times. By default, file access times are preserved whenever possible.

e
Preserve everything (default mode), the user ID, group ID, file mode bits, file access time, and file modification time.

Note: The e flag is the sum of the o and p flags.

m
Do not preserve file modification times. By default, file modification times are preserved whenever possible.

o
Preserve the user ID and group ID.

p
Preserve the file mode bits. This specification character is intended for a user with regular privileges who wants to preserve all aspects of the file other than the ownership. The file times are preserved by default, but two other flags are offered to disable this and use the time of extraction instead.

In the preceding list, preserve indicates that an attribute stored in the archive is given to the extracted file, subject to the permissions of the invoking process. Otherwise, the attribute of the extracted file is determined as part of the normal file creation action. If neither the e nor the o specification character is specified, or the user ID and group ID are not preserved for any reason, server_archive will not set the S_ISUID (setuid) and S_ISGID (setgid) bits of the file mode. If the preservation of any of these items fails for any reason, server_archive writes a diagnostic message to standard error.

Note: Failure to preserve these items will affect the final exit status, but will not cause the extracted file to be deleted.

If the file characteristic letters in any of the string option-arguments are duplicated, or in conflict with one another, the ones given last will take precedence. For example, if you specify -p eme, file modification times are still preserved.

-s <replstr>

Modifies the file or archive member names specified by the pattern or <file> operand according to the substitution expression <replstr> using the syntax of the ed utility regular expressions.

Note: The ed 1 manual page provides information.

Multiple -s expressions can be specified. The expressions are applied in the order they are specified on the command line, terminating with the first successful substitution. The optional trailing g continues to apply the substitution expression to the pathname substring, which starts with the first character following the end of the last successful substitution.

The optional trailing p causes the final result of a successful substitution to be written to standard error in the following format:

<original pathname> >> <new pathname>

File or archive member names that substitute the empty string are not selected and are skipped.

-t
Resets the access times of any file or directory read or accessed by server_archive to be the same as they were before being read or accessed by server_archive.

-u
Ignores files that are older (having a less recent file modification time) than a pre-existing file, or archive member with the same name. During read, an archive member with the same name as a file in a file system is extracted if the archive member is newer than the file. During copy, the file in the destination hierarchy is replaced by the file in the source hierarchy, or by a link to the file in the source hierarchy if the file in the source hierarchy is newer.

-v
During a list operation, produces a verbose table of contents using the format of the ls 1 utility with the -l option. For pathnames representing a hard link to a previous member of the archive, the output has the format:

<ls -l listing> == <link name>

For pathnames representing a symbolic link, the output has the format:

<ls -l listing> => <link name>

where <ls -l listing> is the output format specified by the ls 1 utility when used with the -l option. Otherwise, for all the other operational modes (read, write, and copy), pathnames are written and flushed a standard error without a trailing <newline> as soon as processing begins on that file or archive member. The trailing <newline> is not buffered, and is written only after the file has been read or written.

-x format
Specifies the output archive format, with the default format being ustar. The server_archive command currently supports the following formats:

cpio
The extended cpio interchange format specified in the -p1003.2 standard. The default blocksize for this format is 5120 bytes.

Inode and device information about a file (used for detecting file hard links by this format) which may be truncated by this format is detected by server_archive and is repaired.

Note: To be readable by server_archive, the archive must be built on another machine with the option -c (write header information in ASCII).

bcpio
The old binary cpio format. The default blocksize for this format is 5120 bytes.

Note: This format is not very portable and should not be used when other formats are available.

Inode and device information about a file (used for detecting file hard links by this format) which may be truncated by this format is detected by server_archive and is repaired.

sv4cpio
The System V release 4 cpio. The default blocksize for this format is 5120 bytes. Inode and device information about a file (used for detecting file hard links by this format) which may be truncated by this format is detected by server_archive and is repaired.

sv4crc
The System V release 4 cpio with file crc checksums. The default blocksize for this format is 5120 bytes. Inode and device information about a file (used for detecting file hard links by this format) which may be truncated by this format is detected by server_archive and is repaired.

tar
The old BSD tar format as found in BSD4.3. The default blocksize for this format is 10240 bytes. Pathnames stored by this format must be 100 characters or less in length. Only regular files, hard links, soft links, and directories will be archived (other file system types are not supported).

ustar
The extended tar interchange format specified in the -p1003.2 standard. The default blocksize for this format is 10240 bytes.

Note: Pathnames stored by this format must be 250 characters or less in length (150 for basename and 100 for <file_name>).

emctar
This format is not compatible with -p1003.2 standard. It allows archiving to a file greater than 8 GB. Pathnames stored by this format are limited to 3070 characters. The other features of this format are the same as ustar.

server_archive detects and reports any file that it is unable to store or extract as the result of any specific archive format restrictions. The individual archive formats may impose additional restrictions on use.

Note: Typical archive format restrictions include (but are not limited to) file pathname length, file size, link pathname length, and the type of the file.

-B bytes
Limits the number of bytes written to a single archive volume to bytes. The bytes limit can end with m, k, or b to specify multiplication by 1048576 (1M), 1024 (1K) or 512, respectively. A pair of bytes limits can be separated by x to indicate a product.

Note: The limit size will be rounded up to the nearest block size.

-C [d|i|m]
When performing a restore, this allows you to choose PAX behaviors on CIFS collision names.
d: delete
i: ignore
m: mangle

-D
Ignores files that have a less recent file inode change time than a pre-existing file, or archive member with the same name. The -u option provides information.

Note: This option is the same as the .u option, except that the file inode change time is checked instead of the file modification time. The file inode change time can be used to select files whose inode information (such as uid, gid, and so on) is newer than a copy of the file in the destination directory.

-E limit
Has the following two goals:

. In case of medium error, to limit the number of consecutive read faults while trying to read a flawed archive to limit. With a positive limit, server_archive attempts to recover from an archive read error and will continue processing starting with the next file stored in the archive. A limit of 0 (zero) will cause server_archive to stop operation after the first read error is detected on an archive volume. A limit of "NONE" will cause server_archive to attempt to recover from read errors forever.

. In case of no medium error, to limit the number of consecutive valid header searches when an invalid format detection occurs. With a positive value, server_archive will attempt to recover from an invalid format detection and will continue processing starting with the next file stored in the archive. A limit of 0 (zero) will cause server_archive to stop operation after the first invalid header is detected on an archive volume. A limit of "NONE" will cause server_archive to attempt to recover from invalid format errors forever. The default limit is 10 retries.

CAUTION
Using this option with NONE requires extreme caution as server_archive may get stuck in an infinite loop on a badly flawed archive.

-J
Backs up, restores, or displays CIFS extended attributes.
        p: displays the full pathnamefor alternate names (for listing and archive only)
        u: specifies UNIX name for pattern search
        w: specifies M256 name for pattern search
        d: specifies M83 name for pattern search

-L
Follows all symbolic links to perform a logical file system traversal.

-N
Used with the -e archive_name option, prevents the tape from rewinding at the end of command execution.

-P
Does not follow symbolic links.

Note: Performs a physical file system traversal. This is the default mode.

-T [from_date][,to_date][/[c][m]]
Allows files to be selected based on a file modification or inode change time falling within a specified time range of from_date to to_date (the dates are inclusive). If only a from_date is supplied, all files with a modification or inode change time equal to or less than are selected. If only a to_date is supplied, all files with a modification or inode change time equal to or greater than will be selected. When the from_date is equal to the to_date, only files with a modification or inode change time of exactly that time will be selected.

When server_archive is in the write or copy mode, the optional trailing field [c][m] can be used to determine which file time (inode change, file modification or both) is used in the comparison. If neither is specified, the default is to use file modification time only. The m specifies the comparison of file modification time (the time when the file was last written). The c specifies the comparison of inode change time (the time when the file inode was last changed; for example, a change of owner, group, mode, and so on). When c and m are both

specified, then the modification and inode change times are both compared. The inode change time comparison is useful in selecting files whose attributes were recently changed, or selecting files which were recently created and had their modification time reset to an older time (as what happens when a file is extracted from an archive and the modification time is preserved). Time comparisons using both file times are useful when server_archive is used to create a time-based incremental archive (only files that were changed during a specified time range will be archived).

A time range is made up of six different fields and each field must contain two digits. The format is:

[yy[mm[dd[hh]]]]mm[ss]
Where yy is the last two digits of the year, the first mm is the month (from 01 to 12), dd is the day of the month (from 01 to 31), hh is the hour of the day (from 00 to 23), the second mm is the minute (from 00 to 59), and ss is seconds (from 00 to 59). The minute field mm is required, while the other fields are optional, and must be added in the following order: hh, dd, mm, yy. The ss field may be added independently of the other fields. Time ranges are relative to the current time, so -T 1234/cm selects all files with a modification or inode change time of 12:34 P.M. today or later. Multiple -T time range can be supplied, and checking stops with the first match.

-X
When traversing the file hierarchy specified by a pathname, does not allow descending into directories that have a different device ID. The st_dev field as described in stat 2 for more information about device IDs.

-Y
Ignores files that have a less recent file inode change time than a pre-existing file, or archive member with the same name.
Note: This option is the same as the -D option, except that the inode change time is checked using the pathname created after all the filename modifications have completed.

-Z
Ignores files that are older (having a less recent file modification time) than a pre-existing file, or archive member with the same name.
Note: This option is the same as the -u option, except that the modification time is checked using the pathname created after all the filename modifications have completed.

The options that operate on the names of files or archive members (-c, -i, -n, -s, -u, -v, -D, -T, -Y, and -Z) interact as follows.

When extracting files during a read operation, archive members are selected, based only on the user-specified pattern operands as modified by the -c, -n, -u, -D, and -T options. Then any -s and -i options will modify, in that order, the names of those selected files. Then the -Y and -Z options will be applied based on the final pathname. Finally, the -v option will write the names resulting from these modifications.

When archiving files during a write operation, or copying files during a copy operation, archive members are selected, based only on the user specified pathnames as modified by the -n, -u, -D, and -T options (the -D option applies only during a copy operation). Then any -s and -i options will modify, in that order, the names of these selected files. Then during a copy operation, the -Y and the -Z options will be applied based on the final pathname. Finally, the -v option will write the names resulting from these modifications.

When one or both of the -u or -D options are specified along with the -n option, a file is not considered selected unless it is newer than the file to which it is compared.

SEE ALSO

Using the server_archive Utility on VNX.

EXAMPLE #1

To archive the contents of the root directory to the device rst0, type:

$ server_archive <movername> -w -e rst0

EXAMPLE #2

To display the verbose table of contents for an archive stored in <file_name>, type:

$ server_archive <movername> -v -f <file_name>

EXAMPLE #3

To copy the entire olddir directory hierarchy to newdir, type:

$ server_archive <movername> -rw <olddir newdir>

EXAMPLE #4

To interactively select the files to copy from the current directory to dest_dir, type:

$ server_archive <movername> -rw -i <olddir dest_dir>

EXAMPLE #5

To extract all files from the archive stored in <file_name>, type:

$ server_archive <movername> -r -f <file_name>

EXAMPLE #6

To update (and list) only those files in the destination directory /backup that are older (less recent inode change or file modification times) than files with the same name found in the source file tree home, type:

$ server_archive <movername> -r -w -v -Y -Z home /backup

STANDARDS

The server_archive utility is a superset of the -p1003.2 standard.

Note: The archive formats bcpio, sv4cpio, sv4crc, and tar, and the flawed archive handling during list and read operations are extensions to the POSIX standard.

ERRORS

The server_archive command exits with one of the following system messages:

        All files were processed successfully.
        or
        An error occurred.

Whenever server_archive cannot create a file or a link when reading an archive, or cannot find a file when writing an archive, or cannot

preserve the user ID, group ID, or file mode when the -p option is specified, a diagnostic message is written to standard error, and a non-zero exit status is returned. However, processing continues.

In the case where server_archive cannot create a link to a file, this command will not create a second copy of the file.

If the extraction of a file from an archive is prematurely terminated by a signal or error, server_archive may have only partially extracted a file the user wanted. Additionally, the file modes of extracted files and directories may have incorrect file bits, and the modification and access times may be wrong.

If the creation of an archive is prematurely terminated by a signal or error, server_archive may have only partially created the archive which may violate the specific archive format specification.
If while doing a copy, server_archive detects a file is about to overwrite itself, the file is not copied, a diagnostic message is written to standard error and when server_archive completes, it exits with a non-zero exit status.

Last modified: May 12, 2011 1:15 pm.

server_arp

Manages the Address Resolution Protocol (ARP) table for the Data Movers.

SYNOPSIS
----
server_arp {<movername>|ALL}
  <ip_addr>
| -all
| -delete <ip_addr>
| -set <ip_addr> <physaddr>

DESCRIPTION
----

server_arp displays and modifies the IP-to-MAC address translation tables used by the ARP for the specified Data Mover.

The ALL option executes the command for all Data Movers.

OPTIONS

<ip_addr>
Displays the ARP entry for the specified IP address.

-all
Displays the first 64 of the current ARP entries.

-delete <ip_addr>
Deletes an ARP entry.

-set <ip_addr> <physaddr>
Creates an ARP entry with an IP address and physical address.

EXAMPLE #1

To create an ARP entry, type:

```
$ server_arp server_2 -set 172.24.102.20
00:D0:B7:82:98:E0
server_2 : added: 172.24.102.20 at 0:d0:b7:82:98:e0
```

EXAMPLE #2

To display all ARP entries for a specified Data Mover, type:

```
$ server_arp server_2 -all
server_2 :
172.24.102.254 at 0:d0:3:f9:37:fc
172.24.102.20 at 0:d0:b7:82:98:e0
172.24.102.24 at 0:50:56:8e:1d:5
128.221.253.100 at 0:4:23:a7:b1:35
```

EXAMPLE #3

To display an ARP entry specified by IP address, type:

```
$ server_arp server_2 172.24.102.20
server_2 : 172.24.102.20 at 0:d0:b7:82:98:e0
```

EXAMPLE #4

To delete an ARP entry, type:

```
$ server_arp server_2 -delete 172.24.102.24
server_2 : deleted: 172.24.102.24 at 0:50:56:8e:1d:5
```

Last Modified: March 31, 2010 11:15 am

server_cdms

Provides File Migration Service for VNX functionality for the specified Data Movers.

SYNOPSIS

```
server_cdms {<movername>|ALL}
  -connect <mgfs> -type {nfsv2|nfsv3} -path <localpath>
   -source <srcName>:/<srcPath>[-option <options>]
| -connect <mgfs> -type cifs -path <localpath> -netbios <netbios> -source
   \\<srcServer>[.<domain>]\<srcShare>[\<srcPath>] -admin
   [<domain>\]<admin_name> [-wins <wins>]
| -disconnect <mgfs> {-path <localpath>|-path <cid>|-all}
| -verify <mgfs> [-path {<localpath>|<cid>}]
| -Convert <mgfs>
| -start <mgfs> -path <localpath> [-Force] -log <logpath>
   [-include <include_path>][-exclude <exclude_path>]
| -halt <mgfs> -path <localpath>
| -info [<mgfs>][-state {START|STOP|ON_GOING|ERROR|SUCCEED|FAIL}]
```

DESCRIPTION
----

server_cdms establishes and removes connections to remote systems, and allows users to start on-access migration.

server_cdms creates an auto-migration process on the Data Mover to ensure that all data has been migrated from the remote system. server_cdms also checks the state of the migrated file system (MGFS), all auto-migration processes, and the connection, and reports if all data has been migrated successfully.

CDMS supports NFSv2 and NFSv3 only.

The ALL option executes the command for all Data Movers.

OPTIONS

-connect <mgfs> -type {nfsv2|nfsv3} -path <localpath> -source <srcName>:/<srcPath>
Provides a connection for the VNX with the remote NFS server. The -type option specifies the protocol type to be used for communication with the remote NFS server. The directory <localpath> in the file system must be unique for that file system.

The -source option specifies the source file server name or IP address of the remote server as the <srcName> and the export path for migration. For example, nfs_server:/export/path

Note: After the -connect command completes, the file system must be exported.

[-option <options>]
Specifies the following comma-separated options:

[useRootCred={true|false}]
When the file system is mounted, true ensures that the MGFS reads from the source file server using root access UID=0, GID=0. This assumes that the source file server path is exported to allow root access from the specified Data Mover. When false (default), the MGFS uses the owner.s UID and GID to access data.

[proto={TCP|UDP}]
Sets the connection protocol type. The default is TCP.

[nfsPort=<port>]
Sets a remote NFS port number in case the Portmapper or RPC bind is not running, and the port is not the default of 2049.

[mntPort=<port>]
Sets a remote mount port number in case Portmapper or RPC bind is not running.

[mntVer={1|2|3}]
Sets the version used for mount protocol. By default, NFSv2 uses mount version 2, unless user specified version 1; NFSv3 uses mount version 3.

[localPort=<port>]
Sets the port number used for NFS services, if it needs to be different from the default. The default port number is always greater than 1024.

-connect <mgfs> -type cifs -path <localpath>
-netbios <netbios> -source \\<srcServer>[.<domain>]
\<srcShare>[\<srcPath>] -admin [<domain>\]
<admin_name>[-wins <wins>]
Provides a connection for the VNX with the remote CIFS server as specified by its NetBIOS name. The directory <localpath> in the file system must be unique for that file system. The -source option specifies the source file server name of the remote server as the <srcName> and the share path for migration that is not at the root of the share. For example, \\share\dir1...

The -source and -admin option strings must be enclosed by quotes when issued in a Linux shell.

The -admin option specifies an administrator for the file system. A password is asked interactively when the command is issued. The -wins option specifies an IP address for the WINS server.

Note: This is required only for Windows NT 4.0.

-disconnect <mgfs> {-path <localpath>|-path <cid>|-all}
Removes a connection without migrating the data. The <localpath> is not removed nor is any partially migrated data.

The administrator should manually remove this data before attempting a -verify or -Convert command. It may require the administrator to handle a partial migration of old data as well as potentially new data created by users.

It is recommended not to use the -disconnect option if the administrator has exported this directory for user access.

-verify <mgfs>
Checks that all data has completed the migration for the <mgfs>.

[-path {<localpath>|<cid>}]
If the -path option is provided, it can check on a communication basis. If no path is provided, the system defaults to checking all connections on the file system.

-Convert <mgfs>
Performs a verify check on the entire file system, then changes the file system type from MGFS to UxFS. After the -Convert option succeeds, no data migration can be done on that file system.

-start <mgfs> -path <localpath> [-Force] -log
<logpath>
Directs the Data Mover to migrate all files from the source file server to the VNX. The -log option provides detailed information on the state of the migration, and any failures that might occur. The <localpath> is the path where the migration thread is started. The -Force option is used if you need to start a migration thread a second time on the same <localpath> where a previous migration thread had already finished. For example, -Force would be needed to start a thread which had no include file (that is, to migrate all remaining files) on <localpath> where a thread with an include file had already

been run.

[-include <include_path>]
Starts the thread in the <include_path> which is the path of the file containing the specified directories.

[-exclude <exclude_path>]
Excludes files or directories from migration. The <include_path> is the path of the file containing the specified directories.

-halt <mgfs> -path <localpath>
Stops a running thread, and halts its execution on the Data Mover. The <mgfs> is the name of the migration file system and the <localpath> is the full path where the migration thread was started.

The -start option resumes thread execution.

-info
Displays a status on the migration file system and the threads.

[<mgfs>]
Specifies the migration file system.

[-state {START|STOP|ON_GOING|ERROR|SUCCEED|FAIL}]
Displays only the threads that are in the state that is specified.

SEE ALSO

VNX CDMS Version 2.0 for NFS and CIFS, server_export, server_mount, and server_setup.

EXAMPLE #1

To provide a connection for the migration file system to communicate with the remote NFS server, type:

```
$ server_cdms server_2 -connect ufs1 -type nfsv3 -path
/nfsdir -source 172.24.102.144:/srcdir -option proto=TCP
server_2 : done
```

EXAMPLE #2

To provide a connection for the migration file system to communicate with the remote CIFS server, type:

```
$ server_cdms server_2 -connect ufs1 -type cifs -path
/dstdir -netbios dm112-cge0 -source
"\\\winserver1.nasdocs.emc.com\srcdir" -admin
"nasdocs.emc.com\administrator" -wins 172.24.102.25
server_2 : Enter Password:*******
done
```

EXAMPLE #3

To display a status on the migration file system, type:

```
$ server_cdms server_2
server_2 :
CDMS enabled with 32 threads.

ufs1:
path = /nfsdir
cid = 0
type = NFSV3
source = 172.24.102.144:/srcdir
options= proto=TCP
path = /dstdir
cid = 1
```

```
type = CIFS
source = \\winserver1.nasdocs.emc.com\srcdir\
netbios= DM112-CGE0.NASDOCS.EMC.COM
admin = nasdocs.emc.com\administrator

When migration is started:

$ server_cdms server_2
server_2 :
CDMS enabled with 32 threads.

ufs1:
path = /nfsdir
cid = 0
type = NFSV3
source = 172.24.102.144:/srcdir
options= proto=TCP
path = /dstdir
cid = 1
type = CIFS
source = \\winserver1.nasdocs.emc.com\srcdir\
netbios= DM112-CGE0.NASDOCS.EMC.COM
admin = nasdocs.emc.com\administrator
threads:
path = /dstdir
state = ON_GOING
log = /
cid = NONE

Where:
 Value                Definition

 ufs1              Migration file system

 path              Directory in the local file system

 cid               Connection ID (0 through 1023)

 type              Protocol type to be used to communicate with the remote server

 source            Source file server name or IP address of the remote server and
                   the export path for migration

 options           Connection protocol type

 netbios           NetBIOS name of the remote CIFS server

 admin             Administrator for the file system

 threads           Currently existing migration threads

 state             Current status of migration threads

 log               Location of the log file that provides detailed information

EXAMPLE #4
----------

To direct server_2 to migrate all files from the source file server to the
VNX, type:

$ server_cdms server_2 -start ufs1 -path /dstdir -log /
server_2 : done

EXAMPLE #5
----------

To display information about migration with the specified status, type:

$ server_cdms server_2 -info ufs1 -state ON_GOING
server_2 :
```

```
ufs1:
path = /nfsdir
cid = 0
type = NFSV3
source = 172.24.102.144:/srcdir
options= proto=TCP
path = /dstdir
cid = 1
type = CIFS
source = \\winserver1.nasdocs.emc.com\srcdir\
netbios= DM112-CGE0.NASDOCS.EMC.COM
admin = nasdocs.emc.com\administrator
threads:
path = /dstdir
state = ON_GOING
log = /
cid = NONE
```

EXAMPLE #6

To stop data migration on server_2 for ufs1, type:

```
$ server_cdms server_2 -halt ufs1 -path /dstdir
server_2 : done
```

EXAMPLE #7

To check that all data has completed the migration, type:

```
$ server_cdms server_2 -verify ufs1 -path /dstdir
server_2 : done
```

EXAMPLE #8

To disconnect the path on server_2 for data migration, type:

```
$ server_cdms server_2 -disconnect ufs1 -path /nfsdir
server_2 : done
```

EXAMPLE #9

To disconnect all paths for data migration, type:

```
$ server_cdms server_2 -disconnect ufs1 -all
server_2 : done
```

EXAMPLE #10

To perform a verify check on ufs1, and then convert it to a uxfs, type:

```
$ server_cdms server_2 -Convert ufs1
server_2 : done
```

Last Modified: March 31, 2010 05:00 pm

server_cepp

Manages the Common Event Publishing Agent (CEPA) service on the specified Data Mover

SYNOPSIS

```
server_cepp {<movername>|ALL}
  -service {-start|-stop|-status|-info}
| -pool {-info|-stats}
```

DESCRIPTION
----

server_cepp starts or stops the CEPA service on the specified Data Mover or all Data Movers and displays information on the status, configuration, and statistics for the service and the pool. The CEPA service is set up in the cepp.conf configuration file. The CEPA configuration is displayed using -service -status, but changes can only be made by directly editing the file with a text editor.

ALL executes the command for all Data Movers.

OPTIONS

-service {-start|-stop|-status|-info}
The -start option starts the CEPA service on the specified Data Mover. -stop stops the CEPA service, -status returns a message indicating whether the CEPA service has started or been stopped, and -info displays information about the CEPA service including key properties of the configured pool.

-pool {-info|-stats}
Displays properties or statistics for the CEPA pool on the specified Data Mover.

SEE ALSO

Using VNX Event Enabler

EXAMPLE #1

To start the CEPA service on a Data Mover, type:

```
$ server_cepp server_2 .service .start
server_2 : done
```

EXAMPLE #2

To display the status of the CEPA service, type:

```
$ server_cepp server_2 .service .status
server_2 : CEPP Stopped
```

EXAMPLE #3

To display the configuration of the CEPA service, type:

To display the configuration of the CEPA service, type:

```
$ server_cepp server_2 .service .info
server_2 :
CIFS share name = \\DVBL\CHECK$
cifs_server = DVBL
heartbeat_interval = 15 seconds
ft level = 1
```

```
ft size = 1048576
ft location = /.etc/cepp
msrpc user = OMEGA13$
msrpc client name = OMEGA13.CEE.LAB.COM
pool_name   server_required access_checks_ignored req_timeout retry_timeout
pool_1            no              0                 5000          25000
```

Where

Value	Definition
CIFS share name	The name of the shared directory and CIFS server used to access files in the Data Movers.
cifs_server	CIFS server to access files.
heartbeat_interval	The time taken to scan each CEPA server.
ft level	Fault tolerance level assigned. This option is required. 0 (continue and tolerate lost events; default setting), 1 (continue and use a persistence file as a circular event buffer for lost events), 2 (continue and use a persistence file as a circular event buffer for lost events until the buffer is filled and then stop CIFS), or 3 (upon heartbeat loss of connectivity, stop CIFS).
ft location	Directory where the persistence buffer file resides relative to the root of a file system. If a location is not specified, the default location is the root of the file system.
ft size	Maximum size in MB of the persistence buffer file. The default is 1 MB and the range is 1 MB to 100 MB.
msrpc user	Name assigned to the user account that the CEPA service is running under on the CEE machine. For example, ceeuser.
msrpc client name	Domain name assigned if the msrpc user is a member of a domain. For example, domain.ceeuser.
pool_name	Name assigned to the pool that will use the specified CEPA options.
server_required	Displays availability of the CEPA server. If a CEPA server is not available and this option is yes, an error is returned to the requestor that access is denied. If a CEPA server is not available and this option is no, an error is not returned to the requestor and access is allowed.
access_checks_ignored	The number of CIFS requests processed when a CEPA server is not available and the server_required option is set to "no." This option is reset when the CEPA server becomes available.
req_timeout	Time out in ms to send a request that allows access to the CEPA server.
retry_timeout	Time out in ms to retry the access request sent to the CEPA server.

EXAMPLE #4

To display information about the CEPA pool, type:

```
$ server_cepp server_2 -pool -info
```

```
server_2 :
pool_name = pool1
server_required = yes
access_checks_ignored = 0
req_timeout = 5000 ms
retry_timeout = 25000 ms
pre_events = OpenFileNoAccess, OpenFileRead
post_events = CreateFile,DeleteFile
post_err_events = CreateFile,DeleteFile
CEPP Servers:
IP = 10.171.10.115, state = ONLINE, vendor = Unknown
...
```

Where

Value	Definition
pre_events	Sends notification before selected event occurs. An empty list indicates that no pre-event messages are generated.
post_events	Sends notification after selected event occurs. An empty list indicates that no post-event messages are generated.
post_err_events	Sends notification if selected event generates an error. An empty list indicates that no post-error-event messages are generated.
CEPP Servers	IP addresses of the CEPA servers; state of the CEPA servers; vendor software installed on CEPA servers.

EXAMPLE #5

To display statistics for the CEPA pool, type:

```
$ server_cepp server_2 -pool -stats
server_2 :
pool_name = pool1
Event Name           Requests     Min(us)       Max(us)       Average(us)
OpenFileWrite        2              659           758             709
CloseModified        2              604           635             620
Total Requests = 4
Min(us) = 604
Max(us) = 758
Average(us) = 664
```

Last Modified: April 05 2010, 11:15 am

server_certificate

Manages VNX for file system's Public Key Infrastructure (PKI) for the specified Data Movers.

SYNOPSIS

```
server_certificate {<movername>|ALL}
    -ca_certificate
        [-list]
        | -info {-all|<certificate_id>}
        | -import [-filename <path>]
        | -delete {-all|<certificate_id>}}
    -persona
        [-list]
        | -info {-all|<persona_name>|id=<persona_id>}
        | -generate {<persona_name>|id=<persona_id>} -key_size {2048|4096}
            [-cs_sign_duration <# of months>]
            {-cn|-common_name} <common_name>[;<common_name>]
            [-ou <org_unit>[;<org_unit>]]
            [-organization <organization>]
            [-location <location>]
            [-state <state>]
            [-country <country>]
            [-filename <output_path>]
        | -clear {<persona_name>|id=<persona_id>}{-next|-current|-both}
        | -import {<persona_name>|id=<persona_id>} [-filename <path>]
```

DESCRIPTION
----

server_certificate manages the use of public key certificates between Data Movers acting as either clients or servers. server_certificate -ca_certificate manages the Certificate Authority (CA) certificates the VNX uses to confirm a server's identity when the Data Mover is acting as a client. server_certificate -persona manages the certificates presented by the Data Mover to a client application when the Data Mover is acting as a server as well as the certificates presented by the Data Mover to a server configured to require client authentication.

OPTIONS

-ca_certificate
Lists the CA certificates currently available on the VNX. The ouput from this command is identical to the output from the -list option.

-ca_certificate -list
Lists the CA certificates currently available on the VNX.

-ca_certificate -info {-all|<certificate_id>}
Displays the properties of a specified CA certificate or all CA certificates.

-ca_certificate -import [-filename <path>]
Imports a CA certificate. You can only paste text in PEM format at the command prompt. Specify -filename and provide a path to import a CA certificate in either DER or PEM format.

-ca_certificate -delete {-all|<certificate_id>}
Deletes a specified CA certificate or all CA certificates.

-persona
Lists the key sets and associated certificates currently available on the VNX. The ouput from this command is identical to the output from the -list option.

-persona -list
Lists the key sets and associated certificates currently available on the VNX.

-persona -info {-all|<persona_name>|id=<persona_id>}
Displays the properties of the key sets and associated certificates, including the text of a pending certificate request, of a specified persona or all personas.

-persona -generate {<persona_name>|id=<persona_id>}
-key_size <bits> {-cn|-common_name} <common_name>
[;<common_name>]
Generates a public/private key set along with a request to sign the certificate. Specify either the persona name or ID. The ID is automatically generated when the persona is created. You can determine the ID using the -list or -info options. The key size can be either 2048 or 4096 bits. Use either -cn or -common_name to specify the commonly used name. The common name is typically a hostname that describes the Data Mover with which the persona is associated. Multiple common names are allowed but must be separated by semicolons.

[-cs_sign_duration <# of months>]
Specifies the number of months the certificate is valid. A month is defined as 30 days. This option is valid only if the certificate will be signed by the Control Station. If this option is specified, you cannot save the request to a file using the -filename option.

[-ou <org_unit>[;<org_unit>]]
Identifies the organizational unit. Multiple organizational units are allowed but must be separated by semicolons.

[-organization <organization>]
Identifies the organization.

[-location <location>]
Identifies the physical location of the organizational unit.

[-state <state>]
Identifies the state where the organizational unit is located.

[-country <country>]
Identifies the country where the organization unit is located. This value is limited to two characters.

[-filename <output_path>]
Provides a path to where the request should be saved to a file. This option is valid only if the certificate will be signed by an external CA. If this option is specified, you cannot specify the number of months the certificate is valid using the -cs_sign_duration option.

-persona -clear {<persona_name>|id=<persona_id>}{-next|-current|-both}
Deletes a key set and the associated certificate. You can delete the current key set and certificate, the next key set and certificate, or both.

-persona -import {<persona_name>|id=<persona_id>}[-filename <path>]
Imports a CA-signed certificate. You can only paste text in PEM format at the command prompt. Specify -filename and provide a path to import a CA-signed certificate in either DER or PEM format.

SEE ALSO
----

nas_ca_certificate

EXAMPLE #1

To import a CA certificate, specifying a filename and path, type:

$ server_certificate server_2 -ca_certificate -import -filename "/tmp/ca_cert.pem"

```
done

EXAMPLE #2
----------

To list all the CA certificates currently available on the VNX, type:

$ server_certificate ALL -ca_certificate -list
server_2 :

id=1
subject=O=Celerra Certificate Authority;CN=sorento
issuer=O=Celerra Certificate Authority;CN=sorento
expire=201203180326392

id=2
subject=C=US;O=VeriSign, Inc.;OU=Class 3 Public Primary Certification Author
issuer=C=US;O=VeriSign, Inc.;OU=Class 3 Public Primary Certification Author
expire=202808012359592

server_3 :
id=1
subject=O=Celerra Certificate Authority;CN=zeus-cs
issuer=O=Celerra Certificate Authority;CN=zeus-cs
expire=201206061812152

EXAMPLE #3
----------

To list the properties of the CA certificate identified by certificate ID
2, type:

$ server_certificate server_2 -ca_certificate -info 2

server_2 :
id=2
subject = C=US;O=VeriSign, Inc.;OU=Class 3 Public Primary Certification
Authority
issuer = C=US;O=VeriSign, Inc.;OU=Class 3 Public Primary Certification
Authority
start = 19960129000000Z
expire = 202808012359592
signature alg. = md2WithRSAEncryption
public key alg. = rsaEncryption
public key size = 1024 bits
serial number = 70ba e41d 10d9 2934 b638 ca7b 03cc babf
version = 1

EXAMPLE #4
----------

To generate a key set and certificate request to be sent to an external
CA for the persona identified by the persona name default, type:

$ server_certificate server_2 -persona -generate default
-key_size 2048 -common_name division.xyz.com

server_2 :
Starting key generation. This could take a long time ...
done

EXAMPLE #5
----------

To list all the key sets and associated certificates currently available
on the VNX, type:

$ server_certificate ALL -persona -list
server_2 :
id=1
name=default
```

```
next state=Request Pending
request subject=CN=name;CN=1.2.3.4

server_3 :
id=1
name=default
next state=Not Available
CURRENT CERTIFICATE:
id=1
subject=CN=test;CN=1.2.3.4
expire=20070706183824Z
issuer=O=Celerra Certificate Authority;CN=eng173100
```

EXAMPLE #6

To list the properties of the key set and certificate identified by persona ID 1, type:

```
$ server_certificate server_2 -persona -info id=1
server_2 :
id=1
name=default
next state=Request Pending
request subject=CN=name;CN=1.2.3.4
Request:
-----BEGIN CERTIFICATE REQUEST-----
MIIEZjCCAk4CAQAwITENMAsGA1UEAxMEbmFtZTEQMA4GA1UEAxMHMS4yLjMuNDCC
AiIwDQYJKoZIhvcNAQEBBQADggIPADCCAgoCggIBANKW3Q/F6eMqIxrCO5IeXLET
bWkm5RzrbI5lHxLNuhobR5S9G2o+k47X0QZFkGzq/2F7kR06vVIH7CPH9X2gGAzV
7GmZaFO0wPcktPJYzjQE8guNhcL1qZpPl4IZrbnSGEAWcAAE0nvNwLp9aN0WSC+N
TDJZY4A9yTURiUc+Bs8plhQh16wLLL0zjUKIvKjAqiTE0F3RApVJEE/9y6N+Idsb
Vwf/rvzP6/z0wZW5Hl84HKXInJaHTBDK59G+e/Y2JgvUY1UNBZ5SODunOakHabex
k6COFYjDu7Vd+yHpvcyTalHJ2RcIavpQuM02o+VVpxgUyX7M1+VXJXTJm0yb4j4g
tZITOSVZ2FqEpOkoIpzqoAL7A9B69WpFbbpIX8danhReafDh4oj4yWocvSwMKYv1
33nLak3+wpMQNrwJ2L9FIHP2fXClnvThBgupm7uqqHP3TfNBbBPTYY3qkNPZ78wx
/njUrZKbfWd81Cc+ngUi33hbMuBR3FFsQNASYZUzgl5+JexALH5jhBahd2aRXBag
itQLhvxYK0dEqIEwDfdDedx7i+yro2gbNxhLLdtkuBtKrmOnuT5g2WWXNKzNa/H7
KWv8JSwCv1mW1N/w7V9aEbDizBBfer+ZdMPkGLbyb/EVXZnHABeWH3iKC6/ecnRd
4Kn7KO9F9qXVHlzzTeYVAgMBAAGgADANBgkqhkiG9w0BAQUFAAOCAgEAzSS4ffYf
2WN0vmZ0LgsSBcVHPVEVg+rP/aU9iNM9KDJ4P4OK41UDU8tOGy09Kc8EvklBUm59
fyjt2T/3RqSgvvkCEHJsVW3ZMnSsyjDo6Ruc0HmuY4q+tuLl+dilSQnZGUxt8asw
dhEpdEzXA6o9cfmVZMSt5QicfAmmBNr4BaO96+VAlg59fu/chU1pvKWWMGXz4I2s
7z+UdMBYO4pEfyG1i34Qof/z4K0SVNICn3CEkW5TIsSt8qA/E2JXXlLhbMYWKYuY
9ur/gspHuWzkIXZFx4SmTK9/RsE1Vy7fBztIoN8myFN0nma84D9pyqls/yhvXZ/D
iDF6Tgk4RbNzuanRBSYiJFu4Tip/nJlK8uv3ZyFJ+3DK0c8ozlBLuQdadxHcJglt
m/T4FsHa3JS+D8CdA3uDPfIvvVNcwP+4RBK+Dk6EyQe8uKrVL7ShbacQCUXn0AAd
Ol+DQYFQ7Mczcm84L98srhov3JnIEKcjaPseB7S9KtHvHvvs4q1lQ5U2RjQppykZ
qpSFnCbYDGjOcqOrsqNehV9F4h9fTszEdUY1UuLgvtRj+FTT2Ik7nMK641wfVtSO
LCial6kuYsZg16SFxncnH5gKHtQMWxd9nv+UyJ5VwX3aN12N0ZQbaIDcQp75Em2E
aKjd28cZ6FEavimn69sz0B8PHQV+6dPwywM=
-----END CERTIFICATE REQUEST-----
```

EXAMPLE #7

To generate a key set and certificate request that is automatically received by the Control Station for the persona identified by the persona name default, type:

```
$ server_certificate server_2 -persona -generate default
-key_size 2048 -cs_sign_duration 12 -comon_name
division.xyz.com
server_2 :
Starting key generation. This could take a long time ...
done
```

EXAMPLE #8

To generate a key set and certificate request to be sent to an external

CA specifying subject information, type:

```
$ server_certificate server_2 -persona -generate default
-key_size 2048 -common_name division.xyz.com -ou QA
-organization XYZ -location Bethesda -state Maryland
-country US -filename /tmp/server_2.1.request.pem
server_2 :
Starting key generation. This could take a long time ...
done
```

EXAMPLE #9

To import a signed certificate and paste the certificate text, type:

```
$ server_certificate server_2 -persona -import default
server_2 : Please paste certificate data. Enter a carriage return and on the
new
line type .end of file. or .eof. followed by another carriage return.
```

Last Modified: March 31, 2010 12:45 pm

server_checkup

Checks the configuration parameters, and state of a Data Mover and its dependencies.

SYNOPSIS

```
server_checkup {<movername>|ALL}
  [-test <component> [-subtest <dependency>][-quiet][-full]]
| -list
| -info {<component>|all}
```

DESCRIPTION
----

server_checkup performs a sanity check of a specific Data Mover component and its dependencies by checking configuration parameters, and the current state of the component and dependencies.

A component is any basic feature that is available on the Data Mover,for example, CIFS. A dependency is a configuration component of aData Mover that the proper operation of a Data Mover functionality(like CIFS) is depending upon. This configuration component can be owned by multiple Data Mover components. For example, proper operation of a CIFS service depends on correctly specified DNS,WINS, Antivirus, and so on.

server_checkup displays a report of errors and warnings detected in the specified Data Mover component and its dependencies.

OPTIONS

No arguments
Performs a sanity check of all the components and all their dependencies on the specified Data Mover or all Data Movers.

-test <component>
Performs a sanity check of a specific component and all of its dependencies.

[-subtest <dependency>]
Performs a sanity check of a specific component and its specified dependency only. If the dependency is not defined, executes the command for all the dependencies of the component.

[-quiet]
Displays only the number of errors and warnings for the sanity check.

[-full]
Provides a full sanity check of the specified Data Movers.

-list
Lists all available components that can be checked on a Data Mover.

-info <component>
Lists all dependencies of the specified component, with details of checks that can be performed on each dependency.

EXAMPLE #1

To list the available component in the Data Mover, type:

```
$ server_checkup server_2 -list
server_2 : done

REPV2
HTTPS
CIFS
FTPDS
```

EXAMPLE #2

To execute the check of the CIFS component, type:

```
$ server_checkup server_2 -test CIFS
server_2 :

-----------------------------------Checks-------------------------------------
Component CIFS :
ACL : Checking the number of ACL per file system.....................*Pass
Connection: Checking the load of TCP connections of CIFS...................Pass
Credential: Checking the validity of credentials...........................Pass
DC : Checking the connectivity and configuration of the DCs.........*Fail
DFS : Checking the DFS configuration files and DFS registry.......... Pass
DNS : Checking the DNS configuration and connectivity to DNS servers. Pass
EventLog : Checking the configuration of Windows Event Logs...............Pass
FS_Type : Checking if all file systems are all DIR3 type................. Pass
GPO : Checking the GPO configuration................................. Pass
HomeDir : Checking the configuration of home directory share.............. Pass
I18N : Checking the I18N mode and the Unicode/UTF8 translation tables. Pass
Kerberos : Checking machine password update for Kerberos..................Fail
LocalGrp : Checking the local groups database configuration...............Fail
NIS : Checking the connectivity to the NIS servers, if defined....... Pass
NTP : Checking the connectivity to theNTP servers, if defined........ Pass
Ntxmap : Checking the ntxmap configuration file.......................... Pass
Security : Checking the CIFS security settings............................Pass
Server : Checking the CIFS files servers configuration................... Pass
Share : Checking the network shares database............................ Pass
SmbList : Checking the range availability of SMB ID.....................*Pass
Threads : Checking for CIFS blocked threads.............................. Pass
UM_Client : Checking for the connectivity to usermapper servers, if any....Pass
UM_Server : Checking the consistency of usermapper database, if primary....*Pass
UnsupOS : Checking for unsupported client network OS..................... Pass
UnsupProto: Checking for unsupported client network protocols..............Pass
VC : Checking the configuration to Virus Checker servers............. Pass
WINS : Checking for the connectivity to WINS servers, if defined...... Pass
NB: a result with a '*' means that some tests were not executed. use -full to
run them
------------------------------------------------------------------------------

----------------------------CIFS : Kerberos Warnings----------------------------
Warning 17451974742: server_2 : No update of the machine password of server
'DM102-CGE1'. hold.
--> Check the log events to find out the reason of this issue.
Warning 17451974742: server_2 : No update of the machine password of server
'DM102-CGE0'. hold.
--> Check the log events to find out the reason of this issue.

----------------------------CIFS : LocalGrp Warnings----------------------------
Warning 17451974726: server_2 : The local group 'Guests' of server
'DM102-CGE1'
contains an unmapped member: S-1-5-15-60415a8a-335a7a0d-6b635f23-202.The
access
to some network resources may be refused.
--> According the configured resolver of your system (NIS, etc config files,
usermapper, LDAP...),add the missing members.
------------------------------------------------------------------------------

--------------------------------CIFS : DC Errors--------------------------------
Error 13160939577: server_2 : pingdc failed due to NT error ACCESS_DENIED at
step
SAMR lookups
--> check server configuration and/or DC policies according to reported error.
Error 13160939577: server_2 : pingdc failed due to NT error ACCESS_DENIED at
step
SAMR lookups
--> check server configuration and/or DC policies according to reported error.
------------------------------------------------------------------------------
```

EXAMPLE #3

To execute only the check of the DNS dependency of the CIFS component, type:
$ server_checkup server_2 -test CIFS -subtest DNS
server_2 :

-----------------------------------Checks-----------------------------------

Component CIFS :
DNS : Checking the DNS configuration and connectivity to DNS servers. Pass

---

EXAMPLE #4

To list the available dependencies of the CIFS component, type:

$ server_checkup server_2 -info CIFS
server_2 :
done

COMPONENT : CIFS
DEPENDENCY : ACL
DESCRIPTION : Number of ACL per file system.
TESTS :
In full mode, check if the number of ACL per file system doesn't exceed 90% of the maximum limit.

COMPONENT : CIFS
DEPENDENCY : Connection
DESCRIPTION : TCP connection number
TESTS :
Check if the number of CIFS TCP connections doesn't exceed 80% of the maximum number.

COMPONENT : CIFS
DEPENDENCY : Credential
DESCRIPTION : Users and groups not mapped
TESTS :
Check if all credentials in memory are mapped to a valid SID.

COMPONENT : CIFS
DEPENDENCY : DC
DESCRIPTION : Connectivity to the domain controllers
TESTS :
Check the connectivity to the favorite DC (DCPing),
In full mode, check the connectivity to all DC of the domain,
Check if DNS site information are defined for each computer name,
Check if the site of each computer name has an available DC,
Check if trusted domain of each computer name can be reached,
Check the ds.useDCLdapPing parameter is enabled,
Check the ds.useADSite parameter is enabled.

COMPONENT : CIFS
DEPENDENCY : DFS
DESCRIPTION : DFS service configuration on computer names
TESTS :
Check the DFS service is enabled in registry if DFS metadata exists,
Check the DFS metadata of each share with DFS flag are correct,
Check if share names in DFS metadata are valid and have the DFS flag,
Check if each DFS link is valid and loaded,
Check in the registry if the WideLink key is enabled and corresponds to a valid share name.

COMPONENT : CIFS
DEPENDENCY : DNS
DESCRIPTION : DNS domain configuration
TESTS :
Check if each DNS domain has at least 2 defined servers,
Check the connectivity to each DNS server of each DNS domain,

Check if each DNS server of each DNS domain supports really the DNS service,
Check the ds.useDSFile parameter (automatic discovery of DC),
Check the ds.useDSFile parameter is enabled if the directoryservice file exists.

COMPONENT : CIFS
DEPENDENCY : EventLog
DESCRIPTION : Event Logs parameters on servers
TESTS :
Check if the pathnames of each event logs files are valid (application, system and security),
Check if the maximum file size of each event logs file doesn't exceed 1GB,
Check if the retention time of each event logs file doesn't exceed 1 month.

COMPONENT : CIFS
DEPENDENCY : FS_Type
DESCRIPTION : DIR3 mode of filesystems
TESTS :
Check if each file system is configured in the DIR3 mode.

COMPONENT : CIFS
DEPENDENCY : GPO
DESCRIPTION : GPO configuration on Win2K servers
TESTS :
Check if the size of the GPO cache file doesn't exceed 10% of the total size of the root file system,
Check the last modification date of the GPO cache file is up-to-date,
Check the cifs.gpo and cifs.gpoCache parameters have not been changed,

COMPONENT : CIFS
DEPENDENCY : HomeDir
DESCRIPTION : Home directory shares configuration
TESTS :
Check if the home directory shares configuration file exists, the feature is enabled,
Check if the home directory shares configuration file is optimized (40 lines maximum),
Check the syntax of the home directory shares configuration file.

COMPONENT : CIFS
DEPENDENCY : I18N
DESCRIPTION : Internationalization and translation tables
TESTS :
Check if computer name exists, the I18N mode is enabled,
Check the .etc_common file system is correctly mounted,
Check the syntax of the definition file of the Unicode characters,
Check the uppercase/lowercase conversion table of Unicode character is valid.

COMPONENT : CIFS
DEPENDENCY : Kerberos
DESCRIPTION : Kerberos configuration
TESTS :
Check the machine password update is enabled and up-to-date.

COMPONENT : CIFS
DEPENDENCY : LocalGrp
DESCRIPTION : Local groups and local users
TESTS :
Check the local group database doesn't contain more than 80% of the maximum number of servers,
Check if the servers in the local group database are all valid servers,
Check the state of the local group database (initialized and writable),
Check if the members of built-in local groups are all resolved in the domain,
Check the number of built-in local groups and built-in local users,
Check if the number of defined local users doesn't exceed 90% of the maximum number.

COMPONENT : CIFS
DEPENDENCY : NIS
DESCRIPTION : Network Information System (NIS) configuration
TESTS :

If NIS is configured, check at least 2 NIS servers are defined (redundancy check),
Check if each NIS server can be contacted on the network,
Check if each NIS server really supports the NIS service.

COMPONENT : CIFS
DEPENDENCY : NTP
DESCRIPTION : Network Time Protocol (NTP) configuration
TESTS :
If NTP is configured, check at least 2 NTP servers are defined (redundancy check),
Check if each NIS server can be contacted on the network,
If computer names exist, check if NTP is configured and is running.

COMPONENT : CIFS
DEPENDENCY : Ntxmap
DESCRIPTION : Checking the ntxmap.conf file.
TESTS :
Check the data consistency of the ntxmap configuration file.

COMPONENT : CIFS
DEPENDENCY : Security
DESCRIPTION : Security settings
TESTS :
If the I18N mode is enabled, check the share/unix security setting is not in use,
Discourage to use the share/unix security setting,
Check the cifs.checkAcl parameter is enabled if the security setting is set to NT.

COMPONENT : CIFS
DEPENDENCY : Server
DESCRIPTION : Files servers
TESTS :
Check if each CIFS server is configured with a valid IP interface,
Check if each computer name has joined its domain,
Check if each computer name is correctly registered in their DNS servers,
Check if the DNS servers have the valid IP addresses of each computer name,
Check if a DNS domain exists if at least one computer name exists,

COMPONENT : CIFS
DEPENDENCY : Share
DESCRIPTION : Network shares
TESTS :
Check the available size and i-nodes on the root file system are at least 10% of the total size,
Check the size of the share database doesn’t exceed 30% of the total size of the root file system,
Check if the pathname of each share is valid and is available,
Check if each server in the share database really exists,
Check if the I18N mode is enabled, all the share names are UTF-8 compatible,
Check the list of ACL of each share contains some ACE,
Check the length of each share name doesn’t exceed 80 Unicode characters.

COMPONENT : CIFS
DEPENDENCY : SmbList
DESCRIPTION : 64k UID, TID and FID limits
TESTS :
In full mode, check the 3 SMB ID lists (UID, FID and TID) don’t exceed 90% of the maximum ID number.

COMPONENT : CIFS
DEPENDENCY : Threads
DESCRIPTION : Blocked threads and overload
TESTS :
Check CIFS threads blocked more than 5 and 30 seconds,
Check the maximum number of CIFS threads in use in the later 5 minutes doesn’t exceed 90% of the total number,
Check the number of threads reserved for Virus Checker doesn’t exceed 20% of the total number of CIFS threads.

```
COMPONENT : CIFS
DEPENDENCY : UM_Client
DESCRIPTION : Connectivity to the usermapper server
TESTS :
If usermapper servers are defined, check each server can be contacted,
Check if usermapper servers are defined, NIS is not simultaneously activated.

COMPONENT : CIFS
DEPENDENCY : UM_Server
DESCRIPTION : Primary usermapper server
TESTS :
If a primary usermapper is defined locally, check its database size doesn't
exceed 30% of the total size,
Check if configuration file is in use, the filling rate of the ranges doesn't
exceed 90%,
Check if configuration file is in use, 2 ranges do not overlap,
Check if secmap is enabled,
In full mode, check the SID/UID and SID/GID mappings and reverses are correct
and coherent.

COMPONENT : CIFS
DEPENDENCY : UnsupOS
DESCRIPTION : Client OS not supported
TESTS :
Check for unsupported client network OS.

COMPONENT : CIFS
DEPENDENCY : UnsupProto
DESCRIPTION : Unsupported protocol commands detected
TESTS :
Check for unsupported client network protocol commands.

COMPONENT : CIFS
DEPENDENCY : VC
DESCRIPTION : Virus checker configuration
TESTS :
If VC is enabled, check the syntax of the VC configuration file,
Check if the VC 'enable' file and the VC configuration are compatible,
Check the number of VC servers. Make sure at least 2 servers are defined, for
redundancy,
Check if there are offline VC servers,
Check if the VC high watermark has not been reached,
Check the connection of VC servers to the Data Mover.

COMPONENT : CIFS
DEPENDENCY : WINS
DESCRIPTION : WINS servers.
TESTS :
If NetBIOS names are defined, check if at least one WINS server is defined,
Check the number of WINS servers. check if two servers are defined for
redundancy,
Check if each WINS server can be contacted on the network,
Check these servers are really WINS servers,
Check if the NetBIOS are correctly registered on the servers.

EXAMPLE #5
----------

To execute additional tests, type:

$ server_checkup server_2 -full
server_2 :


------------------------------------Checks-------------------------------------

Component REPV2 :

F_RDE_CHEC: Checking the F-RDE compatibilty of Repv2 sessions........ Fail

Component HTTPS :
```

```
HTTP: Checking the configuration of HTTP applications..... Pass
SSL : Checking the configuration of SSL applications..... Pass

Component CIFS :

ACL : Checking the number of ACL per file system..... Pass
Connection: Checking the load of TCP connections of CIFS..... Pass

Credential: Checking the validity of credentials..... Pass

DC : Checking the connectivity and configuration of the DCs..... Fail
DFS : Checking the DFS configuration files and DFS registry..... Pass
DNS : Checking the DNS configuration and connectivity to DNS servers. Pass
EventLog : Checking the configuration of Windows Event Logs..... Pass
FS_Type : Checking if all file systems are all DIR3 type..... Pass
GPO : Checking the GPO configuration..... Pass
HomeDir : Checking the configuration of home directory share..... Pass
I18N : Checking the I18N mode and the Unicode/UTF8 translation tables Pass
Kerberos : Checking machine password update for Kerberos..... Fail
LocalGrp : Checking the local groups database configuration..... Fail
NIS : Checking the connectivity to the NIS servers, if defined..... Pass
NTP : Checking the connectivity to theNTP servers, if defined..... Pass
Ntxmap : Checking the ntxmap configuration file..... Pass
Security : Checking the CIFS security settings..... Pass
Server : Checking the CIFS files servers configuration..... Pass
Share : Checking the network shares database..... Pass
SmbList : Checking the range availability of SMB ID..... Pass
Threads : Checking for CIFS blocked threads..... Pass
UM_Client : Checking for the connectivity to usermapper servers, if any....
Pass

UM_Server : Checking the consistency of usermapper database, if primary....
Pass

UnsupOS : Checking for unsupported client network OS..... Pass
UnsupProto: Checking for unsupported client network protocols..... Pass

VC : Checking the configuration to Virus Checker servers..... Pass
WINS : Checking for the connectivity to WINS servers, if defined..... Pass


Component FTPDS :

FS_Type   : Checking if all file systems are in the DIR3 format..... Pass
FTPD      : Checking the configuration of FTPD..... Fail
NIS       : Checking the connectivity to the NIS servers..... Pass
NS        : Checking the naming services configuration..... Fail
NTP       : Checking the connectivity to the NTP servers..... Fail
SSL       : Checking the configuration of SSL applications..... Fail
--------------------------------------------------------------------------------

--------------------------HTTPS : SSL Warnings----------------------------

Warning 17456169084: server_2 :  The SSL feature 'DHSM' can not get
certificate from the persona default. Because this feature needs a certificate
and a private key, it can not start,
--> Run the server_certificate command to generate a new key set and
certificate for this persona. Or run the appropriate command (like server_http
for instance) to set a correct persona for this SSL feature.

Warning 17456169084: server_2 :  The SSL feature 'DIC' can not get certificate
from the persona default. Because this feature needs a certificate and a
private key, it can not start,
--> Run the server_certificate command to generate a new key set and
certificate for this persona. Or run the appropriate command (like server_http
for instance) to set a correct persona for this SSL feature.

Warning 17456169084: server_2 :  The SSL feature 'DIC_S' can not get
certificate from the persona default. Because this feature needs a certificate
and a private key, it can not start,
```

--> Run the server_certificate command to generate a new key set and certificate for this persona. Or run the appropriate command (like server_http for instance) to set a correct persona for this SSL feature.

Warning 17456169084: server_2 : The SSL feature 'DIC_L' can not get certificate from the persona default. Because this feature needs a certificate and a private key, it can not start,
--> Run the server_certificate command to generate a new key set and certificate for this persona. Or run the appropriate command (like server_http for instance) to set a correct persona for this SSL feature.

Warning 17456169084: server_2 : The SSL feature 'DBMS_FILE_TRANSFER' can not get certificate from the persona default. Because this feature needs a certificate and a private key, it can not start,
--> Run the server_certificate command to generate a new key set and certificate for this persona. Or run the appropriate command (like server_http for instance) to set a correct persona for this SSL feature.

----------CIFS : Credential Warnings----------

Warning 17456168968: server_2 : The CIFS service is currently stopped. Many CIFS sanity check tests cannot be done as all CIFS servers are currently disabled on this Data Mover.
--> Start the CIFS server by executing the 'server_setup' command, and try again.

----------CIFS : DC Warnings----------

Warning 17456168968: server_2 : The CIFS service is currently stopped. Many CIFS sanity check tests cannot be done as all CIFS servers are currently disabled on this Data Mover.
--> Start the CIFS server by executing the 'server_setup' command, and try again.

----------CIFS : DFS Warnings----------

Warning 17456168968: server_2 : The CIFS service is currently stopped. Many CIFS sanity check tests cannot be done as all CIFS servers are currently disabled on this Data Mover.
--> Start the CIFS server by executing the 'server_setup' command, and try again.

----------CIFS : EventLog Warnings----------

Warning 17456168968: server_2 : The CIFS service is currently stopped. Many CIFS sanity check tests cannot be done as all CIFS servers are currently disabled on this Data Mover.
--> Start the CIFS server by executing the 'server_setup' command, and try again.

----------CIFS : HomeDir Warnings----------

Warning 17456168968: server_2 : The CIFS service is currently stopped. Many CIFS sanity check tests cannot be done as all CIFS servers are currently disabled on this Data Mover.
--> Start the CIFS server by executing the 'server_setup' command, and try again.

----------CIFS : I18N Warnings----------

Warning 17456168968: server_2 : The CIFS service is currently stopped. Many CIFS sanity check tests cannot be done as all CIFS servers are currently disabled on this Data Mover.
--> Start the CIFS server by executing the 'server_setup' command, and try again.

----------CIFS : Kerberos Warnings----------

Warning 17456168968: server_2 : The CIFS service is currently stopped. Many CIFS sanity check tests cannot be done as all CIFS servers are currently disabled on this Data Mover.

--> Start the CIFS server by executing the 'server_setup' command, and try again.

----------CIFS : LocalGrp Warnings----------

Warning 17456168968: server_2 : The CIFS service is currently stopped. Many CIFS sanity check tests cannot be done as all CIFS servers are currently disabled on this Data Mover.
--> Start the CIFS server by executing the 'server_setup' command, and try again.

----------CIFS : NTP Warnings----------

Warning 17456169044: server_2 : The Network Time Protocol subsystem (NTP) has been stopped or is not connected to its server. It may cause potential errors during Kerberos authentication (timeskew).
--> If the NTP service is not running, start it using the server_date command. If it is not connected, check the IP address of the NTP server and make sure the NTP service is up and running on the server. If needed, add another NTP server in the configuration of the Data Mover. Use the server_date command to manage the NTP service and the parameters on the Data Mover.

----------CIFS : Secmap Warnings----------

Warning 17456168968: server_2 : The CIFS service is currently stopped. Many CIFS sanity check tests cannot be done as all CIFS servers are currently disabled on this Data Mover.
--> Start the CIFS server by executing the 'server_setup' command, and try again.

----------CIFS : Server Warnings----------

Warning 17456168968: server_2 : The CIFS service is currently stopped. Many CIFS sanity check tests cannot be done as all CIFS servers are currently disabled on this Data Mover.
--> Start the CIFS server by executing the 'server_setup' command, and try again.

----------CIFS : Share Warnings----------

Warning 17456168968: server_2 : The CIFS service is currently stopped. Many CIFS sanity check tests cannot be done as all CIFS servers are currently disabled on this Data Mover.
--> Start the CIFS server by executing the 'server_setup' command, and try again.

----------CIFS : SmbList Warnings----------

Warning 17456168968: server_2 : The CIFS service is currently stopped. Many CIFS sanity check tests cannot be done as all CIFS servers are currently disabled on this Data Mover.
--> Start the CIFS server by executing the 'server_setup' command, and try again.

----------CIFS : WINS Warnings----------

Warning 17456168968: server_2 : The CIFS service is currently stopped. Many CIFS sanity check tests cannot be done as all CIFS servers are currently disabled on this Data Mover.
--> Start the CIFS server by executing the 'server_setup' command, and try again.

----------FTPDS : NTP Warnings----------

Warning 17456169044: server_2 : The Network Time Protocol subsystem (NTP) has been stopped or is not connected to its server. It may cause potential errors during Kerberos authentication (timeskew).
--> If the NTP service is not running, start it using the server_date command. If it is not connected, check the IP address of the NTP server and make sure the NTP service is up and running on the server. If needed, add another NTP server in the configuration of the Data Mover. Use the server_date command to

manage the NTP service and the parameters on the Data Mover.

----------FTPDS : SSL Warnings----------

Warning 17456169084: server_2 : The SSL feature 'DHSM' can not get certificate from the persona default. Because this feature needs a certificate and a private key, it can not start,
--> Run the server_certificate command to generate a new key set and certificate for this persona. Or run the appropriate command (like server_http for instance) to set a correct persona for this SSL feature.

Warning 17456169084: server_2 : The SSL feature 'DIC' can not get certificate from the persona default. Because this feature needs a certificate and a private key, it can not start,
--> Run the server_certificate command to generate a new key set and certificate for this persona. Or run the appropriate command (like server_http for instance) to set a correct persona for this SSL feature.

Warning 17456169084: server_2 : The SSL feature 'DIC_S' can not get certificate from the persona default. Because this feature needs a certificate and a private key, it can not start,
--> Run the server_certificate command to generate a new key set and certificate for this persona. Or run the appropriate command (like server_http for instance) to set a correct persona for this SSL feature.

Warning 17456169084: server_2 : The SSL feature 'DIC_L' can not get certificate from the persona default. Because this feature needs a certificate and a private key, it can not start,
--> Run the server_certificate command to generate a new key set and certificate for this persona. Or run the appropriate command (like server_http for instance) to set a correct persona for this SSL feature.

Warning 17456169084: server_2 : The SSL feature 'DBMS_FILE_TRANSFER' can not get certificate from the persona default. Because this feature needs a certificate and a private key, it can not start,
--> Run the server_certificate command to generate a new key set and certificate for this persona. Or run the appropriate command (like server_http for instance) to set a correct persona for this SSL feature.

----------REPV2 : F_RDE_CHECK Errors----------

Error 13160415855: server_2 : For the Replication session: rep1,
Data Mover version on the source fs: 5.6.47
Data Mover version on the destination fs: 5.5.5
Minimum required Data Mover version on the destination fs: 5.6.46

The Data Mover version on the destination file system is incompatible with the Data Mover version on the source file system. After data transfer, the data in the destination file system may appear to be corrupt, even though the data is in fact intact.

Upgrade the Data Mover where the destination file system resides to at least 5.6.46.

Error 13160415855: server_2 : For the Replication session:rsd1,
F-RDE version on the source fs: 5.6.46
F-RDE version on the destination fs: 5.5.5
Minimum required F-RDE version on the destination fs: 5.6.46
The F-RDE versions are incompatible.
After data transfer, the data in the dst FS may appear to be corrupt.
--> Upgrade the DataMover where the dst fs resides to atleast the version on the source.

Error 13160415855: server_2 : For the Replication session:rsd2,
F-RDE version on the source fs: 5.6.46
F-RDE version on the destination fs: 5.5.5
Minimum required F-RDE version on the destination fs: 5.6.46
The F-RDE versions are incompatible.
After data transfer, the data in the dst FS may appear to be corrupt.

--> Upgrade the DataMover where the dst fs resides to atleast the version on the source.

Error 13160415855: server_2 : For the Replication session:rsd3, F-RDE version on the source fs: 5.6.46
F-RDE version on the destination fs: 5.5.5 Minimum required
F-RDE version on the destination fs: 5.6.46
The F-RDE versions are incompatible.
After data transfer, the data in the dst FS may appear to be corrupt.
--> Upgrade the DataMover where the dst fs resides to atleast the version on the source.

----------HTTPS : SSL Errors----------

Error 13156876314: server_2 : The persona 'default' contains nor certificate neither private keys sets. So, this persona can not be used by a SSL feature on the Data Mover.
--> Run the server_certificate command to generate a new key set and certificate for this persona.

----------CIFS : DNS Errors----------

Error 13161070637: server_2 : The DNS service is currently stopped and does not contact any DNS server. The CIFS clients may not be able to access the Data Mover on the network.
--> Start the DNS service on the Data Mover, using the 'server_dns' command.

----------CIFS : NS Errors----------

Error 13156352011: server_2 : None of the naming services defined for the entity 'host' in nsswitch.conf is configured.
--> Make sure each entity (e.g. host, passwd..) in the nsswitch.conf file contains naming services, (e.g. local files, NIS or usermapper), and make sure these services are configured. Use the corresponding commands like server_nis, server_dns or server_ldap to make sure they are configured.

Error 13156352011: server_2 : None of the naming services defined for the entity 'group' in nsswitch.conf is configured.
--> Make sure each entity (e.g. host, passwd..) in the nsswitch.conf file contains naming services, (e.g. local files, NIS or usermapper), and make sure these services are configured. Use the corresponding commands like server_nis, server_dns or server_ldap to make sure they are configured.

Error 13156352011: server_2 : None of the naming services defined for the entity 'netgroup' in nsswitch.conf is configured.
--> Make sure each entity (e.g. host, passwd..) in the nsswitch.conf file contains naming services, (e.g. local files, NIS or usermapper), and make sure these services are configured. Use the corresponding commands like server_nis, server_dns or server_ldap to make sure they are configured.

----------FTPDS : FTPD Errors----------

Error 13156876314: server_2 : The persona 'default' contains nor certificate neither private keys sets. So, this persona can not be used by a SSL feature on the Data Mover.
--> Run the server_certificate command to generate a new key set and certificate for this persona.

----------FTPDS : NS Errors----------

Error 13156352011: server_2 : None of the naming services defined for the entity 'host' in nsswitch.conf is configured.
--> Make sure each entity (e.g. host, passwd..) in the nsswitch.conf file contains naming services, (e.g. local files, NIS or usermapper), and make sure these services are configured. Use the corresponding commands like server_nis, server_dns or server_ldap to make sure they are configured.

Error 13156352011: server_2 : None of the naming services defined for the entity 'group' in nsswitch.conf is configured.
--> Make sure each entity (e.g. host, passwd..) in the nsswitch.conf file contains naming services, (e.g. local files, NIS or usermapper), and make sure

```
these services are configured. Use the corresponding commands like server_nis,
server_dns or server_ldap to make sure they are configured.

Error 13156352011: server_2 :  None of the naming services defined for the
entity 'netgroup' in nsswitch.conf is configured.
--> Make sure each entity (e.g. host, passwd..) in the nsswitch.conf file
contains naming services, (e.g. local files, NIS or usermapper), and make sure
these services are configured. Use the corresponding commands like server_nis,
server_dns or server_ldap to make sure they are configured.

--------------------------FTPDS : SSL Errors-----------------------------

Error 13156876314: server_2 :  The persona 'default' contains nor certificate
neither private keys sets. So, this persona can not be used by a SSL feature
on the Data Mover.
--> Run the server_certificate command to generate a new key set and
certificate for this persona.

------------------------------------------------------------------------------
Total :   14 errors, 25 warnings

------------------------------------------------------------------------------
```

EXAMPLE #6

To display only the number of errors and warnings for a Data Mover and dependency, type:

```
$ server_checkup server_2 -quiet
server_2 :

-----------------------------------Checks-------------------------------------

Component REPV2 :

F_RDE_CHEC: Checking the F-RDE compatibilty of Repv2 sessions..... Fail

Component HTTPS :

HTTP      : Checking the configuration of HTTP applications..... Pass
SSL       : Checking the configuration of SSL applications..... Pass

Component CIFS :

ACL       : Checking the number of ACLs per file system.....*Pass
Connection: Checking the load of CIFS TCP connections..... Pass
Credential: Checking the validity of credentials..... Fail
DC        : Checking the connectivity and configuration of Domain Controlle Fail
DFS       : Checking the DFS configuration files and DFS registry..... Fail
DNS       : Checking the DNS configuration and connectivity to DNS servers.Fail
EventLog  : Checking the configuration of Windows Event Logs..... Fail
FS_Type   : Checking if all file systems are in the DIR3 format..... Pass
GPO       : Checking the GPO configuration..... Pass
HomeDir   : Checking the configuration of home directory shares..... Fail
I18N      : Checking the I18N mode and the Unicode/UTF8 translation tables. Fail
Kerberos  : Checking password updates for Kerberos..... Fail
LDAP      : Checking the LDAP configuration..... Pass
LocalGrp  : Checking the database configuration of local groups..... Fail
NIS       : Checking the connectivity to the NIS servers..... Pass
NS        : Checking the naming services configuration..... Fail
NTP       : Checking the connectivity to the NTP servers..... Fail
Ntxmap    : Checking the ntxmap configuration file..... Pass
Secmap    : Checking the SECMAP database..... Fail
Security  : Checking the CIFS security settings..... Pass
Server    : Checking the CIFS file servers configuration..... Fail
Share     : Checking the network shares database..... Fail
SmbList   : Checking the range availability of SMB IDs.....*Pass
Threads   : Checking for CIFS blocked threads..... Pass
UM_Client : Checking the connectivity to usermapper servers..... Pass
UM_Server : Checking the usermapper server database.....*Pass
```

```
UnsupOS   : Checking for unsupported client network operating systems.. Pass
UnsupProto: Checking for unsupported client network protocols.......... Pass
VC        : Checking the configuration of Virus Checker servers........ Pass
WINS      : Checking the connectivity to WINS servers.................. Fail

Component FTPDS :

FS_Type   : Checking if all file systems are in the DIR3 format........ Pass
FTPD      : Checking the configuration of FTPD.......................... Fail
NIS       : Checking the connectivity to the NIS servers............... Pass
NS        : Checking the naming services configuration................. Fail
NTP       : Checking the connectivity to the NTP servers............... Fail
SSL       : Checking the configuration of SSL applications.............. Pass

NB: a result with a '*' means that some tests were not executed. use -full
to run them

------------------------------------------------------------------------------

Total :   12 errors, 14 warnings

-----------------------------------Checks-------------------------------------

------------------------------------------------------------------------------
Last Modified: April 05, 2010 12:30 pm
```

server_cifs

Manages the CIFS configuration for the specified Data Movers or Virtual Data Movers (VDMs).

SYNOPSIS

```
server_cifs {<movername>|ALL} [<options>]
'options' can be one of the following:
| -option {{audit [,user=<user_name>][,client=<client_name>][,full]}
    | homedir[=NO]}
| -add netbios=<netbios_name>,domain=<domain_name>[,alias=<alias_name>...]
    [,hidden={y|n} [[,interface=<if_name>[,wins=<ip>[:<ip>]]]...]
    [,local_users][-comment <comment>]
| -add compname=<comp_name>,domain=<full_domain_name>[,alias=<alias_name>...]
    [,hidden={y|n}][,authentication={kerberos|all}]
    [,netbios=<netbios_name>][[,interface=<if_name>[,wins=<ip>[:<ip>]]
    [,dns=<if_suffix>]]...][,local_users][-comment <comment>]
|-add standalone=<netbios_name>,workgroup=<workgroup_name>
    [,alias=<alias_name>...][,hidden={y|n}]
    [[,interface=<if_name>[,wins=<ip>[:<ip>]]...][,local_users]
    [-comment <comment>]
| -rename -netbios <old_name> <new_name>
| -Join compname=<comp_name>,domain=<full_domain_name>,admin=<admin_name>
    [,ou=<organizational_unit>]
    [-option {reuse|resetserverpasswd|addservice=nfs}]
| -Unjoin compname=<comp_name>,domain=<full_domain_name>,admin=<admin_name>
| -add security={NT|UNIX|SHARE} [,dialect=<dialect_name>]
| -add wins=<ip_addr>[,wins=<ip_addr>...]
| -add usrmapper=<ip_addr>[,usrmapper=<ip_addr>...]
| -Disable <interface>[,<interface>...]
| -Enable <interface>[,<interface>...]
| -delete netbios=<netbios_name> [-remove_localgroup]
    [,alias=<alias_name>...][,interface=<if_name>]
| -delete compname=<comp_name> [-remove_localgroup]
    [,alias=<alias_name>...][,interface=<if_name>]
| -delete wins=<ip_addr>[,wins=<ip_addr>...]
| -delete usrmapper=<ip_addr>[,usrmapper=<ip_addr>...]
| -delete standalone=<netbios_name> [-remove_localgroup]
    [,alias=<alias_name>...][,interface=<if_name>]
| -update {<share_name>|<path>} [mindirsize=<size>][force]
| -Migrate {<fs_name> -acl|<netbios_servername> -localgroup}
    <src_domain>{:nb=<netbios>|:if=<interface>}
    <dst_domain>{:nb=<netbios>|:if=<interface>}
| -Replace {<fs_name> -acl|<netbios_servername> -localgroup}
    {:nb=<netbios>|:if=<interface>}
| -smbhash
        {-hashgen <path> [-recursive] [-minsize <size>]
        | -hashdel <path> [-recursive]
        | -abort <id>
        | -info
        | -fsusage <fs_name>
        | -exclusionfilter <filter>
        | -audit {enable|disable} [-task] [-service] [-access]
        | -service {enable|disable}
        | -cleanup <fs_name> [-all |-unusedfor <days>|-unusedsince
    <date>}}
| -setspn {-list [server=<full_comp_name>]
| -add <SPN> compname=<comp_name>,domain=<full_domain_name>,
      admin=<admin_name>
| -delete <SPN> compname=<comp_name>,domain=<full_domain_name>,
      admin=<admin_name>}
      }
}
```

DESCRIPTION
----

server_cifs manages the CIFS configuration for the specified <movername> which can be the physical Data Mover or VDMs.

Most command options are used with both VDMs and physical Data Movers, whereas others are only used with physical Data Movers. Options available for physical Data Movers only are:

- -add security/dialect
- -add/delete usrmapper
- -enable/disable interface

The ALL option executes the command for all Data Movers.

OPTIONS

No arguments
Displays the CIFS protocol configuration. Certain inputs are not casesensitive; however, variables may be automatically converted to uppercase.

<options>
CIFS options include:

-option audit
Audits the CIFS configuration by testing for live connections to a Data Mover.

[,user=<user_name>][,client=<client_name>][,full]
Audits the live connections created when the session is initiated by the specified <client_name> or audits the live connections for those owned by the specified <user_name>. The full option can be used to identify open files. The <client_name> can be a string or an IPV4 address and the <user_name> can be a string of maximum 20 characters.

-option homedir[=NO]
Enables and disables (default) the home directory feature. The Data Mover reads information from the homedir map file.

-add netbios=<netbios_name>, domain=<domain_name>
Configures a Windows NT 4.0-like CIFS server on a Data Mover, assigning the specified <netbios_name> and <domain_name> to the server. The domain name is limited to 15 bytes.

Caution: Each NetBIOS name must be unique to the domain and the Data Mover.

[,alias=<alias_name>...]
Assigns a NetBIOS alias to the <netbios_name> associated with the NetBIOS name. The <alias_name> must:

- Be unique on a Data Mover
- Be limited to 15 bytes
- Not begin with an @ (at sign) or - (dash) character
- Not include spaces, tab characters, or the following symbols: / \ : ; , = * +|[] ? < > "

[,hidden={y|n}]
By default, the <netbios_name> is displayed in the Network Neighborhood. If hidden=y is specified, the <netbios_name> does not appear.

[[,interface=<if_name>[,wins=<ip>[:<ip>]]]...]
Specifies a logical IP interface for the CIFS server in the Windows NT 4.0 domain and associates up to two WINS IP addresses with each interface. The interface name is case-sensitive.

Note: When configuring a CIFS server without any interfaces for a Data Mover, it becomes the default CIFS server and is available on all interfaces not used by other CIFS servers. The default CIFS server can be

deleted at any time. It is recommended that IP interfaces should always be specified. VDMs do not have default CIFS servers.

[,local_users]
Enables local user support that allows the creation of a limited number of local user accounts on the CIFS server. When this command executes, type and confirm a password that is assigned to the local Administrator account on the CIFS server. In addition to the Administrator account, a Guest account is also created. The Guest account is disabled by default. The Administrator account password must be changed before the Administrator can log in to the CIFS server.

After initial creation of the stand-alone server, the local_users option resets the local Administrator account password. The password can only be reset if it has not been changed through a Windows client. If the password has already been changed through Windows, the reset will be refused.

[-comment <comment>]
Assigns a comment to the configuration. The comment is delimited by quotes. Comment length is limited to 48 bytes (represented as 48 ASCII characters or a variable number of Unicode multibyte characters) and cannot include colons since they are recognized as delimiters.

-add compname=<comp_name>,domain=<full_domain_name>
Configures a CIFS server as the <comp_name> in the specified Windows Active Directory workgroup. A default NetBIOS name is automatically assigned to the <comp_name>. Since the default for <netbios_name> is derived from the <comp_name>, the <comp_name> must not contain any characters that are invalid for a <netbios_name>.

In the case of disjointed namespaces, you must use the fully qualified domain name for the <comp_name>. For example, for a disjointed namespace, you must always specify the fully qualified domain name (FQDN) with the computer name when joining a CIFS server to a domain, that is, dm112-cge0.emc.com, not just dm112-cge0.

The <comp_name> is limited to 63 bytes. The fully qualified domain name is limited to 155 bytes. The <full_domain_name> must contain a dot (.). There cannot be a @ (at sign) or - (dash) character. The name also cannot include spaces, tab characters, or the symbols: / \ : ; , = * +|[] ? < > "

Caution: Each computer name must be unique to the domain and the Data Mover.

Note: Using International Character Sets for File provides details. Only Windows NT security mode can be configured when UTF-8 is enabled.

[,alias=<alias_name>...]
Assigns an alias to the NetBIOS name. The <alias_name> must:

* Be unique on a Data Mover
* Be limited to 15 bytes
* Not begin with an @ (at sign) or - (dash) character
* Not include spaces, tab characters, or the following symbols: / \ : ; , = * +|[] ? < > "

[,hidden={y|n}]
By default, the computer name appears in the Network Neighborhood. If hidden=y is specified, then the computer name does not appear.

[,authentication={kerberos|all}]
Specifies the type of user authentication. The kerberos option limits the server usage to Kerberos authentication; the all option (default) allows both Kerberos and NTLM authentication.

[,netbios=<netbios_name>]
Specifies a <netbios_name> for the <comp_name> in place of the default. The default for <netbios_name> is assigned automatically and is derived from the first 15 bytes of the <comp_name>. The <netbios_name> cannot begin with an @ (at sign) or - (dash) character. The name also cannot include spaces, tab characters, or the symbols: / \ : ; , = * +|[] ? < > "

[[,interface=<if_name>[,wins=<ip>[:<ip>]]]...]
Specifies a logical IP interface for the CIFS server in the Active Directory domain and associates up to two WINS IP addresses with each interface. The interface name is case-sensitive.

Note: When configuring a CIFS server without any interfaces for a Data Mover, it becomes the default CIFS server and is available on all interfaces not used by other CIFS servers. The default CIFS server can be deleted at any time. It is recommended that IP interfaces should always be specified. VDMs do not have default CIFS servers.

[,dns=<if_suffix>]
Specifies a different DNS suffix for the interface for DNS updates. By default, the DNS suffix is derived from the domain. This DNS option does not have any impact on the DNS settings of the Data Mover.

[,local_users]
Enables local user support that allows the creation of a limited number of local user accounts on the CIFS server. When this command executes, type and confirm a password that is assigned to the local Administrator account on the CIFS server. In addition to the Administrator account, a Guest account is also created. The Guest account is disabled by default. The Administrator account password must be changed before the Administrator account can log in to the CIFS server.

After initial creation of the stand-alone server, the local_users option resets the local Administrator account password. The password can only be reset if it has not been changed through a Windows client. If the password has already been changed through Windows, the reset will be refused.

[-comment <comment>]
Assigns a comment to the configuration. The comment is delimited by quotes. Comment length is limited to 48 bytes (represented as 48 ASCII characters or a variable number of Unicode multibyte characters) and cannot include colons, since they are recognized as delimiters.

-add standalone=<netbios_name>, workgroup=<workgroup_name>
Creates or modifies a stand-alone CIFS server on a Data Mover, assigning the specified <netbios_name> and <workgroup_name> to the server. The NetBIOS and workgroup names are limited to 15 bytes. When creating a stand-alone CIFS server for the first time, the ,local_users option must be typed, or the command will fail. It is not required when modifying the CIFS server. A stand-alone CIFS server does not require any Windows domain infrastructure. A stand-alone server has local user accounts on the Data Mover and NTLM is used to authenticate users against the local accounts database.

Caution: Each NetBIOS name must be unique to the workgroup and the Data Mover.

[,alias=<alias_name>...]
Assigns an alias to the NetBIOS name. The <alias_name> must:

* Be unique on a Data Mover
* Be limited to 15 bytes
* Not begin with an @ (at sign) or - (dash) character
* Not include spaces, tab characters, or the following symbols: / \ : ; , = * +|[] ? < > "

[,hidden={y|n}]
By default, the <netbios_name> is displayed in the Network Neighborhood. If hidden=y is specified, the <netbios_name> does not appear.

[[,interface=<if_name>[,wins=<ip>[:<ip>]]]...]
Specifies a logical IP interface for the CIFS server and associates up to two WINS IP addresses with each interface. The interface name is case-sensitive.

Note: When configuring a CIFS server without any interfaces for a Data Mover, it becomes the default CIFS server and is available on all interfaces not used by other CIFS servers. The default CIFS server can be deleted at any time. It is recommended that IP interfaces should always be specified. VDMs do not have default CIFS servers.

[,local_users]
Enables local user support that allows the creation of a limited number of local user accounts on the CIFS server. When this command executes, type and confirm a password that is assigned to the local Administrator account on the CIFS server. In addition to the Administrator account, a Guest account is also created. The Guest account is disabled by default. The Administrator account password must be changed before the Administrator can log in to the CIFS server.

After initial creation of the stand-alone server, the local_users option resets the local Administrator account password. The password can only be reset if it has not been changed through a Windows client. If the password has already been changed through Windows, the reset will be refused.

[-comment <comment>]
Assigns a comment to the configuration. The comment is delimited by quotes. Comment length is limited to 48 bytes (represented as 48 ASCII characters or a variable number of Unicode multibyte characters) and cannot include colons since they are recognized as delimiters.

-rename -netbios <old_name> <new_name>
Renames a NetBIOS name. For Windows Server, renames a Compname after the CIFS server is unjoined from the domain.

Note: Before performing a rename, the new NetBIOS name must be added to the domain using the Windows Server Users and Computers MMC snap-in.

-Join compname=<comp_name>,domain=
<full_domain_name>,admin=<admin_name>
Creates an account for the CIFS server in the Active Directory. By default, the account is created under the domain root as ou=Computers,ou=EMC VNX.

Caution: Before performing a -Join, CIFS service must be started using server_setup.

The <comp_name> is limited to 63 bytes and represents the name of the server to be registered in DNS. The <full_domain_name> is the full domain name to which the server belongs. This means the name must contain at least one period (.). The <admin_name> is the logon name of the user with the right to create and manage computer accounts in the Organizational Unit that the CIFS server is being joined to. If a domain is given as part of the admin username it should be of the form: admin@FQDN. If no domain is given the admin user account is assumed to be part of the domain the CIFS Server is being joined to. The user is prompted to type a password for the admin account.

An Active Directory and a DNS can have the same domain name, or a different domain name (disjoint namespace). For each type of Active Directory and DNS domain relationship, specific VNX parameters

and command values must be used. For example, for a disjoint namespace, you must always specify the fully qualified domain name (FQDN) with the computer name when joining a CIFS server to a domain, that is, dm112-cge0.emc.com, not just dm112-cge0.

Caution: Time services must be synchronized using server_date.

[,ou=<organizational_unit>]
Specifies the organizational unit or container where computer accounts are created in the Active Directory. By default, computer accounts are created in an organizational unit called Computers. The name must be in a valid distinguished name format, for example, ou="cn=My_mover". The name may contain multiple nested elements, such as ou="cn=comp:ou=mach". The colon (:) must be used as a separator for multiple elements. By default, ou=Computers,ou=EMC VNX is used. The organizational unit name is limited to 256 bytes.

[-option {reuse|resetserverpasswd|addservice=nfs}]
The reuse option reuses the existing computer account with the original principal or joins a CIFS server to the domain where the computer account has been created manually.

The resetserverpasswd option resets the CIFS server password and encryption keys on a domain controller. This option could be used for security reasons, such as changing the server password in the Kerberos Domain Controller.

The addservice option adds the NFS service to the CIFS server, making it possible for NFS users to access the Windows Kerberos Domain Controller. Before adding NFS service, the <comp_name> must already be joined to the domain, otherwise the command will fail.

-Unjoin compname=<comp_name>,domain=
<full_domain_name>,admin=<admin_name>
Deletes the account for the CIFS server as specified by its <comp_name> from the Active Directory database. The user is prompted to type a password for the admin account.

-add security={NT|UNIX|SHARE}
Defines the user authentication mechanism used by the Data Mover for CIFS services. NT (default) security mode uses standard Windows domain based user authentication. The local password and group files, NIS, EMC Active Directory UNIX users and groups extension, or UserMapper are required to translate Windows user and group names into UNIX UIDs and GIDs. NT security mode is required for the Data Mover to run Windows 2000 or later native environments. Unicode should be enabled for NT security mode.

Caution: EMC does not recommend the use of UNIX or SHARE security modes.

For UNIX security mode, the client supplies a username and a plain-text password to the server. The server uses the local (password or group) file or NIS to authenticate the user. To use UNIX security mode, CIFS client machines must be configured to send user passwords to the Data Mover unencrypted in plain text. This requires a registry or security policy change on every CIFS client machine. For VDM, UNIX and SHARE security modes are global to the Data Mover and cannot be set for each VDM. Unicode must not be enabled.

For SHARE security mode, clients supply a read-only or read-write password for the share. No user authentication is performed using SHARE security. Since this password is sent through the network in clear text, you must modify the Client Registry to allow for clear text passwords.

Caution: Before adding or changing a security mode, CIFS service must be stopped using server_setup, then restarted once options have been

set.

[,dialect=<dialect_name>]
Specifies a dialect. Optimum dialects are assigned by default. Options include CORE, COREPLUS, LANMAN1 (default for UNIX and SHARE security modes), LANMAN2, and NT1 (which represents SMB1 and is the default for NT security mode), SMB2 and SMB3.

. SMB1 dialect is NT1 dialect.
. SMB2 dialect means max dialect in SMB2 which is SMB2.1. SMB2.0 or SMB2.1 can be specified explicitly to refine the dialect revision.
. SMB3 dialect means max dialect in SMB3 which is SMB3.0. SMB3.0 can be specified explicitly.

Note: SMB3 is enabled by default.

-add wins=<ip_addr>[,wins=<ip_addr>...]
Adds the WINS servers to the CIFS configuration. The list of WINS servers is processed in the order in which they are added. The first one is the preferred WINS server. If after 1500 milliseconds, the first WINS server times out, the next WINS server on the list is used.

-add usrmapper=<ip_addr>[,usrmapper=<ip_addr>...]
Adds the IP address(es) of a secondary Usermapper hosts to the CIFS configuration. A single IP address can point to a primary or secondary Usermapper host. If you are using distributed Usermappers, up to eight subsequent IP addresses can point to secondary Usermapper hosts.

-Disable <interface> [<interface>,...]
Disables the specified IP interfaces for CIFS service. Interface names are case-sensitive. All unused interfaces should be disabled.

-Enable <interface> [<interface>,...]
Enables the specified IP interfaces for CIFS service. Interface names are case-sensitive.

-delete standalone=<netbios_name>
[-remove_localgroup][,alias=<alias_name>...][,interface=<if_name>]
Deletes the stand-alone CIFS server as identified by its NetBIOS name from the CIFS configuration of the Data Mover.

-delete netbios=<netbios_name>
[-remove_localgroup][,alias=<alias_name>...][,inter
face=<if_name>]
Deletes the CIFS server as identified by its NetBIOS name from the CIFS configuration of the Data Mover.

-delete compname=<comp_name> [-remove_localgroup]
[,alias=<alias_name>...][,interface=<if_name>]
Deletes the CIFS server as identified by its compname from the CIFS configuration of the Data Mover. This does not remove the account from the Active Directory. It is recommended that an -Unjoin be executed prior to deleting the computer name.

Caution: The -remove_localgroup option permanently deletes the local group information of the CIFS server from the permanent storage of the Data Mover. The alias and interface options delete the alias and the interface only, however, the CIFS server exists. The alias and interface options can be combined in the same delete command.

Deletes the WINS servers from the CIFS configuration.

-delete usrmapper=<ip_addr>[,usrmapper=<ip_addr>...]
Deletes the IP addresses of a secondary Usermapper hosts from the CIFS configuration.

-update {<share_name>|<path>}

Updates the attributes and their CIFS names for COMPAT file systems. For every file system, CIFS maintains certain attributes for which there are no NFS equivalents. Updating CIFS attributes updates file attributes and CIFS names by searching the subdirectories of the defined share or path, generating a listing of Microsoft clients filenames (M8.3 and M256), and converting them to a format that CIFS supports. It is not necessary to use this command for DIR3 file systems. Options include:

[mindirsize=<size>]
Updates the directories with the minimum size specified. Size must be typed in multiples of 512 bytes. A value of 0 ensures that all directories are rebuilt.

[force]
Forces a previous update to be overwritten.

Caution: The initial conversion of a directory can take considerable time when the directory contains a large number of files. Although the process is designed to take place in the background, an update should be run only during periods of light system usage.

-Migrate {<fs_name> -acl|<netbios_servername>
-localgroup}<src_domain>{:nb=<netbios>|:if=<interfa
ce>}<dst_domain>{:nb=<netbios>|:if=<interface>}
Updates all security IDs (SIDs) from a <src_domain> to the SIDs of a <dst_domain> by matching the user and group account names in the source domain to the user and group account names in the destination domain. The interface that is specified in this option queries the local server, then its corresponding source and target Domain Controllers to look up each object.s SID.

If -acl is specified, all secure IDs in the ACL database are migrated for the specified file system.

The -localgroup option must be used to migrate the SID members of local group defined for the specified NetBIOS name.

On the source domain, an interface specified to issue a lookup of the SID is defined by either the NetBIOS name or the interface name. On the destination domain, an interface specified to issue a lookup of the SID is defined by either the NetBIOS name or the interface name.

-Replace {<fs_name> -acl|<netbios_servername>
-localgroup}{:nb=<netbios>|:if=<interface>}
Replaces the history SIDs from the old domain with the new SIDS in the new domain. An interface that can be specified to issue a lookup of the SIDs is defined by the interface name or the NetBIOS name.

The -localgroup option must be used to migrate the SID members of the local group defined for the specified NetBIOS name. When the -Replace option is used, the user or group migrated in the new domain keeps their old SID in addition to the new SID created in the new domain.

The -localgroup option does the same kind of migration for a specified NetBIOS name in the local groups (instead of the ACL in a file system for the history argument).

-smbhash -hashgen <path> [-recursive] [-minsize <size>]
Triggers the generation of all SMB Hash Files for this path.Both BranchCache V1 and BranchCache V2 hash files are generated. This path is an absolute path from the root of the VDM.
If the path is a file, only the SMB Hash File for this file will be generated. If the path is a directory, then SMB Hash File for all files will be generated in this directory. Additionally, if the -recursive option is specified, then the SMB Hash File for all files will be generated recursively inside the sub-directories.

By default, only files greater than 64KB are considered. If -minsize option is specified, then all files greater or equal to the specified size in KB will be

considered. Any size specified smaller than 64 KB will be ignored. SMB Hash Files are generated only if they are missing or obsolete.

The hash file generation is asynchronous, so the command will reply immediately. Use -info or check the system event log to monitor if the request has been completed.

-smbhash -hashdel <path> [-recursive]
Triggers the deletion of all SMB Hash Files for this path. Both BranchCache V1 and BranchCache V2 hash files are deleted. This path is an absolute path from the root of the VDM.

If the path is a file, only the SMB Hash File for this file will be deleted.

If the path is a directory, then SMB Hash File for all files will be deleted in this directory. Additionally, if the -recursive option is specified, then the SMB Hash File for all files will be deleted recursively inside the sub-directories.

The hash file deletion is asynchronous, so the command will reply immediately. Use -info or check the system event log to monitor if the request has been completed.

-smbhash -abort <id>
Cancels the pending or ongoing request (generation or deletion) provided its ID is given. Request ID is received from the output of the command -info.

-smbhash -info
Get all kinds of information relative to the hash generation service:
* The list of pending requests with their ID.
* The list of under processing requests with their ID.
* Values of the parameters which are actually in use.
* Value of the GPO setting taken into account for each server.
* Statistics

-smbhash -fsusage <fs_name>
Displays the SMB Hash File disk usage of the specified file system. The return values are:
* Total size in bytes of the file system
* Usage in bytes of the SMB Hash Files of the file system
* Usage in percentage of the file system of the SMB Hash Files

-smbhash -exclusionfilter <filter>
Files which match the exclusion filter will not have a SMB Hash File generated. This is to avoid waste of resources spent on files that frequently change like temporary files.

> This command directly modifies the parameter ExclusionFilter as defined with the specified format:
> Type: REG_STRING
>
> Meaning: Hash files are not generated for files which match one of the specified filters. The comparison between this parameter and the filename is done case less. Any change is taken into account immediately.
>
> Values: Default is no filter. A filter is a list of items separated by a character ":". Each item is made of:
> - Any valid character for a filename
> - *: means any string
> - ?:means any character

-smbhash -audit { enable | disable } [-service] [-task] [-access]
Enables the generation of audits in the smbhash event log. By default, it is not validated. The parameters are one of the following:

* enable: Enables generation of specified event. If no event is specified in the optional list, all events are enabled.

* disable: Disable generation of specified event. If no event is specified in the optional list, all events are disabled.

Optional list of event's category is :
- service: Generate service events
- task: Generate task events
- access: Generate SMB Hash access events.

-smbhash -service {enable | disable}
Enables or disables the SMB hash generation service (default is started).If CIFS service is started, this command is taken into account immediately. If CIFS is not running, this command is executed at the next "cifs start".

-smbhash -cleanup <fs_name> [-all | -unusedfor <days> | -unusedsince <date>]
Cleans up the SMB Hash Files of the specified file system.

* If no option is specified, only obsolete SMB Hash Files are removed.
* If -all option is specified, the entire "smbhash" directory is removed.
* If -unusedfor <days> option is specified, obsolete SMB Hash Files plus SMB Hash Files not accessed since the specified number of days are removed.
* If -unusedsince <date> option is specified, SMB Hash Files not accessed since the specified date are removed. The format of the date is <YYMMDDHHMM>.

-setspn {-list [server=<full_comp_name>]
| -add <SPN>compname=<comp_name>,
domain=<full_domain_name>,admin=<admin_name>
| -delete <SPN> compname=<comp_name>,
domain=<full_domain_name>,admin=<admin_name}

Displays all SPNs for the specified FQDN server, both for the Data Mover and for the KDC Windows Active Directory entry. If no server is specified, then the SPNs for all joined CIFS Servers for the specified movername is displayed. The command fails if an error occurs. For example, unable to connect to the Active Directory, the specified server is not joined to the domain.

When the -add and -delete sub-options are used, the user is prompted for the password associated with the admin name. The SPN must be the full value to use, including the realm.

The -add sub-option attempts to add the specified SPN to both the Data Mover and Active Directory. The operation succeeds if the SPN is added to both the Data Mover and Active Directory. In an entry already exists in one of these places, it is not duplicated. Otherwise, the operation fails if an error occurs. For example, unable to connect to the Active Directory, the specified server is not joined to the domain, or incorrect admin password.

The -delete sub-option attempts to remove the specified SPN from both the Data Mover and Active Directory. The operation succeeds if the SPN is removed from both the Data Mover and Active Directory. If the entry has already been deleted, it is not considered an error. Otherwise, the operation fails if an error occurs. For example, unable to connect to the Active Directory, the specified server is not joined to the domain, or incorrect admin password.

SEE ALSO

Using EMC Utilities for the CIFS Environment, Managing a Multiprotocol Environment on VNX, Using VNX Replicator, Using International Character Sets on VNX for File, server_date, server_export, server_mount, and server_setup.

OUTPUT NOTE

The network interface that appears in the output is dependent on the type of network interface cards that are installed. Dates appearing in the output are in UTC format.

EXAMPLE #1

To display the number and names of open files on server_2, type:

```
$ server_cifs server_2 -o audit,full

AUDIT Ctx=0xdffcc404, ref=2, Client(fm-main07B60004) Port=36654/139
NS40_1[BRCSLAB] on if=cge0_new
CurrentDC 0xceeab604=W2K3PHYAD
Proto=NT1, Arch=UNKNOWN, RemBufsz=0xfefb, LocBufsz=0xffff, popupMsg=1
0 FNN in FNNlist NbUsr=1 NbCnx=0
Uid=0x3f NTcred(0xcf156a04 RC=1 NTLM Capa=0x401) 'BRCSLAB\gustavo' CHECKER
AUDIT Ctx=0xde05cc04, ref=2, XP Client(BRCSBARREGL1C) Port=1329/445
NS40_1[BRCSLAB] on if=cge0_new
CurrentDC 0xceeab604=W2K3PHYAD
Proto=NT1, Arch=Win2K, RemBufsz=0xffff, LocBufsz=0xffff, popupMsg=1
0 FNN in FNNlist NbUsr=1 NbCnx=2
Uid=0x3f NTcred(0xceeabc04 RC=3 NTLMSSP Capa=0x11001) 'BRCSLAB\gustavo'
CHECKER
Cnxp(0xceeaae04), Name=IPC$, cUid=0x3f Tid=0x3f, Ref=1, Aborted=0
readOnly=0, umask=22, opened files/dirs=0
Cnxp(0xde4e3204), Name=gustavo, cUid=0x3f Tid=0x41, Ref=1, Aborted=0
readOnly=0, umask=22, opened files/dirs=2
Fid=64, FNN=0x1b0648f0(FREE,0x0,0), FOF=0x0 DIR=\
Notify commands received:
Event=0x17, wt=0, curSize=0x0, maxSize=0x20, buffer=0x0
Tid=0x41, Pid=0xb84, Mid=0xec0, Uid=0x3f, size=0x20
Fid=73, FNN=0x1b019ed0(FREE,0x0,0), FOF=0xdf2ae504 (CHECK) FILE=\New Wordpad
Document.doc
```

EXAMPLE #2

To configure CIFS service on server_2 with a NetBIOS name of dm110-cge0, in the NT4 domain NASDOCS, with a NetBIOS alias of dm110-cge0a1, hiding the NetBIOS name in the Network Neighborhood, with the interface for CIFS service as cge0, the WINS server as 172.24.102.25, and with the comment string EMC VNX, type:

```
$ server_cifs server_2 -add
netbios=dm110-cge0,domain=NASDOCS,alias=dm110-cge0a1,hidden=y,interface=cge0,wins
=172.24.102.25
-comment "EMC Celerra"
server_2 : done
```

EXAMPLE #3

To enable the home directory on server_2, type:

```
$ server_cifs server_2 -option homedir
server_2 : done
```

EXAMPLE #4

To add the WINS servers, 172.24.103.25 and 172.24.102.25, type:

```
$ server_cifs server_2 -add wins=172.24.103.25,wins=172.24.102.25
server_2 : done
```

EXAMPLE #5

To rename the NetBIOS name from dm110-cge0 to dm112-cge0, type:

```
$ server_cifs server_2 -rename -netbios dm110-cge0 dm112-cge0
server_2 : done
```

EXAMPLE #6

To display the CIFS configuration for NT4 with Internal Usermapper, type:

```
$ server_cifs server_2
server_2 :
256 Cifs threads started
Security mode = NT
Max protocol = NT1
I18N mode = UNICODE
Home Directory Shares ENABLED, map=/.etc/homedir
Usermapper auto broadcast enabled

Usermapper[0] = [127.0.0.1] state:active port:14640 (auto discovered)

Default WINS servers = 172.24.103.25:172.24.102.25
Enabled interfaces: (All interfaces are enabled)

Disabled interfaces: (No interface disabled)

DOMAIN NASDOCS RC=3
SID=S-1-5-15-99589f8d-9aa3a5f-338728a8-ffffffff
>DC=WINSERVER1(172.24.102.66) ref=2 time=0 ms

CIFS Server DM112-CGE0[NASDOCS] RC=2 (Hidden)
Alias(es): DM110-CGE0A1
Comment='EMC Celerra'
if=cge0 l=172.24.102.242 b=172.24.102.255 mac=0:60:16:4:35:4f
wins=172.24.102.25
Password change interval: 0 minutes
```

Where:

Value	Definition
Cifs threads started	Number of CIFS threads used when the CIFS service was started.
Security mode	User authorization mechanism used by the Data Mover.
Max protocol	Maximum dialect supported by the security mode.
I18N mode	I18N mode (unicode or ASCII).
Home Directory Shares	Whether Home Directory shares are enabled.
map	Home directory used by the Data Mover.
Usermapper auto broadcast enabled	Usermapper is using its broadcast mechanism to discover its servers. This only displays when the mechanism is active. It is disabled when you manually set the Usermapper server addresses.
Usermapper	IP address of the servers running the Usermapper service.
state	Current state of Usermapper.
Default WINS servers	Addresses of the default WINS servers.
Enabled interfaces	Data Mover's enabled interfaces.
Disabled interfaces	Data Mover's disabled interfaces.
Unused Interfaces	Interfaces not currently used by the Data Mover.
RC	Reference count indicating the number of internal objects (such as client contexts) using the CIFS server.
SID	Security ID of the domain.
DC	Domain controllers used by the Data Mover. Depending on the number of DCs in the domain, this list may be large.
ref	Number of internal objects using the Domain Controller.
time	Domain Controller response time.
Aliases	Alternate NetBIOS names assigned to the CIFS server configuration.
if	Interfaces used by the CIFS server.
Password change interval	The amount of time between password changes.

EXAMPLE #7

To display the CIFS configuration for NT4, type:

```
$ server_cifs server_2
server_2 :
256 Cifs threads started
Security mode = NT
Max protocol = NT1
I18N mode = UNICODE
Home Directory Shares ENABLED, map=/.etc/homedir
Usermapper auto broadcast suspended

Usermapper[0] = [172.24.102.20] state:available

Default WINS servers = 172.24.103.25:172.24.102.25
Enabled interfaces: (All interfaces are enabled)

Disabled interfaces: (No interface disabled)

DOMAIN NASDOCS RC=3
SID=S-1-5-15-99589f8d-9aa3a5f-338728a8-ffffffff
>DC=WINSERVER1(172.24.102.66) ref=2 time=0 ms

CIFS Server DM112-CGE0[NASDOCS] RC=2 (Hidden)
Alias(es): DM110-CGE0A1
Comment='EMC Celerra'
if=cge0 l=172.24.102.242 b=172.24.102.255 mac=0:60:16:4:35:4f
wins=172.24.102.25
Password change interval: 0 minutes
```

EXAMPLE #8

To add a Windows server using the compname dm112-cge0, in the Active Directory domain nasdocs.emc.com, with a NetBIOS alias of dm112-cge0a1, hiding the NetBIOS name in the Network Neighborhood, with the interface for CIFS service as cge0, the WINS servers as 172.24.102.25 and 172.24.103.25, in the DNS domain nasdocs.emc.com, and with the comment string EMC VNX, type:

```
$ server_cifs server_2 -add
compname=dm112-cge0,domain=nasdocs.emc.com,alias=dm112-cge0a1,hidden=y,
interface=cge0,wins=172.24.102.25:172.24.103.25,dns=nasdocs.emc.com
-comment "EMC Celerra"

server_2 : done
```

EXAMPLE #9

To join dm112-cge0 into the Active Directory domain nasdocs.emc.com, using the Administrator account, and to add this server to Engineering\Computers organizational unit, type:

```
$ server_cifs server_2 -Join
compname=dm112-cge0,domain=nasdocs.emc.com,admin=administrator,ou="ou=Computers:o
u=Engineering"

server_2 : Enter Password:********
done
```

EXAMPLE #10

To add the NFS service to the CIFS server in order to make it possible for NFS users to access the Windows KDC, type:

```
$ server_cifs server_2 -Join
```

```
compname=dm112-cge0,domain=nasdocs.emc.com,admin=administrator -option
addservice=nfs
server_2 : Enter Password:********
done

EXAMPLE #11
-----------

To enable the cge1 interface, type:

$ server_cifs server_2 -Enable cge1
server_2 : done

EXAMPLE #12
-----------

To display CIFS information for a Data Mover in a Windows domain
 with internal usermapper, type:

$ server_cifs server_2
server_2 :
256 Cifs threads started
Security mode = NT
Max protocol = NT1
I18N mode = UNICODE
Home Directory Shares ENABLED, map=/.etc/homedir
Usermapper auto broadcast enabled

Usermapper[0] = [127.0.0.1] state:active (auto discovered)

Default WINS servers = 172.24.103.25:172.24.102.25
Enabled interfaces: (All interfaces are enabled)

Disabled interfaces: (No interface disabled)

Unused Interface(s):
if=cge1 l=172.24.102.243 b=172.24.102.255 mac=0:60:16:4:35:4e

DOMAIN NASDOCS FQDN=nasdocs.emc.com SITE=Default-First-Site-Name RC=3
SID=S-1-5-15-99589f8d-9aa3a5f-338728a8-ffffffff
>DC=WINSERVER1(172.24.102.66) ref=3 time=1 ms (Closest Site)

CIFS Server DM112-CGE0[NASDOCS] RC=2 (Hidden)
Alias(es): DM112-CGEA1
Full computer name=dm112-cge0.nasdocs.emc.com realm=NASDOCS.EMC.COM
Comment='EMC Celerra'
if=cge0 l=172.24.102.242 b=172.24.102.255 mac=0:60:16:4:35:4f
wins=172.24.102.25:172.24.103.25
FQDN=dm112-cge0.nasdocs.emc.com (Updated to DNS)
Password change interval: 30 minutes
Last password change: Thu Oct 27 15:59:17 2005
Password versions: 2

EXAMPLE #13
-----------
To display CIFS information for a Data Mover in a Windows domain, type:

$ server_cifs server_2
server_2 :
256 Cifs threads started
Security mode = NT
Max protocol = NT1
I18N mode = UNICODE
Home Directory Shares ENABLED, map=/.etc/homedir
Usermapper auto broadcast suspended

Usermapper[0] = [172.24.102.20] state:available

Default WINS servers = 172.24.103.25:172.24.102.25
Enabled interfaces: (All interfaces are enabled)
```

```
Disabled interfaces: (No interface disabled)

Unused Interface(s):
if=cge1 l=172.24.102.243 b=172.24.102.255 mac=0:60:16:4:35:4e

DOMAIN NASDOCS FQDN=nasdocs.emc.com SITE=Default-First-Site-Name RC=3
SID=S-1-5-15-99589f8d-9aa3a5f-338728a8-ffffffff
>DC=WINSERVER1(172.24.102.66) ref=3 time=1 ms (Closest Site)

CIFS Server DM112-CGE0[NASDOCS] RC=2 (Hidden)
Alias(es): DM112-CGEA1
Full computer name=dm112-cge0.nasdocs.emc.com realm=NASDOCS.EMC.COM
Comment='EMC Celerra'
if=cge0 l=172.24.102.242 b=172.24.102.255 mac=0:60:16:4:35:4f
wins=172.24.102.25:172.24.103.25
FQDN=dm112-cge0.nasdocs.emc.com (Updated to DNS)
Password change interval: 30 minutes
Last password change: Thu Oct 27 16:29:21 2005
Password versions: 3, 2

EXAMPLE #14
-----------

To display CIFS information for a Data Mover when CIFS service is not
started, type:

$ server_cifs server_2
server_2 :
Cifs NOT started
Security mode = NT
Max protocol = NT1
I18N mode = UNICODE
Home Directory Shares ENABLED, map=/.etc/homedir
Usermapper auto broadcast suspended

Usermapper[0] = [172.24.102.20] state:available

Default WINS servers = 172.24.103.25:172.24.102.25
Enabled interfaces: (All interfaces are enabled)

Disabled interfaces: (No interface disabled)

Unused Interface(s):
if=cge1 l=172.24.102.243 b=172.24.102.255 mac=0:60:16:4:35:4e

CIFS Server DM112-CGE0[NASDOCS] RC=2 (Hidden)
Alias(es): DM112-CGEA1
Full computer name=dm112-cge0.nasdocs.emc.com realm=NASDOCS.EMC.COM
Comment='EMC Celerra'
if=cge0 l=172.24.102.242 b=172.24.102.255 mac=0:60:16:4:35:4f
wins=172.24.102.25:172.24.103.25
FQDN=dm112-cge0.nasdocs.emc.com (Updated to DNS)
Password change interval: 30 minutes
Last password change: Thu Oct 27 16:29:21 2005
Password versions: 3, 2

EXAMPLE #15
-----------

To add a Windows server named, dm112-cge0, in the Active
Directory domain nasdocs.emc.com, with the interface for CIFS service as
cge0, and enable local users support, type:

$ server_cifs server_2 -add
compname=dm112-cge0,domain=nasdocs.emc.com,interface=cge0,local_users
server_2 : Enter Password:********
Enter Password Again:********
done

EXAMPLE #16
-----------
```

To set a security mode to NT for a Data Mover, type:

```
$ server_cifs server_2 -add security=NT
server_2 : done
```

EXAMPLE #17

To disable a CIFS interface, type:

```
$ server_cifs server_2 -Disable cge1
server_2 : done
```

EXAMPLE #18

To display CIFS audit information for a Data Mover, type:

```
$ server_cifs server_2 -option audit
server_2 :

|||| AUDIT Ctx=0xad3d4820, ref=1, W2K3 Client(WINSERVER1) Port=1638/139
||| DM112-CGE0[NASDOCS] on if=cge0
||| CurrentDC 0xad407620=WINSERVER1
||| Proto=NT1, Arch=Win2K, RemBufsz=0xffff, LocBufsz=0xffff
||| 0 FNN in FNNlist NbUsr=1 NbCnx=1
||| Uid=0x3f NTcred(0xad406a20 RC=2 KERBEROS Capa=0x2)
'NASDOCS\administrator'
|| Cnxp(0xad3d5420), Name=IPC$, cUid=0x3f Tid=0x3f, Ref=1, Aborted=0
| readOnly=0, umask=22, opened files/dirs=1

|||| AUDIT Ctx=0xad43c020, ref=1, W2K3 Client(172.24.102.67) Port=1099/445
||| DM112-CGE0[NASDOCS] on if=cge0
||| CurrentDC 0xad407620=WINSERVER1
||| Proto=NT1, Arch=Win2K, RemBufsz=0xffff, LocBufsz=0xffff
||| 0 FNN in FNNlist NbUsr=1 NbCnx=1
||| Uid=0x3f NTcred(0xad362c20 RC=2 KERBEROS Capa=0x2) 'NASDOCS\user1'
|| Cnxp(0xaec21020), Name=IPC$, cUid=0x3f Tid=0x3f, Ref=1, Aborted=0
| readOnly=0, umask=22, opened files/dirs=2
```

Where:

Value	Definition
Ctx	Address in memory of the Stream Context.
ref	Reference counter of components using this context at this time.
Port	The client port and the Data Mover port used in the current TCP connection.
CurrentDC	Specifies the address of the Domain Controller that is currently used.
Proto	Dialect level that is currently used.
Arch	Type of the client OS.
RemBufsz	Max buffer size negotiated by the client.
LocBufsz	Max buffer size we have negotiated.
FNN/FNNlist	Number of blocked files that has not yet been checked by Virus Checker.
NbUsr	Number of sessions connected to the stream context (TCP connection).
NbCnx	Number of connections to shares for this TCP connection.
Uid/NTcred	User Id(this number is not related to the Unix UID used to create a file), the credential address, and the type of authentication.
Cnxp/Name	Share connection address and the name of the share the user is connecting to.
cUid	User Id who has opened the connection first.
Tid	Tree Id (number which represents the share connection in any protocol request).
Aborted	Status of the connection.
readOnly	If the share connection is readonly.
umask	A user file-creation mask.
opened files/dirs	Number of files or directories opened on this share

connection.

EXAMPLE #19

To unjoin the computer dm112-cge0 from the nasdocs.emc.com domain, type:

```
$ server_cifs server_2 -Unjoin
compname=dm112-cge0,domain=nasdocs.emc.com,admin=administrator
server_2 : Enter Password:********
done
```

EXAMPLE #20

To delete WINS servers, 172.24.102.25, and 172.24.103.25, type:

```
$ server_cifs server_2 -delete wins=172.24.102.25,wins=172.24.103.25
server_2 : done
```

EXAMPLE #21

To delete a NetBIOS name, dm112-cge0, type:

```
$ server_cifs server_2 -delete netbios=dm112-cge0
server_2 : done
```

EXAMPLE #22

To delete the compname, dm112-cge0, type:

```
$ server_cifs server_2 -delete compname=dm112-cge0
server_2 : done
```

EXAMPLE #23

To delete the usermapper, 172.24.102.20, type:

```
$ server_cifs server_2 -delete usrmapper=172.24.102.20
server_2 : done
```

EXAMPLE #24

To add and join a Windows server in disjoint DNS and Windows domains, type:

```
$ server_cifs server_2 -add
compname=dm112-cge0,domain=nasdocs.emc.com,interface=cge0,dns=eng.emc.com
-comment "EMC Celerra"

$ server_cifs server_2 -Join
compname=dm112-cge0.eng.emc.com,domain=nasdocs.emc.com,admin=Administrator
```

EXAMPLE #25

To add a Windows server using a delegated account from a trusted domain, type:

```
$ server_cifs server_2 -Join
compname=dm112-cge0,domain=nasdocs.emc.com,admin=delegateduser@it.emc.com
server_2 : Enter Password:********
done
```

EXAMPLE #26

To add a Windows server in the Active Directory domain using a pre-created computer account, type:

```
$ server_cifs server_2 -Join
compname=dm112-cge0,domain=nasdocs.emc.com,admin=administrator -option reuse

server_2 : Enter Password:********
done
```

EXAMPLE #27

```
To update the directory /ufs1/users with a new minimum directory size of
8192, type:

$ server_cifs server_2 -update /ufs1/users mindirsize=8192
server_2 : done
```

EXAMPLE #28

```
To migrate all SIDs in the ACL database for file system, ufs1, from the
<src_domain>, eng.emc.com:nb=dm112-cge1:if=cge1 to the <dst_domain>,
nasdocs.emc.com:nb=dm112-cge0:if=cge0, type:

$ server_cifs server_2 -Migrate ufs1 -acl eng.emc.com:nb=dm112-cge1:if=cge1
nasdocs.emc.com:nb=dm112-cge0:if=cge0
server_2 : done
```

EXAMPLE #29

```
To migrate SIDs of members of the local group defined for the specified
NetBIOS name, from the <src_domain>, eng.emc.com:nb=dm112-cge1:if=cge1 to
the <dst_domain>, nasdocs.emc.com:nb=dm112-cge0:if=cge0, type:

$ server_cifs server_2 -Migrate dm112-cge1 -localgroup
eng.emc.com:nb=dm112-cge1:if=cge1 nasdocs.emc.com:nb=dm112-cge0:if=cge0
server_2 : done
```

EXAMPLE #30

```
To replace the SIDs for ufs1, type:

$ server_cifs server_2 -Replace ufs1 -acl :nb=dm112-cge0:if=cge0
server_2 : done
```

EXAMPLE #31

```
To configure a stand-alone CIFS server on server_2 with a NetBIOS
name of dm112-cge0, in the workgroup NASDOCS, with a NetBIOS
alias of dm112-cge0a1, hiding the NetBIOS name in the Network
Neighborhood, with the interface for CIFS service as cge0, the WINS
servers as 172.24.102.25 and 172.24.103.25, and with enabled local
users support, type:

$ server_cifs server_2 -add
standalone=dm112-cge0,workgroup=NASDOCS,alias=dm112-cge0a1,hidden=y,interface=cge
0,
wins=172.24.102.25:172.24.103.25,local_users

server_2 : Enter Password:********
Enter Password Again:********
done
```

EXAMPLE #32

```
To delete the standalone CIFS server, dm112-cge0, type:

$ server_cifs server_2 -delete standalone=dm112-cge0
```

```
server_2 : done

EXAMPLE #33
-----------

To display a summary of SMB statistics, type:
$ server_cifs server_2 -stats -summary

server_2 :
State info:
Open connection Open files
2               2

SMB total requests:
totalAllSmb     totalSmb        totalTrans2Smb  totalTransNTSmb
10038           6593            3437            8

EXAMPLE #33
-----------

To display all non-zero CIFS statistics, type:

$ server_cifs server_2 -stats

server_2 :
SMB statistics:
proc            ncalls          %totcalls       maxTime         ms/call
Close           1305            7.96            46.21           2.16
Rename          2               0.01            0.81            0.50
Trans           314             1.91            0.77            0.08
Echo            21              0.13            0.01            0.00
ReadX           231             1.41            0.03            0.00
WriteX          3697            22.54           39.96           0.98
Trans2Prim      9375            57.16           34.27           0.46
TreeDisco       10              0.06            0.06            0.00
NegProt         29              0.18            0.42            0.24
SessSetupX      47              0.29            60.55           5.81
UserLogoffX     9               0.05            0.01            0.00
TreeConnectX    13              0.08            0.39            0.23
TransNT         8               0.05            0.01            0.00
CreateNTX       1338            8.16            47.11           0.81
CancelNT        1               0.01            0.03            0.00

Trans2 SMBs:
proc            ncalls          %totcalls       maxTime         ms/call
FindFirst       22              0.23            0.22            0.09
QFsInfo         3154            33.65           0.08            0.05
QPathInfo       1113            11.87           6.73            0.15
QFileInfo       2077            22.16           0.04            0.02
SetFileInfo     3007            32.08           34.26           1.28

NT SMBs:
proc            ncalls          %totcalls       maxTime         ms/call
NotifyChange    8               100.00          0.01            0.00

Performance info:
Read            Re/s            Write           Wr/s            All
Ops/sec
231             231000.00       3697            1021.27         25783
1575.40
State info:
Open connection Open files
2               2

Shadow info:
Reads           Writes          Splits          Extinsert       Truncates
0               0               0               0               0

SMB total requests:
totalAllSmb     totalSmb        totalTrans2Smb  totalTransNTSmb (unsupported)
25783           16400           9375            8               2
```

Where:

Value	Definition
proc	Name of CIFS requests received.
ncalls	Number of requests received.
%totcalls	Percentage of this type of request compared to all requests.
maxTime	Maximum amount of time used.
ms/call	Average time in milliseconds taken to service calls.
failures	Number of times the call has failed.
Read	Total number of read operations.
Re/s	Number of read operations per second.
Write	Total number of write operations.
Wr/s	Number of write operations per second.

EXAMPLE #35

To reset to zero the values for all SMB statistics, type:

```
$ server_cifs server_2 -stats -zero
server_2 : done
```

EXAMPLE #36

To configure CIFS service in a language that uses multibyte characters, type:

```
$ server_cifs server_2 -add compname=<computer_name_in_local_language_text>,
domain=nasdocs.emc.com, -comment <comment_in_local_language_text>
server_2 : done
```

EXAMPLE #37

To enable the SMB3 protocol, type:

```
$ server_cifs server_2 -add security=NT,dialect=SMB3
server_2 :
256 Cifs threads started
Security mode = NT
Max protocol = SMB3.0
I18N mode = UNICODE
```

EXAMPLE #38

To disable both SMB2 and SMB3, type:

```
$ server_cifs server_2 -add security=NT,dialect=NT1
server_2 : done
```

Last Modified: September 28, 2011 12:10 pm

server_cifssupport

Provides support services for CIFS users.

SYNOPSIS

```
server_cifssupport {<movername>|ALL}
  -accessright
   {-name <name> [-domain <domain_name>]
    | -sid <SID>|-uname <unix_name>|-uid <user_id>}
    {-path <pathname>|-share <sharename>}
    [-policy {mixed|native|secure|nt|unix}]
    [-build [-admin <admin_name>]]
    [-netbios <netbios_servername>|-compname <comp_name>
    | -standalone <netbios_ name>]
        | -acl {-path <pathname>|-share <sharename>} [-verbose]
 | -cred
   {-name <name> [-domain <domain_name>]
     | -sid <SID>|-uname <unix_name>|-uid <user_id>}
     [-build [-ldap][-admin <admin_name>]]
     [-netbios <netbios_servername>|-compname <comp_name>
     | -standalone <netbios_name>]
 | -pingdc
   {-netbios <netbios_servername>|-compname <comp_name>}
     [-dc <netbios_Dcname>]
     [-verbose]}
 | -secmap
    -list
     [-name <name> [-domain <domain_name>
     | -domain <domain_name>
     | -sid <SID>
     | -uid <user_id>
     | -gid <group_id>]
 | -create
   {-name <name> [-domain <domain_name>]}
 | -sid <SID>}
| -verify
  {-name <name> [-domain <domain_name>]}
    | -sid <SID>}
 |-update
   {-name <name> [-domain <domain_name>]}
     | -sid <SID>}
 | -delete
   {-name <name> [-domain <domain_name>]}
 | -sid <SID>}
 | -export [-file <filename>]
 | -import -file <filename>
 | -report
 | -migration
```

DESCRIPTION
----

server_cifssuport checks network connectivity between a CIFS server and domain controller, manages access rights, generates credentials,and manages secure mapping cache.

The -accessright option:

* Displays user access rights to a file, directory, or share in a Windows permission mask.

* Rebuilds and displays a credential for users of a file, directory, or share, who do not have a session opened in one of the CIFS servers.

* Without impact on the actual access-checking policy of a file system object, shows how user permissions are reset if you were to change the policy on that object.

The -acl option displays the access control list (ACL) of files, directories, or shares in plain text form.

The -cred option generates a credential containing all groups to which a user belongs, including local groups, without the user being connected to a CIFS server. This allows you to verify if user's SIDs are being correctly mapped to UNIX UIDs and GIDs and to troubleshoot any user access control issues.

The -pingdc option checks the network connectivity between a CIFS server and a domain controller then verifies that a CIFS server can access and use the following domain controller services:

* IPC$ share logon
* Secure Channel when verifying domain users during NT LAN Manager (NTLM) authentication
* Local Security Authority (LSA) pipe information when mapping Windows SIDs to UNIX UIDs and GIDs
* SAMR (Remote Security Account Manager) pipe when merging a user's UNIX and Windows groups together to create a credential
* Trusted domain information
* Privilege names for internationalization: pingdc

The -secmap option manages the secure mapping (secmap) cache. Secmap contains all mapping between SIDs and UID/GIDs used by a Data Mover or Virtual Data Mover (VDM). The Data Mover permanently caches all mappings it receives from any mapping mechanism (local files, NIS, iPlanet, Active Directory, and Usermapper) in the secmap database, making the response to subsequent mapping requests faster and less susceptible to network problems. Reverse mapping provides better quota support.

ACCESS RIGHT OPTIONS

-accessright {-name <name> [-domain <domain_name>]|
-sid <SID>|-uname <unix_name>|-uid <user_id>}
{-path <pathname>|-share <sharename>}
Displays user access rights to a file, directory, or share in a Windows permission mask for the specified:

* Windows username and the optional domain to which the user belongs

or

* <SID> which is the the user.s Windows security identifier

or

* <unix_name>

or

* <user_id> which is the user's UNIX identifier

The -path option specifies the path of the file or directory to check for user permissions, or the absolute path of the share to check for user permissions.

[-policy {mixed|native|secure|nt|unix}]
Specifies an access-checking policy for the specified file, directory, or share. This does not change the current access-checking policy, instead it helps you anticipate any access problems before actually resetting the policy on a file system object. server_cifs provides more information.

[-build [-admin <admin_name>]]
Rebuilds a credential for a user of a file, directory, or share, who does not have a session opened in one of the CIFS servers. If -build is not specified, the system searches the known user credentials in cache. If none are found, an error message is generated. The -admin option specifies the name of an administrative user to use for creating the access right list. The password of the admin_name user is prompted when executing the command.

[-netbios <netbios_servername>|-compname <comp_name>|-standalone <netbios_name>]}
Indicates the CIFS server, as specified by its NetBIOS name or computer name to use when rebuilding the user credential.

The -standalone option specifies the stand-alone CIFS server, as specified by its name, to use when rebuilding a user credential.

Note: If no CIFS server is specified, the system uses the default CIFS server, which uses all interfaces not assigned to other CIFS servers on the Data Mover.

ACL OPTIONS

-acl {-path <pathname>|-share <sharename>}[-verbose]
Displays the ACL of a file, directory, or a share in plain text form. Windows or UNIX access control data are both displayed in their native forms. The -verbose option displays the ACE access rights mask in plain text form in addition to their native forms.

CREDENTIAL OPTIONS

-cred {-name <name> [-domain <domain_name>]|-sid <SID>|-uname <unix_name>|-uid <user_id>}
Generates a credential containing all of the groups to which a user belongs without being connected to a CIFS server. The credential is specifies by the user.s:

* Windows username and the domain to which the user belongs

or

* Windows security identifier

or

* UNIX name

or

* UNIX identifier

[-build [-ldap][-admin <admin_name>]]
Rebuilds a user credential. If -build is not specified, the system searches the known user credentials in cache. If none are found, an error message is generated. The -ldap option retrieves the user.s universal groups to be included in the credential. If none are found, no universal groups are incorporated into the credential. The -admin option indicates the name of an administrative user for creating the credential. The password of the <admin_name> is prompted when executing the command.

[-netbios <netbios_servername>|-compname <comp_name>|-standalone <netbios_ name>]}
Indicates the CIFS server, as specified by its NetBIOS name or computer name to use when rebuilding the user credential.

The -standalone option specifies the stand-alone CIFS server to use when rebuilding a user credential.

Note: If no CIFS server is specified, the system uses the default CIFS server, which uses all interfaces not assigned to other CIFS servers on the Data Mover.

PINGDC OPTIONS

-pingdc {-netbios <netbios_servername>|-compname <comp_ name>}
Checks the network connectivity for the CIFS server as specified by its NetBIOS name or by its computer name with a domain controller. Once connectivity is established, it verifies that a CIFS server can access and use the domain controller services.

Note: An IP address can be used for the <netbios_servername> and the <comp_name>.

[-dc <netbios_Dcname>]
Indicates the domain controller to ping for network and resource connectivity with the CIFS server. If not specified, the command checks the domain controllers currently in use by the CIFS server.

Note: An IP address can be used for the <netbios_Dcname>.

[-verbose]
Adds troubleshooting information to the command output.

SECMAP OPTIONS

-secmap -list
Lists the secmap mapping entries.

-secmap -list -name <name> -domain <domain_name>
Lists the secmap mapping entries with the specified name and domain name.

-secmap -list -domain <domain_name>
Lists the secmap mapping entries with the specified domain name.

-secmap -list -sid <SID>
Lists the secmap mapping entries with the specified SID.

-secmap -list -uid <user_id>
Lists the secmap mapping entries with the specified UID (reverse mapping).

-secmap -list -gid <group_id>
Lists the secmap mapping entries with the specified GID (reverse mapping).

-secmap -create {-name <name> [-domain<domain_name>]}
Creates the secmap mapping entry with the specified name and domain name.

-secmap -create -sid <SID>
Creates the secmap mapping entry with the specified SID.

-secmap -verify {-name <name> [-domain<domain_name>]}
Checks the mapping entry stored in secmap with the specified name and optional domain name with what is currently available in the mapping sources. If a mapping has changed, it is marked.

-secmap -verify -sid <SID>
Checks the secmap mapping entry with the specified SID.

-secmap -update {-name <name> [-domain<domain_name>]}
Updates the specified mapping entry stored in secmap with the mappings currently available in the mapping sources. Once this option is performed, force an update of the Data Mover.s file systems ACLs so that the new mappings are recognized.

-secmap -update -sid <SID>
Updates the secmap mapping entry with the specified SID.

-secmap -delete -name <name> [-domain<domain_name>]
Deletes the secmap mapping entry with the specified name and domain name.

-secmap -delete -sid <SID>
Deletes the secmap mapping entry with the specified SID.

-secmap -export [-file <filename>]
Exports the secmap mapping entry to the specified file.

Note: If no filename is specified, the secmap database is displayed on the screen.

-secmap -import -file <filename>
Imports secmap mapping entries from the specified file.

-secmap -report
Displays current secmap status, including database state, domains handled by secmap, and resource usage.

-secmap -migration
Displays secmap database migration information like start date and end date of the operation, and migration status.

SEE ALSO

EXAMPLE #1

To display user access rights to a file for user1, type:

```
$ server_cifssupport server_2 -accessright -name user1 -domain NASDOCS
-path /ufs1/test/test.txt
server_2 : done

ACCOUNT GENERAL INFORMATIONS

Name                    : user1
Domain                  : NASDOCS
Path                    : /ufs1/test/test.txt
Allowed mask            : 0x200a9
Action                  : List Folder / Read data
Action                  : Read Extended Attributes
Action                  : Traverse Folder / Execute File
Action                  : Read Attributes
Action                  : Read Permissions
```

EXAMPLE #2

To rebuild a credential for a user to a file using the SID, type:

```
$ server_cifssupport server_2 -accessright -sid
S-1-5-15-b8e641e2-33f0942d-8f03a08f-1f4 -path /ufs1/test/test.txt -build
-compname dm102-cge0
server_2 : done

ACCOUNT GENERAL INFORMATIONS
```

```
Path                     : /ufs1/test/test.txt
Allowed mask             : 0x301ff
Action                   : List Folder / Read data
Action                   : Create Files / Write data
Action                   : Create Folders / Append Data
Action                   : Read Extended Attributes
Action                   : Write Extended Attributes
Action                   : Traverse Folder / Execute File
Action                   : Delete Subfolders and Files
Action                   : Read Attributes
Action                   : Write Attributes
Action                   : Delete
Action                   : Read Permissions
```

EXAMPLE #3

To display user access rights to a file for user1 with access-checking policy UNIX, type:

```
$ server_cifssupport server_2 -accessright -name user1 -domain NASDOCS -path
/ufs1/test/test.txt -policy unix
server_2 : done

ACCOUNT GENERAL INFORMATIONS

Name                     : user1
Domain                   : NASDOCS
Path                     : /ufs1/test/test.txt
Allowed mask             : 0x20089
Action                   : List Folder / Read data
Action                   : Read Extended Attributes
Action                   : Read Attributes
Action                   : Read Permissions
```

EXAMPLE #4

To rebuild a credential for user1 to a file using an administrative account, type:

```
$ server_cifssupport server_2 -accessright -name user1 -domain NASDOCS -path
/ufs1/test/test.txt -build -admin administrator
server_2 : Enter Password:*******
done

ACCOUNT GENERAL INFORMATIONS

Name                     : user1
Domain                   : NASDOCS
Path                     : /ufs1/test/test.txt
Allowed mask             : 0x200a9
Action                   : List Folder / Read data
Action                   : Read Extended Attributes
Action                   : Traverse Folder / Execute File
Action                   : Read Attributes
Action                   : Read Permissions
```

EXAMPLE #5

To display the verbose ACL information of a file, type:

```
$ server_cifssupport server_2 -acl -path /ufs1/test/test.txt -verbose
server_2 : done

ACL DUMP REPORT

Path         : /ufs1/test/test.txt
UID          : 32770
GID          : 32797
```

```
Rights       : rw-r--r--
acl ID       : 0x4
acl size     : 174
owner SID    : S-1-5-20-220
group SID    : S-1-5-15-b8e641e2-33f0942d-8f03a08f-201

DACL

Owner        : USER 32770 S-1-5-15-b8e641e2-33f0942d-8f03a08f-1f4
Access       : ALLOWED 0x0 0x1f01ff RWXPDO
Rights       : List Folder / Read data
               Create Files / Write data
               Create Folders / Append Data
               Read Extended Attributes
               Write Extended Attributes
               Traverse Folder / Execute File
               Delete Subfolders and Files
               Read Attributes
               Write Attributes
               Delete
               Read Permissions
               Change Permissions
               Take Ownership
               Synchronize

Owner        : USER 32771 S-1-5-15-b8e641e2-33f0942d-8f03a08f-a59
Access       : ALLOWED 0x0 0x1200a9 R-X---
Rights       : List Folder / Read data
               Read Extended Attributes
               Traverse Folder / Execute File
               Read Attributes
               Read Permissions
               Synchronize

EXAMPLE #6
----------

To display the access control level of a share, type:

$ server_cifssupport server_2 -acl -share ufs1
server_2 : done

ACL DUMP REPORT

Share        : ufs1
UID          : 0
GID          : 1
Rights       : rwxr-xr-x

EXAMPLE #7
----------

To generate a credential for user1, type:

$ server_cifssupport server_2 -cred -name user1 -domain NASDOCS
server_2 : done

ACCOUNT GENERAL INFORMATIONS

Name                     : user1
Domain                   : NASDOCS
Primary SID              : S-1-5-15-b8e641e2-33f0942d-8f03a08f-a59
UID                      : 32771
GID                      : 32768
Authentification         : KERBEROS
Credential capability    : 0x2
Privileges               : 0x8
System privileges        : 0x2
Default Options          : 0x2
NT administrator         : False
Backup administrator     : False
```

```
Backup                   : False
NT credential capability : 0x2

ACCOUNT GROUPS INFORMATIONS

Type UNIX ID    Name                Domain              SID
NT   32797
S-1-5-15-b8e641e2-33f0942d-8f03a08f-201
NT   32798
S-1-5-15-b8e641e2-33f0942d-8f03a08f-e45
NT   4294967294                                         S-1-1-0
NT   4294967294                                         S-1-5-2
NT   4294967294                                         S-1-5-b
NT   2151678497                                         S-1-5-20-221
UNIX 32797
UNIX 32798
UNIX 4294967294
UNIX 2151678497

EXAMPLE #8
----------

To rebuild a user credential including the user's universal groups for a
user using SID, type:

$ server_cifssupport server_2 -cred -sid
S-1-5-15-b8e641e2-33f0942d-8f03a08f-1f4 -build -ldap -compname dm102-cge0
server_2 : done

ACCOUNT GENERAL INFORMATIONS

Name                     :
Domain                   : NASDOCS
Server                   : dm102-cge0
Primary SID              : S-1-5-15-b8e641e2-33f0942d-8f03a08f-1f4
UID                      : 32770
GID                      : 32768
Authentification         : NTLM
Credential capability    : 0x0
Privileges               : 0x7f
System privileges        : 0x1
Default Options          : 0xe
NT administrator         : True
Backup administrator     : True
Backup                   : False
NT credential capability : 0x0

ACCOUNT GROUPS INFORMATIONS

Type UNIX ID    Name                Domain              SID
NT   32794      Group Policy Cre    NASDOCS
S-1-5-15-b8e641e2-33f0942d-8f03a08f-208
NT   32795      Schema Admins       NASDOCS
S-1-5-15-b8e641e2-33f0942d-8f03a08f-206
NT   32796      Enterprise Admin    NASDOCS
S-1-5-15-b8e641e2-33f0942d-8f03a08f-207
NT   32797      Domain Users        NASDOCS
S-1-5-15-b8e641e2-33f0942d-8f03a08f-201
NT   32793      Domain Admins       NASDOCS
S-1-5-15-b8e641e2-33f0942d-8f03a08f-200
NT   4294967294 Everyone                                S-1-1-0
NT   4294967294 NETWORK             NT AUTHORITY        S-1-5-2
NT   4294967294 ANONYMOUS LOGON     NT AUTHORITY        S-1-5-7
NT   2151678496 Administrators      BUILTIN             S-1-5-20-220
NT   2151678497 Users               BUILTIN             S-1-5-20-221
NT   1          UNIX GID=0x1 &ap                        S-1-5-12-2-1
UNIX 32794
UNIX 32795
UNIX 32796
UNIX 32797
UNIX 32793
```

EXAMPLE #9

To check the network connectivity for the CIFS server with netbios dm102-cge0, type:

```
$ server_cifssupport server_2 -pingdc -netbios dm102-cge0
server_2 : done

PINGDC GENERAL INFORMATIONS

DC SERVER:
Netbios name  : NASDOCSDC

CIFS SERVER :
Compname      : dm102-cge0
Domain        : nasdocs.emc.com
```

EXAMPLE #10

To check the network connectivity between the domain controller and the CIFS server with compname dm102-cge0, type:

```
$ server_cifssupport server_2 -pingdc -compname dm102-cge0 -dc NASDOCSDC
-verbose
server_2 : done

PINGDC GENERAL INFORMATIONS

DC SERVER:
Netbios name  : NASDOCSDC

CIFS SERVER :
Compname      : dm102-cge0
Domain        : nasdocs.emc.com
```

EXAMPLE #11

To display the secmap mapping entries, type:

```
$ server_cifssupport server_2 -secmap -list
server_2 : done

SECMAP USER MAPPING TABLE

UID         Origin      Date                      Name
SID
32772       usermapper  Tue Sep 18 19:08:40 2007 NASDOCS\user2
S-1-5-15-b8e641e2-33f0942d-8f03a08f-452
32771       usermapper  Tue Sep 18 17:56:53 2007 NASDOCS\user1
S-1-5-15-b8e641e2-33f0942d-8f03a08f-a59
32770       usermapper  Sun Sep 16 07:50:39 2007 NASDOCS\Administrator
S-1-5-15-b8e641e2-33f0942d-8f03a08f-1f4

SECMAP GROUP MAPPING TABLE

GID         Origin      Date                      Name
SID
32793       usermapper  Wed Sep 12 14:16:18 2007 NASDOCS\Domain Admins
S-1-5-15-b8e641e2-33f0942d-8f03a08f-200
32797       usermapper  Sun Sep 16 07:50:40 2007 NASDOCS\Domain Users
S-1-5-15-b8e641e2-33f0942d-8f03a08f-201
32799       usermapper  Mon Sep 17 19:13:16 2007 NASDOCS\Domain Guests
S-1-5-15-b8e641e2-33f0942d-8f03a08f-202
32800       usermapper  Mon Sep 17 19:13:22 2007 NASDOCS\Domain Computers
S-1-5-15-b8e641e2-33f0942d-8f03a08f-203
32795       usermapper  Sun Sep 16 07:50:40 2007 NASDOCS\Schema Admins
S-1-5-15-b8e641e2-33f0942d-8f03a08f-206
```

```
32796       usermapper  Sun Sep 16 07:50:40 2007 NASDOCS\Enterprise Admins
S-1-5-15-b8e641e2-33f0942d-8f03a08f-207
32794       usermapper  Sun Sep 16 07:50:40 2007 NASDOCS\Group Policy Creator
Owners S-1-5-15-b8e641e2-33f0942d-8f03a08f-208
32798       usermapper  Mon Sep 17 19:13:15 2007 NASDOCS\CERTSVC_DCOM_ACCESS
S-1-5-15-b8e641e2-33f0942d-8f03a08f-e45
32801       usermapper  Tue Sep 18 19:08:41 2007 NASDOCS\NASDOCS Group
S-1-5-15-b8e641e2-33f0942d-8f03a08f-45b
```

EXAMPLE #12

To display the secmap mapping entry for a user user1 in a domain NASDOCS, type:

```
$ server_cifssupport server_2 -secmap -list -name user1 -domain NASDOCS
server_2 : done

SECMAP USER MAPPING TABLE

UID         Origin      Date                      Name
SID
32771       usermapper  Tue Sep 18 17:56:53 2007 NASDOCS\user1
S-1-5-15-b8e641e2-33f0942d-8f03a08f-a59
```

EXAMPLE #13

To display the secmap mapping entry for a user with UID 32771, type:

```
$ server_cifssupport server_2 -secmap -list -uid 32771
server_2 : done

SECMAP USER MAPPING TABLE

UID         Origin      Date                      Name
SID
32771       usermapper  Tue Sep 18 17:56:53 2007 NASDOCS\user1
S-1-5-15-b8e641e2-33f0942d-8f03a08f-a59
```

EXAMPLE #14

To create the secmap mapping entry for user3 in a domain NASDOCS, type

```
$ server_cifssupport server_2 -secmap -create -name user3 -domain NASDOCS
server_2 : done

SECMAP USER MAPPING TABLE

UID         Origin      Date                      Name
SID
32773       usermapper  Tue Sep 18 19:21:59 2007 NASDOCS\user3
S-1-5-15-b8e641e2-33f0942d-8f03a08f-a3d
```

EXAMPLE #15

To check the secmap mapping for user1 in a domain NASDOCS, type:

```
$ server_cifssupport server_2 -secmap -verify -name user1 -domain NASDOCS
server_2 : done
```

EXAMPLE #16

To update the secmap mapping entry for a user using SID, type:

```
$ server_cifssupport server_2 -secmap -update -sid
S-1-5-15-b8e641e2-33f0942d-8f03a08f-a3d
server_2 : done
```

EXAMPLE #17

To delete the secmap mapping entry for user3, type:

```
$ server_cifssupport server_2 -secmap -delete -name user3 -domain NASDOCS
server_2 : done
```

EXAMPLE #18

To display current secmap status, type:

```
$ server_cifssupport server_2 -secmap -report
server_2 : done
SECMAP GENERAL INFORMATIONS

Name             : server_2
State            : Enabled
Fs               : /
Used nodes       : 12
Used blocks      : 8192


SECMAP MAPPED DOMAIN

Name                     SID
NASDOCS                  S-1-5-15-b8e641e2-33f0942d-8f03a08f-ffffffff
```

EXAMPLE #19

To export the secmap mapping entries to the display, type:

```
$ server_cifssupport server_2 -secmap -export
server_2 : done

SECMAP MAPPING RECORDS

S-1-5-15-b8e641e2-33f0942d-8f03a08f-200:2:96:8019:8019:NASDOCS\Domain Admins
S-1-5-15-b8e641e2-33f0942d-8f03a08f-201:2:96:801d:801d:NASDOCS\Domain Users
S-1-5-15-b8e641e2-33f0942d-8f03a08f-202:2:96:801f:801f:NASDOCS\Domain Guests
S-1-5-15-b8e641e2-33f0942d-8f03a08f-203:2:96:8020:8020:NASDOCS\Domain Computers
S-1-5-15-b8e641e2-33f0942d-8f03a08f-206:2:96:801b:801b:NASDOCS\Schema Admins
S-1-5-15-b8e641e2-33f0942d-8f03a08f-207:2:96:801c:801c:NASDOCS\Enterprise
Admins
S-1-5-15-b8e641e2-33f0942d-8f03a08f-208:2:96:801a:801a:NASDOCS\Group Policy
Creator Owners
S-1-5-15-b8e641e2-33f0942d-8f03a08f-e45:2:96:801e:801e:NASDOCS\CERTSVC_DCOM_ACCES
S
S-1-5-15-b8e641e2-33f0942d-8f03a08f-452:1:96:8004:8000:NASDOCS\user2
S-1-5-15-b8e641e2-33f0942d-8f03a08f-a59:1:96:8003:8000:NASDOCS\user1
S-1-5-15-b8e641e2-33f0942d-8f03a08f-45b:2:96:8021:8021:NASDOCS\NASDOCS Group
S-1-5-15-b8e641e2-33f0942d-8f03a08f-1f4:1:96:8002:8000:NASDOCS\Administrator
```

EXAMPLE #20

To export the secmap mapping entries to a file, type

```
$ server_cifssupport server_2 -secmap -export -file exportfile.txt
server_2 : done
```

EXAMPLE #21

To import the secmap mapping entries from a file, type:

```
$ server_cifssupport server_2 -secmap -import -file exportfile.txt
server_2 :
Secmap import in progress : #
```

done

Last Modified: April 06, 2011 3:00 pm

server_cpu

Performs an orderly, timed, or immediate halt or reboot of a Data Mover.

SYNOPSIS
----

server_cpu {<movername>|ALL}
 {-halt|-reboot[cold|warm]} [-monitor] <time>

DESCRIPTION
----

server_cpu performs an orderly halt or reboot of the specified Data Mover.

The ALL option executes the command for all Data Movers.

OPTIONS

-halt
Performs an orderly shutdown of a Data Mover for the VNX. To restart a Data Mover, perform a -reboot. For the NS series, a -halt causes a system reboot.

-reboot
Performs an orderly shutdown, and restarts a Data Mover. The default parameter of the -reboot option will be the warm parameter. In case the warm rebooting fails, the -reboot option will use the cold parameter to reboot the Data Mover.

[cold]
A cold reboot or a hardware reset shuts down the Data Mover completely before restarting, including a Power on Self Test (POST).

[warm]
A warm reboot or a software reset performs a partial shutdown of the Data Mover, and skips the POST after restarting. A software reset is faster than the hardware reset.

Caution: Performing a reboot for ALL Data Movers can be time consuming relative to the size of the mounted file system configuration.

-monitor
Polls and displays the boot status until completion of the halt or reboot.

<time>
Specifies the time when the Data Mover is to be halted or rebooted. Time is specified as {now|+<min>|<hour>:<min>}. The now option is used for an immediate shutdown or reboot.

After a power fail and crash recovery, the system reboots itself at power-up unless previously halted.

SEE ALSO

VNX System Operations.

EXAMPLE #1

To monitor an immediate reboot of server_2, type:

$ server_cpu server_2 -reboot -monitor now
server_2 : reboot in progress 0.0.0.0.0.0.0.0.0.0.3.3.3.3.3.3.4.done

Where:

```
Value Definition
0     Reset
1     DOS booted
2     SIB failed
3     Loaded
4     Configured
5     Contacted
7     Panicked
9     Reboot pending

EXAMPLE #2
----------

To immediately halt server_2, type:

$ server_cpu server_2 -halt now
server_2 : done

EXAMPLE #3
----------

To immediately reboot server_2, type:

$ server_cpu server_2 -reboot now
server_2 : done

EXAMPLE #4
----------

To monitor a reboot of server_2, that is set to take place in one minute, type:

$ server_cpu server_2 -reboot -monitor +1
server_2 : reboot in progress ........3.3.3.3.3.done

---------------------------------------
Last Modified: April 06, 2011 6:00 pm
```

server_date

Displays or sets the date and time for a Data Mover, and synchronizes time between a Data Mover and an external time source.

SYNOPSIS

```
server_date {<movername>|ALL}
        [+<format>][<yymmddhhmm>[<ss>]]
        | timesvc start ntp [-sync_delay][-interval <hh>[:<mm>]][<host>[<host>...
]]
        | timesvc update ntp
        | timesvc stop ntp
        | timesvc delete ntp
        | timesvc set ntp
        | timesvc stats ntp
        | timesvc
        | timezone [<timezonestr>]
        | timezone -name <timezonename>
```

DESCRIPTION
----

server_date sets and displays the current date and time for the specified Data Movers.

The server_date timesvc commands control the synchronization of the Data Mover with external timing sources and gets and sets the time zone.

The ALL option executes the command for all Data Movers.

OPTIONS

No arguments
Displays the current date and time for the specified Data Mover.

+<format>
Displays the date information in the format specified by each field descriptor. Each field descriptor is preceded by percent and is replaced in the output by its corresponding value. A single percent is encoded by double percent (%%).

If the argument contains embedded blanks, it must be quoted.

The complete listing of all field descriptors can be viewed using the Linux strftime (3C) man page.

<yymmddhhmm>[<ss>]
Sets a two-digit number for the year, month, day, hour, minutes, and seconds in this order where <yy> is the year; the first <mm> is the month; <dd> is the day; <hh> is the hour (in 24-hour system); and the second <mm> is the minute, and <ss> is the second.

timesvc start ntp <host> [<host>...]
Starts time synchronization immediately between a Data Mover and a host, which is the IP address of the time server hosts, and adds an entry to the database. The host must be running the NTP protocol. Only four host entries are allowed.

Other options include:

-sync_delay
Indicates that the clock should not be synchronized when the time server is activated. Instead, when the first poll is taken, latency adjustments are handled slowly. This option is generally used if time service is started after the Data Mover has already started, or if synchronization is starting after other services have already started.

Note: If -sync_delay is not typed, by default, the clock is set at Data Mover startup. The clock is synchronized after the first poll.

-interval <hh>[:<mm>]
Sets the delay in hours (or hours and minutes) between polls (default=1 hour which is entered 01 or 00:60). Interval is displayed in minutes.

timesvc update ntp
Immediately polls the external source and synchronizes the time on the Data Mover.

timesvc stop ntp
Stops timing synchronization between the Data Mover and an external timing host for the NTP protocol, and does not remove the entry from the database.

Note: A stop of time services takes about 12 seconds. If time service is restarted within this time, a "busy" status message is returned.

timesvc delete ntp
Stops time synchronization and deletes the NTP protocol from the database.

timesvc set ntp
Immediately polls the external source and synchronizes the time on the Data Mover without slewing the clock.

timesvc stats ntp
Displays the statistical information of time synchronization for the Network Time Protocol such as time differences between the Data Mover and the time server. Also provides information about the current state of NTP service on the Data Mover.

timesvc
Displays the current time service configuration.

timezone
Displays the current time zone on the specified Data Mover.

[<timezonestr>]
Sets the current time zone on the specified Data Mover. The <timezonestr> is a POSIX style time zone specification with the following formats:

<std><offset> (no daylight savings time)
<std><offset><dst>[offset],start[/time],end[/time]
Adjusts for daylight savings time.

Note: The Linux man page for tzset provides information about the format.

timezone -name <timezonename>
Sets the time zone on the Data Mover to the specified <timezonename>. The <timezonename> is in Linux style time zone specification. A list of valid Linux timezones is located in the /usr/share/zoneinfo directory. The third column in the table in this file lists the valid timezones.

Note: The timezone -name option does not reset time on the Data Mover to the specified <timezonename> time.

SEE ALSO

Configuring Time Services on VNX, server_dns, and server_nis.

EXAMPLE #1

To display the current date and time on a Data Mover, type:

```
$ server_date server_2

server_2 : Thu Jan 6 16:55:09 EST 2005

EXAMPLE #2
----------

To customize the display of the date and time on a Data Mover, type:

$ server_date server_2 "+%Y-%m-%d %H:%M:%S"

server_2 : 2005-01-06 16:55:58

EXAMPLE #3
----------

To start time synchronization between a Data Mover and an external
source, type:

$ server_date server_2 timesvc start ntp -interval 06:00 172.24.102.20

server_2 : done

EXAMPLE #4
----------

To set the time service without slewing the clock, type:

$ server_date server_2 timesvc set ntp

server_2 : done

EXAMPLE #5
----------

To display statistical information, type:

$ server_date server_2 timesvc stats ntp

server_2 :
Time synchronization statistics since start:
hits= 2, misses= 0, first poll hit= 2, miss= 0
Last offset: 0 secs, 0 usecs
Current State: Running, connected, interval=360
Time sync hosts:
0 1 172.24.102.20

Where:

Value                   Definition
-----                   ----------

hits                    When a client sends a request to the server requesting
                        the current time, if there is a reply, that is a
                        hit.
misses                  No reply from any of the time servers.
first poll hit          First poll hit which sets the first official time for
                        the Data Mover.
miss                    First poll miss.
Last offset             Time difference between the time server and the Data Mov
er.
Current State           State of the time server.
Time sync hosts         IP address of the time server.

EXAMPLE #6
----------

To update time synchronization between a Data Mover and an external source, type:

$ server_date server_2 timesvc update ntp
```

```
server_2 : done

EXAMPLE #7
----------

To get the time zone on the specified Data Mover, type:

$ server_date server_2 timezone

server_2 : Local timezone: GMT

EXAMPLE #8
----------

To set the time zone to Central Time for a Data Mover when you do
not have to adjust for daylight savings time, type:

$ server_date server_2 timezone CST6

server_2 : done

EXAMPLE #9
----------

To set the time zone to Central Time and adjust for daylight
savings time for a Data Mover, type:

$ server_date server_2 timezone CST6CDT5,M4.1.0,M10.5.0

server_2 : done

EXAMPLE #10
-----------

To set the time zone to Central Time and adjust the daylight
savings time for a Data Mover using the Linux method, type:

$ server_date server_2 timezone -name America/Chicago

server_2 : done

EXAMPLE #11
-----------

To display the time service configuration for a Data Mover, type:

$ server_date server_2 timesvc

server_2 :

Timeservice State
time: Thu Jan 6 17:04:28 EST 2005
type: ntp
sync delay: off
interval: 360
hosts: 172.24.102.20,

Where:

Value                   Definition
-----                   ----------

time                    Date and time known to the Data Mover.
type                    Time service protocol configured on the Data Mover.
sync delay              Whether sync delay is on or off.
interval                Time interval between polls.
hosts                   Specifies the IP address of the time server.

EXAMPLE #12
-----------
```

To stop time services for a Data Mover, type:

```
$ server_date server_2 timesvc stop ntp

server_2 : done
```

EXAMPLE #13

To delete the time service configuration for a Data Mover, type:

```
$ server_date server_2 timesvc delete ntp

server_2 : done
```

EXAMPLE #14

To set the timezone on a Data Mover to Los Angeles, type:

```
$ server_date server_2 timezone -n America/Los_Angeles

server_2 : done
```

Last modified: Feb 20, 2013 4:36 pm.

server_dbms

Enables backup and restore of databases, displays database environment statistics.

SYNOPSIS

```
server_dbms {<movername>|ALL}
{-db
  {-list [<db_name>]
  | -delete <db_name>
  | -check [<db_name>]
  | -repair [<db_name>]
  | -compact [<db name>]
  | -fullbackup -target <pathname>
  | -incrbckup -previous <pathname> -target <pathname>
  | -restore [<db_name>] -source <pathname>
  | -stats [<db_name> [-table <name>]][-reset]}
| -service -stats [transaction|memory|log|lock|mutex][-reset]
}
```

DESCRIPTION
----

server_dbms provides recovery of media failure or application corruption, displays database information, checks application database consistency, and fixes inconsistencies.

The ALL option executes the command for all Data Movers.

OPTIONS

-db -list [<db_name>]
Gets the list of all application databases and their status. If <db_name> is specified, displays the list of all tables belonging to that database.

-db -delete <db_name>
Deletes the target application database.

Note: This command will fail if the target database is not closed.

-db -check [<db_name>]
Checks the consistency of the target database at application level.

-db -repair [<db_name>]
Fixes the application level inconsistencies in the database.

-db -compact [<db_name> [-table <name>]]
Frees up disc space by compacting the target environment or database.

-db -fullbackup -target <pathname>
Performs an online full backup of the VDM database environment. The target parameter specifies the location to copy the database files.

The <pathname> specifies the local path of the database environment on the Control Station.

-db -incrbackup -previous <pathname> -target <pathname>
Downloads the transactional logs from the VDM and replays them on a copy of the previous VDM backup specified by previous <pathname>.

The -target option specifies the location to which the database files are copied.

-db -restore [<db_name>] -source <pathname>
Restores the environment or database specified by <db_name>.
source <pathname> specifies the location for the backup of the
environment/database to be restored.

Note: The database must be closed before the command is executed.

-db -stats [<db_name> [-table <name>]][-reset]
Displays statistics related to the specified databases and tables. If
-reset is specified, resets the statistics.

-service -stats [transaction|memory|log|lock|
mutex][-reset]
Displays transaction, memory, logging, locking or mutex statistics of
the VDM database environment. If -reset is specified, resets all or
specified statistics.

Note: For this command to be executed, the VDM on which the target
environment resides, must be up.

EXAMPLE #1

To get the list of all application databases and their status, type:

$ server_dbms server_3 -db -list
server_3 : done
BASE NAME : Secmap
Version : 1
Comment : CIFS Secure mapping database.
This is a cache of the sid to uid/gid mapping of the VDM.
This database is part of the CIFS application.
It can closed with the command server_setup
Size : 16384
Modification time : Fri May 25 09:58:21 2007
Creation time : Fri May 25 09:58:21 2007
TABLE NAME : Mapping
Version : 1
Comment : Sid to uid/gid mapping table with one secondary key on xid ((1,uid)
& (2,gid))
Size : 16384
Modification time : Fri May 25 09:58:21 2007
Creation time : Fri May 25 09:58:21 2007
BASE NAME : V4NameSpace
Version : 1
Comment : NFSv4 namespace database, this represents the pseudofs and referrals.
Size : 8192
Modification time : Tue Jun 5 08:57:12 2007
Creation time : Tue Jun 5 08:57:12 2007
TABLE NAME : pseudofs
Version : 1
Comment : Pseudofs-table, this holds the export tree heirarchy
Size : 8192
Modification time : Mon Jun 11 11:06:23 2007
Creation time : Mon Jun 11 11:06:23 2007
BASE NAME : Usermapper
Version : 1
Comment : Usermapper database. It allows to assign a new uid or gid to a
given SID.
Size : 57344
Modification time : Tue Jun 12 09:14:31 2007
Creation time : Tue Jun 12 09:14:31 2007
TABLE NAME : aliases
Version : 1
Comment : This table allows to retrieve a domain name from one of his aliases
Size : 8192
Modification time : Tue Jun 12 09:14:31 2007
Creation time : Tue Jun 12 09:14:31 2007
TABLE NAME : usrmapc
Version : 1

```
        Comment : Store the uid &amp;amp; gid ranges allocations for domains.
        Size : 8192
        Modification time : Tue Jun 12 09:14:31 2007
        Creation time : Tue Jun 12 09:14:31 2007
        TABLE NAME : idxname
        Version : 1
        Comment : Store the reverse mapping uid/gid to sid.
        Size : 8192
        Modification time : Tue Jun 12 09:14:31 2007
        Creation time : Tue Jun 12 09:14:31 2007
        TABLE NAME : usrmapusrc
        Version : 1
        Comment : Store the mapping SID -&gt; (uid, name).
        Size : 8192
        Modification time : Tue Jun 12 09:14:31 2007
        Creation time : Tue Jun 12 09:14:31 2007
        TABLE NAME : usrgrpmapnamesid
        Version : 1
        Comment : Store the mapping user.domain -&gt; SID.
        Size : 8192
        Modification time : Tue Jun 12 09:14:31 2007
        Creation time : Tue Jun 12 09:14:31 2007
        TABLE NAME : usrmapgrpc
        Version : 1
        Comment : Store the mapping SID -&gt; (gid, name).
        Size : 8192
        Modification time : Tue Jun 12 09:14:31 2007
        Creation time : Tue Jun 12 09:14:31 2007
        TABLE NAME : groupmapnamesid
        Version : 1
        Comment : Store the mapping group.domain -&gt; SID.
        Size : 8192
        Modification time : Tue Jun 12 09:14:31 2007
        Creation time : Tue Jun 12 09:14:31 2007

        EXAMPLE #2
        ----------

        To display Secmap statistics, type:

        $ server_dbms server_3 -db -stats Secmap
        server_3 : done
        STATISTICS FOR DATABASE : Secmap
        TABLE : Mapping
server_dbms
        magic 340322 Magic number.
        version 9 Table version number.
        metaflags 0 Metadata flags.
        nkeys 14 Number of unique keys.
        ndata 14 Number of data items.
        pagesize 4096 Page size.
        minkey 2 Minkey value.
        re_len 0 Fixed-length record length.
        re_pad 32 Fixed-length record pad.
        levels 1 Tree levels.
        int_pg 0 Internal pages.
        leaf_pg 1 Leaf pages.
        dup_pg 0 Duplicate pages.
        over_pg 0 Overflow pages.
        empty_pg 0 Empty pages.
        free 0 Pages on the free list.
        int_pgfree 0 Bytes free in internal pages.
        leaf_pgfree 2982 Bytes free in leaf pages.
        dup_pgfree 0 Bytes free in duplicate pages.
        over_pgfree 0 Bytes free in overflow pages.

        EXAMPLE #3
        ----------

        To display statistics of the VDM database environment, type:
```

```
        $ server_dbms server_3 -service -stats
        STATISTICS FOR MODULE : LOG

server_dbms
        magic               264584                  Log file magic number.
        version             12                      Log file version number.
        mode                0                       Log file mode.
        lg_bsize            32768                   Log buffer size.
        lg_size             5242880                 Log file size.
        record              96                      Records entered into the log.
        w_bytes             16001                   Bytes to log.
        w_mbytes            0                       Megabytes to log.
        wc_bytes            0                       Bytes to log since checkpoint.
        wc_mbytes           0                       Megabytes to log since checkpoint.
        wcount              31                      Total writes to the log.
        wcount_fill         0                       Overflow writes to the log.
        rcount              137                     Total I/O reads from the log.
        scount              31                      Total syncs to the log.
        region_wait         0                       Region lock granted after wait.
        region_nowait       0                       Region lock granted without wait.
        cur_file            3                       Current log file number.
        cur_offset          16001                   Current log file offset.
        disk_file           3                       Known on disk log file number.
        disk_offset         16001                   Known on disk log file offset.
        regsize             98304                   Region size.
        maxcommitperflush   1                       Max number of commits in a flush.
        mincommitperflush   1                       Min number of commits in a flush.

        STATISTICS FOR MODULE : LOCK

server_dbms
        last_id             91                      Last allocated locker ID.
        cur_maxid           2147483647              Current maximum unused ID.
        maxlocks            1000                    Maximum number of locks in table.
        maxlockers          1000                    Maximum num of lockers in table.
        maxobjects          1000                    Maximum num of objects in table.
        nmodes              9                       Number of lock modes.
        nlocks              20                      Current number of locks.
        maxnlocks           21                      Maximum number of locks so far.
        nlockers            49                      Current number of lockers.
        maxnlockers         49                      Maximum number of lockers so far.
        nobjects            20                      Current number of objects.
        maxnobjects         21                      Maximum number of objects so far.
        nrequests           65711                   Number of lock gets.
        nreleases           65691                   Number of lock puts.
        nupgrade            0                       Number of lock upgrades.
        ndowngrade          20                      Number of lock downgrades.
        lock_wait           0                       Lock conflicts w/ subsequent wait.
        lock_nowait         0                       Lock conflicts w/o subsequent wait.
        ndeadlocks          0                       Number of lock deadlocks.
        locktimeout         0                       Lock timeout.
        nlocktimeouts       0                       Number of lock timeouts.
        txntimeout          0                       Transaction timeout.
        ntxntimeouts        0                       Number of transaction timeouts.
        region_wait         0                       Region lock granted after wait.
        region_nowait       0                       Region lock granted without wait.
        regsize             352256                  Region size.

        STATISTICS FOR MODULE : TXN

server_dbms
        last_ckp            3/15945                  lsn of the last checkpoint.
        time_ckp            Fri Aug 3 09:38:36 2007 time of last checkpoint.
        last_txnid          0x8000001a               last transaction id given out.
        maxtxns             20                       maximum txns possible.
        naborts             0                        number of aborted transactions.
        nbegins             26                       number of begun transactions.
        ncommits            26                       number of committed transactions.
        nactive             0                        number of active transactions.
        nsnapshot           0                        number of snapshot transactions.
        nrestores           0                        number of restored transactions
```

```
        after recovery.
        maxnactive          2                    maximum active transactions.
        maxnsnapshot        0                    maximum snapshot transactions.
        region_wait         0                    Region lock granted after wait.
        region_nowait       0                    Region lock granted without wait.
        regsize             16384                Region size.

        STATISTICS FOR MODULE : MPOOL

server_dbms
        gbytes              0                    Total cache size: GB.
        bytes               10487684             Total cache size: B.
        ncache              1                    Number of caches.
        regsize             10493952             Region size.
        mmapsize            0                    Maximum file size for mmap.
        maxopenfd           0                    Maximum number of open fd's.
        maxwrite            0                    Maximum buffers to write.
        maxwrite_sleep      0                    Sleep after writing max buffers.
        map                 0                    Pages from mapped files.
        cache_hit           65672                Pages found in the cache.
        cache_miss          36                   Pages not found in the cache.
        page_create         0                    Pages created in the cache.
        page_in             36                   Pages read in.
        page_out            2                    Pages written out.
        ro_evict            0                    Clean pages forced from the cache.
        rw_evict            0                    Dirty pages forced from the cache.
        page_trickle        0                    Pages written by memp_trickle.
        pages               36                   Total number of pages.
        page_clean          36                   Clean pages.
        page_dirty          0                    Dirty pages.
        hash_buckets        1031                 Number of hash buckets.
        hash_searches       65744                Total hash chain searches.
        hash_longest        1                    Longest hash chain searched.
        hash_examined       65672                Total hash entries searched.
        hash_nowait         0                    Hash lock granted with nowait.
        hash_wait           0                    Hash lock granted after wait.
        hash_max_nowait     0                    Max hash lock granted with nowait
        hash_max_wait       0                    Max hash lock granted after wait.
        region_nowait       0                    Region lock granted with nowait.
        region_wait         0                    Region lock granted after wait.
        mvcc_frozen         0                    Buffers frozen.
        mvcc_thawed         0                    Buffers thawed.
        mvcc_freed          0                    Frozen buffers freed.
        alloc               123                  Number of page allocations.
        alloc_buckets       0                    Buckets checked during allocation.
        alloc_max_buckets   0                    Max checked during allocation.
        alloc_pages         0                    Pages checked during allocation.
        alloc_max_pages     0                    Max checked during allocation.
        io_wait             0                    Thread waited on buffer I/O.

        STATISTICS FOR MODULE : MUTEX

server_dbms
        mutex_align         4                    Mutex alignment.
        mutex_tas_spins     1                    Mutex test-and-set spins.
        mutex_cnt           3254                 Mutex count.
        mutex_free          1078                 Available mutexes.
        mutex_inuse         2176                 Mutexes in use.
        mutex_inuse_max     2176                 Maximum mutexes ever in use.
        region_wait         0                    Region lock granted after wait.
        region_nowait       0                    Region lock granted without wait.
        regsize             278528               Region size.

        ---------------------------------------
        Last Modified: April 07, 2011 12:45 PM
```

server_devconfig

Queries, saves, and displays the SCSI over Fibre Channel device configuration connected to the specified Data Movers.

SYNOPSIS

```
server_devconfig {<movername>|ALL}
| -create -scsi [<chain_number>] {-disks|-nondisks|-all}
[-discovery {y|n}][-monitor {y|n}][-Force {y|n}]
| -list -scsi [<chain_number>] {-disks|-nondisks|-all}
| -probe -scsi [<chain_number>] {-disks|-nondisks|-all}
| -rename <old_name> <new_name>
```

DESCRIPTION
----

server_devconfig queries the available storage system device and tape device configuration, and saves the device configuration into the Data Mover.s database. server_devconfig renames the device name, and lists SCSI devices.

Caution: It is recommended that all Data Movers have the same device configuration. When adding devices to the device table for a single Data Mover only, certain actions such as standby failover will not be successful unless the standby Data Mover has the same disk device configuration as the primary Data Mover.

The ALL option executes the command for all Data Movers.

OPTIONS

-create -scsi [<chain_number>] {-disks|-nondisks|-all}
Queries SCSI devices and saves them into the device table database on the Data Mover. The <chain_number> specifies a SCSI chain number.

The -disks option limits operations to disks. The -nondisks option limits operations to non-disks such as tapes, juke boxes, and gatekeeper devices. The -all option permits all SCSI devices (disks and non-disks).

Note: The -create option modifies VNX for lock LUN names to the VNX_<vnx-hostname>_<lun-id>_<vnx-dvol-name> format, if the LUNs use the default Unisphere name.

Caution: The time taken to complete this command might be lengthy, dependent on the number and type of attached devices.

[-discovery {y|n}]
Enables or disables the storage discovery operation.

Caution: Disabling the -discovery option should only be done under the direction of an EMC Customer Service Engineer.

[-monitor {y|n}]
Displays the progress of the query and discovery operations.

[-Force {y|n}]
Overrides the health check failures and changes the storage configuration.

Caution: High availability can be lost when changing the storage configuration. Changing the storage configuration should only be done under the direction of an EMC Customer Service Engineer.

-list -scsi [<chain_number>] {-disks|-nondisks|-all}
Lists the SCSI device table database that has been saved on the Data Mover. The <chain_number> specifies a SCSI chain number.

Note: Fibre Channel devices appear as SCSI devices. Therefore, chain numbers might be different for Fibre Channel.

The -disks option limits operations to disks. The -nondisks option limits operations to non-disks such as tapes, juke boxes, and gatekeeper devices. The -all option permits all SCSI devices (disks and non-disks).

-probe -scsi [<chain_number>] {-disks|-nondisks|-all}
Queries and displays the SCSI devices without saving them into the database. The <chain_number> specifies a SCSI chain number.

Note: Fibre Channel devices appear as SCSI devices, therefore, chain numbers may be different for Fibre Channel.

The -disks option limits operations to disks. The -nondisks option limits operations to non-disks such as tapes, juke boxes, and gatekeeper devices. The -all option permits all SCSI devices (disks and non-disks).

-rename <old_name> <new_name>
Renames the specified non-disk from the <old_name> to <new_name>. The -rename option is available for non-disks only.

SEE ALSO

VNX System Operations, nas_disk, and nas_storage.

STORAGE SYSTEM OUTPUT

The number associated with the storage device is dependent on the attached storage system. VNX for block displays a prefix of APM before a set of integers, for example, APM00033900124-0019. For example, Symmetrix storage systems appear as 002804000190-003C.

EXAMPLE #1

For the VNX storage system, to list all devices, type:

```
$ server_devconfig server_2 -list -scsi -all
server_2:
Scsi Disk Table
                                Director Port

name         addr        num      type  num  sts   stor_id       stor_dev
root_disk   c0t0l0                                APM00043807043   0000
root_disk   c16t0l0                               APM00043807043   0000
root_ldisk  c0t0l1                                APM00043807043   0001
root_ldisk  c16t0l1                               APM00043807043   0001
d3          c0t0l2                                APM00043807043   0002
d3          c16t0l2                               APM00043807043   0002
d4          c0t0l3                                APM00043807043   0003
d4          c16t0l3                               APM00043807043   0003
d5          c0t0l4                                APM00043807043   0004
d5          c16t0l4                               APM00043807043   0004
d6          c0t0l5                                APM00043807043   0005
d6          c16t0l5                               APM00043807043   0005
d7          c0t1l0                                APM00043807043   0010
d7          c16t1l0                               APM00043807043   0010
d8          c16t1l1                               APM00043807043   0011
d8          c0t1l1                                APM00043807043   0011
```

```
                        Scsi Device Table
name         addr              type            info
gk01         c0t0l             disk        5 0207000000000APM00043807043
ggk01        c0t1l0            disk        5 0207100001000APM00043807043
gk161        c16t1l1           disk        5 0207110011100APM00043807043
```

For the VNX with a Symmetrix storage system, to list all the devices in the SCSI table, type:

```
$ server_devconfig server_2 -list -scsi -all
server_2 :
                        Scsi Disk Table
                        Director Port
name         addr       num      type   num      sts     stor_id      stor_dev
root_disk    c0t0l0     16C      FA      0        On    000187940268  0000
root_disk    c16t0l0    01C      FA      0        On    000187940268  0000
root_ldisk   c0t0l1     16C      FA      0        On    000187940268  0001
root_ldisk   c16t0l1    01C      FA      0        On    000187940268  0001
d3           c0t1l0     16C      FA      0        On    000187940268  0006
d3           c16t1l0    01C      FA      0        On    000187940268  0006
d4           c0t1l1     16C      FA      0        On    000187940268  0007
d4           c16t1l1    01C      FA      0        On    000187940268  0007
d5           c0t1l2     16C      FA      0        On    000187940268  0008
d5           c16t1l2    01C      FA      0        On    000187940268  0008
d6           c0t1l3     16C      FA      0        On    000187940268  0009
d6           c16t1l3    01C      FA      0        On    000187940268  0009
d7           c0t1l4     16C      FA      0        On    000187940268  000A
d7           c16t1l4    01C      FA      0        On    000187940268  000A
<... removed ...>
d377         c1t8l6     16C      FA      0        On    000187940268  017C
d377         c17t8l6    01C      FA      0        On    000187940268  017C
rootd378     c1t8l7     16C      FA      0        On    000187940268  0180
rootd378     c17t8l7    01C      FA      0        On    000187940268  0180
rootd379     c1t8l8     16C      FA      0        On    000187940268  0181
rootd379     c17t8l8    01C      FA      0        On    000187940268  0181
rootd380     c1t8l9     16C      FA      0        On    000187940268  0182
rootd380     c17t8l9    01C      FA      0        On    000187940268  0182
rootd381     c1t8l10    16C      FA      0        On    000187940268  0183
rootd381     c17t8l10   01C      FA      0        On    000187940268  0183

                        Scsi Device Table
name         addr       type            info
gk01         c0t0l15    disk     56706817D480 000187940268
gk161        c16t0l15   disk     56706817D330 000187940268
```

Note: This is a partial display due to the length of the output.

Where:

Value	Definition
name	A unique name for each device in the chain
addr	SCSI chain, target, and LUN information
Director num	Director number. This output is applicable for Symmetrix storage systems only.
type	Device type, as specified in the SCSI specification for peripherals. This output is applicable for Symmetrix storage systems only.
Port num	Port number. This output is applicable for Symmetrix storage systems only.
sts	Indicates the port status. Possible values are: On, Off, WD (write disabled), and NA. This output is applicable for Symmetrix storage systems only.
stor_id	Storage system ID

```
stor_dev        Storage system device ID

EXAMPLE #2
-----------

For the VNX, to list all SCSI-attached non-disk devices, type:

$ server_devconfig server_2 -list -scsi -nondisks
server_2 :
                Scsi Device Table
name     addr         type              info
gk01    c0t0l0        disk       5 0207000000000APM00043807043
ggk01   c0t1l0        disk       5 0207100001000APM00043807043
gk161   c16t1l1       disk       5 0207110011100APM00043807043

For the VNX with a Symmetrix storage system, to list all
SCSI-attached non-disk devices, type:

$ server_devconfig server_2 -list -scsi -nondisks
server_2 :
                Scsi Device Table
name    addr          type              info
gk01    c0t0l15       disk       56706817D480   000187940268
gk161   c16t0l15      disk       56706817D330   000187940268

For info=56706817D480, the following breakdown applies:

5670            Symmcode
68              Last 2 digits in the Symm S/N
17D             Symm Device ID#
48              Symm SA #
0               SA Port # (0=a, 1=b)

EXAMPLE #3
----------

To rename a device, type:

$ server_devconfig server_2 -rename gk161 gk201
server_2 : done

EXAMPLE #4
----------

For the VNX, to discover SCSI disk devices, without saving them to
the database table, type:

$ server_devconfig server_2 -probe -scsi -disks
server 2 :
SCSI disk devices :
chain= 0, scsi-0
stor_id= APM00043807043 celerra_id= APM000438070430000
tid/lun= 0/0 type= disk sz= 11263 val= 1 info= DGC RAID 5 02070000000000NI
tid/lun= 0/1 type= disk sz= 11263 val= 2 info= DGC RAID 5 02070100010001NI
tid/lun= 0/2 type= disk sz= 2047 val= 3 info= DGC RAID 5 02070200020002NI
tid/lun= 0/3 type= disk sz= 2047 val= 4 info= DGC RAID 5 02070300030003NI
tid/lun= 0/4 type= disk sz= 2047 val= 5 info= DGC RAID 5 02070400040004NI
tid/lun= 0/5 type= disk sz= 2047 val= 6 info= DGC RAID 5 02070500050005NI
tid/lun= 1/0 type= disk sz= 245625 val= 7 info= DGC RAID 5 02071000100010NI
tid/lun= 1/1 type= disk sz= 0 val= -5 info= DGC RAID 5 02071100110011NI
tid/lun= 1/2 type= disk sz= 273709 val= 9 info= DGC RAID 5 02071200120012NI
tid/lun= 1/3 type= disk sz= 0 val= -5 info= DGC RAID 5 02071300130013NI
tid/lun= 1/4 type= disk sz= 273709 val= 10 info= DGC RAID 5 02071400140014NI

tid/lun= 1/5 type= disk sz= 0 val= -5 info= DGC RAID 5 02071500150015NI
tid/lun= 1/6 type= disk sz= 273709 val= 11 info= DGC RAID 5 02071600160016NI

tid/lun= 1/7 type= disk sz= 0 val= -5 info= DGC RAID 5 02071700170017NI
tid/lun= 1/8 type= disk sz= 273709 val= 12 info= DGC RAID 5 02071800180018NI

tid/lun= 1/9 type= disk sz= 0 val= -5 info= DGC RAID 5 02071900190019NI
```

```
chain= 1, scsi-1 : no devices on chain
chain= 2, scsi-2 : no devices on chain
chain= 3, scsi-3 : no devices on chain
chain= 4, scsi-4 : no devices on chain
chain= 5, scsi-5 : no devices on chain
chain= 6, scsi-6 : no devices on chain
chain= 7, scsi-7 : no devices on chain
chain= 8, scsi-8 : no devices on chain
chain= 9, scsi-9 : no devices on chain
chain= 10, scsi-10 : no devices on chain
chain= 11, scsi-11 : no devices on chain
chain= 12, scsi-12 : no devices on chain
chain= 13, scsi-13 : no devices on chain
chain= 14, scsi-14 : no devices on chain
chain= 15, scsi-15 : no devices on chain
```

For the VNX with a Symmetrix storage system, to discover SCSI disk devices, without saving them to the database table, type:

```
$ server_devconfig server_2 -probe -scsi -disks
server_2 :
SCSI disk devices :
chain= 0, scsi-0 : no devices on chain
chain= 1, scsi-1 : no devices on chain
chain= 2, scsi-2
stor_id= 000190102173 celerra_id= 0001901021730041
tid/lun= 0/0 type= disk sz= 11507 val= 1 info= 577273041291SI00041
tid/lun= 0/1 type= disk sz= 11507 val= 2 info= 577273042291SI00042
tid/lun= 1/0 type= disk sz= 11501 val= 3 info= 57727304F291SI0004F
tid/lun= 1/1 type= disk sz= 11501 val= 4 info= 577273050291SI00050
tid/lun= 1/2 type= disk sz= 11501 val= 5 info= 577273051291SI00051
tid/lun= 1/3 type= disk sz= 11501 val= 6 info= 577273052291SI00052
tid/lun= 1/4 type= disk sz= 11501 val= 7 info= 577273053291SI00053
tid/lun= 1/5 type= disk sz= 11501 val= 8 info= 577273054291SI00054
tid/lun= 1/6 type= disk sz= 11501 val= 9 info= 577273055291SI00055
tid/lun= 1/7 type= disk sz= 11501 val= 10 info= 577273056291SI00056
tid/lun= 1/8 type= disk sz= 11501 val= 11 info= 577273057291SI00057
tid/lun= 1/9 type= disk sz= 11501 val= 12 info= 577273058291SI00058
tid/lun= 1/10 type= disk sz= 11501 val= 13 info= 577273059291SI00059
tid/lun= 1/11 type= disk sz= 11501 val= 14 info= 57727305A291SI0005A
tid/lun= 1/12 type= disk sz= 11501 val= 15 info= 57727305B291SI0005B
tid/lun= 1/13 type= disk sz= 11501 val= 16 info= 57727305C291SI0005C
tid/lun= 1/14 type= disk sz= 11501 val= 17 info= 57727305D291SI0005D
tid/lun= 1/15 type= disk sz= 11501 val= 18 info= 57727305E291SI0005E
tid/lun= 2/0 type= disk sz= 11501 val= 19 info= 57727305F291SI0005F
tid/lun= 2/1 type= disk sz= 11501 val= 20 info= 577273060291SI00060
tid/lun= 2/2 type= disk sz= 11501 val= 21 info= 577273061291SI00061
<... removed ...>
tid/lun= 7/6 type= disk sz= 11501 val= 105 info= 577273517291SI00517
tid/lun= 7/7 type= disk sz= 11501 val= 106 info= 577273518291SI00518
tid/lun= 7/8 type= disk sz= 11501 val= 107 info= 577273519291SI00519
tid/lun= 7/9 type= disk sz= 11501 val= 108 info= 57727351A291SI0051A
tid/lun= 7/10 type= disk sz= 11501 val= 109 info= 57727351B291SI0051B
tid/lun= 7/11 type= disk sz= 11501 val= 110 info= 57727351C291SI0051C
tid/lun= 7/12 type= disk sz= 11501 val= 111 info= 57727351D291SI0051D
tid/lun= 7/13 type= disk sz= 11501 val= 112 info= 57727351E291SI0051E
tid/lun= 7/14 type= disk sz= 11501 val= 113 info= 57727351F291SI0051F
tid/lun= 7/15 type= disk sz= 11501 val= 114 info= 577273520291SI00520

chain= 3, scsi-3 : no devices on chain
chain= 4, scsi-4 : no devices on chain
chain= 5, scsi-5 : no devices on chain
chain= 6, scsi-6 : no devices on chain
<... removed ...>

chain= 18, scsi-18
stor_id= 000190102173 celerra_id= 0001901021730041
tid/lun= 0/0 type= disk sz= 11507 val= 1 info= 577273041201SI00041
tid/lun= 0/1 type= disk sz= 11507 val= 2 info= 577273042201SI00042
tid/lun= 1/0 type= disk sz= 11501 val= 3 info= 57727304F201SI0004F
tid/lun= 1/1 type= disk sz= 11501 val= 4 info= 577273050201SI00050
```

```
tid/lun= 1/2 type= disk sz= 11501 val= 5 info= 577273051201SI00051
tid/lun= 1/3 type= disk sz= 11501 val= 6 info= 577273052201SI00052
tid/lun= 1/4 type= disk sz= 11501 val= 7 info= 577273053201SI00053

Note: This is a partial listing due to the length of the output.
```

EXAMPLE #5

To discover and save all SCSI devices, type:

```
$ server_devconfig server_2 -create -scsi -all
Discovering storage (may take several minutes)
server_2 : done
```

EXAMPLE #6

To discover and save all non-disk devices, type:

```
$ server_devconfig server_2 -create -scsi -nondisks
Discovering storage (may take several minutes)
server_2 : done
```

EXAMPLE #7

To save all SCSI devices with the discovery operation disabled, and display information regarding the progress, type:

```
$ server_devconfig ALL -create -scsi -all -discovery n -monitor y
server_2 :
server_2:
chain 0 ..........
chain 16 .....
done
server_3 :
server_3:
chain 0 ..........
chain 16 .....
done
server_4 :
server_4:
chain 0 ..........
chain 16 .....
done
server_5 :
server_5:
chain 0 ..........
chain 16 .....
done
```

Last Modified: April 07, 2011 03:25 pm

server_df

Reports free and used disk space and inodes for mounted file systems on the specified Data Movers.

SYNOPSIS
----

server_df {<movername>|ALL}
[-inode][<pathname>|<fs_name>]

DESCRIPTION
----

server_df reports the amount of used and available disk space for a Data Mover or file system, how much of a file system's total capacity has been used, and the number of used and free inodes.

The ALL option executes the command for all Data Movers.

OPTIONS

No arguments
Displays the amount of disk space in kilobytes used by file systems.

-inode
Reports used and free inodes.

[<pathname>|<fs_name>]
Gets file system information. If <fs_name> specified, gets information for file system only.

SEE ALSO

Managing Volumes and File Systems for VNX Manually, nas_disk, and nas_fs.

EXAMPLE #1

To display the amount of used and available disk space on a Data Mover, type:

```
$ server_df server_2
server_2 :
Filesystem          kbytes         used        avail capacity Mounted on
ufs1             1075686032       477816   1075208216    0%    /ufs1
ufs4              101683184          584    101682600    0%    /nmfs1/ufs4
ufs2              206515184          600    206514584    0%    /nmfs1/ufs2
nmfs1             308198368         1184    308197184    0%    /nmfs1
root_fs_common        13624         5264         8360   39%    /.etc_common
root_fs_2            114592          760       113832    1%    /
```

Where:

Value	Definition
Filesystem	Name of the file system.
kbytes	Total amount of space in kilobytes for the file system.
used	Amount of kilobytes used by the file system.

avail	Amount of space in kilobytes available for the file system.
capacity	Percentage capacity that is used.
Mounted on	Mount point of the file system.

EXAMPLE #2

To display the amount of disk space and the amount of free and unused inodes on a Data Mover, type:

```
$ server_df server_2 -inode
server_2 :
Filesystem          inodes         used        avail capacity Mounted on
ufs1             131210494          140    131210354    0%    /ufs1
ufs4              25190398           10     25190388    0%    /nmfs1/ufs4
ufs2              25190398           11     25190387    0%    /nmfs1/ufs2
nmfs1             50380796           21     50380775    0%    /nmfs1
root_fs_common       21822           26        21796    0%    /.etc_common
root_fs_2           130942           66       130876    0%    /
```

EXAMPLE #3

To display the amount of disk space and the amount of free and unused inodes on a file system, type:

```
$ server_df server_2 -inode ufs1
server_2 :
Filesystem          inodes         used        avail capacity Mounted on
ufs1             131210494          140    131210354    0%    /ufs
```

Last Modified: April 07, 2011 03:35 pm

server_dns

Manages the Domain Name System (DNS) lookup server configuration for the specified Data Movers.

SYNOPSIS

```
server_dns {<movername>|ALL}
          [[-protocol {tcp|udp}] <domainname> {<ip_addr>,...}]
        | [-delete <domainname>]
        | [-option {start|stop|flush|dump}]
```

DESCRIPTION
----

server_dns provides connectivity to the DNS lookup servers for the specified Data Movers to convert hostnames and IP addresses. Up to three DNS lookup servers are supported for each domain on the Data Mover.

server_dns also provides the ability to clear the cache that has been saved on the Data Mover as a result of the DNS lookup servers.

The ALL option executes the command for all Data Movers.

OPTIONS

No arguments
Displays the DNS configuration.

-protocol {tcp|udp} <domainname> {<ip_addr>,...}
Sets the protocol for the DNS lookup servers (udp is the default).

<domainname> {<ip_addr>,...}
Creates list of up to three IP addresses to be used as the DNS lookup servers for the specified <domainname>.

-delete <domainname>
Deletes the DNS lookup servers in the DNS domain name.

-option {start|stop|flush|dump}
The start option activates the link for the DNS lookup servers. The stop option halts access to the DNS lookup servers. After DNS service has been halted, the flush option can be used to clear the cache that has been saved on the Data Mover, and the dump option displays the DNS cache.

SEE ALSO

Configuring VNX Naming Services and server_nis.

EXAMPLE #1

To connect to a DNS lookup server, type:

```
$ server_dns server_2 prod.emc.com 172.10.20.10

server_2 : done
```

EXAMPLE #2

To display the DNS configuration, type:

```
$ server_dns server_2

server_2 :
```

```
DNS is running.
prod.emc.com
proto:udp server(s):172.10.20.10
```

EXAMPLE #3

To change the protocol to TCP from UDP, type:

```
$ server_dns server_2 -protocol tcp prod.emc.com 172.10.20.10

server_2 : done
```

EXAMPLE #4

To halt access to the DNS lookup servers, type:

```
$ server_dns server_2 -option stop

server_2 : done
```

EXAMPLE #5

To flush the cache on a Data Mover, type:

```
$ server_dns server_2 -option flush

server_2 : done
```

EXAMPLE #6

To dump the DNS cache, type:

```
$ server_dns server_2 -option dump

server_2 :
DNS cache size for one record type: 64
DNS cache includes 6 item(s):
dm102-cge0.nasdocs.emc.com
Type:A TTL=184 s dataCount:1
172.24.102.202 (local subnet)
---
winserver1.nasdocs.emc.com
Type:A TTL=3258 s dataCount:1
172.24.103.60
---
_ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.nasdocs.emc.com
Type:SRV TTL=258 s dataCount:1
priority:0 weight:100 port:389 server:winserver1.nasdocs.emc.com
---
_kerberos._tcp.Default-First-Site-Name._sites.dc._msdcs.nasdocs.emc.com
Type:SRV TTL=258 s dataCount:1
priority:0 weight:100 port:88 server:winserver1.nasdocs.emc.com
---
Expired item(s): 2
```

EXAMPLE #7

To delete the DNS lookup servers, type:

```
$ server_dns server_2 -delete prod.emc.com

server_2 : done
```

Last modified: May 12, 2011 9:30 am.

server_export

Exports file systems and manages access on the specified Data Movers for NFS and CIFS clients.

SYNOPSIS

```
server_export {<movername>|ALL}
  operations on all cifs and/or nfs entries:
| [-Protocol {cifs|nfs}] -list -all
| [-Protocol {cifs|nfs}] -all
| [-Protocol {cifs|nfs}] -unexport [-perm] -all

  nfs operations per entry:
| -list <pathname>
| [-Protocol nfs [-name <name>]][-ignore][-option <options>]
    [-comment <comment>] <pathname>
| -unexport [-perm] <pathname>

  cifs operations per entry:
| -list -name <sharename> [-option <options>]
| -name <sharename> [-ignore][-option <options>][-comment <comment>]
   <pathname>
| -unexport -name <sharename> [-option <options>]
-option type = {
  CA [:] Encrypted [:][ABE [:] HASH [:][OCAutoI|OCVDO|OCNONE]]|NONE
               }
```

DESCRIPTION
----

server_export provides user access by exporting an NFS pathname, or creating a CIFS share. Allows specification of multiple clients identified by hostnames or network and subnet addresses separated by a colon.

server_export removes access by unexporting an NFS pathname, deleting a CIFS share, and displaying the exported entries and available shares for the specified Data Mover.

The ALL option executes the command for all of the Data Movers.

Note: NFSv4 does not support the -name option.

GENERAL OPTIONS FOR CIFS AND NFS OPERATIONS

No arguments
Displays all exported NFS entries and CIFS shares.

[-Protocol {cifs|nfs}] -list -all
Lists all exported entries as defined by the protocol. The default is NFS.

[-Protocol {cifs|nfs}] -all
Exports all entries on a Data Mover as defined by the protocol. The default is NFS.

[-Protocol {cifs|nfs}] -unexport [-perm] -all
Unexports all entries as defined by the protocol. By default, unexports are permanent for CIFS, and temporary for NFS, unless -perm is specified. If -perm is specified, it removes all entries from the export table. When the entry is temporarily unexported, clients are denied access to the entry until it is re-exported or the system is rebooted, but the entries are not removed from the export table. The default is NFS.

FOR NFS OPERATIONS

-list <pathname>
Lists a specific NFS entry. If there are extra spaces in the <pathname>, the entire pathname must be enclosed by quotes. By using the server_export command, IPv6 addresses can be specified and the hosts configured with these addresses can mount and access file systems over NFS.

Note: If you are configuring an IPv6 address for ro, rw, access, and root, it must be enclosed in [ ] or square brackets. This is to properly handle the colon used to separate entries. Link local addresses are not supported.

-Protocol nfs [-name <name>] <pathname>
Exports an NFS <pathname> by default as read-write for everyone. If specified, assigns an optional filesystem name for the <name>. Pathname length is limited to 1024 bytes (represented as 1024 ASCII characters or a variable number of Unicode multibyte characters), and must be enclosed by quotes, if spaces are used. Name length is limited to 255 bytes.

Note: In a nested mount file system hierarchy, users can export the mount point path of the component file system. Subdirectories of the component file system cannot be exported. In a multilevel file system hierarchy, users can export any part of a file system independent of existing exports.

[-ignore] <pathname>
Overwrites previous options and comments in the export table for the entry.

[-comment <comment>] <pathname>
Adds a comment for the specified NFS export entry. The comment is displayed when listing the exported entries.

[-option <options>] <pathname>
Specifies the following comma-separated options:

sec=[sys|krb5|krb5i|krb5p]:<mode> [,<mode>,...]
Specifies a user authentication or security method with an access mode setting. The sys (default) security option specifies AUTH_SYS security. The access mode can be one, or a combination of the following: ro, rw=, ro=, root=, access=, anon=, webroot, public.

If the sec option is specified, it must always be the first option specified in the string.

krb5 security specifies Kerberos user and data authentication. krb5i checks for the integrity of the data by adding a signature to each NFS packet and krb5p encrypts the data before sending it over the network.

For krb5, krb5i, and krb5p security, the access mode can be one, or a combination of the following: ro, rw=, ro=, root=, access=.

ro
Exports the <pathname> for all NFS clients as read-only.

ro=<client>[:<client>]...
Exports the <pathname> for the specified NFS clients as read-only.

Note: If <client> is an IPv6 address, it must be enclosed in square brackets or [ ].

ro=<-client>[:<-client>]...
Excludes the specified NFS clients from ro privileges. Clients must be preceded with dash (-) to specify exclusion.

Note: If <client> is an IPv6 address, it must be enclosed in square brackets or [ ].

rw=<client>[:<client>]...
Exports the <pathname> as read-mostly for the specified NFS clients. Read-mostly means exported read-only to most machines, but read-write to those specified. The default is read-write to all.

Note: If <client> is an IPv6 address, it must be enclosed in square brackets or [ ].

rw=<-client>[:<-client>]...
Excludes the specified NFS clients from rw privileges. The description of read-mostly provides information. Clients must be preceded with - (dash) to specify exclusion.

Note: If <client> is an IPv6 address, it must be enclosed in square brackets or [ ].

root=<client>[:<client>]...
Provides root privileges for the specified NFS clients. By default, no host is granted root privilege.

Note: If <client> is an IPv6 address, it must be enclosed in square brackets or [ ].

root=<-client>[:<-client>]...
Excludes the specified NFS clients from root privileges. Clients must be preceded with - (dash) to specify exclusion.

Note: If <client> is an IPv6 address, it must be enclosed in square brackets or [ ].

anon=<uid>
If a request comes from an unknown user, the UID should be used as the effective user ID. Root users (UID =0) are considered "unknown" by the NFS server unless they are included in the root option. The default value for anon=<uid> is the user "nobody". If the user "nobody" does not exist, then the value 65534 is used.

Caution: Using anon=0 is not recommended for security concerns.

access=<client>[:<client>]...
Provides mount access for the specified NFS clients.

Note: If <client> is an IPv6 address, it must be enclosed in square brackets or [ ].

access=<-client>[:<-client>]...
Excludes the specified NFS clients from access even if they are part of a subnet or netgroup that is allowed access. Clients must be preceded with - (dash) to specify exclusion.

Note: If <client> is an IPv6 address, it must be enclosed in square brackets or [ ].

nfsv4only
Specifies that the NFS export can be accessed only when a client is using NFSv4.

vlan=<vlanid>[,<vlanid>,...]
Specifies that all hosts belonging to the VLAN ID will have access to the exported filesystem. Hosts on other VLANs will be denied.

The VLANs are seperated by : or colon, just as any other server_export option values.

Clients can be a hostname, netgroup, subnet, or IP address and must be colon-separated, without spaces. A subnet is an IP address/netmask (for example, 168.159.50.0/255.255.255.0). A hostname is first checked for in the Data Mover.s local hosts database, then on the NIS (host database) or DNS server (if enabled). A netgroup is searched in the local netgroup database

and then on the NIS server's netgroup database. If the client name does not exist in any case, then access is denied.

Note: Netgroups are supported. The hosts and netgroup files can be created on the Control Station using your preferred method (for example, with an editor, or by copying from another node), then copied to the Data Mover.

nosuid=<client>[:<client>]... OR
nosuid=<-client>[:<-client>]...
When the nosuid NFS export option is used with a list of client names, the setuid and setgid bits are cleared from the permissions before setting the permissions on any file on the exported pathname for those clients.

When the nosuid NFS export option is used with a dash (-) before each client name, the setuid and setgid bits are cleared from the permissions before setting the permissions on any file on the exported pathname for all clients except for the clients listed.

-unexport [-perm] <pathname>
Temporarily unexports a <pathname> unless -perm is specified. If -perm is specified, removes the entry from the export table.

FOR CIFS OPERATIONS

-list -name <sharename>
Displays the specified CIFS share.

[-option <options>]
Specifies the following comma-separated list of options:

[netbios=<netbios_name>]
When the share has an associated NetBIOS name, that name is required to locate the entry. Multiple CIFS entries can have same <sharename> when belonging to a different NetBIOS name.

-name <sharename> [-ignore] [-option <options>]
[-comment <comment>] <pathname>
Creates a CIFS share. Share name length is limited to 12 ASCII characters unless Unicode is enabled, in which case the limit is 80 multibyte characters. Share names cannot include the following characters: /, \, %, ", NUL (Null character), STX (start of header), SOT (start of text), and LF (line feed). Share names can contain spaces and other nonalphanumeric characters, but must be enclosed by quotes if spaces are used. Share names cannot begin with a - (hyphen). Share names are case-sensitive.

Comment length is limited to 256 bytes (represented as 256 ASCII characters or a variable number of Unicode multibyte characters). A comment cannot include the following characters: NUL (Null character), STX (start of header), and SOT (start of text). Comments can contain spaces and other nonalphanumeric characters, but must be enclosed by quotes if spaces are used. Pathname length is limited to 1024 bytes.

The -ignore option overwrites the previous options and comment in the export table.

[-option <options>]
Specifies the following comma-separated options:

ro
Exports the <pathname> for CIFS clients as read-only.

rw=<client>[:<client>]...
Creates the share for CIFS clients as read-mostly. Read-mostly means shared read-only to most clients, but read-write to those specified. By default, the <pathname> is shared read-write to all. A client may be

either a <user_name> or <group_name>.

Note: If <client> is an IPv6 address, it must be enclosed in [ ] or square brackets.

umask=<mask>
Specifies a user file-creation mask for the umask allowing NFS permissions to be determined for the share.

user=<default_user>
When using share level access (server_checkup provides information), specifies a <default_user> which must be entered as a character string. The user must be defined in the Data Mover.s password file. There is a 20 character limit for the username.
group=<default_group>

When using share level access (server_checkup provides information), indicates a <default_group> which must be entered as a character string. There is a 256 character limit for group names.

ropasswd=<share_passwd>
When using share level access (server_checkup provides information), creates a read-only password to allow clients access to the share. Passwords can be viewed in the list of shared entries.

rwpasswd=<share_rw_passwd>
When using share level access (server_checkup provides information), creates a read-write password to allow clients access to the share. Passwords are displayed in the list of shared entries.

Note: Users from any client machine who know the value of the ropasswd or rwpasswd can access the share for read-only and read-write operations.

maxusr=<maxusr>
Sets the maximum number of simultaneous users permitted for a share.

netbios=<netbiosName>[,netbios=<netbiosName>]...
Associates a share on a single domain with one or more NetBIOS names created with server_checkup. By default, if a NetBIOS name is not specified for a share, the share is visible to all NetBIOS names.

-comment
Adds a comment for the specified CIFS share. The comment is displayed when listing the shared entries.

-unexport -name <sharename>
Permanently removes access to a share by removing the entry from the export table.

[-option <options>]
Specifies the following comma-separated options:

netbios=<netbios_name>
When the share has an associated NetBIOS name, the NetBIOS name is required to locate the entry. This is required because multiple CIFS entries can have same <sharename> when belonging to different NetBIOS name.

-option type={CA[:]Encrypted[:][ABE[:]HASH[:][OCAutoI|OCVDO|OCNONE]]|NONE}

Specifies the following colon-separated list of options:

* Continuous Availability (CA): Indicates continuous availability of data on the specific share.

* Encrypted: The server requires encrypted messages to access the share.
* Access Based Enumeration (ABE): Only files and directories to which the user has read access are visible (Access Based Enumeration).
* HASH: Indicates that the share supports hash generation for BranchCache retrieval.
* Offline Caching Attributes (OC): User MUST allow only manual caching for the files open from this share by default.
- CAutoI: The user MAY cache every file that it opens from this share.
- OCVDO: The user MAY cache every file that it opens from this share. Also, the user MAY satisfy the file requests from its local cache.
- OCNone: Indicates that no files or programs from the shared folder are available offline.

SEE ALSO

Configuring NFS on VNX, Managing Volumes and File Systems for VNX Manually, server_checkup, and server_mount.

EXAMPLE #1

To export a specific NFS entry, type:

```
$ server_export server_2 -Protocol nfs /ufs1
server_2 : done
```

EXAMPLE #2

To export an NFS entry and overwrite existing settings, type:

```
$ server_export server_2 -Protocol nfs -ignore -option
access=172.24.102.0/255.255.255.0,root=172.24.102.240 -comment 'NFS Export
for ufs1' /ufs1
server_2 : done
```

EXAMPLE #3

To export NFS entry dir1, a subdirectory of the exported entry /ufs1 in a multilevel file system hierarchy, type:

```
$ server_export server_2 -Protocol nfs /ufs1/dir1
server_2 : done
```

EXAMPLE #4

To assign a name to a NFS export, type:

```
$ server_export server_2 -Protocol nfs -name nasdocsfs /ufs1
server_2 : done
```

EXAMPLE #5

To export an NFS entry using Kerberos authentication, type:

```
$ server_export server_2 -Protocol nfs -option
sec=krb5:ro,root=172.24.102.240,access=172.24.102.0/255.255.255.0 /ufs2
server_2 : done
```

EXAMPLE #6

To export an NFS entry for NFSv4 only, type:

```
$ server_export server_2 -Protocol nfs -option nfsv4only /ufs1
```

```
server_2 : done

EXAMPLE #7
----------

To list all NFS entries, type:

$ server_export server_2 -Protocol nfs -list -all
server_2 :
export "/ufs2" sec=krb5 ro root=172.24.102.240
access=172.24.102.0/255.255.255.0
export "/ufs1" name="/nasdocsfs" access=172.24.102.0/255.255.255.0
root=172.24.102.240 nfsv4only comment="NFS Export for ufs1"
export "/" anon=0
access=128.221.252.100:128:221.253.100:128.221.252.101:128.221.253.101

EXAMPLE #8
----------

To list NFS entries for the specified path, type:

$ server_export server_2 -list /ufs1
server_2 :
export "/ufs1" name="/nasdocsfs" access=172.24.102.0/255.255.255.0
root=172.24.102.240 nfsv4only comment="NFS Export for ufs1"

EXAMPLE #9
----------

To temporarily unexport an NFS entry, type:

$ server_export server_2 -Protocol nfs -unexport /ufs2
server_2 : done

EXAMPLE #10
-----------

To export all NFS entries, type:

$ server_export server_2 -Protocol nfs -all
server_2 : done

EXAMPLE #11
-----------

To export a specific NFS entry in a language that uses multibyte
characters, type:
$ server_export server_2 -Protocol nfs
/<nfs_entry_in_local_language_text>
server_2 : done

EXAMPLE #12
-----------

To permanently unexport an NFS entry, type:

$ server_export server_2 -unexport -perm /ufs1
server_2 : done

EXAMPLE #13
-----------

To permanently unexport all NFS entries, type:

$ server_export server_2 -Protocol nfs -unexport -perm -all
server_2 : done

EXAMPLE #14
-----------

To provide access to a CIFS share, type:
```

```
$ server_export server_2 -name ufs1 /ufs1
server_2 : done

EXAMPLE #15
-----------

To create a CIFS share and overwrite existing settings, type:

$ server_export server_2 -name ufs1 -ignore -option
ro,umask=027,maxusr=200,netbios=dm112-cge0 -comment 'CIFS share' /ufs1
server_2 : done

EXAMPLE #16
-----------

To create a CIFS share in a language that uses multibyte characters, type:

$ server_export server_2 -P cifs -name <name_in _local _language_text>
-comment <comment_in local_language_text> /accounting

server_2 : done

EXAMPLE #17
-----------

To list all CIFS entries, type:

$ server_export server_2 -Protocol cifs -list
server_2 :
share "ufs1" "/ufs1" ro umask=027 maxusr=200 netbios=DM112-CGE0
comment="CIFS share"
share "ufs2" "/ufs2" umask=022 maxusr=4294967295

EXAMPLE #18
-----------

To display a specific CIFS share, type:

$ server_export server_2 -list -name ufs1 -option netbios=dm112-cge0
server_2 :
share "ufs1" "/ufs1" ro umask=027 maxusr=200 netbios=DM112-CGE0
comment="CIFS share"

EXAMPLE #19
-----------

To export all CIFS entries, type:

$ server_export server_2 -Protocol cifs -all
server_2 : done

EXAMPLE #20
-----------

To list all NFS and CIFS entries, type:

$ server_export server_2
server_2 :
export "/ufs2" sec=krb5 ro root=172.24.102.240
access=172.24.102.0/255.255.255.0
export "/ufs1" nfsv4only
export "/" anon=0
access=128.221.252.100:128.221.253.100:128.221.252.101:128.221.253.101
share "ufs2" "/ufs2" umask=022 maxusr=4294967295
share "ufs1" "/ufs1" ro umask=027 maxusr=200 netbios=DM112-CGE0
comment="CIFS share"

Where:
 Value                          Definition
 export         A file system entry to be exported.
```

```
 sec            Security mode for the file system.
 ro             File system is to be exported as read-only.
 root           IP address with root access.
 access         Access is permitted for those IP addresses.
 share          Entry to be shared.
 ro             Filesystem is to be shared as read-only.
 umask          User creation mask.
 maxuser        Maximum number of simultaneous users.
 netbios        NetBIOS name for the share.
 comment        Comment specified for the share.
```

EXAMPLE #21

To permanently unexport all CIFS and NFS entries, type:

```
$ server_export server_2 -unexport -perm -all
server_2 : done
```

EXAMPLE #22

To delete a CIFS share, type:

```
$ server_export server_2 -unexport -name ufs1 -option netbios=dm112-cge0
server_2 : done
```

EXAMPLE #23

To delete all CIFS shares, type:

```
$ server_export server_2 -Protocol cifs -unexport -all
server_2 : done
```

EXAMPLE #24

To export a file system for NFS that specifies an IPv4 and IPv6 address, type:

```
$ server_export server_2 -Protocol nfs -option
access=172.24.108.10:[1080:0:0:0:8:800:200C:417A] /fs1
server_2 : done
```

EXAMPLE #25

To export a file system for NFS that specifies two IPv6 addresses, type:

```
$ server_export server_2 -Protocol nfs -option rw=[1080:0:0:0:8:80:200C:417A]:[10
80:0:0:0:8:800:200C:417B] /fs1
server_2 : done
```

EXAMPLE #26

To verify that the file system was exported, type:

```
$ server_export server_2 -list /fs1
server_2 :
export "/fs1" rw=[1080:0:0:0:8:80:200C:417A]:[1080:0:0:0:8:800:200C:417B]
```

EXAMPLE #27

To export the fs42 file system of the VDM vdm1, type:

```
$ server_export vdm1 -P nfs /fs42
done
```

EXAMPLE #28

To create a share foo on the server PALIC with HASH and ABE enabled, type:

```
$ server_export server_3 -name foo -option netbios=PALIC,
type=ABE:HASH /fs3/foo
server_3 : done
```

EXAMPLE #29

To change attributes to this share to ABE only, type:

```
$ server_export server_3 -name foo -option netbios=PALIC,
type=ABE /fs3/foo
server_3 : done
```

EXAMPLE #30

To remove all the attributes, type:

```
server_export server_3 -name foo -ignore -option netbios=PALIC,type=None
/fs3/fro
server_3 : done
```

EXAMPLE #31

To view the attributes, type:

```
server_export server_3 share "foo" "/fs3/fro" type=ABE:HASH umask=022
maxusr=4294967295 netbios=PALIC
server_3 : done
```

EXAMPLE #32

To create a share foo on the server palic with CA and ABE enabled,
type:

```
$ server_export server_3 -name foo -option netbios=PALIC,
type=CA:ABE /fs3/foo
server_3 : done
```

EXAMPLE #33

To change attributes of the share foo to CA only, type:

```
$ server_export server_3 -name foo -option netbios=PALIC,
type=CA /fs3/foo
server_3 : done
```

EXAMPLE #34

To view the attributes, type:

```
$ server_export server_3 share "foo" "/fs3/fro" type=CA umask=022
maxusr=4294967295 netbios=PALIC
server_3 : done
```

EXAMPLE #35

To create a share share10 accessible only through encrypted SMB
messages, type:

```
$ server_export vdm1 -P cifs -name share10 -o
type=Encrypted /fs42/protected_dir1
server_3 : done
```

EXAMPLE #36

```
$ server_export server_2 -Protocol nfs -option
nosuid=host10:host11 /users/gary
server_2 : done
```

EXAMPLE #37

```
$ server_export ALL -option nosuid=host123 /production1
server_2 : done
```

EXAMPLE #38

```
$ server_export server_2 -Protocol nfs -option
root=10.241.216.239,nosuid=-10.241.216.239 /fs1
server_2 : done
```

Last Modified: November 20, 2012 11:55 a.m.

server_file

Copies files between the Control Station and the specified Data Movers.

SYNOPSIS
----

server_file {<movername>|ALL}
{-get|-put} <src_file> <dst_file>

DESCRIPTION
----

server_file copies the source file from the specified Data Mover (or Control Station) to the destination file on the Control Station (or specified Data Mover). The <src_file> indicates the source file, and the name <dst_file> indicates destination file. By default, if a directory is not specified on the Data Mover, the /.etc directory is used.

The ALL option executes the command for all Data Movers.

OPTIONS

-get <src_file> <dst_file>
Copies the source file on Data Mover to destination file on Control Station. Both the <src_file> and <dst_file> may be full pathnames.

-put <src_file> <dst_file>
Copies source file on the Control Station to the destination file on the Data Mover. Both the <src_file> and <dst_file> must be full pathnames.

Caution: This command overwrites existing files of the same name without notification. Use care when copying files.

EXAMPLE #1

To copy a file from the Control Station to a Data Mover, type:

```
$ server_file server_2 -put passwd passwd
server_2 : done
```

EXAMPLE #2

To copy a file from the Data Mover to the Control Station, type:

```
$ server_file server_2 -get passwd /home/nasadmin/passwd
server_2 : done
```

Last Modified: April 11, 2011 01:35 pm

server_fileresolve

Starts, deletes, stops, checks, and displays the fileresolve service
for the specified Data Mover. Filereseolve service facilitates
inode-to-filename translation. This translation is required when administrator
monitors the 'fs.qtreeFile' and 'fs.filesystem' statistics.

SYNOPSIS

```
server_fileresolve <movername>
       -service { -start [-maxlimit <1M>]
                | -stop
                | -delete
                | -status }
        | -list
        | -add <path_name> [,...]
        | -drop <path_name> [,...]
        | -lookup { -filesystem <fs_name> -inode <inode>[,...]
                 | -qtree <qt_name> -inode <inode>[,...] } [...]
```

DESCRIPTION

Controls and manages the fileresolve service, which crawls through filesystems
specified by the user. To have the fileresolve service started at boot time,
it is recommended that this command be added to the eof config file for the
Data Mover.

OPTIONS

-service {-start [-maxlimit <1M>]
Starts the fileresolve service on the specified Data Mover. By default, the
fileresolve service caches upto 1 million files (this takes about 32MB of
memory on the Data Mover). By increasing the maximum limit of the
inode-to-filename translation cache, from 1M to 2M, it would increase memory
consumed by the service to 64M.

To change the maxlimit, use the following command:
server_fileresolve <movername> -service -start -maxlimit <new_value>

This new limit will be preserved across Data Mover reboots. However, when a
new limit is applied, the entire inode-to-filename cache will be flushed and
rebuilt. The Filesystem crawler adds files to its cache in the order they are
traversed. Hence, the first 1 million files traversed (by default) go in the
cache.

-stop
Flushes the inode-to-filename cache and stops the service. Deleting the
service also would free up the memory consumed by the fileresolve service and
deletes the configuration files created by the service.

-delete
Deletes the fileresolve service on the specified Data Mover.

-status
Checks the status of the files that are added to the cache
on the specified Data Mover.

-list
Displays the filesystems/ directories that are in the configuration and used
for crawling.

-add <path_name> [,...]
Adds specified path to the server_fileresolver configuration.
Crawls the specified path and buildsthe inode-to-filename cache.

To add a specific file that should be included in the
inode-to-filename map,the following command should be used:

server_fileresolve server_X -add <path for file>

-drop <path_name>[,...]
Drops specified path to the server_fileresolver configuration.
Inode-to-filename cache for the specified path is not cleared
until the service is restarted.

-lookup {-filesystem <fs_name> -inode <inode> [,...]
Performs an on demand crawl of the specified filesystem to translate the
inode to a pathname. If the pathname is not found for the inode, the inode
value is returned. For example, server_stats displays this inode value instead
of a path name in its output.

The user can do a 'deep, non-cached' lookup of the inode to discover the
pathname (if it still exists). However, this could take time (in the order of
minutes). Hence, server_stats only attempts to lookup in the cache and does
not attempt a full Filesystem crawl.

Note: If the file name is successfully resolved, full pathname is returned.
Even if the file name is the same as the inode path is appended.

-lookup -qtree <qt_name> -inode <inode> [,...] } [...]
Performs an on demand crawl of the specified quota tree to
translate the inode to a pathname.

EXAMPLES

EXAMPLE #1

To display the new paths added,type:

$ server_fileresolve server_2 -add /server_2/ufs_0
server_2 :
New paths are added

EXAMPLE #2

To list the specified file paths that are included in the
inode-to-filename map,type:

$ server_fileresolve server_2 -list
server_2 :
PATH
/server_2/ufs_5
/server_2/ufs_4
/server_2/ufs_3
/server_2/ufs_2
/server_2/ufs_1
/server_2/ufs_0

EXAMPLE #3

To check the status of the fileresolve services, type:
$ server_fileresolve server_2 -service -status
server_2 :
FileResolve service is running :Max Limit of the cache:1000000 Entries used:10
Dropped entries:0

EXAMPLE #4

To drop the specified path to the server_fileresolver configuration,
type:

$ server_fileresolve server_2 -drop /server_2/ufs_0
server_2 :

Paths are dropped
Warning: Restart service to remove the cached entries of dropped paths.

EXAMPLE #5

To lookup multiple inodes within the same filesystem, type:

```
$ server_fileresolve  server_2 -lookup -filesystem ufs_0  -inode
61697,61670,61660
server_2 :
Filesystem/QTree                    Inode          Path
ufs_0                               61660         /server_2/ufs_0/dir00000/
                                                  testdir/yYY_0000039425.tmp
ufs_0                               61670        /server_2/ufs_0/dir00000/
                                                 testdir/kNt_0000028175.tmp
ufs_0                               61697        /server_2/ufs_0/dir00000/
                                                 testdir/gwR_0000058176.tmp
```

EXAMPLE #6

To lookup multiple inodes within a Quota Tree, type:

```
$ server_fileresolve  server_2 -lookup -qtree dir00000 -inode 61697
server_2 :
Filesystem/QTree                        Inode      Path
dir00000                                61697      /server_2/ufs_0/dir00000/
                                                   testdir/gwR_0000058176.tmp
```

Date updated: June 04, 2012 12:15 p.m.

server_ftp

Configures the FTP server configuration for the specified Data Movers.

SYNOPSIS

```
server_ftp {<movername>|ALL}
-service {-status|-start|-stop|{-stats [-all|-reset]}}
| -info
| -modify
    [-controlport <controlport>]
    [-dataport <dataport>]
    [-defaultdir <path>]
    [-homedir {enable|disable}]
    [-keepalive <keepalive>]
    [-highwatermark <highwatermark>]
    [-lowwatermark <lowwatermark>]
    [-deniedusers [<path>]]
    [-welcome [<path>]]
    [-motd [<path>]]
    [-timeout <timeout>]
    [-maxtimeout <maxtimeout>]
    [-readsize <readsize>]
    [-writesize <writesize>]
    [-maxcnx <maxcnx>]
    [-umask <umask>]
    [-sslcontrol {no|allow|require|requireforauth}]
    [-ssldata {allow|require|deny}]
    [-sslpersona {anonymous|default|<persona_name>}]
    [-sslprotocol {default|ssl3|tls1|all}]
    [-sslcipher {default|<cipherlist>}]
    [-sslcontrolport <sslcontrolport>]
    [-ssldataport <ssldataport>]
```

DESCRIPTION
----

server_ftp configures the ftp daemon. Optional SSL security support is available. The modifications are performed when the ftp daemon is stopped and are reflected after restart of the ftp daemon. There is no need to reboot the Data Mover for the changes to be reflected.

OPTIONS
----

server_ftp {<movername>|ALL}
Sends a request to the Data Mover to get all the parameters of the ftp daemon.

ALL option executes the command for all Data Movers.

-service {-status|-start|-stop|{-stats [-all|-reset]}}

-status
Retrieves the current status of the ftp daemon.

-start
Starts the ftp daemon. The start option persists after the daemon is rebooted.

-stop
Stops the ftp daemon.

-stats [all|reset]
Displays the statistics of the ftp daemon. The reset option resets all the ftp server statistics. The all option displays detailed statistics.

-info
Retrieves all the parameters for the ftp daemon along with its current

status.

-modify
Modifies the ftp daemon configuration. The ftp daemon has to be stopped to carry out the changes. The modifications are taken into account when the service is restarted.

-controlport <controlport>
Sets the local tcp port for control connections. By default, the port is 21. When control port is set to 0, it disables unsecure ftp usage and validates port for implicit secure connection on SSL port (default 990).

Note: This default port can be changed using the sslcontrolport option.

-dataport <dataport>
Sets the local tcp port for active data connections. By default, the port is 20. When <dataport> is set to 0, the port is allocated dynamically by the server in active mode.

-defaultdir <path>
Sets the default user directory when the user home directory is not accessible. This option replaces "ftpd.defaultdir.. By default, "/" is used.

-homedir {enable|disable}
Restricts or allows user access to their home directory tree. When enabled the user is allowed access to their home directory only. If the user home directory is not accesible, access is denied. During connection the user is denied access to data outside of their home directory space. By default, this feature is disabled.

Note: Using FTP on VNX provides more information about how the home directory of a user is managed.

-umask <umask>
Defines the mask to set the mode bits on file or directory creation. By default the mask is 027, which means that xrw-r---- mode bits are assigned.

-keepalive <keepalive>
Sets TCP keepalive value for the ftp daemon. This value is given in seconds. By default, the value is 60. The value 0 disables the TCP keepalive option. The maximum value is 15300 (255 minutes).

-highwatermark <highwatermark>
Sets TCP high watermark value (amount of data stored without knowledge of the client) for the ftp daemon. By default, the value is 65536. The minimum value is 8192, and the maximum value is 1048576 (1 MB).

Caution: Do not modify this parameter without a thorough knowledge of the impact on FTP client behavior.

-lowwatermark <lowwatermark>
Sets TCP low watermark value (amount of the data to be added, after the highwatermark has been reached and new data can be accepted from the client) for the ftp daemon. The minimum value is 8192, maximum value is 1048576 (1 MB), and default value is 32768.

Caution: Do not modify this parameter without a thorough knowledge of the impact on FTP client behavior.

-deniedusers <deniedusers_file>
Denies FTP access to specific users on a Data Mover. Specifies the path and text file containing the list of usernames to be denied access. Places each username on a separate line. By default, all users are allowed.

-welcome <welcome_file>

Specifies path of the file to be displayed on the welcome screen. For example, this file can display a login banner before the user is requested for authentication data. By default, no welcome message is displayed.

-motd <motd_file>
Specifies path of the file to be displayed on the welcome screen. Users see a welcome screen ("message of the day") after they successfully log in. By default, no message of the day is displayed.

-timeout <timeout>
Specifies the default inactivity time-out period (when not set by the client). The value is given in seconds. After the specified time if there is no activity, the client is disconnected from the server and will have to re-open a connection. By default, the <timeout> value is 900 seconds. The minimum value is 10 seconds, and the maximum value is 7200.

-maxtimeout <maxtimeout>
Sets the maximum time-out period allowed by the client. The value is given in seconds and any value larger than maximum time-out period is not allowed. By default, the <maxtimeout> value is 7200 seconds. The minimum value is 10 seconds, and the maximum value is 7200.

-readsize <readsize>
Sets the size for reading files from the disk. The value must be greater than 8192, and a multiple of 8K. By default, the <readsize> is 8192 bytes. The minimum value is 8192, and the maximum value is 1048576 (1 MB).

-writesize <writesize>
Sets the size for writing files from the disk. The value must be greater than 8192, and a multiple of 8K. By default, the <writesize> is 49152 (48 KB). The minimum value is 8192, and the maximum value is 1048576 (1 MB).

-maxcnx <maxcnx>
Sets the maximum number of control connections the ftp daemon will support. By default, the <maxcnx> value is set to 65535 (64K-1). The minimum value is 1, and the maximum value is 65535 (64K-1).

-sslcontrol {no|allow|require|requireforauth}
Uses SSL for the ftp control connection depending on the attributes specified. By default, SSL is disabled. The no option disables SSL control. The allow option specifies that SSL is enabled, but the user can still connect without SSL. The require option specifies that SSL is required for the connection. The requireforauth option specifies that SSL is required for authentication.The control path goes back to unsecure after this authentication. When the client is behind a firewall, this helps the firewall to filter the ftp commands requiring new port access.

Note: Before the server can be configured with SSL, the Data Mover must be set up with a private key and a public certificate. This key and certificate are identified using a persona. In addition, the necessary Certificate Authority (CA) certificates used to identify trusted servers must be imported into the Data Mover. Use the system.s PKI feature to manage the use of certificates prior to configuring SSL operation.

-ssldata {no|allow|require}
Uses SSL for the data connection depending on the attributes specified. The no option disables SSL. The allow option specifies that SSL is enabled, but the user can also transfer data without SSL. The require option specifies that SSL is required for data connection.The ssldata value cannot be set to allow or require if sslcontrol is set to no. By default, SSL is disabled.

Note: These options are set on the server but are dependent on ftp client capabilities. Some client capabilities may be incompatible with server settings. Using FTP on VNX provides information on validating compatibility.

-sslpersona {anonymous|default|<persona_name>}
Specifies the persona associated with the Data Mover. Personas are used to identify the private key and public certificate used by SSL. The default value specified is default (each Data Mover is configured with a persona named default). The anonymous value specifies that SSL can operate without using a certificate. This implies that the communication between client and server is encrypted and data integrity is guaranteed.

Note: Use server_certificate to configure the persona before using server_ftp.

-sslprotocol {default|ssl3|tls1|all}
Specifies the SSL protocol version that the ftp daemon on the server accepts:

* ssl3 - Only SSLv3 connections
* tls1 - Only TLSv1 connections
* all - Both SSLv3 and TLSv1 connections
* default - Uses the value set in the ssl.protocol parameter which, by default, is 0 (SSLv3 and TLSv1)

-sslcipher {default|<cipherlist>}
Specifies the SSL cipher suite. The value of default is the value set in the ssl.cipher parameter. This value means that all ciphers are supported by VNX except the Anonymous Diffie-Hellman, NULL, and SSLv2 ciphers and that the supported ciphers are sorted by the size of the encryption key.

-sslcontrolport <sslcontrolport>
Sets the implicit control port for FTP connections over SSL. By default, the port is 990. To disable implicit FTP connections over SSL, the <sslcontrolport> must be set to 0.

-ssldataport <ssldataport>
Sets the local tcp port for active data connections using implicit FTP connections over SSL. By default, the port is 899. If the ssldataport is set to 0, the Data Mover will use a port allocated by the system.

SEE ALSO : server_certificate.

EXAMPLE #1

To retrieve all the parameters for the ftp daemon and its status, type:

```
$ server_ftp server_2 -info
FTP started
=========
controlport                             21
dataport                                20
defaultdir /                            .etc/ftpd/pub
homedir                                 disable
umask                                   027

tcp keepalive                           1 minute
tcp high watermark                      65536 bytes
tcp low watermark                       32768 bytes
readsize                                8192 bytes
writesize                               49152 bytes
denied users file path                  /.etc/ftpd/conf/ftpusers
welcome file path                       /.etc/ftpd/conf/welcome
motd file path                          /.etc/ftpd/conf/motd
```

```
session timeout                          900 seconds
max session timeoutQ                     7200 seconds

Security Options
=============
sslpersona                               default
sslprotocol                              default
sslcipher                                default

FTP over TLS explicit Options
-----------------------------------------
sslcontrol                               SSL require for authentication
ssldata                                  allow SSL

FTP over SSL implicit Options
-----------------------------------------
sslcontrolport                           990
ssldataport                              989
```

EXAMPLE #2

To display the statistics of the ftp daemon, type:

```
$ server_ftp server_2 -service -stats
Login Type       Successful     Failed
==========       ==========     =======
 Anonymous       10             0
 Unix            3              2
 CIFS            7              1

                                Throughput (MBytes/sec)
Data transfers  Count   min     average                  max
==============  =====   ====    ======================== =====
Write Bin        10     10.00   19.00                    20.00
Read Bin         0      ----     ----                    ----
Write ASCII      2      1.00    1.50                     2.00
Read ASCII       0      ----    ----                     ----
SSL Write Bin    5      5.00    17.00                    18.00
SSL Read Bin     15     7.00    25.00                    35.00
SSL Write ASCII  0      ----    ----                     ----
SSL Read ASCII   0      ----    ----                     ----
```

Where:

Value	Definition
Throughput (MBytes/sec)	Throughput is calculated using the size of the file (Mbytes) divided by the duration of the transfer (in seconds).
average	Average is the average of the throughputs (sum of the throughputs divided by the number of transfers).
Data transfers	Defines the type of transfer.
Count	Number of operations for a transfer type.
min	Minimum time in milliseconds required to execute the operation (with regards to Data mover).
max	Maximum time in milliseconds required to execute the operation (with regards to Data mover).

EXAMPLE #3

To display the statistics of the ftp daemon with details, type:

```
$ server_ftp server_2 -service -stats -all
Commands        Count
```

```
========        =====
USER            23
PASS            23
QUIT            23
PORT            45
EPRT            10
....            ....
FEAT            23

SITE Commands   Count
=============   =====
UMASK           0
IDLE            10
CHMOD           0
HELP            0
BANDWIDTH       0
KEEPALIVE       10
PASV            56

OPTS Commands   Count
=============   =====
UTF8            10

Login Type      Successful      Failed
==========      ==========      =======
Anonymous       10              0
Unix            3               2
CIFS            7               1

Connections     Count
===========     =====
Non secure
----------
Control         10
Data            44

Explicit SSL
------------
Control Auth    3
Control         8
Data            20

Implicit SSL
------------
Control         0
Data            0

                                Throughput (MBytes/sec)
Data transfers  Count   min     average                 max
==============  =====   ======== ====================   =======
Write Bin       10      10.00    19.00                  20.00
Read Bin        0       ----     ----                   ----
Write ASCII     2       1.00     1.50                   2.00
Read ASCII      0       ----     ----                   ----
SSL Write Bin   5       5.00     17.00                  18.00
SSL Read Bin    15      7.00     25.00                  35.00
SSL Write ASCII 0       ----     ----                   ----
SSL Read ASCII  0       ----     ----                   ----

Where:

Value           Definition

Commands        FTP protocol command name.

Count           Number of commands received by Data mover.

SITE Commands   Class of command in FTP protocol.

OPTS Commands   Class of command in FTP protocol.
```

EXAMPLE #4

To retrieve the status of the ftp daemon, type:

```
$ server_ftp server_3 -service -status
server_3 : done

State                       : running
```

EXAMPLE #5

To start the ftp daemon , type:

```
$ server_ftp server_2 -service -start
server_2 : done
```

EXAMPLE #6

To stop the ftp daemon, type:

```
$ server_ftp server_2 -service -stop
server_2 : done
```

EXAMPLE #7

To set the local tcp port for the control connections, type:

```
$ server_ftp server_2 -modify -controlport 256
server_2 :done

FTPD CONFIGURATION
==================
State                       : stopped
Control Port                : 256
Data Port                   : 20
Default dir                 : /
Home dir                    : disable
Keepalive                   : 1
High watermark              : 65536
Low watermark               : 32768
Timeout                     : 900
Max timeout                 : 7200
Read size                   : 8192
Write size                  : 49152
Umask                       : 27
Max connection              : 65535

SSL CONFIGURATION
=================
Control channel mode        : disable
Data channel mode           : disable
Persona                     : default
Protocol                    : default
Cipher                      : default
Control port                : 990
Data port                   : 989
```

EXAMPLE #8

To set the local tcp port for active data connections, type:

```
$ server_ftp server_2 -modify -dataport 257
server_2 : done
```

```
FTPD CONFIGURATION
==================
State                     : stopped
Control Port              : 256
Data Port                 : 257
Default dir               : /
Home dir                  : disable
Keepalive                 : 1
High watermark            : 65536
Low watermark             : 32768
Timeout                   : 900
Max timeout               : 7200
Read size                 : 8192
Write size                : 49152
Umask                     : 27
Max connection            : 65535

SSL CONFIGURATION
=================
Control channel mode      : disable
Data channel mode         : disable
Persona                   : default
Protocol                  : default
Cipher                    : default
Control port              : 990
Data port                 : 989
```

EXAMPLE #9

To change the default directory of a user when his home directory is not accessible, type:

```
$ server_ftp server_2 -modify -defaultdir /big
server_2 : done

FTPD CONFIGURATION
==================
State                     : stopped
Control Port              : 256
Data Port                 : 257
Default dir               : /big
Home dir                  : disable
Keepalive                 : 1
High watermark            : 65536
Low watermark             : 32768
Timeout                   : 900
Max timeout               : 7200
Read size                 : 8192
Write size                : 49152
Umask                     : 27
Max connection            : 65535

SSL CONFIGURATION
=================
Control channel mode      : disable
Data channel mode         : disable
Persona                   : default
Protocol                  : default
Cipher                    : default
Control port              : 990
Data port                 : 989
```

EXAMPLE #10

To allow users access to their home directory tree, type:

```
$ server_ftp server_2 -modify -homedir enable

server_2 : done
```

```
FTPD CONFIGURATION
==================
State                     : stopped
Control Port              : 256
Data Port                 : 257
Default dir               : /big
Home dir                  : enable
Keepalive                 : 1
High watermark            : 65536
Low watermark             : 32768
Timeout                   : 900
Max timeout               : 7200
Read size                 : 8192
Write size                : 49152
Umask                     : 27
Max connection            : 65535

SSL CONFIGURATION
=================
Control channel mode      : disable
Data channel mode         : disable
Persona                   : default
Protocol                  : default
Cipher                    : default
Control port              : 990
Data port                 : 989
```

EXAMPLE #11

To restrict user access to their home directory tree, type:

```
$ server_ftp server_2 -modify -homedir disable
server_2 : done

FTPD CONFIGURATION
==================
State                     : stopped
Control Port              : 256
Data Port                 : 257
Default dir               : /big
Home dir                  : disable
Keepalive                 : 1
High watermark            : 65536
Low watermark             : 32768
Timeout                   : 900
Max timeout               : 7200
Read size                 : 8192
Write size                : 49152
Umask                     : 27
Max connection            : 65535

SSL CONFIGURATION
=================
Control channel mode      : disable
Data channel mode         : disable
Persona                   : default
Protocol                  : default
Cipher                    : default
Control port              : 990
Data port                 : 989
```

EXAMPLE #12

To set the default umask for creating a file or a directory by means of the ftp daemon, type:

```
$ server_ftp server_2 -modify -umask 077
server_2 : done
```

```
FTPD CONFIGURATION
==================
State                     : stopped
Control Port              : 256
Data Port                 : 257
Default dir               : /big
Home dir                  : disable
Keepalive                 : 1
High watermark            : 65536
Low watermark             : 32768
Timeout                   : 900
Max timeout               : 7200
Read size                 : 8192
Write size                : 49152
Umask                     : 77
Max connection            : 65535

SSL CONFIGURATION
=================
Control channel mode      : disable
Data channel mode         : disable
Persona                   : default
Protocol                  : default
Cipher                    : default
Control port              : 990
Data port                 : 989
```

EXAMPLE #13

To set the TCP keepalive for the ftp daemon, type:

```
$ server_ftp server_2 -modify -keepalive 120
server_2 : done

FTPD CONFIGURATION
==================
State                     : stopped
Control Port              : 256
Data Port                 : 257
Default dir               : /big
Home dir                  : disable
Keepalive                 : 120
High watermark            : 65536
Low watermark             : 32768
Timeout                   : 900
Max timeout               : 7200
Read size                 : 8192
Write size                : 49152
Umask                     : 77
Max connection            : 65535

SSL CONFIGURATION
=================
Control channel mode      : disable
Data channel mode         : disable
Persona                   : default
Protocol                  : default
Cipher                    : default
Control port              : 990
Data port                 : 989
```

EXAMPLE #14

To set the TCP highwatermark for the ftp daemon, type:

```
$ server_ftp server_2 -modify -highwatermark 90112
server_2 : done
```

```
FTPD CONFIGURATION
==================
State                     : stopped
Control Port              : 256
Data Port                 : 257
Default dir               : /big
Home dir                  : disable
Keepalive                 : 120
High watermark            : 90112
Low watermark             : 32768
Timeout                   : 900
Max timeout               : 7200
Read size                 : 8192
Write size                : 49152
Umask                     : 77
Max connection            : 65535

SSL CONFIGURATION
=================
Control channel mode      : disable
Data channel mode         : disable
Persona                   : default
Protocol                  : default
Cipher                    : default
Control port              : 990
Data port                 : 989
```

EXAMPLE #15

To set the TCP lowwatermark for the ftp daemon, type:

```
$ server_ftp server_2 -modify -lowwatermark 32768
server_2 : done

FTPD CONFIGURATION
==================
State                     : stopped
Control Port              : 256
Data Port                 : 257
Default dir               : /big
Home dir                  : disable
Keepalive                 : 120
High watermark            : 90112
Low watermark             : 32768
Timeout                   : 900
Max timeout               : 7200
Read size                 : 8192
Write size                : 49152
Umask                     : 77
Max connection            : 65535

SSL CONFIGURATION
=================
Control channel mode      : disable
Data channel mode         : disable
Persona                   : default
Protocol                  : default
Cipher                    : default
Control port              : 990
Data port                 : 989
```

EXAMPLE #16

To restrict FTP server access to specific users, type:

```
$ server_ftp server_2 -modify -deniedusers /.etc/mydeniedlist
server_2 : done
```

```
FTPD CONFIGURATION
==================
State                       : stopped
Control Port                : 256
Data Port                   : 257
Default dir                 : /big
Home dir                    : disable
Keepalive                   : 120
High watermark              : 90112
Low watermark               : 32768
Denied users conf file      : /.etc/mydeniedlist
Timeout                     : 900
Max timeout                 : 7200
Read size                   : 8192
Write size                  : 49152
Umask                       : 77
Max connection              : 65535

SSL CONFIGURATION
=================
Control channel mode        : disable
Data channel mode           : disable
Persona                     : default
Protocol                    : default
Cipher                      : default
Control port                : 990
Data port                   : 989
```

EXAMPLE #17

To set the path of the file displayed before the user logs in, type:

```
$ server_ftp server_2 -modify -welcome /.etc/mywelcomefile
server_2 : done

FTPD CONFIGURATION
==================
State                       : stopped
Control Port                : 256
Data Port                   : 257
Default dir                 : /big
Home dir                    : disable
Keepalive                   : 120
High watermark              : 90112
Low watermark               : 32768
Welcome file                : /.etc/mywelcomefile
Timeout                     : 900
Max timeout                 : 7200
Read size                   : 8192
Write size                  : 49152
Umask                       : 77
Max connection              : 65535

SSL CONFIGURATION
=================
Control channel mode        : disable
Data channel mode           : disable
Persona                     : default
Protocol                    : default
Cipher                      : default
Control port                : 990
Data port                   : 989
```

Last Modified Date: April 12, 2011. Time: 11:20 am

server_http

Configures the HTTP configuration file for independent services, such as VNX FileMover, for the specified Data Movers.

SYNOPSIS

```
server_http {<movername>|ALL}
  -info [<feature>]
| -service <feature> {-start|-stop}
| -service [<feature>] -stats [-reset]
| -modify <feature>
    [-threads <threads>]
    [-users {valid|<user>[,<user>,<user>...]}]
    [-hosts <ip>[,<ip>,<ip>...]]
    [-port <port_number>]
    [-timeout <max_idle_time>]
    [-maxrequests <maxrequests>]
    [-authentication {none|basic|digest}]
    [-realm <realm_name>]
    [-ssl {required|off}]
    [-sslpersona {anonymous|default|<persona_name>}]
    [-sslprotocol {default|ssl3|tls1|all}]
    [-sslcipher {default|<cipherlist>}]
| -append <feature>
    [-users {valid|<user>[,<user>,<user>...]}]
    [-hosts <ip>[,<ip>,<ip>...]]
| -remove <feature>
    [-users {valid|<user>[,<user>,<user>...]}]
    [-hosts <ip>[,<ip>,<ip>...]]
```

DESCRIPTION

Server_http manages user and host access to HTTP servers for independent services such as FileMover.

The ALL option executes the command for all of the Data Movers.

OPTIONS

[-info <feature>]
Displays information about the specified feature or all features including server status, port, threads, requests allowed, timeout, access control, and SSL configuration.

-service <feature> {-start |-stop}
Stops or starts the HTTP server for the specified feature.

-service [<feature>] -stats [-reset}
Lists the usage statistics of the HTTP server for the specified feature or all features. If -reset is specified, statistics are reset to zero.

-modify <feature>
Displays the current HTTP protocol connection for the specified feature. When issued with options, -modify sets the HTTP protocol connection for the specified option. Any options previously set will be overwritten.

[-threads <threads>]
Sets the number of threads (default=20) for incoming service requests. The minimum value is 4, the maximum 99. The HTTP threads are started on the Data Mover at boot time.

[-users [valid|<user>[,<user>,<user>.]]]
Allows the users who correctly authenticate as defined in the Data Mover passwd file (server_user provides more information) to execute commands for th
e specified
<feature>.

If valid is entered, all users in the passwd file are allowed to use digest authentication. A comma-separated list of users can also be given. If no users are given, digest authentication is turned off.

[-hosts <ip>[,<ip>,<ip>.]]
Specifies hosts by their IP addresses that are allowed to execute commands for the specified <feature>.

[-port <port_number>]
Specifies the port on which the HTTP server listens for incoming service requests. By default, the HTTP server instance for FileMover listens on port 5080.

[-timeout <max_idle_time>]
Specifies the maximum time the HTTP server waits for a request before disconnecting from the client. The default value is 60 seconds.

[-maxrequests <max_requests>]
Specifies the maximum number of requests allowed. The default value is 300 requests.

[-authentication {none|basic|digest}]
Specifies the authentication method. none disables user authentication, allowing for anonymous access (that is, no authentication). basic authentication uses a clear text password. digest authentication uses a scripted password. The default value is digest authentication.

[-realm <realm_name>]
Specifies the realm name. This information is required when authentication is enabled (that is, the -authentication option is set to basic or digest). The default realm name for FileMover is DHSM_authorization.

[-ssl {required|off}]
Specifies whether the HTTP server runs in secure mode, that is, only accepts data received on encrypted SSL sessions. The default value is off.

Note: Before the HTTP server can be configured with SSL, the Data Mover must be set up with a private key and public certificate. This key and certificate are identified using a persona. In addition, the necessary Certificate Authority (CA) certificates to identify trusted servers must be imported into the Data Mover. Use Celerra.s PKI feature to manage the use of certificates prior to configuring SSL operation.

[-sslpersona {default|anonymous|<persona_name>}]
Specifies the persona associated with the Data Mover. Personas are used to identify the private key and public certificate used by SSL. The default value is default (each Data Mover is currently configured with a single persona named default). anonymous specifies that SSL can operate without using a certificate.

[-sslprotocol {default|ssl3|tls1|all}]
Specifies the SSL protocol version the HTTPS server accepts.

* ssl3 - Only SSLv3 connections
* tls1 - Only TLSv1 connections
* all - Both SSLv3 and TLSv1 connections
* default - Uses the value set in the ssl.protocol parameter which, by default, is 0 (SSLv3 and TLSv1)

[-sslcipher {default|<cipherlist>}]
Specifies the SSL cipher suite. The value of default is the value set in the ssl.cipher parameter which, by default, is ALL:!ADH:!SSLv2:@STRENGTH. This value means that all ciphers are supported by Celerra except the Anonymous Diffie-Hellman, NULL, and SSLv2 ciphers and that the supported ciphers are sorted by the size of the encryption key.

-append <feature> [-users {valid|<user>[,<user>,<user>.]}]
[-hosts <ip>[,<ip>,<ip>.]}]
Adds the specified users or hosts to the list of those who can execute commands for the specified <feature> without having to re-enter the existing list.The users and hosts descriptions provide information. If users or hosts are not

specified, displays the current HTTP configuration.

-remove <feature> [-users {valid|<user>[,<user>,<user>.]}]
[-hosts <ip>[,<ip>,<ip>.]}]
Removes the specified users and hosts from the list of those who can execute commands for the specified <feature> without impacting others in the list. The users and hosts descriptions provide information. If users or hosts are not specified, displays the current HTTP configuration.

SEE ALSO

Using VNX FileMover, Security Configuration Guide for File, fs_dhsm, server_certificate, and nas_ca_certificate.

EXAMPLE #1

To display information about the HTTP protocol connection for the FileMover service, type:

```
$ server_http server_2 -info dhsm
server_2 : done
DHSM FACILITY CONFIGURATION
  Service name       : EMC File Mover service
  Comment            : Service facility for getting DHSM attributes
  Active             : False
  Port               : 5080
  Threads            : 16
  Max requests       : 300
  Timeout            : 60 seconds
  ACCESS CONTROL
    Allowed IPs        : any
    Authentication     : digest ,Realm : DHSM_Authorization
    Allowed user       : nobody
  SSL CONFIGURATION
    Mode               : OFF
    Persona            : default
    Protocol           : default
    Cipher             : default
```

Where:

Value	Definition
Service name	Name of the File Mover service.
active	Whether VNX FileMover is enabled or disabled on the file system.
port	TCP port of the File Mover service.
threads	Number of threads reserved for the File Mover service.
max requests	Maximum number of HTTP requests the service allows, to keep the connection alive.
timeout	The time in seconds until which the service is kept alive after a period of no activity.
allowed IPs	List of client IP addresses that are allowed to connect to the service.
authentication	The HTTP authentication method used by the service.
allowed user	Users allowed to connect to the service.
mode	The SSL mode.
persona	Name of the persona associated with the Certificate for establishing a secure connection.

protocol The level of SSL protocol used for the service.

cipher The cipher suite the service is negotiating, for establishing a secure connection with the client.

EXAMPLE #2

To display statistical information about the HTTP protocol connection for the FileMover service, type:

```
$ server_http server_2 -service dhsm -stats

server_2 : done

Statistics report for HTTPD facility DHSM :
   Thread activity
      Maximum in use count          : 0
   Connection
      IP filtering rejection count  : 0
   Request
      Authentication failure count  : 0
   SSL
      Handshake failure count       : 0
```

EXAMPLE #3

To configure an HTTP protocol connection for FileMover using SSL, type:

```
$ server_http server_2 -modify dhsm -ssl required
server_2 : done
```

EXAMPLE #4

To modify the threads option of the HTTP protocol connection for FileMover, type:

```
$ server_http server_2 -modify dhsm -threads 40
server_2 : done
DHSM FACILITY CONFIGURATION
  Service name       : EMC File Mover service
  Comment            : Service facility for getting DHSM attributes
  Active             : False
  Port               : 5080
  Threads            : 40
  Max requests       : 300
  Timeout            : 60 seconds
  ACCESS CONTROL
    Allowed IPs        : any
    Authentication     : digest ,Realm : DHSM_Authorization
    Allowed user       : nobody
  SSL CONFIGURATION
    Mode               : OFF
    Persona            : default
    Protocol           : default
    Cipher             : default
```

EXAMPLE #5

To allow specific users to manage the HTTP protocol connection for FileMover, type:

```
$ server_http server_2 -modify dhsm -users valid -hosts 10.240.12.146
server_2 : done
```

EXAMPLE #6

To add specific users who can manage the existing HTTP protocol connection for FileMover, type:

```
$ server_http server_2 -append dhsm -users user1,user2,user3
server_2 : done
```

EXAMPLE #7

To add a specific user who can manage the existing HTTP protocol connection for FileMover, type:

```
$ server_http server_2 -append dhsm -users user4 -hosts
172.24.102.20,172.24.102.21
server_2 : done
```

EXAMPLE #8

To remove the specified users and hosts so they can no longer manage the HTTP connection for FileMover, type:

```
$ server_http server_2 -remove dhsm -users user1,user2 -hosts 10.240.12.146
server_2 : done
```

Last Modified: April 12, 2011 12:45 pm

server_ifconfig

Manages the network interface configuration for the specified Data Movers.

SYNOPSIS

```
server_ifconfig { <movername> | ALL }
          -all [ -ip4 | -ip6 ]
        | -delete <if_name>
        | -create -Device <device_name> -name <if_name>
          -protocol { IP <ipv4_addr> <ipmask> <ipbroadcast>
                    | IP6 <ipv6_addr>[/PrefixLength] }
                    [ mtu=<MTUbytes>] [ vlan=<vlanID>]
                    [ down ]
        | <if_name> [ up | down
                    | [ mtu=<MTUbytes>] [vlan=<vlanID>] ]
    | <if_name> [ sync=<ID> lrdfd=<device,local_ctl> rrdfd=<device,remote_ctl> ]
```

DESCRIPTION
----

server_ifconfig creates a network interface, assigns an IP address to a network interface, enables and disables an interface, sets the MTU size and the VLAN ID, and displays network interface parameters for the specified Data Mover.

server_ifconfig is used to define the network address of each interface existing on a machine, to delete and recreate an interface's address and operating parameters.

The ALL option executes the command for all Data Movers.

OPTIONS

-all [ip4|ip6]
Displays parameters for all configured interfaces. The -ip4 option displays all ipv4 only interfaces and the -ip6 option displays all ipv6 only interfaces.

-delete <if_name>
Deletes a network interface configuration. However, the autogenerated link local interfaces cannot be deleted.

-create -Device <device_name> -name <if_name> -protocol IP <ipv4_addr> <ipmask> <ipbroadcast>|IP6 <ipv6_addr> [/PrefixLength]} [mtu=<MTUbytes>] [vlan=<vlanID>] [down]
Creates a network interface configuration on the specified device with the specified name and assigns a protocol to the interface. The <if_name> must not contain a colon (:).

Available protocols are:

IP <ipv4_addr> <ipmask> <ipbroadcast>|IP6 <ipv6_addr> [/PrefixLength]}
IPv4 assigns the IP protocol with the specified IP address, mask, and broadcast address. The IP address is the address of a particular interface. Multiple interfaces are allowed for each device, each identified by a different IP address. The IP mask includes the network part of the local address and the subnet, which is taken from the host field of the address. For example, 255.255.255.0 would be a mask for a Class C network. The IP broadcast is a special destination address that specifies a broadcast message to a network. For example, x.x.x.255 is the broadcast address for a Class C network.

IP6 assigns the IPv6 address and prefix length. When prefix length is not specified, the default value of 64 is used. It also assigns the maximum transmission unit (MTU) size in bytes, the ID for the virtual LAN (VLAN)(valid inputs are 0 (default) to 4094).

When creating the first IPv6 interface with a global unicast address on a broadcast domain, the system automatically creates an associated IPv6 link-local interface. Similarly, when deleting the last remaining IPv6 interface on a broadcast domain, the system automatically deletes the associated IPv6 link-local interface.

The down option can be specified for both IPv4 and IPv6. If specified, the network interface will be set to the down state; otherwise, the network interface is up by default.

For CIFS users, when an interface is created, deleted, or marked up or down, use the server_setup command to stop and then restart the CIFS service in order to update the CIFS interface list.

<if_name> up
Allows the interface to receive and transmit data, but does not enable the physical port. Interfaces are marked up automatically when initially setting up the IP address.

<if_name> down
Stops data from being transmitted through that interface. If possible, the interface is reset to disable reception as well. This does not automatically disable routes using the interface.

<if_name> mtu=<MTUbytes>
Resets the maximum transmission unit (MTU) size in bytes for the specified interface. By default, the MTU is automatically set depending on the type of network interface card installed.

Regardless of whether you have Ethernet or Gigabit Ethernet, the initial default MTU size is 1500 bytes. To take advantage of the capacity of Gigabit Ethernet, the MTU size can be increased up to 9000 bytes if your switch supports jumbo frames. Jumbo frames should be used only when the entire infrastructure, including client NICs, supports them.

For UDP, it is important that both the client and server use the same MTU size. TCP negotiates the MTU size when the connection is initialized. The switch.s MTU must be greater than or equal to the host.s MTU.

Note: The MTU size specified here is for the interface. The MTU size specified in server_netstat applies to the device and is automatically set.

<if_name> vlan=<vlanID>
Sets the ID for the virtual LAN (VLAN). Valid inputs are 0 (default) to 4094. When a VLAN ID other than 0 is set, the interface only accepts packets tagged with that specified ID. Outbound packets are also tagged with the specified ID.

Note: IEEE 802.1Q VLAN tagging is supported. VLAN tagging is not supported on ana interfaces.

<if_name> sync=<ID> lrdfd=<device,local_ctl> rrdfd=<device,remote_ctl>
Resets the VDM Sync Replication session properties for an interface.

For sync, valid inputs are 0 (indicates the interface is not DR enabled anymore) to 65,536. Any other non-zero value indicates the VDM Sync Session is using this interface.

For lrdfd, pass the local device name and one of its SCSI CTL path.

For rrdfd, pass the remote device name and one of its SCSI CTL path.

SEE ALSO

Configuring and Managing Networking on VNX and Configuring and Managing Network High Availability on VNX, server_netstat, server_setup, and server_sysconfig.

FRONT-END OUTPUT

The network device name is dependent on the front end of the system (for example, NS series Data Mover, 514 Data Movers, 510 Data Movers, and so on) and the network device type. NS series and 514 Data Movers network device name display a prefix of cge, for example, cge0. 510 or earlier Data Movers display a prefix of ana or ace, for example, ana0, ace0. Internal network devices on a Data Mover are displayed as el30, el31.

EXAMPLE #1

To display parameters of all interfaces on a Data Mover, type:

```
$ server_ifconfig server_2 -all

server_2 :
loop protocol=IP device=loop
        inet=127.0.0.1 netmask=255.0.0.0 broadcast=127.255.255.255
        UP, loopback, mtu=32768, vlan=0, macaddr=0:0:0:0:0:0 netname=localhost
cge0 protocol=IP device=cge0
        inet=172.24.102.238 netmask=255.255.255.0 broadcast=172.24.102.255
        UP, ethernet, mtu=1500, vlan=0, macaddr=0:60:16:4:29:87
el31 protocol=IP device=cge6
        inet=128.221.253.2 netmask=255.255.255.0 broadcast=128.221.253.255
        UP, ethernet, mtu=1500, vlan=0, macaddr=0:60:16:4:11:a6 netname=localhost
el30 protocol=IP device=fxp0
        inet=128.221.252.2 netmask=255.255.255.0 broadcast=128.221.252.255
        UP, ethernet, mtu=1500, vlan=0, macaddr=8:0:1b:43:7e:b8 netname=localhost
```

EXAMPLE #2

To create an IP interface for Gigabit Ethernet, type:

```
$ server_ifconfig server_2 -create -Device cge1 -name cge1 -protocol IP
172.24.102.239 255.255.255.0 172.24.102.255

server_2 : done
```

EXAMPLE #3

To create an interface for network device cge0 with an IPv6 address with a nondefault prefix length on server_2, type:

```
$ server_ifconfig server_2 -create -Device cge0 -name cge0_int1 -protocol IP6
3ffe:0000:3c4d:0015:0435:0200:0300:ED20/48

server_2 : done
```

EXAMPLE #4

To create an interface for network device cge0 with an IPv6 address on server_2, type:

```
$ server_ifconfig server_2 -create -Device cge0 -name cge0_int1 -protocol IP6
3ffe:0000:3c4d:0015:0435:0200:0300:ED20

server_2 : done
```

EXAMPLE #5

To verify that the settings for the cge0_int1 interface for server_2 are correct, type:

```
$ server_ifconfig server_2 cge0_int1
```

```
server_2 :
cge0_int1 protocol=IP6 device=cge0
inet=3ffe:0:3c4d:15:435:200:300:ed20 prefix=48
UP, Ethernet, mtu=1500, vlan=0, macaddr=0:60:16:c:5:5
```

Note: The bold item in the output highlights the nondefault 48-bit prefix.

EXAMPLE #6

To verify that the interface settings for server_2 are correct, type:

```
$ server_ifconfig server_2 -all

server_2 :
el30 protocol=IP device=mge0
        inet=128.221.252.2 netmask=255.255.255.0 broadcast=128.221.252.255
    UP, Ethernet, mtu=1500, vlan=0, macaddr=0:60:16:d:30:b1 netname=localhost
el31 protocol=IP device=mge1
        inet=128.221.253.2 netmask=255.255.255.0 broadcast=128.221.253.255
    UP, Ethernet, mtu=1500, vlan=0, macaddr=0:60:16:d:30:b2 netname=localhost
loop6 protocol=IP6 device=loop
        inet=::1 prefix=128
    UP, Loopback, mtu=32768, vlan=0, macaddr=0:0:0:0:0:0 netname=localhost
loop protocol=IP device=loop
        inet=127.0.0.1 netmask=255.0.0.0 broadcast=127.255.255.255
    UP, Loopback, mtu=32768, vlan=0, macaddr=0:0:0:0:0:0 netname=localhost
cge0_int1 protocol=IP6 device=cge0
        inet=3ffe:0:3c4d:15:435:200:300:ed20 prefix=64
    UP, Ethernet, mtu=1500, vlan=0, macaddr=0:60:16:c:2:5
cge0_0000_ll protocol=IP6 device=cge0
        inet=fe80::260:16ff:fe0c:205 prefix=64
    UP, Ethernet, mtu=1500, vlan=0, macaddr=0:60:16:c:2:5
```

Note: The first bold item in the output highlights the default 64-bit prefix. The second and third bold items highlight the link-local name and address that are automatically generated when you configure a global address for cge0. The automatically created link local interface name is made by concatinating the device name with the four digit VLAN ID between 0 and 4094. Note that the interface you configured with the IPv6 address 3ffe:0:3c4d:15:435:200:300:ed20 and the address with the link-local address fe80::260:16ff:fe0c:205 share the same MAC address. The link-local address is derived from the MAC address.

EXAMPLE #7

To verify that the interface settings for server_2 are correct, type:

```
$ server_ifconfig server_2 -all

server_2 :
cge0_int2 protocol=IP device=cge0
        inet=172.24.108.10 netmask=255.255.255.0 broadcast=172.24.108.255
        UP, Ethernet, mtu=1500, vlan=0, macaddr=0:60:16:c:2:5
cge0_int1 protocol=IP6 device=cge0
        inet=3ffe:0:3c4d:15:435:200:300:ed20 prefix=64
        UP, Ethernet, mtu=1500, vlan=0, macaddr=0:60:16:c:2:5
cge0_0000_ll protocol=IP6 device=cge0
        inet=fe80::260:16ff:fe0c:205 prefix=64
        UP, Ethernet, mtu=1500, vlan=0, macaddr=0:60:16:c:2:5
el30 protocol=IP device=mge0
        inet=128.221.252.2 netmask=255.255.255.0 broadcast=128.221.252.255
        UP, Ethernet, mtu=1500, vlan=0, macaddr=0:60:16:d:30:b1 netname=localhost
el31 protocol=IP device=mge1
        inet=128.221.253.2 netmask=255.255.255.0 broadcast=128.221.253.255
        UP, Ethernet, mtu=1500, vlan=0, macaddr=0:60:16:d:30:b2 netname=localhost
loop6 protocol=IP6 device=loop
        inet=::1 prefix=128
        UP, Loopback, mtu=32768, vlan=0, macaddr=0:0:0:0:0:0 netname=localhost
```

```
loop protocol=IP device=loop
        inet=127.0.0.1 netmask=255.0.0.0 broadcast=127.255.255.255
        UP, Loopback, mtu=32768, vlan=0, macaddr=0:0:0:0:0:0 netname=localhost
```

Note: The bold items in the output highlight the IPv4 interface, cge0_int2, and the IPv6 interface, cge0_int1.

EXAMPLE #8

To disable an interface, type:

```
$ server_ifconfig server_2 cge0_int2 down

server_2 : done
```

EXAMPLE #9

To enable an interface, type:

```
$ server_ifconfig server_2 cge0_int2 up

server_2 : done
```

EXAMPLE #10

To reset the MTU for Gigabit Ethernet, type:

```
$ server_ifconfig server_2 cge0_int2 mtu=9000

server_2 : done
```

EXAMPLE #11

To set the ID for the Virtual LAN, type:

```
$ server_ifconfig server_2 cge0_int1 vlan=40

server_2 : done
```

EXAMPLE #12

To verify that the VLAN ID in the interface settings for server_2 are correct, type:

```
$ server_ifconfig server_2 -all

server_2 :
cge0_int1 protocol=IP6 device=cge0
        inet=3ffe:0:3c4d:15:435:200:300:ed20 prefix=64
        UP, Ethernet, mtu=1500, vlan=40, macaddr=0:60:16:c:2:5
cge0_0040_ll protocol=IP6 device=cge0
        inet=fe80::260:16ff:fe0c:205 prefix=64
        UP, Ethernet, mtu=1500, vlan=40, macaddr=0:60:16:c:2:5
cge0_int2 protocol=IP device=cge0
        inet=172.24.108.10 netmask=255.255.255.0 broadcast=172.24.108.255
        UP, Ethernet, mtu=1500, vlan=20, macaddr=0:60:16:c:2:5
el30 protocol=IP device=mge0
        inet=128.221.252.2 netmask=255.255.255.0 broadcast=128.221.252.255
        UP, Ethernet, mtu=1500, vlan=0, macaddr=0:60:16:d:30:b1 netname=localhost
el31 protocol=IP device=mge1
        inet=128.221.253.2 netmask=255.255.255.0 broadcast=128.221.253.255
        UP, Ethernet, mtu=1500, vlan=0, macaddr=0:60:16:d:30:b2 netname=localhost
loop6 protocol=IP6 device=loop
        inet=::1 prefix=128
        UP, Loopback, mtu=32768, vlan=0, macaddr=0:0:0:0:0:0 netname=localhost
loop protocol=IP device=loop
```

```
        inet=127.0.0.1 netmask=255.0.0.0 broadcast=127.255.255.255
        UP, Loopback, mtu=32768, vlan=0, macaddr=0:0:0:0:0:0 netname=localhost
```

Note: The bold items in the output highlight the VLAN tag.
Note that the link-local address uses the VLAN tag as part of its name.

EXAMPLE #13

To delete an IP interface, type:

```
$ server_ifconfig server_2 -delete cge1_int2

server_2 : done
```

Note: The autogenerated link local interfaces cannot be deleted.

Last modified: May 12, 2011 1:40 pm.

server_ip

Manages the IPv6 neighbor cache and route table for VNX.

SYNOPSIS

```
server_ip {ALL|<mover>}
  -neighbor {
       -list [<v6addr> [.interface <ifname>]]
     | -create <v6addr> -lladdress <macaddr> [-interface <ifname>]
     | -delete {-all|<v6addr> [-interface <ifname>]}
    }
  | -route {
       -list
     | -create {
          -destination <destination> -interface <ifname>
     | -default -gateway <v6gw> [-interface <ifname>]
       }
      | -delete {
           -destination <destination>
       | -default -gateway <v6gw> [-interface <ifname>]
       | -all
      }
    }
```

DESCRIPTION
----

server_ip creates, deletes, and lists the neighbor cache and route tables.

OPTIONS
----

server_ip {<movername>|ALL}
Sends a request to the Data Mover to get IPv6 parameters related to the IPv6 routing table and neighbor cache.

ALL option executes the command for all of the Data Movers.

-neighbor {-list|-create|-delete}
Lists, creates, or deletes the neighbor cache entries from the neighbor cache table.

-list
Displays the neighbor cache entries.

-create
Creates a neighbor cache table entry with the specified details.

-delete
Deletes the specified neighbor cache table entries or all entries.

-route {-list|-create|-delete}
Lists, creates, or deletes entries in the IPv6 route table.

-list
Displays the IPv6 route table.

-create
Creates a route table entry with the specified details.

-delete
Deletes the specified route table entries.

EXAMPLE #1

To view a list of neighbor cache entries on the Data Mover server_2, type:

```
$ server_ip server_2 -neighbor -list
```

```
server_2:
Address                   Link layer address    Interface    Type    State
fe80::204:23ff:fead:4fd4    0:4:23:ad:4f:d4  cge1_0000_ll   host    STALE
fe80::216:9cff:fe15:c00     0:16:9c:15:c:0   cge1_0000_ll   router  STALE
fe80::216:9cff:fe15:c00     0:16:9c:15:c:0   cge4_0000_ll   router  STALE
fe80::216:9cff:fe15:c00     0:16:9c:15:c:0   cge3_2998_ll   router  STALE
fe80::216:9cff:fe15:c00     0:16:9c:15:c:0   cge2_2442_ll   router  STALE
3ffe::1                     0:16:9c:15:c:10  cge3_0000_ll   router  REACHABLE

Where:
Value                   Definition
Address                 The neighbor IPv6 address.
Link layer address      The link layer address of the neighbor.
Interface               Interface name of the interface connecting to the
                        neighbor.
Type                    Type of neighbor. The neighbor can be either host
                        or router.
State                   The state of the neighbor such as REACHABLE,
                        INCOMPLETE, STALE, DELAY, or PROBE.
```

EXAMPLE #2

To view a list of neighbor cache entries for a specific IP address on the Data Mover server_2, type:

```
$ server_ip server_2 -neighbor -list fe80::216:9cff:fe15:c00
server_2:
Address                     Link layer address    Interface     Type    State
fe80::216:9cff:fe15:c00     0:16:9c:15:c:0       cge1_0000_ll   router  STALE
fe80::216:9cff:fe15:c00     0:16:9c:15:c:0       cge4_0000_ll   router  STALE
fe80::216:9cff:fe15:c00     0:16:9c:15:c:0       cge3_2998_ll router    STALE
fe80::216:9cff:fe15:c00     0:16:9c:15:c:0       cge2_2442_ll router    STALE
```

EXAMPLE #3

To view a list of neighbor cache entries for a specific IP address and interface type, on the Data Mover server_2, type:

```
$ server_ip server_2 -neighbor -list fe80::216:9cff:fe15:c00 -interface
cge1_0000_ll
server_2:
Address                   Link layer address    Interface   Type  State
fe80::216:9cff:fe15:c00     0:16:9c:15:c:0 cge1_0000_ll router  STALE
```

EXAMPLE #4

To add an entry to the neighbor cache for a global unicast IPv6 address, on the Data Mover server_2, type:

```
$ server_ip server_2 -neighbor -create 2002:8c8:0:2310::2 -lladdress
0:16:9c:15:c:15
OK
```

EXAMPLE #5

To add an entry to the neighbor cache for a link local IPv6 address, on the Data Mover server_2, type:

```
$ server_ip server_2 -neighbor -create fe80::2 -lladdress 0:16:9c:15:c:12
-interface cge1v6
OK
```

EXAMPLE #6

To delete an entry from the neighbor cache for a global unicast IPv6 address, on the Data Mover server_2, type:

```
$ server_ip server_2 -neighbor -delete 2002:8c8:0:2310:0:2:ac18:f401
OK
```

EXAMPLE #7

To delete an entry from the neighbor cache for a link local IPv6 address, on all the Data Movers, type:

```
$ server_ip ALL -neighbor -delete fe80::1 -interface cge1v6
OK
```

EXAMPLE #8

To delete entries from the neighbor cache on the Data Mover server_2 type:

```
$ server_ip server_2 -neighbor -delete -all
OK
```

EXAMPLE #9

To view a list of route table entries on the Data Mover server_2, type:

```
$ server_ip server_2 -route -list
server_2:
Destination                          Gateway    Interface Expires (secs)
2002:8c8:0:2310::/64                             cge1v6             0
2002:8c8:0:2311::/64                             cge1v6             0
2002:8c8:0:2312::/64                             cge1v6             0
2002:8c8:0:2313::/64                             cge1v6             0
default              fe80::260:16ff:fe05:1bdd cge1_0000_ll          1785
default              fe80::260:16ff:fe05:1bdc cge1_0000_ll          1785
default                    2002:8c8:0:2314::1       cge4v6             0
selected default     fe80::260:16ff:fe05:1bdd cge1_0000_ll          1785
```

Where:

Value	Definition
Destination	The prefix of the destination or the default route entry. There can be multiple default routes, but only one is active and shown as selected default. The default sorting of the destination column displays the default routes at the bottom of the list and the selected default at the end of the list.
Gateway	The default gateway for default route entries. This value is blank for prefix destination entries.
Interface	Interface name of the interface used for the route.
Expires	The time until the route entry is valid. Zero denotes route is permanent and does not have an expiry.

EXAMPLE #10

To add a default route table entry on the Data Mover server_2 to the destination network with the specified prefix, type:

```
$ server_ip server_2 -route -create -destination 2002:8c8:0:2314::/64
-interface cge4v6
OK
```

EXAMPLE #11

To add a default route table entry on the Data Mover server_2 through the specified gateway, type:

```
$ server_ip server_2 -route -create -default -gateway 2002:8c8:0:2314::1
OK
```

EXAMPLE #12

To add a default route table entry on the Data Mover server_2 through the specified gateway using the link-local interface, type:

```
$ server_ip server_2 -route -create -default -gateway fe80::1 -interface cge1v6
OK
```

EXAMPLE #13

To delete an entry from the route table with an IPv6 prefix route destination for all the Data Movers, type:

```
$ server_ip ALL -route -delete -destination 2002:8c8:0:2314::/64
OK
```

EXAMPLE #14

To delete an entry from the route table for a global unicast IPv6 address, on the Data Mover server_2, type:

```
$ server_ip server_2 -route -delete -default -gateway 2002:8c8:0:2314::1
OK
```

EXAMPLE #15

To delete an entry from the route table for a link local IPv6 address, on the Data Mover server_2, type:

```
$ server_ip server_2 -route -delete -default -gateway fe80::1 -interface cge1v6
OK
```

EXAMPLE #16

To delete all entries from the IPv6 route table on the Data Mover server_2 type:

```
$ server_ip server_2 -route -delete -all
OK
```

Last modified: April 12, 2011 1:30 pm

server_kerberos

Manages the Kerberos configuration within the specified Data Movers.

SYNOPSIS

```
server_kerberos {<movername>|ALL}
 -add realm=<realm_name>,kdc=<fqdn_kdc_name>[:<port>]
  [,kdc=<fqdn_kdc_name>[:<port>]...]
  [,kpasswd=<fqdn_kpasswd_server_name>]
  [,kadmin=<kadmin_server>]
  [,domain=<domain_name>][,defaultrealm]
| -delete realm=<realm_name>
| -keytab
| -ccache [-flush]
| -list
| -kadmin [<kadmin_options>]
```

DESCRIPTION
----

server_kerberos adds, deletes, lists the realms within the Kerberos configuration of a Data Mover, and manages the Data Movers service principals and keys.

server_kerberos displays the key table content, and specifies a kadmin server.

OPTIONS

-add realm=<realm_name>,kdc=<fqdn_kdc_name>
Adds the specified realm to the Kerberos configuration on the specified Data Mover. The <realm_name> is the fully qualified domain name of the Kerberos realm to be added to the key distribution center (KDC) configuration. The <fqdn_kdc_name> is the fully qualified domain name of the KDC for the specified realm.

Note: The -add option is only relevant if you are using a UNIX/Linux Kerberos KDC.

[:<port>]
Specifies a port that the KDC listens on.

[,kdc=<fqdn_kdc_name[:<port>]...]
Specifies additional KDCs with ports that KDCs listen on.

[,kpasswd=<fqdn_kpasswd_server_name>]
Specifies a password server for the KDC. The <fqdn_kpasswd_server_name> must be a fully qualified domain name for the server.

[,kadmin=<kadmin_server>]
Specifies the kadmin server.

[,domain=<domain_name>]
The <domain_name> is the full name of the DNS domain for the realm.

[,defaultrealm]
Indicates that the default realm is to be used.

-delete realm=<realm_name>
Deletes the specified realm from the Kerberos configuration for the specified Data Mover.

Note: The -delete option is only relevant if you are using a UNIX/Linux Kerberos KDC.

-keytab
Displays the principal names for the keys stored in the keytab file.

-ccache
Displays the entries in the Data Movers Kerberos credential cache.

Note: The -ccache option can also be used to provide EMC Customer Support with information for troubleshooting user access problems.

[-flush]
Flushes the Kerberos credential cache removing all entries. Credential cache entries are automatically flushed when they expire or during a Data Mover reboot.

Once the cache is flushed, Kerberos obtains new credentials when needed. The repopulation of credentials may take place immediately, over several hours, or be put off indefinitely if no Kerberos activity occurs.

-list
Displays a listing of all configured realms on a specified Data Mover or on all Data Movers.

-kadmin [<kadmin_options>]
Invokes the kadmin tool with the following specified options:

[-r <realm>]
Specifies a realm as the default database realm.

[-p <principal>]
Specifies the principal for authentication. Otherwise, kadmin will append "/admin" to the primary principal name of the default cache, the value of the USER environment variable, or the username as obtained with getpwuid, in order of preference.

[-q <query>]
Runs kadmin in non-interactive mode. This passes the query directly to kadmin, which performs the query, then exits.

[-w <password>]
Uses a specified password instead of prompting for a password.

[-s <admin_server> [:<port>]]
Specifies the kadmin server with its associated port.

Note: The kadmin tool is only relevant if you are using a UNIX/Linux Kerberos KDC. You must be root to execute the -kadmin option.

SEE ALSO

Configuring NFS on VNX, server_checkup, and server_nfs.

OUTPUT

Dates appearing in output are in UTC format.

EXAMPLE #1

To add a realm to the Kerberos configuration of a Data Mover, type:

```
$ server_kerberos server_2 -add
realm=nasdocs.emc.com,kdc=winserver1.nasdocs.emc.com,domain=nasdocs.emc.com
server_2 : done
```

EXAMPLE #2

```
----------

To list the keytabs, type:

$ server_kerberos server_2 -keytab
server_2 :
Dumping keytab file

keytab file major version = 0, minor version 0

-- Entry number 1 --
principal: DM102-CGE0$@NASDOCS.EMC.COM
realm: NASDOCS.EMC.COM
encryption type: rc4-hmac-md5
principal type 1, key version: 332
key length: 16, key: b1c199a6ac11cd529df172e270326d5e
key flags:(0x0), Dynamic Key, Not Cached
key cache hits: 0

-- Entry number 2 --
principal: DM102-CGE0$@NASDOCS.EMC.COM
realm: NASDOCS.EMC.COM
encryption type: des-cbc-md5
principal type 1, key version: 332
key length: 8, key: ced9a23183619267
key flags:(0x0), Dynamic Key, Not Cached
key cache hits: 0

-- Entry number 3 --
principal: DM102-CGE0$@NASDOCS.EMC.COM
realm: NASDOCS.EMC.COM
encryption type: des-cbc-crc
principal type 1, key version: 332
key length: 8, key: ced9a23183619267
key flags:(0x0), Dynamic Key, Not Cached
key cache hits: 0

-- Entry number 4 --
principal: host/dm102-cge0@NASDOCS.EMC.COM
realm: NASDOCS.EMC.COM
encryption type: rc4-hmac-md5
principal type 1, key version: 332
key length: 16, key: b1c199a6ac11cd529df172e270326d5e
key flags:(0x0), Dynamic Key, Not Cached
key cache hits: 0

<... removed ...>

-- Entry number 30 --
principal: cifs/dm102-cge0.nasdocs.emc.com@NASDOCS.EMC.COM
realm: NASDOCS.EMC.COM
encryption type: des-cbc-crc
principal type 1, key version: 333
key length: 8, key: d95e1940b910ec61
key flags:(0x0), Dynamic Key, Not Cached
key cache hits: 0

End of keytab entries. 30 entries found.

This is a partial listing due to the length of the output.

Where:
Value            Definition
principal type   Type of the principal as defined in the GSS-API. Reference
                 to RFC 2743.
key version      Every time a key is regenerated its version changes.

EXAMPLE #3
----------

To list all of the realms on a Data Mover, type:
```

```
$ server_kerberos server_2 -list
server_2 :

Kerberos common attributes section:
Supported TGS encryption types: rc4-hmac-md5 des-cbc-md5 des-cbc-crc
Supported TKT encryption types: rc4-hmac-md5 des-cbc-md5 des-cbc-crc
Use DNS locator: yes

End of Kerberos common attributes.

Kerberos realm configuration:

realm name: NASDOCS.EMC.COM
kdc: winserver1.nasdocs.emc.com
admin server: winserver1.nasdocs.emc.com
kpasswd server: winserver1.nasdocs.emc.com
default domain: nasdocs.emc.com

End of Kerberos realm configuration.

Kerberos domain_realm section:
DNS domain = Kerberos realm
.nasdocs.emc.com = NASDOCS.EMC.COM

End of Krb5.conf domain_realm section.
```

EXAMPLE #4

```
# server_kerberos server_2 -add
realm=eng.nasdocs.emc.com,kdc=winserver1.nasdocs.emc.com,kadmin=172.24.102.67

server_2 : done
```

Note: You must be root to execute the -kadmin option. Replace $ with # as the root login is a requirement.

EXAMPLE #5

```
$ server_kerberos server_2 -delete realm=eng.nasdocs.emc.com
server_2 : done
```

EXAMPLE #6

```
$ server_kerberos server_2 -ccache
server_2 :
Dumping credential cache

Names:
Client: DM102-CGE0$@NASDOCS.EMC.COM
Service: WINSERVER1.NASDOCS.EMC.COM
Target: HOST/WINSERVER1.NASDOCS.EMC.COM@NASDOCS.EMC.COM
Times:
Auth: 09/12/2005 07:15:04 GMT
Start: 09/12/2005 07:15:04 GMT
End: 09/12/2005 17:15:04 GMT
Flags: PRE_AUTH,OK_AS_DELEGATE
Encryption Types:
Key: rc4-hmac-md5
Ticket: rc4-hmac-md5

Names:
```

```
Client: DM102-CGE0$@NASDOCS.EMC.COM
Service: winserver1.nasdocs.emc.com
Target: ldap/winserver1.nasdocs.emc.com@NASDOCS.EMC.COM
Times:
Auth: 09/12/2005 07:15:04 GMT
Start: 09/12/2005 07:15:04 GMT
End: 09/12/2005 17:15:04 GMT
Flags: PRE_AUTH,OK_AS_DELEGATE
Encryption Types:
Key: rc4-hmac-md5
Ticket: rc4-hmac-md5

Names:
Client: DM102-CGE0$@NASDOCS.EMC.COM
Service: NASDOCS.EMC.COM
Target: krbtgt/NASDOCS.EMC.COM@NASDOCS.EMC.COM
Times:
Auth: 09/12/2005 07:15:04 GMT
Start: 09/12/2005 07:15:04 GMT
End: 09/12/2005 17:15:04 GMT
Flags: INITIAL,PRE_AUTH
Encryption Types:
Key: rc4-hmac-md5
Ticket: rc4-hmac-md5

End of credential cache entries.

Where:
Value     Definition
client    Client name and its realm.
service   Domain controller and its realm.
target    Target name and its realm.
auth      Time of the initial authentication for the named principal.
start     Time after which the ticket is valid.
end       Time after which the ticket will not be honored (its expiration
          time).
flags     Options used or requested when the ticket was issued.
key       Key encryption type.
ticket    Ticket encryption type.

EXAMPLE #7
----------

To flush the credential cache on a Data Mover, type:

$ server_kerberos server_2 -ccache flush
server_2 :
Purging credential cache.
Credential cache flushed.

---------------------------------------
Last Modified: April 13, 2011 11:35 am
```

server_ldap

Manages the LDAP-based directory client configuration and LDAP over SSL for the specified Data Movers.

SYNOPSIS

```
server_ldap {<movername>|ALL}
 {-set|-add} [-p] {-domain <FQDN>|-basedn
<attribute_name>=<attribute_value>[,...]}
     [-servers {<IPv4_addr>[:<port>]|<IPv6_addr>|<\[IPv6_addr\]:port>}[,...]]
     [-profile <profile_name>]|{-file <file_name>}
     [-nisdomain <NIS_domain>]
     [-binddn <bind_DN>|{-kerberos -kaccount <account_name> [-realm
      <realm_name>]}]
     [-sslenabled {y|n}]
     [-sslpersona {none|<persona_name>}]
     [-sslcipher {default|<cipher_list>}]
| -clear [-all|-domain <FQDN>|-basedn
<attribute_name>=<attribute_value>[,...]]
| -info [-all | -domain <FQDN> | -basedn
<attribute_name>=<attribute_value>[,...]][-verbose]
| -service {-start|-stop|-status}
| -lookup [-domain <FQDN> | -basedn
<attribute_name>=<attribute_value>[,...]]{-user <username>
           | -group <groupname>
           | -uid <uid>
           | -gid <gid>
           | -hostbyname <hostname>
           | -netgroup <groupname>}
```

DESCRIPTION
----

server_ldap configures, starts, stops, deletes, and displays the status of the LDAP-based directory client configuration, and queries the LDAP-based directory server.

OPTIONS

{-set|-add} [-p] {-domain <FQDN>|-basedn <attribute_name>=<attribute_value>[,...]
}

Specifies the LDAP-based directory client domain for the specified Data Mover and starts the service.The -add and -set options can be used to configure one initial LDAP-based directory client domain for the specified Data Mover and start the service. The -add option supersedes the ï¿½set option as the preferred method to configure one initial LDAP-based directory client domain for the specified Data Mover. The -add option must be used to add domains and extend the configuration if multiple domains are required. Domains must be configured or added one at a time. The -p option requests a prompt for the password. A password is required in conjunction with a bind distinguished name in order to specify the use of simple authentication. The -basedn option specifies the Distinguished Name (DN) of the directory base, an x509 formatted name that uniquely identifies the directory base. For example: ou=abc,o=def,c=ghi. If a base distinguished name contains space characters, enclose the entire string within double quotation marks and enclose the name with a backslash and double quotation mark. For example, "\"cn=abc,cn=def ghi,dc=com\"".

It is recommended configuring an LDAP-based directory client by using the -basedn option instead of the -domain option. The DN provides the root position for:

* Searching for iPlanet profiles
* Defining default search containers for users, groups, hosts, and netgroups according to RFC 2307. An iPlanet profile and

OpenLDAP or Active Directory with SFU or IdMU ldap.conf file are only required for customized setups.

Note: In the case in which the DN of the directory base contains dots and the client is configured using the domain name, the default containers may not be set up correctly. For example, if the name is dc=my.company,dc=com and it is specified as domain name my.company.com, VNX incorrectly defines the default containers as dc=my,dc=company,dc=com.

[-servers {<IPv4_addr>[:<port>]|<IPv6_addr>| <\[IPv6_addr\]:port>}[,...]]

Specifies the IP addresses of the LDAP-based directory client servers. <IPv4_addr> or <IPv6_addr> indicates the IP address of the LDAP-based directory servers. IPv6 addresses need to be enclosed in square brackets if a port is specified; the brackets do not signify optional content. The <port> option specifies the LDAP-based directory server TCP port number. If the port is not specified, the default port is 389 for LDAP and 636 for SSL-based LDAP. It is recommended that at least two LDAP servers are defined, so that DART can switch to the second server in case the first cannot be reached.

Note: IP addresses of the LDAP-based directory servers do not have to be included every time with the server_ldap command once you have indicated the configuration server, and if configuring the same LDAP-based directory service.

[-profile <profile>]
Specifies the profile name or the profile distinguished name which provides the iPlanet client with configuration information about the directory service. For example, both the following values are allowed: -profile vnx_profile and -profile cn=vnx_profile,ou=admin,dc=mycompany,dc=com.

Note: It is recommended that unique profile names be used in the Directory Information Tree (DIT). The specified profile is searched for by scanning the entire tree and if it is present in multiple locations, the first available profile is used unless the profile distinguished name is specified.

{-file <file_name>}
Allows to specify a LDAP configuration file per domain:

* The various LDAP domains may have different schemas (OpenLDAP, IdMU, and so on) or different customizations (non-standard containers).
* All LDAP domains can share the same /.etc/ldap.conf setup file or even no file if all the domains comply with the RFC2307.
* The configuration files must be put in /.etc using server_file. In order to prevent collisions with other system files, it is required that the LDAP configuration is prefixed by "ldap" and suffixed by ".conf", i.e. "ldap<anything>.conf".
* The default value of the -file option is "ldap.conf".
* server_ldap -service -status lists all the configured domains, and their configuration source (default, file or profile). Several LDAP domains can be configured using the same LDAP configuration file.

[-nisdomain <NIS_domain>]
Specifies the NIS domain of which the Data Mover is a member since an LDAP-based directory domain can host more than one NIS domain.

[-binddn <bind_DN>|{-kerberos -kaccount <account_name> [-realm <realm_name>]}]
Specifies the distinguished name (DN) or Kerberos account of the identity used to bind to the service. Active Directory with SFU or IdMU requires an authentication method that uses simple authentication, SSL, or Kerberos.

Simple authentication requires that a DN be specified along with

a password. For SSL-based client authentication to succeed, the Data Mover certificate Subject must match the distinguished name for an existing user (account) at the directory server.

Note: To configure a LDAP-based directory service for authentication, -binddn is not required if the -sslpersona option is specified. In this case
,
SSL-based client authentication will be used.

The Kerberos account name must be the CIFS server computer name known by the KDC. The account name must terminate with a $ symbol.

By default, the Data Mover assumes that the realm is the same as the LDAP domain provided in the -domain or -basedn options. But a different realm name can be specified, if necessary.

[-sslenabled {y|n}]
Enables (y) or disables (n) SSL. SSL is disabled by default.

[-sslpersona {none|<persona_name>}]
Specifies the key and certificate of the directory server. If a persona has been previously configured, none disables the user of a client key and certificate. The -sslpersona option without the -binddn option indicates that the user wants to authenticate using the client (persona) certificate. To authenticate using the client certificate, the LDAP server must be configured to always request (or require) the persona certificate during the SSL transaction, or the authentication will fail. If authentication using the client certificate is not desired, then the -binddn option must be used. The configuration rules are explained in the table below.

Note: The -sslpersona option does not automatically enable SSL, but configures the specified value. The value remains persistent and is used whenever SSL is enabled.

Configuration rules

Description	Data Mover Configuration
SSL enabled on Data Mover, LDAP server accept SSL, anonymous authentication is used.	server_ldap -sslenabled y
SSL enabled, password-based authentication, is used.	server_ldap -p -binddn cn=foo -sslenabled y
SSL enabled, SSL certificate authentication is used, LDAP server should be configured to request client certificate.	server_ldap -sslenabled y -sslpersona default (use server_certificate to verify that the certificate for the Data Mover's default persona exists)

Note: The user should refer to the LDAP server documentation for information about configuring the server to request the client certificate.

[-sslcipher {default|<cipher_list>}]
Specifies default or the cipher list.

Note: The -sslcipher option does not automatically enable SSL, but configures specified value. The value remains persistent and is used whenever SSL is enabled.

-clear
Deletes the LDAP-based directory client configuration for the specified Data Mover and stops the service.

-info
Displays the service status as well as the static and dynamic configuration.

[-verbose]
Adds troubleshooting information to the output.

-service {-start|-stop|-status}
The -start option enables the LDAP-based directory client service. The LDAP-based directory client service is also restarted when the VNX is rebooted. The -stop option disables the LDAP-based directory client service, and the -status option displays the status of the LDAP-based directory service.

-lookup {user=<username>|group=<groupname>|uid=<uid>|gid=<gid>|hostbyname=<hostname>|netgroup=<groupname>}
Provides lookup information about the specified resource for troubleshooting purposes.

Note: The server_ldap requires the user to specify the domain name when more than one domain is configured for the -clear, -info, and -lookup options. Other options are not changed and they are applicable for each different domain.

SEE ALSO

Configuring VNX Naming Services.

EXAMPLE #1

To configure the use of an LDAP-based directory by a Data Mover, type:

```
$ server_ldap server_4 -set -domain nasdocs.emc.com -servers 172.24.102.62
server_4 : done
```

EXAMPLE #2

To configure the use of an LDAP-based directory by a Data Mover using the Distinguished Name of the server at IPv4 address 172.24.102.62 with the default port, type:

```
$ server_ldap server_2 -set -basedn dc=nasdocs,dc=emc,dc=com -servers
172.24.102.62
server_2 : done
```

EXAMPLE #3

To configure the use of an LDAP-based directory by a Data Mover using the Distinguished Name of the server at IPv6 address 2002:c8c::24:172:63 with the default port, type:

```
$ server_ldap server_2 -set -basedn dc=nasdocs,dc=emc,dc=com -servers
2002:c8c::24:172:63
server_2 : done
```

EXAMPLE #4

To configure the use of an LDAP-based directory by a Data Mover and specify the use of the client profile, type:

```
$ server_ldap server_4 -set -domain nasdocs.emc.com -servers 172.24.102.62
-profile celerra_profile -nisdomain nasdocs -sslenabled y
server_4 : done
```

EXAMPLE #5

To configure the use of an LDAP-based directory by a Data Mover and specify the use of the client profile using its distinguished name, type:

```
$ server_ldap server_4 -set -domain nasdocs.emc.com -servers 172.24.102.62
-profile cn=celerra_profile,dc=nasdocs,dc=emc,dc=com -nisdomain nasdocs
-sslenabled y
server_4 : done
```

EXAMPLE #6

To specify the NIS domain to which the Data Mover is a member, type:

```
$ server_ldap server_2 -set -domain nasdocs.emc.com -servers 172.24.102.62
-nisdomain nasdocs
server_2 : done
```

EXAMPLE #7

To configure the use of simple authentication by specifying a bind Distinguished Name (DN) and password, type:

```
$ server_ldap server_2 -set -p -domain nasdocs.emc.com -servers 172.24.102.10
-binddn "cn=admin,cn=users,dc=nasdocs,dc=emc"
server_2 : Enter Password:********
done
```

EXAMPLE #8

To configure the use of an LDAP-based directory by a Data Mover using SSL, type:

```
$ server_ldap server_4 -set -basedn dc=nasdocs,dc=emc,dc=com -servers
172.24.102.62 -sslenabled y
server_4 : done
```

EXAMPLE #9

To configure the use of an LDAP-based directory by a Data Mover using SSL and user key and certificate, type:

```
$ server_ldap server_4 -set -basedn dc=nasdocs,dc=emc,dc=com -servers
172.24.102.62 -sslenabled y -sslpersona default
server_4 : done
```

EXAMPLE #10

To configure the use of an LDAP-based directory by a Data Mover using SSL and using specified ciphers, type:

```
$ server_ldap server_4 -set -basedn dc=nasdocs,dc=emc,dc=com -servers
172.24.102.62 -sslenabled y -sslcipher "RC4-MD5,RC4-SHA"
server_4 : done
```

EXAMPLE #11

To display informaton about the LDAP-based directory configuration on a Data Mover, type:

```
$ server_ldap server_4 -info

server_4 :
LDAP domain: nasdocs.emc.com
     base DN: dc=nasdocs,dc=emc,dc=com
     State: Configured - Connected
NIS domain: nasdocs.emc.com
```

```
     No client profile nor config. file provided (using default setup)
Connected to LDAP server address: 172.24.102.62 - port 636
SSL enabled/disabled by Command line, cipher suites configured by Command line
```

EXAMPLE #12

To configure the use of Kerberos authentication by specifying a Kerberos account, type:

```
$ server_ldap server_2 -set -basedn dc=nasdocs,dc=emc,dc=com -servers
172.24.102.62 -kerberos -kaccount cifs_compname$
server_2 : done
```

EXAMPLE #13

To display detailed informaton about the LDAP-based directory configuration on a Data Mover, type:

```
$ server_ldap server_2 -info -verbose
server_2 :
LDAP domain: devldapdom1.lcsc
State: Configured - Connected
Schema: OpenLDAP
Base dn: dc=devldapdom1,dc=lcsc
Bind dn: <anonymous>
Configuration: RFC-2307 defaults
Global warnings & errors
{
The LDAP cache is disabled.
}
LDAP server: 192.168.67.11 - Port: 389 - Active
SSL: Not enabled
Naming ctx: (baseDn is ticked)
[x] dc=devldapdom1,dc=lcsc
Containers: (no [scope] means ignored, unless parent container with sub scope
is valid)
Passwd: Class: posixAccount - Attributes: uid, uidNumber, gidNumber,
userPassword, homeDirectory
[one] ou=People,dc=devldapdom1,dc=lcsc - prefix=uid
Group: Class: posixGroup - Attributes: gidNumber, memberUid
memberUid syntax is DN (Windows)
[one] ou=Group,dc=devldapdom1,dc=lcsc - prefix=cn
Hosts: Class: ipHost - Attributes: ipHostNumber
[one] ou=Hosts,dc=devldapdom1,dc=lcsc - prefix=cn
Netgroup: Class: nisNetgroup - Attributes: nisNetgroupTriple,
memberNisNetgroup
[one] ou=netgroup,dc=devldapdom1,dc=lcsc - prefix=cn
LDAP server: 10.64.220.148 - Port: 389 - Spare
SSL: Not enabled
```

EXAMPLE #14

To display lookup information about the user nasadmin, type:

```
$ server_ldap server_4 -lookup -user nasadmin
server_4:
user: nasadmin, uid: 1, gid: 201, gecos: nasadmin, home dir: /home/nasadmin,
shell: /bin/csh
```

EXAMPLE #15

To display the status of the LDAP-based directory service, type:

```
$ server_ldap server_2 -service -status
server_2 :
LDAP domain "devldapdom1.lcsc" is active - Configured with RFC-2307 defaults
```

EXAMPLE #16

To stop the LDAP-based directory service, type:

```
$ server_ldap server_4 -service -stop
server_4 : done
```

EXAMPLE #17

To delete the LDAP configuration for the specified Data Mover and stop the service, type:

```
$ server_ldap server_4 -clear
server_4 : done
```

EXAMPLE #18

To check if any ldap domain is configured, type:

```
server_ldap server_3 -service -status
server_3 :
LDAP domain is not configured yet.
```

EXAMPLE #19

To configure a domain for openLdap with standard schema, type:

```
server_ldap server_3 -set -domain devldapdom1.lcsc
-servers 192.168.67.114, 192.168.67.148
server_3 : done
```

Note: Since this is the first domain, you can use -set or -add option.

EXAMPLE #20

To configure a domain for Fedora Directory Service (same as openLdap), type:

```
server_ldap server_3 -add -p -basedn dc=389-ds,dc=lcsc
-servers 192.168.67.10.64.223.182 -binddn
"\"cn=Directory Manager\""
server_3 : Enter Password:********
done
```

Note: Since a domain is already set up, you must use -add option.

EXAMPLE #21

To configure a domain for iPlanet using specific configuration profile, type:

```
server_ldap server_3 -add -domain dvt.emc -servers
192.168.67.140 -profile profilecad3
server_3 : done
```

EXAMPLE #22

To configure a domain for IDMU using specific configuration file, type:

```
server_ldap server_3 -add -p -basedn dc=eng,dc=lcsc
-servers 192.168.67.82 -binddn
cn=administrator,cn=Users,dc=eng,dc=lcsc -file
ldap.conf
```

```
server_3 : Enter Password:******
done

EXAMPLE #23
-----------

To check if the domains are ok, type:

server_ldap server_3 -service -status
server_3 :
LDAP domain "dev.lcsc" is active - Configured with RFC-2307 defaults
LDAP domain "ds.lcsc" is inactive - Configured with RFC-2307 defaults
LDAP domain "dvt.emc" is active - Configured with profile "profilecad3"
LDAP domain "eng.lcsc" is active - Configured with file "ldap.conf"

EXAMPLE #24
-----------

To get the details about the domain ds.lcsc, type:

server_ldap server_3 -info -verbose -domain ds.lcsc
server_3 :
LDAP domain: ds.lcsc
State: Uninitialized - Disconnected
Schema: Unknown yet (must succeed to connect)
Base dn: dc=ds,dc=lcsc
Bind dn: cn=Directory Manager
Configuration: RFC-2307 defaults
Global warnings & errors
{
Only one LDAP server is configured for LDAP domain ds.lcsc.
}
LDAP server: 192.168.67.182 - Port: 389 - Spare
SSL: Not enabled
Last error: 91 / Connect error
Server warnings & errors
{
LDAP server 192.168.67.182: LDAP protocol error: LDAP is unable to connect to
the
specified port.
LDAP server 192.168.67.182: LDAP protocol error: Connect error.

EXAMPLE #25
-----------

To delete the domain ds.lcsc, type:

server_ldap server_3 -clear -domain ds.lcsc
server_3 : done
server_ldap server_3 -service -status
server_3 :
LDAP domain "dev.lcsc" is active - Configured with RFC-2307 defaults
LDAP domain "dvt.emc" is active - Configured with profile "profilecad3"
LDAP domain "eng.lcsc" is active - Configured with file "ldap.conf"

EXAMPLE #26
-----------

To lookup a user in a given domain, type:

server_ldap server_3 -lookup -user cad -domain eng.lcsc
server_3 :
user: cad, uid: 33021, gid: 32769, homeDir: /emc/cad

EXAMPLE #27
-----------

To get info on all domains, type:

server_ldap server_3 -info -all
server_3 :
```

```
LDAP domain: dev.lcsc
State: Configured - Connected
Schema: OpenLDAP
Base dn: dc=devldapdom1,dc=lcsc
Bind dn: <anonymous>
Configuration: RFC-2307 defaults
LDAP server: 192.168.67.114 - Port: 389 - Active
SSL: Not enabled
LDAP server: 192.168.67.148 - Port: 389 - Spare
SSL: Not enabled
LDAP domain: dvt.emc
State: Configured - Connected
Schema: Sun Directory Server (iPlanet) (Sun-ONE-Directory/5.2)
Base dn: dc=dvt,dc=emc
Bind dn: <anonymous>
Configuration: Profile name: profilecad3 - TTL: 11 s
LDAP conf server: 192.168.67.140 - Port: 389
SSL: Not enabled
LDAP default servers:
LDAP server: 192.168.67.140 - Port: 389 - Active
SSL: Not enabled
LDAP domain: eng.lcsc
State: Configured - Connected
Schema: Active Directory
Base dn: dc=eng,dc=lcsc
Bind dn: cn=administrator,cn=Users,dc=eng,dc=lcsc
Configuration: File: ldap.conf - TTL: 1200 s
LDAP server: 192.168.67.82 - Port: 389 - Active
SSL: Not enabled
```

EXAMPLE #28

To clear all the domains, type:

```
server_ldap server_3 -clear -all
server_3 : done
server_ldap server_3 -service -status
server_3 :
LDAP domain is not configured yet.
```

Last Modified: October 20, 2011 12:30 pm.

server_log

Displays the log generated by the specified Data Mover.

SYNOPSIS
----

server_log <movername>
[-a][-f][-n][-s][-v|-t]

DESCRIPTION
----

server_log reads and displays the log generated by the Data Mover. Information in the log file is read from oldest to newest. To view that most recent log activity, add|tail to the end of your command line.

OPTIONS

No arguments
Displays the contents of the log added since the last reboot.

-a
Displays the complete log.

-f
Displays the contents of the log added since the last reboot. Additionally monitors the growth of the log by entering into an endless loop, pausing, reading the log being generated. The output is updated every second. To exit, press Ctrl-C together.

-n
Displays the log without the time stamp.

-s
Displays the time in yyyy-mm-dd format when each command in the log was executed.

-v|-t
Displays the log files in verbose form or terse form.

EXAMPLE #1

To monitor the growth of the current log, type:

```
$ server_log server_2 -f
--------------------
NAS LOG for slot 2:
--------------------
0 keys=0 h=0 nc=0

1200229390: VRPL: 6: 122: Allocating chunk:3 Add:50176 Chunks:24
1200229390: SVFS: 6: Merge Start FsVol:118 event:0x0
1200229390: SVFS: 6: D113118_736: hdr:82944 currInd:6, Destpmdv:D114118_503
1200229390: CFS: 6: Resuming fs 24
1200229390: SVFS: 6: 118:D113118_736:Merge hdr=82944 prev=99328 id=113 chunk=0
s
tableEntry=7
1200229390: UFS: 6: Volume name:Sh122113
1200229390: UFS: 6: starting gid map file processing.
1200229390: UFS: 6: gid map file processing is completed.
1200229390: DPSVC: 6: DpRequest::done() BEGIN
reqType:DpRequest_VersionInt_SchSr
cRefresh reqCaller:DpRequest_Caller_Scheduler status:0
1200229390: DPSVC: 6:
```

```
SchedulerSrc=118_APM00062400708_0000_253_APM00062400708_00
00, curState=active, input=refreshDone
1200229390: DPSVC: 6: DpVersion::getTotalBlocksVolume enter
1200229390: DPSVC: 6: DpVersion::getTotalBlocksVolume found newV 118.ckpt003,
bl
ocks 17534
1200229390: DPSVC: 6: DpVersion::getTotalBlocksVolume 0 blocks for vnumber
1038
totalB 0
1200229390: DPSVC: 6: DpVersion::getTotalBlocksVolume found oldV 118.ckpt004
1200229390: DPSVC: 6: DpVersion::getTotalBlocksVolume exit
1200229390: DPSVC: 6: DpVersion::getTotalBytes 0 blocks 0 bytes
1200229390: DPSVC: 6:
SchedulerSrc=118_APM00062400708_0000_253_APM00062400708_00
00, newState=active
1200229390: SVFS: 6: D113118_736: After Merge err:4 full:0 mD:0
1200229390: SVFS: 6: D113118_736: prev !full release ch:82944 newPrev:99328
1200229390: SVFS: 6: D113118_737: Chunk:0 hdrAdd:50176 ==> prevChunk:82944
befor
e changePrevChunk
1200229390: SVFS: 6: D113118_737: Ch:0 hdr:50176 : prevCh:99328 after
changePrev
1200229510: DPSVC: 6: refreshSnap: cur=1200229510, dl=1200229520, kbytes=0,
setu
p=0, rate=1000
1200229510: DPSVC: 6:
SchedulerSrc=199_APM00062400708_0000_258_APM00062400708_00
00, curState=active, input=refresh
1200229510: DPSVC: 6: DpRequest::execute() BEGIN
reqType:DpRequest_VersionInt_Sc
hSrcRefresh reqCaller:DpRequest_Caller_Scheduler reqMode:0
1200229510: DPSVC: 6: DpRequest::execute() END
reqType:DpRequest_VersionInt_SchS
rcRefresh reqCaller:DpRequest_Caller_Scheduler status:0 reqMode:0
1200229510: DPSVC: 6:
SchedulerSrc=199_APM00062400708_0000_258_APM00062400708_00
00, newState=active
--More--
```

Note: This is a partial listing due to the length of the output.

EXAMPLE #2

To display the current log, type:

```
$ server_log server_2
NAS LOG for slot 2:
--------------------
0 keys=0 h=0 nc=0

0 keys=0 h=0 nc=0
2008-01-13 08:03:10: VRPL: 6: 122: Allocating chunk:3 Add:50176 Chunks:24
2008-01-13 08:03:10: SVFS: 6: Merge Start FsVol:118 event:0x0
2008-01-13 08:03:10: SVFS: 6: D113118_736: hdr:82944 currInd:6,
Destpmdv:D114118
_503
2008-01-13 08:03:10: CFS: 6: Resuming fs 24
2008-01-13 08:03:10: SVFS: 6: 118:D113118_736:Merge hdr=82944 prev=99328
id=113
chunk=0 stableEntry=7
2008-01-13 08:03:10: UFS: 6: Volume name:Sh122113
2008-01-13 08:03:10: UFS: 6: starting gid map file processing.
2008-01-13 08:03:10: UFS: 6: gid map file processing is completed.
2008-01-13 08:03:10: DPSVC: 6: DpRequest::done() BEGIN
reqType:DpRequest_Version
Int_SchSrcRefresh reqCaller:DpRequest_Caller_Scheduler status:0
2008-01-13 08:03:10: DPSVC: 6:
SchedulerSrc=118_APM00062400708_0000_253_APM00062
400708_0000, curState=active, input=refreshDone
2008-01-13 08:03:10: DPSVC: 6: DpVersion::getTotalBlocksVolume enter
```

```
2008-01-13 08:03:10: DPSVC: 6: DpVersion::getTotalBlocksVolume found newV
118.ck
pt003, blocks 17534
2008-01-13 08:03:10: DPSVC: 6: DpVersion::getTotalBlocksVolume 0 blocks for
vnum
ber 1038 totalB 0
2008-01-13 08:03:10: DPSVC: 6: DpVersion::getTotalBlocksVolume found oldV
118.ck
pt004
2008-01-13 08:03:10: DPSVC: 6: DpVersion::getTotalBlocksVolume exit
2008-01-13 08:03:10: DPSVC: 6: DpVersion::getTotalBytes 0 blocks 0 bytes
2008-01-13 08:03:10: DPSVC: 6:
SchedulerSrc=118_APM00062400708_0000_253_APM00062
400708_0000, newState=active
2008-01-13 08:03:10: SVFS: 6: D113118_736: After Merge err:4 full:0 mD:0
2008-01-13 08:03:10: SVFS: 6: D113118_736: prev !full release ch:82944
newPrev:9
9328
2008-01-13 08:03:10: SVFS: 6: D113118_737: Chunk:0 hdrAdd:50176 ==>
prevChunk:82
944 before changePrevChunk
2008-01-13 08:03:10: SVFS: 6: D113118_737: Ch:0 hdr:50176 : prevCh:99328
after c hangePrev
2008-01-13 08:05:10: DPSVC: 6: refreshSnap: cur=1200229510, dl=1200229520,
kbyte
s=0, setup=0, rate=1000
2008-01-13 08:05:10: DPSVC: 6:
SchedulerSrc=199_APM00062400708_0000_258_APM00062
400708_0000, curState=active, input=refresh
2008-01-13 08:05:10: DPSVC: 6: DpRequest::execute() BEGIN
reqType:DpRequest_Vers
ionInt_SchSrcRefresh reqCaller:DpRequest_Caller_Scheduler reqMode:0
```

Note: This is a partial listing due to the length of the output.

EXAMPLE #3

To display the log file without the time stamp, type:

```
$ server_log server_2 -n
NAS LOG for slot 2:
--------------------
0 keys=0 h=0 nc=0

VRPL: 6: 122: Allocating chunk:3 Add:50176 Chunks:24
SVFS: 6: Merge Start FsVol:118 event:0x0
SVFS: 6: D113118_736: hdr:82944 currInd:6, Destpmdv:D114118_503
CFS: 6: Resuming fs 24
SVFS: 6: 118:D113118_736:Merge hdr=82944 prev=99328 id=113 chunk=0
stableEntry=7
UFS: 6: Volume name:Sh122113
UFS: 6: starting gid map file processing.
UFS: 6: gid map file processing is completed.
DPSVC: 6: DpRequest::done() BEGIN reqType:DpRequest_VersionInt_SchSrcRefresh
req
Caller:DpRequest_Caller_Scheduler status:0
DPSVC: 6: SchedulerSrc=118_APM00062400708_0000_253_APM00062400708_0000,
curState
=active, input=refreshDone
DPSVC: 6: DpVersion::getTotalBlocksVolume enter
DPSVC: 6: DpVersion::getTotalBlocksVolume found newV 118.ckpt003, blocks 17534
DPSVC: 6: DpVersion::getTotalBlocksVolume 0 blocks for vnumber 1038 totalB 0
DPSVC: 6: DpVersion::getTotalBlocksVolume found oldV 118.ckpt004
DPSVC: 6: DpVersion::getTotalBlocksVolume exit
DPSVC: 6: DpVersion::getTotalBytes 0 blocks 0 bytes
DPSVC: 6: SchedulerSrc=118_APM00062400708_0000_253_APM00062400708_0000,
newState
=active
SVFS: 6: D113118_736: After Merge err:4 full:0 mD:0
SVFS: 6: D113118_736: prev !full release ch:82944 newPrev:99328
```

```
SVFS: 6: D113118_737: Chunk:0 hdrAdd:50176 ==> prevChunk:82944 before
changePrev
Chunk
SVFS: 6: D113118_737: Ch:0 hdr:50176 : prevCh:99328 after changePrev
DPSVC: 6: refreshSnap: cur=1200229510, dl=1200229520, kbytes=0, setup=0,
rate=10
00
DPSVC: 6: SchedulerSrc=199_APM00062400708_0000_258_APM00062400708_0000,
curState
=active, input=refresh
DPSVC: 6: DpRequest::execute() BEGIN reqType:DpRequest_VersionInt_SchSrcRefresh
reqCaller:DpRequest_Caller_Scheduler reqMode:0
DPSVC: 6: DpRequest::execute() END reqType:DpRequest_VersionInt_SchSrcRefresh
re
qCaller:DpRequest_Caller_Scheduler status:0 reqMode:0
DPSVC: 6: SchedulerSrc=199_APM00062400708_0000_258_APM00062400708_0000,
newState
=active
VBB: 6: VBB session list empty
CFS: 6: fs 0x78 type = dhfs being unmounted. Waiting for quiesce ...
CFS: 6: fs 0x78 type = dhfs unmounted
--More--
```

Note: This is a partial listing due to the length of the output.

EXAMPLE #4

To display all of the current logs available, type:

```
$ server_log server_2 -a
NAS LOG for slot 2:
--------------------
1200152690: SVFS: 6: D113118_606: prev !full release ch:82944 newPrev:99328
1200152690: SVFS: 6: D113118_607: Chunk:0 hdrAdd:50176 ==> prevChunk:82944
befor
e changePrevChunk
1200152690: SVFS: 6: D113118_607: Ch:0 hdr:50176 : prevCh:99328 after
changePrev
1200152950: DPSVC: 6: refreshSnap: cur=1200152950, dl=1200152960, kbytes=0,
setu
p=0, rate=666
1200152950: DPSVC: 6:
SchedulerSrc=199_APM00062400708_0000_258_APM00062400708_00
00, curState=active, input=refresh
1200152950: DPSVC: 6: DpRequest::execute() BEGIN
reqType:DpRequest_VersionInt_Sc
hSrcRefresh reqCaller:DpRequest_Caller_Scheduler reqMode:0
1200152950: DPSVC: 6: DpRequest::execute() END
reqType:DpRequest_VersionInt_SchS
rcRefresh reqCaller:DpRequest_Caller_Scheduler status:0 reqMode:0
1200152950: DPSVC: 6:
SchedulerSrc=199_APM00062400708_0000_258_APM00062400708_00
00, newState=active
1200152950: VBB: 6: VBB session list empty
1200152950: CFS: 6: fs 0x78 type = dhfs being unmounted. Waiting for
quiesce ...
1200152950: CFS: 6: fs 0x78 type = dhfs unmounted
1200152950: SVFS: 6: pause() requested on fsid:78
1200152950: SVFS: 6: pause done on fsid:78
1200152950: SVFS: 6: Cascaded Delete...
1200152950: SVFS: 6: D120199_1131: createBlockMap PBM root=0 keys=0 h=0 nc=0
1200152950: VRPL: 6: 217: Allocating chunk:4 Add:66560 Chunks:15
1200152950: SVFS: 6: Merge Start FsVol:199 event:0x0
1200152950: SVFS: 6: D120199_1130: hdr:99328 currInd:6, Destpmdv:D119199_1124
1200152950: CFS: 6: Resuming fs 78
1200152950: SVFS: 6: 199:D120199_1130:Merge hdr=99328 prev=82944 id=120 chunk=0
stableEntry=7
1200152950: UFS: 6: Volume name:Sh217120
1200152950: UFS: 6: starting gid map file processing.
1200152950: SVFS: 6: D120199_1130: After Merge err:4 full:0 mD:0
```

```
1200152950: SVFS: 6: D120199_1130: prev !full release ch:99328 newPrev:82944
1200152950: SVFS: 6: D120199_1131: Chunk:0 hdrAdd:66560 ==> prevChunk:99328
befo
re changePrevChunk
1200152950: SVFS: 6: D120199_1131: Ch:0 hdr:66560 : prevCh:82944 after
changePre
v
1200152950: UFS: 6: gid map file processing is completed.
1200152950: DPSVC: 6: DpRequest::done() BEGIN
reqType:DpRequest_VersionInt_SchSr
cRefresh reqCaller:DpRequest_Caller_Scheduler status:0
1200152950: DPSVC: 6:
SchedulerSrc=199_APM00062400708_0000_258_APM00062400708_00
00, curState=active, input=refreshDone
--More--
```

Note: This is a partial listing due to the length of the output.

EXAMPLE #5

To display the current log in terse form, type:

```
$ server_log server_2 -t
NAS LOG for slot 2:
--------------------
0 keys=0 h=0 nc=0

1200229390: 26043285504: 122: Allocating chunk:3 Add:50176 Chunks:24
1200229390: 26042826752: Merge Start FsVol:118 event:0x0
1200229390: 26042826752: D113118_736: hdr:82944 currInd:6, Destpmdv:D114118_503
1200229390: 26040008704: Resuming fs 24
1200229390: 26042826752: 118:D113118_736:Merge hdr=82944 prev=99328 id=113
chunk
=0 stableEntry=7
1200229390: 26042433536: Volume name:Sh122113
1200229390: 26042433536: starting gid map file processing.
1200229390: 26042433536: gid map file processing is completed.
1200229390: 26045513728: DpRequest::done() BEGIN
reqType:DpRequest_VersionInt_Sc
hSrcRefresh reqCaller:DpRequest_Caller_Scheduler status:0
1200229390: 26045513728:
SchedulerSrc=118_APM00062400708_0000_253_APM00062400708
_0000, curState=active, input=refreshDone
1200229390: 26045513728: DpVersion::getTotalBlocksVolume enter
1200229390: 26045513728: DpVersion::getTotalBlocksVolume found newV
118.ckpt003,
 blocks 17534
1200229390: 26045513728: DpVersion::getTotalBlocksVolume 0 blocks for
vnumber 10
38 totalB 0
1200229390: 26045513728: DpVersion::getTotalBlocksVolume found oldV 118.ckpt004
1200229390: 26045513728: DpVersion::getTotalBlocksVolume exit
1200229390: 26045513728: DpVersion::getTotalBytes 0 blocks 0 bytes
1200229390: 26045513728:
SchedulerSrc=118_APM00062400708_0000_253_APM00062400708
_0000, newState=active
1200229390: 26042826752: D113118_736: After Merge err:4 full:0 mD:0
1200229390: 26042826752: D113118_736: prev !full release ch:82944 newPrev:99328
1200229390: 26042826752: D113118_737: Chunk:0 hdrAdd:50176 ==> prevChunk:82944
b
efore changePrevChunk
1200229390: 26042826752: D113118_737: Ch:0 hdr:50176 : prevCh:99328 after
change
Prev
1200229510: 26045513728: refreshSnap: cur=1200229510, dl=1200229520,
kbytes=0, s
etup=0, rate=1000
1200229510: 26045513728:
SchedulerSrc=199_APM00062400708_0000_258_APM00062400708
_0000, curState=active, input=refresh
```

```
1200229510: 26045513728: DpRequest::execute() BEGIN
reqType:DpRequest_VersionInt
_SchSrcRefresh reqCaller:DpRequest_Caller_Scheduler reqMode:0
1200229510: 26045513728: DpRequest::execute() END
reqType:DpRequest_VersionInt_S
--More--
```

Note: This is a partial listing due to the length of the output.

EXAMPLE #6

To display the current log in verbose form, type:

```
$ server_log server_2 -v
DART Work Partition Layout found @ LBA 0x43000 (134MB boundary)

slot 2) About to dump log @ LBA 0xc7800

NAS LOG for slot 2:
-------------------

About to print log from LBA c8825 to c97ff
0 keys=0 h=0 nc=0

logged time         = 2008-01-13 08:03:10
id                  = 26043285504
severity            = INFO
component           = DART
facility            = VRPL
baseid              = 0
type                = STATUS
argument name       = arg0
argument value      = 122: Allocating chunk:3 Add:50176 Chunks:24

argument type       = string (8)
brief description   = 122: Allocating chunk:3 Add:50176 Chunks:24

full description    = No additional information is available.
recommended action  = No recommended action is available. Use the text from the
error message's brief description to search the Knowledgebase on
Powerlink. After logging in to Powerlink, go to Support > Knowledgebase
Search > Support Solutions Search.

logged time         = 2008-01-13 08:03:10
id                  = 26042826752
severity            = INFO
component           = DART
facility            = SVFS
baseid              = 0
type                = STATUS
argument name       = arg0
argument value      = Merge Start FsVol:118 event:0x0

argument type       = string (8)
brief description   = Merge Start FsVol:118 event:0x0

full description    = No additional information is available.
recommended action  = No recommended action is available. Use the text from the
error message's brief description to search the Knowledgebase on
Powerlink. After logging in to Powerlink, go to Support > Knowledgebase Search >
Support
Solutions Search.
--More--
```

Note: This is a partial listing due to the length of the output.

Last Modified: June 2, 2011 2:00 pm

server_mount

Mounts file systems and manages mount options for the specified Data Movers.

SYNOPSIS

```
server_mount {<movername>|ALL}
[-all]
|[-Force][-check][-option options] <fs_name> [mount_point]
options:
[ro|rw][primary=movername]
[nonotify][nooplock]
[notifyonaccess][notifyonwrite]
[accesspolicy={NT|UNIX|SECURE|NATIVE|MIXED|MIXED_COMPAT}]
[nolock|wlock|rwlock]
[cvfsname=<newname>]
[noscan]
[noprefetch]
[uncached]
[cifssyncwrite]
[triggerlevel=<value>]
[ntcredential]
[renamepolicy={CIFS|FULL|NO}]
[cifsnanoroundup]
[ceppcifs|ceppnfs|ceppcifs,ceppnfs|nocepp]]
[nfsv4delegation={NONE|READ|RW}]
[smbca]
```

DESCRIPTION

server_mount attaches a file system to the specified <mount_point> with the specified options, and displays a listing of mounted file systems. server_umount unmounts the file system.

The ALL option executes the command for all of the Data Movers.

Note: The primary=movername option is not used.

OPTIONS

No arguments
Displays a listing of all mounted and temporarily unmounted file systems.

-all
Mounts all file systems in the mount table.

-Force -option rw <fs_name> [mount_point]
Forces a mount of a file system copy (created using fs_timefinder) as read-write. By default, all file system copies are mounted as read-only.

[-check]
Checks if there is a diskmark value mismatch between NAS database and the Data Mover for the file system, and also checks if the diskmark on Data Mover exists. This option is required for SRDF STAR feature.

Note: If the check option is not used, the diskmark mismatch case or missing diskmark case could cause Data Mover panic.

<fs_name> [mount_point]
Mounts a file system to the specified <mount_point>. When a file system is initially mounted, the <mount_point> is required; however, remounting a file system after a temporary unmount, does not require the use of a <mount_point>.

[-option options]
Specifies the following comma-separated options:

[ro|rw]
Specifies the mount as read-write (default), or read-only which is the default for checkpoints and TimeFinder/FS.

Note: MPFS clients do not acknowledge file systems that are mounted read-only and allow their clients to write to the file system.

[accesspolicy={NT|UNIX|SECURE|NATIVE|MIXED|MIXED_COMPAT}]
Indicates the access control policy as defined in the table below:

Note: When accessed from a Windows client, ACLs are only checked if the CIFS user authentication method is set to the recommended default, NT. This is set using the -add security option in the server_cifs command.

Access Policy	CIFS clients	NFS clients
NATIVE (default)	ACL is checked.	UNIX rights are checked.
UNIX	ACL and UNIX rights are checked.	UNIX rights are checked.
NT	ACL is checked.	ACL and UNIX rights are checked.
SECURE	ACL and UNIX rights are checked.	ACL and UNIX rights are checked.
MIXED	ACL is checked. If there is not an ACL, one is created based on the UNIX mode bits. Access is also determined by the ACL. NFSv4 clients can manage the ACL. An ACL modification rebuilds the UNIX mode bits but the UNIX rights are not checked.	ACL is checked. If there is not an ACL, one is created based on the UNIX mode bits. Access is also determined by the ACL. NFSv4 clients can manage the ACL. A modification to the UNIX mode bits rebuilds the ACL permissions but the UNIX rights are not checked.
MIXED_COMPAT	If the permissions of a file or directory were last set or changed by a CIFS client, the ACL is checked and the UNIX rights are rebuilt but are not checked.	If the permissions of a file or directory were last set or changed by an NFS client, the UNIX rights are checked and the ACL is rebuilt but is not checked.
	If the permissions of a file	If the permissions of the file

	or directory were last set or changed by an NFS client, the UNIX rights are checked and the UNIX rights are rebuilt but are not checked. NFSv4 clients can manage the ACL.	or directory were last set or changed by a CIFS client, the UNIX rights are checked and the UNIX rights are rebuilt but are not checked. NFSV4 clients can manage the ACL.

Note: The MIXED policy translates the UNIX ownership mode bits into three ACEs: Owner, Group, and Everyone, which can result in different permissions for the Group ACE and the Everyone ACE. The MIXED_COMPAT policy does not translate a UNIX Group into a Group ACE. The Everyone ACE is generated from the UNIX Group.

[cvfsname=<newname>]
Changes the default name of the checkpoint in each of the .ckpt directories. The default name is the timestamp of when the checkpoint was taken.

[noprefetch]
Turns prefetch processing off. When on (default), performs read ahead processing for file systems.

Caution: Turning the prefetch option to off may affect performance.

[uncached]
Allows well-formed writes (that is, multiple of a disk block and disk block aligned) to be sent directly to the disk without being cached on the server.

For CIFS Clients Only

When mounting a file system, if the default options are not manually entered, the options are active but not displayed in the listing of mounted file systems. Available options are:

[nonotify]
Turns notify off. When on (default), the notify option informs the client of changes made to the directory file structure.

[nooplock]
Turns opportunistic locks (oplocks) off. When oplocks are on (default), they reduce network traffic by enabling clients to cache the file and make changes locally. To turn Windows oplocks off, unmount the file system, then remount with nooplock.

[notifyonaccess]
Provides a notification when a file system is accessed. By default, notifyonaccess is disabled.

[notifyonwrite]
Provides a notification of write access to a file system. By default, the notifyonwrite option is disabled.

[noscan]
Disables the Virus Checker protocol for a file system. The Virus Checker protocol is enabled using server_setup and managed by server_viruschk.

[cifssyncwrite]
Performs an immediate synchronous write on disk independently

of CIFS write protocol option. This can impact write performance.

[triggerlevel=<value>]
Specifies the deepest directory level at which notification occurs. The default is 512. The value -1 disables the notification feature.

[ntcredential]
Enables the VNX to take full account of a user's Windows group memberships when checking an ACL for access through NFS. When a UNIX user intiates a full request for a file system object, the UNIX UID is mapped to the Windows SID, then merges the user's UNIX and Windows groups together to generate a Windows NT Credential. This applies to NT, SECURE, MIXED, and MIXED_COMPAT access-checking policies.

[renamepolicy={CIFS|FULL|NO}]
Enables or disables control if any file or directory is opened on the current directory or any subdirectory, before the current directory is renamed. CIFS (default) stops the renaming of CIFS directories when in use by CIFS clients. FULL denies permission for the renaming of CIFS and NFS directories when in use by CIFS or NFS clients. NO automatically performs the directory rename without checking if a CIFS or NFS client is opening the directory.

Note: The renamepolicy is not supported by NFSv4.

cifsnanoroundup
Rounds up to the next second any date set by a CIFS client.

[ceppcifs]
It enables the CEPA events for CIFS on a file system. This option is enabled by default.

[smbca]
Sets the CA bit on a share, the primary file system must be mounted with the smbca option.
* The lock policy is RWLock (lock checking mandatory)
* The CIFS and NFS access to this file system is denied until the CIFS CA service is started up to the time in seconds defined by the parameters cifs.smb2.maxCaTimeout (default is 2 minutes)
* The CA attribute can be set on a share located on this file system.

For NFS Clients Only

[nolock|wlock|rwlock]
Indicates the impact of locking behavior on NFSv2 and NFSv3 clients against NFSv4 and CIFS file locking. In NFSv2 and NFSv3, locking rules are cooperative, so a client is not prevented from accessing a file locked by another client if it does not use the lock procedure. NFSv2 and NFSv3 locks as advisory. An advisory lock does not affect read and write access to the file, but informs other users that the file is already in use.

Note: NFSv4 and ClFS clients have mandatory locking schemes and do not require a locking policy.

Locking Policy	NFS clients
nolock	This(default) can open and write to a file when it is locked by CIFS or NFSv4 clients.
wlock	This can read but cannot write data to a file locked by CIFS or NFSv4 clients.
rwlock	This (recommended) cannot read or write data to files locke d by CIFS or NFSv4 clients.

[ceppnfs]
It enables the CEPA events for NFS on a file system.

Note: If ceppnfs is used without the ceppcifs option, the CEPA events for CIFS are disabled. To enable CEPA events for NFS and CIFS on a file system, ensure that you add both these options in the command.

nfsv4delegation={NONE|READ|RW}
Indicates that specific actions on a file are delegated to the NFSv4 client. NONE indicates that no file delegation is granted. READ indicates only read delegation is granted. RW (default) indicates write delegation is granted.

SEE ALSO

Managing Volumes and File Systems with VNX Automatic Volume Management, Managing Volumes and File Systems for VNX Manually, Configuring NFS on VNX, Configuring and Managing CIFS on VNX, Using VNX SnapSure, nas_fs, server_checkup, server_export, server_mountpoint, server_nfs, server_setup, server_umount, and server_viruschk.

EXAMPLE #1

To display all mounted file systems on server_2, type:

```
$ server_mount server_2
server_2 :
root_fs_2 on / uxfs,perm,rw
root_fs_common on /.etc_common uxfs,perm,ro
ufs2 on /ufs2 uxfs,perm,rw
```

EXAMPLE #2

To mount all file systems temporarily umounted from the mount table of server_2, type:

```
$ server_mount server_2 -all
server_2 : done
```

EXAMPLE #3

To mount ufs1, on mount point/ufs1, and enable CEPP for both CIFS and NFS, type:

```
$ server_mount server_2 -o ceppcifs,ceppnfs ufs1 /ufs1
server_2 : done
```

EXAMPLE #4

To mount ufs1, on mount point/ufs1, with nonotify, nolock and cifssyncwrite turned on, type:

```
$ server_mount server_2 -option nonotify,nolock,cifssyncwrite ufs1 /ufs1
server_2 : done
```

EXAMPLE #5

To mount ufs1, on mount point/ufs1, with the access policy set to NATIVE, and nooplock turned on, type:

```
$ server_mount server_2 -option accesspolicy=NATIVE,nooplock ufs1 /ufs1
```

```
server_2 : done

EXAMPLE #6
----------

To mount ufs1, on mount point/ufs1, with noscan and noprefetch set
to on, type:

$ server_mount server_2 -option noscan,noprefetch ufs1 /ufs1
server_2 : done

EXAMPLE #7
----------

To mount ufs1, on mount point /ufs1, with notifyonaccess,notifyonwrite
set to on, type:

$ server_mount server_2 -option notifyonaccess,notifyonwrite ufs1 /ufs1
server_2 : done

EXAMPLE #8
----------

To mount a copy of a file system, ufs1_snap1 on mount point/ufs1_snap1
with read-write access, type:

$ server_mount server_2 -Force -option rw ufs1_snap1 /ufs1_snap1
server_2 : done

EXAMPLE #9
----------

To mount ufs1, on mount point/ufs1, with uncached writes turned
on, type:

$ server_mount server_2 -option uncached ufs1 /ufs1
server_2 : done

EXAMPLE #10
-----------

To mount ufs1, on mount point/ufs1, with the trigger level of
notification change set to 256, type:

$ server_mount server_2 -option triggerlevel=256 ufs1 /ufs1
server_2 : done

EXAMPLE #11
-----------

To mount ufs1, on mount point/ufs1, change the default name of the
checkpoint in the ".ckpt" directory, and specify a mount point, type:

$ server_mount server_2 -option cvfsname=test ufs1 /ufs1
server_2 : done

EXAMPLE #12
-----------

To mount ufs1, on mount point/ufs1, with the access policy set
to MIXED, type:

$ server_mount server_2 -option accesspolicy=MIXED ufs1 /ufs1
server_2 : done

EXAMPLE #13
-----------

To mount ufs1, on mount point/ufs1, with the access policy set to
MIXED_COMPAT, type:
```

```
$ server_mount server_2 -option accesspolicy=MIXED_COMPAT ufs1 /ufs1
server_2 : done
```

EXAMPLE #14

To mount ufs1, as a part of the nested file system nmfs1, type:

```
$ server_mount server_2 ufs1 /nmfs1/ufs1
server_2 : done
```

EXAMPLE #15

To mount ufs1, specifying that no file is granted to the NFSv4 client, type:

```
$ server_mount server_2 ufs1 nfsv4delegation=NONE
server_2 : done
```

EXAMPLE #16

To check diskmark value for the file system ufs1632_snap1, type:

```
$ server_mount server_2 -check ufs1632_snap1/ufs1632_snap1
server_2 :
Error 13423542320: server_2 : The marks on disks rootd17 with file system
ufs1632_snap1 are not the same on NAS_DB and the Data Mover.
```

EXAMPLE #17

To check if the diskmark for the file system ufs1632_snap1 exists, type:

```
$ server_mount server_2 -check ufs1632_snap1/ufs1632_snap1
server_2 :
Error 13423542324: server_2 : The marks on disks rootd17 with file system
ufs1632_snap1 cannot be found on the Data Mover.
```

EXAMPLE #18

To mount the file system named "fs 105" on the VDM "vdm1" to the mount point /fs 105, type:

```
$ server_mount vdm1 -o smbca fs105/fs105
vdm1:done
```

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server_mountpoint

Manages mount points for the specified Data Movers.

SYNOPSIS

```
server_mountpoint {<movername>|ALL}
  -list
| {-create|-delete|-exist} <pathname>
```

DESCRIPTION
----

server_mountpoint creates, deletes, lists, or queries a mount point for the specified Data Mover or all Data Movers.

The ALL option executes the command for all Data Movers.

OPTIONS

-list
Lists all mount points for the specified Data Movers.

-create <pathname>
Creates a mount point. A <pathname> must begin with a slash (/).

-delete <pathname>
Deletes a mount point.

-exist <pathname>
Displays whether or not a mount point exists.

SEE ALSO

Managing Volumes and File Systems with VNX Automatic Volume Management, Managing Volumes and File Systems for VNX Manually, nas_fs, server_export, and server_mountpoint.

EXAMPLE #1

To create a mount point on server_2, type:

```
$ server_mountpoint server_2 -create /ufs1
server_2 : done
```

EXAMPLE #2

To list all mount points on a server_2, type:

```
$ server_mountpoint server_2 -list
server_2 :
/.etc_common
/ufs1
/ufs1_ckpt1
/ufs2
/ufs3
```

EXAMPLE #3

To verify that the mount point /ufs1, exists on all of the Data Movers, type:

```
$ server_mountpoint ALL -exist /ufs1
server_2 : /ufs1 : exists
server_3 : /ufs1 : does not exist
```

EXAMPLE #4

To delete the mount point /ufs1, on server_2, type:

```
$ server_mountpoint server_2 -delete /ufs1
server_2 : done
```

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server_mpfs

Sets up and configures MPFS protocol.

SYNOPSIS

```
server_mpfs {<movername>|ALL}
-set <var>=<value>
| -add <number_of_threads>
| -delete <number_of_threads>
| -Stats
| -Default [<var>]
| -mountstatus
```

DESCRIPTION

server_mpfs sets up the MPFS protocol. The configuration values entered with this command are saved into a configuration file on the Data Mover. MPFS is not supported on the NS series.

server_setup provides information to start and stop MPFS for a Data Mover.

The ALL option executes the command for all Data Movers.

OPTIONS

-set <var>=<value>
Sets the specifed value for the specified variable. Currently, the only valid <var> is threads.

If this command is executed before the server_setup -P mpfs -o start command is issued, the system sets the number of threads that will be started with the server_setup -o start command, thereby overriding the default number of threads. If this command is executed after MPFS service is started, threads are to be added and removed dynamically.

-add <number_of_threads>
Increases the previously specified number of MPFS threads (default=16) by <number_of_threads> for the specified Data Movers.

-delete <number_of_threads>
Decreases the number of threads by the <number_of_threads> indicated for the specified Data Movers.

-Stats
Displays the current MPFS server statistics.

-mountstatus
Displays the mountability of file systems for MPFS.

Certain file systems cannot be mounted using MPFS, therefore before attempting to mount a file system on an MPFS client, compatibility should be determined. File systems that are not supported are running quotas, have checkpoints, or are using TimeFinder/FS.

-Default [<var>]
Without a <var> entry, resets all variables to their factory-default values. Currently the only valid <var> is threads.

If a <var> is specified, only the specified value is reset to its factory-default value.

Note: Variable names are case-sensitive.

SEE ALSO

Using VNX Multi-Path File System, server_setup, and server_mt.

EXAMPLE #1

To set a value for a specified MPFS variable, type:

```
$ server_mpfs server_2 -set threads=32
server_2 :done
```

EXAMPLE #2

To display the MPFS stats for server_2, type:

```
$ server_mpfs server_2 -Stats

server_2 :
Server ID=server_2
FMP Threads=32
Max Threads Used=2
FMP Open Files=0
FMP Port=4656
HeartBeat Time Interval=30
```

EXAMPLE #3

To reset all variables back to their factory default value, type:

```
$ server_mpfs server_2 -Default
server_2 :done
```

EXAMPLE #4

To check the mount status of a Data Mover, type:

```
$ server_mpfs server_2 -mountstatus
server_2 :

fs                 mpfs compatible?    reason
--                 ----------------    ------

                   no                  not a ufs file system

testing_renaming   no                  volume structure not FMP compatible
                   no                  not a ufs file system

server2_fs1_ckpt   no                  volume structure not FMP compatible

mpfs_fs2_lockdb_ckpt_5
                   no                  volume structure not FMP compatible

mpfs_fs2_lockdb_ckpt_4
                   no                  volume structure not FMP compatible

mpfs_fs2_lockdb_ckpt_3
                   no                  volume structure not FMP compatible

mpfs_fs2_lockdb_ckpt_2
                   no                  volume structure not FMP compatible

mpfs_fs2_lockdb_ckpt_1
                   no                  volume structure not FMP compatible

mpfs_fs2_lockdb_ckpt_10
```

```
                    no                  volume structure not FMP compatible

mpfs_fs2_lockdb_ckpt_9
                    no                  volume structure not FMP compatible

mpfs_fs2_lockdb_ckpt_8
                    no                  volume structure not FMP compatible

mpfs_fs2_lockdb_ckpt_7
                    no                  volume structure not FMP compatible
                    no                  not a ufs file system

mpfs_fs2_lockdb_ckpt_6
                    no                  volume structure not FMP compatible

root_fs_common      yes

mpfs_fs2            yes

mpfs_fs1            mounted

server2_fs1         yes

root_fs_2           yes
```

EXAMPLE #5

To add 16 threads for server_2, type:

```
$ server_mpfs server_2 -add 16
server_2 : done
```

EXAMPLE #6

To delete 16 threads from server_2, type:

```
$ server_mpfs server_2 -delete 16
server_2 : done
```

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server_mt

Manages the magnetic tape drive for the specified Data Mover.

SYNOPSIS
----

server_mt <movername>
        [-f <device>][-N] <command> [<count>]

DESCRIPTION
----

server_mt sends commands to a magnetic tape drive located on the specified Data Mover.

By default, server_mt performs the requested operation once. Specify <count> to perform multiple operations.

OPTIONS

<command> <count>
Operations for the command are as follows:

eof, weof
Write <count> EOF marks at the current position on the tape.

fsf
Forward space over <count> EOF marks. The tape is positioned on the first block of the file.

bsf
Back space over <count> EOF marks. The tape is positioned on the beginning-of-tape side of the EOF mark.

asf
Absolute space to <count> file number. This is equivalent to a rewind followed by a fsf <count>.

For the following commands, <count> is ignored:

eom
Space to the end of recorded media on the tape. This is useful for appending files onto previously written tapes.

rewind
Rewind the tape.

offline,rewoffl
Rewind the tape and, if appropriate, take the drive unit offline by unloading the tape.

status
Print status information about the tape unit.

erase
Erase the entire tape.

-f <device>
Specifies the magnetic tape device to which the command is applied, and the <device> which is the name of the tape drive.

-N
Indicates no rewind at the end of the tape.

The server_devconfig <movername> -list -scsi -nondisks command displays a listing of devices.

EXAMPLE

To send the rewind command to tape1 (magnetic tape drive) on a Data Mover, type:

```
$ server_mt server_2 -f tape1 rewind

server_2: done
```

Last modified: May 12, 2011 9:33 am.

server_name

Manages the name for the specified Data Movers.

You must delete all user-defined interconnects configured for a Data Mover before you can rename it using this command. After you rename the Data Mover, you must re-create the source and peer interconnects with the new Data Mover name and then restart any associated replication sessions.

SYNOPSIS

```
server_name {<movername>|ALL}
  [<new_name>]
```

DESCRIPTION

server_name displays and renames the current movername for the specified Data Mover.

The ALL option executes the command for all Data Movers.

OPTIONS

No arguments
Displays movernames for all Data Movers.

<new_name>
Changes the current movername to the newly specified movername. The movername is case-sensitive and supports the following characters: a through z, A through Z, 0 through 9, _(underscore), -(hyphen), though names may not start with a hyphen.

The ALL option is not available when renaming Data Movers.

SEE ALSO

nas_rp.

EXAMPLE #1

To change the current movername for a Data Mover, type:

```
$ server_name server_2 my_srv2
server_2 : my_srv2
```

EXAMPLE #2

To view all movernames, type:

```
$ server_name ALL
server_2 : my_srv2
server_3 : my_srv3
server_4 : my_srv4
server_5 : my_srv5
```

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server_netstat

Displays the network statistics for the specified Data Mover.

SYNOPSIS
----

server_netstat {<movername>}
[-a][-i][-r][-s][-p {tcp|udp|icmp|ip}]

DESCRIPTION
----

server_netstat displays network statistics for the specified Data Mover.

OPTIONS

No arguments
Displays the active TCP connections.

-a
Displays the state of all the IP, ICMP, TCP, and UDP sockets.

-i
Displays a summary of the state of all physical interfaces including the total amount of data transferred both ways and the number of errors during this process as well as the MTU size and MAC address.

-r
Displays the routing table.

-s
Displays the statistics of each protocol.

-p {tcp|udp|icmp|ip}
Limits the display of the sockets, or the statistics of each protocol to the specified protocol. tcp is default.

SEE ALSO
----

server_ifconfig, server_route.

EXAMPLE #1

To display all active IP, ICMP, TCP, and UDP sockets, type:

```
$ server_netstat server_2
Proto Local Address             Foreign Address         (state)
*************************************************************
tcp 172.24.102.237.2483         172.24.102.17.445       ESTABLISHED
tcp 172.24.102.237.netbios-ssn  172.24.103.63.3962      ESTABLISHED
```

EXAMPLE #2

To display the states of the IP, ICMP, TCP, and UDP sockets, type:

```
$ server_netstat server_2 -a
Proto Local Address             Foreign Address           (state)
*****************************************************************
tcp 172.24.102.237.2483         172.24.102.17.445        ESTABLISHED
tcp 172.24.102.237.netbios-ssn  172.24.103.63.3962       ESTABLISHED
tcp *.microsoft-ds              *.*                      LISTEN
tcp *.netbios-ssn               *.*                      LISTEN
```

```
tcp *.12345                    *.*                    LISTEN
tcp *.5080                     *.*                    LISTEN
tcp *.2272                     *.*                    LISTEN
tcp *.2271                     *.*                    LISTEN
tcp *.2270                     *.*                    LISTEN
tcp *.ftp                      *.*                    LISTEN
tcp *.10000                    *.*                    LISTEN
tcp *.4658                     *.*                    LISTEN
tcp *.2269                     *.*                    LISTEN
tcp *.2268                     *.*                    LISTEN
tcp *.nfs                      *.*                    LISTEN
tcp *.1234                     *.*                    LISTEN
tcp *.5033                     *.*                    LISTEN
tcp *.8888                     *.*                    LISTEN
tcp *.sunrpc                   *.*                    LISTEN

Proto Local Address
*******************
udp *.sunrpc
udp *.netbios-ns
udp *.netbios-dgm
udp *.snmp
udp *.router
udp *.1024
udp *.1036
udp *.1037
udp *.1038
udp *.1046
udp *.1054
udp *.1065
udp *.1234
udp *.nfs
udp *.2268
udp *.4646
udp *.4647
udp *.4658
udp *.9999
udp *.12345
udp *.31491
udp *.38914
```

EXAMPLE #3

To display a summary of the state of all physical interfaces, type:

```
$ server_netstat server_2 -i

Name   Mtu     Ibytes      Ierror     Obytes      Oerror         PhysAddr
*************************************************************************

fxp0  1500    758568220     0       534867239        0         8:0:1b:43:49:9a
cge0  9000    18014329      0       7195540          0         8:0:1b:42:46:3
cge1  9000    306495706     0       9984             0         8:0:1b:42:46:4
cge2  9000       0          0        0               0         8:0:1b:42:46:2
cge3  9000       0          0        0               0         8:0:1b:42:46:7
cge4  9000       0          0        0               0         8:0:1b:42:46:5
cge5  9000       0          0        0               0         8:0:1b:42:46:6
```

EXAMPLE #4

To display routing table statistics, type:

```
$ server_netstat server_2 -r

  Destination    Gateway         Mask            Type      Proto      Interface
  ***************************************************************************

  0.0.0.0         172.24.102.254  255.255.255.0  DIRECT    RIP        cge0
```

```
  128.221.253.0   128.221.253.2   255.255.255.0  DIRECT    RIP        fxp0

  172.24.102.0    172.24.102.237  255.255.255.0  DIRECT    RIP        cge0

  128.221.252.0   128.221.252.2   255.255.255.0  DIRECT    RIP        fxp0
```

EXAMPLE #5

To display the statistics of each protocol, type:

```
$ server_netstat server_2 -s
ip:
***
2315636 total packets received
0 bad header checksums
0 with unknown protocol
4 fragments received
0 fragments dropped (dup or out of space)
0 fragments dropped after timeout
4 packets reassembled
2 packets forwarded
13046 packets not forwardable
13046 no routes
2302596 packets delivered
2267772 total packets sent
3 packets fragmented
0 packets not fragmentable
6 fragments created

icmp:
*****
162 calls to icmp_error
Output histogram:
   echo reply: 1079145
   destination unreachable: 90
   echo: 1996
Input histogram:
   echo reply: 1993
   destination unreachable: 162
   routing redirect: 0
   echo: 1079145
   time exceeded: 0
   address mask request: 0
   1081300 messages received
1081231 messages sent

tcp:
****
437648 packets sent
2 data packets retransmitted
0 resets
434138 packets received
212 connection requests
19 connections lingered

udp:
****
0 incomplete headers
27048 bad ports
760361 input packets delivered
744999 packets sent
```

EXAMPLE #6

To display TCP protocol statistics, type:

```
$ server_netstat server_2 -s -p tcp
tcp:
****
```

437690 packets sent
2 data packets retransmitted
0 resets
434195 packets received
212 connection requests
19 connections lingered

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server_nfs

Manages the NFS service, including secure NFS and NVSv4, for the specified Data Movers.

SYNOPSIS

```
server_nfs {<movername>|ALL}
-secnfs
  [-service {-start|-stop}
  | -user {-list|-info {principal=<user_principal>|handle=<handle_number>}
     | -release {principal=<user_principal>|handle=<handle_number>}}
  | -principal {-create <service@server>|-delete <service@server>}
  | -mapper
    {-info|-set {-source {auto|nis|file [path=<file_path>]}
      | -passwddb {nis|file}}
   | -mapping {-list [name=<user_name>|uid=<UID>]
      | -create {name=<user_name>|uid=<UID>}
      | -delete {name=<user_name>|uid=<UID>}}}]
  | -v4 {-service {-start|-stop}
    | -client {-list
      | -info {index=<index>|hostname=<host>|ipaddress=<ipaddr>}
      | -release {index=<index>|hostname=<host>|ipaddress=<ipaddr>}}
   | -stats [-zero][-rpc|-nfs]
```

DESCRIPTION

server_nfs manages the NFS service, including secure NFS and NFSv4.

For secure NFS, server_nfs displays the status of the secure NFS service, stops and starts Kerberos authentication for secure NFS, and manages users authenticated through secure NFS. It also manages secure NFS security principals, releases the security context of a particular user, and manages the secure NFS user mapping service. Secure NFS supports UNIX-based, Linux-based, and Windows-based Kerberos KDCs.

For NFSv4, server_nfs stops and starts the NFSv4 service, lists NFSv4 clients, and displays information about those clients.

Note: NFSv2 and v3 services are not started or stopped by this command.

The ALL option executes the command for all Data Movers.

OPTIONS

-secnfs
Displays the status of the secure NFS service.

-service {-start|-stop}
Enables or disables the secure NFS service on the Data Mover.

-user -list
Lists all the NFS users authenticated by secure NFS.

-user -info {principal=<user_principal>|handle=<handle_number>}
Displays the attributes of an authenticated user as specified by the <user_principal> or <handle_number>. The <user_principal> is the Kerberos representation of a user. It is the username followed by the Kerberos realm name. The <handle_number> is the identifier of a secure NFS user authentication context and is displayed using -user -list.

-user -release {principal=<user_principal>|handle=<handle_number>}}

Releases the authentication context of the specified user.

-principal {-create <service@server>|-delete <service@server>}
Creates or deletes a secure NFS service instance.

-mapper -info
Displays the secure NFS mapping service configurations for the local file.

-mapper -set {-source {auto|nis|file [path=<file_path>]}
Sets the type of mapping provider and the file path if it is a local file (the default file is gsscred_db). These options are only relevant when using a Solaris UNIX Kerberos KDC. The default and recommended value is -source auto.

-mapper -set -passwddb {nis|file}
Sets the location of the password database.

-mapper -mapping -list [name=<user_name>|uid=<UID>]
Lists one or more user-mapping records as specified by <user_name> or <UID>. This option is only relevant if you are using a local mapping file and a Solaris UNIX Kerberos KDC.

-mapper -mapping -create {name=<user_name>|uid=<UID>}
Creates a new mapping record. This option is only relevant if you are using a local mapping file and a Solaris UNIX Kerberos KDC.

-mapper -mapping -delete {name=<user_name>|uid=<UID>}]
Deletes the mapping record from the database. This option is only relevant if you are using a local mapping file and a Solaris UNIX Kerberos KDC.

-v4
Displays the status of the NFSv4 service.

-service {-start|-stop}
Enables or disables the NFSv4 service on the Data Mover. To start the NFSv4 service, the /nas/server/slot_<x>/config file, where <x> is the slot number of the Data Mover, must be configured so that the hivers argument is set to 4. If the hivers is not set to 4, append the nfs start line and ensure that the hivers value is set to 4. Restart the Data Mover to apply the change. Other versions of NFS can continue to run even if NFSv4 is disabled.

-client -list
Lists all NFSv4 client systems that have established state with the VNX NFSv4 server.

-client -info {index|hostname=<host>|ipaddress=<ipaddr>}
Displays the attributes of the client (clientID) as specified by the index, <host>, or <ipaddr>. The index value is obtained from the client list.

-client -release {index|hostname=<host>|ipaddress=<ipaddr>}
Releases the state of the client (clientID) as specified by the index, <host>, or <ipaddr>. The index value is obtained from the client list.

-stats [-zero]
Displays all statistics or resets to zero all NFS and RPC statistics.

[-rpc|-nfs]
Displays RPC statistics or displays NFS statistics.

[-zero][-rpc]
Resets to zero all RPC statistics.

[-zero][-nfs]
Resets to zero all NFS statistics.

SEE ALSO

Configuring NFS on VNX and server_kerberos.

EXAMPLE #1

To display the status of the secure NFS service, type:

```
$ server_nfs server_2 -secnfs
server_2 :
RPCSEC_GSS server stats

Credential count: 2
principal: nfs@dm112-cge0.nasdocs.emc.com
principal: nfs@dm112-cge0

Total number of user contexts: 1
Current context handle: 3
```

EXAMPLE #2

To enable secure NFS service on server_2, type:

```
$ server_nfs server_2 -secnfs -service -start
server_2 : done
```

EXAMPLE #3

To disable secure NFS service on server_2, type:

```
$ server_nfs server_2 -secnfs -service -stop
server_2 : done
```

EXAMPLE #4

To display all secure NFS service instances, type:

```
$ server_nfs server_2 -secnfs -user -list
server_2 :
RPCSEC_GSS server stats

Credential count: 2
principal: nfs@dm112-cge0.nasdocs.emc.com
principal: nfs@dm112-cge0

Total number of user contexts: 1
Current context handle: 3

PARTIAL user contexts:
Total PARTIAL user contexts: 0

USED user contexts:
principal=nfsuser1@NASDOCS.EMC.COM, service=nfs@dm112-cge0.nasdocs.emc.com,
handle=3, validity=35914s
Total USED user contexts: 1

EXPIRED user contexts:
Total EXPIRED user contexts: 0
```

EXAMPLE #5

To display the attributes of an authenticated server as specified by the handle, type:

```
$ server_nfs server_2 -secnfs -user -info handle=3
server_2 :
principal: nfsuser1@NASDOCS.EMC.COM
service: nfs@dm112-cge0.nasdocs.emc.com
handle: 3
validity: 35844s
GSS flags: mutl conf intg redy tran
credential: uid=1010, inuid=1010, gid=1000
```

EXAMPLE #6

To release the authentication context of the user specified by the handle, type:

```
$ server_nfs server_2 -secnfs -user -release handle=3
server_2 : done
```

EXAMPLE #7

To create a secure NFS service instance, type:

```
$ server_nfs server_2 -secnfs -principal -create
nfs1@dm112-cge0.nasdocs.emc.com
server_2 : done
```

EXAMPLE #8

To delete a secure NFS service instance, type:

```
$ server_nfs server_2 -secnfs -principal -delete
nfs1@dm112-cge0.nasdocs.emc.com
server_2 : done
```

EXAMPLE #9

To set the mapping provider for the file, type:

```
$ server_nfs server_2 -secnfs -mapper -set -source file
server_2 : done
```

EXAMPLE #10

To set the location of the password database, type:

```
$ server_nfs server_2 -secnfs -mapper -set -passwddb file
server_2 : done
```

EXAMPLE #11

To display the secure NFS mapping service configurations for the local file, type:

```
$ server_nfs server_2 -secnfs -mapper -info
server_2 :
Current NFS user mapping configuration is:
gsscred db = File /.etc/gsscred_db
gsscred db version = Dart_v1
passwd db = File
```

EXAMPLE #12

To create a new mapping record, type:

```
$ server_nfs server_2 -secnfs -mapper -mapping -create name=nfsuser1
server_2 : done
```

EXAMPLE #13

To display a list of the mapping records, type:

```
$ server_nfs server_2 -secnfs -mapper -mapping -list
server_2 :
0401000B06092A864886F7120102020000001A7365636E6673757365723140737461746573 2E656D6
32E636F6D
1000 nfsuser1, kerberos_v5
```

EXAMPLE #14

To delete a mapping record, type:

```
$ server_nfs server_2 -secnfs -mapper -mapping -delete name=nfsuser1
server_2 : done
```

EXAMPLE #15

To enable the NFSv4 service on server_2, type:

```
$ server_nfs server_2 -v4 -service -start
server_2 : done
```

EXAMPLE #16

To start the NFSv4 service, type:

```
$ server_nfs {<movername>|ALL} -v4 -service -start
server_2 : done
```

Note: The pNFS service starts automatically once the NFSv4 starts. To start the NFSv4 service, ensure that the /nas/server/slot_<x>/config file has the hivers set to 4, where <x> is the slot number, for example slot_2, slot_3, and so on. The nfs entry in the above file should look like:
nfs config hivers=4.

EXAMPLE #17

To stop the pNFS service temporarily, type:

```
$ server_nfs {<movername>|ALL} -pnfs -service -stop
server_2 : done
```

Note: When the pNFS service is stopped, all granted layouts are recalled and requests for new layouts is denied.

EXAMPLE #18

To restart the pNFS service while the NFSv4 service is running, type:

```
$ server_nfs {<movername>|ALL} -pnfs -service -start
server_2 : done
```

EXAMPLE #19

To display the status of the NFSv4 service and the pNFS service, type:

```
$ server_nfs server_2 -v4
server_2 :
-------------- nfsv4 server status ---------------
             * Service Started *
             * pNFS service Started *
```

```
             * (yet operating) *

-------- NFSv4 Clients --------
Confirmed Clients : 1
UnConfirmed Clients : 0
Number of users : 0
Number of lock owners : 0
Longest List : 0
Shortest List : 0
Greatest depth to date : 0
Average List length : 0.00
Domain Name : Not Defined

---------------------------------
--------- NFSv4 State --------
Opens : 4
Locks : 0
Delegations: 4
Layouts : 0
Free : 524280

---------------------------------
-------------------------------------------------
```

Where:

Value	Definition
Confirmed Clients	Active client (ready to work).
UnConfirmed Clients	Client in the process to establishing context.
Number of users	To be removed in non-debug images.
Longest List	To be removed in non-debug images.
Shortest List	To be removed in non-debug images.
Greatest depth to date	To be removed in non-debug images.
Average List length	To be removed in non-debug images.
Opens	Number of open files.
Locks	Number of locks being held.
Delegations	Number of granted delegations.
Free	To be removed in non-debug images.

EXAMPLE #20

To display all NFSv4 clients, type:

```
$ server_nfs server_2 -v4 -client -list
server_2 :
------------ nfsv4 server client list ------------
hostname/ip : Index
NFSCLIENT1.nasdocs.emc.com : 0xa5400000
-------------------------------------------------
```

EXAMPLE #21

To display the attributes of the NFSv4 client as specified by the index, type:

```
$ server_nfs server_2 -v4 -client -info index=0xa5400000
server_2 :
NFSCLIENT1.nasdocs.emc.com : 0xa5400000
user: nfsuser1 : inode# 81
```

EXAMPLE #22

To release the client ID of the client specified by the index, type:

```
$ server_nfs server_2 -v4 -client -release index=0xa5400000
server_2 : done
```

EXAMPLE #23

To disable the NFSv4 service on server_2, type:

```
$ server_nfs server_2 -v4 -service -stop
server_2 : done
```

EXAMPLE #24

To display all NFS statistics, type:

```
$ server_nfs {<movername>|ALL} -stats

server_2 :
Server nfs (v2):
proc    ncalls    %totcalls       ms/call      failures
null      10        100.0           0.0           0
getattr    0        0.0             0.0           0
setattr    0        0.0             0.0           0
root       0        0.0             0.0           0
lookup     0        0.0             0.0           0
readlink   0        0.0             0.0           0
read       0        0.0             0.0           0
wrcache    0        0.0             0.0           0
write      0        0.0             0.0           0
create     0        0.0             0.0           0
remove     0        0.0             0.0           0
rename     0        0.0             0.0           0
link       0        0.0             0.0           0
symlink    0        0.0             0.0           0
mkdir      0        0.0             0.0           0
rmdir      0        0.0             0.0           0
readdir    0        0.0             0.0           0
fsstat     0        0.0             0.0           0

Server nfs (v3):
proc       ncalls  %totcalls       ms/call   failures
v3null        0        0.0          0.0        0
v3getattr    233843    78.8         0.0        0
v3setattr     0        0.0          0.0        0
v3lookup     37130     12.5         0.0        0
v3access     11828     4.0          0.0        0
v3readlink    0        0.0          0.0        0
v3read       120       0.0          0.0        0
v3write       0        0.0          0.0        0
v3create      0        0.0          0.0        0
v3mkdir       0        0.0          0.0        0
v3symlink     0        0.0          0.0        0
v3mknod       0        0.0          0.0        0
v3remove      0        0.0          0.0        0
v3rmdir       0        0.0          0.0        0
v3rename      0        0.0          0.0        0
v3link        0        0.0          0.0        0
v3readdir    4005      1.3          0.0        0
v3readdirplus 0        0.0          0.0        0
v3fsstat     4878      1.6          0.0        0
v3fsinfo     4878      1.6          0.0        0
v3pathconf    0        0.0          0.0        0
v3commit      0        0.0          0.0        0

Server nfs (v4):
proc      ncalls   %totcalls      ms/call    failures
v4null      4         0.0          3.0          0
v4compound 33645      48.8         0.1          0
v4reserved 0          0.0          0.0          0
v4access   217        0.3          0.0          0
v4close    44         0.1          0.0          0
v4commit   0          0.0          0.0          0
v4create   0          0.0          0.0          0
v4delegPrg 0          0.0          0.0          0
v4delegRet 30         0.0          0.0          0
```

```
v4getAttr 858         1.2          0.1          0
v4getFh   220         0.3          0.0          0
v4link     0          0.0          0.0          0
v4lock     0          0.0          0.0          0
v4lockT    0          0.0          0.0          0
v4lockU    0          0.0          0.0          0
v4lookup 171          0.2          0.0         37
v4lookupp  0          0.0          0.0          0
v4nVerify  0          0.0          0.0          0
v4open    48          0.1          8.2         37
v4openAttr 0          0.0          0.0          0
v4open_Conf 5         0.0          0.0          0
v4open_DG  0          0.0          0.0          0
v4putFh   1305        1.9          0.0          0
v4putpubFh 0          0.0          0.0          0
v4putrootFh3          0.0          0.0          0
v4read     1          0.0          0.0          0
v4readDir 21          0.0          0.6          0
v4readLink 0          0.0          0.0          0
v4remove  30          0.0          2.9          0
v4rename   2          0.0          0.0          0
v4renew  32335        46.9         0.0          2
v4restoreFh 0         0.0          0.0          0
v4saveFh  2           0.0          0.0          0
v4secInfo 0           0.0          0.0          0
v4setAttr 39          0.1          0.7          0
v4setClntid 2         0.0          0.0          0
v4clntid_Conf 2       0.0          0.0          0
v4verify  0           0.0          0.0          0
v4write  24           0.0          5.7          0
v4rel_Lockown 0       0.0          0.0          0
v4backChanCtl 0       0.0          0.0          0
v4bindConn 0          0.0          0.0          0
v4exchangeId  0       0.0          0.0          0
v4createSess 0        0.0          0.0          0
v4destroySess 0       0.0          0.0          0
v4freeStateid 0       0.0          0.0          0
v4getDirDeleg 0       0.0          0.0          0
v4getDevInfo 0        0.0          0.0          0
v4getDevList 0        0.0          0.0          0
v4layoutCmmt 0        0.0          0.0          0
v4layoutGet 0         0.0          0.0          0
v4layoutRet 0         0.0          0.0          0
v4secinfoNoName 0     0.0          0.0          0
v4sequence 0          0.0          0.0          0
v4setSsv  0           0.0          0.0          0
v4testStateid 0       0.0          0.0          0
v4wantDeleg 0         0.0          0.0          0
v4destroyClid 0       0.0          0.0          0
v4reclaimCmpl 0       0.0          0.0          0
v4illegal   0         0.0          0.0          0

Server lookupcache:
nHit nFind nNegadd nChecked
39459 46408 21 39459

Server rpc:
ncalls nBadRpcData nDuplicates nResends nBadAuths
822126

Where:
Value            Definition
ncalls           Number of calls per NFS operation.
%totcalls        Percentage of calls per operation out of total NFS calls
                 received.
ms/calls         Average time taken for the NFS operations.
failures         Number of NFS failures per NFS operation.
nHit             'Directory name lookup cache' hits.
nFind            'Directory name lookup cache' operations.
nNegadd          Number of negative entries added to the 'Directory name
                 lookup cache.'
```

```
nChecked          'Directory name lookup cache' entries searched.
nBadRpcData       Calls with bad RPC header.
nDuplicate        Calls with duplicate XID.
nResends          Number of RPC replies resent.
nBadAuths         Number of replies failing RPC authentication.
```

EXAMPLE #25

To display RPC statistics, type:

```
$ server_nfs server_2 -stats -rpc
server_2 :
Server rpc:
ncalls   nBadRpcData nDuplicates  nResends  nBadAuths
822155        0           0          0          0
```

EXAMPLE #26

To reset statistics counters, type:

```
$ server_nfs {<movername>|ALL} -stats -zero
server_2 : done
server_3 : done
server_4 : done
server_5 : done
```

Last Modified: April 15, 2011 06:05 pm

server_nis

Manages the Network Information Service (NIS) configuration for the specified Data Movers.

SYNOPSIS

```
server_nis {<movername>|ALL}
        [-add] <domainname> {<hostname>|<ip_addr>}[,{<hostname>|<ip_addr>},...} ]
        | -delete [-all | <domainname>]
        | -query [<domainname>]
         {<hostname>|<ip_addr>}[,{<hostname>|<ip_addr>},...]
        | -status [-all | <domainname>]
```

DESCRIPTION
----

server_nis provides connectivity to the NIS lookup servers and supports multiple domain configuration.

The ALL option executes the command for all Data Movers.

OPTIONS

No arguments
Displays the NIS configuration.

[-add] <domainname>
{<hostname>|<ip_addr>}[,{<hostname>|<ip_addr>},...}]
Creates a list of up to 10 IP addresses to be used as NIS lookup servers for the specified <domainname>. Up to 10 NIS servers in a single NIS domain are supported for each Data Mover. A Data Mover can support only one NIS domain until -add option is used.

The -add option allows the user to extend the domain configuration to add more than one domain. The server_nis command's behavior is constant until the -add option is used.

Each time an NIS domain is already configured, the previous configuration is overwritten.

To provide high availability, you must configure the environment with multiple NIS servers. If you configure only a single NIS server, any problem with this NIS server could result in data unavailability.

-delete
Deletes all the NIS lookup servers.

-query [<domainname>]
{<hostname>|<ip_addr>}[,{<hostname>|<ip_addr>},...]
Support of multiple domain allows the application running on the Data Mover to query the NIS client for an object of a specified domain. This option is used when the application or service works for a VDM configured with a specific NIS domain.

The -query option queries the NIS lookup servers using either a <hostname> or IP address. Multiple hostnames or IP addresses can be queried. Prior to performing a query, the NIS server must be activated and operational.

-status [-all | <domainname>]
Displays the status of the NIS lookup server even when the NIS server is not operational.

Note: The server_nis command requires the user to specify the domain name when more than one domain is configured for the -delete and -status options,

and when performing a basic query testing.

SEE ALSO

Configuring VNX Naming Services and server_dns.

EXAMPLE #1

To provide connectivity to the NIS lookup server for the specified domain, type:

```
$ server_nis server_2 nasdocs 172.24.102.30
server_2 : done
```

EXAMPLE #2

To query NIS lookup servers using both a hostname and IP address, type:

```
$ server_nis server_2 -query test40,172.24.102.36,test44
server_2 :
test40 = 172.24.102.30
test46 = 172.24.102.36
test44 = 172.24.102.34
```

EXAMPLE #3

To display the NIS configuration, type:

```
$ server_nis server_2
server_2 : yp domain=nasdocs server=172.24.102.30
```

EXAMPLE #4

To display the status of the NIS lookup servers, type:

```
$ server_nis server_2 -status
server_2 :
NIS default domain: nasdocs
NIS server 172.24.102.30
```

If NIS was not started, the output of this command will appear as:

```
$ server_nis server_2 -status
server_2 :
NIS not started
```

EXAMPLE #5

To delete all of the NIS lookup servers for a Data Mover, type:

```
$ server_nis server_2 -delete
server_2 : done
```

EXAMPLE #6

To configure the first domain, type:

```
$ server_nis server_2 emclab 192.168.67.11
server_2 : done

$ server_nis server_2
server_2 :
yp domain=emclab server=192.168.67.11
```

EXAMPLE #7

To configure the second domain, type:

```
$ server_nis server_2 -add eng 192.168.67.13
server_2 : done
```

Note: This operation requires the usage of -add option, otherwise the first domain is overwritten.

EXAMPLE #8

To query for the current configuration, type:

```
$ server_nis server_2
server_2 :
yp domain=emclab server=192.168.67.11
yp domain=emceng server=192.168.67.13
```

EXAMPLE #9

To query for all domain status, type:

```
$ server_nis server_2 -status -all
server_2 :
NIS emclab context (5):
        Servers:
          192.168.67.11 Online (current server)
RPC failure
NIS eng context (4):
        Servers:
          192.168.67.13 RPC failure (current server)
```

EXAMPLE #10

To delete a domain, type:

```
$ server_nis server_2 -delete emclab
server_2 : done
```

Note: Once multiple domains are configured, the delete operation requires specifying the domain name.

EXAMPLE #11

To perform a single query on a particular domain, type:

```
$ server_nis server_2 -query emclab bbvm
server_2 :
bbvm = 192.168.67.237
```

Last modified: October 24, 2011 10:40 a.m.

server_nsdomains

Allows the user to manage the domain configuration per Data Mover or VDM.

SYNOPSIS

```
server_nsdomains <movername> [options]
'options' can be one of the following:
| {-enable | -disable}
| -set -resolver <resolver>=<value> [-resolver <resolver>=<value>...]
| -unset -resolver <resolver> [-resolver <resolver>...]
```

DESCRIPTION
----

server_nsdomains enables/disables the VDM domain configuration and sets/unsets the resolver.

OPTIONS

{-enable | -disable}
Enables and disables the domain configuration for a physical Data Mover or VDM. By default, the domain configuration is disabled (the VDM works with the Data Mover resolver rules).

Enabling the domain configuration confines the resolution to the physical Data Mover or VDM:

* Local files (nsswitch, hosts, password, group, and netgroups) replace the files located in the Data Mover root filesystem.
* The NIS, LDAP, and DNS resolver are limited to the domains specified for the physical Data Mover or VDM.

The domain configuration includes the NIS, LDAP, and DNS domains specifications. The supported resolver keywords are 'NIS', 'LDAP', 'NFSV4', and 'DNS'. The value for each keyword is the appropriate domain name.

-set -resolver <resolver>=<value> [-resolver <resolver>=<value>...]
The -set option is used to set the resolver value. If the domain is not specified for a resolver, the usage of such revolver is disabled for the physical Data Mover or VDM, when the domain configuration is enabled.

-unset -resolver <resolver> [-resolver <resolver>...]
The -unset option allows clearing a domain name resolver.

EXAMPLE #1

To query the domain configuration, type:

```
$ server_nsdomains vdm1
vdm1 :
NSDOMAINS CONFIGURATION = Disabled
```

Note: By default, the domain configuration is disabled.

EXAMPLE #2

To enable the domain configuration, type:

```
$ server_nsdomains vdm1 -enable
vdm1 : done

Note: By default, there is no domain configured. With such configuration,
any NIS, LDAP, or DNS query will fail. Only local host resolution works
assuming a hosts file exists in the physical Data Mover or VDM root
filesystem.

$ server_nsdomains vdm1
vdm1 :
NSDOMAINS CONFIGURATION = Enabled
NIS domain : <not defined>
LDAP domain : <not defined>
DNS domain : <not defined>
NFSV4 domain : <not defined>

EXAMPLE #3
----------

To set the NIS domain for the VDM vdm1, type:

$ server_nsdomains vdm1 -set -resolver NIS=emclab
vdm1 : done

EXAMPLE #4
----------

To unset the NIS domain for the VDM vdm1, type:

$ server_nsdomains vdm1 -unset -resolver NIS=emclab
vdm1 : done

EXAMPLE #5
----------

To set all domains for the VDM vdm1, type:

$ server_nsdomains vdm1 -set -resolver NIS=emclab -resolver
LDAP=eng.lcsc -resolver DNS=italie.lcsc -resolver
NFSV4=eng.emc.com
vdm1 : done

EXAMPLE #6
----------

To view the domain configuration for the VDM vdm1, type:

$ server_nsdomains vdm1
vdm1:
NSDOMAINS CONFIGURATION = Enabled
NIS domain : emclab
LDAP domain : eng.lcsc
DNS domain : italie.lcsc
NFSV4 domain : eng.emc.com


--------------------------------------------------------------------------------

Date created: October 20, 2011 1:00 p.m.
```

server_param

Manages parameter information for the specified Data Movers.

SYNOPSIS

```
server_param {<movername>|ALL}
  -info
| -facility {<facility_name>|-all}
   {-list|-info {<param_name> [-verbose]|-all}
| -facility <facility_name> -modify <param_name> -value <new_value>
```

DESCRIPTION
----

server_param modifies and displays information on parameters for the VNX.

The ALL option executes the command for all Data Movers.

OPTIONS

-info
Displays the facilities that can be modified.

-facility {<facility_name>|-all} -list
Lists all of the parameters with their default, current, and configured values for the specified facility or facilities.

-facility {<facility_name>|-all} -info{<param_name>
[-verbose]|-all}
Displays information on the specified parameter or parameters for the specified facility or facilities. The -verbose option displays a more detailed description, if one is available, for the specified <param_name>.

-facility <facility_name> -modify <param_name>
-value <new_value>
Modifies the value of the specified <param_name> and changes the value to <new_value>. The <facility_name> is the facility for <param_name>.

SEE ALSO

Parameters Guide for VNX for File.

EXAMPLE #1

To view all facilities that can be modified on server_2, type:

```
$ server_param server_2 -info
server_2 :
facility        description
FLRCompliance   File Level Retention
NDMP            Network Data Management Protocol
PAX             Data Backup/Restore facility for NDMP/server_archive
RCP             Replication Control Protocol
arp             Address Resolution Protocol
cfs             Common File System
cifs            Common internet FS (Microsoft Network File Protocol)
config          Misc. configuration parameters for CIFS
cvfs            Celerra Virtual File System
dedupe          DEDUPE:File-level Data Deduplication
```

```
dns             Domain Name Service
ds              Domain Controller service for CIFS
fcTach          Agilent Fibre Channel Controller
file            Overall file system parameters
filesystem      File system
ftpd            File Transfer Protocol Daemon
http            Hypertext Transfer Protocol
ip              Internet protocol
iscsi           Internet Scsi Protocol
kernel          THREADs deadlock detection
ldap            Lightweight Directory Access Protocol
lockd           Network Lock Manager
lockmgr         CFS Lock Manager
mount           NFS Mount Protocol
nbs             Network Block Service Protocol
nfs             Network File System
nfsv4           NFS version 4 protocol
quota           File system quota management
security        Security/Credential parameters
shadow          Cross Protocol naming support
ssl             SSL security network protocol
statd           Host status demon
statmon         Statistics Framework
streamio        Streaming tape I/O support
tcp             Transmission Control Protocol
tftp            Trivial File Transfer Protocol Daemon
trunk           Network trunking support
ufs             Dart native file system
usrmap          User name mapping support
vbb             Volume Based Backup
vdevice         Virtual IP Device Parameters
viruschk        Virus checking service
```

Where:

Value	Definition
facility	Facility for the parameter.
description	Description of the facility.

EXAMPLE #2

To view the PAX parameters that can be modified, type:

```
$ server_param server_2 -facility PAX -list
server_2 :
param_name                      facility  default     current   configured
checkUtf8Filenames                 PAX          1           1
dump                               PAX          0           0
nPrefetch                          PAX          8           8
nThread                            PAX         64          64
writeToArch                        PAX          1           1
paxReadBuff                        PAX         64          64
writeToTape                        PAX          1           1
filter.numDirFilter                PAX          5           5
paxWriteBuff                       PAX         64          64
filter.numFileFilter               PAX          5           5
filter.dialect                     PAX         ''          ''
nFTSThreads                        PAX          8           8
paxStatBuff                        PAX        128         128
readWriteBlockSizeInKB             PAX         64          64
nRestore                           PAX          8           8
filter.caseSensitive               PAX          1           1
scanOnRestore                      PAX          1           1
noFileStreams                      PAX          0           0
allowVLCRestoreToUFS               PAX          0           0
```

Where:

Value	Definition
param_name	Name of the parameters with the specified facility that can be

```
            modified.
facility    Facility for the parameters.
default     Default value for the parameter.
current     Current value used by the Data Mover.
configured  Value set by the user. If some user action is pending (such as
            a Data Mover reboot), it might not have taken effect.

If the values for current and configured differ, refer to the
user_action field of the -info.

EXAMPLE #3
----------

To view information on the nThread parameter, type:

$ server_param server_2 -facility PAX -info nThread
server_2 :
name                     = nThread
facility_name            = PAX
default_value            = 64
current_value            = 64
configured_value         =
user_action              = none
change_effective         = immediate
range                    = (1,128)
description              = Number of worker threads per backup session

Where:

Value              Definition
facility_name      Facility for the parameter.
default_value      Default value set for the parameter.
current_value      Value set on the Data Mover.
configured_value   Value set by the user. If some user action is pending (such
                   as a Data Mover reboot), it might not have taken effect.
user_action        Action necessary for the parameter to take effect.
change_effective   States when the change will be effective.
range              Range of possible parameter values.
description        Description of what the parameter does.

EXAMPLE #4
----------

To modify the configured nThread parameter, type:

$ server_param server_2 -facility PAX -modify nThread -value 32
server_2 : done

EXAMPLE #5
----------

To modify the configured cipher parameter, type:

$ server_param server_2 -facility PAX -modify cipher
-value foobar

server_2 : done
Warning 17716815750: server_2 : You must reboot server_2 for paxReadBuff
changes
to take effect.
To verify the configured cipher parameter, restart the Data Mover and
type:

$ server_param server_2 -facility ssl -info cipher

server_2 :
name = cipher
facility_name = ssl
default_value = ALL:!ADH:!SSLv2:@STRENGTH
current_value = ALL:!ADH:!SSLv2:@STRENGTH
configured_value = foobar
```

```
user_action = reboot DataMover
change_effective = reboot DataMover
range = *
description = Keyword specifying the default supported SSL cipher suites (e.g:
ALL:!LOW:@STRENGTH)

Note: If the current_value and configured_value parameters differ and if the
user_action and change_effective parameters display the text reboot Data
Mover, restart the Data Mover. After restarting the Data Mover, if the
current_value and configured_value parameters continue to differ, it
indicates that the Data Mover encountered an error after it was restarted.
Check the server_log output to view the error reported.
To view the server_log command output file, type:

$ server_log server_2 | grep param

...
2009-08-25 12:20:59: ADMIN: 3: Command failed: param ssl cipher=foobar
...

EXAMPLE #6
----------

To view the values of the NDMP port ranges on the Data Mover server_2, type:

$ server_param server_2 -facility NDMP -info portRange
server_2 :
name                    = portRange
facility_name           = NDMP
default_value           = 1024-65535
current_value           = 1024-65535
configured_value        =
user_action             = none
change_effective        = immediate
range                   = 1024-65535
description             = Port range for NDMP data connection listening

EXAMPLE #7
----------

To set the values of the NDMP port ranges on the Data Mover server_2, type:

$ server_param server_2 -facility NDMP -modify portRange -value 50000-50100
server_2 : done

EXAMPLE #8
----------

To display the parameters for the SSL facility, type:

$ server_param server_2 -facility ssl -info -all
server_2 :
name = trace
facility_name = ssl
default_value = 0x00000000
current_value = 0x00000000
configured_value =
user_action = none
change_effective = immediate
range = (0x00000000,0xffffffff)
description = Define SSL traces displayed in the server log

name = timeout
facility_name = ssl
default_value = 5
current_value = 5
configured_value =
user_action = reboot DataMover
change_effective = reboot DataMover
range = (1,120)
description = Timeout (in seconds) used to receive SSL packets
```

```
from network during SSL handshake

name = protocol
facility_name = ssl
default_value = 0
current_value = 0
configured_value =
user_action = reboot DataMover
change_effective = reboot DataMover
range = (0,2)
description = Set the default ssl protocol. Possible values are:
0=all ssl/tls protocol are allowed, 1=only sslv3 is allowed, 2=only tlsv1
is allowed

name = threads
facility_name = ssl
default_value = 10
current_value = 10
configured_value =
user_action = reboot DataMover
change_effective = reboot DataMover
range = (4,30)
description = Number of SSL threads

name = cipher
facility_name = ssl
default_value = 'ALL:!ADH:!SSLv2:@STRENGTH'
current_value = 'ALL:!ADH:!SSLv2:@STRENGTH'
configured_value =
user_action = none
change_effective = reboot DataMover
range = '*'
description = Keyword specifying the default supported SSL cipher
suites (e.g: ALL:!LOW:@STRENGTH)

EXAMPLE #9
----------

To display the default SSL parameters on server_2, type:

$ server_param server_2 -facility ssl -list
server_2 :
param_name   facility    default    current    configured
trace             ssl 0x00000000 0x00000000
timeout           ssl          5          5
protocol          ssl          0          0
threads           ssl         10         10
cipher            ssl 'ALL:!ADH:!SSLv2:@STRENGTH'

EXAMPLE #10
-----------

To modify the SSL dedicated threads to 20, type:

$ server_param server_2 -facility ssl -modify threads -value 20
server_2 : done
Warning 17716815750: server_2 : You must reboot server_2 for threads
changes to take effect.

EXAMPLE #11
-----------

To modify the default cipher suite to all (except low-security
algorithms and MD5), type:

$ server_param server_2 -facility ssl -modify cipher -value
'ALL:!LOW:!MD5:@STRENGTH'

server_2 : done
Warning 17716815750: server_2 : You must reboot server_2 for cipher
changes to take effect.
```

EXAMPLE #12

To display the default ftpd parameters, type:

```
# server_param server_2 -facility ftpd -list

server_2 :
param_name      facility        default  current       configured
shortpathdir        ftpd              0        0
defaultdir          ftpd              /        /
wildcharsInDir      ftpd              0        0
bounceAttackChk     ftpd              1        1
```

EXAMPLE #13

To display the parameters for the ftpd facility, type:

```
$ server_param server_2 -facility ftpd -info -all

server_2 :
name = shortpathdir
facility_name = ftpd
default_value = 0
current_value = 0
configured_value =
user_action = none
change_effective = immediate
range = (0,1)
description = Enable return file name instead of full pathname in DIR
command

name = defaultdir
facility_name = ftpd
default_value = /
current_value = /
configured_value =
user_action = none
change_effective = immediate
range = *
description = Sets the default working directory for FTP

name = wildcharsInDir
facility_name = ftpd
default_value = 0
current_value = 0
configured_value =
user_action = none
change_effective = immediate
range = (0,1)
description = Enable wild characters for directory names

name = bounceAttackChk
facility_name = ftpd
default_value = 1
current_value = 1
configured_value =
user_action = none
change_effective = immediate
range = (0,1)
description = Enable bounce attack check
```

EXAMPLE #14

To display the detailed description of the shortpathdir parameter for the ftpd facility, type:

```
$ server_param server_2 -facility ftpd -info shortpathdir -verbose
```

```
server_2 :
name = shortpathdir
facility_name = ftpd
default_value = 0
current_value = 0
configured_value =
user_action = none
change_effective = immediate
range = (0,1)
description = Enable return file name instead of full pathname in DIR
              command

detailed_description
Enable (1) or disable (0) return file name instead of full pathname in the
commands dir or ls. if wild char are used this parameter is inefficient.

------------------------------------------------------------------------------
Last Modified: December 14, 2011 12:40 p.m.
```

server_pax

Displays and resets backup and restore statistics and file system information for a backup session already in progress.

SYNOPSIS

```
server_pax {<movername>|ALL}
 -stats [-reset|-verbose]
```

DESCRIPTION
----

server_pax displays and resets backup and restore statistics and file system information for a backup session already in progress.

OPTIONS

-stats
Displays backup and restore statistics for a backup or restore session in progress.

[-reset|-verbose]
The -reset option resets the statistics for a backup or restore session already in progress. The -verbose option displays backup performance statistics and file size distribution data for a backup or restore session in progress.

SEE ALSO

Parameters Guide for VNX for File.

EXAMPLE #1

To reset the statistics for a backup session already in progress, type:

```
$ server_pax server_2 -stats -reset
server_2 : done
```

EXAMPLE #2

To view the verbose statistics for an active NDMP backup session on server_2, type:

```
$ server_pax server_2 -stat -verbose
************** SUMMARY PAX STATS ****************
---- NASS STATS ----
nass00 is not doing backup
nass01 is not doing backup
nass02 is not doing backup
nass03 is not doing backup
---- NASA STATS ----
** nasa thid 0 (RESTORE) **
Backup root directory: /16m_ok_1_0
Total bytes processed: 12140605440
Total file processed: 738
throughput: 68 MB/sec
average file size: 16065KB
Total nasa wait nass count: 0
Total nasa wait nass time: 0 msec
Total time since last reset: 170 sec
Tape device name: /dev/c80t0l0
0 size file processed: 13
1 -- 8KB size file processed: 1
8KB+1 -- 16KB size file processed: 0
16KB+1 -- 32KB size file processed: 0
```

```
32KB+1 -- 64KB size file processed: 0
64KB+1 -- 1MB size file processed: 0
1MB+1 -- 32MB size file processed: 724
32MB+1 -- 1GB size file processed: 0
1G more size file processed: 0
fs /16m_ok_1_0 size is: 120855445504 Bytes
Estimated time remain is 1524 sec

nasa01 is not doing backup/restore
nasa02 is not doing backup/restore
nasa03 is not doing backup/restore
---- NASW STATS ----
nasw00 RESTORE (in progress)
Session Total Time: 00:02:50 (h:min:sec)
Session Idle Time: 00:00:56 (h:min:sec)
KB Tranferred: 11858820 Block Size: 61440 (60 KB)
Average Transfer Rate: 68 MB/Sec 239 GB/Hour
Average Burst Transfer: 101 MB/Sec 357 GB/Hour
__Point-in-Time__ (over the last 10 seconds):
Rate=69 MB/Sec Burst=96 MB/Sec Idle=283 msec/sec
Get Pool: 17 buffers Put Pool: 29 buffers
Compression Page not available
ReadC=0.00 WriteC=0.00 Read=0 KB Written=0 KB

nasw01 BACKUP (terminated)
nasw02 BACKUP (terminated)
nasw03 BACKUP (terminated)
```

Value	Definition
NASS STATS	Thread responsible for traversing the file system and providing metadata for each directory and/or file.
Total file processed	Total number of files and/or directories for which metadata was processed.
Total NASS wait NASA count	The number of times NASS waited for NASA.
Total NASS wait NASA time	Amount of time NASS waited for NASA.
Total time since last reset	Time since the last reset; a reset occurs automatically when a backup completes.
fts_build time	Time spent building the file system or directory tree.
getstatpool	If the value is consistently zero, then NASA may be slowing down the backup.
putstatpool	If the value is consistently zero, then NASS may be slowing down the backup.
NASA STATS	Thread responsible for writing file header information, reading file data, and writing to the buffer.
Backup root directory	Directory being backed up.
Total bytes processed	Bytes backed up since the last reset or start of the current backup.
Total file processed	Number of files backed up since the start or reset of the current backup.
Throughput	How fast NASA processed data.
Average file size	Average file size for the current backup.
Total nasa wait nass count time	Number of times NASA waited for NASS.
Total nasa wait nass time	Amount of time NASA waited for NASS.
Total time since last reset	Amount of time since the backup statistics were reset; a reset occurs automatically when a backup completes.
Tape device name	Target device for the backup data.
File size statistics	Statistics on the size of files backed up since the start or reset of the current backup.
NASW STATS	Thread responsible for getting data from the buffer pool, writing it to tape or sending it to a remote Data Mover.
Session total time	Total time of the current session.
Session idle time	Idle time for the current session.
KB transferred	Total KB transferred.
Average transfer rate	Per second and per hour transfer rate for the current session’s data.

Average burst transfer	Burst transfer rate in MB/s and GB/s.
Write block counters (List/Direct)	Scatter/gather write count.
Point-in-time_ (over the last 10 seconds)	Information on data processed during a 10 second interval.
Rate	Transfer rate in MB/s.
Burst	Burst transfer rate in MB/s.
Idle	Amount of time NASW was idle in msec.
Get pool	Number of buffers in get pool; if value is consistently 0, then NASA and NASS may be slowing down the backup.
Put pool	Number of buffers in put pool; if value is consistently 0, then the tape may be slowing down the backup.
Compression rate retrieved	Compression rate.
ReadC	Read compression rate at the tape device.
WriteC	Write compression rate at the tape device.
Read	Amount of data read in KB.
Written	Amount of data written in KB.

EXAMPLE #3

To view the verbose statistics for an active NDMP restore session on server_2, type:

```
$ server_pax server_2 -stats -verbose
server_2 :
************** SUMMARY PAX STATS ****************
---- NASS STATS ----
nass00 is not doing backup
nass01 is not doing backup
nass02 is not doing backup
nass03 is not doing backup
---- NASA STATS ----
** nasa thid 0 (non-DAR RESTORE) **
The first five entries of restore name list are:
original name: /filt, destination name /ufsvbbr/r_filter_pax
Total bytes processed: 172326912
Total file processed: 42
throughput: 7 MB/sec
average file size: 4006KB
Total nasa wait nass count: 0
Total nasa wait nass time: 0 msec
Total time since last reset: 21 sec
Tape device name: c0t0l1
dir or 0 size file processed: 17
1 -- 8KB size file processed: 6
8KB+1 -- 16KB size file processed: 18
16KB+1 -- 32KB size file processed: 0
32KB+1 -- 64KB size file processed: 0
64KB+1 -- 1MB size file processed: 1
1MB+1 -- 32MB size file processed: 0
32MB+1 -- 1GB size file processed: 0
1G more size file processed: 0

nasa01 is not doing backup/restore
nasa02 is not doing backup/restore
nasa03 is not doing backup/restore
---- NASW STATS ----
nasw00 RESTORE (in progress)
Session Total Time: 00:00:21 (h:min:sec)
Session Idle Time: 00:00:00 (h:min:sec)
KB Tranferred: 168384 Block Size: 32768 (32 KB)
Average Transfer Rate: 7 MB/Sec 27 GB/Hour
Average Burst Transfer: 7 MB/Sec 27 GB/Hour
__Point-in-Time__ (over the last 10 seconds):
Rate=6 MB/Sec Burst=7 MB/Sec Idle=0 msec/sec
Get Pool: 61 buffers Put Pool: 0 buffers
```

```
nasw01 No session found
nasw02 No session found
nasw03 No session found

--------------------------------------
Last Modified: April 19, 2011 5:50 pm
```

server_ping

Checks the network connectivity for the specified Data Movers.

SYNOPSIS
---

server_ping {<movername>|ALL}
 [-send|-interface <interface>] {<hostname>|<ip_addr>}

DESCRIPTION
---

server_ping sends ICMP ECHO_REQUEST messages to a remote host and receives ICMP ECHO_REPLY messages back that verify and determine network connectivity.

As the transmission time is stamped into the ECHO_REQUEST message, server_ping displays the transmission delay between the Data Mover and the remote host.

The ALL option executes the command for all Data Movers.

OPTIONS

<hostname>|<ip_addr>
Provides a one-shot test to determine if the specified <hostname> or <ip_addr> is accessible. The IP address may be used if NIS or DNS or both are not operational.

-send {<hostname>|<ip_addr>}
Repeats sending/receiving ECHO_REQUEST messages to a specified <hostname> or <ip_addr> for a remote host. The IP address may be used if NIS or DNS or both are not operational.

Press Ctrl+C to quit.

When invoking server_ping with a hostname, the /.etc/hosts file is checked first, then the NIS and the DNS servers (if operational on the Data Mover) are checked. If the name does not exist in either case, an error message is displayed.

-interface {<interface> <hostname>|<ip_addr>}
Uses the address of the specified interface name as the source address. If a route to the destination exists with a matching source interface, the requests will also be transmitted out the specified interface. Otherwise, the routing table will be searched for the most specific route to the destination and the interface associated with this route will be used to transmit the request.

SEE ALSO

Configuring and Managing Networking on VNX, server_dns, server_ifconfig, and server_nis.

EXAMPLE #1

To display connectivity from a Data Mover to the outside world, type:

$ server_ping server_2 172.24.102.2

server_2 : 172.24.102.2 is alive, time= 0 ms

or

$ server_ping server_2 172.24.102.5

server_2 :

```
Error 6: server_2 : No such device or address no answer from 172.24.102.5
```

EXAMPLE #2

To display connectivity for a Data Mover to the outside world while sending continuous ECHO_REQUEST messages, type:

```
$ server_ping server_2 -send 172.24.102.2

server_2 :
172.24.102.2 is alive, time= 0 ms
172.24.102.2 is alive, time= 0 ms
172.24.102.2 is alive, time= 0 ms
172.24.102.2 is alive, time= 0 ms
172.24.102.2 is alive, time= 0 ms
172.24.102.2 is alive, time= 0 ms
172.24.102.2 is alive, time= 0 ms
172.24.102.2 is alive, time= 0 ms
172.24.102.2 is alive, time= 0 ms
172.24.102.2 is alive, time= 3 ms
172.24.102.2 is alive, time= 0 ms
172.24.102.2 is alive, time= 0 ms
172.24.102.2 is alive, time= 0 ms
```

EXAMPLE #3

To display connectivity from a Data Mover to the outside world using the specified interface, type:

```
$ server_ping server_2 -interface cge0 172.24.102.2

server_2 : 172.24.102.2 is alive, time= 0 ms
```

Last modified: April 18, 2011 2:00 pm

server_ping6

Checks the IPv6 network connectivity for the specified Data Movers.

SYNOPSIS
----

server_ping {<movername>|ALL}
[-send|-interface <interface>] {<hostname>|<ip6_addr>}

DESCRIPTION
----

server_ping6 sends ICMP6_ECHO_REQUEST messages to a remote host and receives ICMP6_ECHO_REPLIES messages back that verify and determine network connectivity.

As the transmission time is stamped into the ECHO_REQUEST message, server_ping6 displays the transmission delay between the Data Mover and the remote host.

The ALL option executes the command for all Data Movers.

OPTIONS

<hostname>|<ip6_addr>
Provides a one-shot test to determine if the specified <hostname> or <ip6_addr> is accessible. The IP address may be used if NIS or DNS or both are not operational.

-send {<hostname>|<ip6_addr>}
Repeats sending/receiving ECHO_REQUEST messages to a specified <hostname> or <ip6_addr> for a remote host. The IP address may be used if NIS or DNS or both are not operational. Press Ctrl C to quit.

When invoking server_ping6 with a hostname, the /etc/hosts file is checked first, then the NIS and the DNS servers (if operational on the Data Mover) are checked. If the name does not exist in either case, an error message is displayed.

Note: server_ping6 requests sent to a link-local or multicast destination address require the specification of a '%<intf>' scope identifier.

-interface {<interface> <hostname>|<ip6_addr>}
Provides a one-shot test to determine if the specified <hostname> or

<ip6_addr> is accessible through the specified <interface>. The IP address may be used if NIS or DNS or both are not operational.

EXAMPLE #1

To verify Data Mover connectivity from cge0_int1 on server_2 to a client at 3ffe:0000:3c4d:0015:0435:0200:0300:00aa, type:

```
$ server_ping6 server_2 -interface cge0_int1
3ffe:0000:3c4d:0015:0435:0200:0300:00aa

server_2 : 3ffe:0000:3c4d:0015:0435:0200:0300:00aa is alive, time= 0 ms
```

or

```
$ server_ping6 server_2 -interface cge0_int1

3ffe:0000:3c4d:0015:0435:0200:0300:00aa

server_2 :

Error 6: server_2 : No such device or address no answer from client.
```

EXAMPLE #2

To ping link-local address fe80::260:16ff:fe0c:205%cge0_0000_ll, type:

```
$ server_ping6 server_2 fe80::260:16ff:fe0c:205%cge0_0000_ll

server_2 : fe80::260:16ff:fe0c:205%cge0_0000_ll is alive, time= 0 ms
```

or

```
$ server_ping6 server_2 fe80::260:16ff:fe0c:205%cge0_0000_ll

server_2 :

Error 6: server_2 : No such device or address no answer from client
```

EXAMPLE #3

To ping multicast address ff02::1%cge0_0000_ll, type:

```
$ server_ping6 server_2 ff02::1%cge0_0000_ll

server_2 : ff02::1%cge0_0000_ll is alive, time= 0 ms
```

or

```
$ server_ping6 server_2 ff02::1%cge0_0000_ll

server_2 :

Error 6: server_2 : No such device or address no answer from client.
```

Last modified: April 18, 2011 1:15 pm.

server_rip

Manages the Routing Information Protocol (RIP) configuration for the specified Data Movers.

SYNOPSIS

```
server_rip {<movername>|ALL}
        status
| {{ripin|noripin}{<interface_name>[,...]}}
```

DESCRIPTION
----

server_rip disables and enables RIP processing for the specified interfaces on a specified Data Mover. server_rip displays the status of the RIP configuration for the specified Data Mover. By default, all the interfaces process RIP messages. server_setup provides information to start and stop RIP for a Data Mover.

The ALL option executes the command for all Data Movers.

OPTIONS
----

status
Displays the status of the routed daemon, RIP processing, and the interfaces with RIP processing disabled.

ripin|noripin {<interface_name>[,...]}
ripin enables RIP processing and noripin disables RIP processing for the specified interfaces.

SEE ALSO
----

server_setup

EXAMPLE #1

To disable an interface for RIP processing, type:

```
$ server_rip server_2 noripin cge0

server_2 : done
```

EXAMPLE #2

To display the status of RIP processing, type:

```
$ server_rip server_2 status

server_2 :
routed started, RIP processing is on
        cge0 (172.24.102.237), ifp = 0x8d6a178, disabled
```

EXAMPLE #3

To enable RIP processing for a specified interface, type:

```
$ server_rip server_2 ripin cge0

server_2 : done
```

Last modified: May 12, 2011 12:05 pm.

server_route

Manages the routing table for the specified Data Movers.

SYNOPSIS

```
server_route {<movername>|ALL}
  -flush
| -DeleteAll
| -list
| {-add|-delete} default <gateway>
| {-add|-delete} [host|net] <dest> <gateway> [<netmask>]
```

DESCRIPTION
----

server_route creates, displays, and deletes routing entries for the specified Data Movers.

The ALL option executes the command for all the Data Movers.

OPTIONS

-flush
Temporarily removes all manually configured routing entries for as long as the Data Mover remains up. System generated entries are not removed. A reboot of the Data Mover re-establishes the routing entries that were temporarily removed.

-DeleteAll
Permanently removes all the routing entries.

-list
Lists all routing entries including the subnet mask for each route.

To display the full routing table for a Data Mover, type:
$ server_netstat <movername> -r

{-add|-delete} default <gateway>
Adds or deletes a default gateway for all unspecified destinations. The Data Mover passes all packets not qualifying against a specific route entry to the specified gateway. The keyword default implies a network destination address of 0.0.0.0. A default route uses a fixed mask of 0.0.0.0.

{-add|-delete} net <dest> <gateway> [<netmask>]
Adds or deletes a routing entry to a network host as specified by the optional keyword net (default) to the specified destination and gateway address. The <dest> argument represents the network for the routing entry, while the <gateway> argument indicates the network gateway to which packets should be addressed.

The <netmask> is an optional field for network addresses. In case a netmask is not specified, a netmask will be assigned based on the class of the address and addresses assigned on the system. A <netmask> of 0.0.0.0 or 255.255.255.255 is invalid in the case of net routes. The default for the route is net.

{-add|-delete} host <dest> <gateway>
Adds or deletes a routing entry to a particular host as specified by the optional keyword host (net is the default) to the specified destination and gateway address. The <dest> argument represents the specific host for the routing entry, while the <gateway> argument indicates the network gateway to which packets should be addressed.

By default, a netmask of 255.255.255.255 is assigned to host routes.

SEE ALSO

Configuring and Managing Networking on VNX, server_netstat, and
server_ifconfig.

EXAMPLE #1

To list the routing table for server_2, type:

```
$ server_route server_2 -list
server_2 :
net 128.221.253.0 128.221.253.2 255.255.255.0 el31
net 128.221.252.0 128.221.252.2 255.255.255.0 el30
net 172.24.102.0 172.24.102.238 255.255.255.0 cge0
host 127.0.0.1 127.0.0.1 255.255.255.255 loop
```

Where: The output displayed is as follows:

<routing type> <desintation> <gateway> <mask> <network interface>

EXAMPLE #2

To create a host routing entry, type:

```
$ server_route server_2 -add host 10.10.1.2 172.24.102.77
server_2 : done
```

EXAMPLE #3

To create a default gateway for all unspecified destinations, type:

```
$ server_route server_2 -add default 172.24.102.254
server_2 : done
```

EXAMPLE #4

To add a network routing entry with a netmask, type:

```
$ server_route server_2 -add net 192.162.0.0 172.24.102.238 255.255.0.0
server_2 : done
```

EXAMPLE #5

To delete a routing entry, type:

```
$ server_route server_2 -delete host 10.10.1.2 172.24.102.77
server_2 : done
```

EXAMPLE #6

To flush the routing table for a Data Mover, type:

```
$ server_route server_2 -flush
server_2 : done
```

EXAMPLE #7

To delete all routing entries, type:

```
$ server_route server_2 -DeleteAll
server_2 : done
```

Last Modified: April 20, 2011 1:15 pm

server_security

Manages the security policy settings for the specified Data Movers. The VNX provides support for Group Policy Objects (GPOs) by retrieving and storing a copy of the GPO settings for each CIFS server joined to a Windows domain.

SYNOPSIS

```
server_security {<movername>|ALL}
{{-update|-info} -policy gpo[server=<server_name>|
   domain=<domain_name>]
| {-add|-delete|-modify} -policy chap -name <client_name>
| -info -policy chap [-name <client_name>]}
```

DESCRIPTION

server_security queries or updates the security policy settings on the Data Movers. The VNX stores the GPO settings in a GPO cache on the Data Mover. When you start the CIFS service on a Data Mover, the VNX reads the settings stored in the GPO cache, and then retrieves the most recent GPO settings from the Windows domain controller.

The ALL option executes the command for all Data Movers.

OPTIONS

-update -policy gpo
Forces an update of the specified GPO settings.

-info -policy gpo
Queries and displays the GPO settings.

[server=<server_name>|domain=<domain_name>]
Limits operations to the specified server or domain. The server refers to the configured CIFS servers on the Data Movers. If no server or domain has been specified, the operations are performed to all configured servers.

-add -policy chap -name <client_name>
Adds a new Challenge Handshake Authentication Protocol (CHAP) security for the specified <client_name>.

-delete -policy chap -name <client_name>
Deletes a CHAP security for the specified <client_name>.

-modify -policy chap -name <client_name>
Modifies a CHAP security setting for the specified <client_name>.

-info -policy chap [-name <client_name>]
Displays current CHAP security settings for the Data Mover, or the specified <client_name>.

SEE ALSO

Configuring iSCSI Targets on VNX for File, server_iscsi, and server_checkup.Configuring iSCSI Targets on VNX for File, server_iscsi, and server_checkup.

EXAMPLE #1

To query the GPO settings for all configured CIFS servers, type:

```
$ server_security server_2 -info -policy gpo
server_2 :
Server compname: dm102-cge0
Server NetBIOS: DM102-CGE0
Domain: nasdocs.emc.com
Kerberos Max Clock Skew (minutes): 5
LAN Manager Auth Level: Not defined
Digitally sign client communications (always): Not defined
Digitally sign client communications (if server agrees): Not defined
Digitally sign server communications (always): Not defined
Digitally sign server communications (if client agrees): Not defined
Send unencrypted password to connect to third-party SMB servers: Not defined
Disable machine account password changes: Not defined
Maximum machine account password age: Not defined
Audit account logon events: Not defined
Audit account management: Not defined
Audit directory service access: Not defined
Audit logon events: Not defined
Audit object access: Not defined
Audit policy change: Not defined
Audit privilege use: Not defined
Audit process tracking: Not defined
Audit system events: Not defined
Back up files and directories: Not defined
Restore files and directories: Not defined
Bypass traverse checking: Not defined
Generate security audits: Not defined
Manage auditing and security log: Not defined
Access this computer from the network: Not defined
Deny access to this computer from the network: Not defined
Take ownership of files or other objects: Not defined
EMC Virus Checking: Not defined
Maximum security log size: Not defined
Restrict guest access to security log: Not defined
Retention period for security log: Not defined
Retention method for security log: Not defined
Maximum system log size: Not defined
Restrict guest access to system log: Not defined
Retention period for system log: Not defined
Retention method for system log: Not defined
Maximum application log size: Not defined
Restrict guest access to application log: Not defined
Retention period for application log: Not defined
Retention method for application log: Not defined
Disable background refresh of Group Policy: Not defined
Group Policy Refresh interval (minutes): 90
Refresh interval offset (minutes): Not defined
GPO Last Update time (local): Thu Dec 1 13:49:08 EST 2005
GPO Next Update time (local): Thu Dec 1 15:19:08 EST 2005
```

EXAMPLE #2

To add a new CHAP security for client1, type:

```
$ server_security server_2 -add -policy chap -name client1
server_2 : Enter Secret:****
done
```

EXAMPLE #3

To display CHAP information for client1, type:

```
$ server_security server_2 -info -policy chap -name client1
server_2 :
chapdb name=client1 pass=********
```

EXAMPLE #4

To update the GPO settings for the CIFS server, type:

```
$ server_security server_2 -update -policy gpo server=dm32-cge0
server_2 : done
```

EXAMPLE #5

To modify a password for client1, type:

```
$ server_security server_2 -modify -policy chap -name client1
server_2 : Enter New Secret:****
done
```

EXAMPLE #6

To delete CHAP security for client1, type:

```
$ server_security server_2 -delete -policy chap -name client1
server_2 : done
```

Last Modified: April 20, 2011 1:35 pm

server_setup

Manages the type and protocol component for the specified Data Movers.

SYNOPSIS

```
server_setup {<movername>|ALL}
  -type {nas|standby}
| -Load <image_name>
| -Protocol {cifs|mpfs|viruschk|rip|cdms}
    -option {start[=<n>|stop[,comment=<message>]|delete}
```

DESCRIPTION
----

server_setup sets up the type and protocol component for the specified Data Mover.

The ALL option executes the command for all Data Movers.

OPTIONS

-type {nas|standby}
Reboots the Data Mover as the type specified. When setting up a standby relationship between a primary Data Mover and a standby, make sure that the standby Data Mover has the same disk configuration as the primary.

-Load <image_name>
Loads an image onto a Data Mover. The image may be an emergency patch or debug image provided by EMC. Do not use this option unless directed by EMC customer service.

-Protocol {cifs|mpfs|viruschk|rip|cdms}
Selects the protocol configuration to be managed. Multiple protocol configurations can be enabled for a Data Mover.

Note: NFS and RIP processing are enabled by default. NFS cannot be disabled.

-option
Specifies the following options for the specified protocol:

start
Starts the specified protocol configuration. To verify whether the protocol configuration has started, use the respective server command for the protocol.

[=<n>]
Indicates the number of threads for users of CIFS, virus checker, and CDMS. The recommended number of CIFS threads for each user is based on the amount of memory on the Data Mover. If there is less than 1 GB memory, the default thread number is 96, but if the memory is greater than 1 GB, the default thread number is 256.

Note: The [=<n>] option is not available for RIP protocol configurations.

stop
Stops the protocol configuration.

[,comment=<message>]
Allows a message enclosed with quotes to be entered. The message is delivered directly to CIFS clients.

Note: The [,comment=<message>] option is not available for RIP or virus checker protocol configurations.

delete
Deletes the protocol configuration.

SEE ALSO

Configuring Standbys on VNX, server_cdms, server_checkup, server_mpfs, server_rip, server_ssh, and server_viruschk.

EXAMPLE #1

To set up a Data Mover as a standby, type:

```
$ server_setup server_3 -type standby
server_3 : reboot in progress 0.0.0.0.0.0.0.0.1.3.3.3.3.3.4. done
done
```

EXAMPLE #2

To set up a Data Mover as nas, type:

```
$ server_setup server_3 -type nas
server_3 : reboot in progress 0.0.0.0.0.0.0.0.1.1.3.3.3.3.3.4. done
server_3 : checking root filesystem ... done
done
```

EXAMPLE #3

To start the CIFS protocol for a Data Mover, type:

```
$ server_setup server_2 -Protocol cifs -option start
server_2 : done
```

EXAMPLE #4

To start the virus checker protocol for a Data Mover, type:

```
$ server_setup server_2 -Protocol viruschk -option start
server_2 : done
```

EXAMPLE #5

To stop the CIFS protocol for a Data Mover, type:

```
$ server_setup server_2 -Protocol cifs -option stop,comment="CIFS is going
down"
server_2 : done
```

EXAMPLE #6

To delete the CIFS protocol configuration for a Data Mover, type:

```
$ server_setup server_2 -Protocol cifs -option delete
server_2 : done

Warning 4037: server_2 : CIFS configurations for server_2 and VDM(s)
currently loaded on it
```

EXAMPLE #7

To load a new image onto a Data Mover, type:

```
$ server_setup server_2 -Load nas.exe
server_2: will load : nas.exe
```

Last Modified: April 20, 2011 3:30 pm

server_snmpd

Manages the Simple Network Management Protocol (SNMP)
configuration values for the specified Data Movers.

SYNOPSIS

```
server_snmpd {<movername>|ALL}
    -info
  | -service {
        -status
      | -stop
      | -start
    }
  | -modify [-location <sys_location>]
            [-contact <sys_contact>]
            [-community {-clear|<community>}]
  | -user {
        -list
      | -create <name> -authpw -privpw
      | -delete <name>
      | -modify <name> -authpw -privpw
    }
```

DESCRIPTION
----

server_snmpd controls SNMP server agent and SNMP configuration
for the specified Data Mover. The service can be started/stopped
depending on user preferences; the location and contact can be
configured. SNMPv1 community strings can be created and cleared.
SNMPv3 users can be created or deleted, and modified. The ALL
option executes the command for all Data Movers.

OPTIONS

-info
Displays information about the SNMP server agent and configuration such as
location, contact, and community.

-service {-status|-stop|-start}
Starts, stops, or displays the current SNMP service.

-modify
Modifies the configuration of the SNMP service.

        -location <sys_location>
        Modifies the location value.

        -contact <sys_contact>
        Modifies the syscontact value.

        -community {-clear|<community>}
        Modifies the community name being used by the server SNMPD
        agent and enables the SNMPv1/SNMPv2c access to the agent.
        The -clear option allows the adminstrator to disable the
        community.

-user {-list}
Lists the SNMPv3 users on the Data Mover.

        -create <name> -authpw -privpw
        Creates an SNMPv3 user on the Data Mover.

        -delete <name>
        Deletes an SNMPv3 user on the Data Mover.

-modify <name> -authpw -privpw
Modifies the SNMPv3 passwords of a user on the Data Mover. Both the passwords should be changed together.

EXAMPLE #1

To view the SNMP service properties of the Data Mover server_2, type:

```
$ server_snmpd server_2 -info
server_2:
enabled    = yes
location   = RTP, NC
contact    = John Doe
community  = public
```

EXAMPLE #2

To view the SNMP service properties of all Data movers, type:

```
$ server_snmpd ALL -info
server_2:
enabled    = yes
location   = RTP, NC
contact    = John Doe
community  = public

server_3:
enabled    = no
location   = Bangalore, India
contact    = Pandu
community  = testcommunity
```

EXAMPLE #3

To start the SNMP service on the Data Mover server_2, type:

```
$ server_snmpd server_2 -service -start
server_2:
OK
```

EXAMPLE #4

To stop the SNMP service on the Data Mover server_3, type:

```
$ server_snmpd server_2 -service -stop
server_3:
OK
```

EXAMPLE #5

To view the SNMP service status for all Data Movers, type:

```
$ server_snmpd ALL -service -status
server_2:
SNMP Running

server_3:
SNMP Stopped
```

EXAMPLE #6

To modify the location, contact, and community values on Data Mover server_2, type:

```
$ server_snmpd server_2 -modify -location "RTP, NC"
-contact "Anamika Kadian" -community public
server_2:
OK
```

EXAMPLE #7

To clear the community value on Data Mover server_2, type:

```
$ server_snmpd server_2 -modify -community -clear
server_2:
OK
```

EXAMPLE #8

To display the list of SNMPv3 users on all Data Movers, type:

```
$ server_snmpd ALL -user -list
server_2:
user
gsmith
smith

server_3:
user
clowe
```

EXAMPLE #9

To create a new user John, on Data Mover server_2, type:

```
$ server_snmpd server_2 -user -create john -authpw  -privpw
Enter the authentication password:*********
Confirm the authentication password:*********
Enter the privacy password:*********
Confirm the privacy password:*********
server_2:
OK
```

EXAMPLE #10

To delete the user John, on Data Mover server_2, type:

```
$ server_snmpd server_2 -user -delete John
server_2:
OK
```

EXAMPLE #11

To modify the passwords of the user John, on Data Mover server_2, type:

```
$ server_snmpd server_2 -user -modify John -authpw -privpw
Enter the authentication password:*********
Confirm the authentication password:*********
Enter the privacy password:*********
Confirm the privacy password:*********
server_2:
OK
```

Last modified: April 20, 2011 at 4:00 pm

server_ssh

Manages and configures the SSH server on the specified Data Mover.

SYNOPSIS

```
server_ssh serverX
-info
| -start
server_ssh serverX
      -info
    | -start
    | -stop
    | -modify {
      -banner <banner file> |
      -port <port number> |
      -threads <number> |
      -defaultdir <path> |
      -restricthomedir {enable|disable} |
      -application <application_name> |
      -cipher <keyword> |
      -compression {yes|delayed|no} |
      -mac <keyword> |
      -maxauthtries <number> |
      -passwordauthentication {enable|disable} |
      -permitemptypassword {enable|disable} |
      -pubkeyauthentication {enable|disable} |
      -timeout <seconds> }
    | -append {
      -allowhosts <ip>[,<ip>...]|
      -allowgroups <group>[,<group>...]|
      -allowusers <user>[,<user>...] |
      -denyhosts <ip>[,<ip>...] |
      -denygroups <group>[,<group>...]|
      -denyusers <user>[,<user>...] }
    | -remove {
      -allowhosts <ip>[,<ip>...] |
      -allowgroups <group>[,<group>...] |
      -allowusers <user>[,<user>...] |
      -denyhosts <ip>[,<ip>...] |
      -denygroups <group>[,<group>...]|
      -denyusers <user>[,<user>...] }
    | -generate [-type {rsa|dsa} [-keysize <bits>]]
    | -delete [-type {rsa|dsa}]
```

OPTIONS

-info
Displays the current configuration and state of the SSH server.

-info
Displays the current configuration and state of the SSH server.

-start
Starts the SSH server. Allocates the pools of SSH threads and applications threads, and listens to incoming client requests from the network.

If the SSH server is already active, an error is returned.

-stop
Stops the SSH server. Stops listening to incoming client requests, and releases all the SSH threads.

If the SSH server is inactive, an error is returned.

-modify

Modifies some configuration parameters of the SSH server. The arguments are:

[-banner <banner file>]
The file name specified should contain an absolute path on the Data Mover. The file contains an unformatted text, which is sent to the SSH clients during the authentication step of the connection. It is a welcome message. The file name is saved in the SSH configuration file on the Data Mover. If the file name is " " or an empty string, then the current file name is removed from the configuration. If the file does not exist or is unreadable, an error message is returned.

[-port <number>]
Changes the TCP lists port of the SSH server. The server must be stopped before executing this command. The SSH server listens on only one TCP port. This port number is saved in the SSH configuration file on the Data Mover. If this port is invalid, or already used by another active service of the Data Mover, an error message is returned.

[-threads <number>]
Allocates the number of threads dedicated to the SSH server. If the server is started, the threads are restarted by this action, and all the established SSH connections are disconnected. Ten default threads are started when the server starts. The supported values are from 4 to 128 threads.

[-defaultdir <path>]
Specifies the default directory of an authenticated user when the home directory is undefined or inaccessible. If the given path does not exist or is inaccessible, then an error message is returned. The default value is /or forward slash.

[-restricthomedir {enable|disable}]
Restricts authenticated users to their home directory tree, to prevent them from accessing other user's directories. If their home directory is undefined or unreachable, the login is denied. This feature is enabled, by default, for security reasons.

[-application <application_name>]
Enables or disables the specified SSH application. The names of applications are case insensitive. If an SSH client tries to execute a disabled application, the Data Mover returns an error. The applications supported by the SSH server in VNX version 7.0 are SFTP and SCP. The default applications are disabled. To disable an enabled application, use the same option with the name no<application>. For instance, use the -application nosftp option to disable SFTP.

[-cipher <keyword>]
Specifies the symmetric ciphers allowed to encrypt an SSH connection. The openSSH documentation or http://www.openssh.com provides more information on the supported ciphers algorithms. The default option is undefined, which means all these ciphers are allowed.

-compression {yes|delayed|no}
Specifies whether the compression of data inside an SSH connection is allowed, disallowed, or delayed.

The argument must be:

* yes: always compressed, if possible
* delayed: compressed after the user has authenticated successfully, if possible
* no: compression allowed

The default is delayed.

-mac <keyword>

Specifies the available Message Authentication Code or MAC algorithms to guarantee the integrity of the SSH packets on the network. The default value is undefined, which means all these algorithms are allowed.

-maxauthtries <number>
Specifies the maximum number of authentication attempts permitted per SSH connection. The supported values are between 1 and 99. The default value is 6.

-passwordauthentication {enable|disable}
Specifies whether password authentication is allowed during the establishment of an SSH connection. The parameters passwordAuthentication and pubkeyAuthentication cannot be disabled simultaneously. The default value is enabled.

-permitemptypassword {enable|disable}
When password authentication is allowed, it specifies whether the server allows login to accounts with empty password strings. The supported values are: disabled or not permitted, and enabled or permitted. The default value is disabled.

-pubkeyauthentication {enable|disable}
Specifies whether public key authentication is allowed during the establishment of an SSH connection. The options passwordAuthentication and pubkeyAuthentication cannot be disabled simultaneously.

The default value is enabled.

-timeout <seconds>
Specifies the time out in seconds before a network disconnection in case of inactivity. The default value is 900 seconds or 15 minutes. The range is from 10 to 3600 seconds.

-append
Appends to the current configuration new allowed or denied host/group/user. The arguments are:

-allowhosts <ip>[,<ip>...]
Defines new client IPs or client subnets that are allowed to connect through SSH to the Data Mover. IP is the IPv4 or IPv6 address or subnet. In case of subnets, the CIDR notation must be used (for instance: 192.168.0.0./16 or 2001:DB8::/48). Network names are ignored.

If specified, SSH connections are allowed only for clients if their IP matches one of the defined allowed hosts, that is, any client host not declared on that list receives systematically an access denied error. If a subnet is defined on the allowed hosts list, then the client IP must belong to this subnet to be allowed to connect to the Data Mover.

If no allowed host is defined, then all client hosts are allowed to connect by default.

The list of the allowed host is saved in the SSH configuration file on the Data Mover. Duplicate items of the allowed host are prohibited. The Data Mover can save up to 256 different allowed hosts.

-allowgroups <group>[,<group>...]
Defines new groups of users that are allowed to connect through SSH to the Data Mover. The group should be the name of the allowed group and numerical group ID are ignored.

If specified, SSH connections are allowed only for users whose primary or supplementary group list matches one of the defined allowed groups, which means any user not a member of one of these groups receives an access denied error.

If no allowed group is defined, then all users of the groups are allowed to connect by default.

The list of allowed groups is saved in the SSH configuration file on the Data Mover.

Duplicate names of an allowed group are prohibited.

The Data Mover can save up to 256 different allowed groups.

-allowusers <user>[,<user>...]
Defines new users that are allowed to connect through an SSH server to the Data Mover. User ID should be the name of a user; numerical user IDs are ignored. If the user is a CIFS user, the format should be user@domain or domain\user.
If specified, SSH connections are allowed only for user names that match one of the patterns, which means any user not listed as an allowed user receives an access denied error.

If no allowed user is defined, then all users are allowed to connect by default.

The list of the allowed users is saved in the SSH configuration file on the Data Mover.

Duplicate names of allowed users are prohibited.

The Data Mover can save up to 256 different allowed users.

-denyhosts <ip>[,<ip>...]
Defines new client IPs or client subnets that are disallowed to connect through th SSH server to the Data Mover. IP is the IPv4 or IPv6 address or subnet (CIDR notation) of the client host. Network names are ignored.

If specified, SSH connections are disallowed for clients whose IP address matches one of the defined denied host or IP addresses belonging to a defined subnet, which means any client host declared in that list receives systematically an access denied error.

If no denied host is defined, then all client hosts are allowed to connect. This is the default.

The list of the denied hosts is saved in the SSH configuration file on the Data Mover.

Duplicate items of denied hosts are prohibited.

The Data Mover can save up to 256 different denied hosts.

-denygroups <group>[,<group>...]
Defines new groups of users that are disallowed to connect through SSH to the Data Mover. ’group’ should be the name of the group and numerical group IDs are ignored.

If specified, SSH connections are disallowed for users whose primary group or supplementary group list matches one of the defined "denied group", i.e. any user member of one of these groups receives systematically an "Access denied" error.

If no "denied group" is defined, then all users of all groups are allowed to connect. This is the default.

The list of the denied groups is saved in the SSH configuration file on the Data Mover.

Duplicate names of "denied group" are prohibited.

The Data Mover can save up to 256 different denied groups.

-denyusers <user>

Defines a new user that is disallowed to connect through SSH to the Data Mover. "user" should be the name of the user and numerical user IDs are ignored.
If the user is a CIFS user, his format should be user@domain or domain\user.

If specified, SSH connections are disallowed for user names that match one of the patterns, that is any user listed as a denied user receives systematically an access denied error.

If no denied user is defined, then all users are allowed to connect. This is the default.

The list of the denied users is saved in the SSH configuration file on the DART.

Duplicate names of denied users are prohibited.

The Data Mover can save up to 256 different denied users.

-remove
Removes from the current configuration an allowed/denied group/user. The arguments are:

-allowhosts <ip>[,<ip>...]
Removes the specified IPs or subnets from the allowed hosts list.

-allowgroups <group>[,<group>...]
Removes the specified groups from the allowed groups list.

-allowusers <user>[,<user>...]
Removes the specified users from the allowed users list.

-denyhosts <ip>[,<ip>...]
Removes the specified IPs or subnets from the denied hosts list.

-denygroups <group>[,<group>...]
Removes the specified groups from the denied groups list.

-denyusers <user>[,<user>...]
Removes the specified users from the denied users list.

-generate
Generates new public and private keys and saves it as the host key of the Data Mover. This operation must be the first step during the setup procedure of SSH. It could take some minutes to complete.

By default, two host keys are generated: a 2048-bit RSA key and a 1024-bit DSA key.

If the host keys of the Data Mover do not exist when the SSH server is started the first time, the keys are automatically generated with the default parameters.

The host keys are stored in the /.etc directory of the Data Mover, in an inaccessible directory.

The options are:

-type {rsa|dsa} Specifies the type of key to generate. The two valid types are: rsa and dsa. If not specified, both keys types are generated.

-keysize <bits>: Specifies the number of bits of the host key. Default is 2048 bits for RSA keys and 1024 bits for DSA keys.

Valid range for RSA keys is between 2048 and 8192 bits.

The length of a DSA key must be 1024 bits exactly, as specified by FIPS 186-2.

-delete: Deletes all the generated host keys of the Data Mover. This command is useful if the administrator needs to generate new host keys.

The options are:

-type {rsa|dsa} : This argument specifies the type of key to delete. The two valid types are: rsa and dsa. If not specified, both key types are deleted.

GENERAL NOTES
---

* The allow or deny directives are processed in the following order: denyhost, allowhost, denyusers, allowusers, denygroups and finally allowgroups.

* The allow or deny directives can specify multiple items separated by a , or comma without spaces.

* It is recommended to encapsulate IPv6 address in [ ] or square brackets.

* The host keys are generated automatically when the SSH server is started for the very first time if no host key exists. In that case, both keys are generated with their default size (RSA and DSA keys each).

* VDM are not supported by the server_ssh command.

EXAMPLE #1

To display the current configuration, type:

```
$ server_ssh server_2 -info

server_2 : done

SERVICE CONFIGURATION
Port                       : 22
State                      : running
Thread count               : 4
Banner                     : /server2fs1/banner.txt
Default home directory     : /
Restrict home directory    : disabled
Application                : sftp,scp
Cipher                     : <all>
Compression                : delayed
Mac                        : <all>
Max authentication tries   : 5
Password authentication    : enabled
Permit empty password      : disabled
Public key authentication  : enabled
Timeout                    : 30 seconds
```

EXAMPLE #2

To display the current configuration including allow and deny directives, type:

```
$ server_ssh server_3 -info

server_3 : done
SERVICE CONFIGURATION
Port                       : 22
State                      : running
Thread count               : 16
```

```
Banner                     : /fs40/banner.txt
Default home directory     : /
Restrict home directory    : disabled
Application                : sftp,scp
Cipher                     : <all>
Compression                : no
Mac                        : <all>
Max authentication tries   : 10
Password authentication    : enabled
Permit empty password      : disabled
Public key authentication  : enabled
Timeout                    : 900 seconds

ACCESS CONTROL

Allowed IPs                : <all>
Denied IPs                 : 192.168.34.23,192.168.34.10
Allowed users              : jack,tony,angela,root,sheila,Monica
Denied users               : <none>
Allowed groups             : <all>
Denied groups              : <none>
```

EXAMPLE #3

To start the SSH server, type:

```
$ server_ssh server_3 -start

server_3 : done
```

EXAMPLE #4

To stop the SSH server, type:

```
$ server_ssh server_3 -stop

server_3 : done
```

EXAMPLE #5

To update the number of parameter threads, type:

```
$ server_ssh server_3 -modify -threads 16

server_3 : done
```

EXAMPLE #6

To remove an allowed group, type:

```
$ server_ssh server_3 -remove -allowgroups guests

server_3 : done
```

EXAMPLE #7

To modify the banner file, type:

```
$ server_ssh server_2 -modify -banner /server2fs1/banner.txt

server_2 : done
```

Note: The file /server2fs1/banner.txt must exist for the Data Mover point of view.

EXAMPLE #8

To modify the timeout to five minutes, type:

```
$ server_ssh server_2 -modify -timeout 300

server_2 : done
```

EXAMPLE #9

To add a new allowed user defined in NIS or LDAP, type:

```
$ server_ssh server_2 -append -allowusers john

server_2 : done
```

EXAMPLE #10

To add a new allowed user defined in the dom10 Windows domain, type:

```
$ server_ssh server_2 -append -allowusers dom10\\cindy

server_2 : done
```

EXAMPLE #11

To add a new allowed group of users, type:

```
$ server_ssh server_2 -append -allowgroups admin

server_2 : done
```

EXAMPLE #12

To add a new allowed client IP, type:

```
$ server_ssh server_2 -append -allowhosts 110.171.1.10

server_2 : done
```

EXAMPLE #13

To add new allowed client hosts using their subnet, type:

```
$ server_ssh server_2 -append -allowhosts 110.121.0.0/16

server_2 : done
```

EXAMPLE #14

To add a new denied user, type:

```
$ server_ssh server_2 -append -denyusers john

server_2 : done
```

EXAMPLE #15

To add a new denied group of users, type:

```
$ server_ssh server_2 -append -denygroups guest

server_2 : done
```

EXAMPLE #16

To add a new denied client IP, type:

$ server_ssh server_2 -append -denyhosts 110.171.1.54

server_2 : done

EXAMPLE #17

To generate a new host key for the Data Mover, type:

$ server_ssh server_2 -generate

server_2 : done

Note: This operation may take a long time. The SSH server must be
      stopped as the Data Mover does not have existing host keys.

EXAMPLE #18

To delete the existing host keys of the Data Mover, type:

$ server_ssh server_2 -delete

server_2 : done The SSH server must be stopped.

ERROR CASE #1

To change the banner file to a non-existing file, type:

$ server_ssh server_2 -modify -banner foo

server_2 :
Error 13163823109: server_2 : Invalid SSH configuration: Invalid banner
file name.

ERROR CASE #2

To enable an unknown application on top of SSH, type:

$ server_ssh server_2 -modify -application foo

server_2 :
Error 13163823110: server_2 : Invalid configuration value for the SSH
server: Unknown application 'foo'..

ERROR CASE #3

To change the number of SSHD threads to an unauthorized value, type:

$ server_ssh server_2 -modify -threads 256

server_2 :
Error 13163823110: server_2 : Invalid configuration value for the SSH
server: Bad threads value specified, allowed range is (4-128).

ERROR CASE #4

To change the port of the SSH server to an already used port, type:

$ server_ssh server_2 -modify -port 445

```
server_2 :
Error 13163823111: server_2: The SSH server cannot bind the TCP port 445.

Note: 445 is used by the CIFS server.

ERROR CASE #5
-------------

To regenerate the Data Mover host keys while the SSH server is active, type:

$ server_ssh server_2 -generate

server_2 :
Error 13163823112: server_2 : The SSH server must be stopped before executing
this command.

ERROR CASE #6
-------------

To generate the Data Mover host key, and specify an invalid key type, type:

$ server_ssh server_2 -generate -type foo

server_2 :
Error 13163823109: server_2 : Invalid SSH configuration: Bad KEYTYPE value
attribute.

ERROR CASE #7
-------------

To generate the Data Mover host key, and specify an RSA key with
an invalid size, type:

$ server_ssh server_2 -generate -type RSA -keysize 23

server_2 :
Error 13163823110: server_2 : Invalid configuration value for
the SSH server: Bad keysize value specified.

ERROR CASE #8
-------------

To generate the Data Mover host key as a DSA key when a key of this type
already exists, type:

$ server_ssh server_2 -generate -type dsa

server_2 :
Error 13163823123: server_2 : The command failed as the DSA host
key is already defined.

------------------------------------------------------------------------
Created on: July 13 2011, 04:20 pm
```

server_standby

Manages the standby and RDF relationships for the specified Data Movers.

SYNOPSIS

```
server_standby {<movername>|ALL}
  -activate {mover|rdf}
| -restore {mover|rdf}
| -delete {mover[=<movername>]|rdf}
| -verify {mover}
| -create {mover=<source_movername> [-policy <policy_type>]|
    valid policy_type:manual | retry | aut
    default policy_type: manual}
```

DESCRIPTION
----

server_standby manages the standby and RDF relationship for the specified Data Mover.

The ALL option executes the command for all Data Movers.

Caution: The ALL option should not be used when creating, restoring, or activating standby or RDF Data Movers.

OPTIONS

-activate {mover|rdf}
Activates the standby Data Mover or the RDF Data Mover for the primary Data Mover. Before activating the standby, the reboot caused by creating the standby relationship must be complete.

-delete {mover[=<movername>]|rdf}
Deletes relationship between the standby and primary Data Mover or deletes the RDF Data Mover for the primary Data Mover. If you have more than one standby for a primary, you can specify which standby is to be deleted by entering a <movername>. server_setup provides information to change the type defined for a Data Mover.

-restore {mover|rdf}
Restores the standby Data Mover or the RDF Data Mover for the primary Data Mover once it has been activated.

-verify {mover}
Verifies the standby status of the Data Mover after a hardware replacement. If the Data Mover hardware has been replaced, the reboot cycle may not be displayed.

-create {mover=<source_movername> [-policy<policy_type>]}
Designates the <source_movername> as the standby. The following rules apply when selecting a standby Data Mover:

* The primary Data Mover cannot be a standby Data Mover.
* A standby cannot be created for a standby.
* No file systems can be mounted on the standby.
* The standby must have the same disk configuration as the primary.
* One standby Data Mover can be created for multiple Data Movers. To do this, execute server_standby -create for each primary Data Mover.

-policy <policy_type>
When a fault is detected on the primary Data Mover, the following -policy options apply:

auto
Initiates immediate activation of the standby Data Mover.

retry
Attempts to recover the primary Data Mover first, then if recovery fails, initiates activation of the standby.

manual (default)
Reboots the primary Data Mover. No action on the standby is initiated.

SEE ALSO

Configuring Standbys on VNX and server_setup.

EXAMPLE #1

To create a standby relationship between server_2 (primary) and server_3 (standby), type:

```
$ server_standby server_2 -create mover=server_3
server_2 : server_3 is rebooting as standby
```

Note: Before any other actions can take place, the reboot must be complete.

EXAMPLE #2

To activate the server_3 (standby) to take over for server_2 (primary), type:

```
$ server_standby server_2 -activate mover
server_2 :
server_2 : going offline
server_3 : going active
replace in progress ...done
failover activity complete
commit in progress (not interruptible)...done

server_2 : renamed as server_2.faulted.server_3
server_3 : renamed as server_2
```

EXAMPLE #3

To restore server_3 as the standby Data Mover and server_2.faulted.server_3 as the primary, type:

```
$ server_standby server_2 -restore mover
server_2 :
server_2 : going standby
server_2.faulted.server_3 : going active
replace in progress ...done
failover activity complete
commit in progress (not interruptible)...done

server_2 : renamed as server_3
server_2.faulted.server_3 : renamed as server_2
```

EXAMPLE #4

To verify readiness of the standby Data Mover, type:

```
$ server_standby server_2 -verify mover
server_2 : ok
```

EXAMPLE #5

To delete the standby relationship for server_2, type:

```
$ server_standby server_2 -delete mover
server_2 : done
```

EXAMPLE #6

To create a standby relationship for three Data Movers, type:

```
$ server_standby server_2 -create mover=server_3
server_2 : server_3 is rebooting as standby

$ server_standby server_4 -create mover=server_3
server_4 : done

$ server_standby server_5 -create mover=server_3
server_5 : done
```

Note: Before any other actions can take place, the reboot must be complete.

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server_stats

Displays sets of statistics that are running on the specified Data Mover.

SYNOPSIS

```
server_stats <movername>
   -list
 | -info [-all|<statpath_name>[,...]]
 | -service { -start [-port <port_number>]
           | -stop
           | -delete
           | -status }
 | -monitor -action {status|enable|disable}
 |[
    [{ -monitor {statpath_name|statgroup_name}[,...]
     | -monitor {statpath_name|statgroup_name}
        [-sort <field_name>]
        [-order {asc|desc}]
        [-lines <lines_of_output>]
     }...]
     [-count <count>]
     [-interval <seconds>]
     [-terminationsummary {no|yes|only}]
     [-format {text [-titles {never|once|repeat}]|csv}]
     [-type {rate|diff|accu}]
     [-file <output_filepath> [-overwrite]]
 |-noresolve
 ]
```

DESCRIPTION
----

server_stats provides real-time statistics for the specified Data Mover. Statistics are displayed in a time-series fashion at the end of each interval. The statistics are organized as a hierarchy of families that replaces the summary and table collections. Previously defined summary and table collections have been converted to system-defined statistics groups that can be used as arguments with the new -monitor option:

System-defined statistics group name	User-defined statistics group name
basic-std	-summary basic
caches-std	-summary caches
cifs-std	-summary cifs
nfs-std	-summary nfs
cifsOps-std	-table cifs
diskVolumes-std	-table dvol
metaVolumes-std	-table fsvol
netDevices-std	-table net
nfsOps-std	-table nfs

Note: The -summary and -table options are deprecated and will not be supported in future releases.

IP Resolution

The nfs.client and cifs.client stats are indexed by the client IP address. Customers requested that the client name be shown instead.

To implement this, IP addresses have to be resolved to client name. The server_stats command now does this resolution by default when a users requests these stats.

NIS, DNS, or a local hosts file must be set up for resolution to work.

UID Resolution

The nfs.user stat is indexed by user id (UID). To maintain consistency, UIDs need to be resolved to User Names.

The server_stats command now does this resolution by default when a users requests this stat.

The NIS Service or a local password file must have user information for resolution to work.

GID Resolution

The nfs.group stat is indexed by group id (GID). To maintain consistency, GIDs need to be resolved to Group Names.

The server_stats command now does this resolution by default when a users requests this stat. This is a support stat that requires the -vis support argument.

The NIS Service or a local group file must have group information for resolution to work.

In addition, server_stats manages the Statistics Monitoring service (statmonService) running on Data Movers including the ability to disable and enable statistics.

NEW CORRELATED STATISTICS

The new statistics are:

cifs.branchcache

The cifs.branchcache counters provide the statistics about the SMB2 BranchCache functionality, a new feature introduced with Microsoft Windows 7 and Microsoft Windows 2008 R2. They are divided in two sections: the cifs.branchcache.basic branch and the cifs.branchcache.usage branch.

cifs.branchcache.basic

Provides the counters related to the dialog with the BranchCache client. The statistics contain the following information:

* Hit
* Miss
* hashCount
* hashSize
* hashTransferred
* hashError
* filtered
* taskQueued
* taskRunning

cifs.branchcache.usage

Provides the counters related to the generation of the hash files. The statistics contain the following information:

* hashSizeMax
* hashSizeAvg
* hashSizeMin
* hashTimeMax
* hashTimeAvg
* hashTimeMin
* taskCount
* taskQueueFull
* maxUsedThread

cifs.user

Provides cifs read and write statistics by call and bytes correlated to cifs users. It displays the same data that cifs.client does but correlated to user as opposed to IP address.

This statistic provides the following information:

* Total CIFS Ops/S
* read Ops/S
* write Ops/S
* Suspicious Ops Diff
* Total KiB/Sec
* Read KiB/Sec
* Write KiB/Sec
* avgTime

The default sort field is Total Ops/S.

cifs.server
Displays the same data as cifs.client but correlated to CIFS server name (as opposed to the CIFS client's IP address for cifs.client).

This statistic provides the following information:

* Total CIFS Ops/S
* read Ops/S
* write Ops/S
* Suspicious Ops Diff
* Total KiB/Sec
* Read KiB/Sec
* Write Ki B/Sec
* avgTime

nfs.user
Provides nfs read and write statistics by call and bytes correlated to nfs users. It displays the same data that nfs.client does but correlated to user as opposed to IP address.

This statistic provides the following information:

* Total NFS Ops/S
* read Ops/S
* write Ops/S
* Suspicious Ops Diff
* Total KiB/Sec
* Read KiB/Sec
* Write Ki B/Sec
* avgTime

The default sort field is Total Ops/S.

nfs.export
Displays the same data as nfs.client, but correlated to NFS export (as opposed to the NFS client's IP address).

This statistic provides the following information:

* Total NFS Ops/S
* read Ops/S
* write Ops/S
* Suspicious Ops Diff
* Total KiB/Sec
* Read KiB/Sec
* Write Ki B/Sec
* avgTime

nfs.group

Displays the same data as nfs.client, but correlated to NFS group ID (as opposed to the NFS client's IP address).

This statistic provides the following information:

* Total NFS Ops/S
* read Ops/S
* write Ops/S
* Suspicious Ops Diff
* Total KiB/Sec
* Read KiB/Sec
* Write Ki B/Sec
* avgTime

nfs.vdm
Names of VDMs configured on a Data Mover will be the elements of this set statistic. Physical Data Mover name will also be one of the elements in this set statistic.

nfs.vdm.*.client
Displays the same data as nfs.client, but only for VDMs.

This statistic provides the following information:

- Total NFS Ops/S
- read Ops/S
- write Ops/S
- Suspicious Ops Diff
- Total KiB/Sec
- Read KiB/Sec
- Write Ki B/Sec
- avgTime

nfs.vdm.*.user
Displays the same data as nfs.user, but only for VDMs.

This statistic provides the following information:

- Total NFS Ops/S
- read Ops/S
- write Ops/S
- Suspicious Ops Diff
- Total KiB/Sec
- Read KiB/Sec
- Write Ki B/Sec
- avgTime

nfs.vdm.*.group
Displays the same data as nfs.group, but only for VDMs.

This statistic provides the following information:

- Total NFS Ops/S
- read Ops/S
- write Ops/S
- Suspicious Ops Diff
- Total KiB/Sec
- Read KiB/Sec
- Write Ki B/Sec
- avgTime

nfs.vdm.*.export
Displays the same data as nfs.export, but only for VDMs.

This statistic provides the following information:

- Total NFS Ops/S
- read Ops/S
- write Ops/S
- Suspicious Ops Diff
- Total KiB/Sec
- Read KiB/Sec
- Write Ki B/Sec

- avgTime

fs.filesystem
Displays the most active files within each specified filesystem.

This statistic provides the following information:

* Total KiB/Sec
* readBytes
* writtenBytes
* avgTime
* readAvgTime
* writeAvgTime

Note: File inodes will not be resolved until the filesystem is configured for file resolution using the server_fileresolve command.

fs.qtreeFile
Displays the most active files within each specified Qtree.

This statistic provides the following information:

* Total KiB/Sec
* readBytes
* writtenBytes
* avgTime
* readAvgTime
* writeAvgTime

Note: File inodes will not be resolved until the Quota Tree is configured for file resolution using the server_fileresolve command.

store.volume
Provides Disk Volume read and write statistics by blocks and bytes correlated to FileSystem and Disk Volume. It displays top FileSystems per disk volume. To list filesystems for a specific disk volume (for example, a volume named d133), run the server_stats command as:

```
$ server_stats server_2 -m store.volume.d133
```

This statistic provides the following information:

* totalBlocks
* readBlocks
* writtenBlocks
* Total KiB/Sec
* readBytes
* writeBytes

The default sort field is totalBlocks.

OPTIONS

No arguments
Displays a basic summary of statistics for the specified Data Mover as defined by the basic-std Statistics Group.

-list
Displays all defined statistics starting with the statgroup names followed by statpaths and their types.

-info
Displays the statgroup and statpath information.

-service
Specifies whether to start, stop, delete, or query the status of the statmonService. The statmonService runs on the Data Mover and listens for the server_stats requests.

[-start]
Starts the statmonService on the Data Mover. If the -port argument is specified, it is used by the statmonManager service. These settings are persistent and execute as part of the Data Mover's boot-up configurations.

[-stop]
Shuts down the statmonService on the specified Data Mover.

[-delete]
Deletes the statmonService persistent configurations so it does not execute as part of the Data Mover's boot-up settings. If -delete is executed while the statmonService is running, the service stops and its configuration is deleted.

[-status]
Checks the status of the statmonService on the specified Data Mover.

-monitor [-action]
Enables, disables, or queries the state of the stats collection.

-monitor {statpath_name|statgroup_name}
Takes a comma-separated list of statpath and statgroup names. In cases where stats are available for multiple elements, the user can specify an element name or use 'ALL-ELEMENTS' to refer to all elements at once.

Since server_stats command considers periods within the statpath name as delimiters, statpath names with periods as part of the element name require those periods to be double escaped.

For example, statistics for a filesystem named ufs1.accounting should be requested using the following statpath name:

store.logicalvolume.metavolume.ufs1\\.accounting

Any duplicate statpath or statgroup names is consolidated and reported once. The options below are only applicable to Set and Correlated Set statpath names:

[-sort <field_name>]
Sorts a Set based on <field_name> values.

[-order {asc|desc}]
Sorted Sets can be ordered in ascending or descending order.

[-lines <lines_of_output>]
Sets output can be limited to display the Set statistics <lines_of_output> lines.

Set statistics is not sorted if none of the above three options are specified. If some of these options are specified with a Set statpath name, the omitted options default as follows:

* -sort defaults to the element name
* -lines defaults to all
* -order default value depends on the -sort field. If the Set is being sorted on a numeric field, -order defaults to descending order; otherwise, it defaults to ascending order. For sorting based on the Set element name, -order always defaults to ascending order.

Each Correlated Set is defined with a default sort field and order. If some of these options are specified with a Set statpath name, the omitted options default as follows:

* -sort default field is unique for each Correlated Set
* -lines always defaults to 20 for all Correlated Sets
* -order default value depends on the -sort field. If the

Correlated Set is being sorted on a numeric field, -order defaults to descending order; otherwise, it defaults to ascending order. Correlated Sets cannot be sorted on non numeric fields, including the Correlated Set element ID.

[-count <count>]
Specifies the number of reports that are generated. Reports are produced after each set interval of time. The <count> must be a positive integer. By default, reports are generated indefinitely till the user presses Ctrl-C.

[-interval <seconds>]
Specifies the number of seconds between samples. The <seconds> must be an integer, 1 through 300 or higher. The default is 15.

[-terminationsummary {no|yes|only}]
Enables, disables, or manages the termination summary that appears after the <count> lines have completed or a Ctrl-C interrupt is executed. All items, both active and inactive, are reported. The summary reports average, minimum, and maximum values for numeric fields collected over the complete run. The no and yes choices control the summary message. The only option suppresses the reporting of the time series and reports only the summary. The default value is yes.

[-format {text|csv}]
Sets the format for printing statistics. The text (default) format prints statistics separated by a sequence of blank characters when rendered in a fixed-sized font. The time stamp that appears on the front of each snapshot is local time in the form hh:mm:ss.

The csv format supports the display of elements in a dynamic Set statistic. A dynamic collection is a Set statistic where new elements are reported in any sample. In order to support this dynamic behavior for the csv format and keep the csv output consistent between the title line and the sample data lines, the title line is reprinted with samples where new elements are detected.

To avoid seeing multiple title lines in the csv output, use the .file option that saves the output to a specified file instead of outputting to the standard output.

The advantage of using the -file option is that server_stats prints samples to the file and does not print the title line except once at the end of the session when the title line is complete. In this case, the title line appears once in the output at the top of the file.

The csv format produces output in a comma-separated list and renders the local time in a yyyy/mm/dd hh:mm:ss format.

The csv output and the -file option:

* Without the -file option: The output to stdout may show multiple title lines where new elements were detected.

* With the -file option: The output is printed to the specified file so at the end of the session, the final titles line is printed once at the top of the file. The result is a cleaner output than multiple titles lines where new elements were detected.

[-titles {never|once|repeat}]
Controls the generation of titles. The never option indicates that titles are not printed, the once option indicates that they are only printed at the top of the report, and the repeat option indicates that they are repeated for each time-series snapshot. The default setting is once.

The -titles option cannot be combined with the -format csv option since titles could be dynamically updated based on the dynamic nature of some statistics. The -format option provides more

information.

[-type {rate|diff|accu}]
Specifies the display type of value for statistics with monotonically increasing values. The display type applies to statistics that increase monotonically, for example, network in-bound bytes. Other statistics that represent a point-in-time value, for example, current CIFS connections, are not affected by this option. The rate value displays the rate of change since the previous sample, the diff value displays the change in value since the previous sample, and the accu value displays the change in value since the initial sample. The default display type is rate.

[-file <output_filepath> [-overwrite]]
Allows the users to specify a filename to save the output of session instead of printing to standard output (stdout). This is necessary to provide a cleaner csv output. Since the titles line continues to change each time a new element is detected, it is printed a number of times to stdout. However, with the -file option, the titles line appears once at the top of the output.

In case <output_filepath> exists, the results will be appended to it. The -overwrite argument can be used to change this behavior so that an existing file is deleted before the results are written to it. The argument of the -file option is a valid file name and does not have a default argument.

In case that server_stats is killed abruptly, two temporary files can possibly be left behind depending on whether an append to a previous file was done or not and whether csv format was used or not. These files will be in the same directory as the specified file and have the following formats:

     temp_<output_filepath><timestamp>
     orig_<output_filepath ><timestamp>

Where <timestamp> is the timestamp of when the file was created. The output of these files may not include the most updated titles, but they will contain the statistical data up until program termination.

-noresolve
The -noresolve option is a new server_stats option that prevents any id resolution of any kind. This includes: FS IDs, Volume IDs, MetaVolume IDs, Quota Tree IDs, IP Addresses, UIDs, and GIDs.

SEE ALSO

Managing Statistics for VNX.

EXAMPLE #1

To display the basic-std group by indicating the rate of change since the previous sample, type:

$ server_stats server_2 -monitor basic-std -interval 5 -count 5 -type rate

server_2 Timestamp	CPU Util %	Network In KiB/s	Network Out KiB/s	dVol Read KiB/s	dVol Write KiB/s
02:53:09	33	88618	729	9988	89730
02:53:11	61	142057	1087	18632	167076
02:53:13	29	98110	762	8976	78599
02:53:15	46	109783	836	14288	125717
02:53:17	34	111847	886	10652	95719

server_2 Summary	CPU Util	Network In	Network Out	dVol Read	dVol Write

```
                %      KiB/s       KiB/s      KiB/s       KiB/s
Minimum        29      88618         729       8976       78599
Average        41     110083         860      12507      111368
Maximum        61     142057        1087      18632      167076
```

Where:

Value	Definition
Timestamp	Time the poll was taken.
CPU Util	CPU utilization in percentage in this interval.
Network In KiB/s	Network kibibytes received over all network interfaces.
Network Out KiB/s	Network kibibytes sent over all network interfaces.
dVol Read KiB/s	Storage kibibytes received from all server-storage interfaces.
dVol Write Kib/s	Storage kibibytes sent to all server-storage interfaces.

EXAMPLE #2

To display the basic-std group by indicating the change in value since the previous sample, type:

```
$ server_stats server_2 -monitor basic-std -interval 5 -count 5 -type diff

server_2    CPU     Network     Network        dVol        dVol
Timestamp  Util      In KiB     Out KiB    Read KiB       Write
              %        diff        diff        diff    KiB diff
02:53:29     46      267660        2136       26128      232654
02:53:31     38      200668        1543       23144      211182
02:53:33     46      226761        1749       26488      230558
02:53:35     48      246921        1876       28720      255957
02:53:37     40      212353        1673       23016      210573

server_2    CPU     Network     Network        dVol        dVol
Summary    Util      In KiB     Out KiB    Read KiB       Write
              %        diff        diff        diff    KiB diff
Minimum      38      200668        1543       23016      210573
Average      44      230873        1795       25499      228185
Maximum      48      267660        2136       28720      255957
```

Where:

Value	Definition
Timestamp	Time the poll was taken.
CPU Util %	CPU utilization in percentage in this interval.
Network In KiB diff	Network kibibytes received over all network interfaces per differential value.
Network Out KiB diff	Network kibibytes sent over all network interfaces per differential value.
dVol Read KiB diff	Storage kibibytes received from all server-storage interfaces per differential value.
dVol Write KiB diff	Storage kibibytes sent to all server-storage interfaces per differential value.

EXAMPLE #3

To display the basic-std group by indicating the change in value since the first sample, type:

```
$ server_stats server_2 -monitor basic-std -interval 5 -count 5 -type accu

server_2    CPU  Network In KiB  Network Out KiB  dVol Read KiB dVol Write KiB
Timestamp  Util
              %
```

```
02:53:48     42          236257             1880          25504         224832
02:53:50     54          505640             3983          55760         500538
02:53:52     29          686282             5377          74096         662494
02:53:54     46          922765             7183         101704         908813
02:53:56     41         1125518             8777         126640        1134362

server_2    CPU  Network In KiB  Network Out KiB  dVol Read KiB dVol Write KiB
Summary    Util
            %
Minimum      29          236257             1880          25504         224832
Average      42          695293             5440          76741         686208
Maximum      54         1125518             8777         126640        1134362
```

Where:

Value	Definition
Timestamp	Time the poll was taken.
CPU Util	Disk utilized in percentage in this interval.
Network In KiB	Network kibibytes received over all network interfaces per accumulated value.
Network Out KiB	Network kibibytes sent over all network interfaces per accumulated value.
dVol Read KiB	Storage kibibytes received from all server-storage interfaces per accumulated value.
dVol Write KiB	Storage kibibytes sent to all server-storage interfaces per accumulated value.

EXAMPLE #4

To display a list of statistics group names followed by statpaths and their types, type:

```
$ server_stats server_2 -list

Type              Stat Name
...
Correlated Set    cifs.user
Counter           cifs.user.ALL-ELEMENTS.totalCalls
Counter           cifs.user.ALL-ELEMENTS.readCalls
Counter           cifs.user.ALL-ELEMENTS.writeCalls
Fact              cifs.user.ALL-ELEMENTS.suspectCalls
Counter           cifs.user.ALL-ELEMENTS.totalBytes
Counter           cifs.user.ALL-ELEMENTS.readBytes
Counter           cifs.user.ALL-ELEMENTS.writeBytes
Fact              cifs.user.ALL-ELEMENTS.avgTime
Fact              cifs.user.ALL-ELEMENTS.server
Fact              cifs.user.ALL-ELEMENTS.client
...
Correlated Set    nfs.user
Counter           nfs.user.ALL-ELEMENTS.totalCalls
Counter           nfs.user.ALL-ELEMENTS.readCalls
Counter           nfs.user.ALL-ELEMENTS.writeCalls
Fact              nfs.user.ALL-ELEMENTS.suspectCalls
Counter           nfs.user.ALL-ELEMENTS.totalBytes
Counter           nfs.user.ALL-ELEMENTS.readBytes
Counter           nfs.user.ALL-ELEMENTS.writeBytes
Fact              nfs.user.ALL-ELEMENTS.avgTime
...
Set               store.volume
Correlated Set    store.volume.ALL-ELEMENTS.fileSystem
Counter
store.volume.ALL-ELEMENTS.fileSystem.ALL-ELEMENTS.totalBlocks
Counter           store.volume.ALL-ELEMENTS.fileSystem.ALL-ELEMENTS.readBlocks
Counter
store.volume.ALL-ELEMENTS.fileSystem.ALL-ELEMENTS.writeBlocks
Counter           store.volume.ALL-ELEMENTS.fileSystem.ALL-ELEMENTS.totalBytes
Counter           store.volume.ALL-ELEMENTS.fileSystem.ALL-ELEMENTS.readBytes
Counter           store.volume.ALL-ELEMENTS.fileSystem.ALL-ELEMENTS.writeBytes
```

```
...

EXAMPLE #5
----------

To display the statgroup information, type:

$ server_stats server_2 -info statsb

server_2 :
name                          = statsB
description                   = My group # 2
type                          = Group - user-defined
member_stats                  = nfs.basic,cifs.basic,iscsi.basic
member_elements               =
member_of                     = statsA

EXAMPLE #6
----------

To display information of statistics group names followed by statpaths, type:

$ server_stats server_2 -info

server_2 :
name                          = statsA
description                   = My group # 1
type                          = Group - user-defined
member_stats                  = statsB
member_elements               =
member_of                     =
...
name                          = cifs
description                   = The CIFS-protocol service
type                          = Family
member_stats                  =
member_elements               =
member_of                     =
...

EXAMPLE #7
----------

To start the statmonService, type:

$ server_stats server_2 -service -start -port 7777

statmonService started on port: 7777.

EXAMPLE #8
----------

To stop the statmonService, type:

$ server_stats server_2 -service -stop

server_2: done.

EXAMPLE #9
----------

To delete the statmonService configurations from the Data Mover's boot-up
settings, type:

$ server_stats server_2 -service -delete

server_2: done.

EXAMPLE #10
-----------
```

To query the status of the statmonService, type:

```
$ server_stats server_2 -service -status

server_2: The statmonService has started.
Interface=INTERNAL
Port=7777
Allow=128.221.252.100:128.221.252.101:128.221.253.100:128.221.253.101
The statmonService is listing for incoming network connections
Max Connections: 32, Current: 0
```

EXAMPLE #11

To enable stats collection, type:

```
$ server_stats server_2 -monitor -action enable

server_2: done.
```

EXAMPLE #12

To query the state of the stats collection, type:

```
$ server_stats server_2 -monitor -action status

server_2 : Statistics are enabled.
```

EXAMPLE #13

To display five iterations of the of the cifs-std statistics group with a three second interval, type:

```
$ server_stats server_2 -monitor cifs-std -i 3 -c 5

server_2   CIFS   CIFS   CIFS  CIFS Avg   CIFS    CIFS CIFS Avg        CIFS  CIFS
Timestamp Total   Read   Read      Read  Write   Write    Write       Share  Open
          Ops/s  Ops/s  KiB/s  Size KiB  Ops/s   KiB/s Size KiB Connections Files
02:54:31   2133      0      0         -   1947  110600        57          96   587
02:54:34   1895      0      0         -   1737   99057        57          96   631
02:54:37   2327      0      0         -   2104  119556        57          96   649
02:54:40   2109      0      0         -   1864  106081        57          96   653
02:54:43   2439      0      0         -   2172  123578        57          96   639

server_2   CIFS   CIFS   CIFS  CIFS Avg   CIFS    CIFS  CIFS Avg        CIFS  CIF
S
Summary   Total   Read   Read      Read  Write   Write     Write       Share  Ope
n
          Ops/s  Ops/s  KiB/s  Size KiB  Ops/s   KiB/s  Size KiB Connections File
s
Minimum    1895      0      0         -   1737   99057        57          96   58
7
Average    2180      0      0         -   1965  111775        57          96   63
2
Maximum    2439      0      0         -   2172  123578        57          96   65
3
```

Where:

Value	Definition
Timestamp	Time the poll was taken.
CIFS Total Ops/s	Total operations per second.
CIFS Read Ops/s	CIFS read operations per second in the interval.
CIFS Read KiB/s	CIFS read data response in kibibytes per second.
CIFS Avg Size Read KiB	Average read data response.
CIFS Write Ops/s	CIFS write operations per second.
CIFS Write KiB/s	CIFS write data response in kibibytes per second.

```
CIFS Avg Size Write KiB Average write data size.
CIFS Share Connections  Number of CIFS protocol connections.
CIFS Open Files         Number of open CIFS files.

EXAMPLE #14
-----------

To display five iterations of the nfs-std statistics group with a one
second interval, type:

$ server_stats server_2 -monitor nfs-std -i 1 -c 5
server_2  Total   NFS   NFS   NFS Avg   NFS   NFS    NFS Avg      NFS
Timestamp   NFS  Read  Read Read Size Write Write Write Size   Active
          Ops/s Ops/s KiB/s     Bytes Ops/s KiB/s      Bytes  Threads

13:44:53  20650  4121 67506     16774  2214 29737      13754      648
13:44:54  11663  2318 37140     16407  1238 17307      14316      648
13:44:55   8678  1790 30761     17597   945 12511      13557      648
13:44:56  17655  3543 56382     16296  1967 27077      14096      648
13:44:57  20302  4033 63822     16205  2271 31469      14189      648

server_2  Total   NFS   NFS   NFS Avg   NFS   NFS    NFS Avg     NFS
Summary     NFS  Read  Read Read Size Write Write Write Size  Active
          Ops/s Ops/s KiB/s     Bytes Ops/s KiB/s      Bytes Threads

Minimum    8678  1790 30761     16205   945 12511      13557     648
Average   15790  3161 51122     16656  1727 23620      13982     648
Maximum   20650  4121 67506     17597  2271 31469      14316     648

Where:

Value                   Definition
-----                   ----------

Timestamp               Time the poll was taken.
Total NFS Ops/s         Total number of operations per second.
NFS Read Ops/s          NFS read operations per second in the interval.
NFS Read KiB/s          NFS read data response in kibibytes per second.
NFS Avg Read Size Bytes  Average read data response.
NFS Write Ops/s         NFS write operations per second.
NFS Write KiB/s         NFS write data response in kibibytes per second.
NFS Avg Write Size Bytes Average write data size.
NFS Active Threads      Number of NFS active threads.

--------------------------------------------------------------------------------
Note: Part of the accuracy of statistics can be linked to how often
server_stats
reports results. For example, statistics was used to monitor NFS write bytes
to a
Data Mover. The NFS client, swiftest, wrote a single by each second for five
minutes. When server_stats was run with an interface of ten minutes, all bytes
written were accounted for. At smaller intervals, such as one second, bytes
were lost. Detailed results are as follows:

      1      2      5     15     30     120    600
S1  0.005  0.045  0.052  0.000  0.050  0.000  0.000
S2  0.002  0.000  0.043  0.050  0.000  0.000  0.000

To review, these numbers are the number of kilobytes per report lost at each
reporting period. The first row (S1) is the result of a single server_stat
session, the second (S2) is two (average to produce a single value). Each
column is the server_stat interval value.

In context to informal numbers, the larger the amount of time between
reporting
periods, the more accurate the server_stat numbers. However, even when loss
was
prevalent at higher reporting periods, the loss rate was still very low.

EXAMPLE #15
-----------
```

To display five iterations of the summary statistics for caches with a three second interval, type:

```
$ server_stats server_2 -monitor caches-std -i 5 -c 5

server_2      DNLC   OF Cache   Buffer
Timestamp      Hit         Hit    Cache
           Ratio %    Ratio %    Hit %
02:55:26         -         100       71
02:55:29         -         100       72
02:55:32         -         100       73
02:55:35         -         100       73
02:55:38         -         100       72

server_2      DNLC   OF Cache   Buffer
Summary        Hit         Hit    Cache
           Ratio %    Ratio %    Hit %
Minimum          -         100       71
Average          -         100       72
Maximum          -         100       73
```

Where:

Value	Definition
Timestamp	Time the poll was taken.
DNLC Hit Ratio %	Directory Name Lookup Cache (DNLC) hit ratio.
OF Cache Hit Ratio %	Open file cache hit ratio.
Buffer Cache Hit %	Kernel buffer cache hit ratio.

EXAMPLE #16

To display the netDevices-std statistics group with a three second interval, type :

```
$ server_stats server_2 -monitor netDevices-std -i 3 -c 3

server_2  device  Network   Network   Network   Network    Network   Network
Timestamp            In        In        In       Out        Out       Out
                   Pkts/s  Errors/s     KiB/s    Pkts/s   Errors/s     KiB/s
02:55:52    mge0        2         0         0         1          0         0
            mge1       17         0        23         9          0         1
            cge0     3593         0     26566      2289          0       203
            cge1     6912         0     50206      4444          0       378
            cge2     3637         0     25570      2342          0       209
02:55:55    mge0        0         0         0         0          0         0
            mge1        7         0         9         4          0         0
            cge0     3444         0     24744      2252          0       204
            cge1     7415         0     53354      4721          0       400
            cge2     3913         0     27796      2502          0       222
02:55:58    mge0        2         0         0         2          0         0
            mge1       32         0        39        19          0         2
            cge0     4029         0     29334      2594          0       230
            cge1     7461         0     54030      4791          0       406
            cge2     3902         0     27319      2505          0       223

server_2  device  Network   Network   Network   Network    Network   Network
Summary              In        In        In       Out        Out       Out
                   Pkts/s  Errors/s     KiB/s    Pkts/s   Errors/s     KiB/s
Minimum     mge0        0         0         0         0          0         0
            mge1        7         0         9         4          0         0
            cge0     3444         0     24744      2252          0       203
            cge1     6912         0     50206      4444          0       378
            cge2     3637         0     25570      2342          0       209
            cge3        0         0         0         0          0         0
Average     mge0        1         0         0         1          0         0
            mge1       19         0        24        11          0         1
            cge0     3689         0     26882      2378          0       213
            cge1     7263         0     52530      4652          0       395
            cge2     3817         0     26895      2450          0       218
            cge3        0         0         0         0          0         0
Maximum     mge0        2         0         0         2          0         0
```

```
            mge1       32         0        39        19          0         2
            cge0     4029         0     29334      2594          0       230
            cge1     7461         0     54030      4791          0       406
            cge2     3913         0     27796      2505          0       223
            cge3        0         0         0         0          0         0
```

Where:

Value	Definition
Timestamp	Time the poll was taken.
Device	Name of the network device.
Network In Pkts/s	Network packets received per second.
Network In Errors/s	Network input errors encountered per second.
Network In KiB/s	Network kibibytes received per second.
Network Out Pkts/s	Network packets sent per second.
Network Out Errors/s	Network output errors encountered per second.
Network Out KiB/s	Network kibibytes sent per second.

EXAMPLE #17

To display the netDevices-std statistics group without the summary and with a three second interval, type:

```
$ server_stats server_2 -monitor netDevices-std -i 3 -c 3 -terminationsummary no

server_2  device   Network   Network   Network   Network   Network     Network
Timestamp             In        In        In       Out       Out         Out
                   Pkts/s  Errors/s     KiB/s    Pkts/s  Errors/s       KiB/s
02:56:11    mge0       16         0         1        19         0          23
            mge1       43         0        60        24         0           2
            cge0     3960         0     29053      2547         0         226
            cge1     6709         0     48414      4296         0         366
            cge2     4829         0     33996      3125         0         281
02:56:14    mge0        0         0         0         0         0           0
            mge1        3         0         3         2         0           0
            cge0     3580         0     25905      2335         0         211
            cge1     6663         0     48212      4273         0         364
            cge2     3970         0     28113      2523         0         222
02:56:17    mge0        2         0         0         2         0           0
            mge1        5         0         6         2         0           0
            cge0     3561         0     25891      2296         0         206
            cge1     7091         0     51721      4564         0         389
            cge2     3931         0     27703      2514         0         223
            cge3        0         0         0         0         0           0
```

EXAMPLE #18

To display the cifsOps-std statistics with a five second interval, type:

```
$ server_stats server_2 -monitor cifsops-std -i 5 -c 3

server_2   SMB Operation          Op       Min       Max       Avg
Timestamp                     Calls/s      uSec      uSec uSec/call

02:57:00   SMB1_Close             89        45    406775     10273
           SMB1_WriteX          1837        30   1618776    144030
           SMB1_CreateNTX         84        51    458090       379
02:57:03   SMB1_Close            122        45    406775     10057
           SMB1_WriteX          1867        30   1618776    133180
           SMB1_CreateNTX        126        51    458090      1826
02:57:06   SMB1_Close            105        45    406775     14663
           SMB1_WriteX          2119        30   1618776    121976
           SMB1_CreateNTX        103        51    458090      1801

server_2   SMB Operation          Op      Min       Max       Avg
Summary                       Calls/s     uSec      uSec  uSec/call
Minimum    SMB1_Mkdir               0        0         0          -
```

SMB1_Rmdir	0	0	0	-
SMB1_Open	0	0	0	-
SMB1_Create	0	0	0	-
SMB1_Close	89	45	406775	10057
SMB1_Flush	0	0	0	-
SMB1_Unlink	0	0	0	-
SMB1_Rename	0	0	0	-
SMB1_GetAttr	0	0	0	-
SMB1_SetAttr	0	0	0	-
SMB1_Read	0	0	0	-
SMB1_Write	0	0	0	-
SMB1_Lock	0	0	0	-
SMB1_Unlock	0	0	0	-
SMB1_CreateTmp	0	0	0	-
SMB1_MkNew	0	0	0	-
SMB1_ChkPath	0	0	0	-
SMB1_Exit	0	0	0	-
SMB1_Lseek	0	0	0	-
SMB1_LockRead	0	0	0	-
SMB1_WriteUnlock	0	0	0	-
SMB1_ReadBlockRaw	0	0	0	-
SMB1_WriteBlockRaw	0	0	0	-
SMB1_SetAttrExp	0	0	0	-
SMB1_GetAttrExp	0	0	0	-
SMB1_LockingX	0	0	0	-
SMB1_Trans	0	0	0	-
SMB1_TransSec	0	0	0	-
SMB1_Copy	0	0	0	-
SMB1_Move	0	0	0	-
SMB1_Echo	0	0	0	-
SMB1_WriteClose	0	0	0	-
SMB1_OpenX	0	0	0	-
SMB1_ReadX	0	0	0	-
SMB1_WriteX	1837	30	1618776	121976
SMB1_CloseTreeDisco	0	0	0	-
SMB1_Trans2Prim	0	0	0	-
SMB1_Trans2Secd	0	0	0	-
SMB1_FindClose2	0	0	0	-
SMB1_FindNotifyClose	0	0	0	-
SMB1_TreeConnect	0	0	0	-
SMB1_TreeDisco	0	0	0	-
SMB1_NegProt	0	44	85	-
SMB1_SessSetupX	0	1088	12058	-
SMB1_UserLogoffX	0	0	0	-
SMB1_TreeConnectX	0	82	499	-
SMB1_DiskAttr	0	0	0	-
SMB1_Search	0	0	0	-
SMB1_FindFirst	0	0	0	-
SMB1_FindUnique	0	0	0	-
SMB1_FindClose	0	0	0	-
SMB1_TransNT	0	0	0	-
SMB1_TransNTSecd	0	0	0	-
SMB1_CreateNTX	84	51	458090	379
SMB1_CancelNT	0	0	0	-
SMB1_SendMessage	0	0	0	-
SMB1_BeginMessage	0	0	0	-
SMB1_EndMessage	0	0	0	-
SMB1_MessageText	0	0	0	-
SMB2_Negotiate	0	0	0	-
SMB2_SessionSetup	0	0	0	-
SMB2_Logoff	0	0	0	-
SMB2_TreeConnect	0	0	0	-
SMB2_TreeDisConnect	0	0	0	-
SMB2_Create	0	0	0	-
SMB2_Close	0	0	0	-
SMB2_Flush	0	0	0	-
SMB2_Read	0	0	0	-
SMB2_Write	0	0	0	-
SMB2_Lock	0	0	0	-
SMB2_Ioctl	0	0	0	-
SMB2_Cancel	0	0	0	-

	SMB2_Echo	0	0	0	-
	SMB2_QueryDirectory	0	0	0	-
	SMB2_ChangeNotify	0	0	0	-
	SMB2_QueryInfo	0	0	0	-
	SMB2_SetInfo	0	0	0	-
	SMB2_OplockBreak	0	0	0	-
Average	SMB1_Mkdir	0	0	0	-
	SMB1_Rmdir	0	0	0	-
	SMB1_Open	0	0	0	-
	SMB1_Create	0	0	0	-
	SMB1_Close	105	45	406775	11664
	SMB1_Flush	0	0	0	-
	SMB1_Unlink	0	0	0	-
	SMB1_Rename	0	0	0	-
	SMB1_GetAttr	0	0	0	-
	SMB1_SetAttr	0	0	0	-
	SMB1_Read	0	0	0	-
	SMB1_Write	0	0	0	-
	SMB1_Lock	0	0	0	-
	SMB1_Unlock	0	0	0	-
	SMB1_CreateTmp	0	0	0	-
	SMB1_MkNew	0	0	0	-
	SMB1_ChkPath	0	0	0	-
	SMB1_Exit	0	0	0	-
	SMB1_Lseek	0	0	0	-
	SMB1_LockRead	0	0	0	-
	SMB1_WriteUnlock	0	0	0	-
	SMB1_ReadBlockRaw	0	0	0	-
	SMB1_WriteBlockRaw	0	0	0	-
	SMB1_SetAttrExp	0	0	0	-
	SMB1_GetAttrExp	0	0	0	-
	SMB1_LockingX	0	0	0	-
	SMB1_Trans	0	0	0	-
	SMB1_TransSec	0	0	0	-
	SMB1_Copy	0	0	0	-
	SMB1_Move	0	0	0	-
	SMB1_Echo	0	0	0	-
	SMB1_WriteClose	0	0	0	-
	SMB1_OpenX	0	0	0	-
	SMB1_ReadX	0	0	0	-
	SMB1_WriteX	1941	30	1618776	133062
	SMB1_CloseTreeDisco	0	0	0	-
	SMB1_Trans2Prim	0	0	0	-
	SMB1_Trans2Secd	0	0	0	-
	SMB1_FindClose2	0	0	0	-
	SMB1_FindNotifyClose	0	0	0	-
	SMB1_TreeConnect	0	0	0	-
	SMB1_TreeDisco	0	0	0	-
	SMB1_NegProt	0	44	85	-
	SMB1_SessSetupX	0	1088	12058	-
	SMB1_UserLogoffX	0	0	0	-
	SMB1_TreeConnectX	0	82	499	-
	SMB1_DiskAttr	0	0	0	-
	SMB1_Search	0	0	0	-
	SMB1_FindFirst	0	0	0	-
	SMB1_FindUnique	0	0	0	-
	SMB1_FindClose	0	0	0	-
	SMB1_TransNT	0	0	0	-
	SMB1_TransNTSecd	0	0	0	-
	SMB1_CreateNTX	104	51	458090	1335
	SMB1_CancelNT	0	0	0	-
	SMB1_SendMessage	0	0	0	-
	SMB1_BeginMessage	0	0	0	-
	SMB1_EndMessage	0	0	0	-
	SMB1_MessageText	0	0	0	-
	SMB2_Negotiate	0	0	0	-
	SMB2_SessionSetup	0	0	0	-
	SMB2_Logoff	0	0	0	-
	SMB2_TreeConnect	0	0	0	-
	SMB2_TreeDisConnect	0	0	0	-
	SMB2_Create	0	0	0	-

	SMB2_Close	0	0	0	-
	SMB2_Flush	0	0	0	-
	SMB2_Read	0	0	0	-
	SMB2_Write	0	0	0	-
	SMB2_Lock	0	0	0	-
	SMB2_Ioctl	0	0	0	-
	SMB2_Cancel	0	0	0	-
	SMB2_Echo	0	0	0	-
	SMB2_QueryDirectory	0	0	0	-
	SMB2_ChangeNotify	0	0	0	-
	SMB2_QueryInfo	0	0	0	-
	SMB2_SetInfo	0	0	0	-
	SMB2_OplockBreak	0	0	0	-
Maximum	SMB1_Mkdir	0	0	0	-
	SMB1_Rmdir	0	0	0	-
	SMB1_Open	0	0	0	-
	SMB1_Create	0	0	0	-
	SMB1_Close	122	45	406775	14663
	SMB1_Flush	0	0	0	-
	SMB1_Unlink	0	0	0	-
	SMB1_Rename	0	0	0	-
	SMB1_GetAttr	0	0	0	-
	SMB1_SetAttr	0	0	0	-
	SMB1_Read	0	0	0	-
	SMB1_Write	0	0	0	-
	SMB1_Lock	0	0	0	-
	SMB1_Unlock	0	0	0	-
	SMB1_CreateTmp	0	0	0	-
	SMB1_MkNew	0	0	0	-
	SMB1_ChkPath	0	0	0	-
	SMB1_Exit	0	0	0	-
	SMB1_Lseek	0	0	0	-
	SMB1_LockRead	0	0	0	-
	SMB1_WriteUnlock	0	0	0	-
	SMB1_ReadBlockRaw	0	0	0	-
	SMB1_WriteBlockRaw	0	0	0	-
	SMB1_SetAttrExp	0	0	0	-
	SMB1_GetAttrExp	0	0	0	-
	SMB1_LockingX	0	0	0	-
	SMB1_Trans	0	0	0	-
	SMB1_TransSec	0	0	0	-
	SMB1_Copy	0	0	0	-
	SMB1_Move	0	0	0	-
	SMB1_Echo	0	0	0	-
	SMB1_WriteClose	0	0	0	-
	SMB1_OpenX	0	0	0	-
	SMB1_ReadX	0	0	0	-
	SMB1_WriteX	2119	30	1618776	144030
	SMB1_CloseTreeDisco	0	0	0	-
	SMB1_Trans2Prim	0	0	0	-
	SMB1_Trans2Secd	0	0	0	-
	SMB1_FindClose2	0	0	0	-
	SMB1_FindNotifyClose	0	0	0	-
	SMB1_TreeConnect	0	0	0	-
	SMB1_TreeDisco	0	0	0	-
	SMB1_NegProt	0	44	85	-
	SMB1_SessSetupX	0	1088	12058	-
	SMB1_UserLogoffX	0	0	0	-
	SMB1_TreeConnectX	0	82	499	-
	SMB1_DiskAttr	0	0	0	-
	SMB1_Search	0	0	0	-
	SMB1_FindFirst	0	0	0	-
	SMB1_FindUnique	0	0	0	-
	SMB1_FindClose	0	0	0	-
	SMB1_TransNT	0	0	0	-
	SMB1_TransNTSecd	0	0	0	-
	SMB1_CreateNTX	126	51	458090	1826
	SMB1_CancelNT	0	0	0	-
	SMB1_SendMessage	0	0	0	-
	SMB1_BeginMessage	0	0	0	-
	SMB1_EndMessage	0	0	0	-

```
           SMB1_MessageText        0        0        0          -
           SMB2_Negotiate          0        0        0          -
           SMB2_SessionSetup       0        0        0          -
           SMB2_Logoff             0        0        0          -
           SMB2_TreeConnect        0        0        0          -
           SMB2_TreeDisConnect     0        0        0          -
           SMB2_Create             0        0        0          -
           SMB2_Close              0        0        0          -
           SMB2_Flush              0        0        0          -
           SMB2_Read               0        0        0          -
           SMB2_Write              0        0        0          -
           SMB2_Lock               0        0        0          -
           SMB2_Ioctl              0        0        0          -
           SMB2_Cancel             0        0        0          -
           SMB2_Echo               0        0        0          -
           SMB2_QueryDirectory     0        0        0          -
           SMB2_ChangeNotify       0        0        0          -
           SMB2_QueryInfo          0        0        0          -
           SMB2_SetInfo            0        0        0          -
           SMB2_OplockBreak        0        0        0          -
```

Where:

Value	Definition
Timestamp	Time the poll was taken.
SMB Operation	Name of the SMB operation.
Op Calls/s	Number of calls to this SMB operation per second.
Min uSec	Minimum time in microseconds per call.
Max uSec	Maximum time in microseconds per call.
Avg uSec/Call	Average time in microseconds consumed per call.

EXAMPLE #19

To display the cifsOps-std statistics group without the summary and with a five second interval, type:

```
$ server_stats server_2 -m cifsops-std -i 5 -c 3 -te no

server_2   SMB Operation        Op    Min      Max        Avg
Timestamp                   Calls/s   uSec     uSec  uSec/call

02:57:24   SMB1_Close           56     45   552768      25299
           SMB1_WriteX        1360     29  1618776     161125
           SMB1_CreateNTX       46     51   458090        971
02:57:27   SMB1_Close          130     45   568291      16814
           SMB1_WriteX        1627     29  1618776     182622
           SMB1_CreateNTX      147     51   458090        276
02:57:30   SMB1_Close           50     45   568291      29992
           SMB1_WriteX        1615     29  1618776     151924
           SMB1_CreateNTX       37     51   458090       2850
```

EXAMPLE #20

To display the nfsOps-std statistics group without the summary and with a five second interval, type:

```
$ server_stats server_2 -m nfsops-std -i 5 -c 3 -te no

server_2   NFS Op       NFS       NFS         NFS    NFS Op %
Timestamp               Op        Op          Op
                   Calls/s  Errors/s   uSec/call
03:18:21   v3Read     23442         0       63846          50
           v3Write    23372         0       99156          50
03:18:24   v3Read     23260         0       65756          50
           v3Write    23243         0      101135          50
03:18:27   v3Read     23385         0       66808          50
           v3Write    23323         0      102201          50
```

Where:

Value	Definition
Timestamp	Time the poll was taken.
NFS Op	Name of the NFS operation.
NFS Op Calls/s	Number of calls to this NFS operation per second.
NFS Op Errors/s	Number of times the NFS operation failed per second.
NFS Op uSec/Call	Average time in microseconds consumed per call.
NFS Op %	Percent of total NFS calls attributed to this operation.

EXAMPLE #21

To display the diskVolumes-std statistics group without the summary and with a five second interval, type:

```
$ server_stats server_2 -m diskVolumes-std -i 5 -c 3 -te no
```

server_2 Timestamp	dVol	Queue Depth	Read Ops/s	Read KiB/s	Avg Read Size Bytes/s	Write Ops/s	Write KiB/s	Avg Write Size Bytes/s	Util%
02:58:09	NBS1	0	0	3	8192	1	7	6827	0
	root_ldisk	0	0	0	-	461	490	1090	47
	d7	0	113	904	8192	530	19619	37881	83
	d11	0	249	1995	8192	431	11640	27634	91
	d8	0	68	547	8192	372	11472	31607	79
	d12	33	424	3389	8192	609	20045	33705	99
	d9	0	36	291	8192	592	20339	35161	67
	d13	0	333	2664	8192	347	11925	35158	93
	d10	0	24	189	8192	385	11896	31668	63
	d14	36	573	4581	8192	454	20173	45468	100
02:58:12	root_ldisk	0	0	0	-	401	462	1182	44
	NBS6	0	0	0	-	1	3	3072	0
	d7	0	78	624	8192	388	13851	36523	70
	d11	0	216	1728	8192	470	11147	24268	84
	d8	0	51	411	8192	333	10672	32850	85
	d12	0	301	2408	8192	483	14411	30531	98
	d9	0	24	192	8192	422	14285	34691	50
	d13	0	290	2317	8192	340	10920	32856	87
	d10	0	19	152	8192	346	10944	32389	70
	d14	47	407	3259	8192	342	14288	42822	100
02:58:15	NBS1	0	0	0	-	3	1	512	0
	root_ldisk	0	0	0	-	409	454	1135	43
	NBS5	0	0	0	-	9	83	9070	1
	d7	0	122	976	8192	471	20179	43839	90
	d11	1	144	1149	8192	225	6608	30118	94
	d8	2	33	261	8192	229	6515	29131	48
	d12	41	424	3395	8192	666	20632	31722	93
	d9	0	44	355	8192	577	20848	36999	82
	d13	2	185	1483	8192	201	6768	34423	93
	d10	0	13	101	8192	238	6789	29252	36
	d14	0	583	4667	8192	521	21131	41505	95

Where:

Value	Definition
Timestamp	Time the poll was taken.
dVol	Name of the disk volume.
Queue Depth	Queue depth of the disk volume.
Read Ops/s	Number of read operations per second.
Read KiB/s	Kibibytes read per second.
Avg Read Size Bytes	Average size in bytes of read requests per second.
Write Ops/s	Number of write operations per second.
Write KiB/s	Kibibytes written per second.
Avg Write Size Bytes	Average size in bytes for write requests per second.
Util %	Disk utilized in percetage.

EXAMPLE #22

To display the metaVolumes-std statistics group without the summary and with five second interval, type:

```
$ server_stats server_2 -m metaVolumes-std -i 5 -c 3 -te no

server_2  MetaVol  Read   Read Avg Read  Read      Write  Write Avg Write Write
Timestamp    Requests/s  KiB/s     Size Ops/s Requests/s  KiB/s      Size Ops/s
                                  Bytes                             Bytes
02:58:37  SNBS6       0      0        -     0          1      3      3072     1
          ufs_4       0      0        -     0        160   1285      8209   161
          ufs_5       0      0        -     0        163   1299      8175   162
          ufs_3       0      0        -     0         11   2155    200580    11
          ufs_2     347   2776     8192   347        140  23544    172208   140
          ufs_0     315   2517     8192   315        148  21427    147916   148
          ufs_1     654   5229     8192   654        313  45512    148895   313
          root_fs_3   1     11     8192     1          0      0         -     0
02:58:40  SNBS5       0      0        -     0          3     37     12743     3
          SNBS1       0      0        -     0          3      1       512     3
          ufs_4       0      0        -     0        159   1257      8089   157
          ufs_5       0      0        -     0        160   1273      8158   159
          ufs_3       0      0        -     0          2    511    224695     2
          ufs_2     396   3166     8192   396        195  27326    143200   195
          ufs_0     431   3446     8192   431        187  29574    162161   187
          ufs_1     408   3262     8192   408        159  27782    178784   159
          root_fs_3   1      5     8192     1          0      0         -     0
02:58:43  SNBS5       0      0        -     0          1      5      5461     1
          SNBS6       0      0        -     0          1      3      4608     1
          ufs_4       0      0        -     0        146   1159      8136   145
          ufs_5       0      0        -     0        148   1183      8174   148
          ufs_3       0      0        -     0          8   1965    262144     8
          ufs_2     522   4174     8192   522        219  35546    166238   219
          ufs_0     492   3933     8192   492        222  33356    153886   222
          ufs_1     467   3736     8192   467        188  31955    173819   188
```

Where:

Value	Definition
MetaVol	Name of the meta volume associated with the file system.
Read Request/s	Number of read requests per second to this volume.
Read KiB/s	Kibibytes read per second.
Avg Read Size Bytes	Average size for read requests to this volume.
Read Ops/s	Number of read operations per second.
Write Requests/s	Number of write requests per second.
Write KiB/s	Number of kibibytes written per second to this volume.
Avg Write Size Bytes	Average size in bytes for write requests.
Write Ops/s	Number of write operations per second.

EXAMPLE #23

To display the nfsOps-std statistics group sorted by the percentage of all the NFS operations for a five second interval, type:

```
$ server_stats server_2 -monitor nfsOps-std -sort opPct -i 5 -c 3 -te

server_2   NFS Op      NFS      NFS        NFS  NFS Op %
Timestamp               Op       Op         Op
                   Calls/s Errors/s  uSec/call
03:18:57   v3Read    23263        0      81632        50
           v3Write   23352        0     116645        50
03:19:00   v3Read    23431        0      82443        50
           v3Write   23345        0     118088        50
03:19:03   v3Read    23176        0      84759        50
           v3Write   23326        0     119608        50
```

EXAMPLE #24

To display the nfsOps-std statistics group sorted by the average time in microseconds used for a five second interval, in ascending order, type:

```
$ server_stats server_2 -m nfsops-std -sort avgTime -order asc -i 5 -c 3 -te no
server_2    NFS Op       NFS       NFS         NFS    NFS Op %
Timestamp                 Op        Op          Op
                       Calls/s  Errors/s   uSec/call
04:05:27   v3Write        605         0     8022318         100
          v3Create          2         0    25304786           0
04:05:30  v3Create          8         0     7722823           1
           v3Write        579         0     8435543          99
04:05:33  v3Create         41         0     1468883           7
           v3Write        567         0     8690860          93
```

EXAMPLE #25

To display the nfsOps-std statistics group sorted by the average time in microseconds used for a five second interval, in descending order, and including the three counts of data output, type:

```
$ server_stats server_2 -m nfsops-std -sort avgTime -order desc -lines 3 -i 5
-c 3 -te no

server_2     NFS Op      NFS         NFS         NFS    NFS Op %
Timestamp                 Op          Op          Op
                       Calls/s    Errors/s   uSec/call
04:09:39   v3Create        1           0    31657550           0
            v3Write      610           0     6223366         100
04:09:44    v3Write      607           0     6275942          98
           v3Create       11           0     3978054           2
04:09:49    v3Write      574           0     6691264          93
           v3Create       42           0     1073819           7
```

EXAMPLE #26

To display the Correlated Set list, type:

```
$ server_stats server_3 -l

server_3 :
Type                    Stat Name
...
Correlated Set          cifs.user
Counter                 cifs.user.ALL-ELEMENTS.totalCalls
Counter                 cifs.user.ALL-ELEMENTS.readCalls
Counter                 cifs.user.ALL-ELEMENTS.writeCalls
Fact                    cifs.user.ALL-ELEMENTS.suspectCalls
Counter                 cifs.user.ALL-ELEMENTS.totalBytes
Counter                 cifs.user.ALL-ELEMENTS.readBytes
Counter                 cifs.user.ALL-ELEMENTS.writeBytes
Fact                    cifs.user.ALL-ELEMENTS.avgTime
Fact                    cifs.user.ALL-ELEMENTS.server
Fact                    cifs.user.ALL-ELEMENTS.client
...
Correlated Set          nfs.user
Counter                 nfs.user.ALL-ELEMENTS.totalCalls
Counter                 nfs.user.ALL-ELEMENTS.readCalls
Counter                 nfs.user.ALL-ELEMENTS.writeCalls
Fact                    nfs.user.ALL-ELEMENTS.suspectCalls
Counter                 nfs.user.ALL-ELEMENTS.totalBytes
Counter                 nfs.user.ALL-ELEMENTS.readBytes
Counter                 nfs.user.ALL-ELEMENTS.writeBytes
Fact                    nfs.user.ALL-ELEMENTS.avgTime
...
Set                     store.volume
```

```
        Correlated Set          store.volume.ALL-ELEMENTS.fileSystem
        Counter                 store.volume.ALL-ELEMENTS.fileSystem.ALL-ELEMENTS.totalBl
ocks
        Counter                 store.volume.ALL-ELEMENTS.fileSystem.ALL-ELEMENTS.readBlo
cks
        Counter                 store.volume.ALL-ELEMENTS.fileSystem.ALL-ELEMENTS.writeBl
ocks
        Counter                 store.volume.ALL-ELEMENTS.fileSystem.ALL-ELEMENTS.totalBy
tes
        Counter                 store.volume.ALL-ELEMENTS.fileSystem.ALL-ELEMENTS.readByt
es
        Counter                 store.volume.ALL-ELEMENTS.fileSystem.ALL-ELEMENTS.writeBy
tes
        ...
```

EXAMPLE #27

To display cifs.client information with IP resolution, type:

```
$ server_stats server_2 -i 2 -m cifs.client -l 10

        server_2 IP address     CIFS CIFS CIFS CIFS CIFS CIFS CIFS CIFS
        Timestamp       Total           Read    Write   Suspicious      Total   Read    W
rite    Avg
        Ops/s           Ops/s           Ops/s   Ops     KiB/s           KiB/s   KiB/s   u
Secs/call
        09:46:49 id=10.103.11.105_20107 28      0       28              0       1627 0  1
627 33106
        id=10.103.11.105_20363          27      0       27              0       1533 0  1
533 27774
        id=10.103.11.105_18571          26      0       26              0       1470 0  1
470 29917
        id=10.103.11.105_13707          25      0       25              0       1439 0  1
439 38483
        id=10.103.11.105_17803          25      0       25              0       1466 0  1
466 46276
        id=10.103.11.105_13195          23      0       23              0       1340 0  1
340 28742
        id=10.103.11.105_16267          23      0       22              0       1277 0  1
277 37569
        id=10.103.11.105_16523          23      0       23              0       1340 0  1
340 28957
        id=10.103.11.105_17291          23      0       22              0       1277 0  1
277 34895
        id=10.103.11.105_19339          23      0       23              0       1313 0  1
313 32875
        09:46:51 p24.perf1.com_15499    27      0       27              0       1568 0  1
568 27840
        p24.perf1.com_16523             26      0       26              0       1507 0  1
507 34868
        p24.perf1.com_19595             26      0       26              0       1507 0  1
507 27609
        p24.perf1.com_20875             25      0       25              0       1441 0  1
441 27752
        p24.perf1.com_14987             25      0       25              0       1410 0  1
410 34752
        p24.perf1.com_15243             24      0       24              0       1348 0  1
348 28965
        p24.perf1.com_19083             23      0       23              0       1317 0  1
317 39723
        p24.perf1.com_19339             23      0       22              0       1256 0  1
256 29662
        p24.perf1.com_20619             23      0       23              0       1317 0  1
317 33112
        p24.perf1.com_13195             23      0       21              0       1194 0  1
194 37954
```

EXAMPLE #28

To display nfs.client information with IP resolution, type:

```
$ server_stats server_2 -monitor nfs.client -te no -c 2

server_2 Client          NFS     NFS     NFS     NFS     NFS     NFS     NFS     N
FS
Timestamp                Total   Read    Write Suspicious Total  Read    Write   A
vg
                         Ops/s   Ops/s   Ops/s   Ops      KiB/s  KiB/s   KiB/s uSe
c/call
09:48:09 id=10.103.11.106 83     0       83      0        42604  0       42604 360
77
id=10.103.11.104          70     0       66      0        28448  0       28448 318
2
id=10.103.11.105          52     0       52      0        26659  0       26659 399
84
09:48:11 p25.perf1.com    97     0       97      0        49868  0       49868 132
44
p23.perf1.com             87     0       82      0        35815  0       35815 254
9
p24.perf1.com             61     0       57      0        29242  0       29242 145
16
09:48:13 p25.perf1.com    116    0       116     0        59576  0       59576 102
01
p23.perf1.com             99     0       91      0        38273  0       38273 130
6
p24.perf1.com             51     0       51      0        26224  0       26224 110
14
09:48:15 p25.perf1.com    85     0       85      0        43591  0       43591 173
85
p23.perf1.com             70     0       62      0        27396  0       27396 551
p24.perf1.com             45     0       45      0        23214  0       23214 145
04
```

EXAMPLE #29

To monitor cifs.user information, type:

```
$ server_stats server_2 -i 1 -m cifs.user

server_2 User name      CIFS  CIFS  CIFS   CIFS  CIFS  CIFS  CIFS   CIFS    CIFS
CIFS
Timestamp              Total  Read Write  Susp- Total  Read Write    Avg   Server
Client
                        Op/s  Op/s  Op/s -cious KiB/s             uSecs/    Name
Name
14:38:52 TESTDOMAIN\admin  1     1     0      0     0     0     0    135    PITTA
-DM2-0  P27
14:38:57 TESTDOMAIN\admin 11    11     0      0     1     1     0   2257    PITTA
-DM2-0  P27
14:39:02

14:39:07 TESTDOMAIN\admin  0     0     0      0     0     0     0     22    PITTA
-DM2-0  P27
14:39:22


server_2 User name      CIFS  CIFS  CIFS   CIFS  CIFS  CIFS  CIFS   CIFS     CIFS
   CIFS
Timestamp              Total  Read Write  Susp- Total  Read Write    Avg    Serve
r Client
                        Op/s  Op/s  Op/s -cious KiB/s             uSecs/     Name
   Name
                                          Ops                     call
Minimum  TESTDOMAIN\admin  0     0     0      0     0     0     0     22       -
     -
Average  TESTDOMAIN\admin  4     4     0      0     0     0     0    634       -
     -
Maximum  TESTDOMAIN\admin 11    11     0      0     1     1     0   2257       -
```

-

EXAMPLE #30

To monitor NFS User information, type:

```
$ server_stats server_2 -i 5 -m nfs.user
server_2  User name        NFS   NFS   NFS        NFS   NFS   NFS   NFS   NFS
Timestamp                 Total  Read Write Suspicious Total  Read Write   Avg
                          Ops/s Ops/s Ops/s        Ops KiB/s KiB/s KiB/s   uSecs/c
all
14:38:52  TESTDOMAIN\admin   3     3     0          0     0     0     0   405
14:38:57  TESTDOMAIN\admin  33    33     0          0     3     3     0   6771
14:39:02

14:39:07  TESTDOMAIN\admin   0     0     0          0     0     0     0   66
14:39:22


server_2  User name         NFS   NFS   NFS        NFS   NFS   NFS   NFS  NFS
Summary                    Total  Read Write Suspicious Total  Read Write  Avg
                           Ops/s Ops/s Ops/s        Ops KiB/s KiB/s KiB/s  uSecs/c
all
Minimum   TESTDOMAIN\admin    0     0     0          0     0     0     0  66
Average   TESTDOMAIN\admin   12    12     0          0     1     1     0  1902
Maximum   TESTDOMAIN\admin   33    33     0          0     3     3     0  6771
```

EXAMPLE #31

To view Correlated Statistics information for Filesystem, type:

```
$ server_stats server_2 -c 2 -i 2 -m fs.filesystem

server_2 Filesystem   File   Total    Read  Written  Average     Read Average   W
riteAverage
Timestamp                    KiB/s    KiB/s  KiB/s   uSecs/Call  uSecs/Call     u
Secs/Call
02:54:49 ufs_2   id=38:7339       512       0      512      43873  0            4
3873
                 id=38:7221       512       0      512      79528  0            7
9528
                 id=38:8056       512       0      512      66702  0            6
6702
                 id=38:8060       512       0      512      50447  0            5
0447
                 id=38:6099       512       0      512      33244  0            3
3244
                 id=38:7338       512       0      512      86104  0            8
6104
                 id=38:6513       512       0      512      45073  0            4
5073
                 id=38:8192       512       0      512      48825  0            4
8825
                 id=38:6640       512       0      512      2417   0            2
417
                 id=38:7332       512       0      512      26889  0            2
6889
                 id=38:6556       512       0      512      88549  0            8
8549
                 id=38:7104       512       0      512      25379  0            2
5379
                 id=38:6136       512       0      512      17293  0            1
7293
                 id=38:6317       512       0      512      76986  0            7
6986
           ufs_0   id=36:6483      512       0      512       11392 0            1
1392
                 id=36:6724       512       0      512      23286  0            2
3286
                 id=36:6701       512       0      512      62777  0            6
```

```
2777
                    id=36:8118       512       0      512      30374  0            3
0374
                    id=36:6054       512       0      512      31020  0            3
1020
              ufs_4 id=40:173        2560      0      2560     2184   0            2
184
                    id=40:156        2560      0      2560     1722   0            1
722
                    id=40:178        2560      0      2560     1980   0            1
980
                    id=40:144        2560      0      2560     2167   0            2
167
                    id=40:166        2560      0      2560     2236   0            2
236
              ufs_1 id=37:10470      512       0      512      29354  0            2
9354
                    id=37:10605      512       0      512      47099  0            4
7099
                    id=37:10968      512       0      512      31437  0            3
1437
                    id=37:10721      512       0      512      68134  0            6
8134
                    id=37:10944      512       0      512      10422  0            1
0422
            ufs_5   /server_2/
                    ufs_5/dir00005/
                    testdir/
                    raN_0000053252.tmp 2560  0        2560     2199   0
2199
                    /server_2/
                    ufs_5/dir00005/
                    testdir/
                    jZE_0000057348.tmp 2560  0        2560     2416   0            2
416
                    /server_2/
                    ufs_5/dir00005/
                    testdir/
                    gCw_0000008196.tmp 2560  0        2560     2426   0            2
426
                    /server_2/
                    ufs_5/dir00005/
                    testdir/
                    KNj_0000002052.tmp 2560  0        2560     2654   0            2
654
                    /server_2/
                    ufs_5/dir00005/
                    testdir/
                    cdd_0000012292.tmp 2560  0        2560     2454   0            2
454
            ufs_3   id=39:169        2304    0        2304     34     0            3
4
                     id=39:184       2304      0      2304     35     0            3
5
                     id=39:176       2304      0      2304     37     0            3
7
                     id=39:188       2304      0      2304     32     0            3
2
                     id=39:172       2048      0      2048     31     0            3
1
                     id=39:189       2048      0      2048     33     0            3
3
      02:54:51 ufs_2 id=38:7238       512       0      512      27145  0            2
7145
                     id=38:5616      512       0      512      201    0            2
01
                     id=38:6468      512       0      512      81317  0            8
1317
                     id=38:6913      512       0      512      78302  0            7
8302
                     id=38:6441      512       0      512      76825  0            7
```

```
6825
                ufs_0   id=36:6565       512       0      512      4537   0            4
537
                        id=36:6937       512       0      512      72659  0            7
2659
                        id=36:7716       512       0      512      11224  0            1
1224
                        id=36:7138       512       0      512      85855  0            8
5855
                        id=36:6233       512       0      512      68502  0            6
8502
               ufs_4    id=40:173        2560      0      2560     3216   0            3
216
                        id=40:185        2560      0      2560     3349   0            3
349
                        id=40:156        2560      0      2560     3268   0            3
268
                        id=40:152        2560      0      2560     2587   0            2
587
                        id=40:201        2560      0      2560     3123   0            3
123
             ufs_1      id=37:9952       512       0      512      30769  0            3
0769
                        id=37:10818      512       0      512      16503  0            1
6503
                        id=37:10053      512       0      512      13284  0            1
3284
                        id=37:10658      512       0      512      35439  0            3
5439
                        id=37:10676      512       0      512      10284  0            1
0284
               ufs_5    /server_2/
                        ufs_5/dir00005/
                        testdir/
                        raN_0000053252.tmp 2560    0     2560      3299   0            3
299
                        /server_2/
                        ufs_5/dir00005/
                        testdir/
                        JeD_0000020484.tmp 2560    0     2560      3267   0            3
267
                        /server_2
                        /ufs_5/dir00005/
                        testdir/
                        jZE_0000057348.tmp 2560   0      2560      2066   0            2
066
                        /server_2/
                        ufs_5/dir00005/
                        testdir/
                        gCw_0000008196.tmp 2560   0      2560      2931   0            2
931
               ufs_3    id=39:169          2560   0      2560      34     0
34
                        id=39:140          2560   0      2560      33     0            3
3
                         id=39:181         2560   0      2560      31     0            3
1
                         id=39:184         2560   0      2560      26     0            2
6
                         id=39:167         2560   0      2560      29     0            2
9

        server_2         Filesystem         File    Total     Read   Written Average Read
 Average  Write Average
        Summary                             KiB/s    KiB/s    KiB/s   uSecs/Call     uSec
s/Call    uSecs/Call
        Minimum  ufs_2  id=38:5616          512        0      512         201
0           201
                        id=38:5787          512        0      512       26397
0           26397
                        id=38:5939          512        0      512       73921
```

```
0           73921
                    id=38:6099          512        0      512       33244
0           33244
                    id=38:6136          512        0      512       17293
0           17293
                    ufs_1.op
                ufs_0   id=36:5924          512        0      512       11537
0           11537
                    id=36:6054          512        0      512       31020
0           31020
                    id=36:6062          512        0      512        8021
0           8021
                    id=36:6140          512        0      512       93666
0           93666
                    id=36:6214          512        0      512       47252
0           47252
                    id=36:6233          512        0      512       68502
0           68502
                   ufs_4    id=40:129           2560       0      2560      2416
0           2416
                    id=40:139           2560       0      2560      2079
0           2079
                    id=40:144           2560       0      2560      2167
0           2167
                    id=40:145           2560       0      2560      2507
0           2507
                    id=40:156           2560       0      2560      1722
0           1722
                   ufs_1    id=37:10053         512        0      512       13284
0           13284
                    id=37:10210         512        0      512       24623
0           24623
                    id=37:10228         512        0      512       11273
0           11273
                    id=37:10385         512        0      512       36637
0           36637
                    id=37:10396         512        0      512       23479
0           23479
                  ufs_5     /server_2/ufs_5/
                            dir00005/testdir/
                            D2K_0000014340.tmp  2560       0     2560       2457            0
             2457
                             /server_2/
                            ufs_5/dir00005/
                            testdir/
                            HNi_0000018436.tmp  2560       0     2560       2727
0           2727
                ufs_3   id=39:140           2560       0     2560       33
0           33
                    id=39:143           2560       0     2560       32
0           32
                    id=39:146           2560       0     2560       31
0           31
                    id=39:177           2560       0     2560       29
0           29
                    id=39:169           2304       0     2304       34
0           34
                   root_fs_common
                   root_fs_2
        Average ufs_2   id=38:5616           512         0     512        201
0           201
                    id=38:5787           512         0     512        26397
0           26397
                    id=38:5939           512         0     512        73921
0           73921
                    id=38:6099           512         0     512        33244
0           33244
                    id=38:6136           512         0     512        17293
0           17293
                    ufs_1.op
                   ufs_0   id=36:5924           512         0     512        11537
```

```
0           11537
                    id=36:6054          512         0     512        31020
0           31020
                    id=36:6062          512         0     512        8021
0           8021
                    id=36:6140          512         0     512        93666
0           93666
                    id=36:6214          512         0     512        47252
0           47252
              ufs_4 id=40:129           2560        0     2560        2416
0           2416
                    id=40:139           2560        0     2560        2558
0           2558
                    id=40:144           2560        0     2560        2167
0           2167
                    id=40:145           2560        0     2560        2507
0           2507
                    id=40:156           2560        0     2560        2495
0           2495
              ufs_1 id=37:10053         512         0     512         13284
0           13284
                    id=37:10210         512         0     512         24623
0           24623
                    id=37:10228         512         0     512         11273
0           11273
                    id=37:10385         512         0     512         36637
0           36637
                    id=37:10396         512         0     512        23479
0           23479
             ufs_5  /server_2/
                    ufs_5/dir00005/
                    testdir/
                    D2K_0000014340.tmp  2560        0     2560        2880
0           2880
                    /server_2/
                    ufs_5/dir00005/
                    testdir/
                    HNi_0000018436.tmp  2560        0     2560        2824
0           2824
            ufs_3   id=39:140           2560        0     2560          33
0           33
                    id=39:143           2560        0     2560          32
0           32
                     id=39:146          2560        0     2560         31
0           31
                     id=39:177          2560        0     2560         29
0           29
                     id=39:169          2432        0     2432         34
0           34
            root_fs_common
             root_fs_2
        Maximum ufs_2   id=38:5616      512         0      512         201
0           201
                     id=38:5787         512         0      512         26397
0           26397
                     id=38:5939         512         0      512         73921
0           73921
                     id=38:6099         512         0      512         33244
0           33244
                     id=38:6136         512         0      512         17293
0           17293
             ufs_1.op
             ufs_0  id=36:5924         512         0     512         11537
0           11537
                     id=36:6054         512         0     512         31020
0           31020
                     id=36:6062         512         0     512         80210
0           8021
                     id=36:6140         512         0     512        93666
0           93666
                     id=36:6214         512         0     512        47252
```

```
0          47252
             ufs_4      id=40:129            2560       0     2560        2416
0          2416
                        id=40:135            2560       0     2560        2656
0          2656
                        id=40:139            2560       0     2560        3037
0          3037
                        id=40:143            2560       0     2560        2896
0          2896
                        id=40:144            2560       0     2560        2167
0          2167
             ufs_1      id=37:10053          512        0     512         13284
0          13284
                        id=37:10210          512        0     512         24623
0          24623
                        id=37:10228          512        0     512         11273
0          11273
                        id=37:10385          512        0     512         36637
0          36637
                        id=37:10396          512        0     512         23479
0          23479
             ufs_5      /server_2/ufs_5/
                        dir00005/testdir/
                        D2K_0000014340.tmp   2560       0    2560         3302
0         3302
                        /server_2/ufs_5/
                        dir00005/testdir/
                        HNi_0000018436.tmp   2560       0    2560          2920
0          2920
            ufs_3       id=39:140           2560       0    2560          33
0          33
                        id=39:143            2560       0     2560         32
0          32
                        id=39:146            2560       0     2560         31
0          31
                        id=39:161            2560       0     2560         28
0          28
                        id=39:167            2560       0     2560         29
0          29
           root_fs_common
           root_fs_2</pre>
```

Note: In order to have proper resolution, perform the following steps:
1.Start the service (server_fileresolve movername -service -start)
2.Register filesystem (server_fileresolve movername -add /filesystem_mount_path). If, however, the service is running, but the filesystem is not registered with it, one can resolve the filename manual
ly:
server_fileresolve movername -lookup -filesystem ufs_2 -inode 38

EXAMPLE #32

To monitor store.volume information, type:

```
server_stats server_2 -i 1 -m store.volume
9:30:06   NBS1             id=0           71762    0     71762    35881        0
   35881
                             root_fs_2     16       0        16        8        0
      8
               d16           ufs_1         2173  2047       126     1087     1024
     63
               d9            ufs_1         362   236        126      181      118
     63
                             ufs_4         47      0         47       24        0
     24
               d10           ufs_2         425   362         63      213      181
     31
               d18           ufs_1         2835  2756        79     1417     1378
     39
               d11           ufs_0         441   378         63      220      189
```

```
      31
                  d19              ufs_2         1465  1339       126      732      669
      63
                  d12              ufs_2         252   142        110      126      71
      55
                                   ufs_5         31     0         31       16       0
      16
                  d20              ufs_0         1559  1433       126      780      717
      63
                  d13              ufs_0         252   157         94      126      79
      47
                                   ufs_3         47     0         47       24       0
      24
                  d21              ufs_2         1921  1827        94      961      913
      47
                  d14              ufs_1         772   646        126      386      323
      63
                  d22              ufs_0         2079  2016        63     1039     1008
      31
```

EXAMPLE #33

To monitor NFS statistics information, type:

```
$ server_stats server_2 -i 1 -m nfs.client -noresolve

        server_2   Client               NFS      NFS      NFS        NFS      NFS    NFS
     NFS        NFS
        Timestamp                       Total     Read    Write Suspicious    Total  Read
    Write        Avg
                                        Ops/s    Ops/s    Ops/s        Ops    KiB/s  KiB/
s    KiB/s  uSec/call
        09:31:41   id=10.103.11.106        81        0       81          0    41307   0
     41307      11341
                   id=10.103.11.104        41        0       41          0    20908   0
     20908       1534
                   id=10.103.11.105        40        0       40          0    20398   0
     20398      13981
        09:31:42   id=10.103.11.104        79        0       79          0    40564   0
     40564       1085
                   id=10.103.11.106        74        0       74          0    38091   0
     38091      16159
                   id=10.103.11.105        35        0       35          0    17809   0
     17809      12770
        09:31:43   id=10.103.11.106        87        0       87          0    44384   0
     44384      14268
                   id=10.103.11.104        58        0       58          0    29589   0
     29589        470
                   id=10.103.11.105        31        0       31          0    15851   0
     15851      10026

        $ server_stats server_2 -i 1 -m nfs.user -noresolve
        server_2   NFS User             NFS      NFS      NFS        NFS     NFS  NFS
      NFS       NFS
        Timestamp                     Total     Read    Write Suspicious   Total  Read
     Write       Avg
                                      Ops/s    Ops/s    Ops/s        Ops   KiB/s  KiB/s
      KiB/s uSec/call
        09:32:51   id=0                 144        0      144          0    73841   0
     73841      11514
                   id=550                 4        0        1          0        8   0
       8       4219
                   id=553                 2        0        0          0        0   0
       0       4696
                   id=555                 2        0        0          0        0   0
       0       4369
                   id=558                 2        0        0          0        0   0
       0        286
                   id=563                 2        0        0          0        0   0
       0       2231
                   id=569                 2        0        0          0        0   0
```

```
0        228
                id=585                2        0        0          0        0   0
0        247
                id=588                2        0        0          0        0   0
0        282
                id=589                2        0        2          0        8   0
8      25418
                id=591                2        0        0          0        0   0
0        214
                id=595                2        0        0          0        0   0
0       3700
                id=551                1        0        1          0        0   0
0       8535
                id=556                1        0        0          0        0   0
0        238
                id=557                1        0        1          0        0   0
0        312
                id=564                1        0        0          0        0   0
0      12921
                id=582                1        0        1          0        0   0
0       5748
```

EXAMPLE #34

To display NFS statistics correlated by file system and NFS operation attributes, type:

```
$ server_stats server_2 -i 1 -m nfs.filesystem
server_2   Filesystem          Client       NFS Op    NFS Op      NFS
Timestamp                                              Avg        Op
                                                     uSecs/Call  Calls/s
16:50:42   ufs_5          id=10.103.11.18  v3Write         522        1
16:50:43
16:50:44   ufs_5          l18.perf1.com    v3Lookup         13       11
16:50:45   ufs_5          l18.perf1.com    v3Write        1810       17
                                           v3Read           49      133
                                           v3Lookup         10      180
           ufs_4          l17.perf1.com    v3Write        1311       18
                                           v3Read           47      115
                                           v3Lookup         10      220
16:50:46   ufs_5          l18.perf1.com    v3Write        7026      137
                                           v3Read           52      248
           ufs_4          l17.perf1.com    v3Write        6347      131
                                           v3Read          121      354
           ufs_3          id=10.103.11.16  v3Write         297      122
                                           v3Read           91      390
16:50:47   ufs_5          l18.perf1.com    v3Write        6701      161
           ufs_4          l17.perf1.com    v3Write        4754      159
                                           v3Read           47       39
           ufs_3          l16.perf1.com    v3Write         129      160
                                           v3Read          123       38

server_2  Filesystem         Client         NFS Op    NFS Op      NFS
Summary                                                Avg        Op
                                                     uSecs/Call  Calls/s
Minimum    ufs_5          l18.perf1.com    v3Write         522        1
                                           v3Read           49      133
                                           v3Lookup         10       11
           ufs_4          l17.perf1.com    v3Write        1311       18
                                           v3Read           47       39
                                           v3Lookup         10      220
           ufs_1
           ufs_2
           ufs_0
           ufs_3          l16.perf1.com    v3Write         129      122
                                           v3Read           91       38
           root_fs_common
           root_fs_2
Average    ufs_5          l18.perf1.com    v3Write        4015       79
                                           v3Read           50      190
```

```
                                          v3Lookup         12       95
          ufs_4          l17.perf1.com    v3Write        4137      102
                                          v3Read           72      170
                                          v3Lookup         10      220
          ufs_1
          ufs_2
          ufs_0
          ufs_3          l16.perf1.com    v3Write         213      141
                                          v3Read          107      214
          root_fs_common
          root_fs_2
Maximum   ufs_5          l18.perf1.com    v3Write        7026      161
                                          v3Read           52      248
                                          v3Lookup         13      180
          ufs_4          l17.perf1.com    v3Write        6347      159
                                          v3Read          121      354
                                          v3Lookup         10      220
          ufs_1
          ufs_2
          ufs_0
          ufs_3          l16.perf1.com    v3Write         297      160
                                          v3Read          123      390
          root_fs_common
          root_fs_2
```

EXAMPLE #35

To display a summary of NFS filesystem statistics correlated by single file system, type:

```
$ server_stats server_2 -m nfs.filesystem.ufs_4
server_2   Filesystem          Client       NFS Op    NFS Op      NFS
Timestamp                                              Avg        Op
                                                     uSecs/Call  Calls/s
02:46:00   ufs_4          l23.perf1.com   v3Write      2569      132
                                         v3Create     38         0
02:46:15   ufs_4          l23.perf1.com    v3Write     3313      132

server_2  Filesystem         Client         NFS Op    NFS Op      NFS
Summary                                                Avg        Op
                                                     uSecs/Call  Calls/s
Minimum    ufs_4          l23.perf1.com    v3Write      2569     132
                                         v3Create     38         0
Average    ufs_4          l23.perf1.com    v3Write      2941     132
                                         v3Create     38         0
Maximum    ufs_4          l23.perf1.com    v3Write      3313      132
                                         v3Create     38         0
                                                      123       390
```

EXAMPLE #36

To display a summary of NFS filesystem statistics correlated by a specific filesystem and specific client, type:

```
$ server_stats server_2 -i 2 -m nfs.filesystem.ufs_4.client.10.103.11.23
server_2   Filesystem          Client       NFS Op    NFS Op      NFS
Timestamp                                              Avg        Op
                                                     uSecs/Call  Calls/s
02:41:36   ufs_4           l23.perf1.com   v3Write      2083        120
02:41:38   ufs_4           l23.perf1.com   v3Write      4318        132

02:41:40   ufs_4           l23.perf1.com   v3Write      2660        116

server_2  Filesystem         Client         NFS Op    NFS Op      NFS
Summary                                                Avg        Op
                                                      uSecs/Call  Calls/s
Minimum    ufs_4           l23.perf1.com    v3Write      2083        116
Average    ufs_4           l23.perf1.com    v3Write      3020        123

Maximum    ufs_4           l23.perf1.com    v3Write      4318        132
```

EXAMPLE #37

To display a summary of NFS filesystem statistics for a specific client and operation, type:
$ server_stats server_2 -i 2 -m nfs.filesystem.ufs_4.client.10.103.11.23.op.v3Write

```
server_2   Filesystem          Client       NFS Op    NFS Op        NFS
Timestamp                                              Avg          Op
                                                     uSecs/Call     Calls/s
02:42:39   ufs_4          l23.perf1.com   v3Write      2335         123
02:42:41   ufs_4          l23.perf1.com   v3Write      4836         134
02:42:43   ufs_4          l23.perf1.com   v3Write      5093         142
02:42:45   ufs_4          l23.perf1.com   v3Write      2129         142

server_2  Filesystem         Client         NFS Op    NFS Op          NFS
Summary                                                Avg           Op
                                                     uSecs/Call      Calls/s
Minimum    ufs_4          l23.perf1.com    v3Write      2129         123
Average    ufs_4          l23.perf1.com    v3Write      3598         135

Maximum    ufs_4           l23.perf1.com   v3Write      5093         142
```

EXAMPLE #38

To monitor the BranchCache information while a SMB2 BranchCache client is reading a tree, type:
$ server_stats server_3 -i 3 -m cifs.branchcache.basic

```
server_3   Filtered  Generated   Fail     Hash    Transf    Hits     Miss Queued
  Running

Timestamp         Hash      Hash      Hash     Files    Hash     Req/s    Req/s T
asks    Tasks
                Files/s    Files/s   Files/s   kB/s     kB/s
  16:27:13          0          0         0        0        0        0        0   0
      1
  16:27:16          0          0         0        0        0        0        2   0
      2
  16:27:19          0          0         0        0        0        0        0   0
      2
  16:27:22          0          0         0        0        0        0        0   0
      2
  16:27:34          0          0         0        0        0        0        2   0
      3
  16:27:37          0          0         0        0        0        0        1   1
      3
  16:27:40          0          0         0        0        0        0        2   2
      3
  16:27:43          0          0         0        0        0        0        2   3
      3
  16:28:07          0          2         0       35        0        0        0   0
      1
  16:28:10          0          0         0        0        0        0        0   0
      1
  16:28:13          0          0         0       30        0        0        0   0
      0
  16:28:16          0          0         0        0        0        0        0   0
      0
  server_stats server_3 -i 3 -m cifs.branchcache.usage

  server_3     Avg      Max       Min      Avg       Max       Min       Max   Coun
t   Rejected
  Timestamp   Hash      Hash     Hash      Hash      Hash      Hash    Threads Task
s    Tasks
              Files B  Files kB  Files B  Files ms  Files ms  Files ms
  18:16:39     24934        90     1268     20431     71831       132        3   8
  0
  18:16:42     24934        90     1268     20431     71831       132        3   8
```

```
0
18:16:45     24934        90     1268     20431     71831       132        3   8
0
```

EXAMPLE #39

To monitor NFS group statistics, type:
$ server_stats server_2 -m nfs.group

```
erver_2 NFS Group NFS      NFS      NFS       NFS        NFS      NFS     NFS
 NFS
Timestamp         Total    Read     Write   Suspicious   Total    Read   Write
 Avg
Ops/s             Ops/s    Ops/s    Ops diff    KiB/s    KiB/s    KiB/s  uSec/cal
l
02:47:14   id=0    264        0      213           0     108919     0   108919
1683
02:47:29   id=0    416        0      416           0     212821     0   212821
2184
02:47:44   id=0    432        0      432           0     221252     0   221252
5206

server_2 NFS Group NFS      NFS      NFS       NFS        NFS      NFS    NFS
   NFS
Summary            Total    Read     Write   Suspicious   Total    Read   Write
   Avg
                   Ops/s    Ops/s    Ops/s    Ops diff    KiB/s    KiB/s  KiB/s
   uSec/call
Minimum    id=0         264        0      213           0   108919   0     108919
   1683
Average    id=0         370        0      354           0   180998   0     180998
   3024
Maximum    id=0         432        0      432           0   221252   0     221252
   5206
```

EXAMPLE #40

To monitor NFS export statistics, type:
$ server_stats server_2 -m nfs.export

```
server_2 NFS Export               NFS   NFS   NFS    NFS   NFS   NFS  NFS
NFS
Timestamp                         Total Read Write Susp- Total Read   Write
Avg
                                  Op/s  Op/s  Op/s -cious KiB/s KiB/s KiB/s
uSecs/call                                          Ops
call
02:48:14 /server_2/ufs_5/dir00005 157      0   157    0    80213  0  80213
6272
DM2-0
         /server_2/ufs_3/dir00003 139      0   139    0    71305  0  71305
462
         /server_2/ufs_4/dir00004 130      0   130    0    66662  0  66662
3700
02:48:29 /server_2/ufs_5/dir00005 158      0   158    0    80828  0  80828
3454
         /server_2/ufs_3/dir00003 140      0   140    0    71646  0  71646  29
         /server_2/ufs_4/dir00004 133      0   133    0    68233  0  68233
3040

server_2 NFS Export              NFS    NFS   NFS    NFS   NFS  NFS    NFS
NFS
Timestamp                        Total  Read Write  Susp-  Total  Read Write
Avg
                                 Op/s  Op/s  Op/s -cious  KiB/s   KiB/s KiB/s
uSecs/call
                                                Ops
Minimum  /server_2/ufs_5/dir00005 157    0    157     0    80213   0   80213
3454
```

```
                /server_2/ufs_3/dir00003 139    0    139     0    71305   0   71305
        29
                /server_2/ufs_4/dir00004 130    0    130     0    66662   0   66662
        3040
        Average  /server_2/ufs_5/dir00005 157    0    157     0    80521   0   80521
        4863
                /server_2/ufs_3/dir00003 140    0    140     0    71475   0    71475
        246
                /server_2/ufs_4/dir00004 132    0    132     0    67447   0    67447
        3370
        Maximum  /server_2/ufs_5/dir00005 158    0    158     0    80828   0    80828
        6272
                /server_2/ufs_3/dir00003 140    0    140     0    71646   0    71646
        462
                /server_2/ufs_4/dir00004 133    0    133     0    68233   0    68233
        3700


        EXAMPLE # 41
        ------------

        To monitor CIFS server statistics, type:
        $ server_stats server_2 -m cifs.server

        server_2 Server name      CIFS  CIFS  CIFS   CIFS  CIFS  CIFS  CIFS   CIFS
        Timestamp                 Total  Read Write    Susp- Total  Read Write Avg
                                  Op/s  Op/s  Op/s   -cious  KiB/s             uSecs/
                                                        Ops                     call
        02:50:29 RAVEN-DM2-2     2176     0    1957      0     0     0     0    135

        14:38:57 TESTDOMAIN\admin 11     11     0        0     1     1     0   2257

        14:39:02

        14:39:07 TESTDOMAIN\admin  0     0      0      0     0     0     0     22

        14:39:22


        server_2 Server name      CIFS  CIFS  CIFS   CIFS  CIFS  CIFS  CIFS   CIFS
        Timestamp              Total  Read Write  Susp- Total  Read Write    Avg
                                Op/s  Op/s  Op/s -cious KiB/s             uSecs/
                                                  Ops                     call
        Minimum  TESTDOMAIN\admin  0     0     0      0     0     0     0     22
        Average  TESTDOMAIN\admin  4     4     0      0     0     0     0    634
        Maximum  TESTDOMAIN\admin 11    11     0      0     1     1     0   2257


        EXAMPLE #42
        -----------

        To monitor FS qtreefile statistics, type:
        $ server_stats server_2 -i 1 -c 2 -m fs.qtreefile

        server_2  Quota Tree File  File        Total    Read   Written  Average   Read Av
erage  Write Average
        Timestamp                             KiB/s    KiB/s  KiB/s    uSecs/Call  uSecs
/Call    uSecs/Call


        02:55:57 ufs_2:/dir00002  id=38:7339   512       0      512      43873       0
        43873
                                  id=38:10137  1024      0      1024     49557       0
        49557
                ufs_0:/dir00000   id=36:10769  1024      0      1024     26188       0
        26188
                                  id=36:11712  1024      0      1024     45377       0
        45377
                ufs_4:/dir00004   id=40:251    2560      0      2560     1538        0
        1538
                                  id=40:256    2560      0      2560     1280        0
```

```
        1280
                 ufs_1:/dir00001  id=37:17393  1024      0      1024     54210       0
        54210
                                  id=37:17572  1024      0      1024     39708       0

        02:55:58 ufs_2:/dir00002  id=38:10221   1024       0     1024     51350       0
        51350
                                  id=38:9981    1024       0     1024     37275       0
        37275
                 ufs_0:/dir00000  id=36:10155   1024       0     1024     60618       0
        60618
                                  id=36:10453   1024       0     1024     32847       0
        32847
                 ufs_4:/dir00004  id=40:183     2560       0     2560     3332        0
        3332
                                  id=40:256     2560       0     2560     1391        0
        1391
                 ufs_1:/dir00001  id=37:17129   1024       0     1024     77310       0
        77310
                                  id=37:17453   1024       0     1024     17741       0
        17741
                 ufs_5:/dir00005  /server_2/
                                  ufs_5/dir00005/testdir/
                                  iv9_0000055303.tmp 2560 0     2560     1982        0
        1982
                                  /server_2/
                                  ufs_5/dir00005/testdir/
                                 KRc_0000008199.tmp  2560 0     2560     2019        0
        2019
                ufs_3:/dir00003   id=39:243      2560     0     2560     26          0
        26
                                  id=39:248      2560     0     2560     29          0
        29

          server_2  Quota Tree File  File        Total    Read   Written  Average  Read Ave
rage  Write Average
          Summary                               KiB/s    KiB/s  KiB/s    uSecs/Call uSecs/
Call    uSecs/Call
          Minimum   ufs_2:/dir00002  id=38:10063  1024      0     1024     32177      0
         32177
                                     id=38:10066  1024      0     1024     18897     0
        18897
                    ufs_2:/dir00008
                    ufs_2:/dir00014
                    ufs_2:/dir00020
                    ufs_2:/dir00026
                    ufs_0:/dir00000  id=36:10151  1024       0    1024      42949     0
         42949
                                     id=36:10155  1024       0    1024      60618     0
         60618
                    ufs_0:/dir00006
                    ufs_0:/dir00012
                    ufs_0:/dir00018
                    ufs_0:/dir00024
                    ufs_4:/dir00004  id=40:183    2560        0     2560     3332      0
        3332
                                     id=40:194    2560        0     2560     1488      0
        1488
                    ufs_4:/dir00010
                    ufs_4:/dir00016
                    ufs_4:/dir00022
                    ufs_4:/dir00028
                    ufs_1:/dir00001  id=37:15343  1024        0     1024     533       0
       533
                                     id=37:16235  1024        0     1024     2197      0
       2197
                    ufs_1:/dir00007
                    ufs_1:/dir00013
                    ufs_1:/dir00019
                    ufs_1:/dir00025
                    ufs_5:/dir00005  /server_2/
```

```
                                ufs_5/dir00005/testdir/
                                72n_0000028679.tmp 2560   0     2560     1580    0
1580
                                /server_2/
                                ufs_5/dir00005/testdir/
                                74u_0000022535.tmp 2560   0     2560     1547    0
1547
                    ufs_5:/dir00011
                    ufs_5:/dir00017
                    ufs_5:/dir00023
                    ufs_5:/dir00029
                    ufs_3:/dir00003  id=39:165     2560        0     2560     32     0
32
                                     id=39:174     2560        0     2560     29     0
29
                    ufs_3:/dir00009
                    ufs_3:/dir00015
                    ufs_3:/dir00021
                    ufs_3:/dir00027
    Average         ufs_2:/dir00002  id=38:10063   1024        0     1024      32177  0
32177
                                     id=38:10066   1024        0     1024      18897  0
18897
                    ufs_2:/dir00008
                    ufs_2:/dir00014
                    ufs_2:/dir00020
                    ufs_2:/dir00026
                    ufs_0:/dir00000  id=36:10151   1024        0     1024       42949  0
 42949
                                     id=36:10155   1024        0     1024       60618  0
 60618
                    ufs_0:/dir00006
                    ufs_0:/dir00012
                    ufs_0:/dir00018
                    ufs_0:/dir00024
                    ufs_4:/dir00004  id=40:183     2560        0     2560        3332  0
 3332
                                     id=40:194     2560        0     2560        1488  0
 1488
                    ufs_4:/dir00010
                    ufs_4:/dir00016
                    ufs_4:/dir00022
                    ufs_4:/dir00028
                    ufs_1:/dir00001  id=37:15343   1024        0     1024        533   0
 533
                                     id=37:16235   1024        0     1024        2197  0
 2197
                    ufs_1:/dir00007
                    ufs_1:/dir00013
                    ufs_1:/dir00019
                    ufs_1:/dir00025
                    ufs_5:/dir00005 /server_2/ufs_5
                                    /dir00005/testdir/
                                    72n_0000028679.tmp 2560     0     2560        1724  0
 1724
                                    /server_2/ufs_5/
                                    dir00005/testdir/
                                    74u_0000022535.tmp 2560     0     2560        1627  0
 1627
                    ufs_5:/dir00011
                    ufs_5:/dir00017
                    ufs_5:/dir00023
                    ufs_5:/dir00029
                    ufs_3:/dir00003 id=39:165       2560        0     2560         32   0
  32
                                    id=39:174       2560        0     2560         29   0
  29
                    ufs_3:/dir00009
                    ufs_3:/dir00015
                    ufs_3:/dir00021
                    ufs_3:/dir00027
```

```
   Maximum    ufs_2:/dir00002 id=38:10063     1024        0     1024       32177  0
32177
                              id=38:10066     1024        0     1024       18897  0
18897
              ufs_2:/dir00008
              ufs_2:/dir00014
              ufs_2:/dir00020
              ufs_2:/dir00026
              ufs_0:/dir00000 id=36:10151     1024        0     1024       42949  0
42949
                              id=36:10155     1024        0     1024       60618  0
60618
              ufs_0:/dir00006
              ufs_0:/dir00012
              ufs_0:/dir00018
              ufs_0:/dir00024
              ufs_4:/dir00004 id=40:183       2560        0     2560        3332  0
3332
                              id=40:191       2560        0     2560        1969  0
1969
              ufs_4:/dir00010
              ufs_4:/dir00016
              ufs_4:/dir00022
              ufs_4:/dir00028
              ufs_1:/dir00001 id=37:15343     1024        0     1024        533   0
 533
                              id=37:16235     1024        0     1024        2197  0
 2197
              ufs_1:/dir00007
              ufs_1:/dir00013
              ufs_1:/dir00019
              ufs_1:/dir00025
              ufs_5:/dir00005 /server_2/
                              ufs_5/dir00005/testdir/
                              72n_0000028679.tmp  2560     0    2560        1867  0
1867
                              /server_2/
                              ufs_5/dir00005/testdir/
                              74u_0000022535.tmp  2560     0     2560       1964  0
1964
              ufs_5:/dir00011
              ufs_5:/dir00017
              ufs_5:/dir00023
              ufs_5:/dir00029
              ufs_3:/dir00003  id=39:165      2560        0     2560        32   0
32
                               id=39:168      2560        0     2560        29   0
29

   EXAMPLE #43
   -----------

   To monitor NFS VDM client statistics, type:
   $ server_stats server_3 -i 1 -m nfs.vdm.*.client -c 5 -te no

   server_2  VDM name Client      NFS   NFS   NFS   NFS       NFS    NFS   NFS   NFS

   Timestamp                     Total Read Write Suspicious Total  Read Write  Avg
                                 Ops/s Ops/s Ops/s  Ops      KiB/s KiB/s KiB/s  uSe
cs/call
   10:42:43  vdm_1 id=10.103.11.13  2      0   0      1       0       0     0     188
57
             vdm_2 id=10.103.11.14  7      0   1      0       0       0     0     163
78
   10:42:44  vdm_1 id=10.103.11.13  2      0   0      1       0       0     0     788
2
             vdm_2 id=10.103.11.14  8      0   1      0       0       0     0     117
84
   10:42:45  vdm_1 l13.perf1.com    2      0   0      1       0       0     0     976
2
             vdm_2 l14.perf1.com    7      0   1      0       0       0     0     198
```

```
13
        10:42:46  vdm_1 l13.perf1.com    2      0   0      1       0       0     0    697
73
                  vdm_2 l14.perf1.com    7      0   1      0       0       0     0    825
7
        10:42:47  vdm_1 l13.perf1.com    2      0   0      1       0       0     0    104
73
                  vdm_2 l14.perf1.com    8      0   1      0       0       0     0    183
5
```

Example #44

To monitor NFS VDM user statistics, type:

```
$ server_stats server_3 -i 1 -m nfs.vdm.*.user -c 5 -te no

server_2  VDM name NFS  NFS   NFS   NFS   NFS   NFS  NFS   NFS  NFS
Timestamp          User Total  Read Write Suspicious Total Read Avg
                        Ops/s Ops/s Ops/s  Ops KiB/s KiB/s Write uSecs/
                                                          KiB/s call
10:43:20  vdm_1   id=0   2    0      0      1     0    0    0    18791
          vdm_2   id=0   5    0      1      0     0    0    0    6070
10:43:21  vdm_1   id=0   2    0      0      1     0    0    0    15574
          vdm_2   id=0   3    0      0      0     0    0    0    94
10:43:22  vdm_1   id=0   2    0      0      1     0    0    0    16976
          vdm_2   id=0   2    0      1      0     0    0    0    116061
10:43:23  vdm_1   id=0   2    0      0      1     0    0    0    38150
          vdm_2   id=0   5    0      1      0     0    0    0    19183
10:43:24  vdm_1   id=0   2    0      0      1     0    0    0    63362
          vdm_2   id=0   5    0      1      0     0    0    0    53115
```

Example #45

To monitor NFS VDM group statistics, type:

```
$ server_stats server_3 -i 1 -m nfs.vdm.*.group -c 5 -te no

server_2  VDM name NFS  NFS   NFS   NFS    NFS   NFS   NFS  NFS   NFS
Timestamp          Group Total  Read Write Suspicious Total  Read Write Avg
                         Ops/s Ops/s Ops/s    Ops KiB/s KiB/s KiB/s uSecs/call
10:43:46   vdm_1   id=0   2      0     0      1   0     0     0     6381
           vdm_2   id=0   5      0     1      0   0     0     0     7557
10:43:47   vdm_1   id=0   1      0     0      0   0     0     0     10
           vdm_2   id=0   5      0     1      0   0     0     0     29440
10:43:48   vdm_1   id=0   1      0     0      1   0     0     0     198524
           vdm_2   id=0   6      0     1      0   0     0     0     5877
10:43:49   vdm_1   id=0   2      0     0      1   0     0     0     52406
           vdm_2   id=0   5      0     1      0   0     0     0     34691
10:43:50   vdm_1   id=0   2      0     0      1   0     0     0     13695
           vdm_2   id=0   5      0     1      0   0     0     0     2984
```

Example #46

To monitor NFS VDM export statistics, type:

```
$ server_stats server_3 -i 1 -m nfs.vdm.*.export -c 5 -te no

server_2  VDM name NFS     NFS   NFS   NFS    NFS     NFS    NFS  NFS  NFS
Timestamp          Export  Total Read Write Suspicious Total  Read Write Avg
                           Ops/s Ops/s Ops/s  Ops KiB/s KiB/s KiB/s uSecs/call
10:44:10   vdm_1 /demo_0    2      0    0      1       0      0    0    16975
           vdm_2 /demo_1    5      0    1      0       0      0    0    4755
10:44:11   vdm_1 /demo_0    2      0    0      1       0      0    0    42104
           vdm_2 /demo_1    5      0    1      0       0      0    0    75083
10:44:12   vdm_1 /demo_0    2      0    0      0       0      0    0    18991
           vdm_2 /demo_1    3      0    0      0       0      0    0    96
10:44:13   vdm_1 /demo_0    2      0    0      1       0      0    0    14357
           vdm_2 /demo_1    2      0    1      0       0      0    0    138779
10:44:14   vdm_1 /demo_0    2      0    0      1       0      0    0    137153
           vdm_2 /demo_1    5      0    1      0       0      0    0    9511
```

Last modified: August 24, 2012, 11:06 a.m

server_sysconfig

Manages the hardware configuration for the specified Data Mover(s).

SYNOPSIS

```
server_sysconfig {<movername>|ALL}
  -Platform
| -pci [ -fmwr_version [ <device> ] ][ <device> [ -option <options> ] ]
| -virtual -delete [-Force] <device>
| -virtual -info <device>
| -virtual -name <device> -create trk
    -option "device=<device,..> [protocol=lacp][lb=<mac|ip|tcp>]"
| -virtual -name <device> -create fsn
    -option primary=<device> device=<device,..>}
```

DESCRIPTION
----

server_sysconfig displays and modifies the hardware configuration of the Data Movers.

To use Ethernet channels and link aggregations, a complimentary configuration on the switch is required. The switch must support:

* IEEE standard Ethernet, Fast Ethernet, or Gigabit Ethernet
* IEEE 802.3ad Link Aggregation protocol

server_sysconfig implements a virtual device which combines one or more devices on the same Data Mover into a single, logical device addressable as a unit. Fail-safe networks, Ethernet channels, and link aggregations are virtual devices. Both Ethernet channels and link aggregations are trunking devices.

The ALL option executes the command for all Data Movers.

OPTIONS

-Platform
Displays the hardware configuration of the Data Mover, including processor type and speed, the bus speed in MHz, the main memory in MB, and the motherboard type.

-pci [-fmwr_version [<device>]][<device> [-option <options>]]

-pci -fmwr_version [<device>]
Displays the attributes of the specified network adapter card or for all devices.

-pci <device> -option <options>
Sets the attributes of the specified network adapter card installed in the Data Mover. Options must be comma-separated and double quoted.

ETHERNET
----

speed={10|100|1000|auto}
Sets the speed for the port. auto (default) turns on autonegotiation; setting a fixed speed disables autonegotiation.

duplex={full|half|auto}
Sets the duplex to full, half, or auto. The auto (default) turns autonegotiation on; setting a fixed duplex disables autonegotiation.

Caution: The speed and duplex settings on both sides of the physical connection must be the same. Mismatched speed and duplex settings can cause errors and impact network performance. For example, if the duplex is set to half on one end and full on the other, there might be network errors and performance issues.

GIGABIT ETHERNET FIBER

For Gigabit Ethernet Fiber connections, the speed is automatically set to 1000, and since it must remain at that setting, no speed setting is required.

linkneg={enable|disable}
Disables autonegotiation on the network adapter card if it is not supported by the network Gigabit switch. The default is enable.

rxflowctl={enable|disable}
Enables the ability to accept and process pause frames.The default is disable.

txflowctl={enable|disable}
Enables pause frames to be transmited. The default is disable.

GIGABIT ETHERNET COPPER
---

speed={10|100|1000|auto}
Sets the speed for the port. The auto (default) turns on autonegotiation; setting a fixed speed disables autonegotiation.

duplex={full|half|auto}
Sets the duplex to full, half, or auto. The auto (default) turns autonegotiation on; setting a fixed duplex disables autonegotiation.

Caution: The speed and duplex settings on both sides of the physical connection must be the same. Mismatched speed and duplex settings can cause errors and impact network performance. For example, if the duplex is set to half on one end and full on the other, there might be network errors and performance issues.

rxflowctl={enable|disable}
Enables the ability to accept and process pause frames. The default is disable.

txflowctl={enable|disable}
Enables pause frames to be transmited. The default is disable.

-virtual -delete [-Force] <device>
Deletes an Ethernet channel, link aggregation, or fail-safe network device. Valid entries for a <device> are trk or fsn. The -Force option is required if an IP address has been assigned.

Caution: All of the IP interfaces configured over a virtual device are deleted when the -Force option is used. As a result, any applications on the VNX that use these interfaces might get disrupted. Among these applications are CIFS and VNX Replicator. If all of the interfaces used by a specific CIFS server get deleted by the server_sysconfig -Force command, the server does not communicate with clients any further, and Windows users lose access to their data.

-virtual
Information for all devices is displayed.

-virtual -info <device>
Displays information for the specified Ethernet channel, link aggregation, or fail-safe network device.

-virtual -name <device> -create trk -option

device=<device,...>
Creates a virtual device for trk (Ethernet channel or link aggregation). An Ethernet channel can combine two, four, or eight (must be a multiple of two) ports into a single virtual device. A maximum of eight devices for Fast Ethernet is allowed:

* For Fast Ethernet trunking, use ana0 through ana7.
* For Gigabit Ethernet trunking, use ace0 and ace1.
* For Gigabit Ethernet using the NS series, use cge0 through cge5.

The -option string defines one or more devices for the Ethernet channel or link aggregation. The <device,...> specifies the device or devices separated by commas. The -option string must have the options separated with a space, and must be enclosed with double quotes. If not, the command appears to execute, but does not pick up any options not contained within double quotes.

A gigabit trunk can be created on more than two gigabit devices. The VNX has only two gigabit devices (ace0, ace1), so two ports are created. The NS series has a six gigabit port, so a four port gigabit trunk can be created.

For link aggregation there can be one or more number of ports, up to a maximum of 12.

Caution: The virtual device must be created before the IP interfaces can be configured for a trunk network device.

[protocol=lacp]
Initializes link aggregation control protocol for Ethernet devices which allows a non-exponential number of devices. For example, one, three, or five entries.

[lb=mac|ip|tcp]
Specifies the statistical load balancing method that applies to the aggregator/trunk. The mac option indicates that the output link on the trunk/aggregator is chosen based on source and destination MAC addresses. The ip (default) option indicates that the output link on the trunk/aggregator is chosen based on source and destination IP addresses. This is useful in the situation that the Data Mover is connected to a router or to a layer 3 switch. The tcp option indicates that the output link on the trunk/aggregator is chosen based on source and destination IP addresses as well as source and destination TCP ports.

-virtual -name <device> -create fsn -option primary=<device> device=<device,...>
Creates a virtual device for fsn (fail-safe network). The fail-safe network device is a high-availability feature that provides switch-level redundancy. Configuring a failover channel is optional. Specifies the device name of the virtual device.

The -option string indicates a primary device for the fail-safe network device and a second device to be defined as the standby in case the primary device fails. The <device,...> specifies the device or devices separated by commas. The -option string must be defined in sequence, separated with a space, and enclosed with double quotes. If not, the command appears to execute, but does not pick up any options not contained within double quotes.

Caution: The virtual device must be created before the IP interfaces can be configured for the fail-safe network device.

SEE ALSO

Configuring and Managing Networking on VNX, Configuring and Managing Network High Availability on VNX, server_dbms, and server_ifconfig.

INTERFACE OUTPUTS

The network interface cards available are dependent on the type of system used.

For the VNX, the following NICs are available: loop, ace, ana, cge, el30, el31, fpa, and fa2. Note that loop, el30, and el31 are for internal use only.

For the NS series, the following NICs are available: loop, cge, el30, el31, and fge.

VDMs are included in both the CNS and NS series.

EXAMPLE #1

For the NS series, to view the system configuration for a Data Mover, type:

```
server_2 :
Processor = Intel Pentium 4
Processor speed (MHz) = 3100
Total main memory (MB) = 4023
Mother board = Barracuda XP
Bus speed (MHz) = 533
Bios Version = 3.30
Post Version = Rev. 02.14
```

For the CNS series, to view the system configuration for a Data Mover, type:

```
$ server_sysconfig server_2 -Platform
server_2 :
Processor = Intel Pentium 4
Processor speed (MHz) = 1600
Total main memory (MB) = 3967
Mother board = CMB-400
Bus speed (MHz) = 400
Bios Version = No Ver Info
Post Version = No Ver Info
```

EXAMPLE #2

For the NS series, to view the installed PCI configuration for a Data Mover, type:

```
$ server_sysconfig server_2 -pci
server_2 : PCI DEVICES:

On Board:
Agilent Fibre Channel Controller
0: fcp-0 IRQ: 22 addr: 50060160006004f0

0: fcp-1 IRQ: 21 addr: 50060161006004f0

0: fcp-2 IRQ: 18 addr: 50060162006004f0

0: fcp-3 IRQ: 20 addr: 50060163006004f0

Broadcom Gigabit Ethernet Controller
0: fge0 IRQ: 24
linkneg=enable txflowctl=disable rxflowctl=disable

0: fge1 IRQ: 23
linkneg=enable txflowctl=disable rxflowctl=disable

0: cge0 IRQ: 24
speed=auto duplex=auto txflowctl=disable rxflowctl=disable
```

```
0: cge1 IRQ: 23
speed=auto duplex=auto txflowctl=disable rxflowctl=disable

0: cge2 IRQ: 26
speed=auto duplex=auto txflowctl=disable rxflowctl=disable

0: cge3 IRQ: 25
speed=auto duplex=auto txflowctl=disable rxflowctl=disable

0: cge4 IRQ: 28
speed=auto duplex=auto txflowctl=disable rxflowctl=disable

0: cge5 IRQ: 27
speed=auto duplex=auto txflowctl=disable rxflowctl=disable
```

For the CNS series, to view the installed PCI configuration for a Data Mover, type:

```
$ server_sysconfig server_2 -pci
server_2 : PCI DEVICES:

Slot: 1
Emulex LP9000 Fibre Channel Controller
0: fcp-0 IRQ: 23 addr: 10000000c92b5a10

1: fcp-1 IRQ: 24 addr: 10000000c92b5a11

Slot: 2
Emulex LP9000 Fibre Channel Controller
0: fcp-2 IRQ: 22 addr: 10000000c92b514e

Slot: 4
Intel 10/100/1K Ethernet Controller
0: cge0 IRQ: 18
speed=auto duplex=auto rxflowctl=disable txflowctl=disable

1: cge1 IRQ: 19
speed=auto duplex=auto rxflowctl=disable txflowctl=disable

2: cge2 IRQ: 20
speed=auto duplex=auto rxflowctl=disable txflowctl=disable

3: cge3 IRQ: 21
speed=auto duplex=auto rxflowctl=disable txflowctl=disable

Slot: 5
Alteon Tigon-2 Gigabit Ethernet Controller
0: ace0 IRQ: 25
linkneg=enable rxflowctl=disable txflowctl=disable
0: ace0 IRQ: 25
linkneg=enable rxflowctl=disable txflowctl=disable
```

Where:

Value	Definition
On Board	The names of each PCI card installed.
0	Port number inside the slot the card is plugged into. If the card put in the slot has 4 ports, the first port is marked as 0, second port as 1, third port as 2 and fourth port as 3.
IRQ	Interrupt vector.
speed	Speed configured. Possible values are: auto, 10, 100, 1000
duplex	Duplex setting configured. Possible values are: auto, half, full
txflowctl	Transmit MAC flow control. Possible values are: disable, enable

rxflowctl Receive MAC flow control. Possible values are: disable, enable

EXAMPLE #3

To view the firmware version for all devices, type:

```
$ server_sysconfig server_2 -pci -fmwr_version
server_2 : PCI DEVICES:
On Board:
VendorID=0x1120 DeviceID=0x1B00 Controller
0: scsi-0 IRQ: 32
0: scsi-16 IRQ: 33
0: scsi-32 IRQ: 34
0: scsi-48 IRQ: 35
Broadcom 10 Gigabit Ethernet Controller
0: fxg-8-0 IRQ: 38
Firmware Version: 6.2.11
0: fxg-8-1 IRQ: 40
Firmware Version: 6.2.11
0: cxg-9-0 IRQ: 44
Firmware Version: 6.2.17
0: cxg-9-1 IRQ: 46
Firmware Version: 6.2.17
```

To view the firmware version for a single device, type:

```
$ server_sysconfig server_2 -pci -fmwr_version fxg-2-1
server_2 :
On Board:
Broadcom 10 Gigabit Ethernet Controller
0: fxg-2-1 IRQ: 38
Firmware Version: 6.2.11
```

EXAMPLE #4

To set the 100 Mbits speed and full duplex setting for cge0 interface, type:

```
$ server_sysconfig server_2 -pci cge0 -option speed=100,duplex=full
server_2 : done
```

EXAMPLE #5

To display the hardware configuration for network device, cge0, type:

```
$ server_sysconfig server_2 -pci cge0
server_2 :
On Board:
Broadcom Gigabit Ethernet Controller
0: cge0 IRQ: 24
speed=100 duplex=full txflowctl=disable rxflowctl=disable
```

EXAMPLE #6

To create an Ethernet channel as a virtual device, type:

```
$ server_sysconfig server_2 -virtual -name trk0_ec -create trk -option
"device=cge2,cge3"
server_2 : done
```

EXAMPLE #7

To display all virtual devices on server_2, type:

```
$ server_sysconfig server_2 -virtual
server_2 :
```

```
Virtual devices:

trk0_ec devices=cge2 cge3
fsn failsafe nic devices :
trk trunking devices : trk0_ec
```

Where:

Value	Definition
Virtual Devices	All the configured virtual devices (trunking or fail safe) on the server.
devices	Lists the virtual or physical device names that in the <virtual device>.
failsafe nic devices	Name of FSN virtual devices configured on the server.
trunking devices	Name of trunking virtual devices configured on the server.

EXAMPLE #8

To view information for the Ethernet channel, trk0_ec, type:

```
$ server_sysconfig server_2 -virtual -info trk0_ec
server_2 :
*** Trunk trk0_ec: Link is Up ***
*** Trunk trk0_ec: Statistical Load Balancing is IP ***
Device     Link Duplex   Speed
-------------------------------------------------------------------
cge2       Up   Full     1000 Mbs
cge3       Up   Full     1000 Mbs
```

Where:

Value	Definition
Devices	Name of devices participating in this trunk.
Link	Status of the link on the specified device (can be up or down).
Duplex	Value of negotiated duplex. Values can be full or half.
Speed	Value of negotiated speed.

EXAMPLE #9

To create a fail-safe network device as a virtual device, type:

```
$ server_sysconfig server_2 -virtual -name fsn0 -create fsn -option
"primary=cge2 device=cge2,cge3"
server_2 : done
```

EXAMPLE #10

To view the fail-safe network device, type:

```
$ server_sysconfig server_2 -virtual -info fsn0
server_2 :
*** FSN fsn0: Link is Up ***
active=cge2 primary=cge2 standby=cge3
```

Where:

Value	Definition
active	Name of the active device in the FSN.
primary	Primary device in the FSN.

standby Standby device in the FSN.

EXAMPLE #11

To create an aggregated link using the LACP protocol with load balancing method set to mac, type:

```
$ server_sysconfig server_2 -virtual -name trk0_la -create trk -option
"device=cge2,cge3 protocol=lacp lb=mac"
server_2 : done
```

EXAMPLE #12

To delete an Ethernet channel, trk0_ec, type:

```
$ server_sysconfig server_2 -virtual -delete -Force trk0_ec
server_2 : done
```

Last Modified: May 12, 2011 01:15 pm

server_sysstat

Displays the operating system statistics for the specified Data Movers.

SYNOPSIS
----

server_sysstat {<movername>|ALL} [-blockmap]

DESCRIPTION
----

server_sysstat displays the operating system statistics and the status of blockmap memory for the specified Data Movers.

The ALL option executes the command for all Data Movers.

OPTIONS

No arguments
Displays all operating system statistics.

[-blockmap]
Displays the status of blockmap memory in the Data Mover. Blockmaps are paged memory used by SnapSure to locate each block of data comprising the checkpoint. Twenty-five percent of physical RAM on the Data Mover is reserved for blockmap memory.

EXAMPLE #1

To display all operating system statistics, type:

```
$ server_sysstat server_2

server_2 :
threads runnable = 6
threads blocked = 1399
threads I/J/Z = 2
memory free(kB) = 2689924
cpu idle_% = 99
```

Where:

Value	Definition
threads runnable	Threads running as well as the ones waiting for a cpu to become available.
threads blocked	Threads waiting on either a condition variable or a Sthread Mutex.
threads I/J/Z	Threads in the following state: IDLE, JOINED, and ZOMBIE.
memory free (kB)	The amount of free memory from the RAM that is available to Data Mover.
cpu idle_%	Idle time of the system averaged across all the processors.

EXAMPLE #2

To display current blockmap memory status, type:

```
$ server_sysstat server_2 -blockmap

server_2 :
total paged in = 0
```

```
total paged out = 1
page in rate = 0
page out rate = 0
block map memory quota = 1048576(KB)
block map memory consumed = 624(KB)

Where:
------

Value                     Definition
-----                     ----------

total paged in            Total number of blockmap pages paged in since the
                          system booted.
total paged out           Total number of blockmap pages paged out since the
                          system booted.
page in rate              Number of blockmap pages paged in per second (over
                          last 180 seconds).
page out rate             Number of blockmap pages paged out per second (over
                          last 180 seconds).
block map memory quota    Current value of the blockmap memory quota.
block map memory consumed Amount of memory consumed for blockmaps.

--------------------------------------------------------------------------------
Last modified: April 26 2011, 06:00 pm.
```

server_tftp

Manages the Trivial File Transfer Protocol (TFTP) for the specified Data Movers.

SYNOPSIS

```
server_tftp {<movername>|ALL}
  -service {-status|-stop|-start|-stats}
| -info
| -set [-path <pathname>][-readaccess {all|none}][-writeaccess {all|none}]
| -clear
```

DESCRIPTION
----

server_tftp provides a service to read and write files to and from a remote server, displays the status of the service, and the number of attempted transfers.

The ALL option executes the command for all Data Movers.

OPTIONS

-service {-status|-stop|-start|-stats}
Displays the status of the TFTP service, stops, and starts the service on the specified Data Mover, or displays TFTP statistics.

-info
Displays the number of attempted TFTP transfers and the status information for ongoing transfers.

-set [-path <pathname>]
Sets the directory <pathname> for the TFTP service to use for file transfers. If the -set option has been executed, re-executing the -set option with a different <pathname> overwrites the original <pathname>.

[-readaccess {all|none}]
Sets read access for all users or no users (default).

[-writeaccess {all|none}]
Sets write access for all users or no users (default).

-clear
Clears the path/readaccess/writeaccess settings the TFTP service for the specified Data Mover.

SEE ALSO

Using TFTP on VNX.

EXAMPLE #1

To start TFTP service for server_2, type:

```
$ server_tftp server_2 -service -start
server_2 : done
```

EXAMPLE #2

To display the status of the TFTP service for server_2, type:

```
$ server_tftp server_2 -service -status
server_2 :
Tftp Running
```

EXAMPLE #3

To modify a path on server_2 for TFTP service with read access for all, and write access for nobody, type:

```
$ server_tftp server_2 -set -path /ufs1 -readaccess all -writeaccess none
server_2 : done
```

EXAMPLE #4

To display TFTP information for server_2, type:

```
$ server_tftp server_2 -info
server_2 :
path="/ufs1/" readaccess=all writeaccess=none
```

EXAMPLE #5

To display statistics for server_2, type:

```
$ server_tftp server_2 -service -stats
server_2 :
Attempted Transfers:28
Successful Transfers:27
createdthrds:28
deletedthrds:28
timedoutthrds:0
TotalBinds:28
TotalUnbinds:28
BindFailures:0
InvalidAttempts:0
AttemptedReadTransfers:19
SuccessfulReadTransfers:19
AttemptedWriteTransfers:9
SuccessfulWriteTransfers:8
```

Where:

Value	Definition
Attempted Transfers	Total TFTP transfers attempted by that time.
Successful Transfers	Total number of successful TFTP transfers.
createdthrds	Total number of TFTP threads created (equal to total transfers).
deletedthrds	Total number of threads deleted (equal to total created threads).
timedoutthrds	Number of timed-out threads. For TFTP transfers, in case of any failures, each thread will time out and free itself.
TotalBinds	Total number of binds.
TotalUnbinds	Total number of unbinds.
BindFailures	Number of bind failures. If the port we try to bind to is in use, the bind fails, and retries with a different port.
InvalidAttempts	Invalid TFTP transfer requests from clients such as trying to transfer a non-existent file.

AttemptedReadTransfers	Total TFTP read transfers initiated.
SuccessfulReadTransfers	Total TFTP read transfers successfully completed.
AttemptedWriteTransfers	Total TFTP write transfers initiated.
SuccessfulWriteTransfers	Total TFTP write transfers successfully completed.

EXAMPLE #6

To stop TFTP service on server_2, type:

```
$ server_tftp server_2 -service -stop
server_2 : done
```

EXAMPLE #7

To delete the settings for the TFTP service on server_2, type:

```
$ server_tftp server_2 -clear
server_2 : done
```

Last Modified: April 26, 2011 3:00 pm

server_umount

Unmounts file systems.

SYNOPSIS
----

server_umount {<movername>|ALL}
[-perm|-temp][-Force]{-all|<fs_name>|<mount_point>}

DESCRIPTION
----

server_umount permanently or temporarily (default) unmounts a mounted file system by its <mount_point> or <fs_name>, or unmounts all file systems.

The ALL option executes the command for all Data Movers.

OPTIONS
----

all|<fs_name>|<mount_point>
Unmounts all file systems, a specific file system, or a file system specified by its mount point.

Note: A mount point always begins with a slash (/).

[-perm]
Permanently deletes the mount entry from mount table.

Note: The export table is not affected by deletes made with server_umount. Export entries remain in the export table regardless of the mount status.

[-temp]
Indicates that the unmount does not persist across reboots. The mount entry reappears after a reboot. Default is -temp.

[-Force]
Forces the unmount of a production file system.

Caution: Unmounting a PFS for which automatic checkpoint scheduling (configured using the VNX Web Manager or VNX Native Manager) has been implemented is not recommended. Doing so will corrupt the schedule and cause checkpoint data to be lost.

SEE ALSO

Managing Volumes and File Systems with VNX Automatic Volume Management,
Managing Volumes and File Systems for VNX Manually, Using VNX SnapSure, server_mount, server_mountpoint, and fs_ckpt.

EXAMPLE #1

To permanently unmount a file system, type:

```
$ server_umount server_2 -perm /bin
server_2: done
```

EXAMPLE #2

To temporarily unmount a file system by specifying its mount point as /bin, type:

```
$ server_umount server_2 -temp /bin
server_2: done
```

EXAMPLE #3

To temporarily unmount a file system by specifying its file system name as ufs1, type:

```
$ server_umount server_2 -temp ufs1
server_2: done
```

Last Modified: April 27, 2011 12:55 pm

server_uptime

Displays the length of time that a specified Data Mover has been running since the last reboot.

SYNOPSIS
----

server_uptime {<movername>|ALL}

DESCRIPTION
----

server_uptime displays the length of time that the specified Data Mover has been running since its last reboot.

The ALL option executes the command for all Data Movers.

OPTIONS

No arguments
Displays the length of time that a Data Mover has been up.

EXAMPLE #1

To display how long a Data Mover has been up, type:

```
$ server_uptime server_2
server_2 : up 28 days 0 hours 15 min 41 secs
```

EXAMPLE #2

To display how long all the Data Movers have been up, type:

```
$ server_uptime ALL
server_2 :  up 28 days 0 hours 15 min 41 secs
server_3 :  up 2 days 5 hours 11 min 31 secs
```

Last Modified: April 27, 2011 12:10 pm

server_user

Manages user accounts for the specified Data Movers.

SYNOPSIS

```
server_user {<movername>|ALL}
  -list
| -add [-md5][-passwd] <name>
| -modify <name>
| -delete <name>
| -passwd [-disable|-lock|-unlock [-force]] <name>
```

DESCRIPTION
----

server_user creates, modifies, or deletes a user account for a specified Data Mover. The server_user adds or disables, locks or unlocks an optional password for a user account. Data Mover user accounts provide administrative access to certain VNX features and third-party applications.

The ALL option executes the command for all Data Movers.

This command must be executed from the /nas/sbin directory. su to root to execute this command.

OPTIONS

-list
Lists the user accounts.

-add [-md5][-passwd] <name>
Adds a new user account with the login <name>.

The -md5 option allows you to specify MD5 password encryption for the new account.

The -passwd option allows you to specify a password for the new user account. Passwords have a six character minimum and cannot be empty.

-modify <name>
Modifies a user account.

-delete <name>
Deletes a user account.

-passwd [-disable|-lock|-unlock [-force]] <name>
Creates, changes, removes, locks, and unlocks the password for a user. If a password lock is applied, the user account is available to root only. The password does not have to be supplied during the execution of the server_user command. To unlock a password, use the -force option.

SEE ALSO

Using FTP on VNX, and Configuring NDMP Backups to Disk on VNX.

EXAMPLE #1

To create a user account for user1, type:

```
# /nas/sbin/server_user server_2 -add user1
Creating new user user1
User ID: 100
Group ID: 101
Comment:
Home directory:
Shell:

Note: Comment, Home directory and Shell are optional, all others are required.

EXAMPLE #2
----------

To create a user account for NDMP connections, with MD5 password encryption
and to configure the password, type:

# /nas/sbin/server_user server_2 -add -md5 -passwd user_name

Creating new user user_name
User ID: User ID: 101
Group ID: 100
Home directory:
Changing password for user user_name
New passwd:
Retype new passwd:

EXAMPLE #3
----------

To list the user accounts, type:

# /nas/sbin/server_user server_2 -list
server_2:
APM000438070430000_APM000420008180000:LNEa7Fjh/43jQ:9000:9000:ftsQgHsc2oMrdysaOnW
eLhN8vB::ndmp_md5
user1:!!:100:101:::
user_name:WX72mBTFp/qV.:101:100:W9z7HIndimdaHs2anCL20EBfNd::ndmp_md5

EXAMPLE #4
----------

To modify account information for user1, type:

# /nas/sbin/server_user server_2 -modify user1
Modifying user account user1
1 User ID (100)
2 Group ID (101)
3 Home directory ()
4 Comment ()
5 Shell ()

Please select a field to modify, "done" to apply
your changes or "quit" to cancel: 2
Group ID: 102
Please select a field to modify, "done" to apply
your changes or "quit" to cancel: quit

EXAMPLE #5
----------

To lock an account password for ndmp, type:

# /nas/sbin/server_user server_2 -passwd -lock user_name
Changing password for user user_name
Locking password for user user_name

EXAMPLE #6
----------

To disable the password for user1, type:
```

```
# /nas/sbin/server_user server_2 -passwd -disable user1
Changing password for user user1
Removing password for user user1
```

EXAMPLE #7

To unlock an account password for user1, type:

```
# /nas/sbin/server_user server_2 -passwd -unlock -force user1
Changing password for user user1
Unlocking password for user user1
```

EXAMPLE #8

To delete a user account for user1, type:

```
# /nas/sbin/server_user server_2 -delete user1
```

Last Modified: April 26, 2011 3:00 pm

server_usermapper

Provides an interface to manage the Internal Usermapper service.

SYNOPSIS

```
server_usermapper {<movername>|ALL}
| -enable [primary=<ip>][config=<path>]
| -disable
| -remove -all
| -Import {-user|-group} [-force] <pathname>
| -Export {-user|-group} <pathname>
```

DESCRIPTION
----

server_usermapper enables, disables, deletes, and displays the status of an Internal Usermapper service running on a Data Mover. Only one primary Usermapper service is recommended for a VNX environment. In a single VNX environment, there should only be one primary instance of the Usermapper service. All the other Data Movers in that environment are clients of the primary or secondary service.

server_usermapper displays whether the Usermapper service is configured as a primary or secondary service. Additionally, it displays information such as the operational status of the service and, if the service is a secondary service, the IP address of the primary Usermapper service used by the secondary.

This command is relevant only for Internal Usermapper, which runs on the Data Mover. It cannot be used to manage External Usermapper, which runs on the Control Station or an external host.

The ALL option executes the command for all Data Movers.

OPTIONS

No arguments
Displays the state of the Internal Usermapper service.

-enable
No arguments
Displays the state of the Internal Usermapper service.

-enable
Enables the Usermapper service. You do not need to issue this option if you are using the default Internal Usermapper configuration. In this case, primary Usermapper is automatically enabled when the NAS software is installed. You only need to issue this option if you are modifying a default Internal Usermapper configuration, or if you are upgrading from External to Internal Usermapper.

Use the -enable command with caution. It changes the relationship between the Data Mover and the Usermapper without confirming the change.

[primary=<ip_address>][config=<pathname>]
The primary option designates the local Usermapper service as secondary by indicating which primary service it depends on. The primary Usermapper is identified by its network IP address.

The config option indicates that the Usermapper service should use an existing Usermapper configuration file to define UID/GID ranges. This option is only relevant if you are upgrading from External to Internal Usermapper.

Note: If there is no specific reason to use particular UID and GID ranges for your environments' domains, EMC encourages you to use the automatic mapping method and let Internal Usermapper automatically assign new UIDs/GIDs based on the next available values. If you need to use an existing Usermapper configuration file, you must specify the config option during the upgrade procedure, that is, before Internal Usermapper has begun issuing default UIDs and GIDs.

-disable
Disables the Usermapper service.

-remove -all
Removes all entries from the Usermapper databases and destroys the database structure. The Usermapper service must be disabled before you can issue this option.

Caution: It is recommended that you consult with Customer Support before issuing the -remove -all option. This option deletes all Usermapper database entries and may result in users losing access to file systems. If you decide to use the -remove -all option, you should first back up your existing Usermapper database file and usrmap.cfg file (if one is in use).

-Import {-user|-group} [-force] <pathname>
Imports Usermapper database information from the file specified by <pathname>.

By default, only new entries are added to the Usermapper database. If an entry in the imported file does not match a similar entry in the existing database, the entry in the imported file is ignored unless the -force option is selected. If the -force option is selected, the existing database is deleted and replaced with new entries.

Caution: It is recommended that you consult with Customer Support before issuing the -force option. This option overwrites the existing Usermapper database file and may result in users losing access to file systems. If you decide to use the -force option, you should first back up your existing Usermapper database file and usrmap.cfg file (if one is in use).

-Export {-user|-group} <pathname>
Exports all the SID, user, and group information from the Usermapper database to the file specified by <pathname>. You can specify any filename but the name should include the suffix .passwd or .group depending on the file type.

The -Export option is relevant only for a primary Usermapper service.

SEE ALSO

Using ntxmap for CIFS User Mapping on VNX.

EXAMPLE #1

To verify the status of Internal Usermapper, type:

```
$ server_usermapper server_2
server_2 : Usrmapper service: Enabled
Service Class: Primary
```

Where:

Value	Definition
Usrmapper service:	The operational status of the service.

Service Class:	If the service is a primary or secondary service.
Primary	The IP address of the primary Usermapper service used by a secondary service. The (c) against the IP address indicates that the primary Usermapper is available and has been connected.

EXAMPLE #2

To enable a secondary Usermapper service, type:

```
$ server_usermapper server_4 -enable primary=172.24.102.238
server_4 : done
```

EXAMPLE #3

To verify the status of Internal Usermapper for the primary Usermapper, type:

```
$ server_usermapper server_2
server_2 : Usrmapper service: Enabled
Service Class: Primary
```

EXAMPLE #4

To verify the status of Internal Usermapper for the secondary Usermapper, type:

```
$ server_usermapper server_4
server_4 : Usrmapper service: Enabled
Service Class: Secondary
Primary = 172.24.102.238 (c)
```

See Example #1 for a description of command outputs.

EXAMPLE #5

To export user information from the Usermapper database, type:

```
$ server_usermapper server_2 -Export -user
/home/nasadmin/users_server_2.passwd
server_2 : done
```

EXAMPLE #6

To export group information from the Usermapper database, type:

```
$ server_usermapper server_2 -Export -group
/home/nasadmin/group_server_2.group
server_2 : done
```

EXAMPLE #7

To import the user file users_server_2.passwd for server_2, type:

```
$ server_usermapper server_2 -Import -user
/home/nasadmin/users_server_2.passwd
server_2 : done
```

EXAMPLE #8

To import the group file group_server_2.group for server_2, type:

```
$ server_usermapper server_2 -Import -group
/home/nasadmin/group_server_2.group
server_2 : done
```

EXAMPLE #9

To disable an Internal Usermapper service, type:

```
$ server_usermapper server_2 -disable
server_2 : done
```

EXAMPLE #10

To remove all entries from the Usermapper database, type:

```
$ server_usermapper server_2 -remove -all
server_2 : Warning: This operation will erase all user/group mappings.
CIFS users may lose access.
Continue(Y/N):
done
```

Last Modified: April 26, 2011 2:30 pm

server_version

Displays the software version running on the specified Data Movers.

SYNOPSIS
----

server_version {<movername>|ALL}

DESCRIPTION
----

server_version displays the version of the software for the specified Data Mover.

The ALL option executes the command for all of the Data Movers.

OPTIONS

No arguments
Displays the software version.

SEE ALSO
----

nas_version

EXAMPLE #1

To display the software version for a Data Mover, type:

```
$ server_version server_2
server_2 : Product: EMC Celerra File Server Version: T5.5.80.0
```

EXAMPLE #2

To display the system software version for all Data Movers, type:

```
$ server_version ALL
server_2 : Product: EMC Celerra File Server Version: T5.5.80.0
server_3 : Product: EMC Celerra File Server Version: T5.5.80.0
server_4 : Product: EMC Celerra File Server Version: T5.5.80.0
server_5 : Product: EMC Celerra File Server Version: T5.5.80.0
```

Last Modified: April 26, 2011, 1:40 pm

server_viruschk

Manages the virus checker configuration for the specified Data Movers.

SYNOPSIS

```
server_viruschk {<movername>|ALL}
  [-audit|-update]
| [-set accesstime={now|none|[[[[<yy>]<mm>]<dd>]<hh>]<mm>[.<ss>]}]
| [-fsscan [<fs_name> {-list|-create [offline]|-delete}]]
```

DESCRIPTION
----

server_viruschk displays, audits, and updates the virus checker configuration file for the specified Data Mover.

The virus checker configuration file is named viruschecker.conf and is located on the Control Station in the /nas/sys directory. This is a template file and should not be edited directly. Copy the viruschecker.conf file to another directory for edits using a text editor, then once the changes have been saved, the file must be copied to a Data Mover using the server_file command.

To start and stop the virus checker for a Data Mover, server_setup provides information.

The ALL option executes the command for all Data Movers.

OPTIONS

No arguments
Displays the virus checker configuration.

-audit
Displays the status of virus checker, including the number of files checked, and their progress.

-update
Integrates the changes made to the virus configuration file without stopping the virus checker.

-set accesstime={now|none|[[[[<yy>]<mm>]<dd>]<hh>]<mm>[.<ss>]}
Sets the reference time on the virus checker configuration file to now, disables the time scan, or sets the access time in the specified format.

-fsscan
Displays the file system scan status for all file systems.

[<fs_name> {-list|-create [offline]|-delete}]]
The -list option displays the scan status for the specified file system. The -create option initializes a full scan on the file system <fs_name> and the offline options allow the file system scan on all offline files. By default, offline file systems are not included. The -delete option stops the scan.

SEE ALSO

Using Common AntiVirus Agent, server_checkup, and server_setup.

EXAMPLE #1

To display the virus checker configuration, type:

```
$ server_viruschk server_2
server_2 :
10 threads started
1 Checker IP Address(es):
172.24.102.18 ONLINE at Mon Jan 31 18:35:43 2005 (GMT-00:00)
RPC program version: 3
CAVA release: 3.3.5, AV Engine: Network Associates
Last time signature updated: Thu Jan 27 19:38:35 2005 (GMT-00:00)
31 File Mask(s):
*.exe *.com *.doc *.dot *.xl? *.md? *.vxd *.386 *.sys *.bin *.rtf *.obd
*.dll
*.scr *.obt *.pp? *.pot *.ole *.shs *.mpp *.mpt *.xtp *.xlb *.cmd *.ovl
*.dev
*.zip *.tar *.arj *.arc *.z
No File excluded
Share \\DM112-CGE0\CHECK$
RPC request timeout=25000 milliseconds
RPC retry timeout=5000 milliseconds
High water mark=200
Low water mark=50
Scan all virus checkers every 60 seconds
When all virus checkers are offline:
Continue to work with Virus Checking and CIFS
Scan on read if access Time less than Thu Jan 27 19:38:35 2005 (GMT-00:00)
Panic handler registered for 65 chunks
```

```
Where:
Value                      Indicates
threads started            The number of threads that have been started.
Checker IP Address(es)     The number of VC servers defined in
                           /.etc/viruschecker.conf
version=2                  CAVA uses RPC program version 2.
Share                      The UNC name used by CAVA to access the Data
                           Mover.
RPC request timeout        Time out for the full CAVA request.
RPC retry timeout          Time out for one unitary CAVA request.
High water mark            A log event is generated when the number of
                           files in the request queue becomes greater than
                           200.
Low water mark             A log event is generated when the number of
                           files in the request queue become less than 50.
Panic handler registered   Panic is used to memorize name of unchecked
for 65 chunks              files.
ERROR_SETUP                List of errors reported by CAVA.
min, max, average          Min, max, and average time for CAVA requests.
```

EXAMPLE #2

To display the status of the virus checker, type:

```
$ server_viruschk server_2 -audit
server_2 :
Total Requests                          : 138
Requests in progress                    : 25

NO ANSWER from the Virus Checker Servers: 0
ERROR_SETUP                             : 0
FILE_NOT_FOUND                          : 0
ACCESS_DENIED                           : 0
FAIL                                    : 0
TIMEOUT                                 : 0
Total Infected Files                    : 875
Deleted Infected Files                  : 64
Renamed Infected Files                  : 0
Modified Infected Files                 : 811
min=70915 uS, max=1164891 uS, average=439708 uS

15 File(s) in the collector queue
10 File(s) processed by the AV threads
```

Read file '/.etc/viruschecker.audit' to display the list of pending requests

Where:

Value	Definition
Total Infected Files	The number of files found that contained viruses. This displays only if infected files are found and remains visible until the Data Mover is rebooted or the CAVA viruschecking service has been restarted.
Deleted Infected Files	The number of files that contained viruses that were deleted. This displays only if infected files are found and remains visible until the Data Mover is rebooted or the CAVA viruschecking service has been restarted.
Renamed Infected Files	The number of files that contained viruses that were renamed. This displays only if infected files are found and remains visible until the Data Mover is rebooted or the CAVA viruschecking service has been restarted.
Modified Infected Files	The number of files that contained viruses that were modified. This displays only if infected files are found and remains visible until the Data Mover is rebooted or the CAVA viruschecking service has been restarted.

EXAMPLE #3

To update the virus checker configuration file that is resident on the Data Mover, type:

```
$ server_viruschk server_2 -update
server_2 : done
```

EXAMPLE #4

To set the access time for the virus checker configuration file, type:

```
$ server_viruschk server_2 -set accesstime=now
server_2 : done
```

EXAMPLE #5

To start a scan on file system, type:

```
$ server_viruschk server_2 -fsscan ufs1 -create
server_2 : done
```

EXAMPLE #6

To check the scan of a file system, type:

```
$ server_viruschk server_2 -fsscan ufs1 -list
server_2 :
FileSystem 24 mounted on /ufs1:
8 dirs scanned and 22 files submitted to the scan engine
firstFNN=0x0, lastFNN=0xe0f34b70, queueCount=0, burst=10
```

EXAMPLE #7

To check the scan status on all file systems, type:

```
$ server_viruschk server_2 -fsscan
server_2 :
FileSystem 24 mounted on /ufs1:
8 dirs scanned and 11 files submitted to the scan engine
firstFNN=0x0, lastFNN=0xe0eba410, queueCount=0, burst=10
```

```
FileSystem 25 mounted on /ufs2:
9 dirs scanned and 11 files submitted to the scan engine
firstFNN=0x0, lastFNN=0xe0010b70, queueCount=0, burst=10

EXAMPLE #8
----------

To stop a scan on a file system, type:

$ server_viruschk server_2 -fsscan ufs1 -delete
server_2 : done

---------------------------------------
Last Modified: April 26, 2011 at 12:30 pm
```

server_vtlu

SYNOPSIS

Configures a virtual tape library unit (VTLU) on the specified Data Movers.

```
server_vtlu {<movername>|ALL}
  -service <service_options>
| -tlu <tlu_options>
| -storage <storage_options>
| -tape <tape_options>
| -drive <drive_options>
| -help
```

DESCRIPTION
----

server_vtlu creates, configures, and displays information on virtual tape library units on the specified Data Movers. At this time, server_vtlu supports only NDMP backups.

To get detailed options for VTLU service, TLU, storage, tape, and drive, type -help for the full usage, or type only -service, -tlu, -tape, and -drive, respectively, after typing the <movername>.

ALL executes the command for all Data Movers.

SERVICE OPTIONS

```
-service {-info|-set -chain <chain_id>}
```

The -info option lists the chain IDs for the VTLU service on the Data Mover. The -set -chain option specifies a <chain_id> for the VTLU device connection. The default chain ID is 0.

TLU OPTIONS

```
 -tlu
  {-list
| -info <tlu_id>
| -new [-robot {[-vendor <vendor>][-product <product>]
   [-revision <revision>]}]
   [-slots <num_slots>]
   [-impexp <num_impexp_slots>][-drives <num_drives>]
   [-drive {[-vendor <vendor>][-product <product>]
   [-revision <revision>]}]
| -modify <tlu_id>
  {[-robot {[-vendor <vendor>][-product <product>][-revision <revision>]}]
  [-slots <num_slots>][-impexp <num_impexp_slots>][-drives <num_drives>]
  [-drive {[-vendor <vendor>][-product <product>][-revision <revision>]}]}
| -delete <tlu_id>}
```

-tlu -list
Lists all of the VTLUs configured on the specified Data Movers.

-tlu -info <tlu_id>
Displays information for the specified VTLU.

-tlu -new [-robot {[-vendor <vendor>][-product<product>][-revision<revision>]}]
Creates a VTLU on the specified Data Mover with the specified robot's vendor, product, and revision information. If this information is not specified, the default values of EMC, svtlRobot, and 1.0 are used. These values have character limits of 8, 16, and 4, respectively.

[-slots <num_slots>]

Configures the number of slots in the VTLU. If no value is defined, then the default value of 32 is used.

[-impexp <num_impexp_slots>]
Configures the number of import/export slots in the VTLU. If no value is defined, the default value of 8 is used.

[-drives <num_drives>]
Configures the number of virtual tape drives in the VTLU. If no value is defined, then the default value of 4 is used.

[-drive {[-vendor <vendor>][-product<product>][-revision <revision>]}]
Specifies the vendor name, product name, and revision number of the virtual drive. If no values are defined, the default values of EMC, svtlDrive, and 1.0 are used. These values have character limits of 8, 16, and 4 respectively.

-tlu -modify <tlu_id> [-robot {[-vendor <vendor>][-product <product>][-revision <revision>]}]
Modifies the vendor name, product name, and revision number for the virtual robot.

[-slots <num_slots>]
Modifies the number of virtual slots in the VTLU.

[-impexp <num_impexp_slots>]
Modifies the number of virtual import/export slots in the VTLU.

[-drives <num_drives>]
Modifies the number of virtual tape drives in the VTLU.

[-drive {[-vendor <vendor>][-product<product>][-revision <revision>]}]
Modifies the vendor name, product name, and revision number for the VTLU drive.

-tlu -delete <tlu_id>
Deletes a configured VTLU. The storage associated with a VTLU must be deleted before the VTLU can be deleted.

STORAGE OPTIONS

-storage
{-list [<tlu_id>]
| -new <fs_name> -tlu <tlu_id> [-tapesize <size>G][-tapes <num_tapes>]
[-barcodeprefix <barcode_prefix>]
[-destination {slot|impexp|vault}]
| -extend <fs_name> -tlu <tlu_id> [-tapesize <size>G][-tapes <num_tapes>]
[-destination {slot|impexp|vault}]
| -import <fs_name> -tlu <tlu_id>
[-destination {slot|impexp|vault}]
| -export <fs_name> -tlu <tlu_id>
| -delete {<fsname>|-barcodeprefix <barcode_prefix>} -tlu <tlu_id>}

-storage -list [<tlu_id>]
Lists storage information on the specified VTLU.

-storage -new <fs_name> -tlu <tlu_id>
Creates storage on an existing VTLU.

[-tapesize <size>G]
Specifies the size of the virtual tapes in the file system. If no value is defined, and the -tape <num_tapes> option is not defined, then the default value of 50 GB is used. If <num_tapes> is defined but the tape size is not, then the tape size is determined by the number of tapes and the file system size.

[-tapes <num_tapes>]
Specifies the number of virtual tapes in the file system. The default value is determined by the file system size and the value

of -tapesize <size>.

[-barcodeprefix <barcode_prefix>]
Specifies the barcode prefix for the virtual tapes. The barcode prefix is the same across a file system. If no value is defined, then a four-character string is assigned automatically.

[-destination {slot|impexp|vault}]
Specifies the destination of the virtual tape as a slot, an import/export slot, or the vault. Slot is the default location.

-storage -extend <fs_name> -tlu <tlu_id>
Adds more tapes to a file system that has been extended.

[-tapesize <size>G]
Specifies the size of the virtual tape. If no value is defined, and the -tapes <num_tapes> is not defined, then the default value of 50 GB is used. If the <num_tapes> is defined but <size> is not, then the tape size is determined by the number of tapes and the file system size.

[-tapes <num_tapes>]
Specifies the number of tapes to be added to a VTLU. The default value is determined by the size of the file system and the tape size.

[-destination {slot|impexp|vault}]
Specifies the destination of the virtual tape as a slot, an import/export slot, or the vault. Slot is the default location.

-storage -import <fs_name> -tlu <tlu_id>
Imports a file system that was exported from another VTLU. The configuration of a file system persists when it is imported.

[-destination {slot|impexp|vault}]
Specifies the destination of the imported storage as a slot, an import/export slot, or the vault. Slot is the default location.

-storage -export <fs_name> -tlu <tlu_id>
Exports storage from a VTLU, which breaks the association between the file system and the VTLU. Tapes must be ejected from the VTLU and placed in the vault before you can export storage.

-storage -delete {<fs_name>|-barcodeprefix <barcode_prefix>} -tlu <tlu_id>
Deletes the association between storage and the VTLU and cleans up related metadata.

TAPE OPTIONS

-tape
  {-list <tlu_id> [-storage <fs_name>]
| -info {<tape_barcode>|-all} -tlu <tlu_id>
| -insert <tape_barcode> -tlu <tlu_id>
| -eject <tape_barcode> -tlu <tlu_id>}
| -drive {-list <tlu_id>
| -info <drive_id> -tlu <tlu_id>
| -unmount <drive_id> -tlu <tlu_id>}

-tape {-list <tlu_id> [-storage <fs_name>]}
Lists the tapes configured on a VTLU, along with their associated file systems.

-tape -info <tape_barcode|-all> -tlu <tlu_id>
Lists information on a particular tape, or on all tapes in a VTLU.

-tape -insert <tape_barcode> -tlu <tlu_id>
Moves a virtual tape from the vault to an import/export slot.

-tape -eject <tape_barcode> -tlu <tlu_id>
Ejects a tape from the import/export slot and places it in the vault.

-drive {-list <tlu_id>}
Lists the virtual tape drives associated with the specified VTLU.

-info <drive_id> -tlu <tlu_id> -id
Displays information on the specified tape drive.

-unmount <drive_id> -tlu <tlu_id>
Unmounts the tape from the drive and places it in its original source slot. If the source slot is unavailable, the tape is placed in another available slot.

SEE ALSO

Configuring NDMP Backups to Disk on VNX, Parameters Guide for VNX for File, and nas_fs.

EXAMPLE #1

To set the chain_ID for server_2, type:

```
$ server_vtlu server_2 -service -set -chain 1
server_2 : done
```

EXAMPLE #2

To list the chain_ID for the VTLU service on server_2, type:

```
$ server_vtlu server_2 -service -info
server_2 :
starting chain id = 1
```

Where:

Value	Definition
starting chain id	Starting chain of the VTLU device name. The device name format is c<x>t<y>l<z>, where <x> is the starting chain. <y> and <z> are the target and LUN, respectively. The target and LUN values cannot be modified.

EXAMPLE #3

To create and configure a new VTLU on server_2, type:

```
$ server_vtlu server_2 -tlu -new -robot -vendor EMCCorp -product vtluRobot
-revision 1.1a -slots 256 -impexp 64 -drives 2 -drive -vendor EMCCorp
-product vtluDrive -revision 2.2a
server_2 :
id = 3
slots = 256
import/export slots = 64
robot vendor = EMCCorp
robot product = vtluRobot
robot revision = 1.1a
robot serial number = P8gIgqs2k5
robot device name = c1t0l0
drives = 2
drive vendor = EMCCorp
drive product = vtluDrive
drive revision = 2.2a
```

Where:

Value	Definition
id	Unique VTLU identifier that is assigned automatically.
slots	Number of virtual slots in the VTLU.
import/export slots	Number of virtual import/export slots in the VTLU.
robot vendor	Vendor name of the virtual robot; maximum length is eight characters.
robot product	Product name of the virtual robot; maximum length is 16 characters.
robot revision	Revision number of the virtual robot; maximum length is four characters.
robot serial number	Serial number of the virtual robot that is assigned automatically.
robot device name	Device name of the virtual robot; only the first number, the starting chain, can be modified.
drives	Number of virtual drives in the VTLU.
drive vendor	Vendor name of the virtual drive; maximum length is eight characters.
drive product	Product name of the virtual drive; maximum length is 16 characters.
drive revision	Revision number of the virtual drive; maximum length is four characters.

EXAMPLE #4

To list all of the VTLUs on server_2, type:

```
$ server_vtlu server_2 -tlu -list
server_2 :
id  vendor   product   revision  serial_number device_name
3   EMCCorp  vtluRobot   1.1a     P8gIgqs2k5    c1t0l0
```

Where:

Value	Definition
id	Unique VTLU identifier that is assigned automatically.
vendor	Vendor name of the virtual robot; maximum length is eight characters.
product	Product name of the virtual robot; maximum length is 16 characters.
revision	VTLU robot's revision number; maximum length is four characters.
serial_number	VTLU serial number that is assigned automatically.
device_name	The device name of the VTLU robot; only the first number, the starting chain, can be modified.

EXAMPLE #5

To display the information for the VTLU on the Data Mover identified by its ID, type:

```
$ server_vtlu server_2 -tlu -info 3
server_2 :
id = 3
slots = 256
import/export slots = 64
robot vendor = EMCCorp
robot product = vtluRobot
robot revision = 1.1a
robot serial number = P8gIgqs2k5
robot device name = c1t0l0
drives = 2
drive vendor = EMCCorp
drive product = vtluDrive
drive revision = 2.2a
```

EXAMPLE #6

To modify vendor, product and revision information for the robot and drive of VTLU 3 for server_2, type:

```
$ server_vtlu server_2 -tlu -modify 3 -robot -vendor EMC -product vRobot
-revision 1.1b -drives 3 -drive -vendor EMC -product vDrive -revision 2.2b
server_2 : done
```

EXAMPLE #7

To modify the number of virtual import/export slots and number of virtual slots of VTLU 1 for server_2, type:

```
$ server_vtlu server_2 -tlu -modify 3 -slots 8 -impexp 4
server_2 : done
```

EXAMPLE #8

To add new storage for VTLU 3 on server_2, with 5 virtual tapes of 1 GB located in slots each with barcode prefix dstpre, using ufs1 file system, type:

```
$ server_vtlu server_2 -storage -new ufs1 -tlu 3 -tape size 1G -tapes 5
-barcodeprefix dstpre -destination slot
server_2 : done
```

EXAMPLE #9

To extend VTLU 3 on server_2 by adding 2 virtual tapes of 1 GB and placing them in the import/export virtual slots, type:

```
$ server_vtlu server_2 -storage -extend ufs1 -tlu 3 -tapesize 1G -tapes 2
-destination impexp
server_2 : done
```

EXAMPLE #10

To export storage from VTLU 3 stored on ufs1 located on server_2 type:

```
$ server_vtlu server_2 -storage -export ufs1 -tlu 3
server_2 : done
```

EXAMPLE #11

To import the ufs2 file system to VTLU 3 and place the virtual tapes in the vault, type:

```
$ server_vtlu server_2 -storage -import ufs1 -tlu 3 -destination vault
server_2 : done
```

EXAMPLE #12

To list the storage on VTLU 3, type:

```
$ server_vtlu server_2 -storage -list 3
server_2 :
tlu_id   filesystem   barcode_prefix
3          ufs1         dstpre
```

Where:

Value	Definition
tlu_id	Unique VTLU identifier that is assigned automatically.
filesystem	Name of the file system associated with the VTLU.
barcode_prefix	Modifiable prefix assigned to virtual tapes that is constant across a file system.

EXAMPLE #13

To list VTLU information on VTLU 3, type:

```
$ server_vtlu server_2 -tape -list 3
server_2 :
barcode filesystem capacity(GB) location source_slot
dstpre0001 ufs1 1 vault
dstpre0002 ufs1 1 vault
dstpre0003 ufs1 1 vault
dstpre0004 ufs1 1 vault
dstpre0005 ufs1 1 vault
dstpre0006 ufs1 1 vault
dstpre0000 ufs1 1 impexp:0
```

Where:

Value	Definition
barcode	Virtual tape barcode, consisting of the modifiable barcode prefix and a four-digit number that is assigned automatically.
filesystem	Name of the file system.
capacity (GB)	Virtual tape capacity in GB.
location	Element type and element ID of the virtual tape; possible element types are slot, drive, import/export, robot, and vault.
source_slot	Slot ID of the tape's previous location.

EXAMPLE #14

To insert the specified tape in a virtual import/export slot on VTLU 3, type:

```
$ server_vtlu server_2 -tape -insert dstpre0001 -tlu 3
server_2 : done
```

EXAMPLE #15

To eject the specified tape from VTLU 3, type:

```
$ server_vtlu server_2 -tape -eject dstpre0001 -tlu 3
```

```
server_2 : done
```

EXAMPLE #16

To list the storage drive on VTLU 3, type:

```
$ server_vtlu server_2 -drive -list 3
server_2 :
drive_id   device_name  serial_number  status   tape_barcode
0            c1t0l1       NXB2w4W000    empty
1            c1t0l2       3u0bx4W000    empty
2            c1t0l3       g0pgy4W000    empty
```

Where:

Value	Definition
drive_id	Unique VTLU drive identifier that is assigned automatically.
device_name	The device name of the VTLU drive.
serial_number	The VTLU serial number that is automatically assigned.
status	Status of the virtual tape drive; possible values are empty, loaded, and in use.
tape_barcode	Barcode of the virtual tape if status is not empty.

EXAMPLE #17

To display information for drive 0 on VTLU 3, type:

```
$ server_vtlu server_2 -drive -info 0 -tlu 3
server_2 :
id = 0
device_name = c1t0l1
serial_number = NXB2w4W000
status = empty
tape_barcode =
```

EXAMPLE #18

To delete storage from VTLU 3, type:

```
$ server_vtlu server_2 -storage -delete ufs1 -tlu 3
server_2 : done
```

EXAMPLE #19

To delete VTLU 3 from server_2, type:

```
$ server_vtlu server_2 -tlu -delete 3
server_2 : done
```

Last Modified: April 26, 2011 1:30 pm

CS Command

This chapter describes the cs_standby command, including its command line syntax (Synopsis), a description of the options, and examples of usage.

```
cs_standby
```

cs_standby

Initiates a takeover and failover of a Control Station on a VNX with dual Control Stations.

SYNOPSIS
----

cs_standby
{-takeover|-failover}

DESCRIPTION

The cs_standby command initiates a Control Station takeover and failover. When a Control Station is activated, the name of the primary Control Station is displayed.

su to root and execute this command from the /nas/sbin or /nasmcd/sbin directory.

Note: EMC SRDF is not supported on the secondary Control Station.

OPTIONS

-takeover
Executed from the standby Control Station, initiates a reboot of the primary Control Station, then changes the state of the standby to that of the primary. The original primary Control Station now becomes the standby Control Station. The -takeover option can be used to failback Control Station 0 to the role of primary Control Station after a failover, or to set Control Station 1 to the role of primary Control Station on demand.

Caution: When executing a takeover or failover, Data Movers performing functions such as RDF, EMC TimeFinder/FS, file system extends, or quotas may be interrupted.

Caution: If a primary Control Station fails over to a standby Control Station, for remote replication, service continues to run but replication management capabilities are no longer available.

Note: After executing a takeover or failover, a few minutes may be needed to stop Linux and other services active on the Control Station.

-failover
Executed from the primary Control Station, initiates a reboot of the primary Control Station, then activates the standby to take over the role of the primary Control Station. The -failover option can be used to complete a failback by forcing a failover from Control Station 1 back to Control Station 0 after Control Station 0 had failed over, or to set Control Station 1 to the role of primary Control Station on demand.

To display the primary Control Station, type:

$ nas/sbin/getreason

EXAMPLE #1

To change the state of the standby Control Station to primary, cd to the /nasmcd/sbin directory of the standby Control Station, then type:

#./cs_standby -takeover
Taking over as Primary Control Station.....done

If the takeover command is executed on the primary Control Station, the following error message appears:

The -takeover option is only valid on a standby Control Station

EXAMPLE #2

To initiate a failover from the primary Control Station to the standby Control Station, cd to the /nas/sbin directory of the primary Control Station, then type:

#./cs_standby -failover

The system will reboot, do you wish to continue [yes or no]: y
Failing over from Primary Control Station

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The migrate Command

This migrate Command Set provided for managing, configuring, and monitoring of Data Movers. The commands are prefixed with migrate, and appear alphabetically.

–	migrate_system_conf

migrate_system_conf

Migrates Data Mover level or cabinet level configurations from source system.

SYNOPSIS

```
migrate_system_conf {
          -mover
                -source_system {<celname> | id=<celId>}
                -source_user <username>
                -source_mover <movername>
                -destination_mover <movername>
                -service
                {-all |{ldap|ftp|sftp|http|nis|dns|ntp|passwd|group|cepp|
                cava|server_param|usermapper_client|netgroup|nsswitch|hosts}[,...
] ] }
                [-overwrite_destination]
          |-cabinet
                -source_system {<celname> | id=<celId>}
                -source_user <username>
                -service usermapper
                [-overwrite_destination]
                }
```

DESCRIPTION

Migrates configurations from source system to destination system.
These configurations can be Data Mover level configuration or cabinet level configurations.

OPTIONS

-mover
Migrates mover-level configuration.

-source_system {<celname | id=<celId>}
Specifies a name or an ID of the source system.

-source_user <username>
Specifies an SSH user of the source system. A hidden option "-p <passwd>" can be specified, or the command prompts for user input by default.The system will not prompt for a password if passwordless login is configured.

-source_mover <movername>
Specifies the name of a Data Mover in the source system.

-destination_mover <movername>
Specifies the name of a Data Mover in the destination system.

-service
Migrates services of different types. The -service option can be executed either by using the -all option or by specifying the following individual services: ldap, ftp, sftp, http, nis, dns, ntp, passwd, group, cepp, cava, server_param, usermapper_client, netgroup, nsswitch, hosts.

When the -all option is used, all mover configurations are migrated. If not, uses comma-separated format to include names of service types to migrate.

Note: Whenever server parameters are migrated, the destination DM will be rebooted, even if no actual parameter values have been changed. The Migrate_system_conf script does not support the migration of server parameters if a file system is mounted as RW on the destination DM.

[-overwrite_destination]

Configures services identical to those of the source system,
if the source system has services configured. Otherwise,
resets the services to the default value, which are customized
for destination VNX system version.

-cabinet
Migrates cabinet-level configuration.

-source_system {<celname | id=<celId>}
Specifies a name or an ID of the source system.

-source_user <username>
Specifies an SSH user of the source system. A hidden option
"-p <passwd>" can be specified, or the command prompts for
user input by default.

-service usermapper
Migrates the usermapper service. For cabinet level configuration,
only usermapper service is available.

[-overwrite_destination]
Configures services identical to those of the source system,
if the source system has services configured. Otherwise,
resets the services to the default value, which are customized
for destination VNX system version.

SEE ALSO
----

nas_cel, server_file, server_ldap, server_ssh, server_http, server_nis,
server_dns, server_setup, server_viruschk, server_cepp, server_param,
server_cifs, server_usermapper, and server_certificate.

EXAMPLE #1

To migrate configuration of DNS and NTP from a source Data Mover, type:

```
migrate_system_conf.pl -mover -source_system 145_16 -source_user nasadmin
-source_mover server_2 -destination_mover server_2 -service dns,ntp

Check network connnection.....................................started
Please Input password for SSH to login nasadmin@10.244.82.52:
Check network connnection.....................................succeeded
Check DNS conflict............................................started
Check DNS conflict............................................succeeded
Check NTP conflict............................................started
Check NTP conflict............................................succeeded
DNS migiration................................................started
DNS DNS migiration............................................succeeded
------------------------------------------------------------------------------

Succeed to copy:        [ntp dns]
```

EXAMPLE #2

To migrate configuration of DNS and NTP from a source Data Mover,
when SSH passwordless is configured, type:

```
migrate_system_conf.pl -mover -source_system 145_16
-source_user nasadmin -source_mover server_2
-destination_mover server_2 -service dns,ntp
Check network connnection.....................................started
Check network connnection.....................................succeeded
Check DNS conflict............................................started
Check DNS conflict............................................succeeded
Check NTP conflict............................................started
Check NTP conflict............................................succeeded
DNS migiration................................................started
DNS migiration................................................succeeded
```

```
        ------------------------------------------------------------------------------

        Succeed to copy:        [ntp dns]

        EXAMPLE #3
        ----------

        To migrate configuration of CAVA from a source Data Mover, when the
        destination overwritten option is specified, type:

        migrate_system_conf.pl -mover -source_system 145_16
        -source_user nasadmin -source_mover server_2
        -destination_mover server_2 -service cava
        -overwrite_destination
        Check network connnection.....................................started
        Check network connnection.....................................succeeded
        CAVA migration................................................started
        CAVA migration................................................succeeded
        ------------------------------------------------------------------------

        Succeed to copy:        [cava]

        [WARNING]: The virus checking rights on the local group of the source
        data mover has not been migrated to the destination.You will need to
        reconfigure  the virus checking rights on the destination using the MMC Snap-in.

        EXAMPLE #4
        ----------

        To migrate usermapper service from a source cabinet, type:
        migrate_system_conf.pl -cabinet -source_system id=1
        -source_user nasadmin -service usermapper
        Check network connection.....................................started
        Check network connection.....................................succeeded
        Check USERMAPPER conflict....................................started
        Check USERMAPPER conflict....................................succeeded
        USERMAPPER mgiration.........................................started
        Backup destination usermapper user database on DataMover [server_2] to /tmp/root/
migrate_system_conf/usermapper_backup/server_2_user_db_2013-Apr-25-09:27:23.gz
        Backup destination usermapper group database on DataMover [server_2] to /tmp/root
/migrate_system_conf/usermapper_backup/server_2_group_db_2013-Apr-25-09:27:23.gz
        Start to import usermapper [USER] database.. Done
        Start to import usermapper [GROUP] database.. Done
        USERMAPPER mgiration.....................................succeeded
        ------------------------------------------------------------------------------

        Succeed to copy:        [usermapper]
```

Roll back script

You can roll back a VDM or FS level migration using a roll back script, which runs the commands for a migration roll back. For information about the commands to run a roll back, see the Using VNX File Migration Technical Notes.

SYNOPSIS

```
migrate_utility
 -migration {<migname> | id=<miglId>}
  -rollback
 |-usermapper
   -rollback -source_system {<celname> | id=<celId>}
   -source_user <username>
```

DESCRIPTION

After the Complete process ends and before a Delete process is initiated during a VDM or FS level migration, you can execute a manual roll back of the VDM or FS level migration. To roll back a migration or a usermapper service using the roll back script, use this command syntax:

OPTIONS

-migration {<migname | id=<migId>}
Specifies a migration roll back and the name or ID of the migration to roll back.

-rollback {<celname | id=<celId>}
Runs the commands to roll back a VDM or a FS level migration.

-usermapper -rollback -source_system {<celname> | id=<celId>}
-source_user <username>
Specifies a usermapper roll back and runs the commands to roll back the usermapper service to the source system given the name or ID of the source system. Also, username on the source is required. You will be prompted subsequently for the related password.

EXAMPLE #1

The following is a sample of running the roll back script:

```
$ /nas/tools/migration_utility -migration vdmMig3001 -rollback

Query information of migration (vdmMig3001) ... succeeded
Rollback migration (vdmMig3001) ...
Check pre-conditions ...
Check migration state ...
Check migration dr solution and state ...
Check migration dr solution and state ... succeeded
Check migration state ... succeeded
Check replication status ...
- Replication (MIGVDM_vdm3001_3001) at remote: SYNCING (destination->source)
- Replication (MIGVDM_vdm3001_3003) at remote: SYNCING (destination->source)
- Replication (MIGVDM_vdm3001_3004) at remote: SYNCING (destination->source)
- Replication (MIGVDM_vdm3001_3002) at remote: SYNCING (destination->source)
- Replication (MIGVDM_vdm3001_3001) at local: SYNCING (destination->source)
- Replication (MIGVDM_vdm3001_3003) at local: SYNCING (destination->source)
- Replication (MIGVDM_vdm3001_3004) at local: SYNCING (destination->source)
- Replication (MIGVDM_vdm3001_3002) at local: SYNCING (destination->source)
Check replication status ... succeeded
Check pre-conditions ... succeeded
Set migration state(ROLLING_BACK) ...
Check migration state ...
Check migration dr solution and state ...
Check migration dr solution and state ... succeeded
Check migration state ... succeeded
Set migration state ... succeeded
Get read-only file systems ...
- 2 Source File Systems to become read-only:fs3002, fs3003
Get read-only file systems ... succeeded
Refresh replications ...
- 4 Replication(s):MIGVDM_vdm3001_3001, MIGVDM_vdm3001_3002, MIGVDM_vdm3001_3003,
MIGVDM_vdm3001_3004
```

```
Refresh replications ... succeeded
Get interfaces attached ...
- 2 Interface(s):eth32, eth33
Get interfaces attached ... succeeded
Cut over the migration ...
--------------------------------------
Cut-over start time: 2014-02-12 02:20:28
Turn down destination interfaces ...
- 2 Interface(s):eth32, eth33
Turn down destination interfaces ... succeeded
Reverse replications(FS) ...
- 3 Replication(s):MIGVDM_vdm3001_3002, MIGVDM_vdm3001_3003, MIGVDM_vdm3001_3004
Reverse replications(FS) ... succeeded
Restore FS back to Read-Only ...
- 2 Source File Systems:fs3002, fs3003
- Retry 1 ...
Restore FS back to Read-Only ... succeeded
Reverse replications(VDM) ...
- 1 Replication(s):MIGVDM_vdm3001_3001
Reverse replications(VDM) ... succeeded
Turn up source interfaces ...
- 2 Interface(s):eth32, eth33
Turn up source interfaces ... succeeded
--------------------------------------
Cut over the migration ... succeeded
- Start time: 2014-02-12 02:20:28
- End time: 2014-02-12 02:21:20
- Duration: 52 secs
Set migration state(READY_TO_COMPLETE) ...
Check migration state ...
Check migration dr solution and state ...
Check migration dr solution and state ... succeeded
Check migration state ... succeeded
Set migration state ... succeeded
Check post-conditions ...
Check replication status ...
- Replication (MIGVDM_vdm3001_3001) at remote: SYNCING (source->destination)
- Replication (MIGVDM_vdm3001_3003) at remote: SYNCING (source->destination)
- Replication (MIGVDM_vdm3001_3004) at remote: SYNCING (source->destination)
- Replication (MIGVDM_vdm3001_3002) at remote: SYNCING (source->destination)
- Replication (MIGVDM_vdm3001_3001) at local: SYNCING (source->destination)
- Replication (MIGVDM_vdm3001_3003) at local: SYNCING (source->destination)
- Replication (MIGVDM_vdm3001_3004) at local: SYNCING (source->destination)
- Replication (MIGVDM_vdm3001_3002) at local: SYNCING (source->destination)
Check replication status ... succeeded
Check post-conditions ... succeeded
Rollback migration ... succeeded
OK
```

THE GET AND SET COMMANDS

This chapter lists the eNAS Command Set provided for managing, configuring, and monitoring of File Movers. The commands are network protocol applications, prefixed with get or set, and appear alphabetically. The command line syntax (Synopsis), a description of the options, and an example of usage are provided for each command.

get_attributes
Reads the attributes of the specified file on the primary storage and verifies the eNAS FileMover API configuration.

SYNOPSIS
get_attributes
[-u <username>] [-p <password>][-d][-h <handle>] [-w] [-a] [-S <CA_cert_file>]
[-V <http_version>] <primary_server> <primary_file>

DESCRIPTION
get_attributes is executed on the Control Station after starting the eNAS FileMover API service to return CIFS, NFS, and all vendor extended attributes in a text format. get_attributes uses the program 'digest' to actually transport the command over the network.
Note: get_attributes is not part of Control Station CLI framework.

PREREQUISITES
Starts the HTTP server for eNAS FileMover by using server_http, and creates user account for the specified eNAS FileMover using server_user unless user authentication is set to none.

OPTIONS
-u <username>
Specifies the account name for digest or basic access authentication. If no authentication is used, just provides dhsm_user.
-p <password>
Specifies the password for the digest or the basic access authentication account. If no authentication is used, just provides dhsm_user.
-d
Displays offline attributes of a deduped file.
-h <handle>
Gets the status of files by handle instead of <primary_file>.
-w
Gets the status of Write Once Read Many (WORM) in file attribute data.
-a
Includes the Access Control List (ACL) information along with the standard file attributes.
-S <CA_cert_file>
Uses SSL to perform server certificate verification with the Certificate Authority (CA) certificated in <CA_cert_file>.
The file name that has the certificate of the CA used to create the Data Movers certificate for SSL authentication. This is only needed if SSL is turned on the FileMover API service.
-V <http_version>
Uses the specified version, HTTP/1.0 or HTTP/1.1. Default is HTTP/1.0.
<primary_server>
Specifies the IP address of the Data Mover, which is hosting the primary file. Must use IP address, not the NW host name.
<primary_file>
Specifies the full path to the file on the Data Mover

EXAMPLE #1
To verify offline status using eNAS FileMover API, type:
$ /nas/tools/dhsm/get_attributes 10.5.8.111 /fs1/pax.tar

```
<?dhsm?>
<DHSM_GET_ATTRS PATH="nfs:/pfs/file1"/>
HTTP/1.0 200 OK
Date: Sat, 20 Aug 2005 16:36:31 GMT
Content-type: text/xml; charset=UTF-8
<?xml version="1.0" encoding="UTF-8"?>
<?dhsm?>
<RESPONSE>
```

```
<OFFLINE_ATTRS
OFFLINE_PATH="http://dhsm-w2k/dir1/File4"
OFFLINE_MTIME="1124555122"
OFFLINE_ETAG="&amp;quot;08c0aea3a5c51:d6a&amp;quot;"
/>
<STANDARD_ATTRS
HANDLE="4294969278-19-1124494767"
ONLINE_CTIME="1124555598000000"
\205 same as before.
/>
<RESULT SUCCESS="True" VALUE=""/>
</RESPONSE>
```

Where:
10.5.8.111 --> Indicates the IP address of the Data Mover which hosts the primary file.
/fs1/pax.tar --> Indicates the path to the primary file.

EXAMPLE #2
To verify offline status of a deduped file with -d option, type:

```
$ /nas/tools/dhsm/get_attributes -d 128.221.252.2

New Command length is 65
spawn telnet 128.221.252.2 5080
Trying 128.221.252.2...
Connected to server_2 (128.221.252.2).
Escape character is '^]'.
POST /dhsm HTTP/1.0
Content-type: text/xml
Content-length: 65
<?dhsm?>
<DHSM_GET_ATTRS PATH="/afs/3-1.log" DEDUPE="True"/>
HTTP/1.1 200 OK
Connection: Close
Content-Type: text/xml; charset=UTF-8
Server: EMC File Mover service
Date: Mon, 28 Jan 2013 02:00:54 GMT
<?xml version="1.0" encoding="UTF-8"?>
<?dhsm?>
<RESPONSE>
<OFFLINE_ATTRS
OFFLINE_PATH="dart://rde/AQAAAA/cd/e1/Adl3gRkNrbbpS7+hdRXX84qYoWvUAAAAAAAAAAA
AAAAAAwAAAAAAACtrGw;100143ab"
OPTIMIZED_HANDLE="29-1359022552"
ORIGINAL_BYTES_USED="2859008"
DEDUPE_LINKCOUNT="1"
DEDUPE_FSIZE="390821"
OFFLINE_BLOCK_COUNT="348"
BLOCKS_NEEDED_FOR_REINGEST="59"
PE_ID="DHSM API"
/>
<STANDARD_ATTRS
HANDLE="4294968227-27-1359022531"
ONLINE_CTIME="1359022551000451"
UID="0"
GID="0"
ATIME="1359022531"
MTIME="1359022531"
CTIME="1359022551"
CREATE_TIME="1359022531"
DOS_ATTRS="2080"
PARENT_INODE="2"
FSIZE="2845467"
BLOCK_SIZE="8192"
BLOCKS="784"
BYTES_USED="401408"
INODE="27"
DEVICE="931"
NLINK="1"
MODE="0644"
```

```
FILE_TYPE="File"
/>
<RESULT SUCCESS="True" VALUE=""/>
</RESPONSE>


EXAMPLE #3
To verify offline status of a deduped file without -d option, type:
$ /nas/tools/dhsm/get_attributes 128.221.252.2
/afs/3-1.log


New Command length is 52
spawn telnet 128.221.252.2 5080
Trying 128.221.252.2...
Connected to server_2 (128.221.252.2).
Escape character is '^]'.
POST /dhsm HTTP/1.0
Content-type: text/xml
Content-length: 52
<?dhsm?>
<DHSM_GET_ATTRS PATH="/afs/3-1.log" />
HTTP/1.1 200 OK
Connection: Close
Content-Type: text/xml; charset=UTF-8
Server: EMC File Mover service
Date: Mon, 28 Jan 2013 02:01:35 GMT
<?xml version="1.0" encoding="UTF-8"?>
<?dhsm?>
<RESPONSE>
<STANDARD_ATTRS
HANDLE="4294968227-27-1359022531"
ONLINE_CTIME="1359022551000451"
UID="0"
GID="0"
ATIME="1359022531"
MTIME="1359022531"
CTIME="1359022551"
CREATE_TIME="1359022531"
DOS_ATTRS="2080"
PARENT_INODE="2"
FSIZE="2845467"
BLOCK_SIZE="8192"
BLOCKS="784"
BYTES_USED="401408"
INODE="27"
DEVICE="931"
NLINK="1"
MODE="0644"
FILE_TYPE="File"
/>
<RESULT SUCCESS="True" VALUE=""/>
</RESPONSE>


EXAMPLE #4
To read the status of a given primary storage, type:
$ /nas/tools/dhsm/get_attributes -u dhsm_user -p
bad_password 10.5.8.111 /


Sending 105 bytes ***
POST /dhsm HTTP/1.0
Content-type: text/xml
Content-length: 38
<?dhsm?>
<DHSM_GET_ATTRS PATH="/" />
HTTP/1.1 401 Unauthorised
Connection: Close
Content-Length: 0
WWW-Authenticate: Basic realm="DHSM_Authorization"
```

```
Server: EMC File Mover service
Date: Mon, 01 Oct 2007 17:34:09 GMT
basic challenge
open_connection: server IP 10.5.8.111
open_connection: streaming socket open
open_connection: bind successful
open_connection: connect successful
open_connection: local port = 55315, local addr = 10.5.8.111
Sending 160 bytes ***
POST /dhsm HTTP/1.0
Authorization: Basic ZGhzbV91c2VyOmJhZF9wYXNzd29yZA==
Content-type: text/xml
Content-length: 38
<?dhsm?>
<DHSM_GET_ATTRS PATH="/" />
***
SENT 160 HEADER
HTTP/1.1 401 Unauthorised
Connection: Close
Content-Length: 0
WWW-Authenticate: Basic realm="DHSM_Authorization"
Server: EMC File Mover service
Date: Mon, 01 Oct 2007 17:34:09 GMT
Did not get response - either the username and password were not presented for
responding to a server challenge OR the response to the challenge was challenged
again
EXAMPLE #1 provides a description of command output.


set_attributes
Changes a file on primary storage into a Stub File or a WORM file.

SYNOPSIS
set_attributes
[-m <migration_method[default|full|partial|read_pass_through]>]
[-u <username>][-p <password>] [-v <offline_mtime>] [-i <info>] [-d <pe_id>]
[-s <file_size>] [-c <online_ctime>][-e <entity_tag>]
[-w <worm_expiration_time>] [-S <CA_cert_file>] [-V <http_version>]
<primary_server> <primary_file>

DESCRIPTION
set_attributes uses the program \223digest\224 to actually transport the command
over the network to set EMC specific attributes, which are not available in
CIFS or NFS.
Note: set_attributes is not part of Control Station CLI framework.


PREREQUISITES
Before running the command, firstly enables eNAS FileMover operations on
a file system by using fs_dhsm, starts the HTTP server for eNAS FileMover
by using pserver_http, and creates user account for the specified eNAS
FileMover using rserver_userunless user authentication is set to none.


OPTIONS
-m <migration_method>
Chooses one of the methods Data Mover will use when a client accesses data
on a stub file.
The methods include the following: Full \226 on the first access to the stub file,
the Data Mover will migrate all of the file\222s data back from the secondary
store. Partial \226 On every access to the stub file, the Data Mover will migrate
128k of data from the secondary store. Read_pass through \226 This is the
default value. The data will be read from the secondary storage device and
given to the client and no data blocks will be migrated back to the Data
Mover.
-u <username>
Specifies the account name for digest or basic access authentication. If no
authentication is used, just provides dhsm_user.
-p <password>
Specifies the password for the digest or the basic access authentication
account. If no authentication is used, just provides dhsm_user.
```

-v <offline_mtime>
Sets <offline_mtime> verifier with a integer value. By default the script will search the mount command to find the stub file\222s verifier value.
-i <info>
Sets offline attribute INFO.
-d <pe_id>
Sets offline attribute PE_ID.
-s <file_size>
Specifies the logical size of the file in bytes.
-c <online_ctime>
Sets <online_ctime> verifier with integer values. By default the script will call get_attributes to find the stub file\222s verifier value.
-e <entity_tag>
Sets offline attribute ETAG. This should be used instead of the -v option for HTTP connections to a secondary server.
-w <worm_expiration_time>
Sets the stub file Write Once Read Memory (WORM) expiration time. <worm_expiration_time> is an integer in seconds since Jan 1, 1970 GMT. A <worm_expiration_time> value of 0 indicates infinite expiration time.
-S <CA_cert_file>
Uses SSL to perform server certificate verification with the Certificate Authority (CA) certificated in <CA_cert_file>.
The file name that has the certificate of the CA used to create the Data Movers\222 certificate for SSL authentication. This is only needed if SSL is turned on the FileMover API service.
-V <http_version>
Uses the specified version, HTTP/1.0 or HTTP/1.1. Default is HTTP/1.0.
<primary_server>
Specifies the IP address of the Data Mover, which is hosting the primary file. Must use IP address, not the NW host name.
<primary_file>
Specifies the full path to the file on the Data Mover.

SEE ALSO
Using VNX FileMover and server_http, server_user, and server_certificate.

EXAMPLE #1
To create a stub file on the primary storage, type:
$ /nas/tools/dhsm/set_attributes -v 1191008770 10.5.8.111 /fs1/pax.tar nfs://io2/fs1ata/pax.tar

```
open_connection: server IP 10.5.8.111
open_connection: streaming socket open
open_connection: bind successful
open_connection: connect successful
open_connection: local port = 55315, local addr = 10.5.8.111
Sending 260 bytes ***
POST /dhsm HTTP/1.0
Content-type: text/xml
Content-length: 192
<?dhsm?>
<DHSM_SET_OFFLINE_ATTRS
ONLINE_CTIME="1191008731000171"
HANDLE="4294967320-16-1191008731"
OFFLINE_PATH="nfs://io2/fs1ata/pax.tar"
OFFLINE_MTIME="1191008770"
/>
***
SENT 260 HEADER
HTTP/1.1 200 OK
Connection: Close
Content-Type: text/xml; charset=UTF-8
Server: EMC File Mover service
Date: Fri, 28 Sep 2007 20:12:03 GMT
<?xml version="1.0" encoding="UTF-8"?>
<?dhsm?>
<RESPONSE>
<RESULT SUCCESS="True" VALUE=""/>
</RESPONSE>
```

Note: Make sure HTTP service for eNAS FileMover is started by using server_http.

EXAMPLE #2
To create a stub file on a secondary server for HTTP connections, type:

```
$ /nas/tools/dhsm/set_attributes -u dhsm_user -p
dhsm_user -e f5040c-14a000-c986cd80 -V HTTP/1.1
10.5.8.111 /fs1/pax.tar http://linc57/pax.tar

FILE == /fs1/pax.tar
open_connection: server IP 10.5.8.111
open_connection: streaming socket open
open_connection: bind successful
open_connection: connect successful
open_connection: local port = 55315, local addr = 10.5.8.111
Sending 275 bytes ***
POST /dhsm HTTP/1.1
Host:10.5.8.111
Content-type: text/xml
Content-length: 190
<?dhsm?>
<DHSM_SET_OFFLINE_ATTRS
ONLINE_CTIME="1191010323000002"
HANDLE="4294967320-16-1191008731"
OFFLINE_PATH="http://linc57/pax.tar"
ETAG="f5040c-14a000-c986cd80"
/>
***
SENT 275 HEADER
HTTP/1.1 200 OK
Connection: Keep-Alive
Content-Type: text/xml; charset=UTF-8
Keep-Alive: max=299, timeout=60
Server: EMC File Mover service
Date: Tue, 02 Oct 2007 13:37:24 GMT
Transfer-Encoding: Chunked
<?xml version="1.0" encoding="UTF-8"?>
<?dhsm?>
<RESPONSE>
<RESULT SUCCESS="True" VALUE=""/>
</RESPONSE>
```

Scripting Guidelines
It is recommended that users follow the guidelines outlined below when invoking eNAS commands withing scripts.
The following The following recomendations should

Scheduling eNAS Database Backups:
The eNAS backs up the NAS database that stores specific configuration information required for each
Data Mover every hour, at one minute after the hour. During part of the backup, the database is
locked, and some commands that rely on the database might not have access. It is recommended that
command scripts avoid starting at one minute after the hour. Note that scripts with complex
commands that run for an extended period may overlap the backup period.
The duration of the backup may vary. Use the following Linux command to check the state of the
backup process prior to executing scripts: ps -ef | grep nasdb_backup. If a lock condition occurs,
wait a few minutes and retry.

Command sequencing:
Some commands must lock the database in order to execute. If multiple user-entered commands or
scripts are active at the same time, some of these commands may lock the database and prevent other
commands from executing. To avoid this, you should arrange commands whenever possible.

Sleep statements
Some processes within a script can take time to complete. Use proper timing
and adequate sleep statements to prevent timing-related issues.

Pipe and grep
Piping script outputs through grep is a helpful tool to check the status
of the script. Use periodic checks to grep for file or database locked messages, timeouts, resource
unavailable warnings, and other failure or success messages, and use this information to check
status, pause the script, or halt it.

Return code check
All commands return a UNIX-style return code (for example: 0 for success or 1
for failure) or a text-based status code (for example, done) which can be used to help determine if
the command completed or if there was an error or a conflict with the NAS database backup, or other
commands being run. If a lock condition occurs, wait a few minutes and retry. If you create and run
scripts, be sure to incorporate return code checks and verify for proper return codes from
individual operations.

If you interrupt a command by issuing Ctrl-C, expect the following messages or traces at the console:
\225 nas_cmd: system execution failed.
\225 nas_cmd: PANIC: caught signal #11 (Segmentation fault) -- Giving up.

NOTE:
Use eNAS CLI to add IPv6 addresses to the NFS export host list.
Enclose the IPv6 address in { } or square brackets in the CLI.
Unisphere displays the IPv6 addresses added to the NFS export list via the CLI as read-only fields.

Scripting examples
The RECOVERABLE variable contains the following errors to retry on:
-> Unable to acquire lock
-> Resource temporarily unavailable
-> Unable to connect to host
-> Socket: All ports in use
-> Database resource vanished

```
-> Connection timed out
-> NAS_DB locked object is stale

An example of what the RECOVERABLE variable looks like is as follows:
RECOVERABLE="unable to acquire lock|Resource
temporarily unavailable|unable to connect to
host|socket: All ports in use|database resource
vanished|Connection timed out|NAS_DB locked object is
stale".
The res variable contains the command output:
#!/bin/sh
##########################################################
# File: nas_cmdrcvr.sh
# Created by: NAS Engineering
# Date: Thursday, May 25, 2006
# Version: 1.0
# Notes:
# 1) script will retry commands for specified period of
time
#2) script will log messages to file only if there's
available disk space
##########################################################
NAS_DB=/nas
export NAS_DB
PATH=$PATH:$NAS_DB/bin:$NAS_DB/sbin:/usr/sbin:.
export PATH
RETRIES=60
SLEEPTIME=30
LOGDIR="/home/nasadmin"
LOGFILE="$0.log"
LOGLAST="$0.last"
DISKFULL=98
RECOVERABLE="Resource temporarily unavailable|\
unable to acquire lock|\
unable to connect to host|\
socket: All ports in use|\
database resource vanished|\
Connection timed out|\
NAS_DB locked object is stale"
#
# function to log messages to a file
#
nas_log()
{
DISKCHK=`df -k $LOGDIR|awk 'NR>1{print $5}'|sed 's/\%//'`
# if there's enough free disk space, append to log
if [ $DISKCHK -lt $DISKFULL ]; then
TDSTAMP=`date +%Y-%m-%d' '%T`
echo $TDSTAMP: $LOGMSG >> $LOGDIR/$LOGFILE
fi
# regardless of available space, always write last error
echo $TDSTAMP: $LOGMSG > $LOGDIR/$LOGLAST
}
#
# function to execute (and potentially retry) commands
#
nas_cmd()
{
# initialize variable(s)
retry_count=0
# loop until either successful of retry count exceeded
while [ $retry_count -le $RETRIES ]; do
# execute command and gather response
RES=`$CMD 2>&1`
# check if response means command is recoverable
if [ `echo "$RES"|egrep -c "$RECOVERABLE"` -ne 0 ]; then
# check retry count
if [ $retry_count -ne $RETRIES ]; then
# retry count has not been exceeded
LOGMSG="Command ($CMD) failed with ($RES)...retrying in
$SLEEPTIME s"
```

```
nas_log
sleep $SLEEPTIME
else
# retry count has been exceeded
LOGMSG="Command ($CMD) failed with ($RES)...exiting
(retry count of $RETRIES exceeded)"
nas_log
exit 1
fi
else
# command was either successful or failed for an unknown
reason
LOGMSG="Command ($CMD) successful with ($RES)"
nas_log
retry_count=$RETRIES
exit 0
fi
#increment counter for retries
retry_count=`expr $retry_count + 1`
done
}
#
# main
#
CMD="nas_volume -d mtv1"
nas_cmd
```

Using the NAS database and query facility
EMC has partially changed the layout or format of eNAS internal databases. This change can impact the use of awk or grep utilities when used in scripts that assume specific positions of fields in databases.
To enable searching of the NAS database, eNAS has developed a new query subsystem that appears as a hidden option on some of the nas_commands. This query subsystem enables you to specify the information you are interested in, allows you to format the output, and is independent of the database format.

CAUTION
Do not use grep and awk to scan the database files. Database positions may change and substrings may return false matches for database objects.

Following is an example of a query to view unused disks:

```
nas_disk -query:inuse==n -format:'%s\n' -Fields:Id
```

To filter root disks, refer to the List all non-root disks that are not in use.
Examples
Use the following commands to view the tags (fields) that you can query:

```
nas_disk -query:tags
nas_fs -query:tags
nas_volume -query:tags
nas_slice -query:tags
```

Following contains a list of examples to help you get started. Note that these commands can be run on the Control Station CLI, so the hardcoded values can be replaced with shell script variables.

Task and query examples

Task	Example
Query the ID of a named file system	nas_fs -query:Name==RLL_fs10 -format:'%s\n' -Fields:Id
Query the ID of a named file system without the new line	nas_fs -query:Name==RLL_fs10 -format:'%s' -Fields:Id
Query the name of a file system ID that corresponds to a particular ID	nas_fs -query:id==20 -format:'%s\n' -Fields:Name

List of all server IDs	nas_server -query:* -format:'%s\n' -Fields:Id
List of all server names	nas_server -query:* -format:'%s\n' -Fields:Name
List of all the checkpoint file systems	nas_fs -query:type==ckpt -fields:name -format:"%s\n"
List type of file system with ID 20	nas_fs -query:id==20 -format:'%s\n' -Fields:Type
List the file systems that are in use	nas_fs -query:inuse==y -format:'%s\n' -Fields:Name or nas_fs -query:inuse==y -format:'%s\n' -Fields:Id
Identify file system of which file system ID 28 is a backup	nas_fs -query:id==28 -format:'%s' -Fields:BackupOf
List the name of the server with ID 2	nas_server -query:id==2 -format:'%s\n' -fields:name`"
View which volume file system is built on	nas_fs -query:Name==my_fs -format:%d -fields:VolumeID
View the block count of meta volume	nas_volume -query:Name==my_meta3 -format:%d -fields:Blocks
View the block size of meta volume	nas_volume -query:Name==JAH_meta3 -format:%d -fields:BlockSize
Find which server IDs use fs123	nas_fs -query:name==fs123 -format:'%s\n' -fields:ServersNumeric
List all non-root disks that are not in use	nas_disk -query:inuse==n:IsRoot==False -format:"%s\n" -fields:name
List unused volumes that contain <93>dc<94> in the volume name	nas_volume -query:inuse==n:IsRoot== False:name=dc -format:"%s\n" -fields:name
List all available disks on a particular storage device (symm_id is a script/env variable)	nas_disk -query:inuse==n:SymmID==$symm_ id:IsRoot==False -format:"%s\n" -fields:name

Query operators
Use the operators in the table below when building your queries:

Operator	Definition
=	Having the pattern (sub-string match)
==	Exact string match
=-	Integer Minimum (not less than)
=+	Integer Maximum (not more than)
=*	Any
=^	NOT having the pattern
=^=	NOT an exact match
=^-	NOT Integer Minimum (is less than)
=^+	NOT Integer Maximum (is more than)
=^*	NOT Any (None)